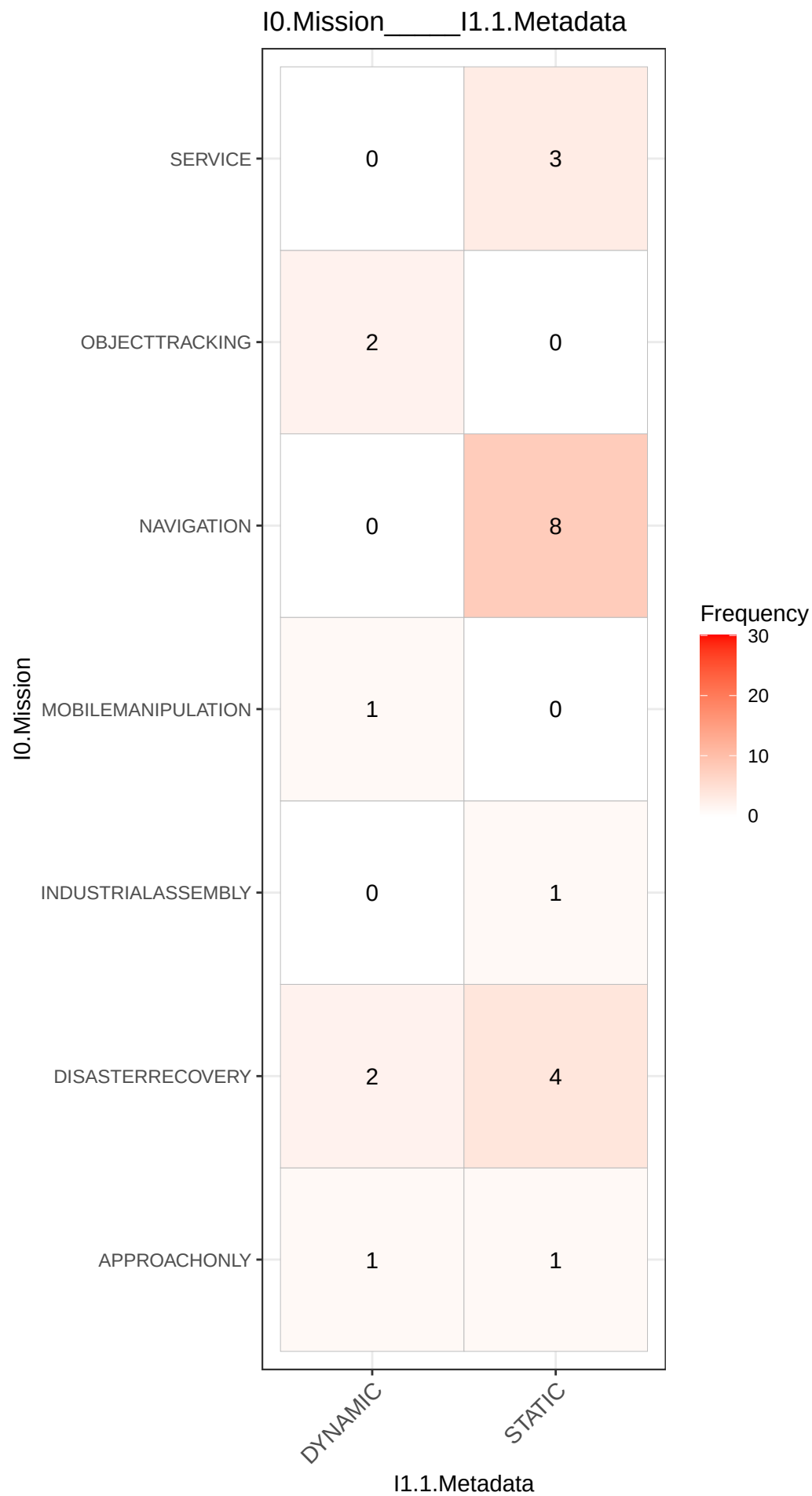
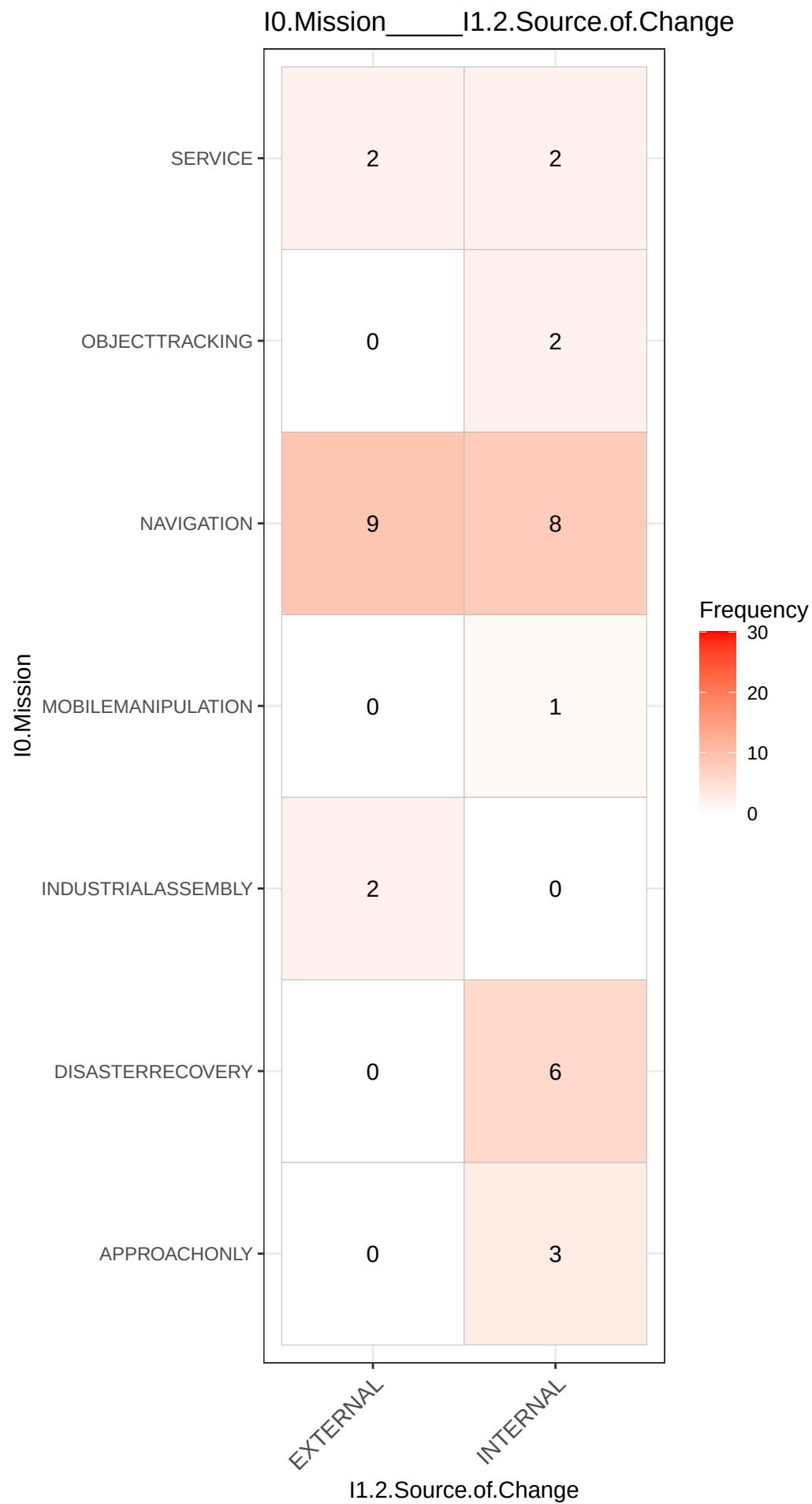
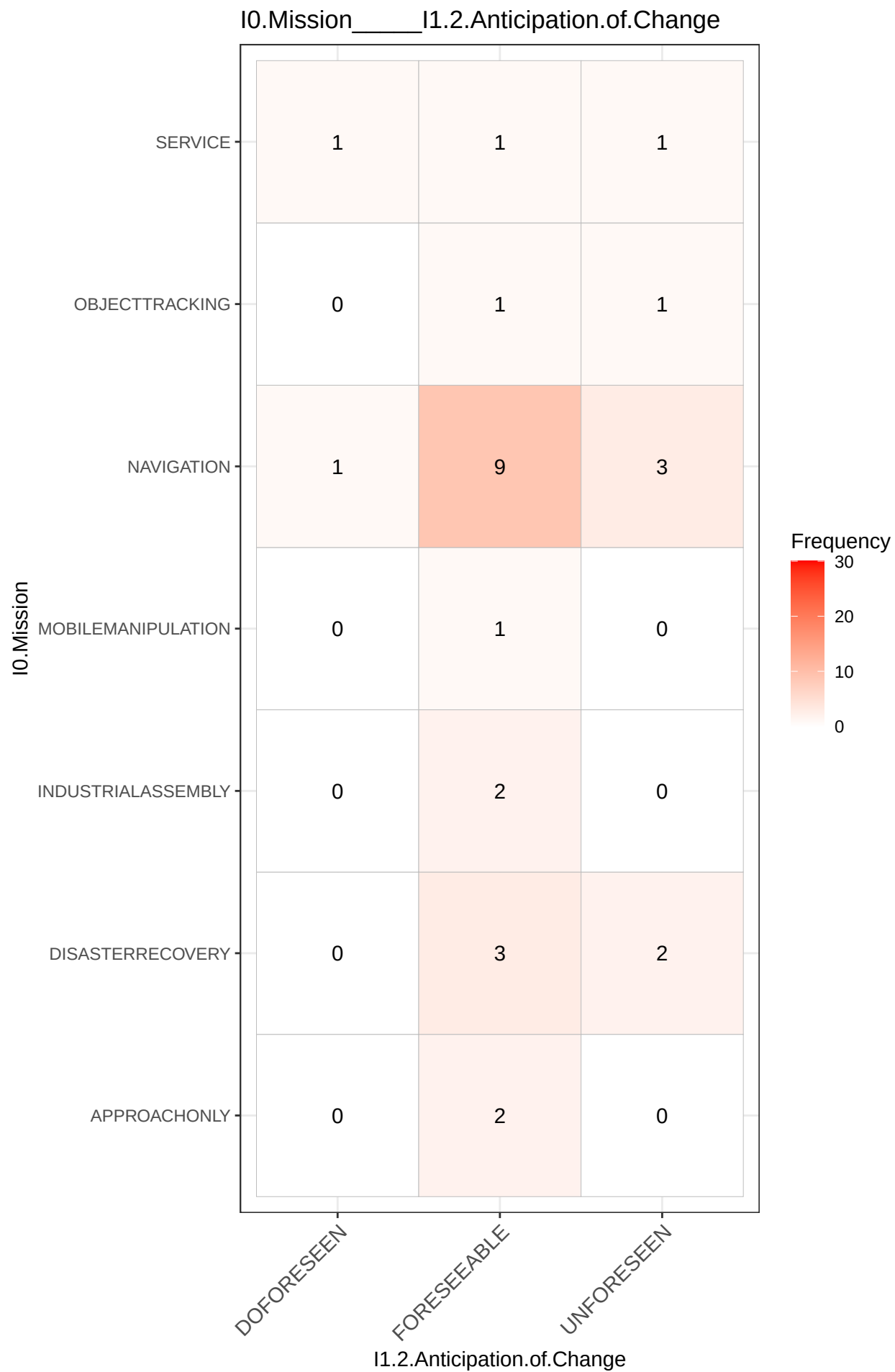
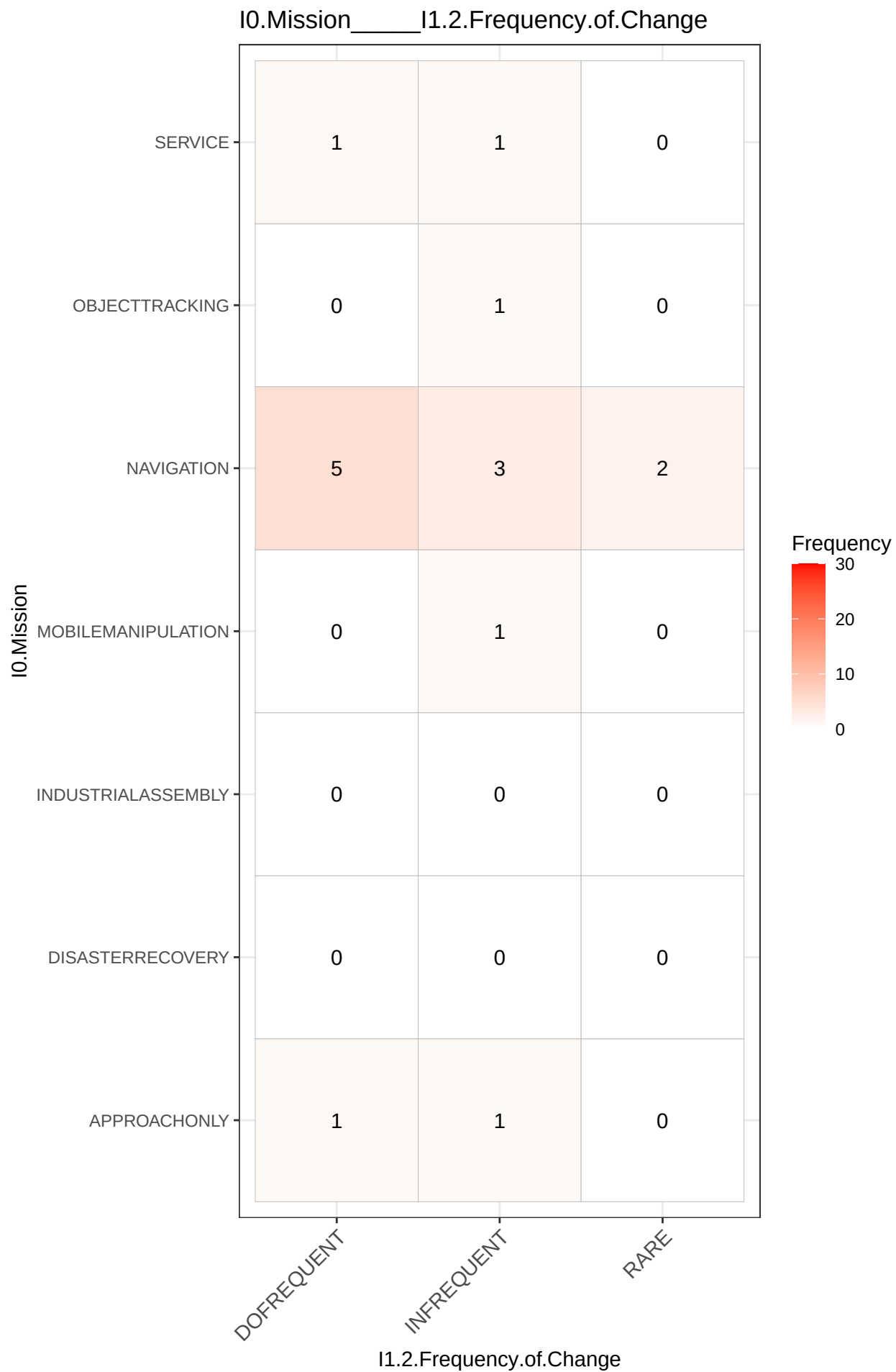


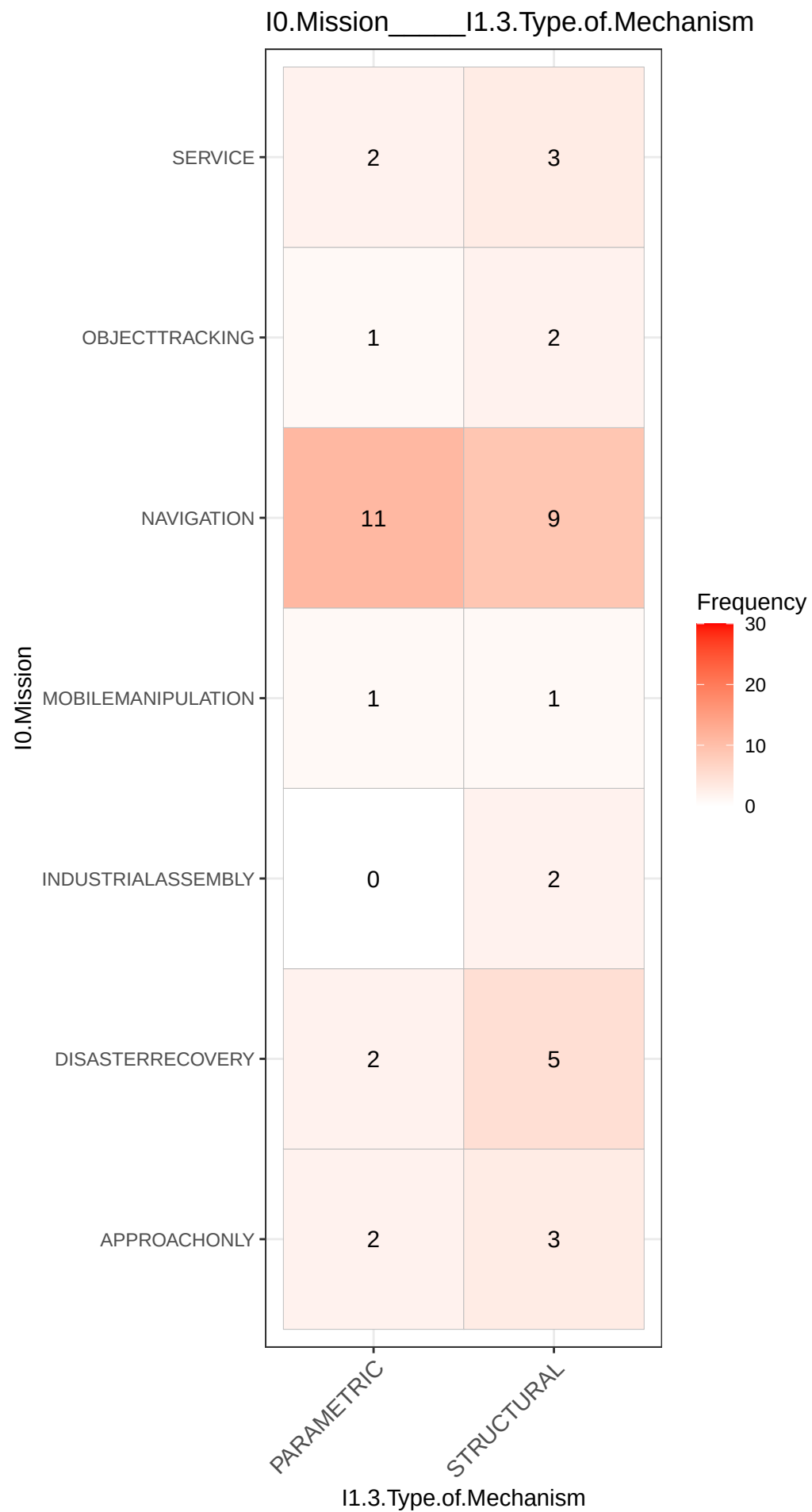


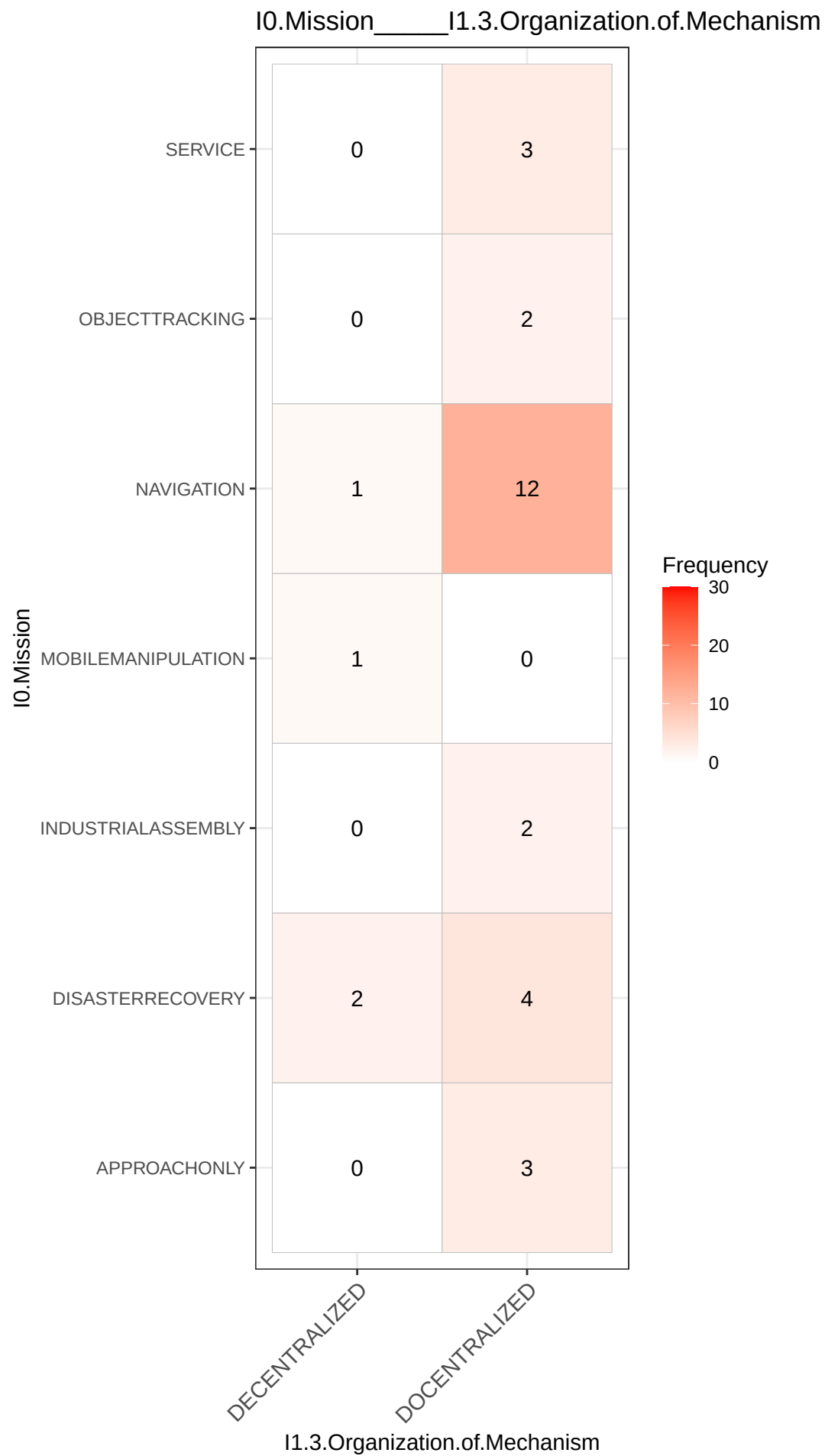


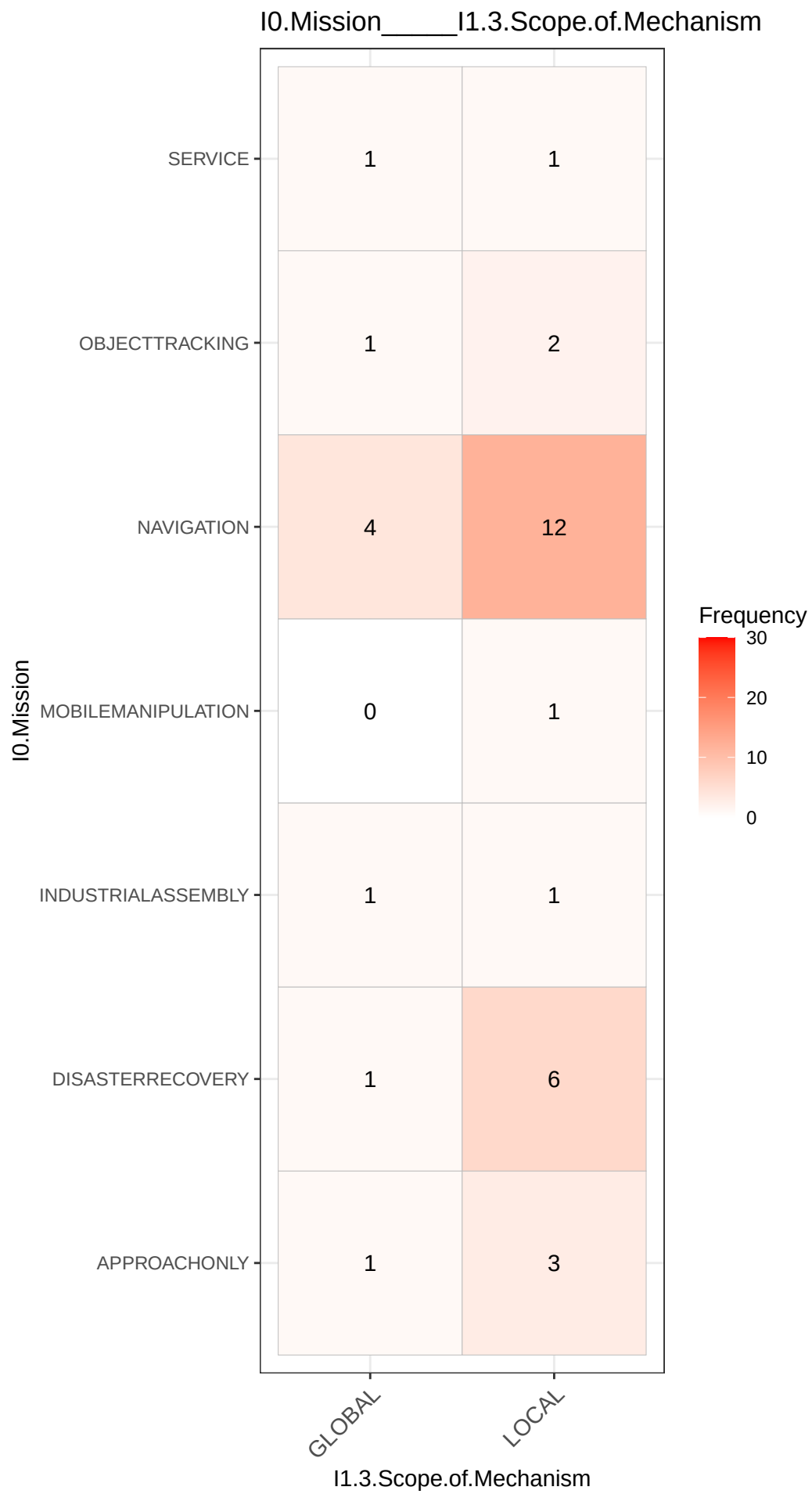


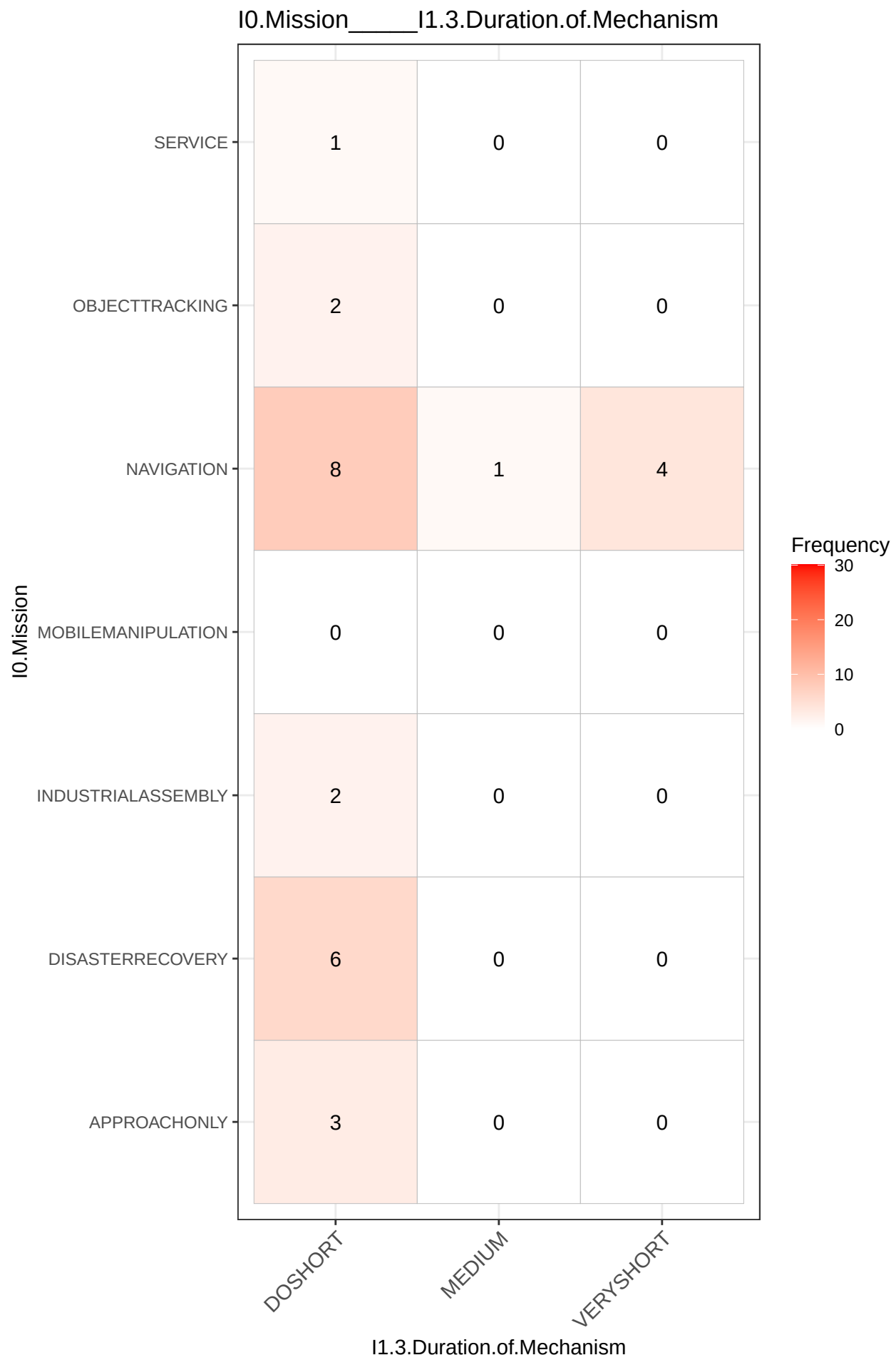


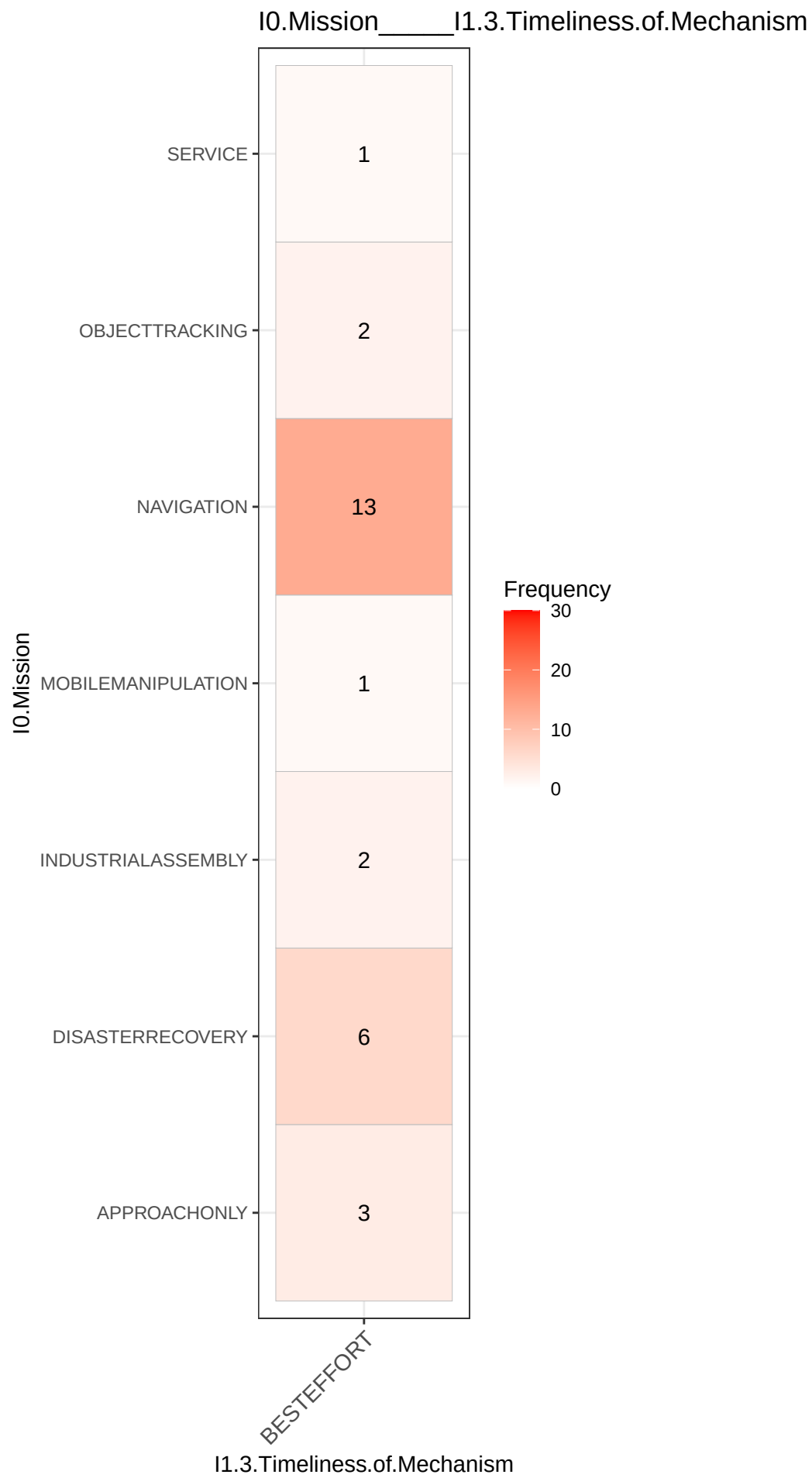


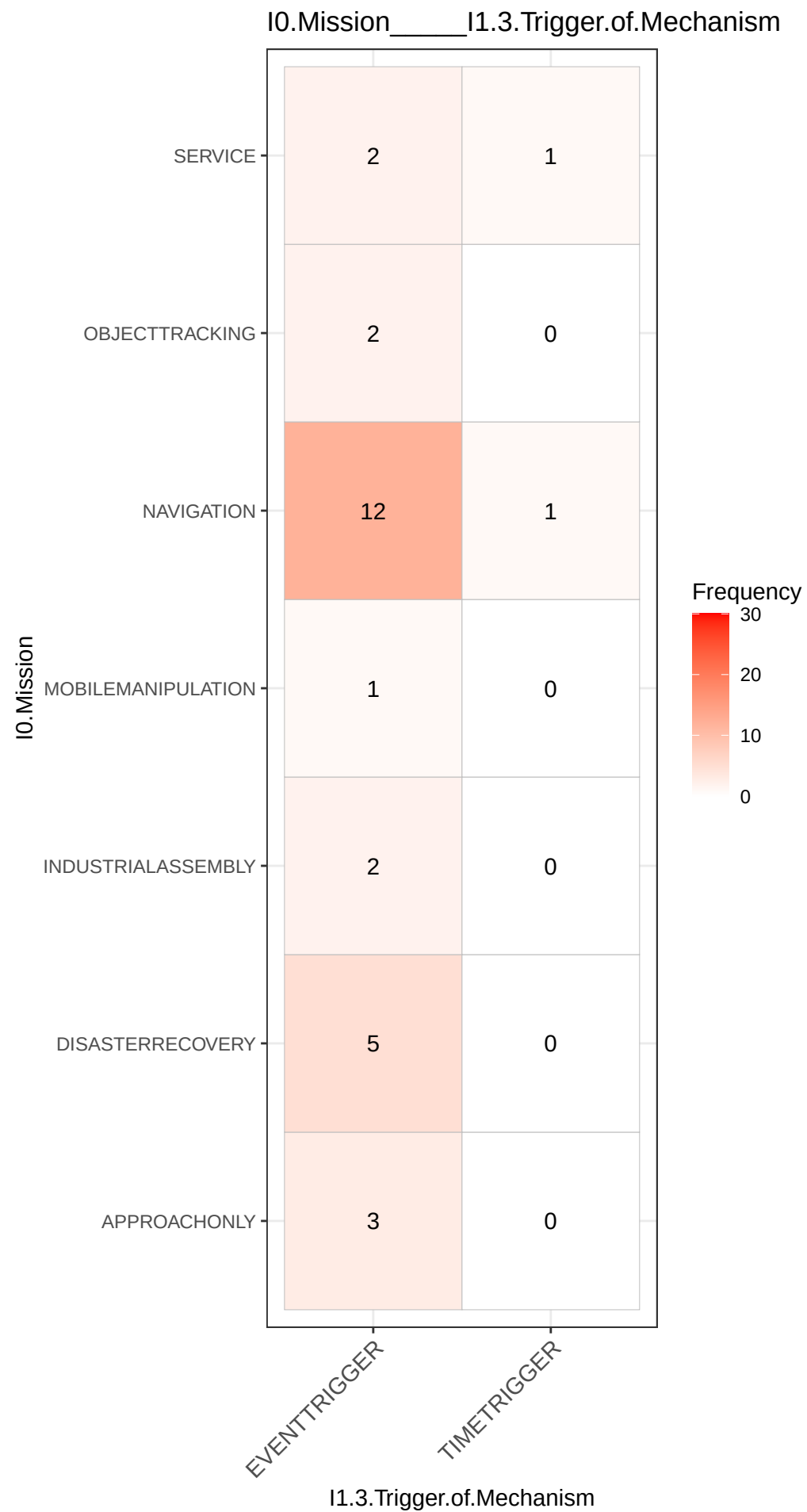


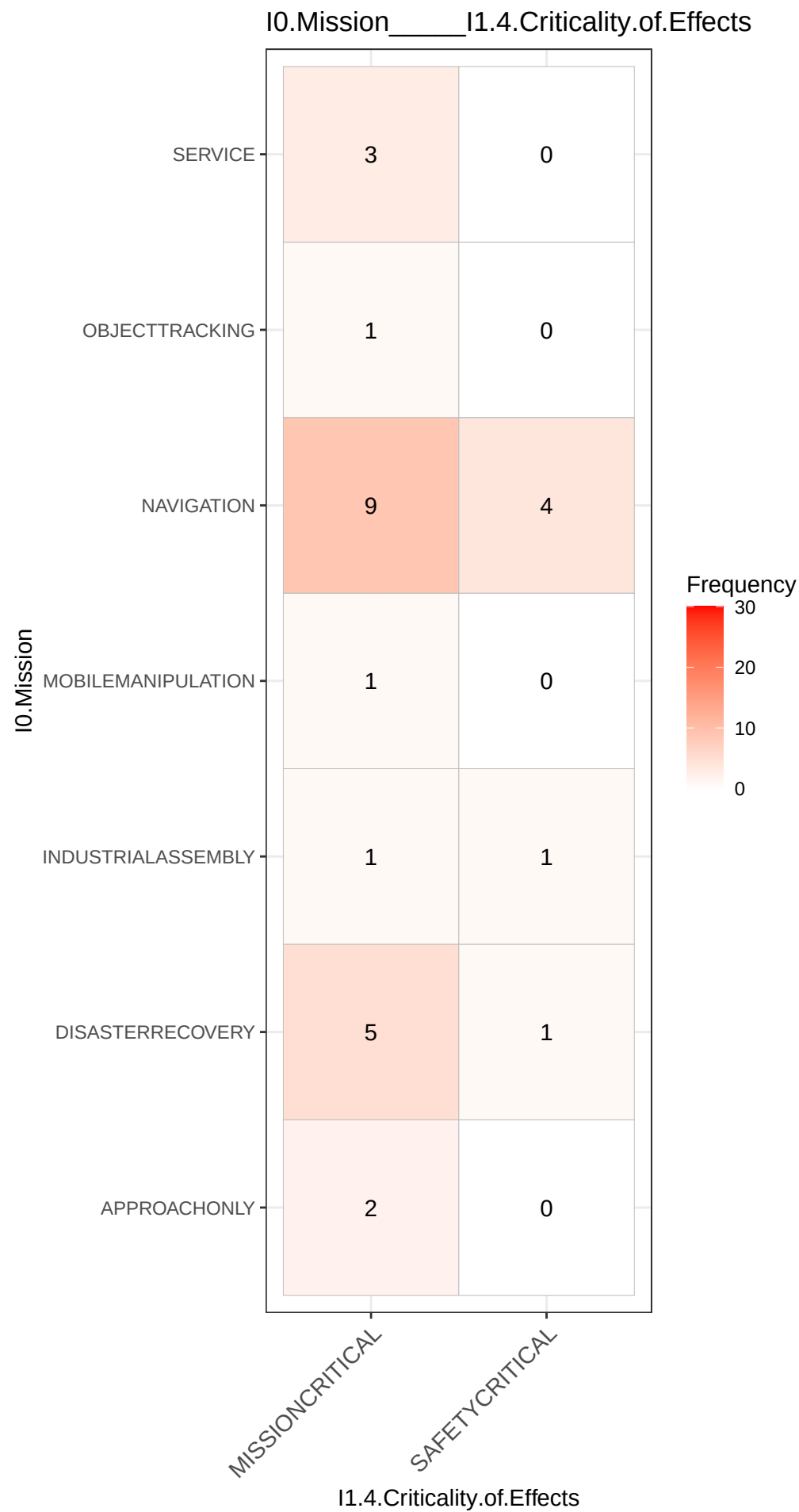


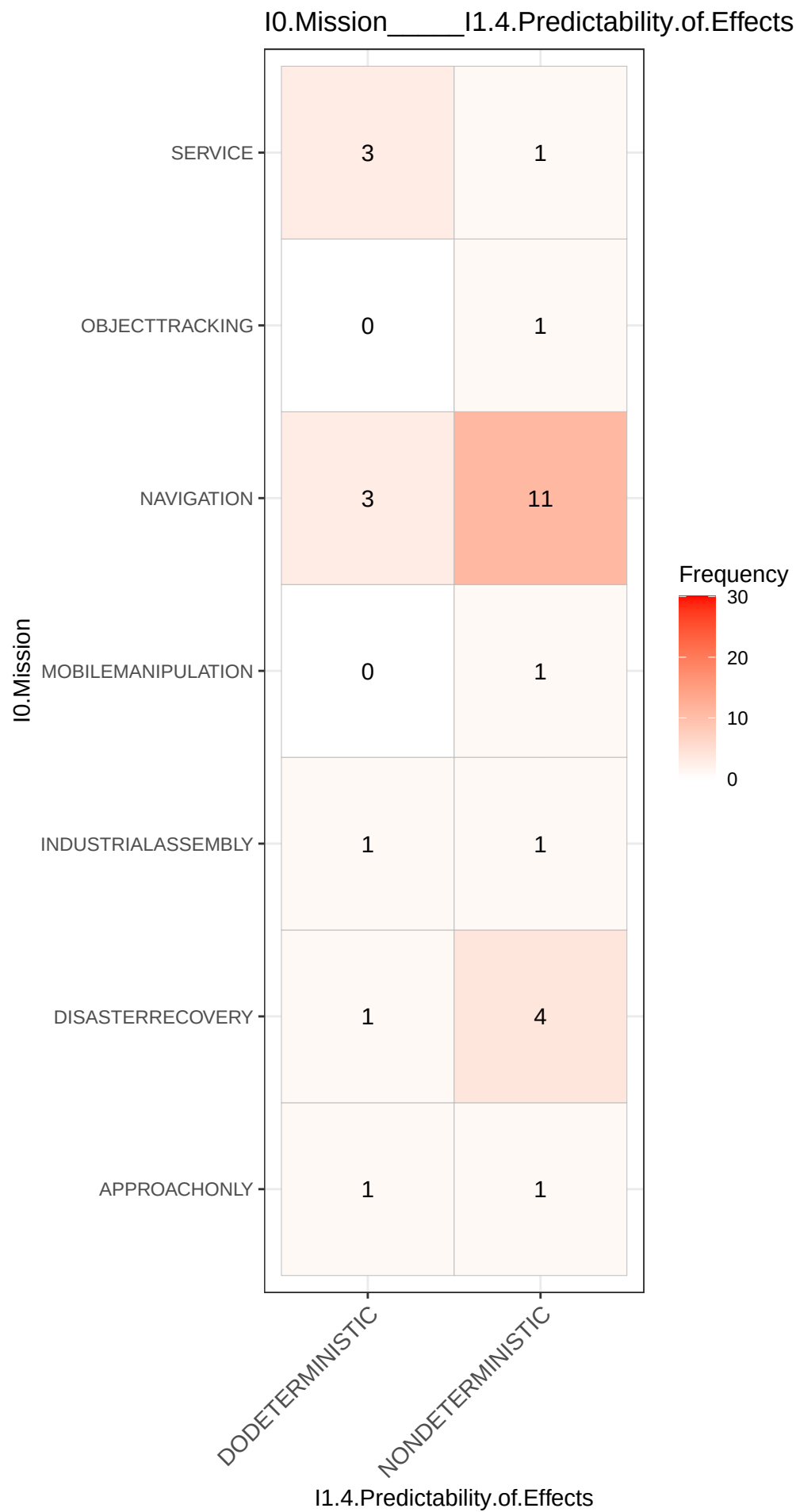


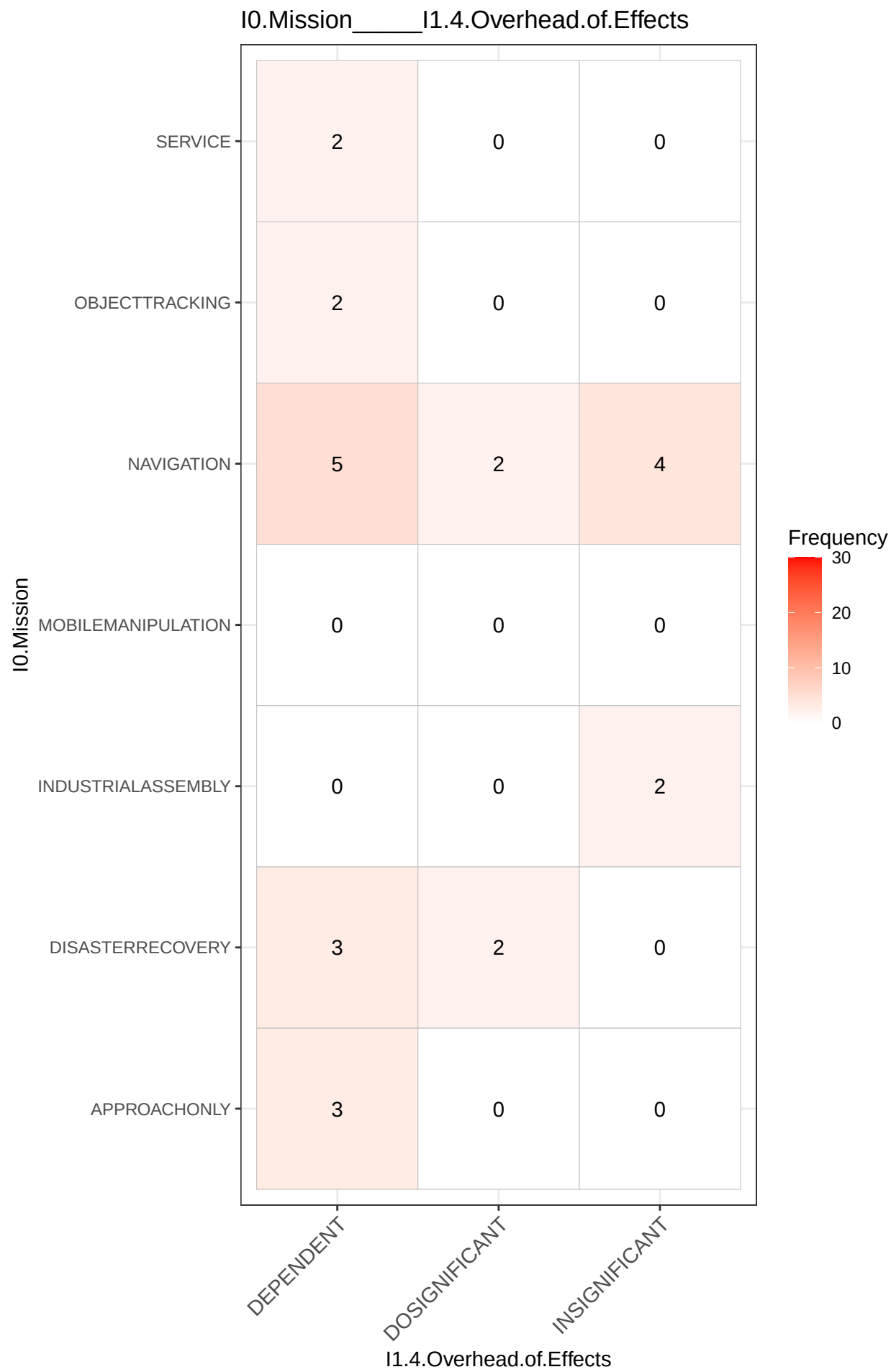


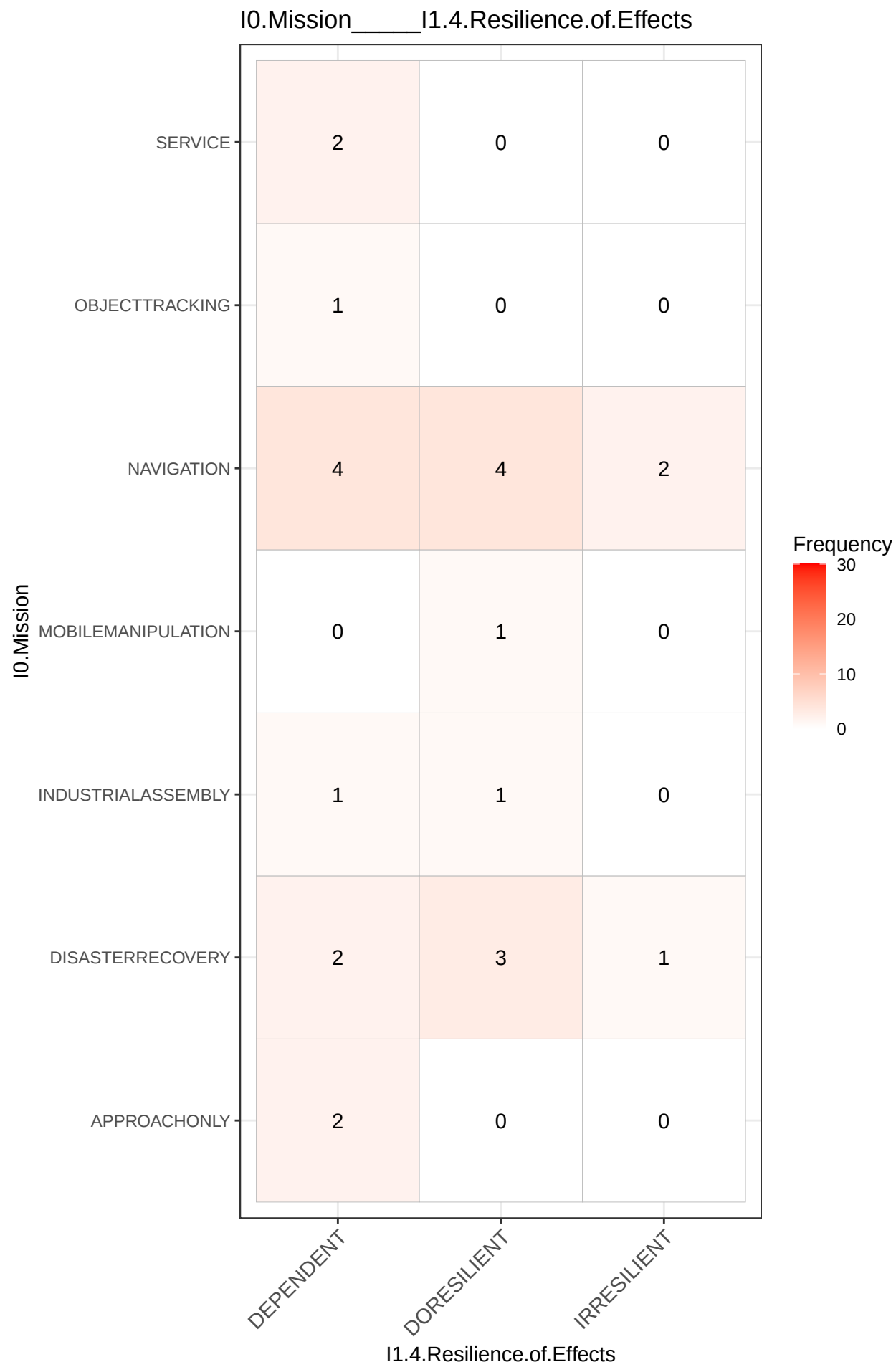


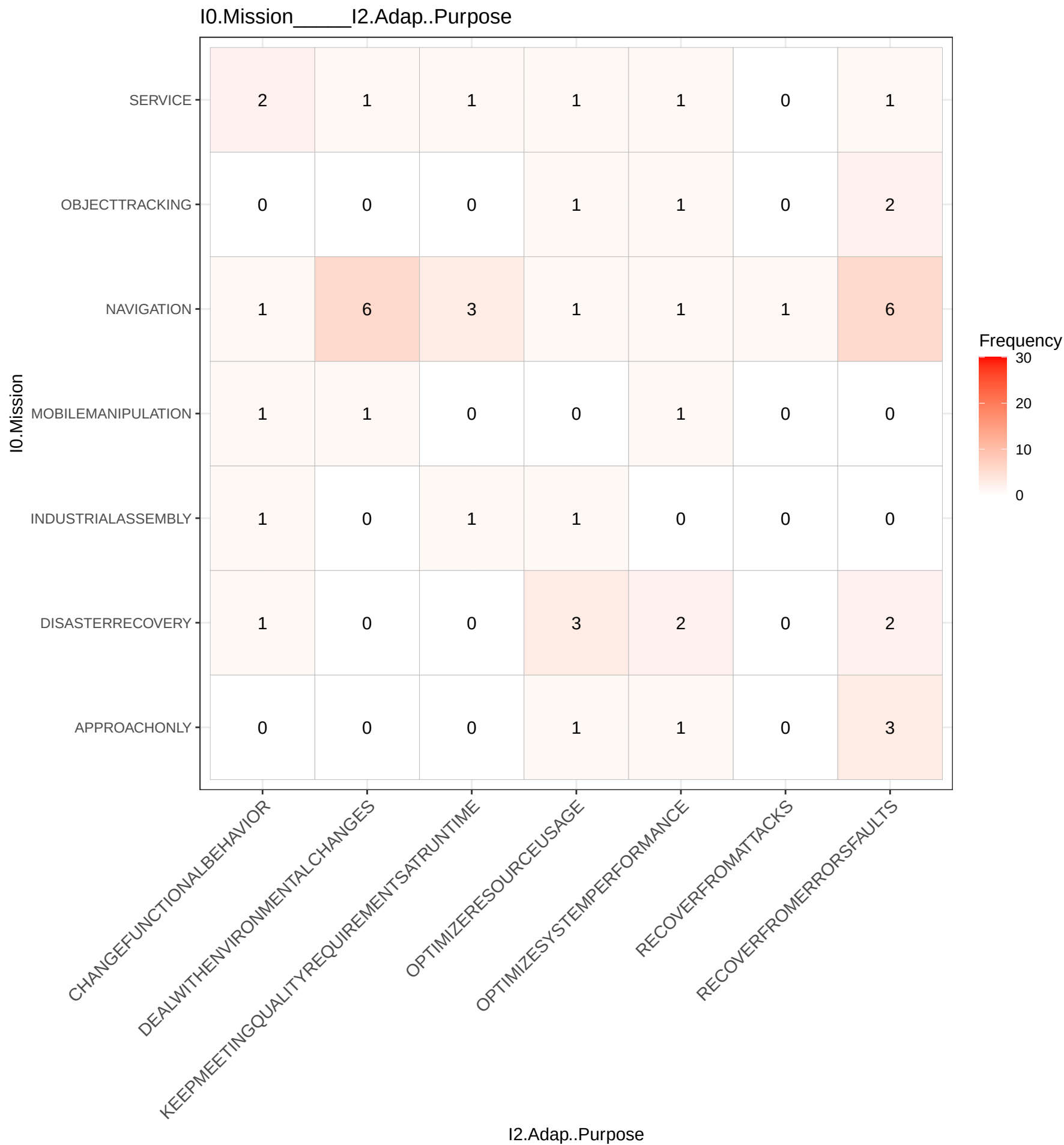


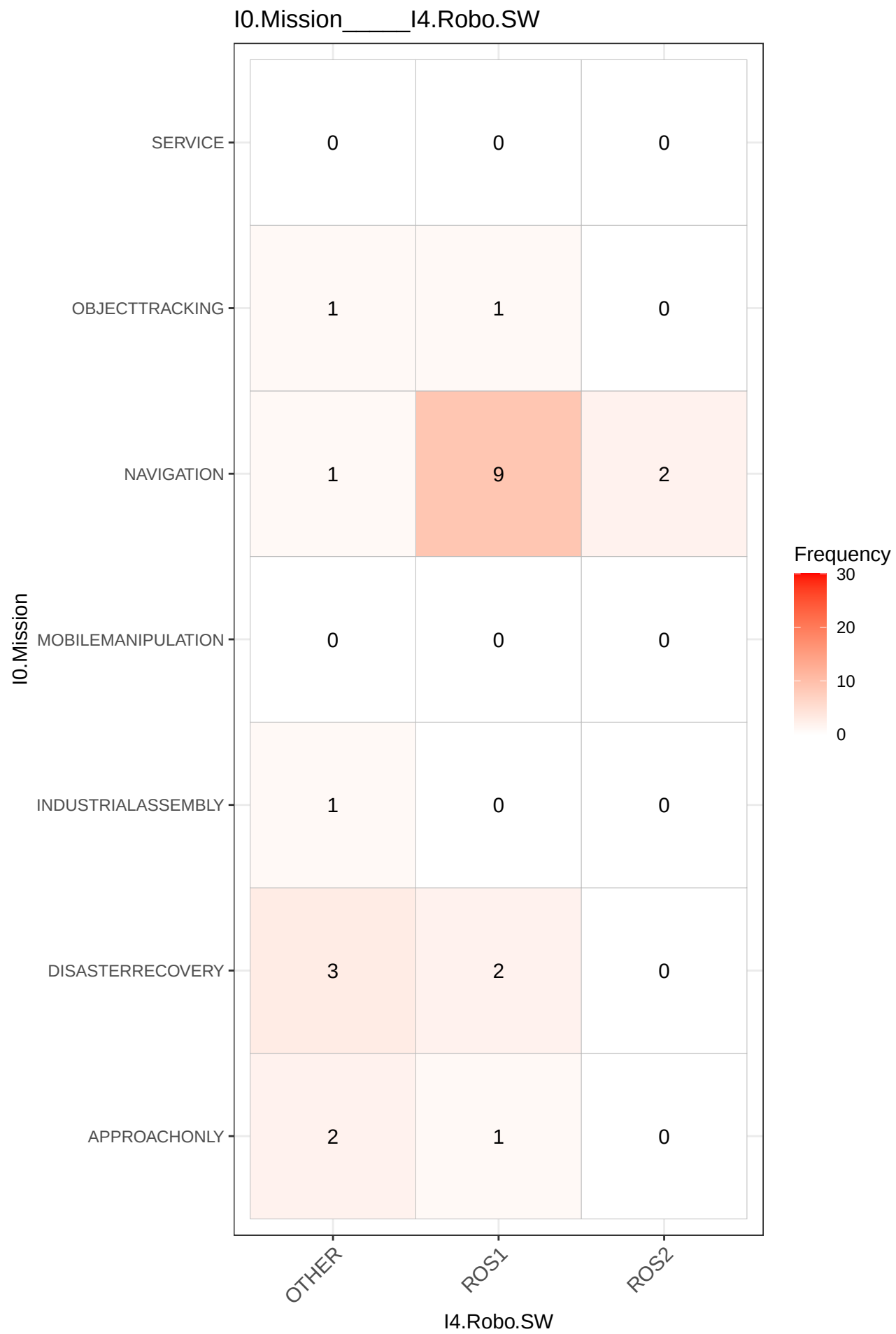




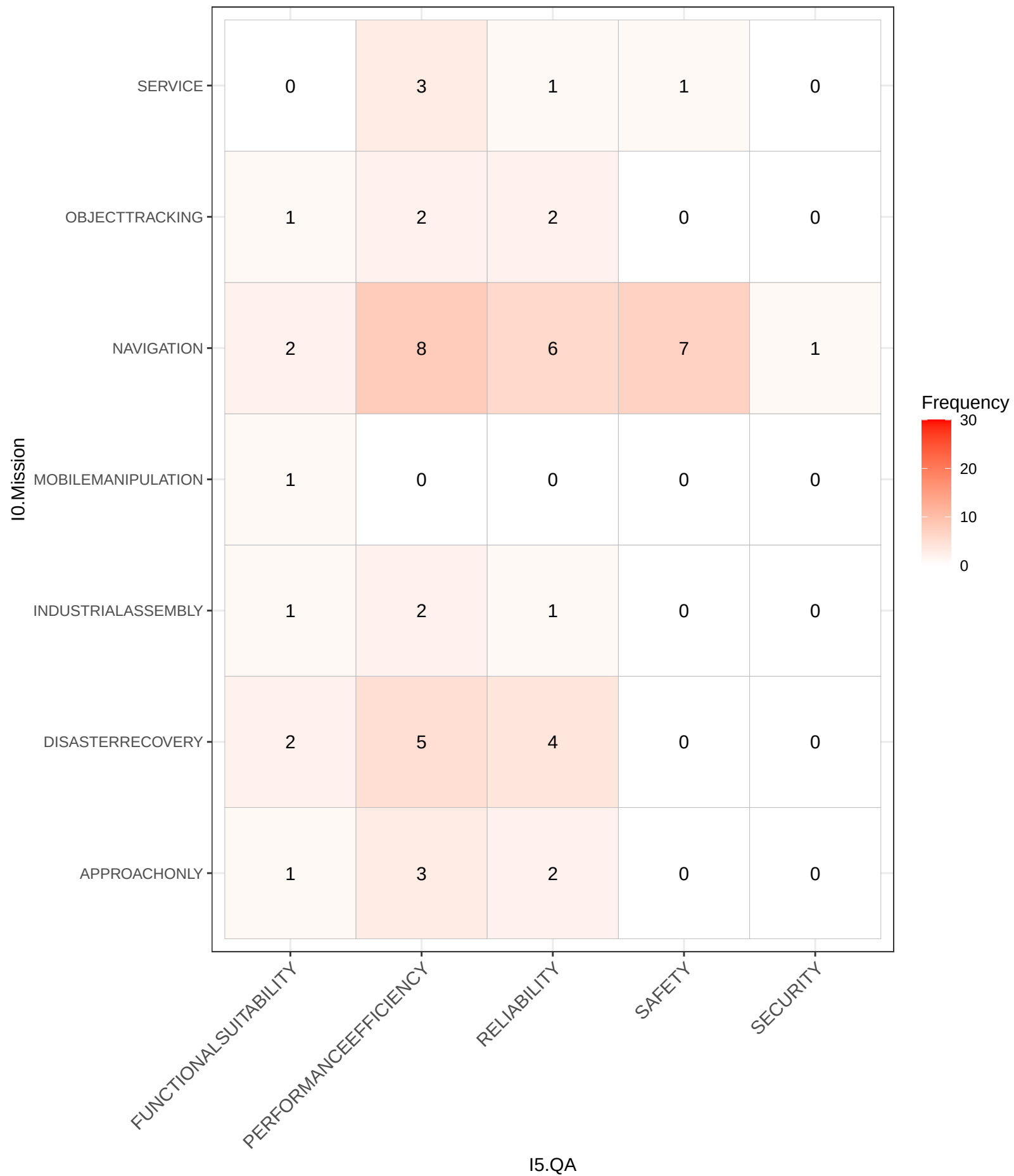




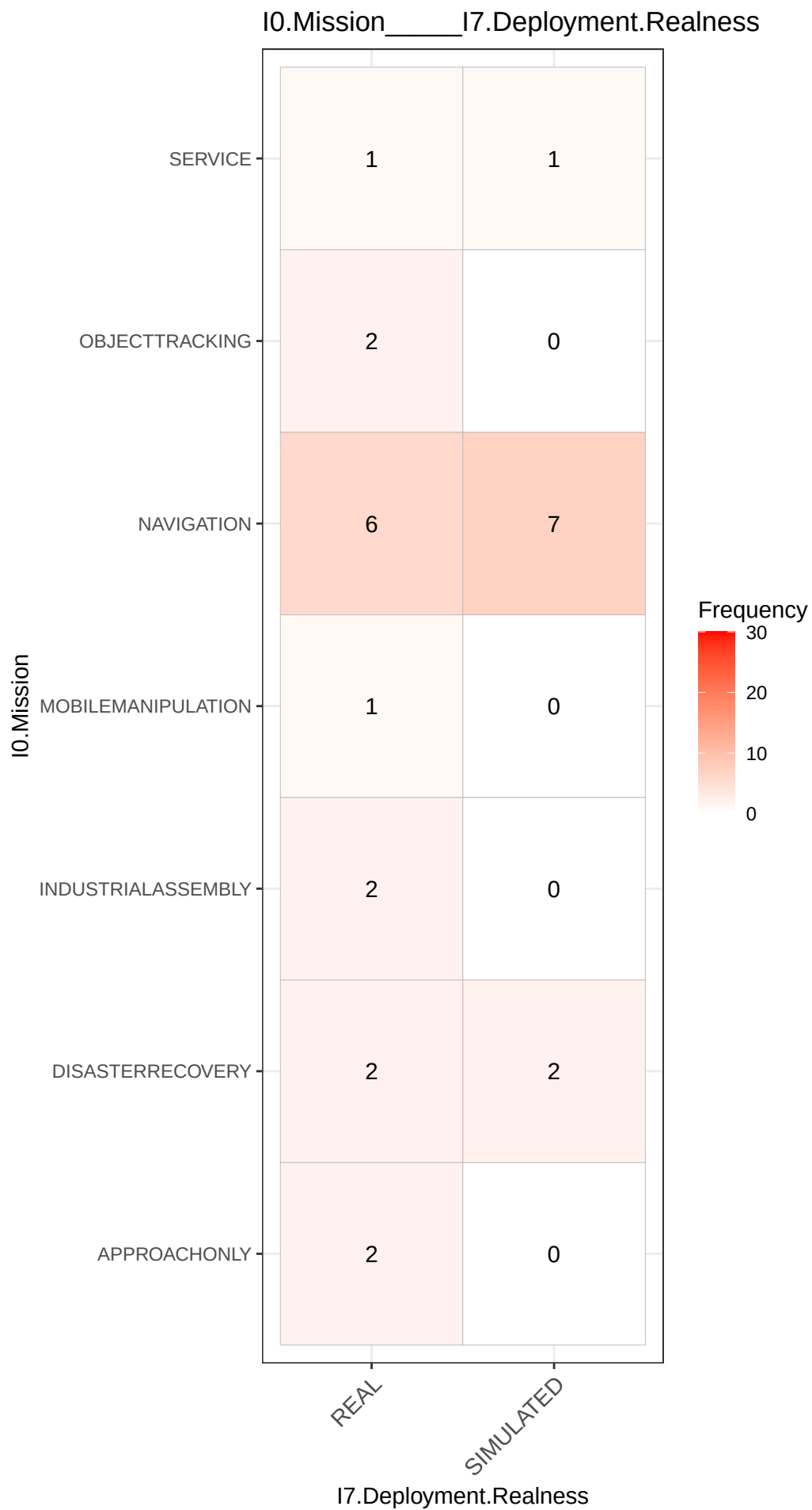


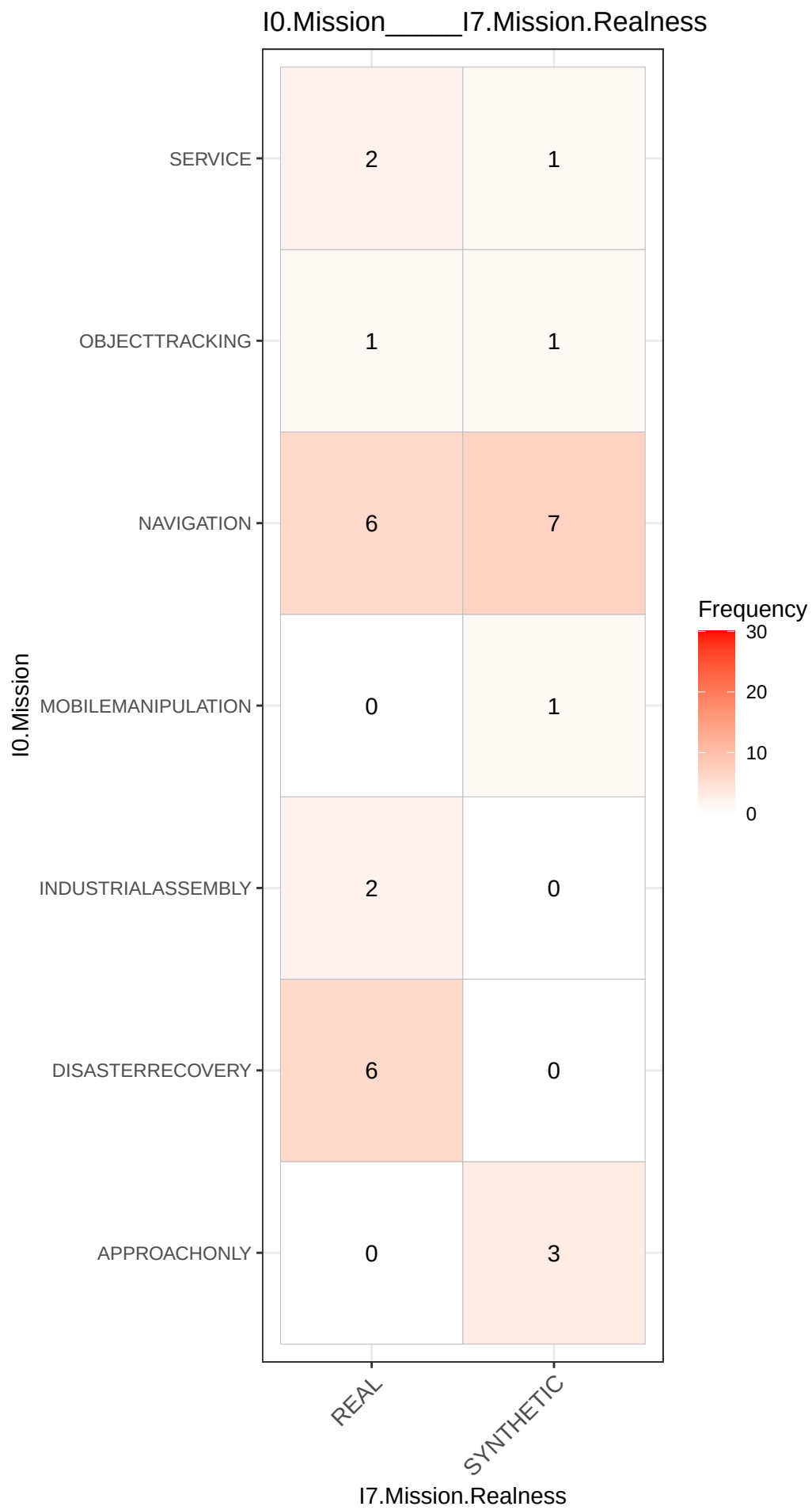


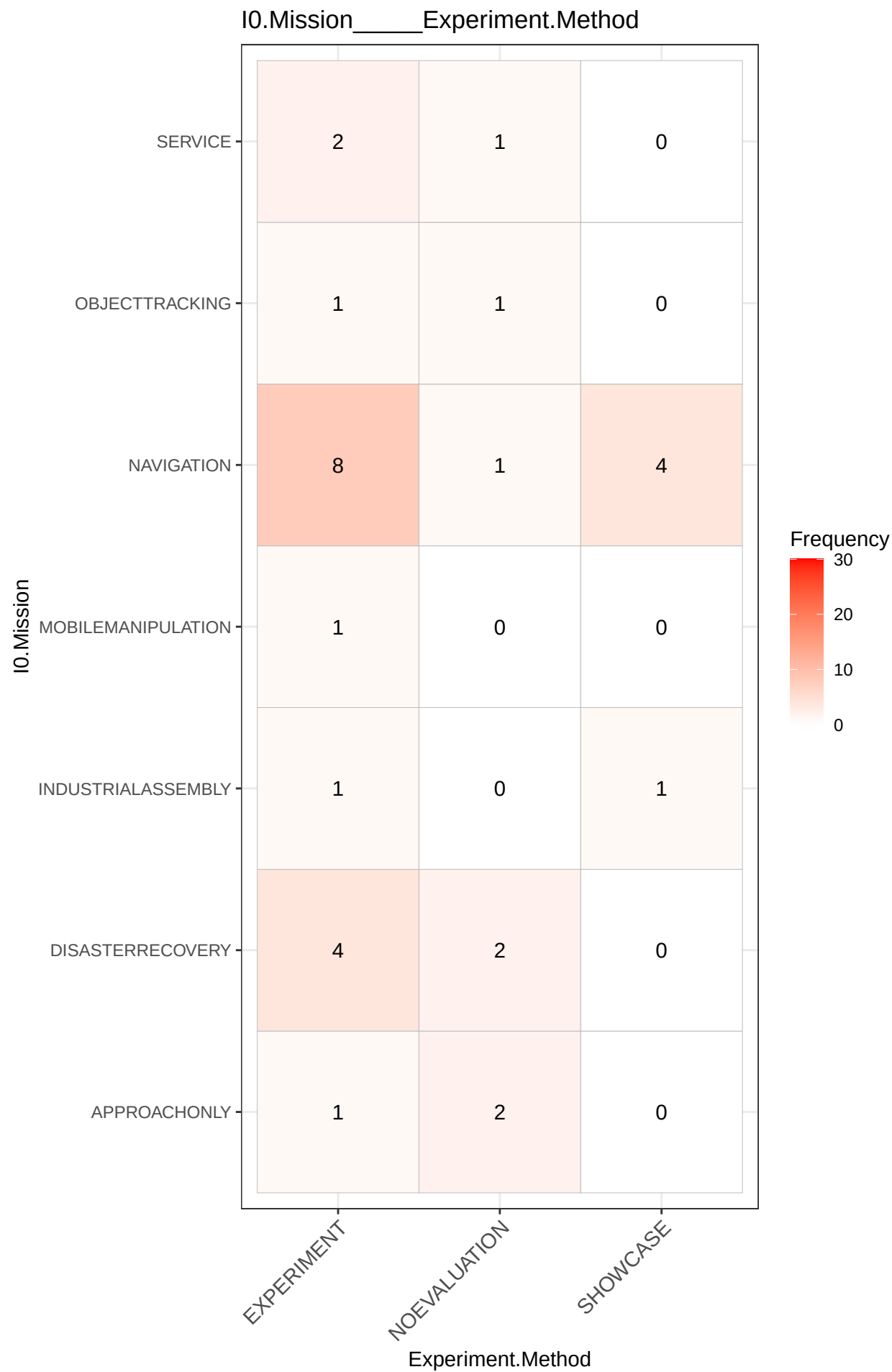
I0.Mission_____I5.QA

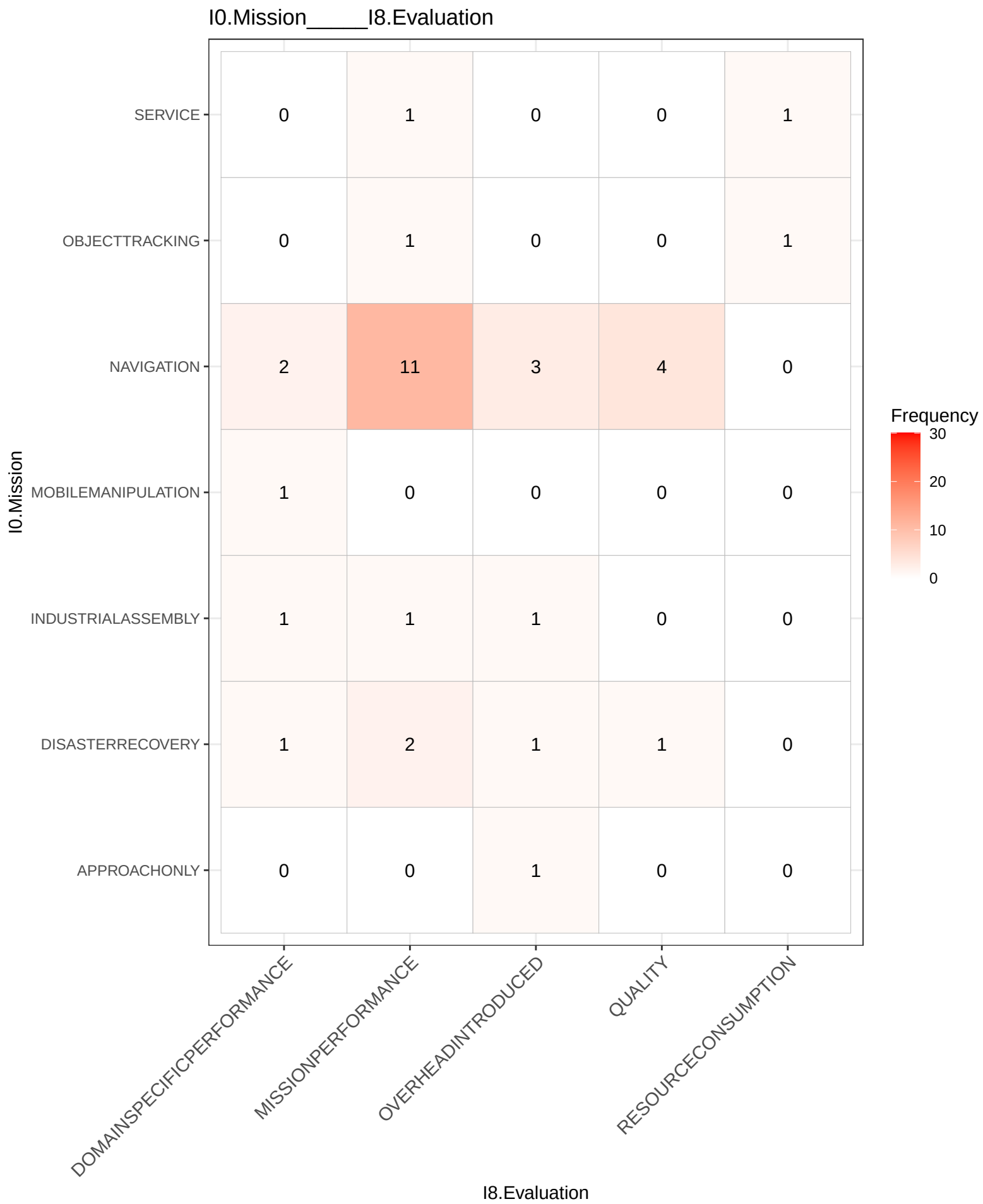






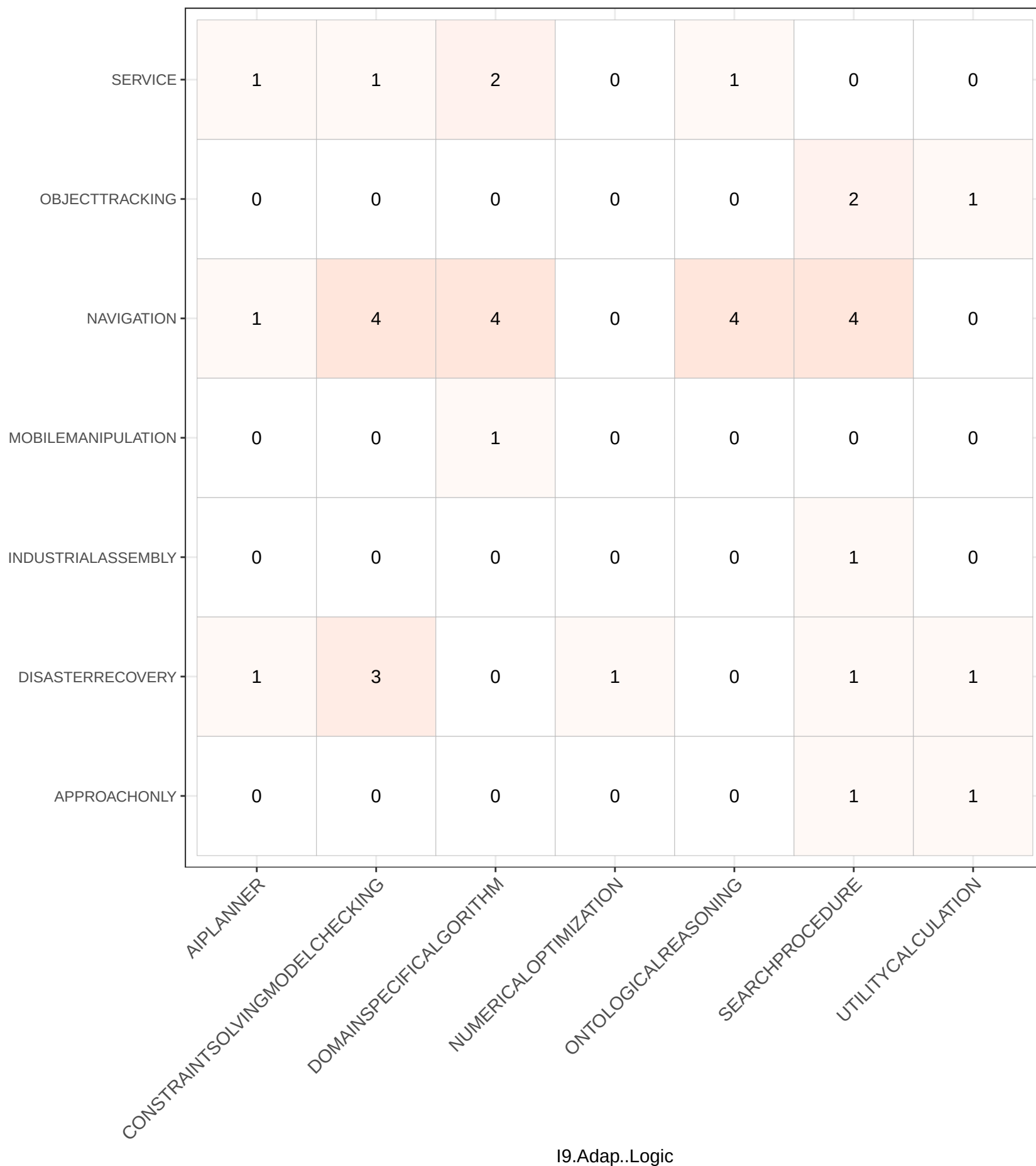




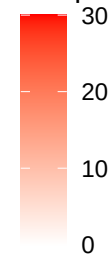


IO.Mission_____I9.Adap..Logic

IO.Mission



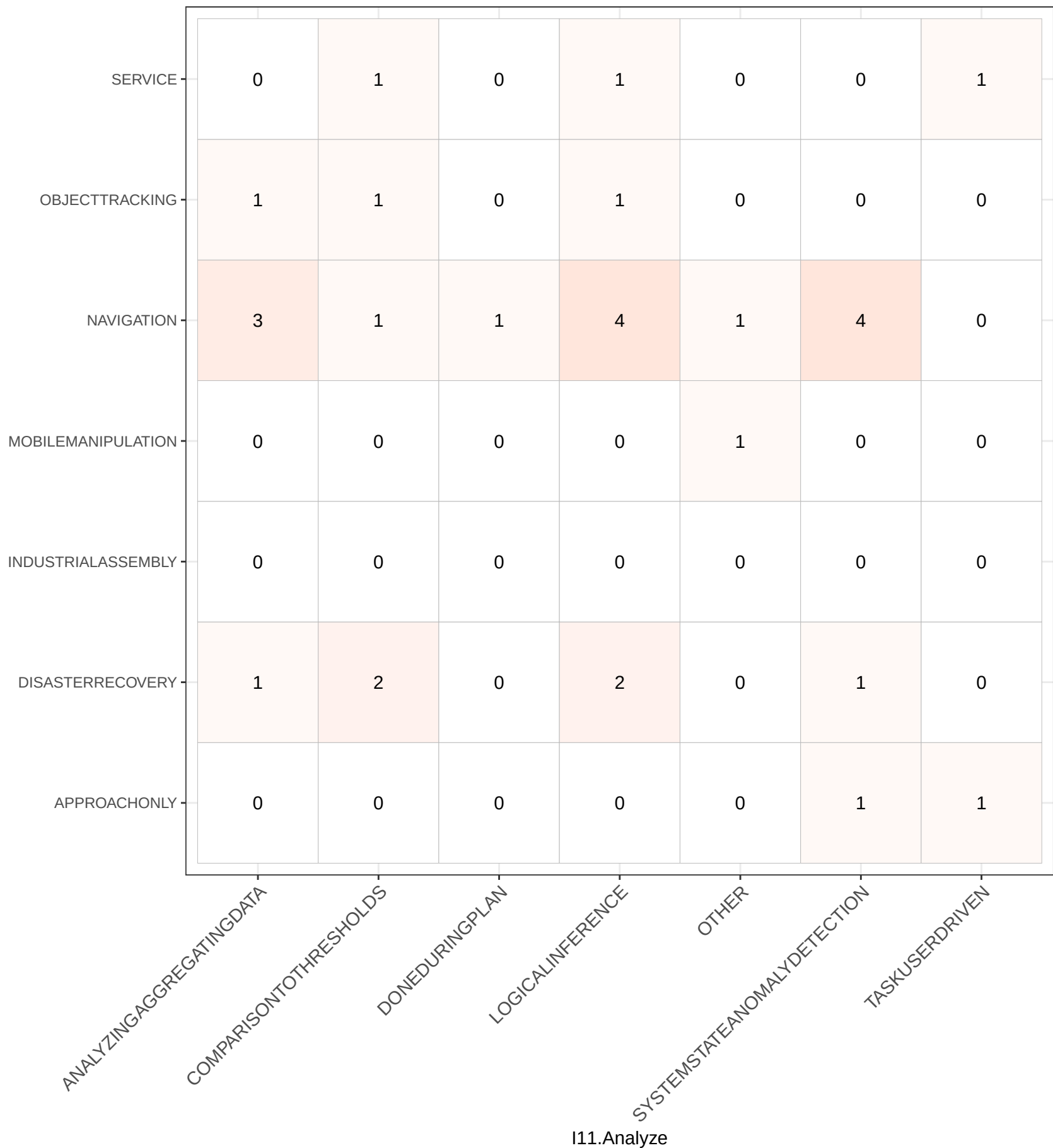
Frequency



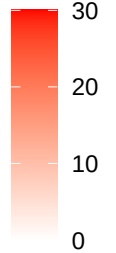


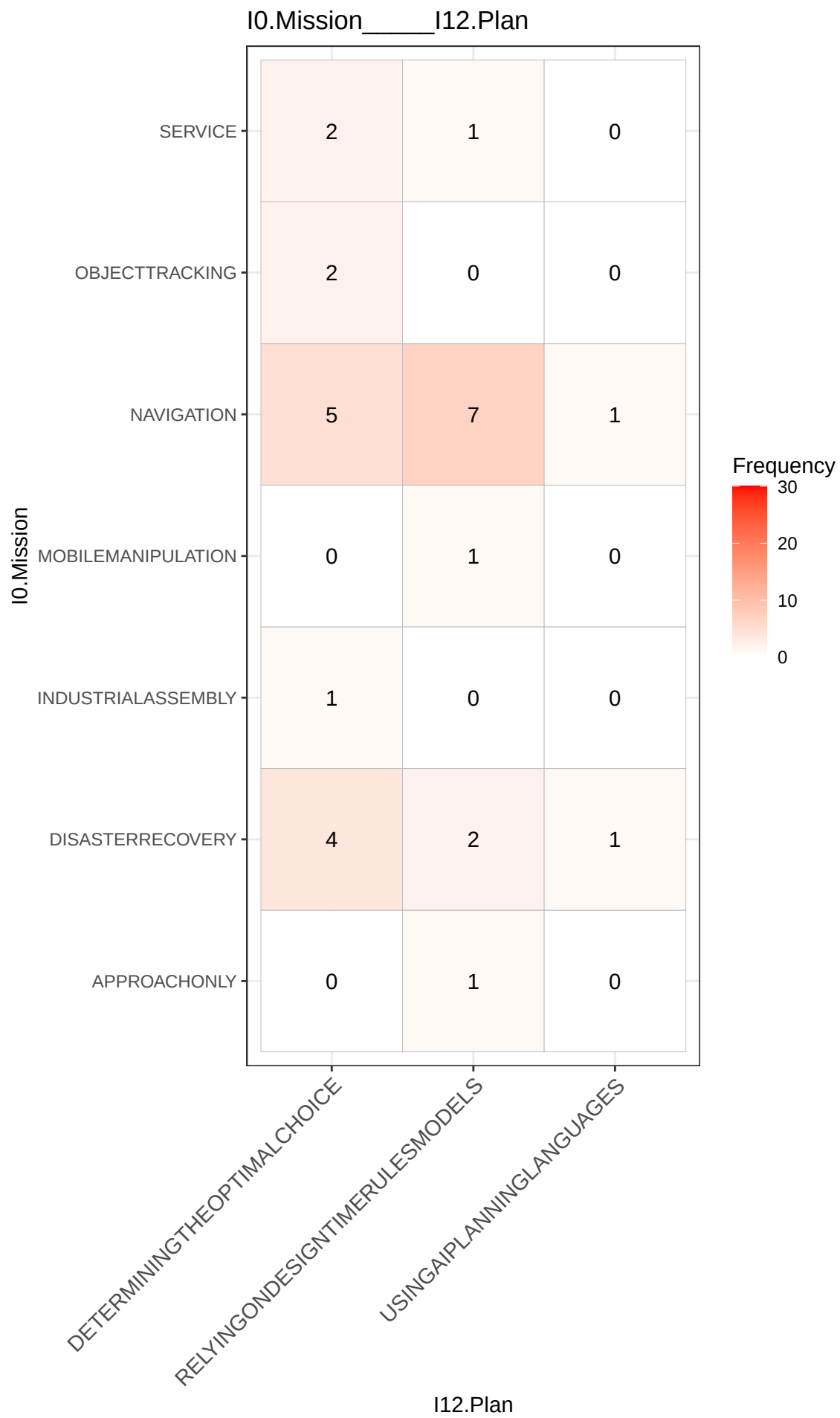
I0.Mission_____I11.Analyze

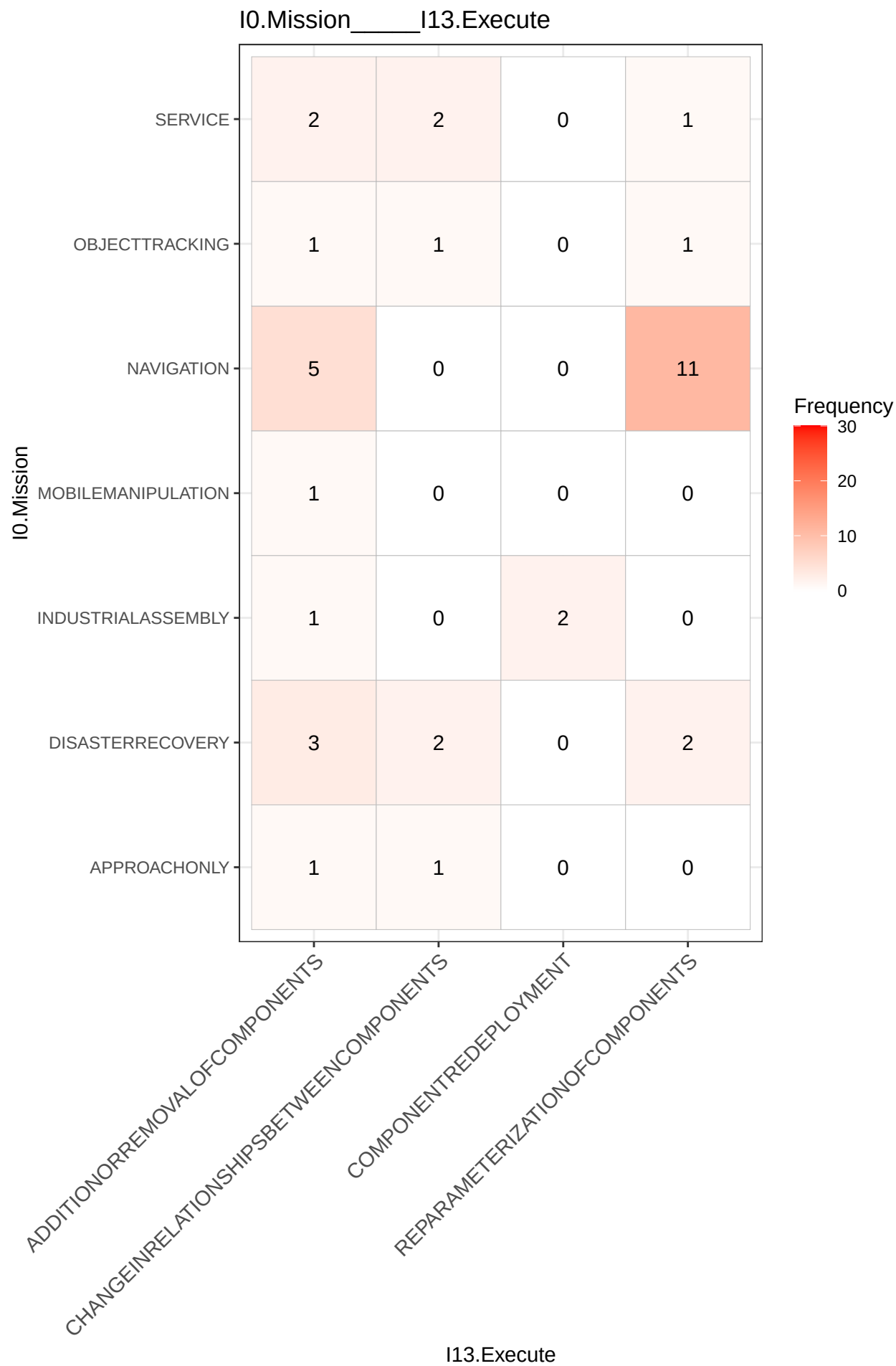
I0.Mission

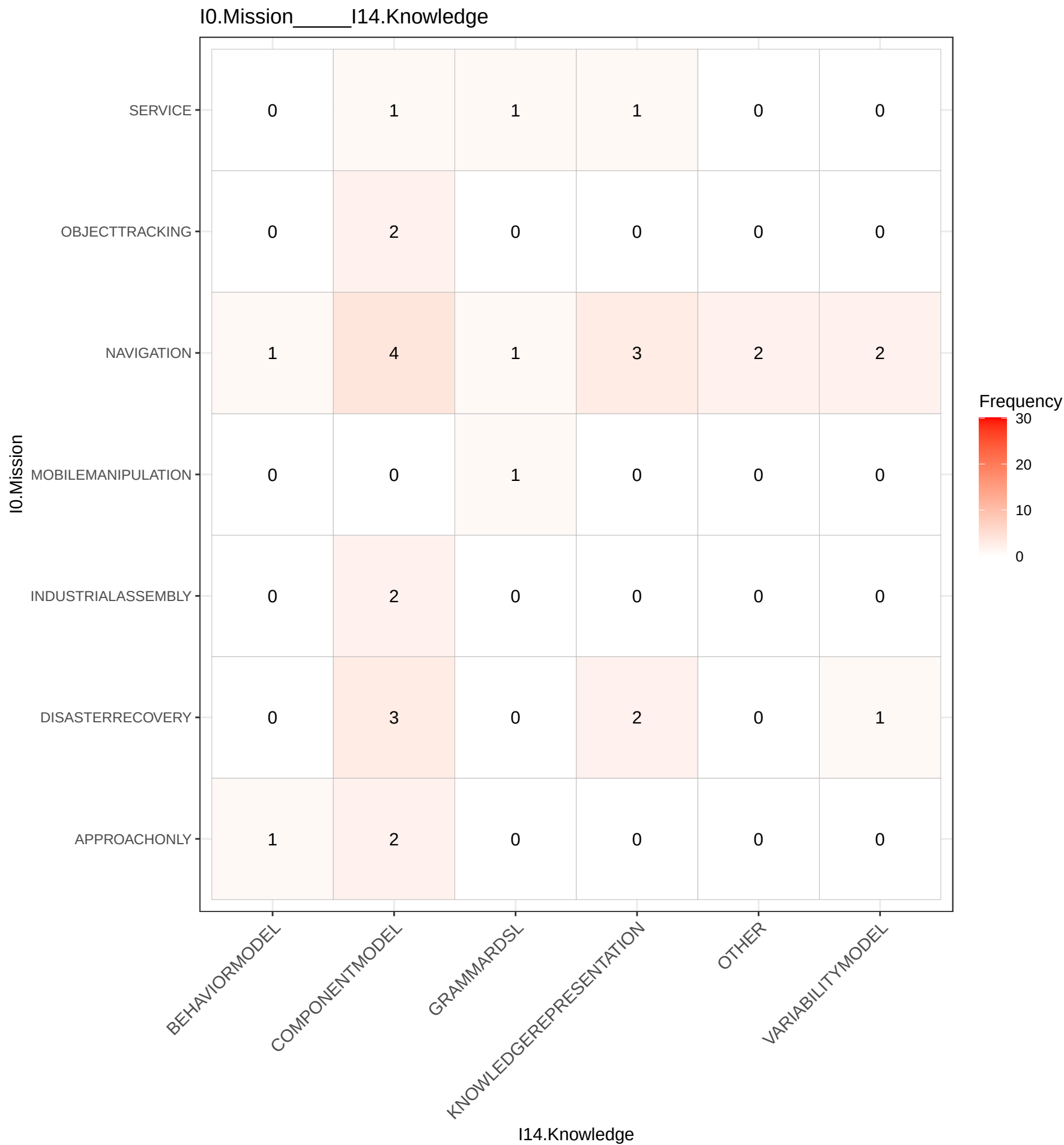


Frequency









I1.1.Metadata_____I1.2.Source.of.Change

I1.1.Metadata

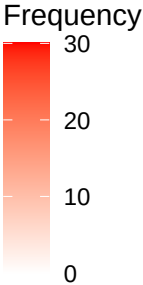
STATIC

DYNAMIC

EXTERNAL

INTERNAL

I1.2.Source.of.Change

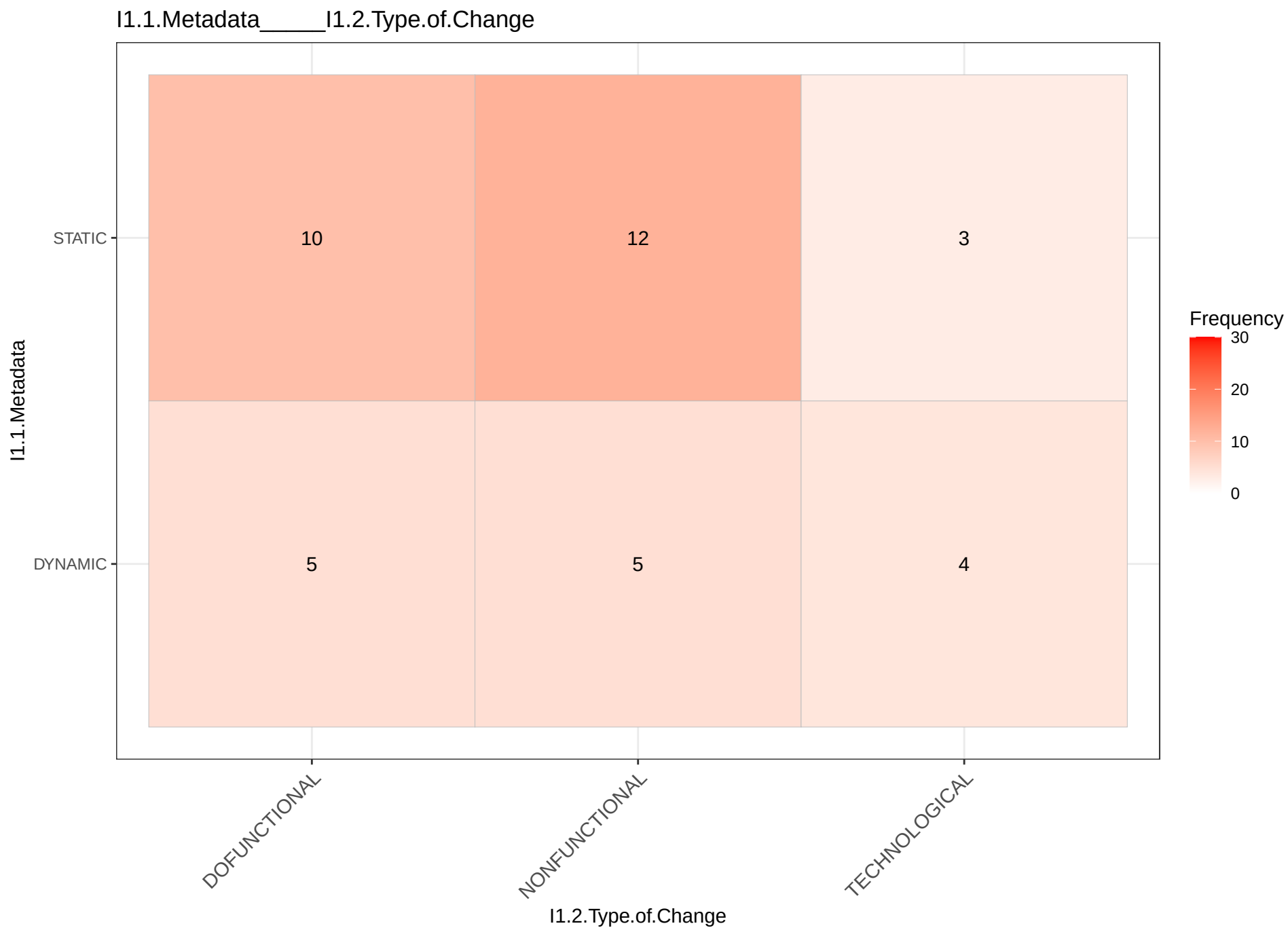


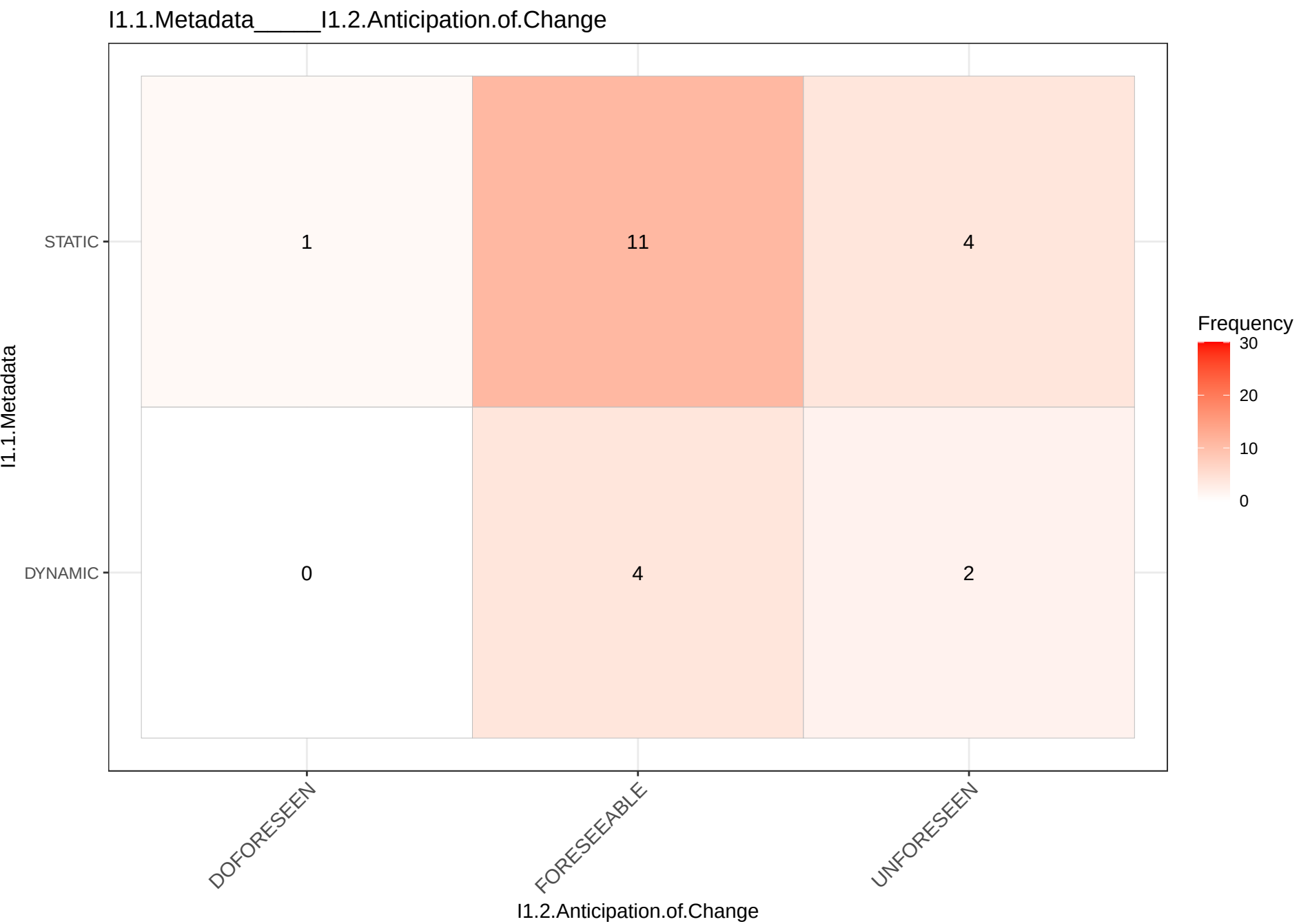
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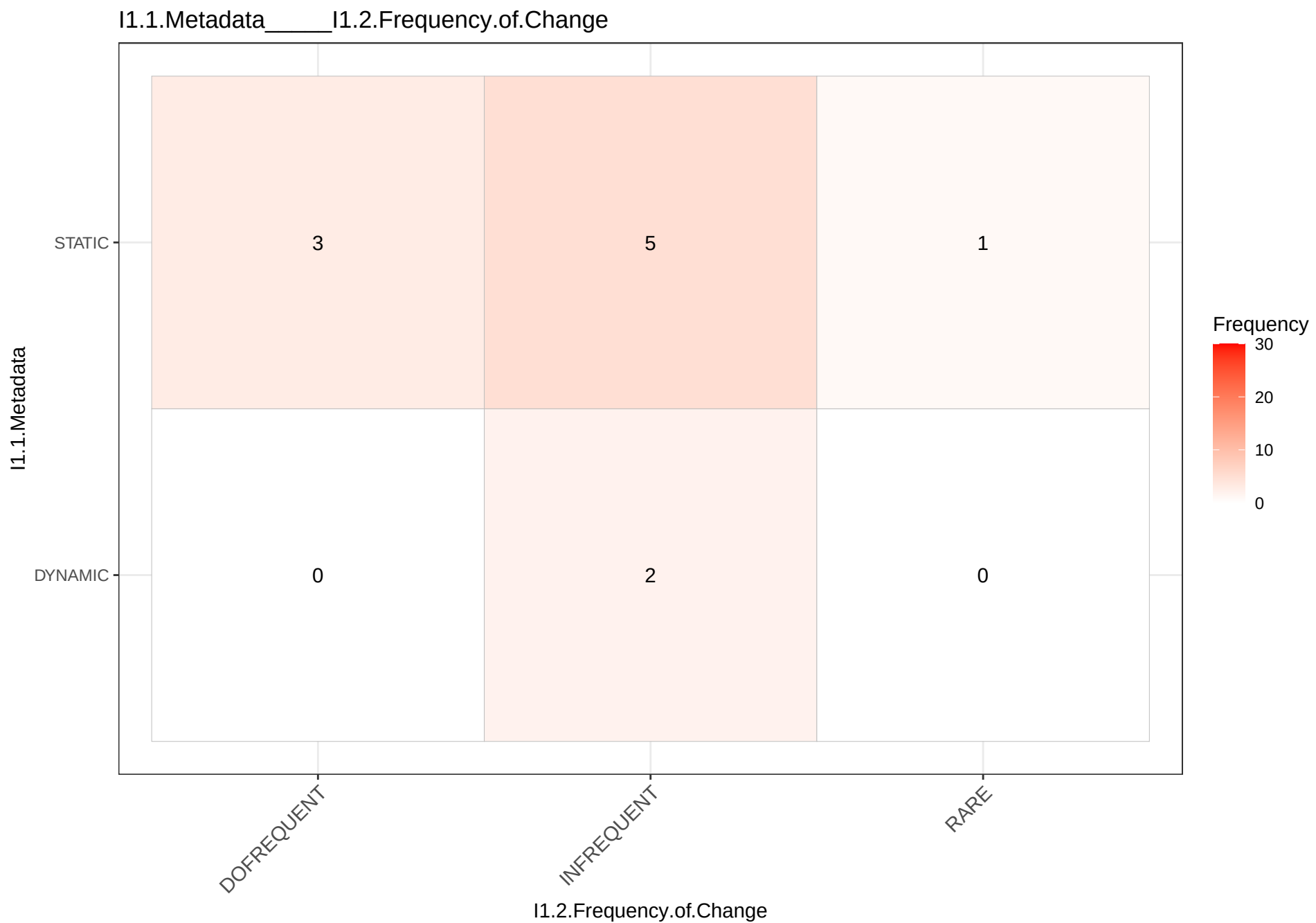
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0

6







I1.1.Metadata_____I1.3.Type.of.Mechanism

I1.1.Metadata

STATIC

12

13

DYNAMIC

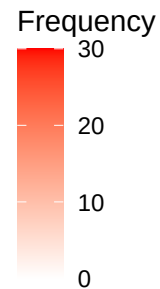
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6

PARAMETRIC

STRUCTURAL

I1.3.Type.of.Mechanism



STATIC

3

14

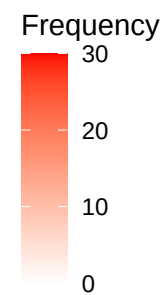
DYNAMIC

1

5

DECENTRALIZED

DECENTRALIZED



I1.1.Metadata_____I1.3.Scope.of.Mechanism

I1.1.Metadata

STATIC

3

15

DYNAMIC

1

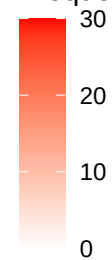
6

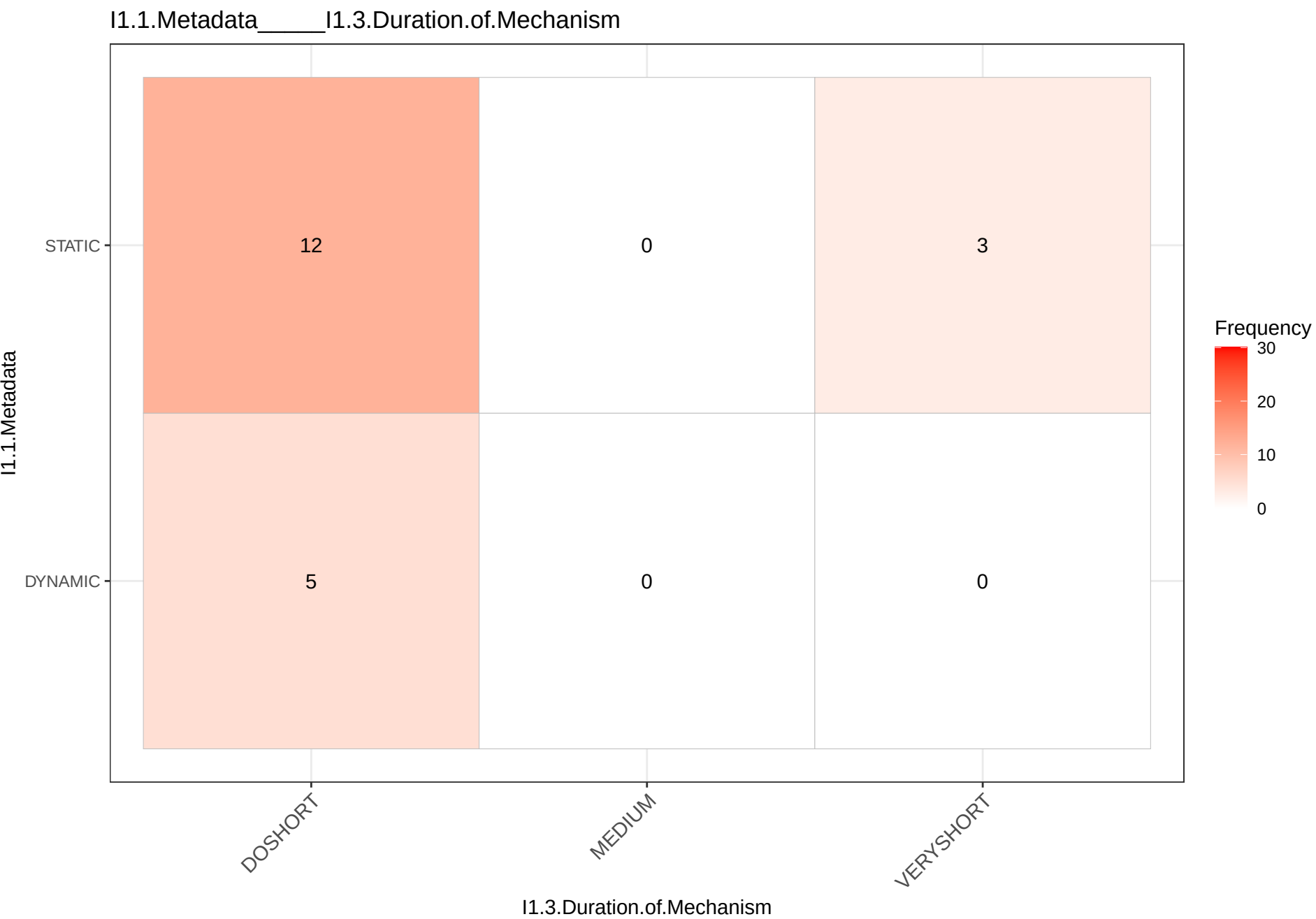
GLOBAL

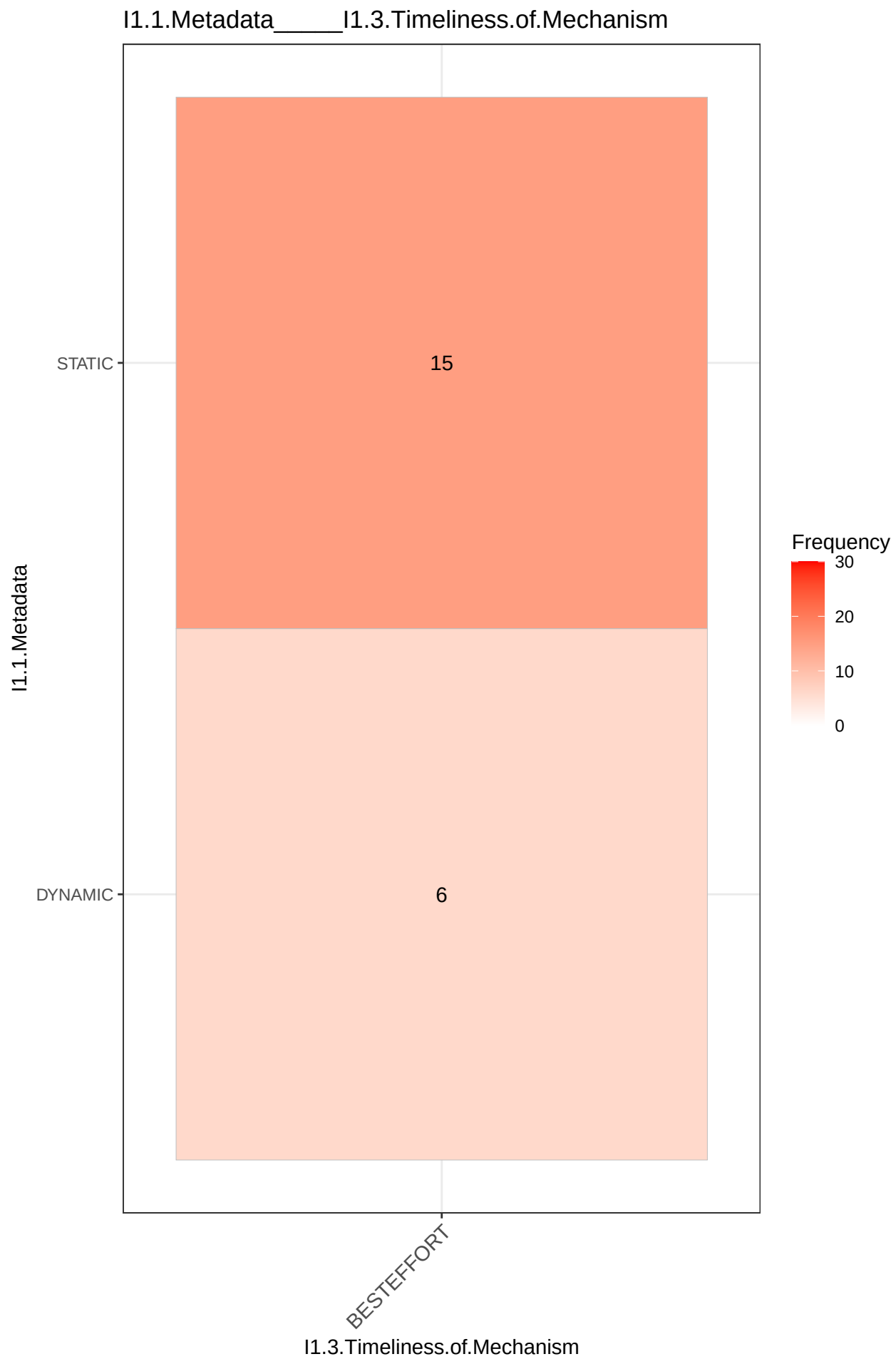
LOCAL

I1.3.Scope.of.Mechanism

Frequency







I1.1.Metadata_____I1.3.Trigger.of.Mechanism

I1.1.Metadata

STATIC

14

2

DYNAMIC

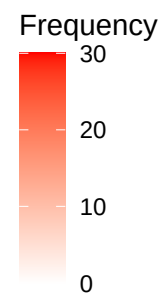
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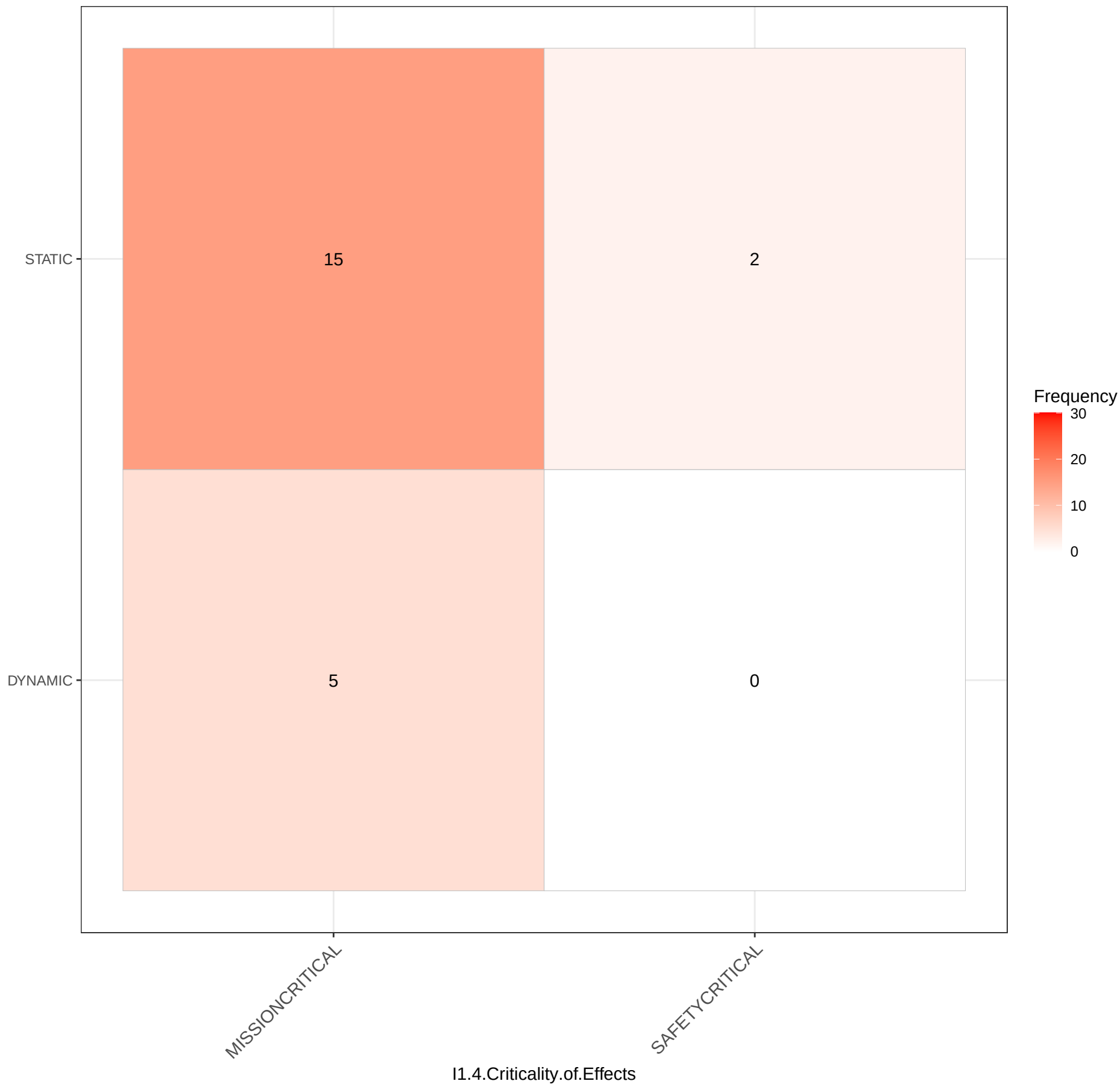
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EVENTTRIGGER

TIMETRIGGER

I1.3.Trigger.of.Mechanism





I1.1.Metadata_____I1.4.Predictability.of.Effects

I1.1.Metadata

STATIC

6

13

DYNAMIC

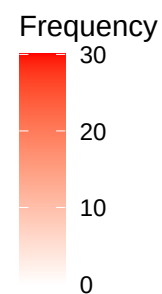
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4

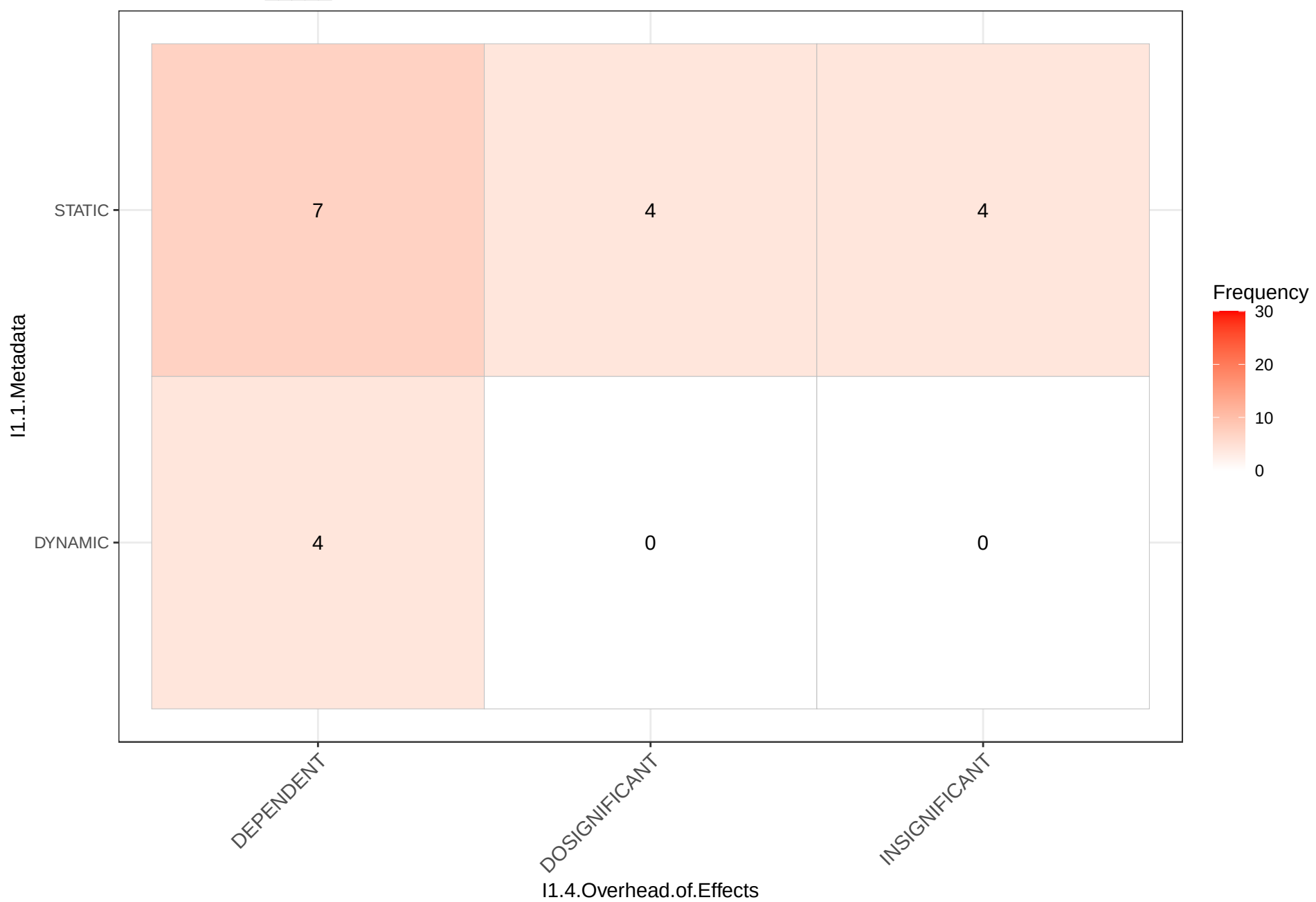
DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects



I1.1.Metadata_____I1.4.Overhead.of.Effects



I1.1.Metadata

I1.4.Resilience.of.Effects

I1.1.Metadata

STATIC

DYNAMIC

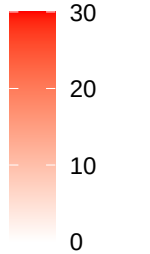
DEPENDENT

DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects

Frequency



8

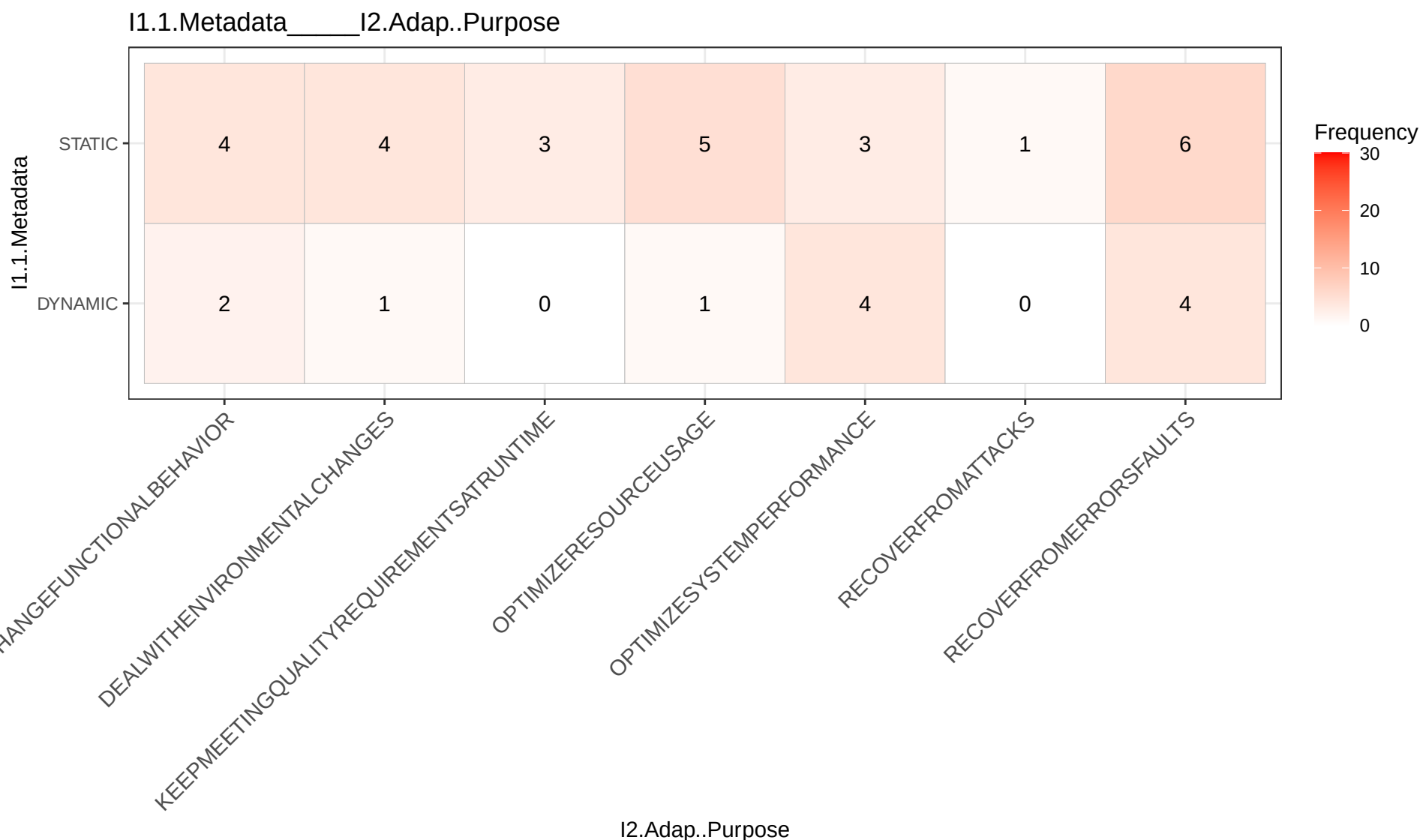
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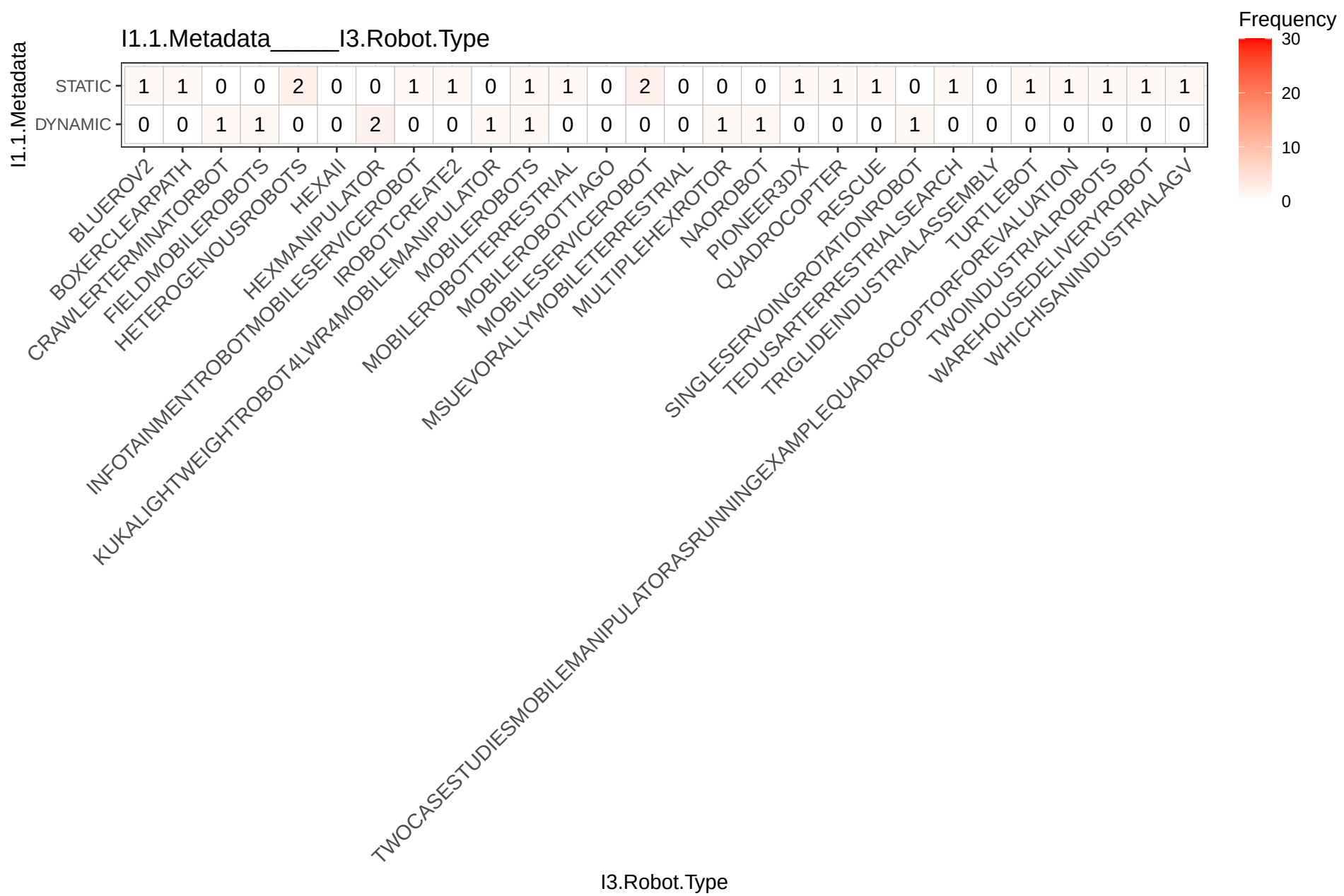
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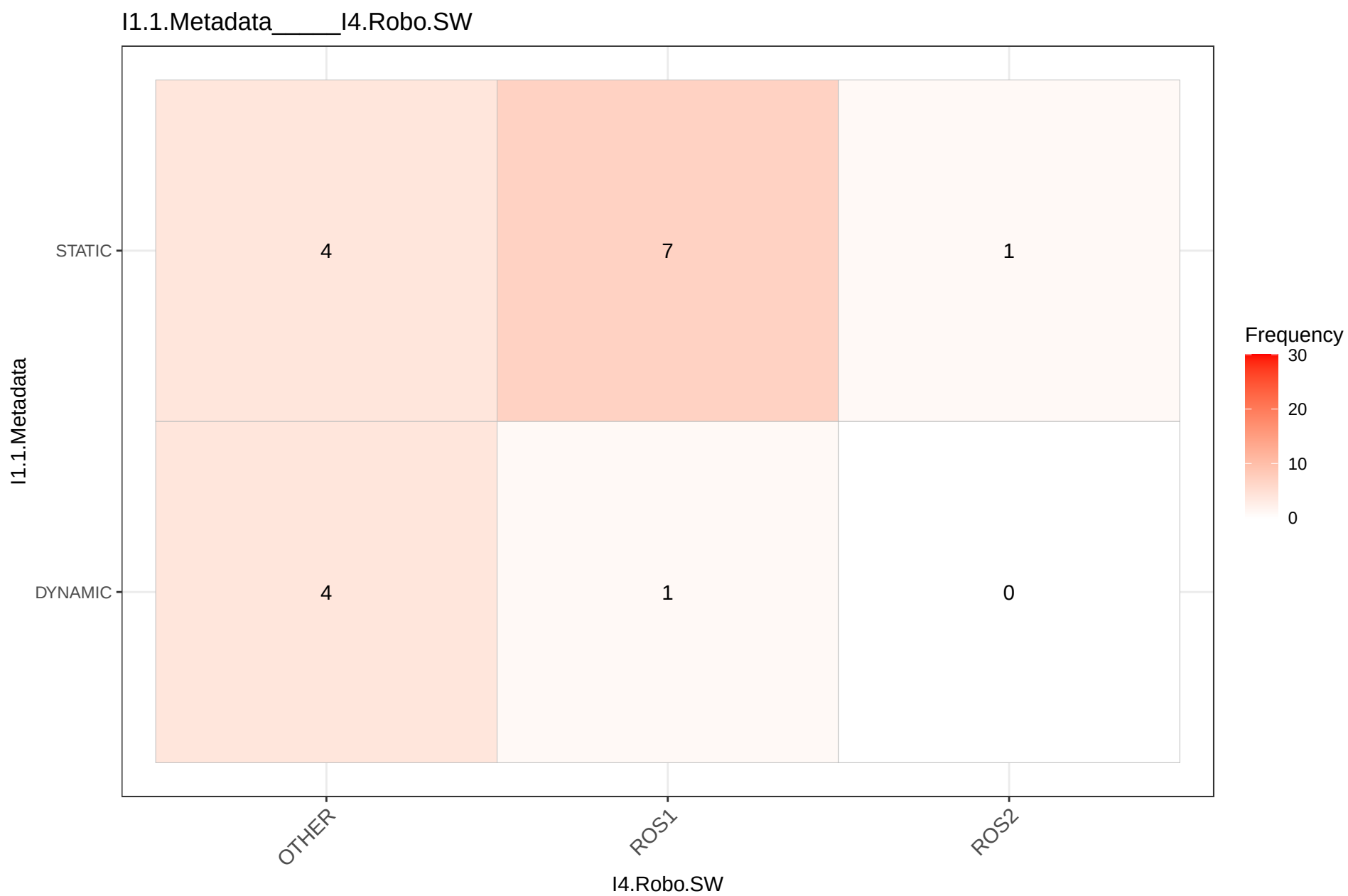
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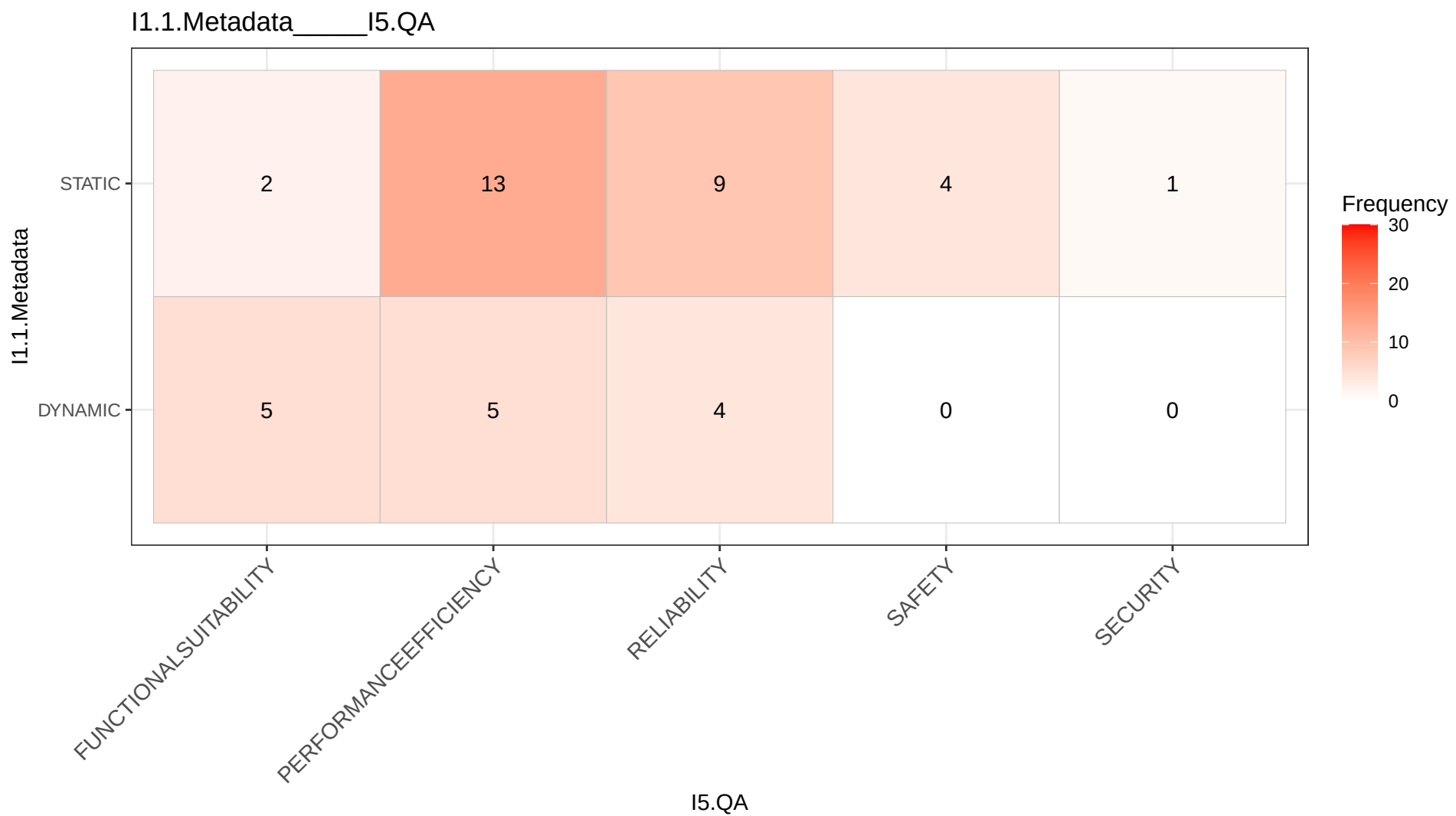
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1

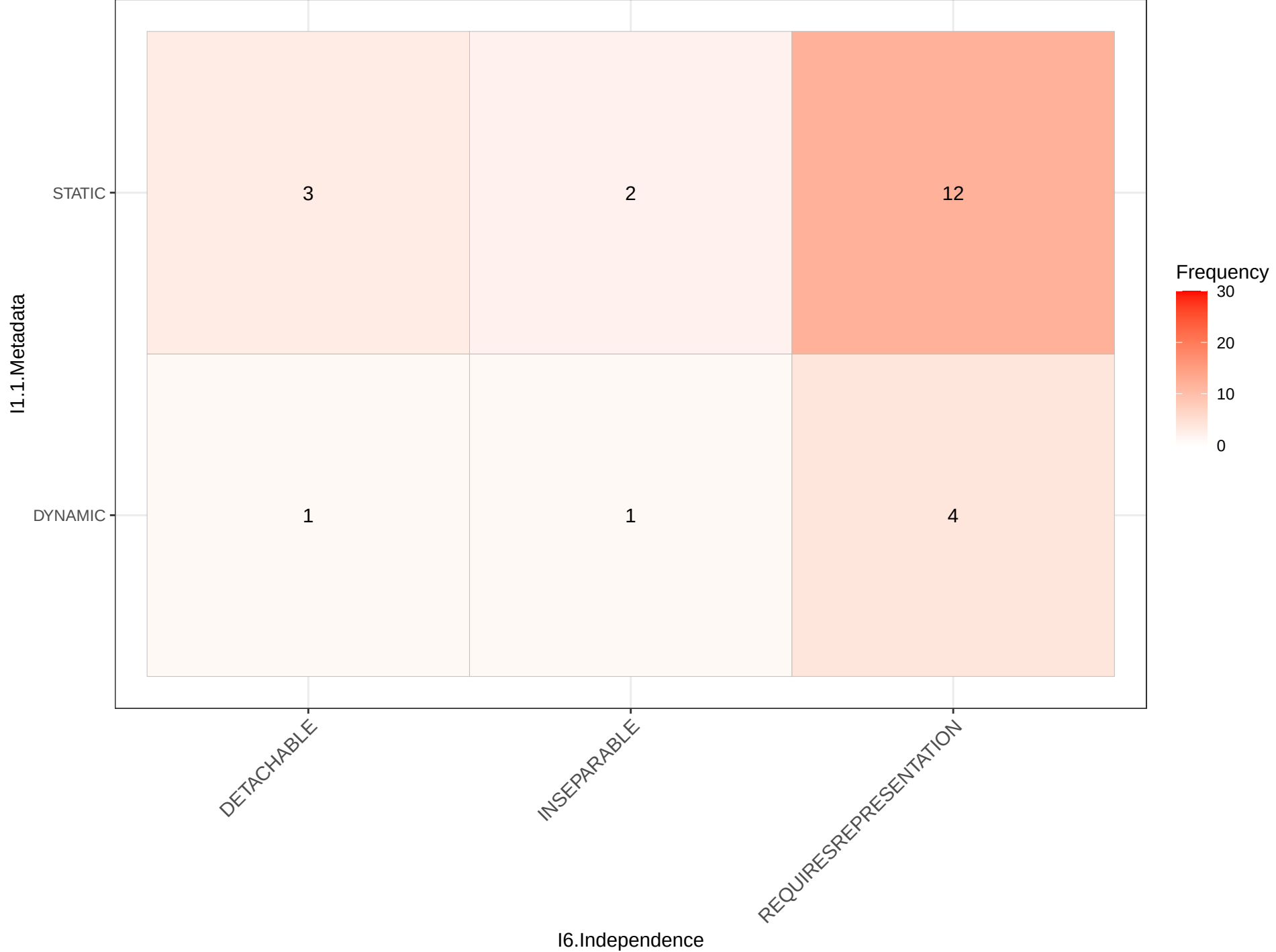








I1.1.Metadata_____I6.Independence



I1.1.Metadata_____I7.Deployment.Realness

I1.1.Metadata

STATIC

8

5

DYNAMIC

4

2

REAL

SIMULATED

I7.Deployment.Realness

Frequency

30

20

10

0

I1.1.Metadata_____I7.Mission.Realness

I1.1.Metadata

STATIC

12

5

DYNAMIC

3

3

REAL

SYNTHETIC

I7.Mission.Realness

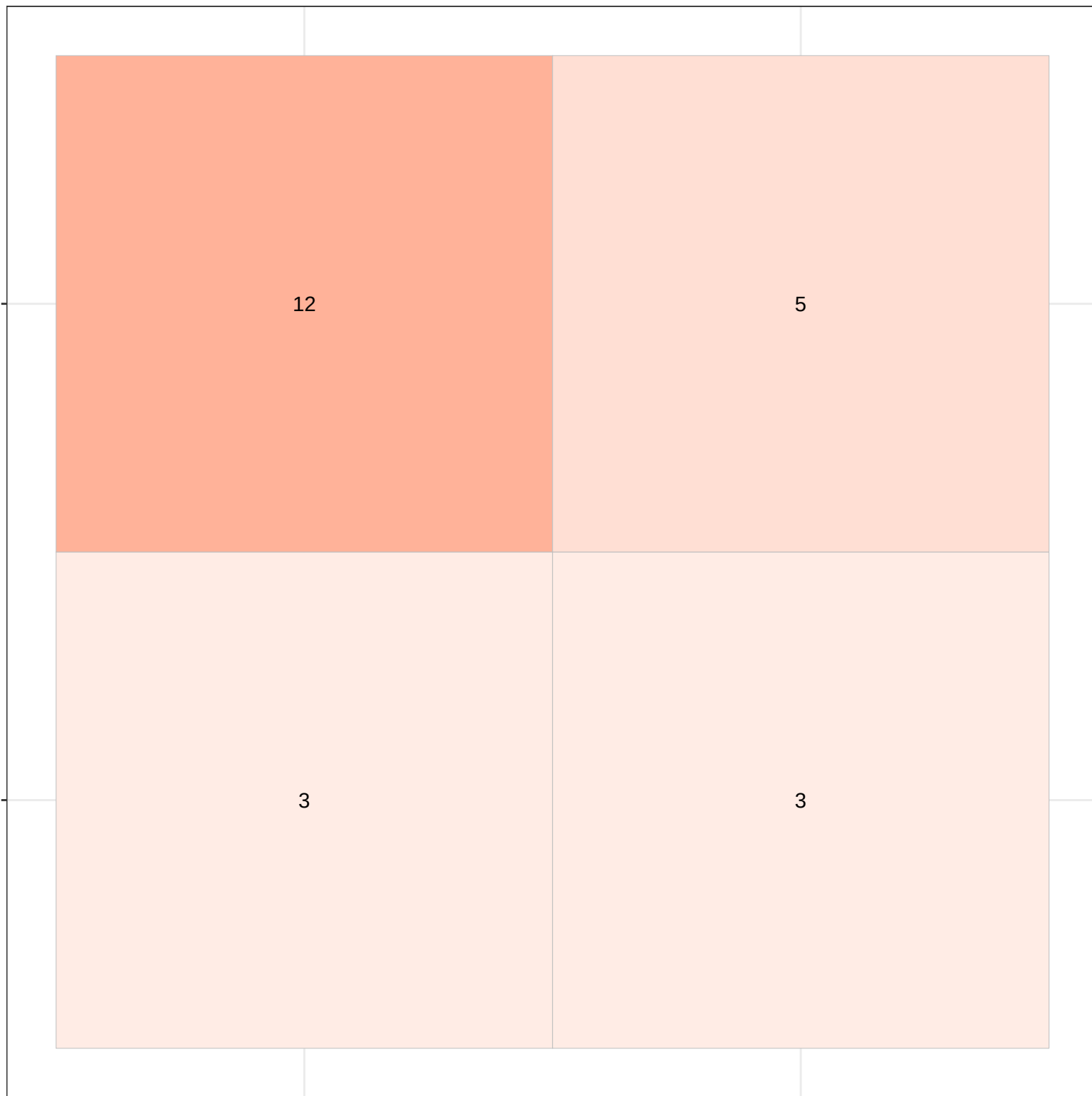
Frequency

30

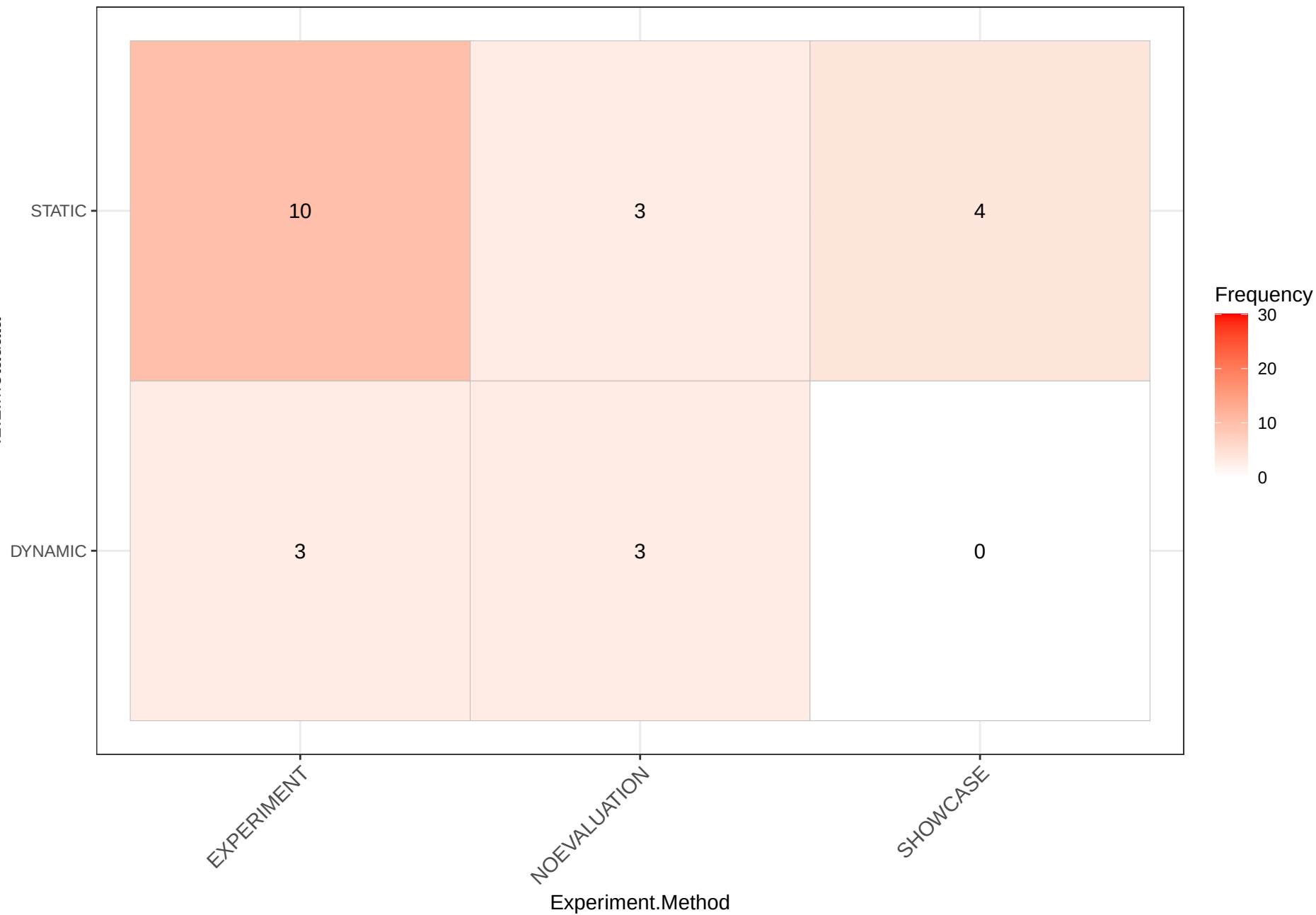
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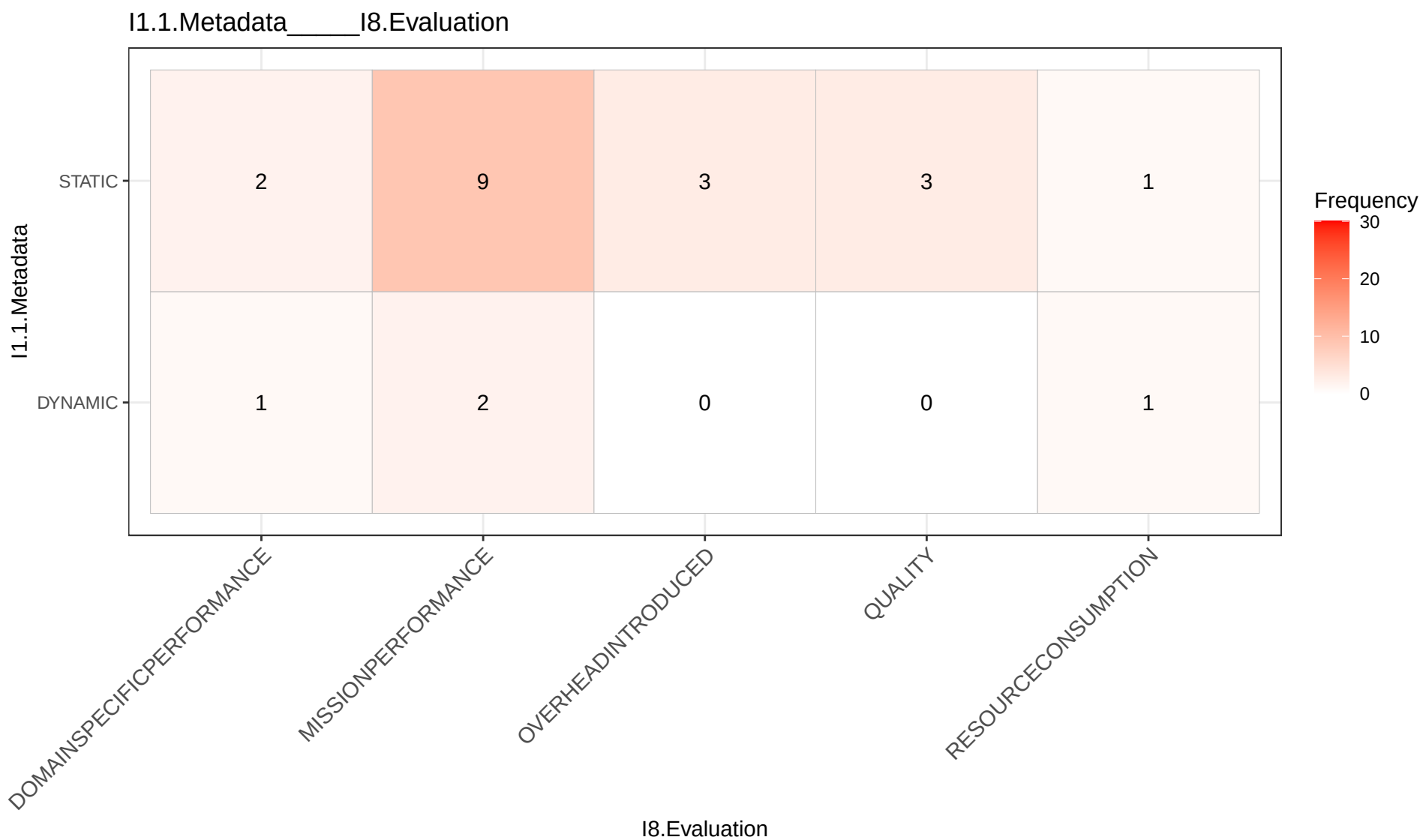
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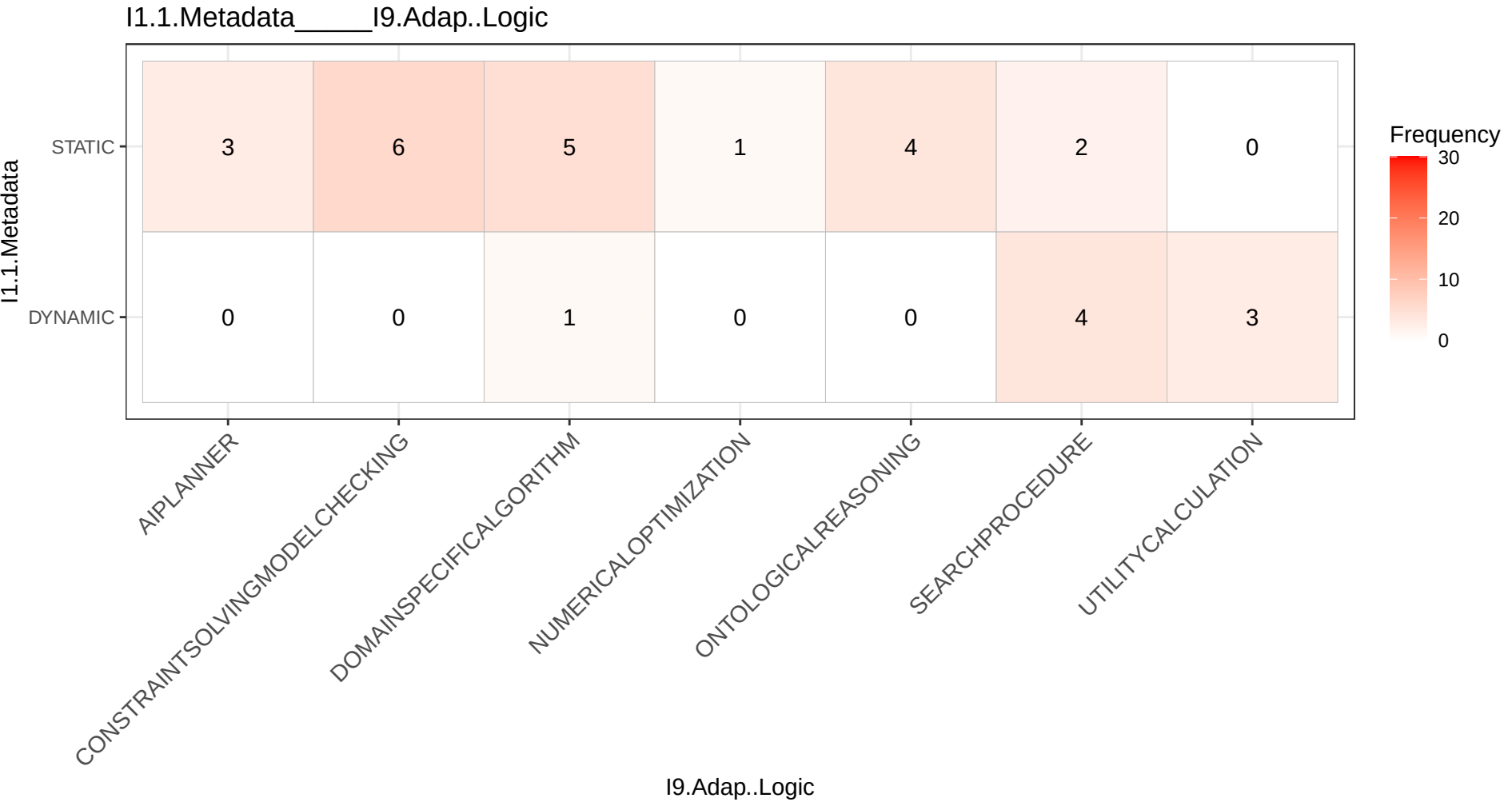
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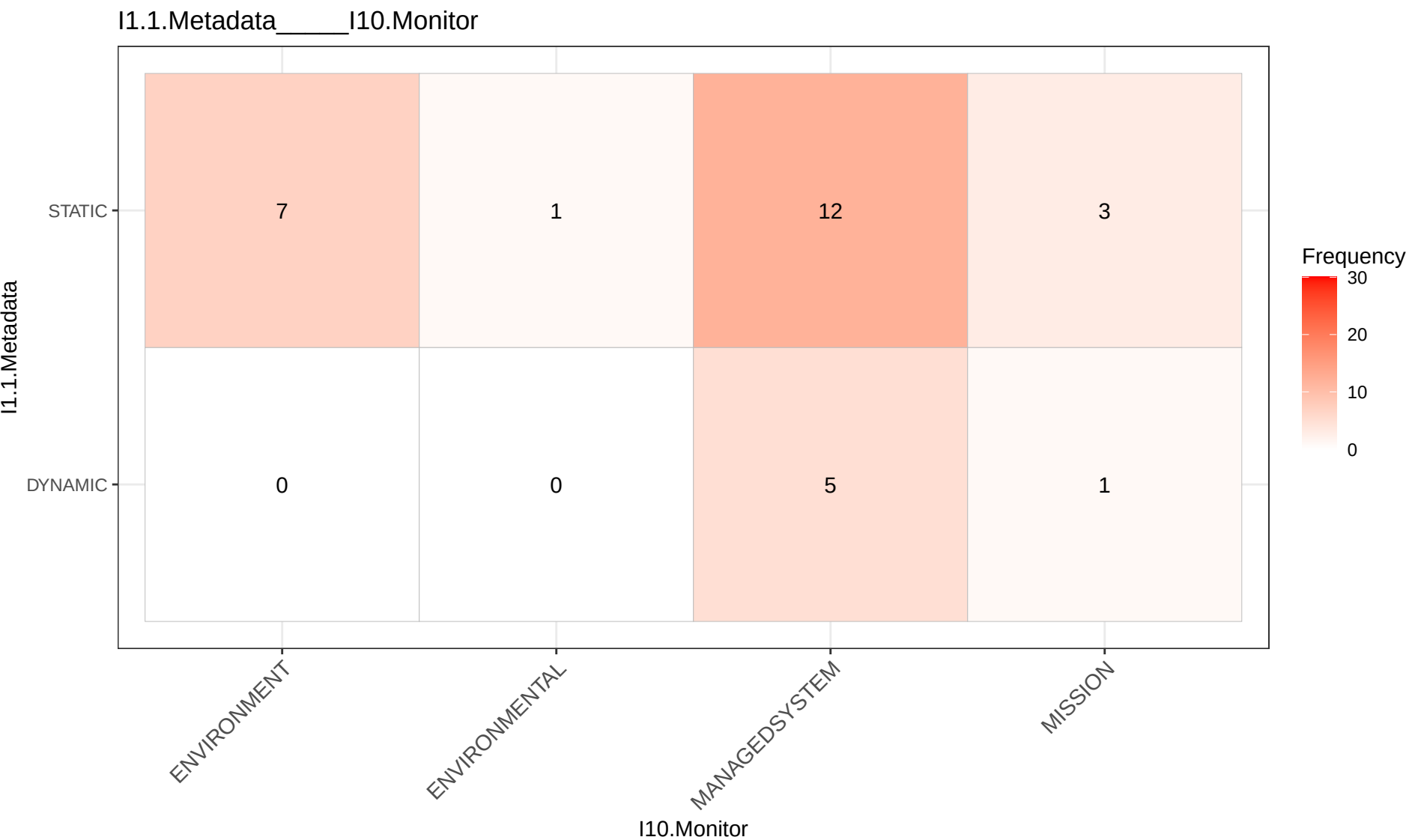


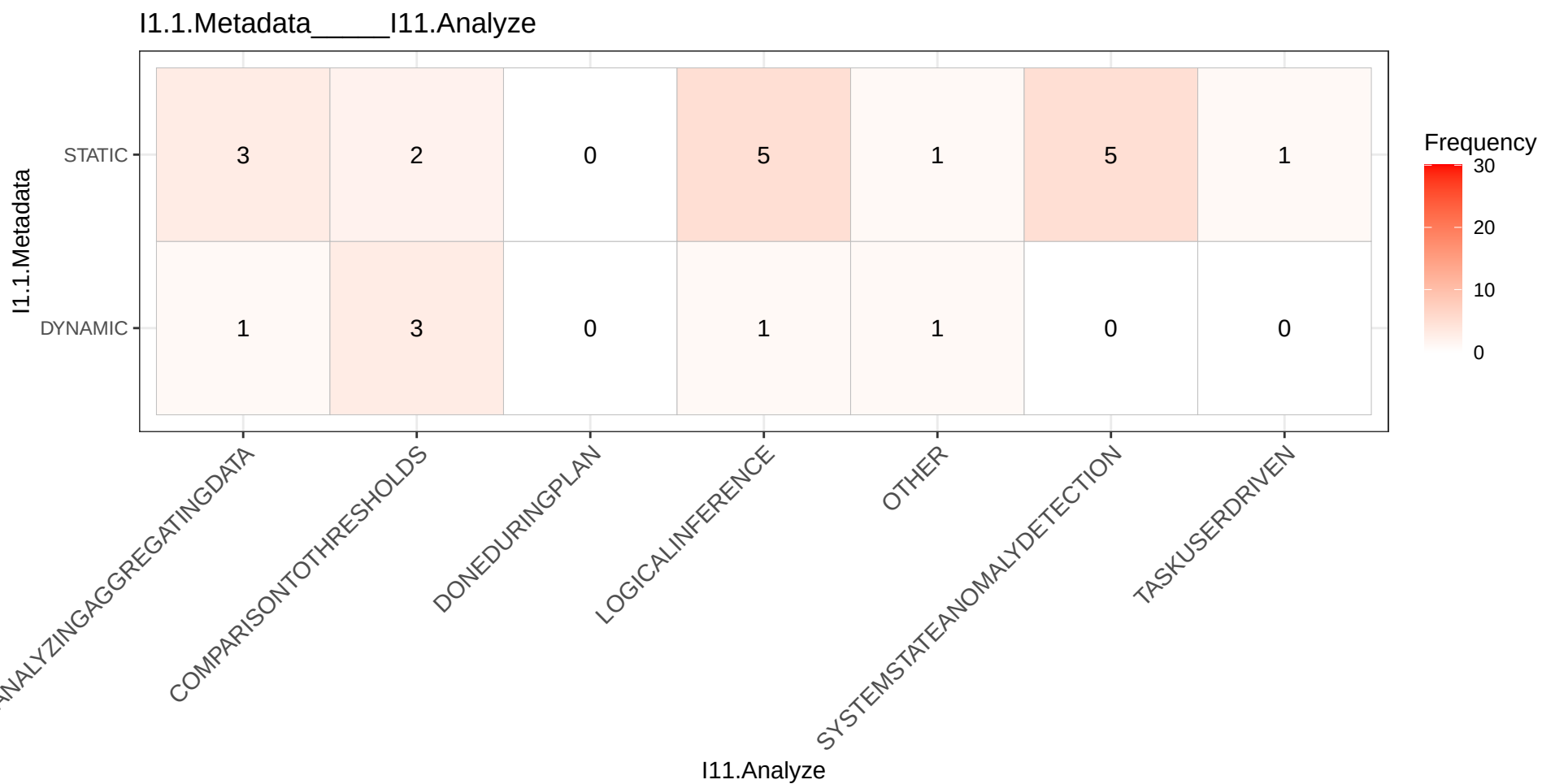
I1.1.Metadata_____Experiment.Method











I1.1.Metadata_____I12.Plan

I1.1.Metadata

STATIC

7

9

2

DYNAMIC

3

1

0

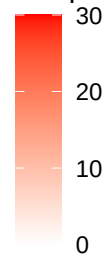
DETERMININGTHEOPTIMALCHOICE

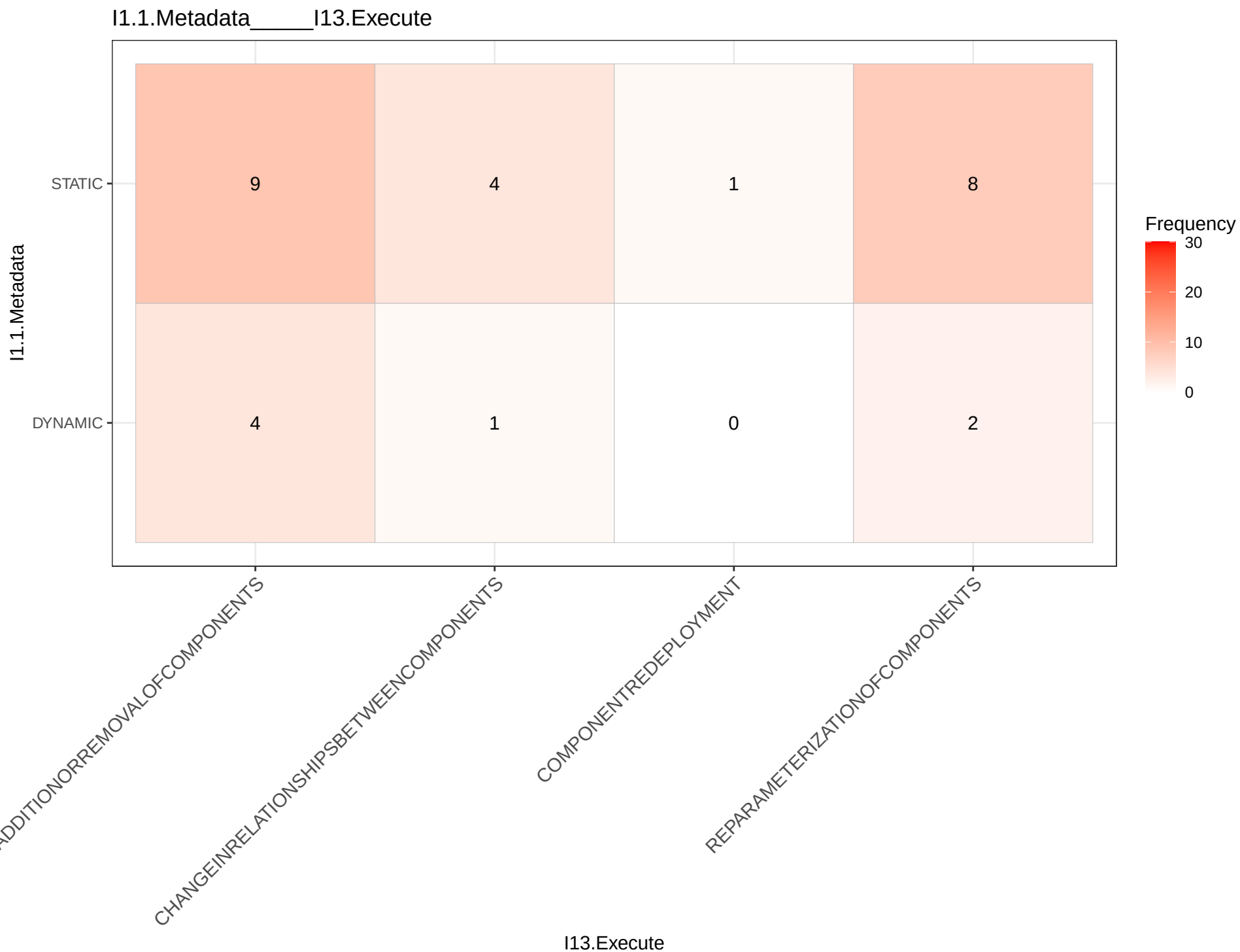
RELYINGONDESIGNTIMERULESMODELS

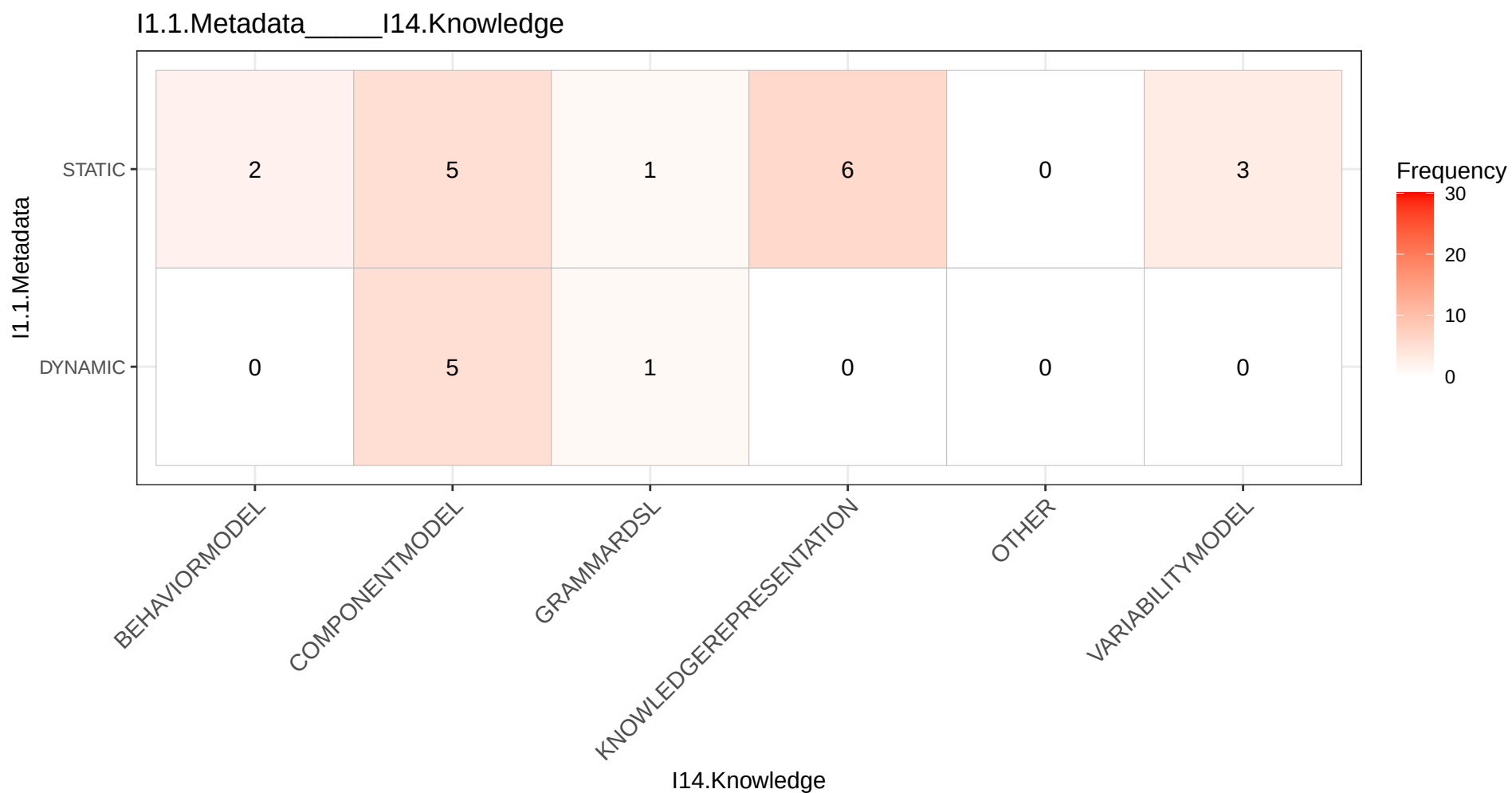
USINGAIPLANNINGLANGUAGES

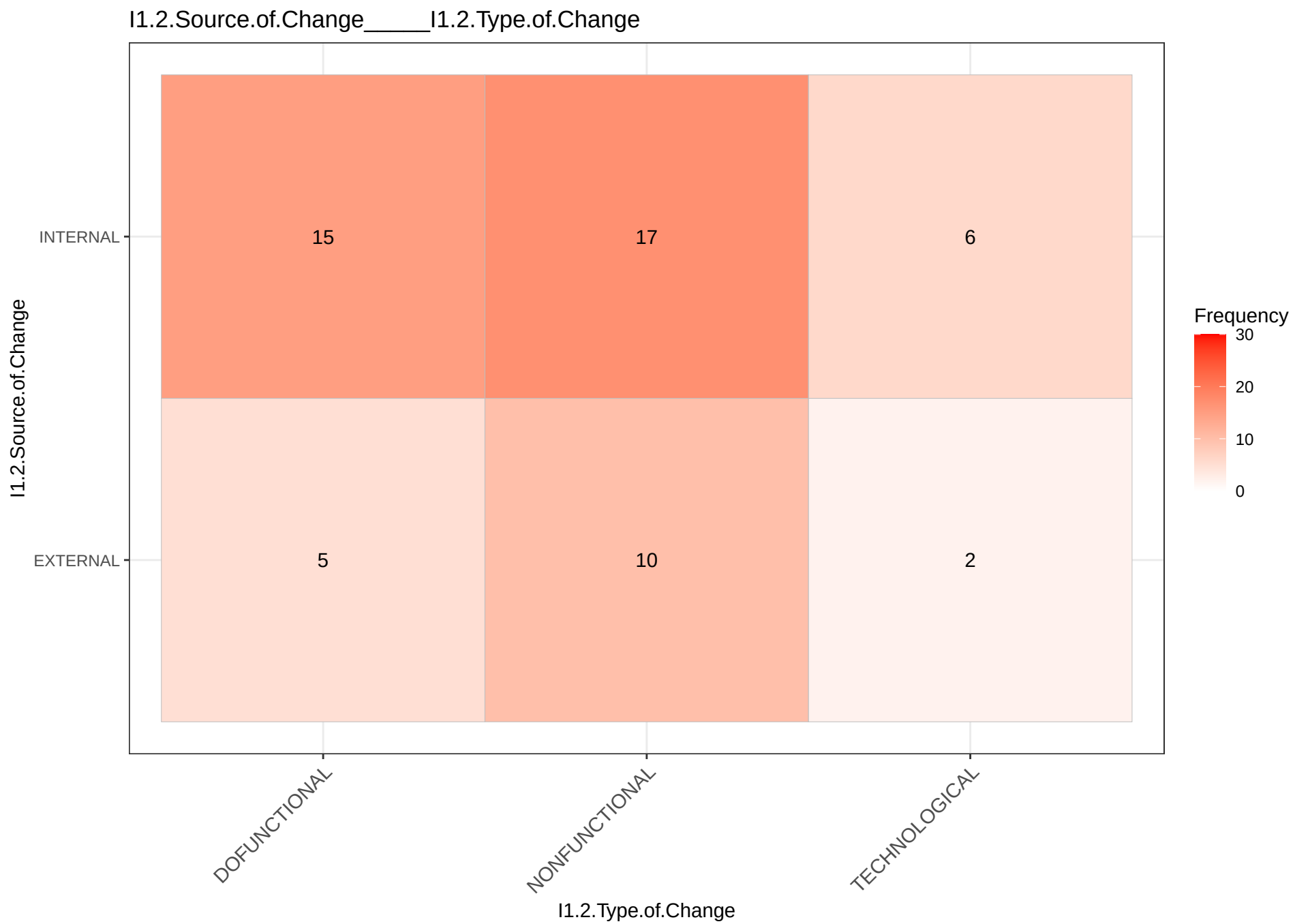
I12.Plan

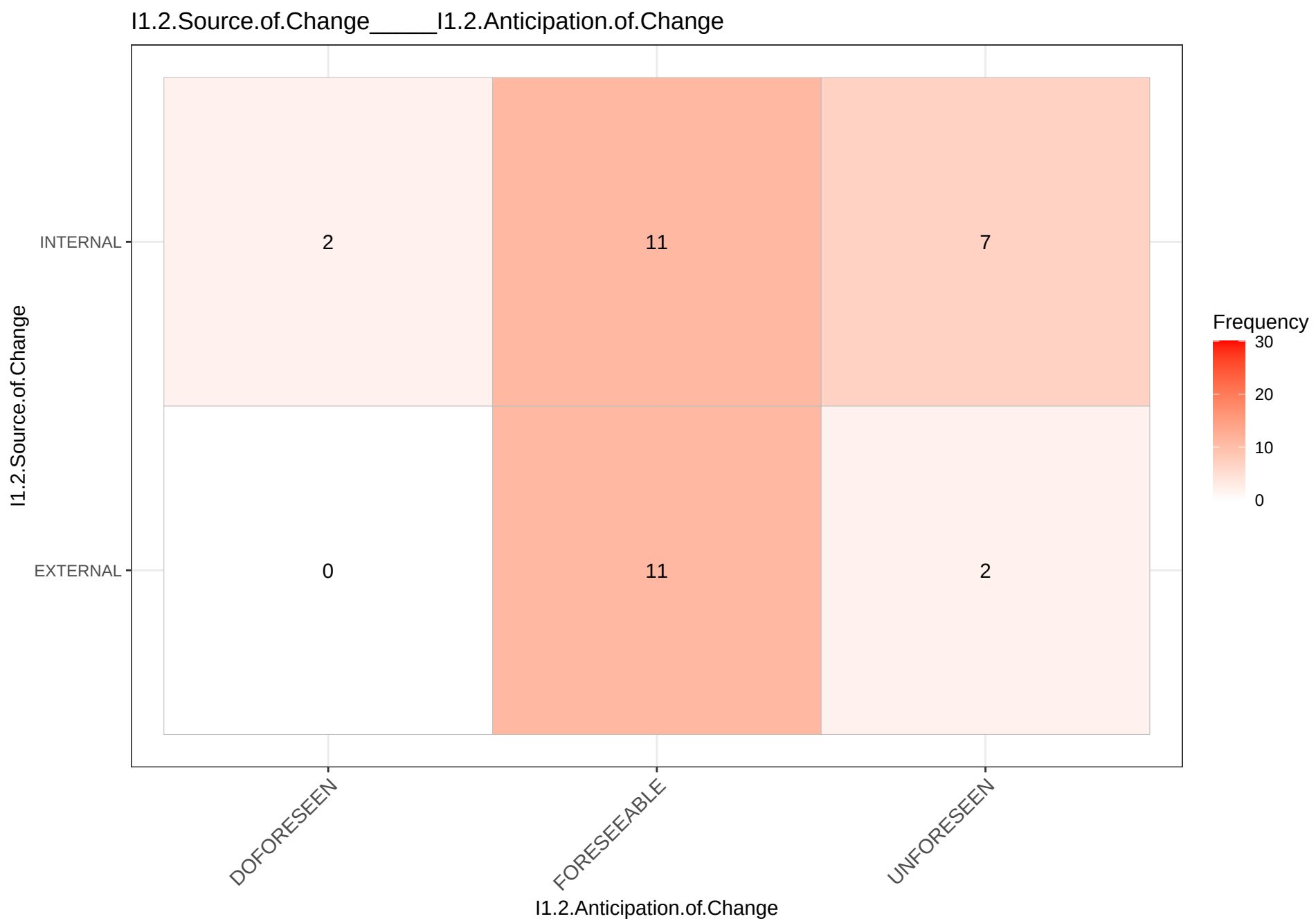
Frequency

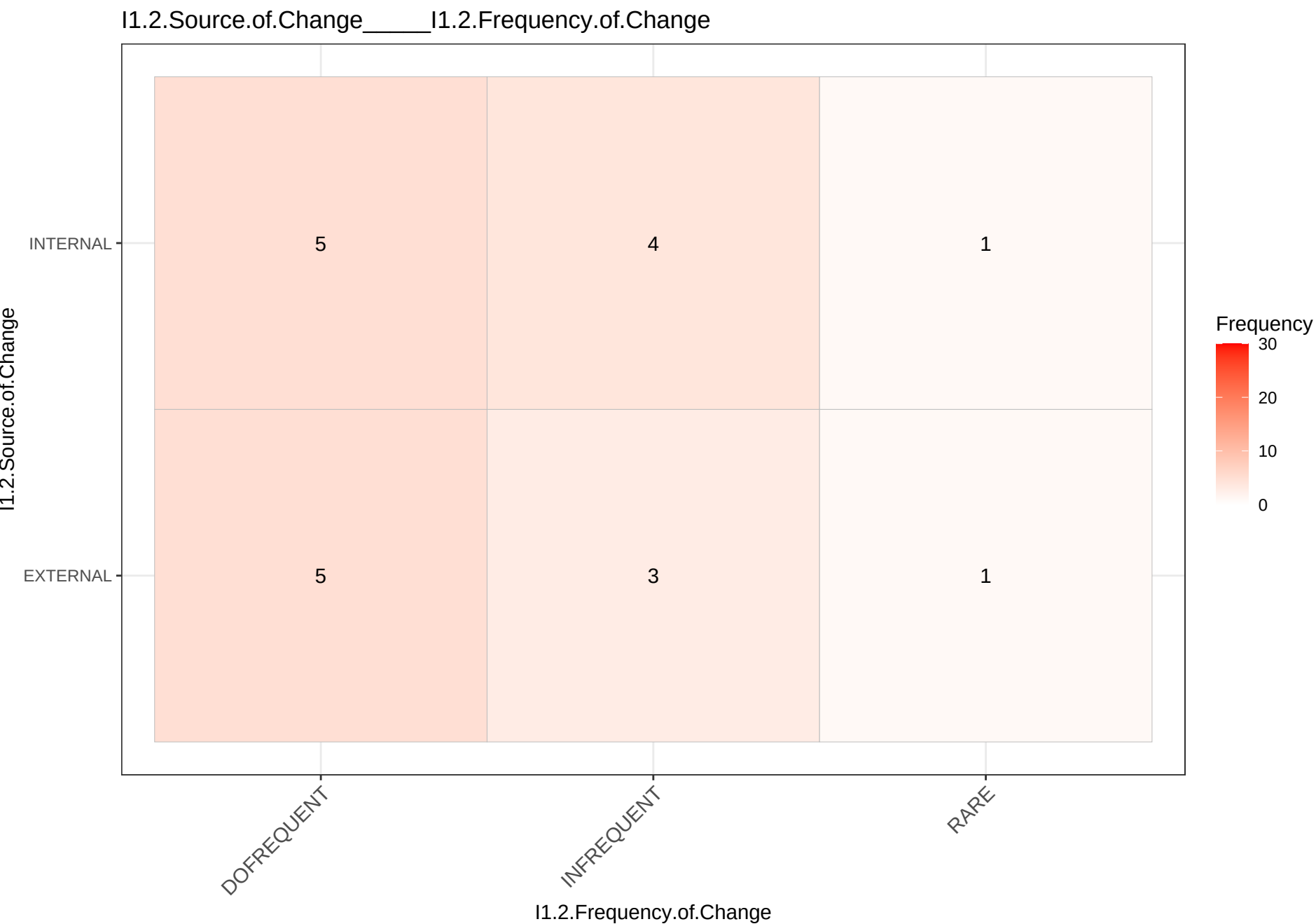












I1.2.Source.of.Change_____I1.3.Type.of.Mechanism

I1.2.Source.of.Change

INTERNAL

14

18

EXTERNAL

10

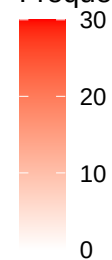
11

PARAMETRIC

STRUCTURAL

I1.3.Type.of.Mechanism

Frequency



I1.2.Source.of.Change_____I1.3.Organization.of.Mechanism

I1.2.Source.of.Change

INTERNAL

4

18

EXTERNAL

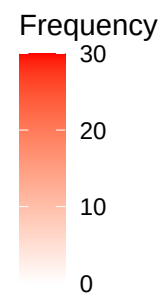
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12

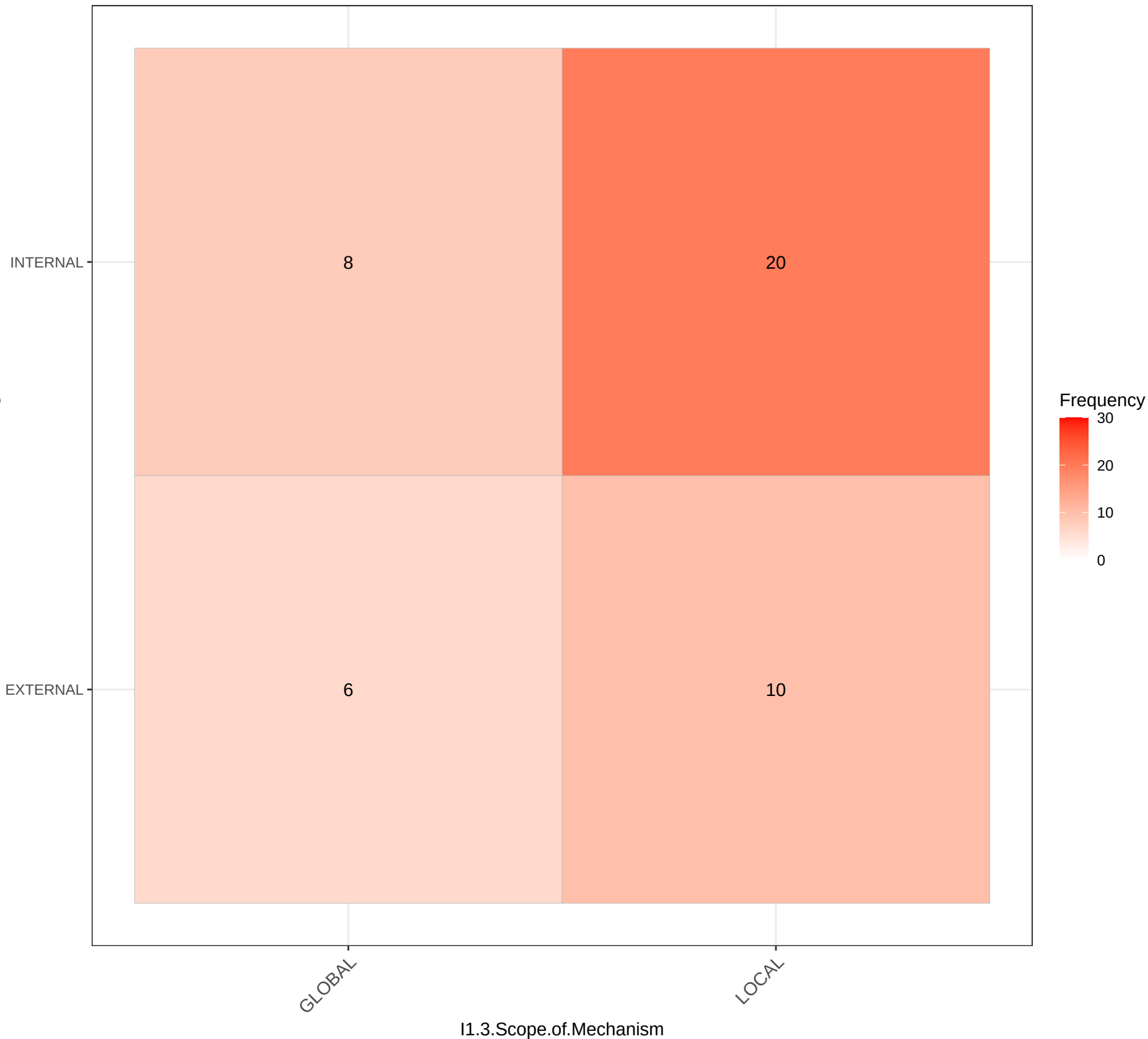
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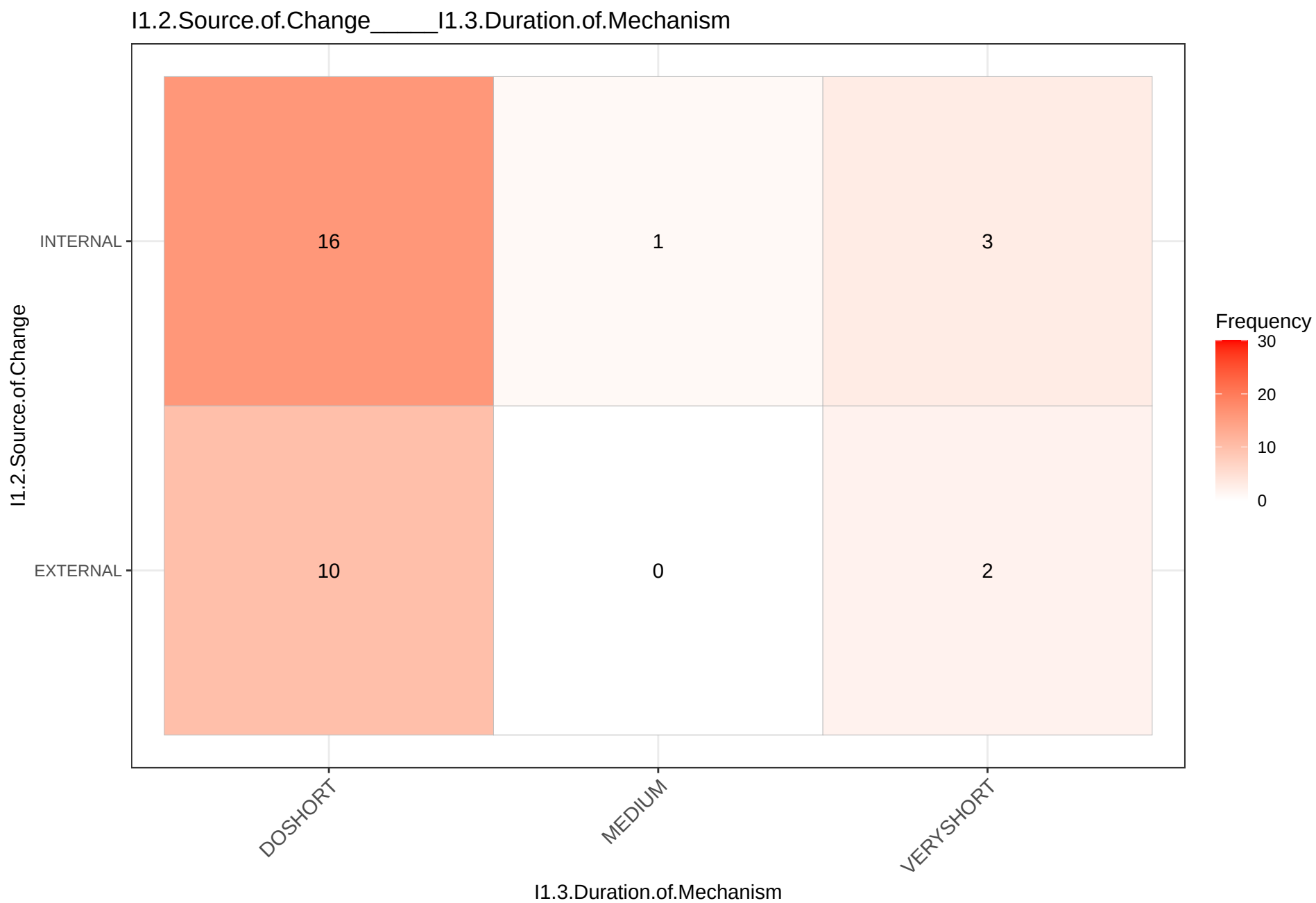
DOCENTRALIZED

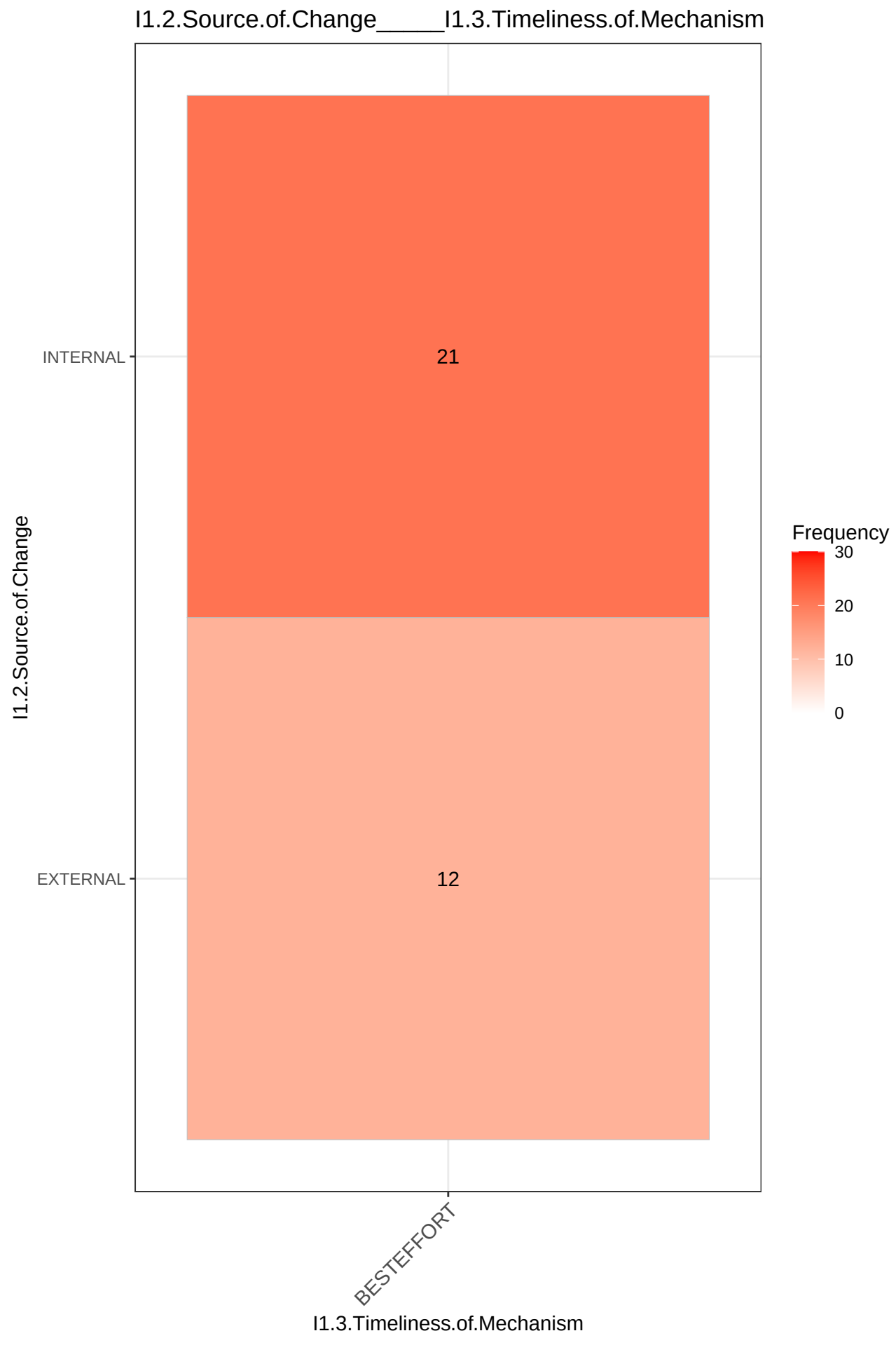
I1.3.Organization.of.Mechanism



I1.2.Source.of.Change_____I1.3.Scope.of.Mechanism







I1.2.Source.of.Change_____I1.3.Trigger.of.Mechanism

I1.2.Source.of.Change

INTERNAL

20

1

EXTERNAL

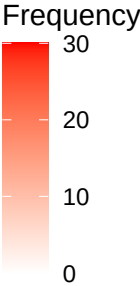
11

2

EVENTTRIGGER

TIMETRIGGER

I1.3.Trigger.of.Mechanism



I1.2.Source.of.Change_____I1.4.Criticality.of.Effects

I1.2.Source.of.Change

INTERNAL

16

4

EXTERNAL

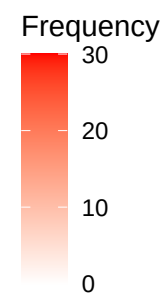
9

4

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects



I1.2.Source.of.Change_____I1.4.Predictability.of.Effects

I1.2.Source.of.Change

INTERNAL

6

14

EXTERNAL

5

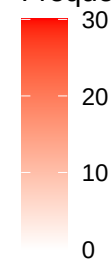
10

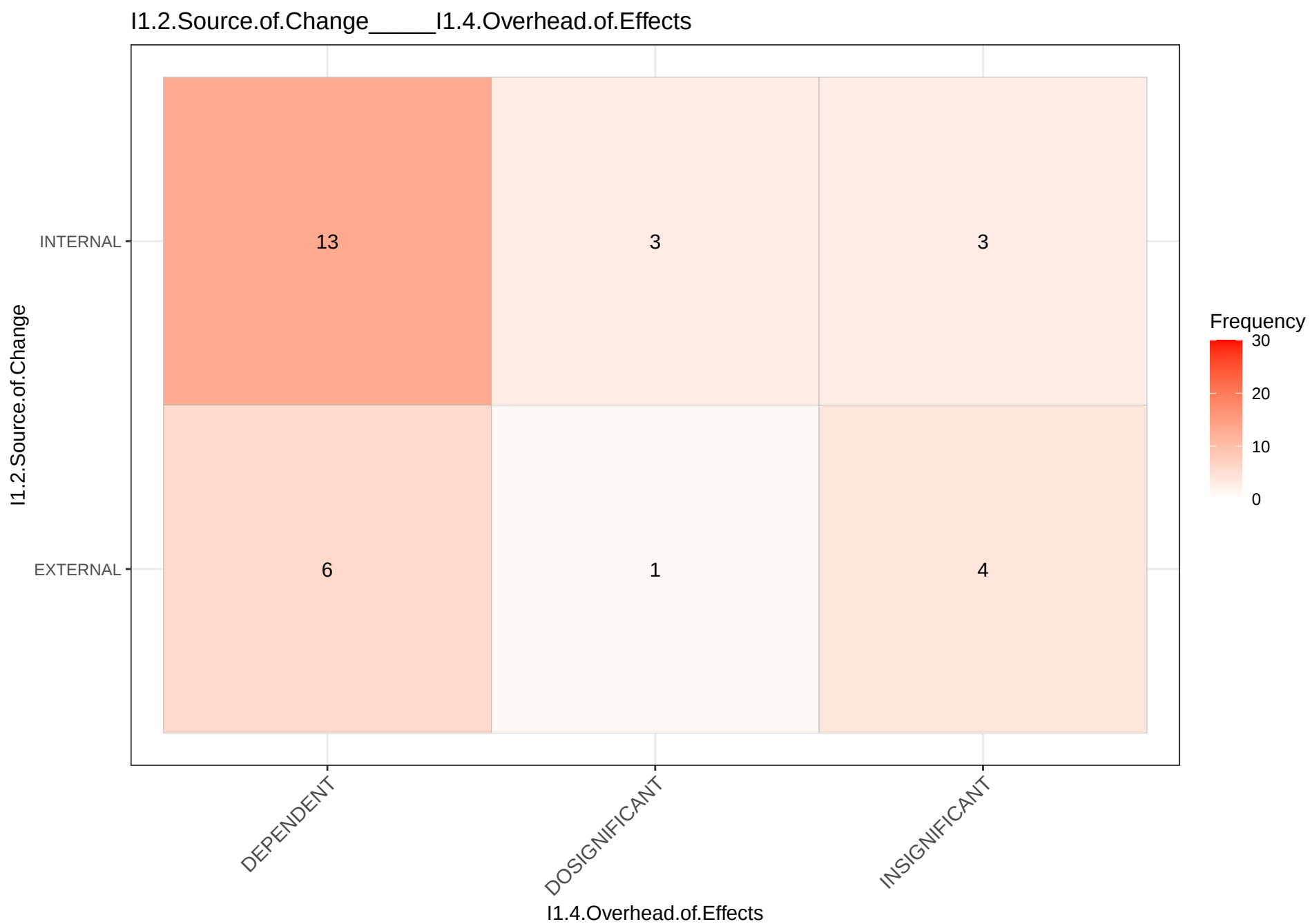
DODETERMINISTIC

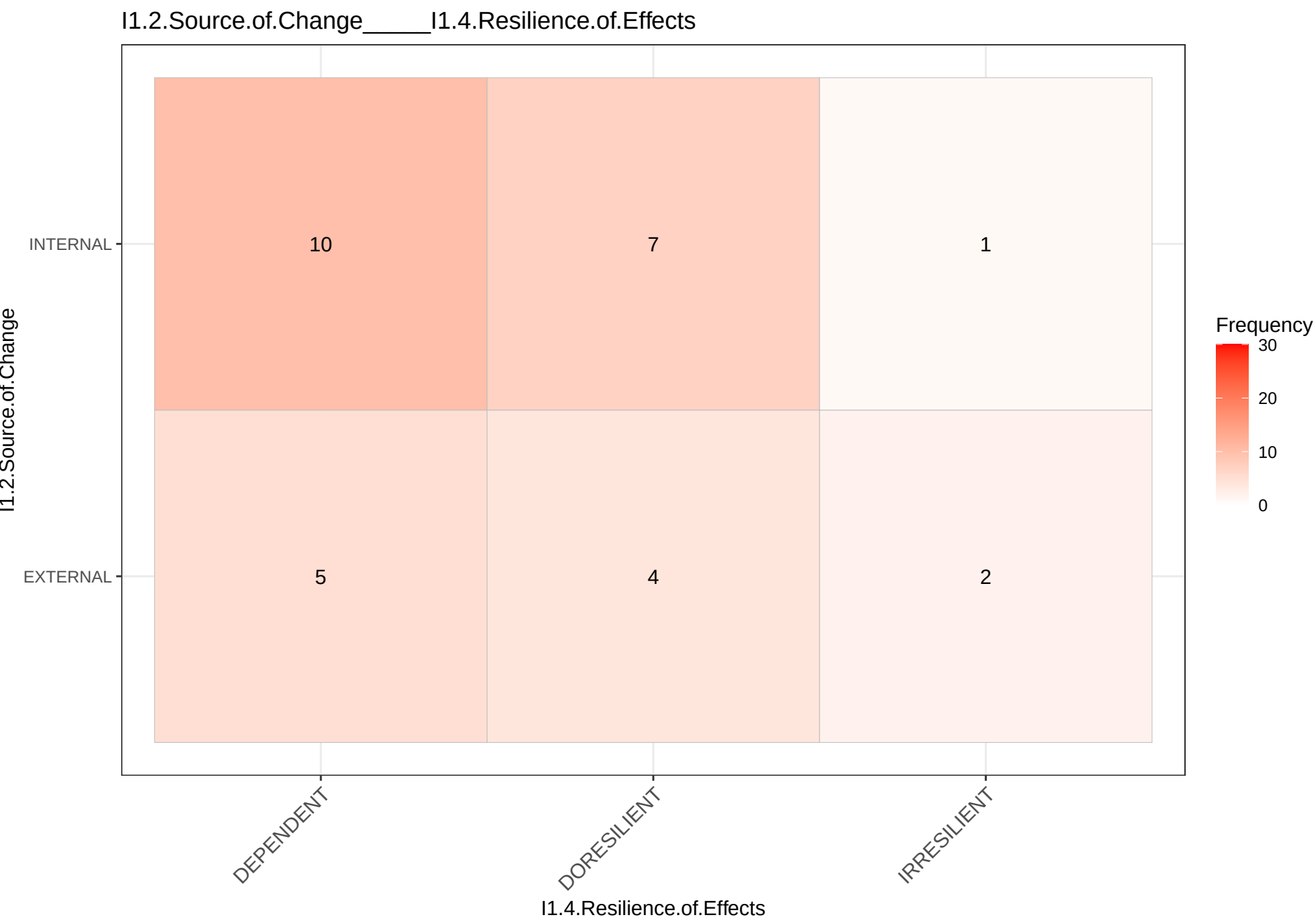
NONDETERMINISTIC

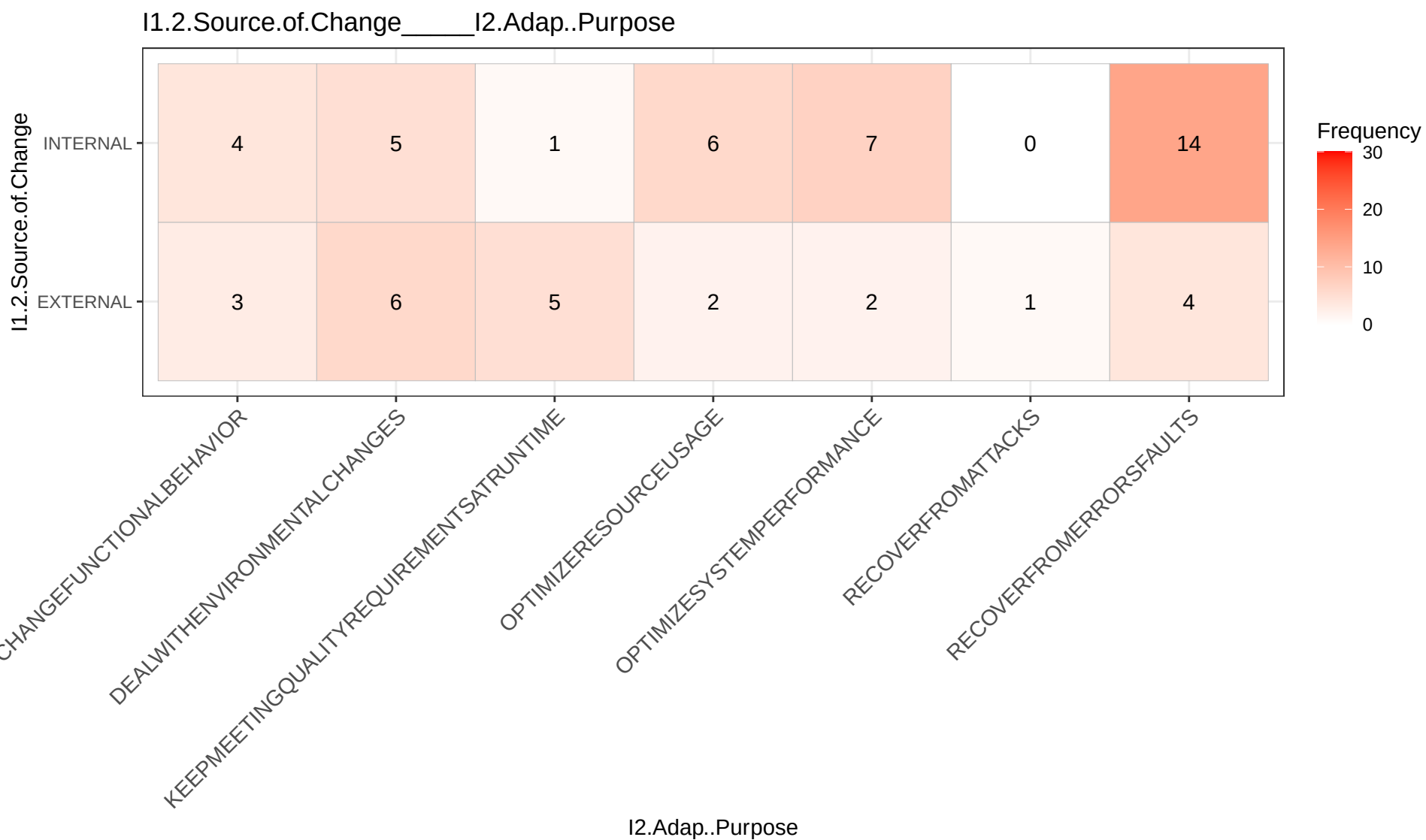
I1.4.Predictability.of.Effects

Frequency



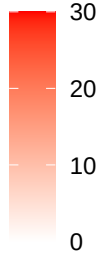






I1.2.Source.of.Change

Frequency



I1.2.Source.of.Change I3.Robot.Type

INTERNAL	1	0	1	1	2	0	2	1	0	1	2	0	1	1	1	1	2	1	0	1	1	1	0	4	0	0	1	0
EXTERNAL	1	1	0	0	0	1	0	0	1	0	0	1	1	1	0	0	0	0	1	0	0	0	1	3	1	1	0	1

BLUE ROV2	BOXER CLEAR PATH	CRAWLER TERMINATOR BOT	FIELD MOBILEROBOTS	HETEROGENOUS ROBOTS	HEXA II	HEX MANIPULATOR	INFOTAINMENT ROBOT	KUKA LIGHTWEIGHT ROBOT 4LR4	MOBILEROBOT	MOBILEROBOT CREATE2	MOBILEROBOTTERRESTRIAL	MOBILEROBOT TIAGO	MSUEVORALLY MOBILEROBOT	MULTIPLE HEX ROTOR	NAO ROBOT	PIONEER 3-DX	QUADRO COPTER	RESCUE	SINGLE SERVOING ROTATION ROBOT	TEDUS ARTERRESTRIAL SEARCH	TRIGLIDE INDUSTRIAL ASSEMBLY	TURTLEBOT	TWO CASE STUDIES MOBILE MANIPULATOR AS RUNNING EXAMPLE	QUADRO COPTOR FOR EVALUATION	TWO INDUSTRIAL ROBOTS	WAREHOUSE DELIVERY ROBOT	WHICH IS AN INDUSTRIAL AGV
-----------	------------------	------------------------	--------------------	---------------------	---------	-----------------	--------------------	-----------------------------	-------------	---------------------	------------------------	-------------------	-------------------------	--------------------	-----------	--------------	---------------	--------	--------------------------------	----------------------------	------------------------------	-----------	--	------------------------------	-----------------------	--------------------------	----------------------------

I3.Robot.Type

I1.2.Source.of.Change_____I4.Robo.SW

I1.2.Source.of.Change

INTERNAL

EXTERNAL

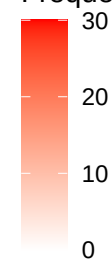
OTHER

ROS1

ROS2

I4.Robo.SW

Frequency



6

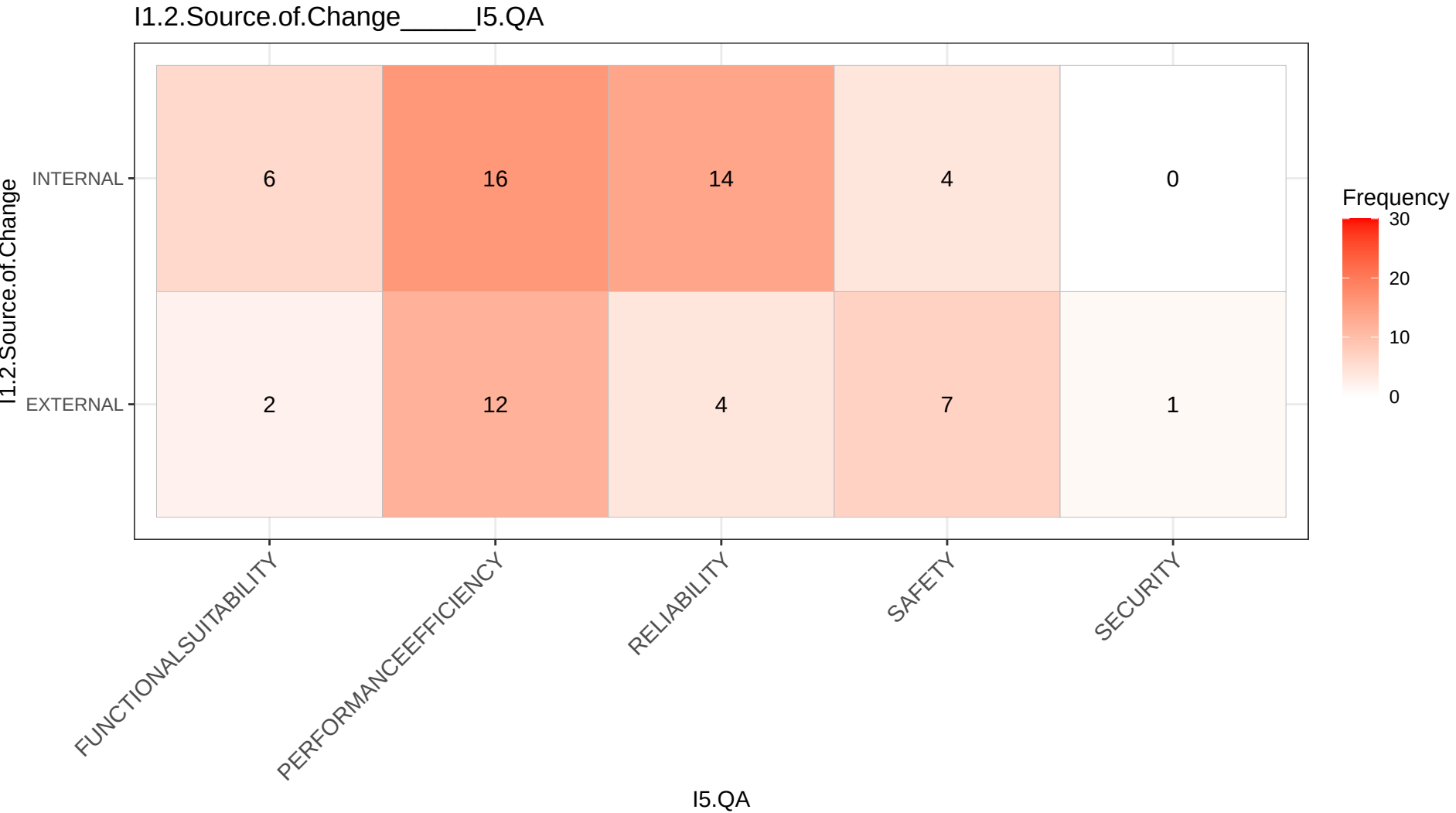
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2

2

5

2



I1.2.Source.of.Change_____I6.Independence

I1.2.Source.of.Change

INTERNAL

EXTERNAL

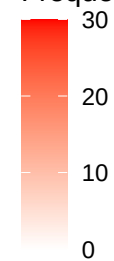
DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



8

2

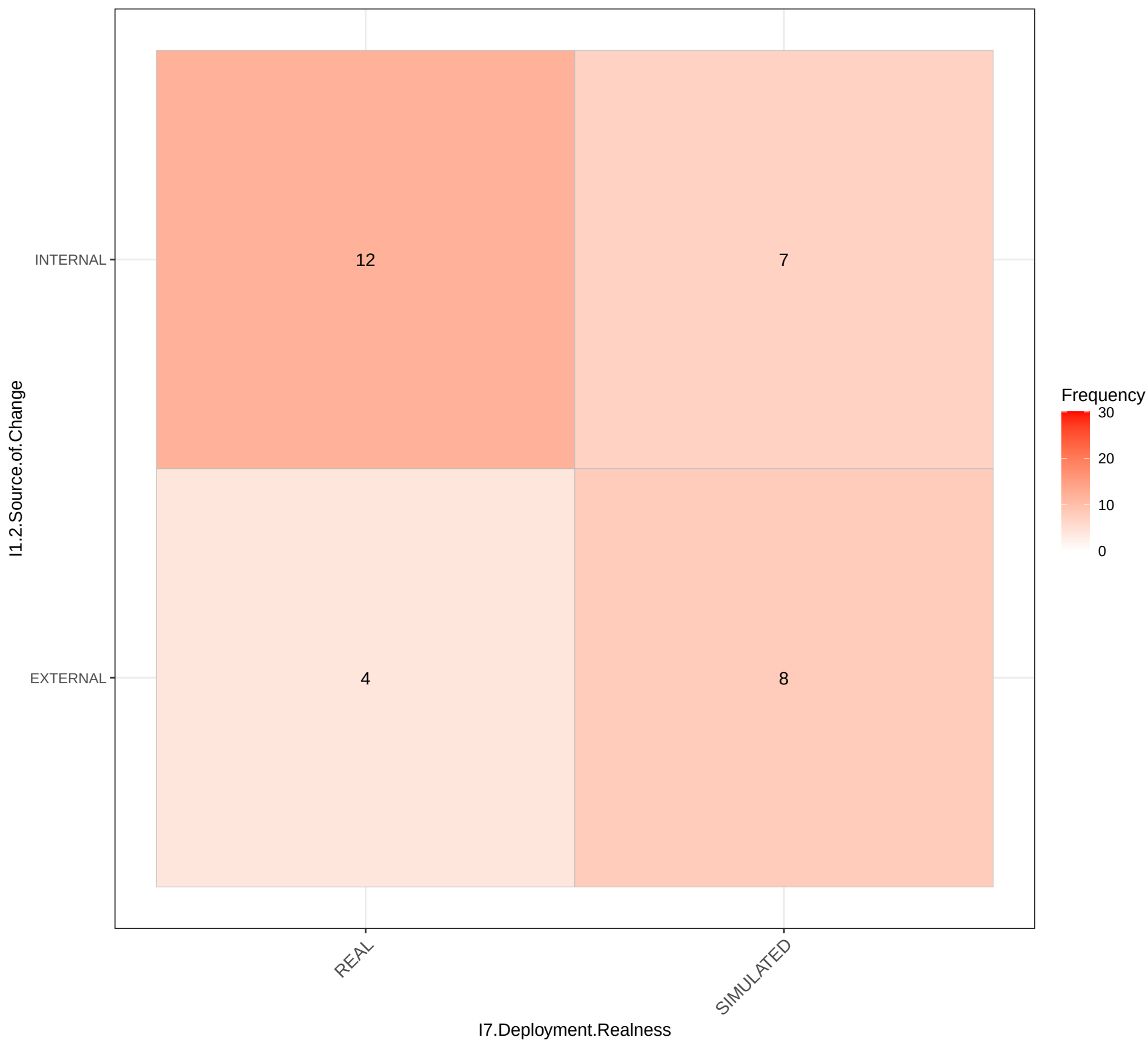
12

3

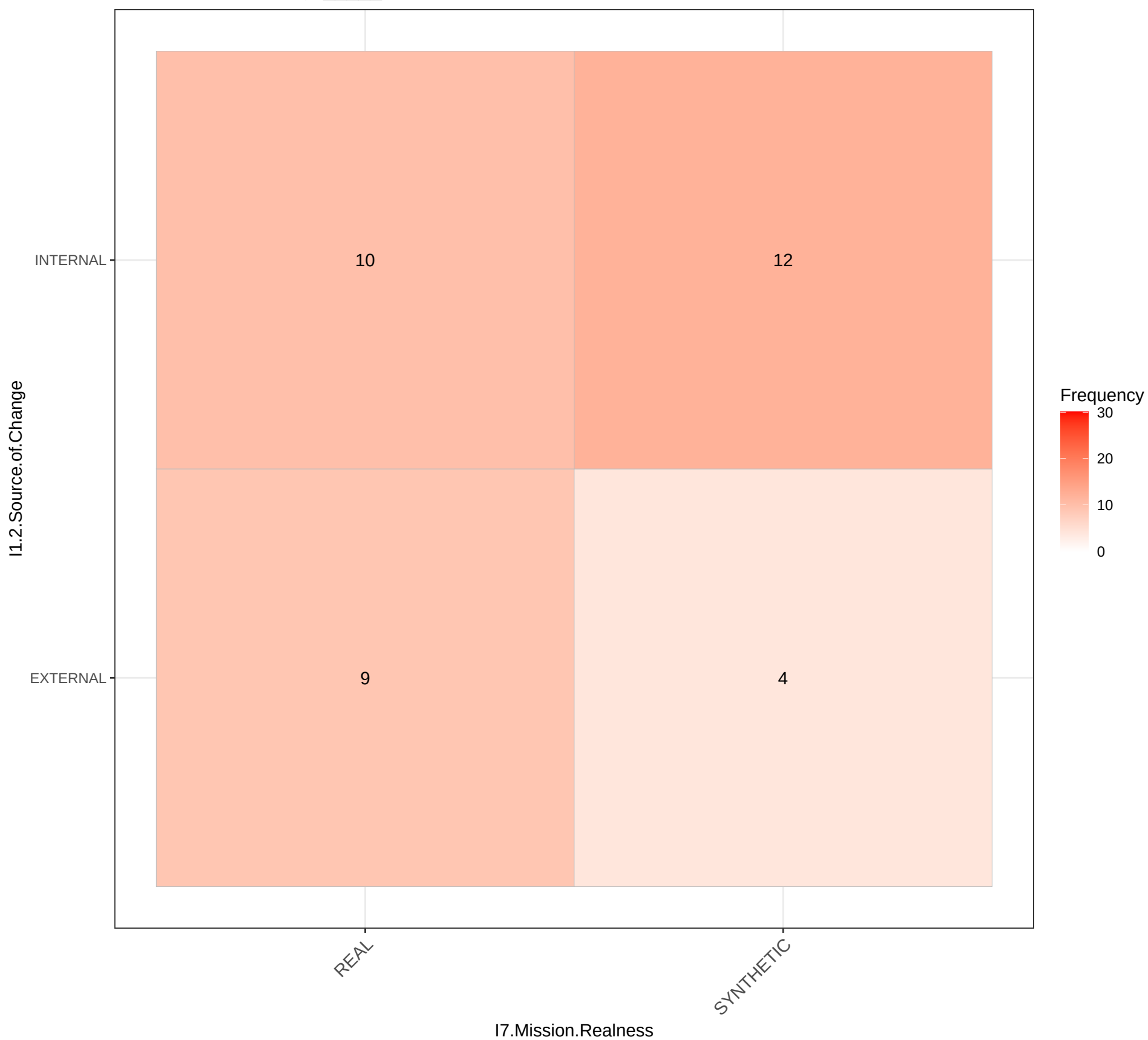
1

9

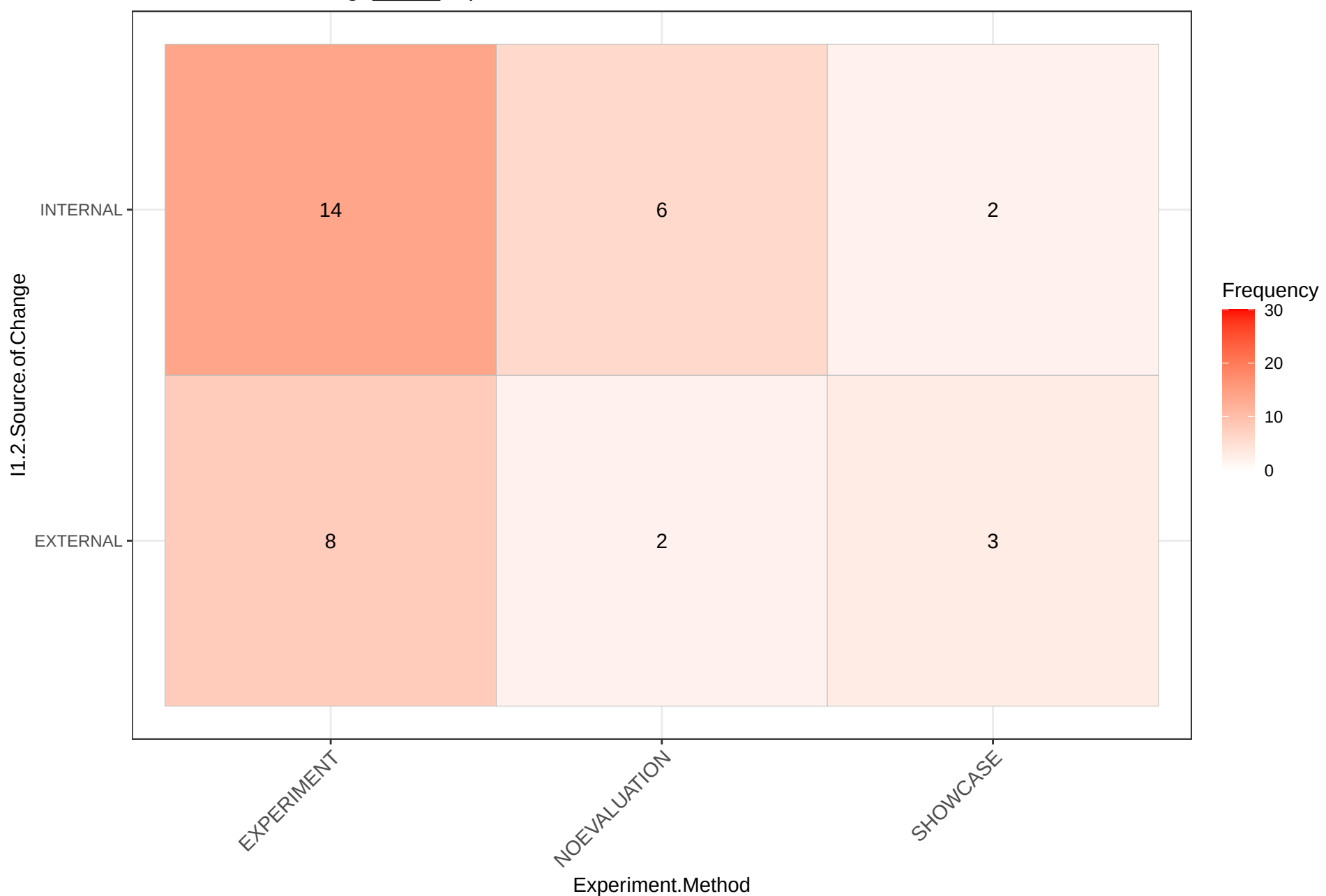
I1.2.Source.of.Change_____I7.Deployment.Realness



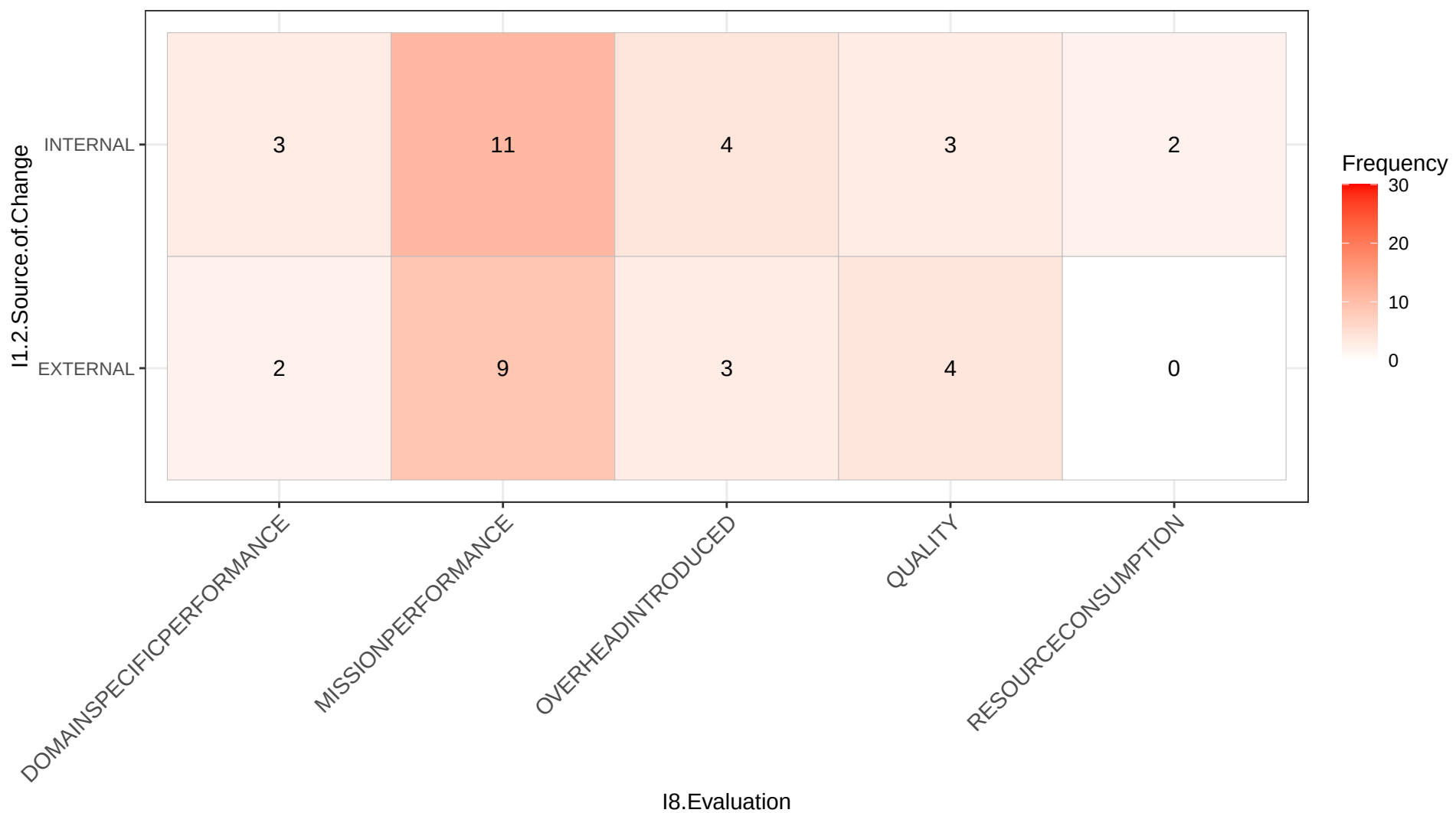
I1.2.Source.of.Change ____ I7.Mission.Realness

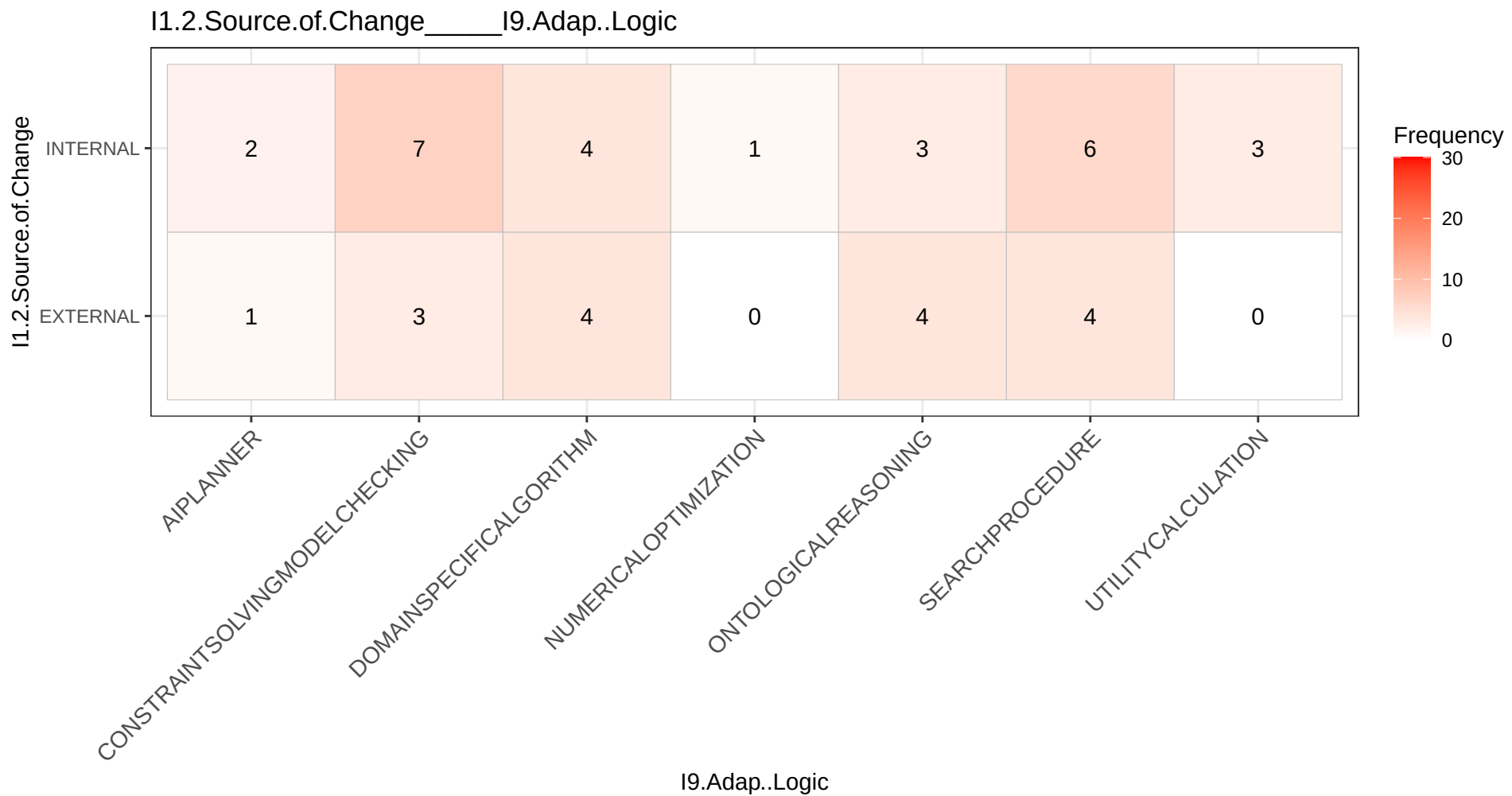


I1.2.Source.of.Change_____Experiment.Method

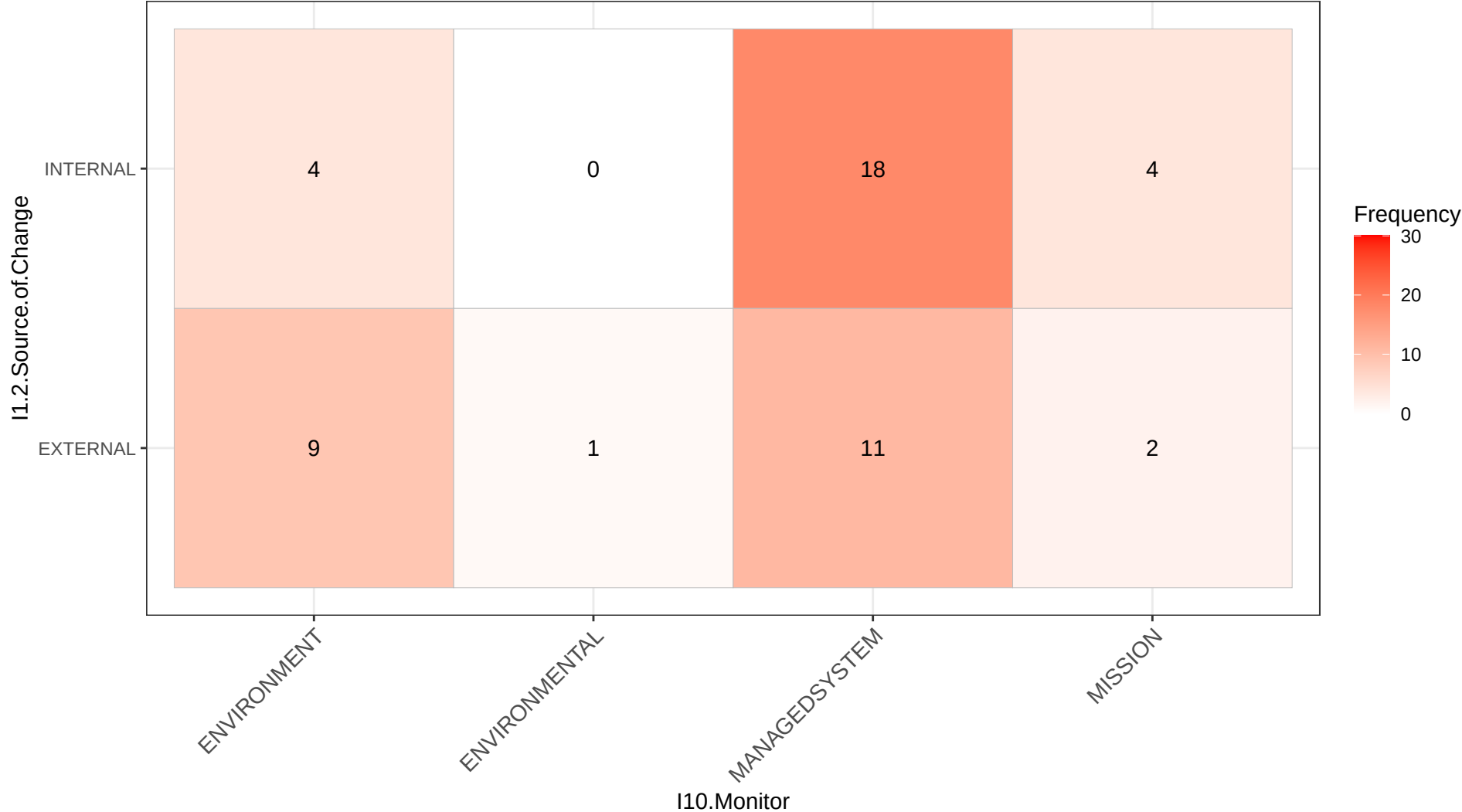


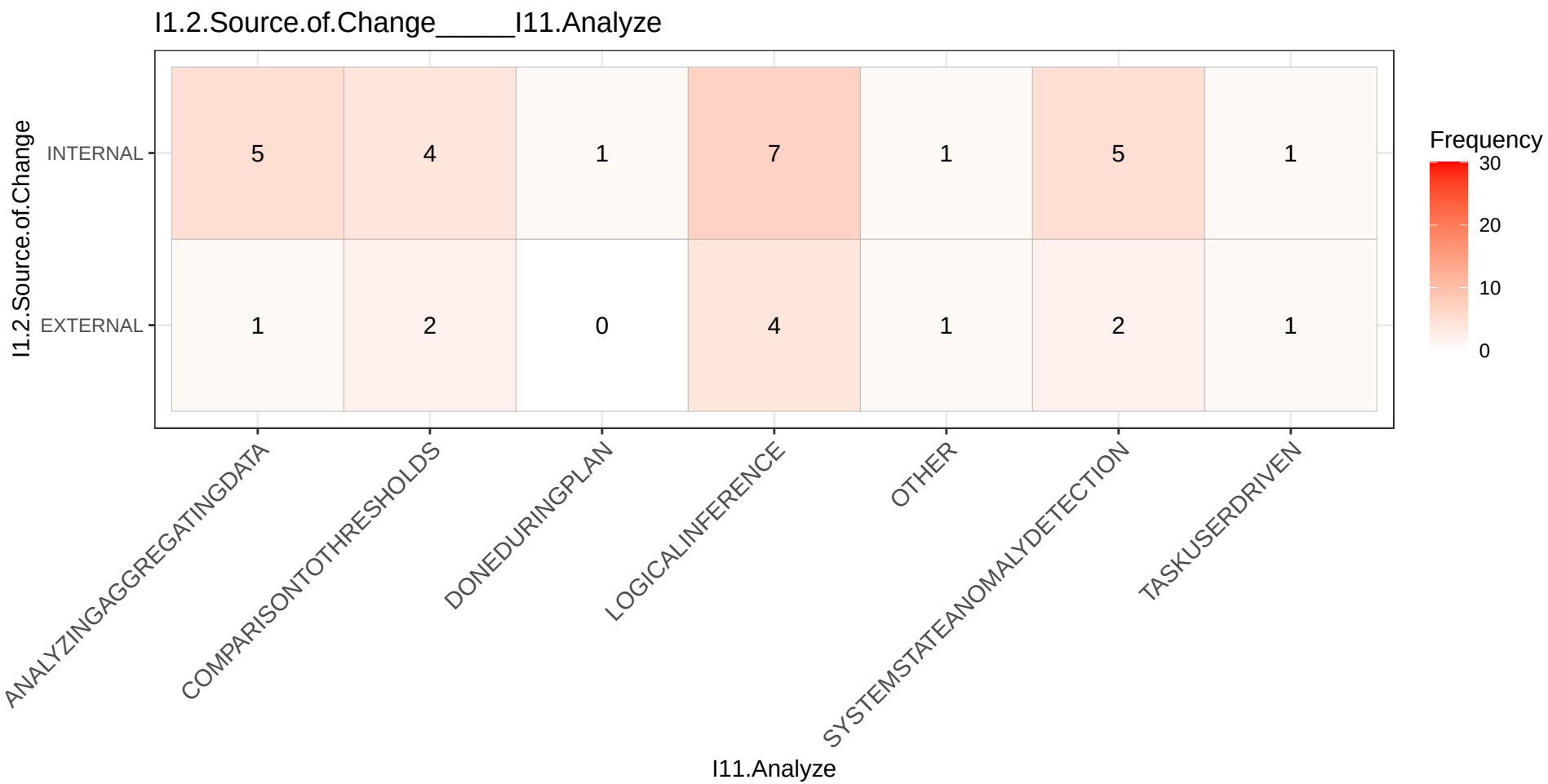
I1.2.Source.of.Change_____I8.Evaluation





I1.2.Source.of.Change_____I10.Monitor





I1.2.Source.of.Change _____ I12.Plan

I1.2.Source.of.Change

INTERNAL

11

8

2

EXTERNAL

6

6

0

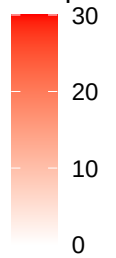
DETERMININGTHEOPTIMALCHOICE

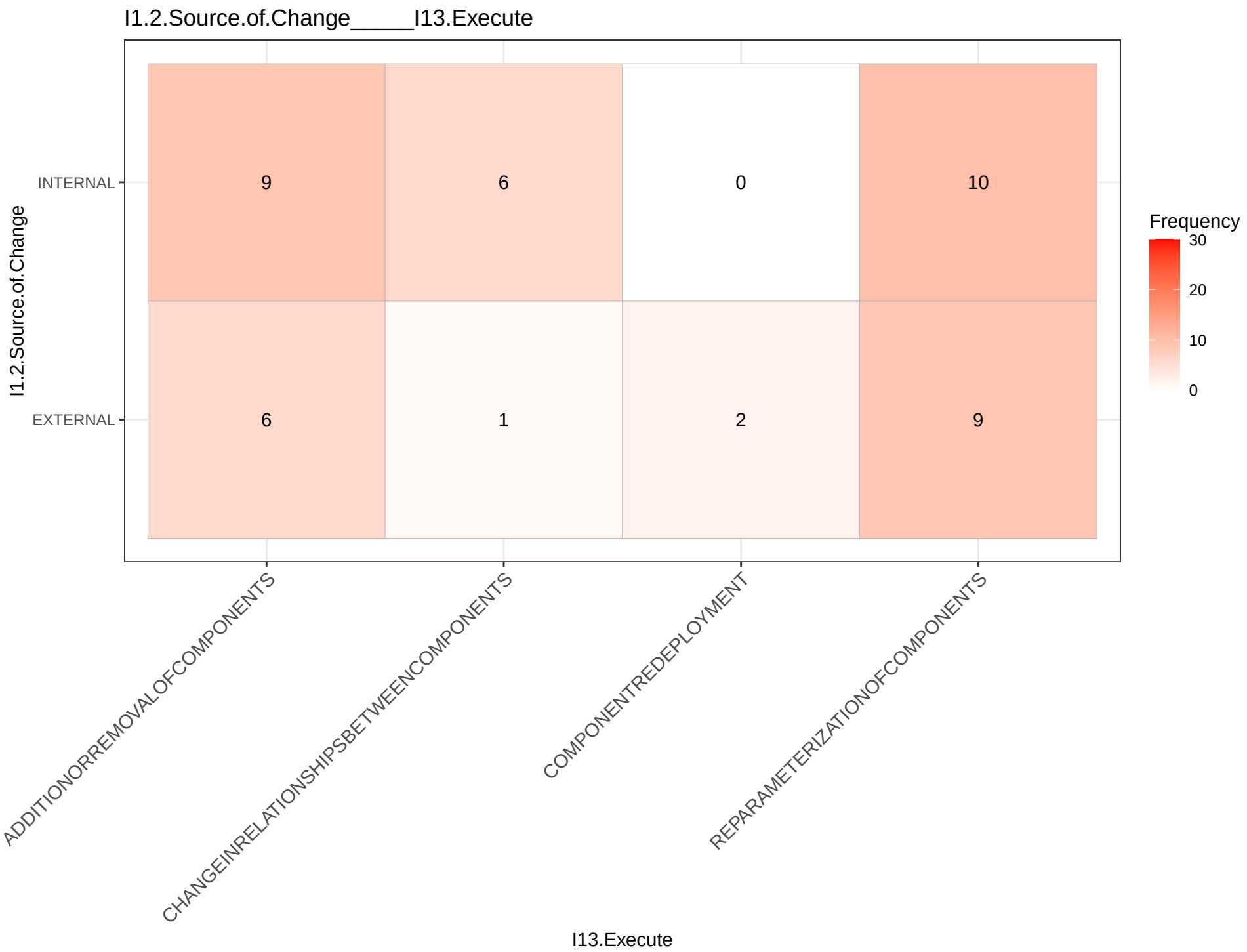
RELYINGONDESIGNTIMERULESMODELS

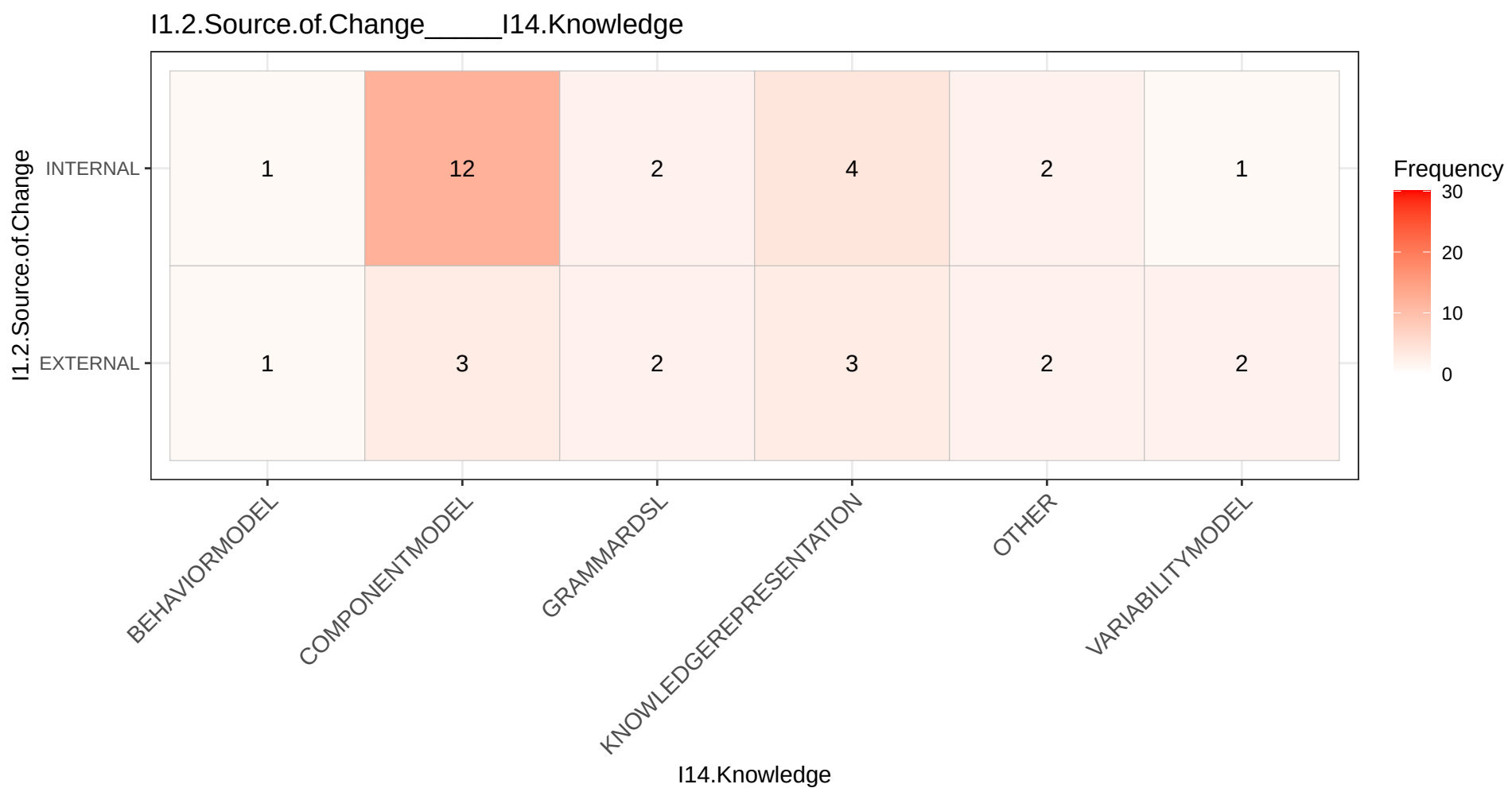
USINGAIPANNINGLANGUAGES

I12.Plan

Frequency







I1.2.Type.of.Change_____I1.2.Anticipation.of.Change

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

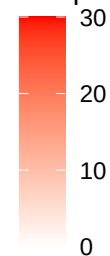
DOFORESEEN

FORESEEABLE

UNFORESEEN

I1.2.Anticipation.of.Change

Frequency



0

5

2

0

15

6

2

9

6

I1.2.Type.of.Change_____I1.2.Frequency.of.Change

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

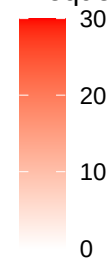
DOFREQUENT

INFREQUENT

RARE

I1.2.Frequency.of.Change

Frequency



0

1

1

6

4

0

3

3

1

I1.2.Type.of.Change_____I1.3.Type.of.Mechanism

I1.2.Type.of.Change

TECHNOLOGICAL

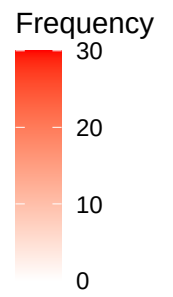
NONFUNCTIONAL

DOFUNCTIONAL

PARAMETRIC

STRUCTURAL

I1.3.Type.of.Mechanism



3

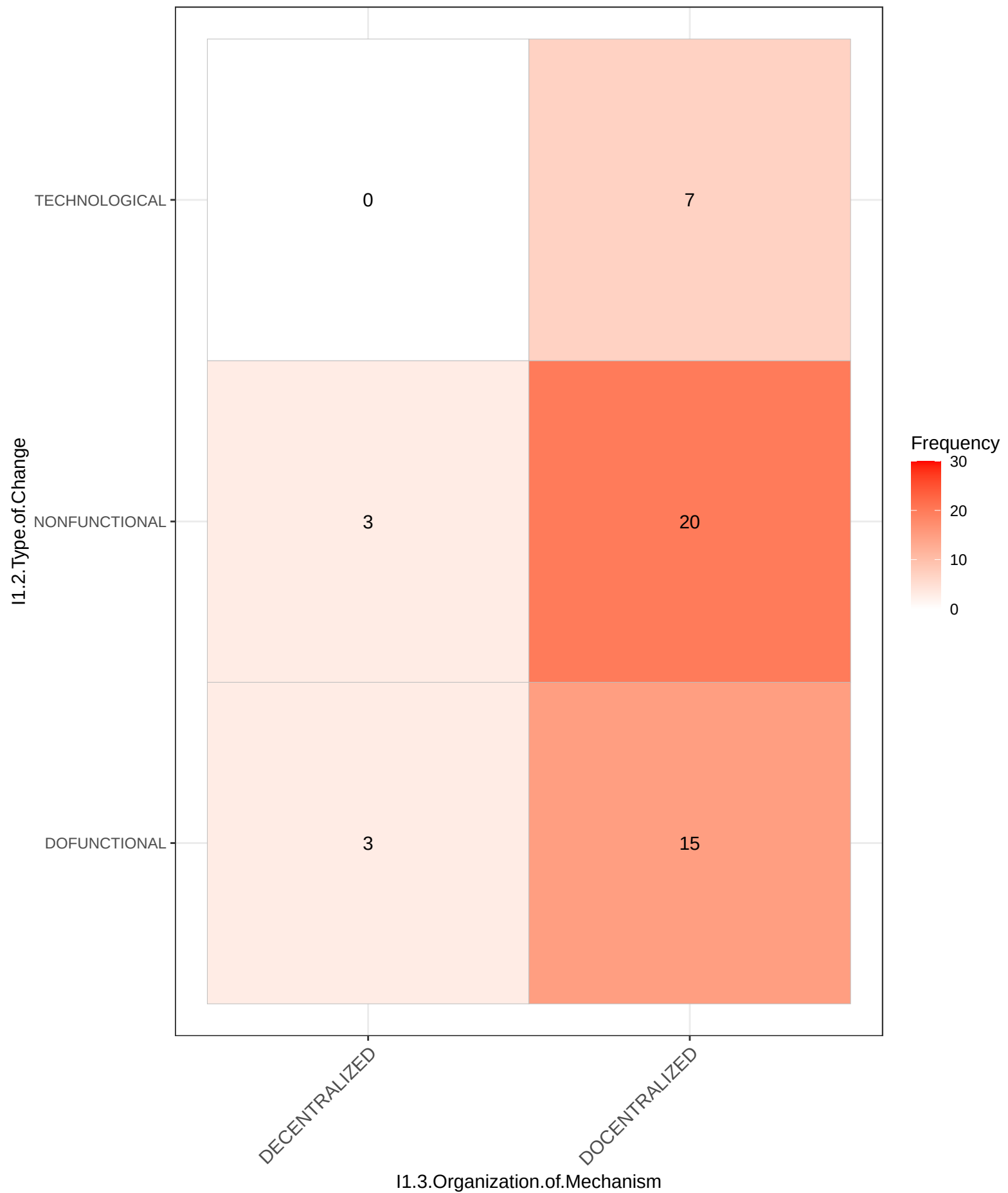
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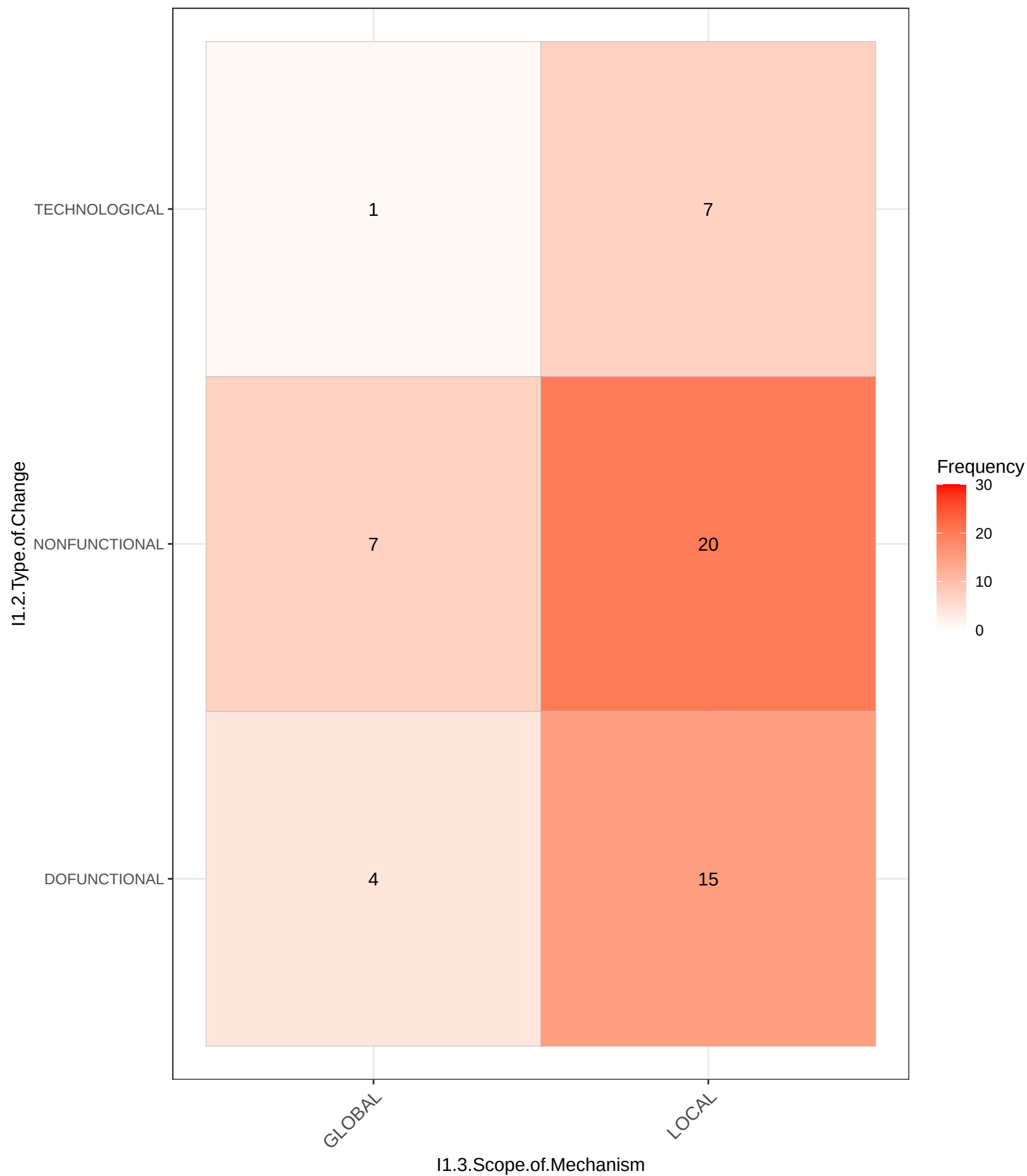
15

20

9

16





I1.2.Type.of.Change_____I1.3.Duration.of.Mechanism

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

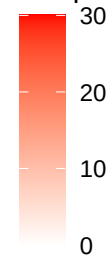
DOSHORT

MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism

Frequency



7

0

0

17

0

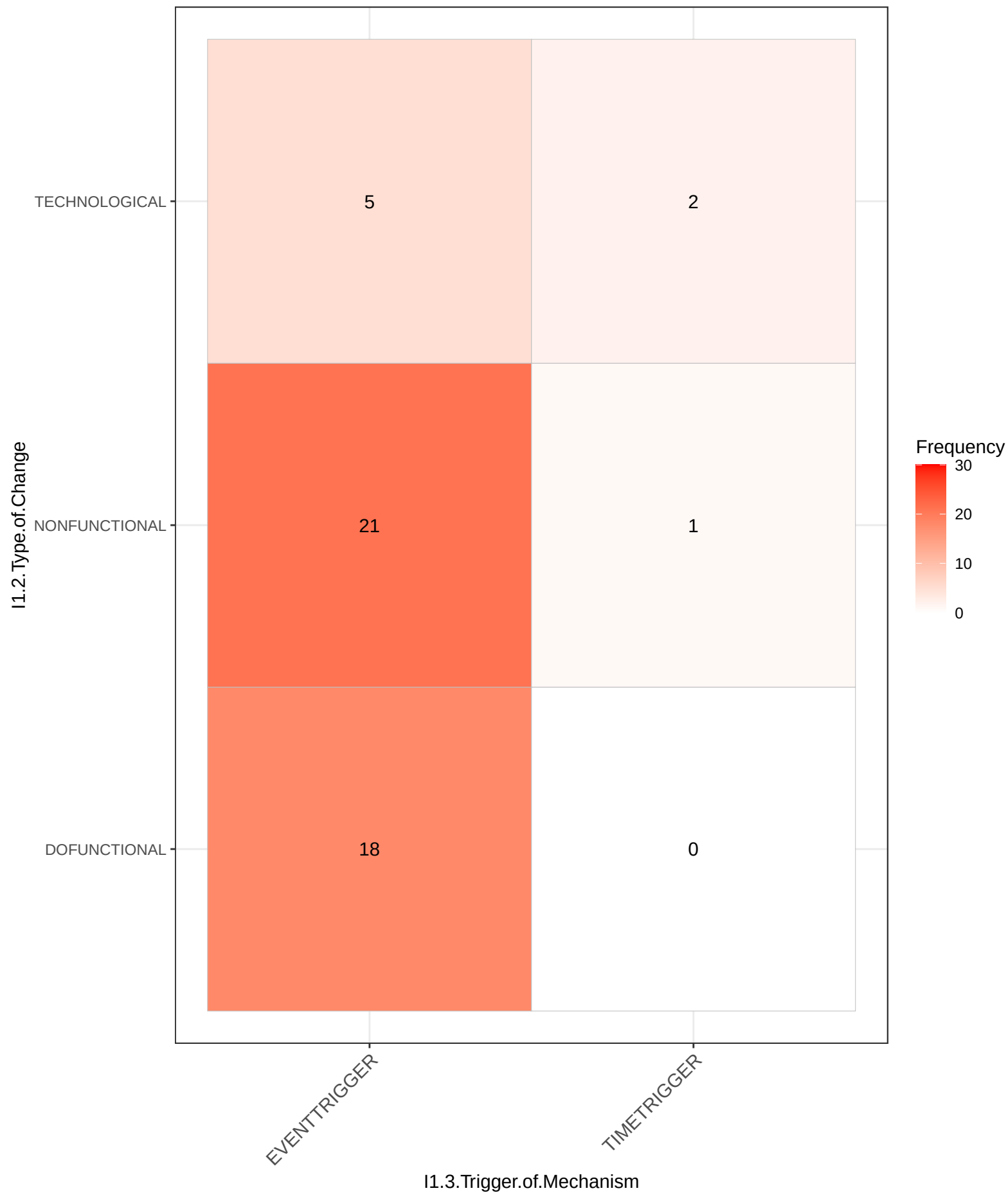
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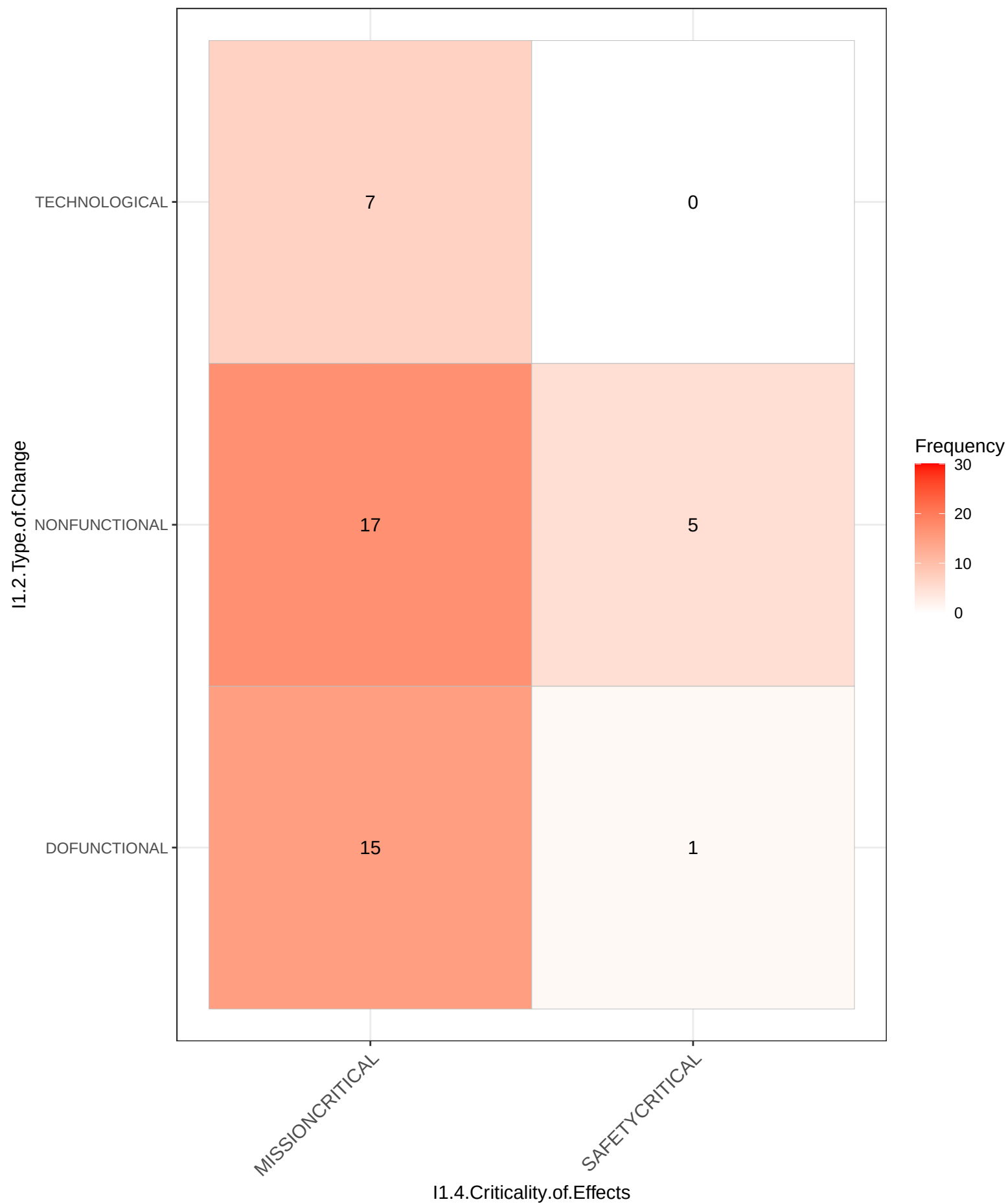
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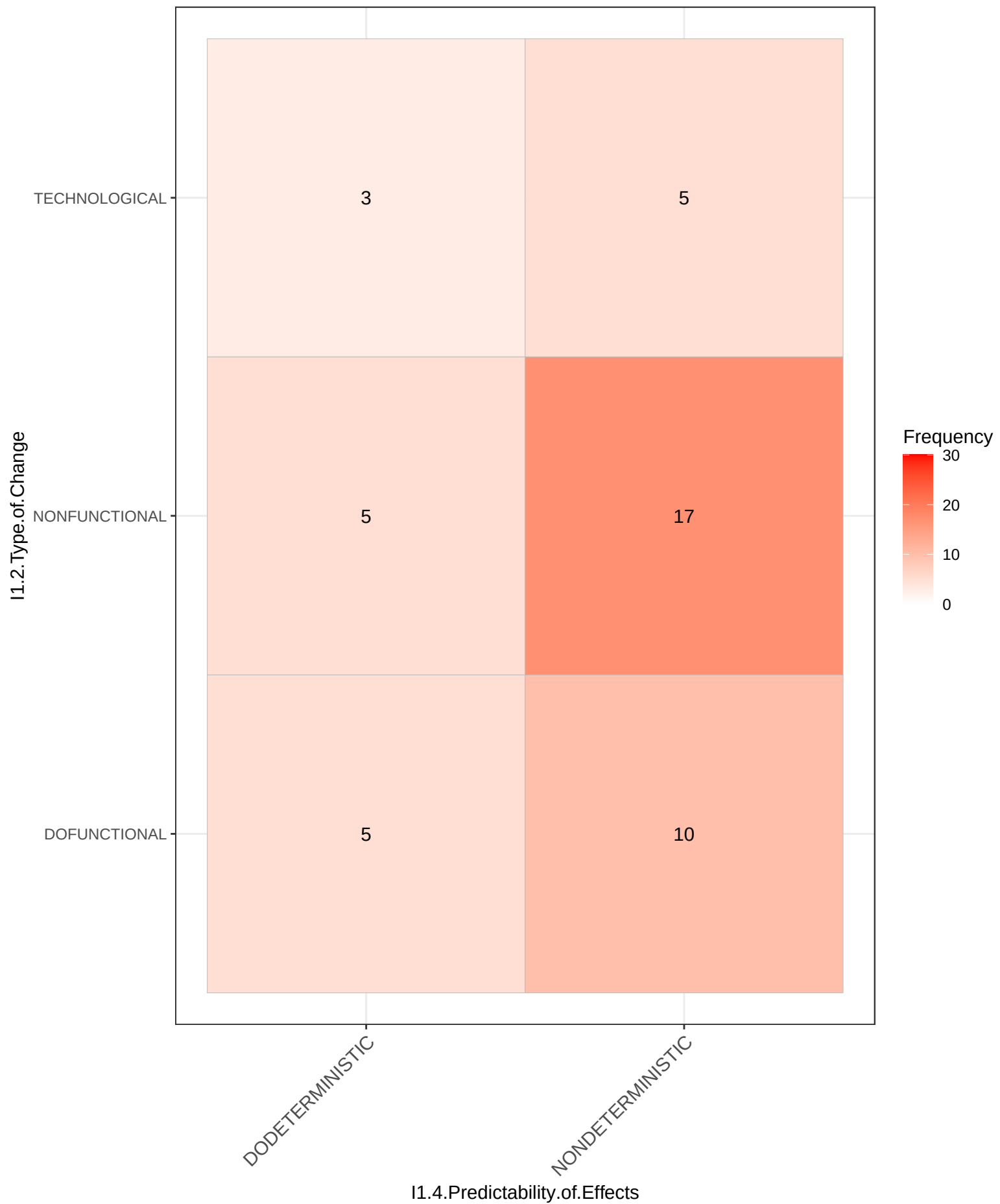
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I1.2.Type.of.Change_____I1.3.Trigger.of.Mechanism





I1.2.Type.of.Change_____I1.4.Predictability.of.Effects



I1.2.Type.of.Change_____I1.4.Overhead.of.Effects

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

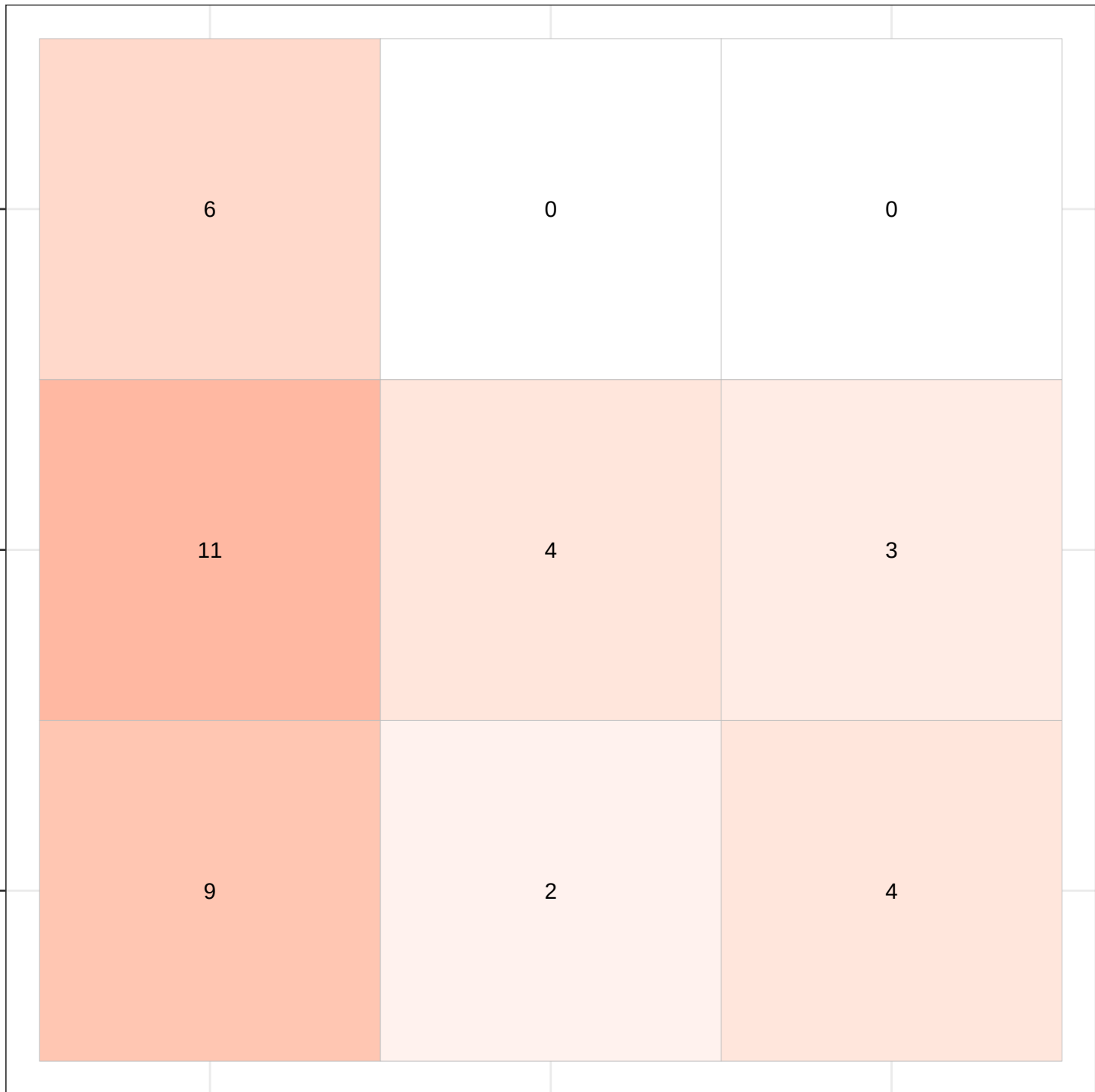
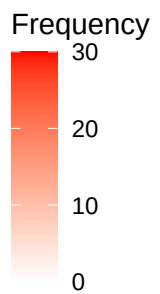
DOFUNCTIONAL

DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.2.Type.of.Change_____I1.4.Resilience.of.Effects

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

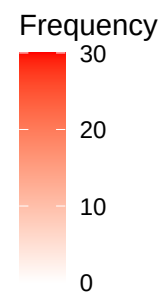
DOFUNCTIONAL

DEPENDENT

DORESILIENT

IRRRESILIENT

I1.4.Resilience.of.Effects



6

0

1

7

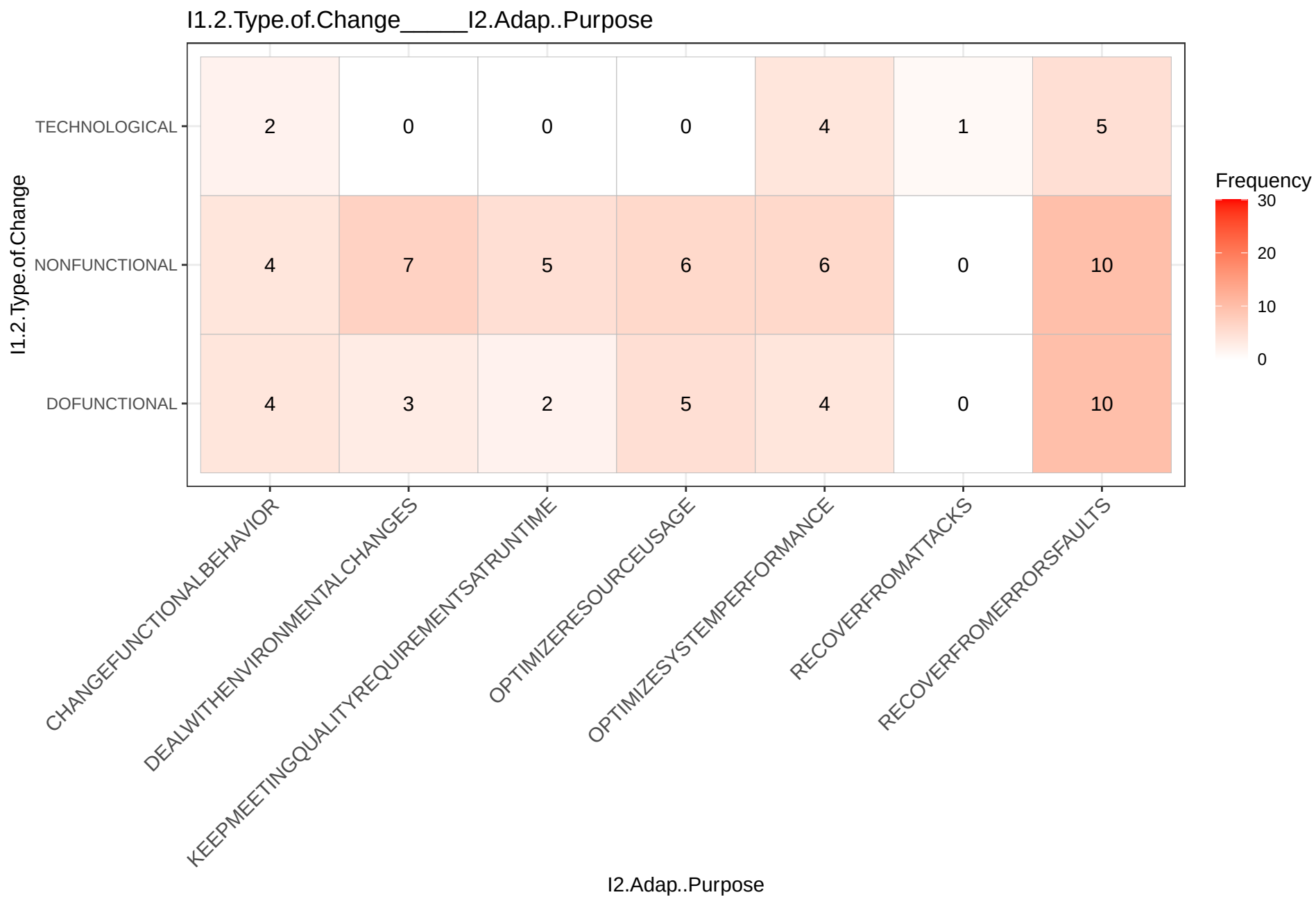
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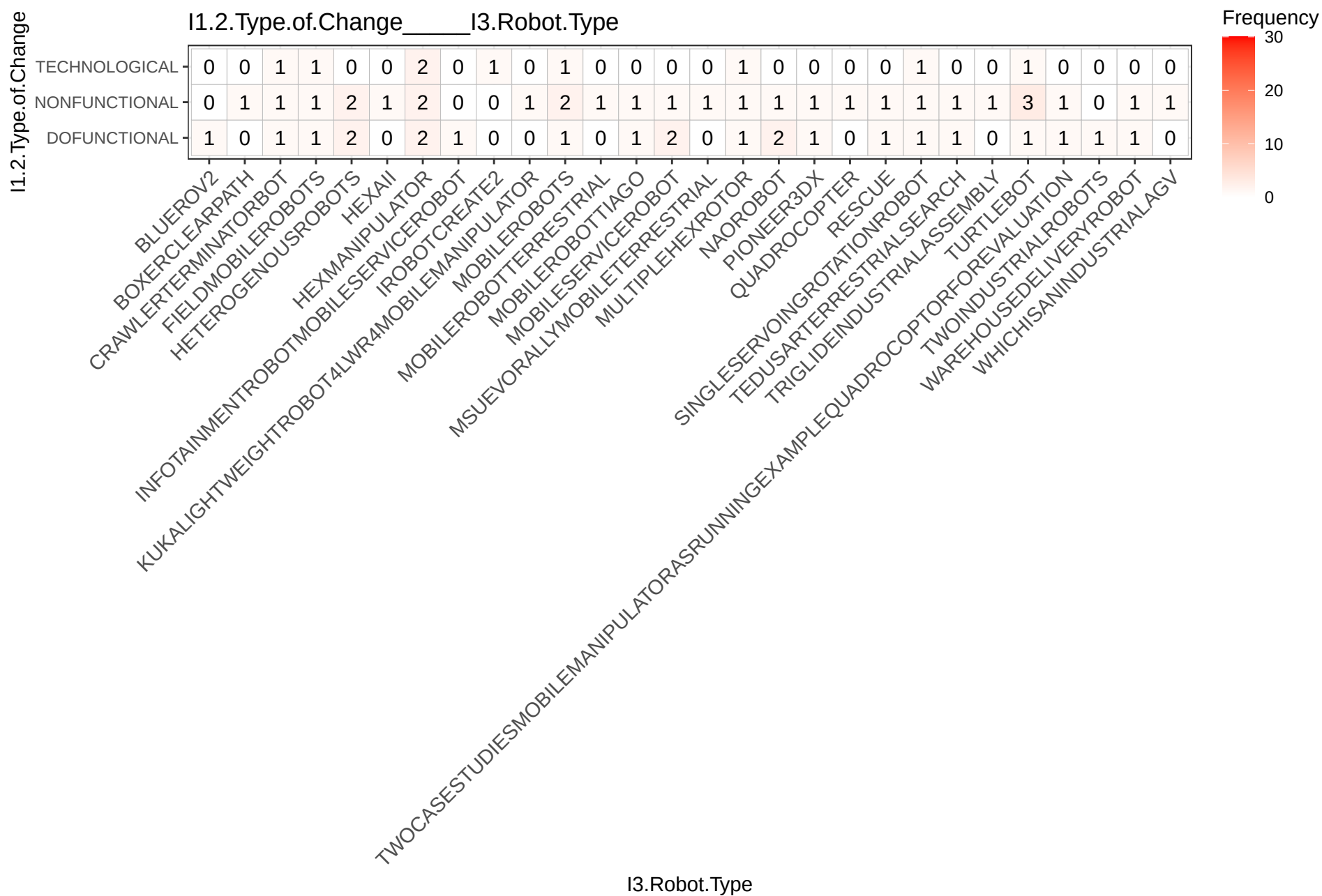
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9

3

1





I1.2.Type.of.Change_____I4.Robo.SW

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

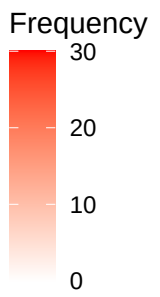
DOFUNCTIONAL

OTHER

ROS1

ROS2

I4.Robo.SW



6

0

0

5

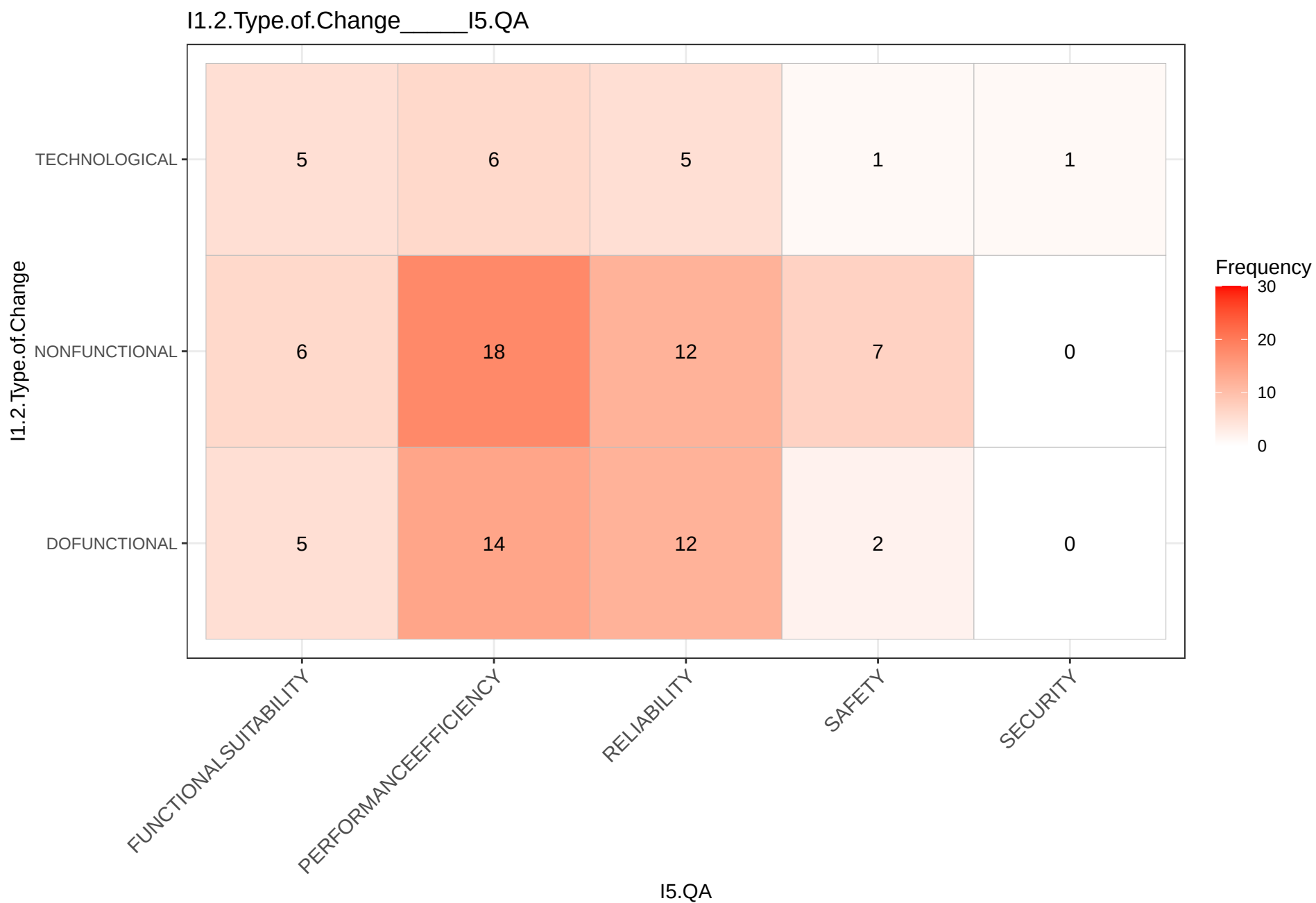
11

1

5

8

2



I1.2.Type.of.Change_____I6.Independence

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

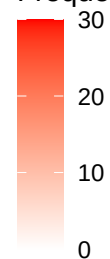
DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



1

1

5

7

2

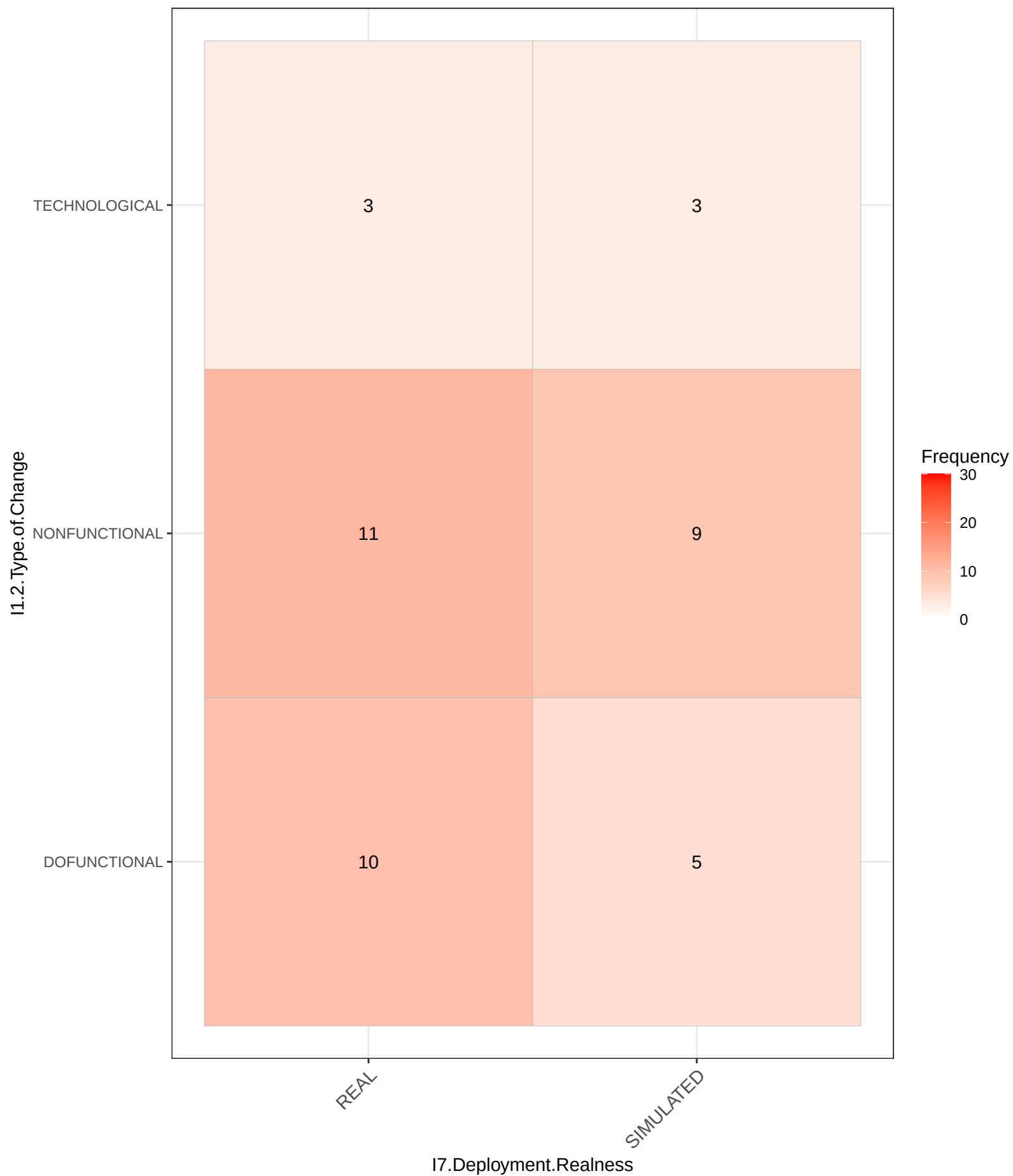
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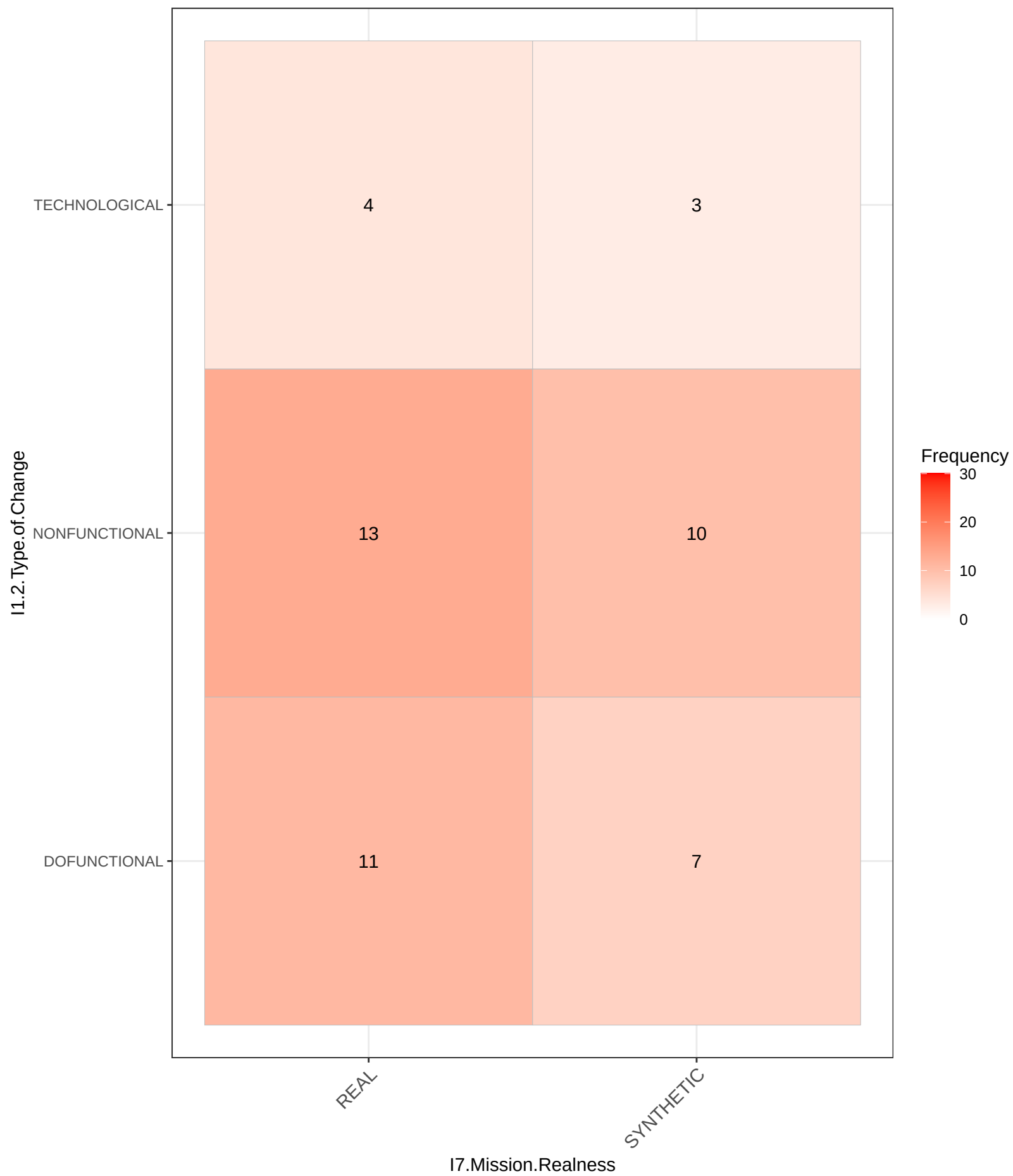
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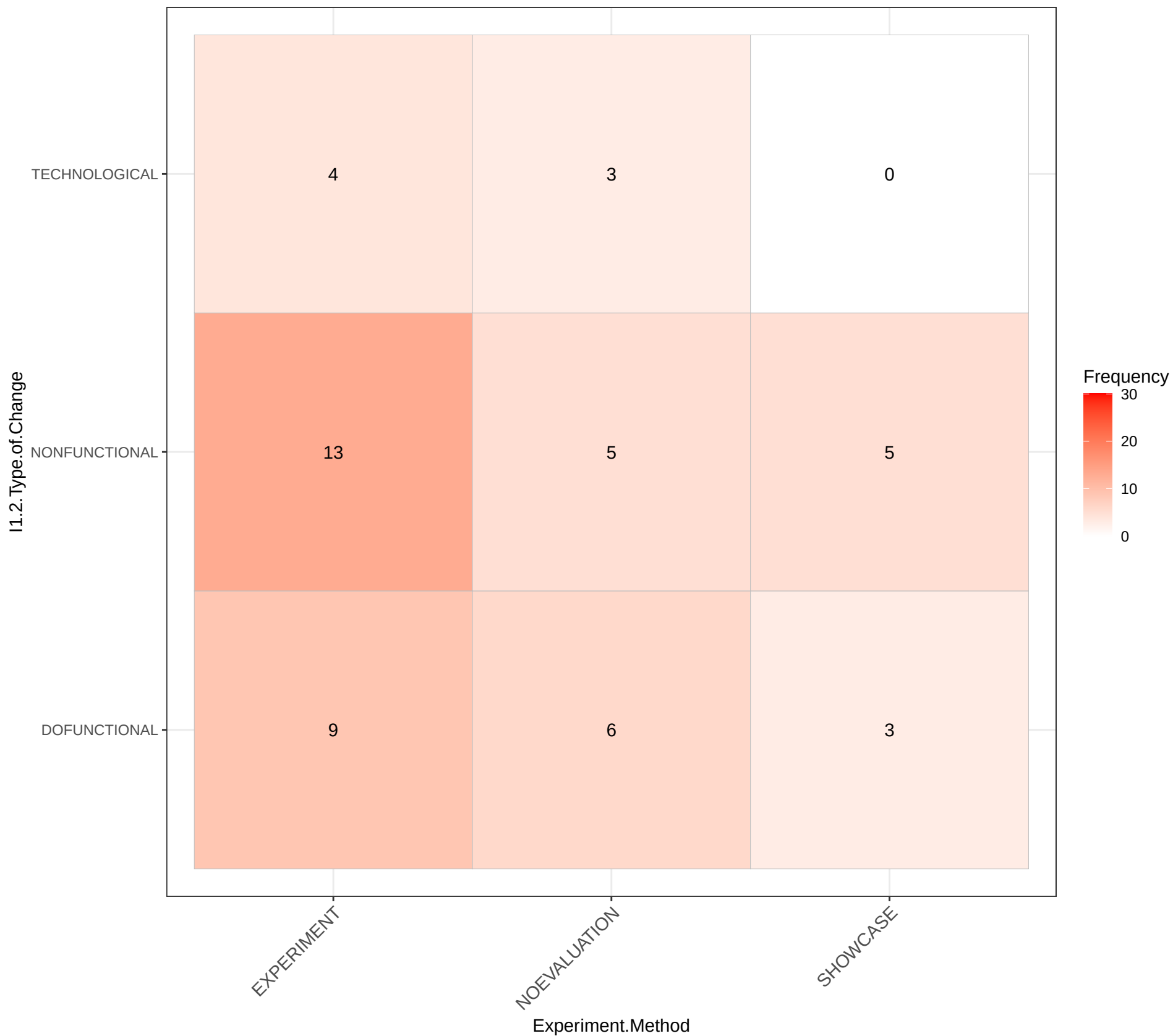
I1.2.Type.of.Change_____I7.Deployment.Realness



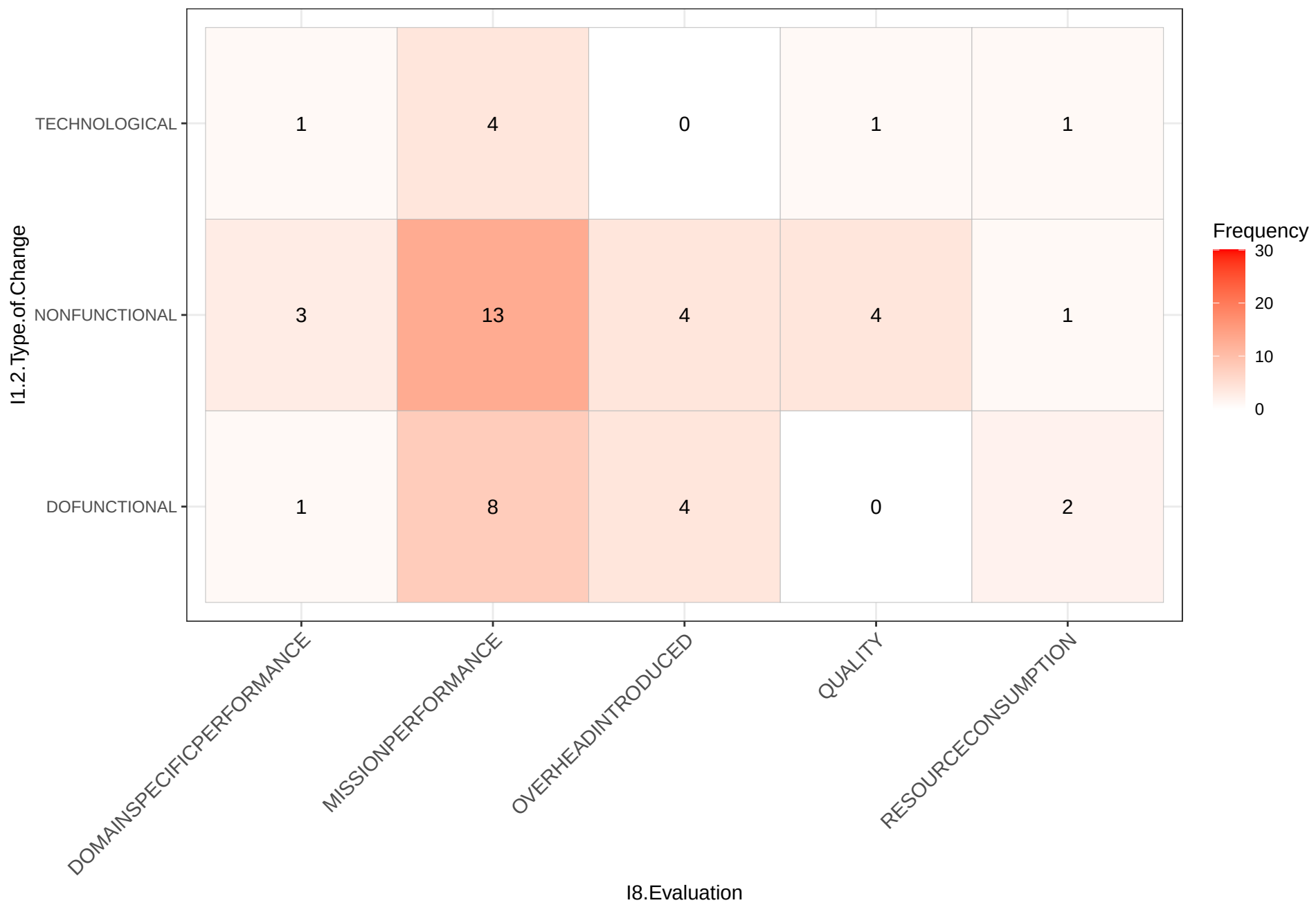
I1.2.Type.of.Change_____I7.Mission.Realness



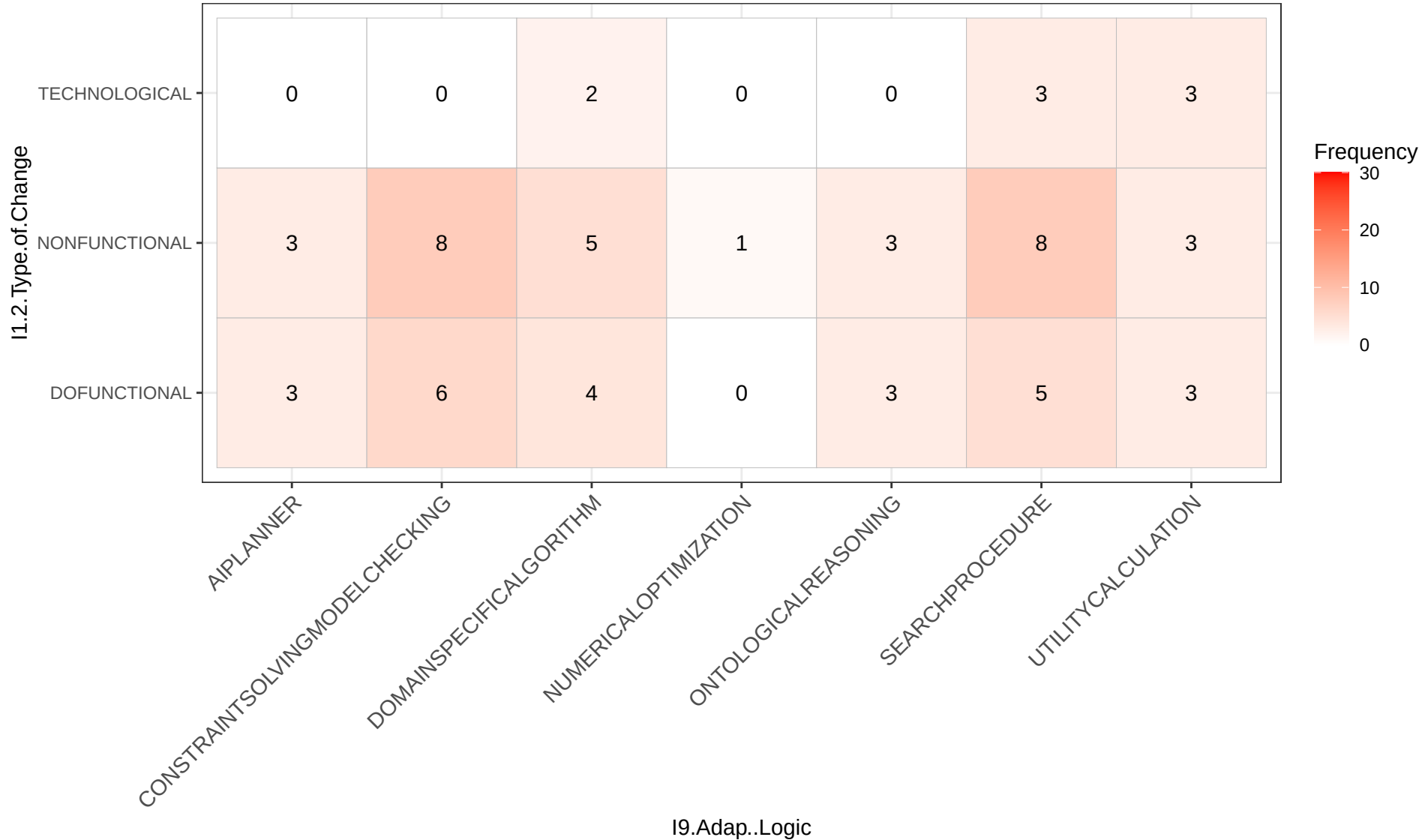
I1.2.Type.of.Change_____Experiment.Method



I1.2.Type.of.Change_____I8.Evaluation



I1.2.Type.of.Change_____I9.Adap..Logic



I1.2.Type.of.Change_____I10.Monitor

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

ENVIRONMENT

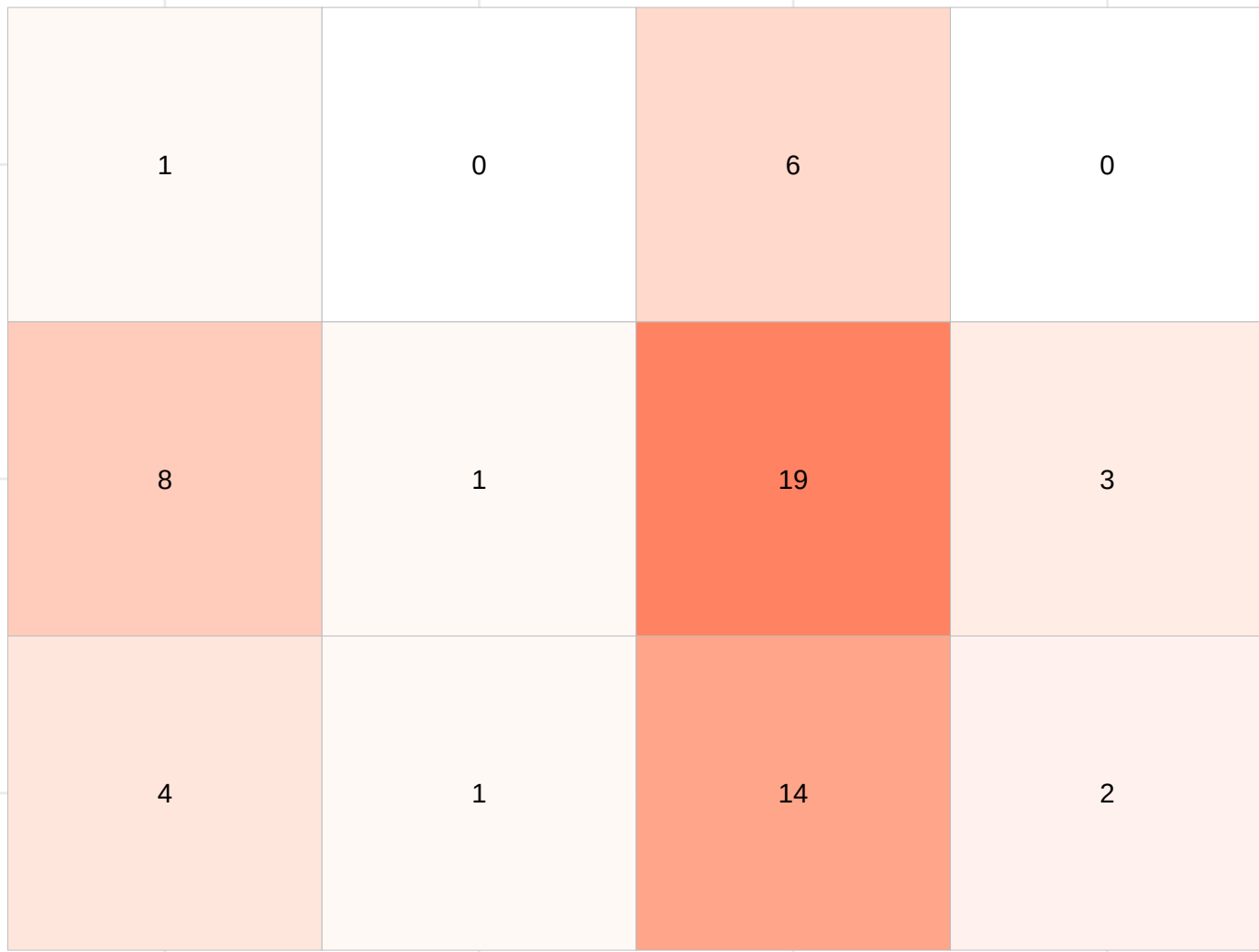
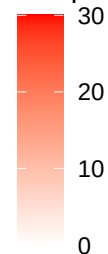
ENVIRONMENTAL

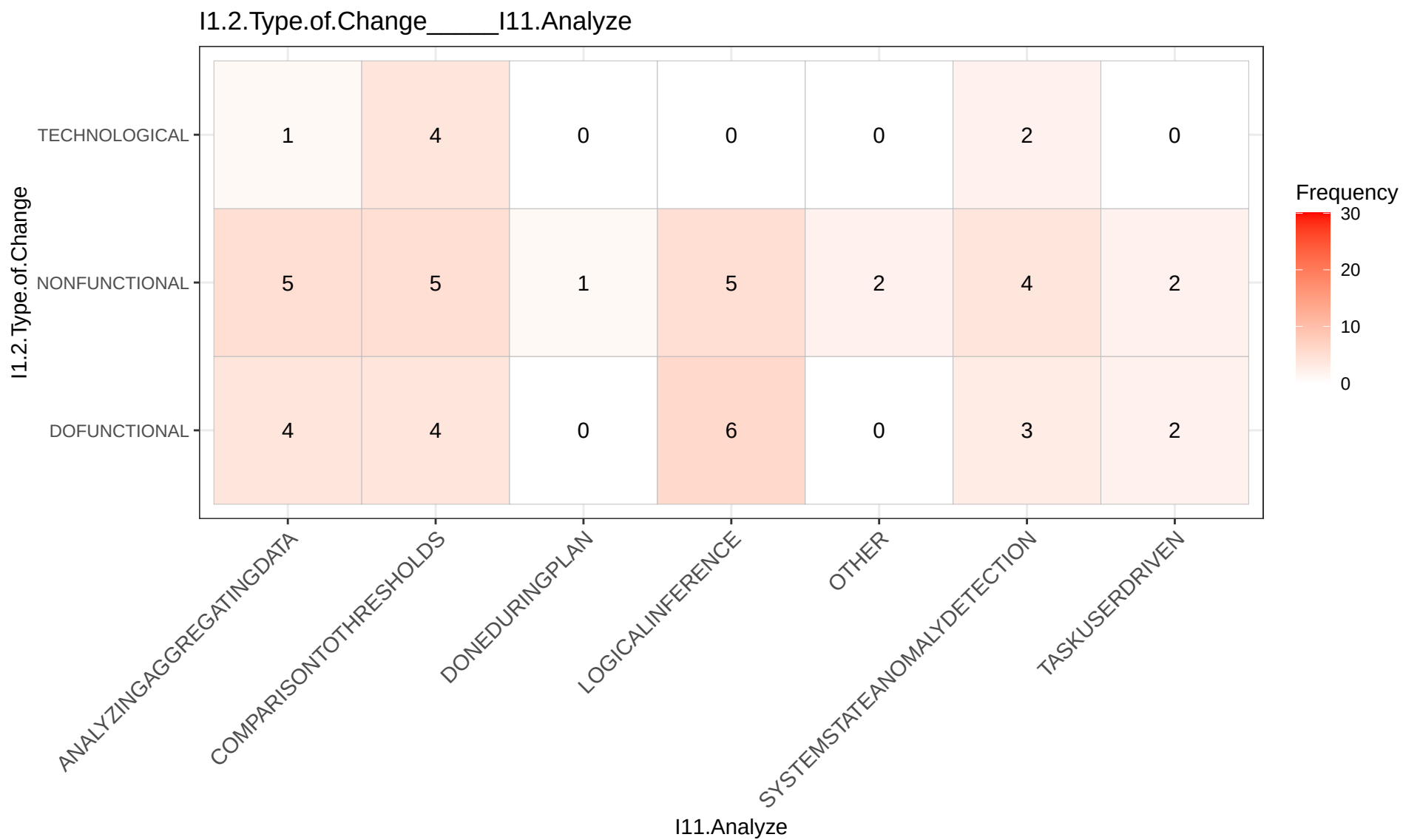
MANAGEDSYSTEM

MISSION

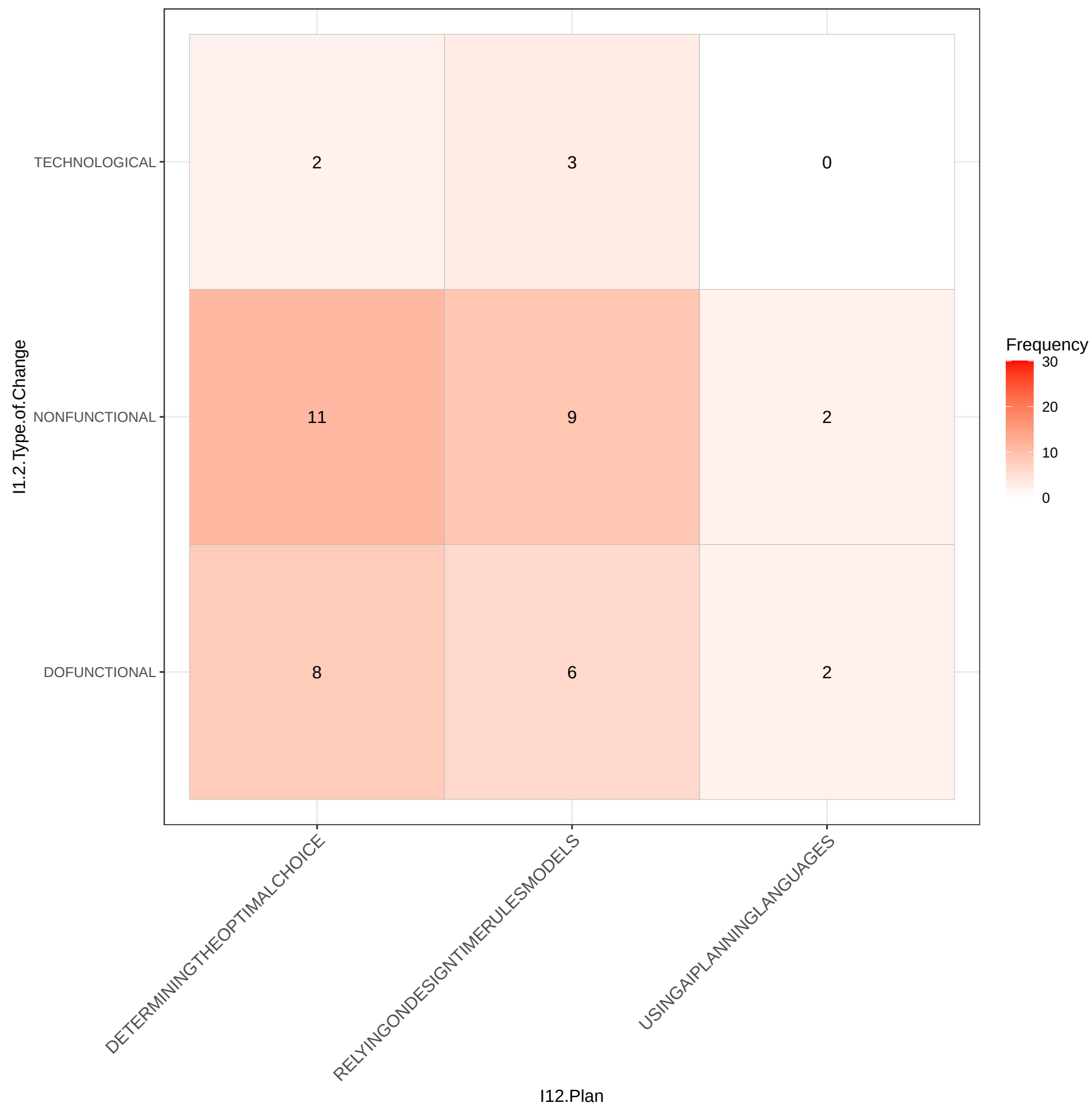
I10.Monitor

Frequency





I1.2.Type.of.Change_____I12.Plan



I1.2.Type.of.Change_____I13.Execute

I1.2.Type.of.Change

TECHNOLOGICAL

NONFUNCTIONAL

DOFUNCTIONAL

Frequency

30

20

10

0

4

1

0

3

11

4

1

12

9

5

1

8

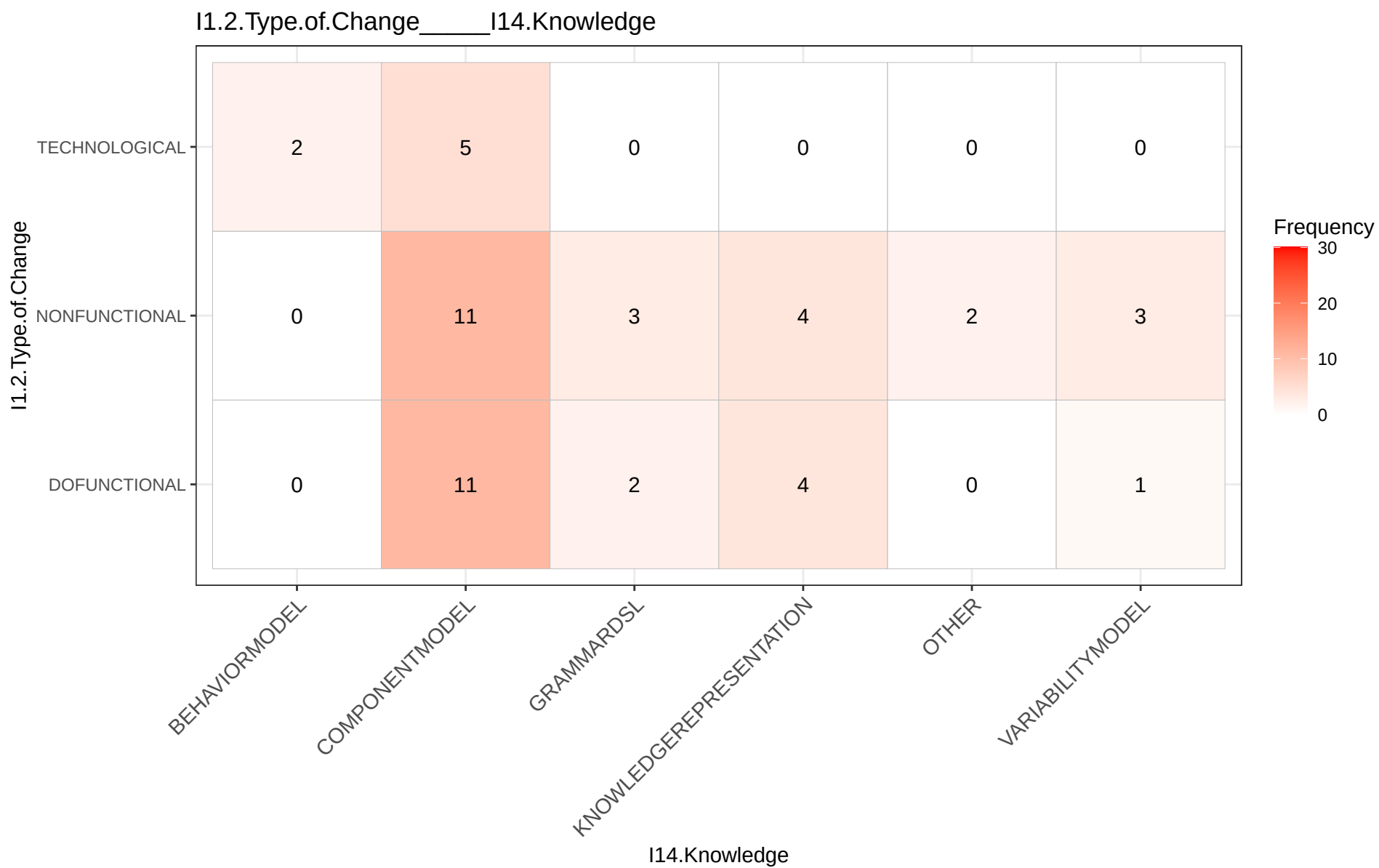
ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPS BETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

I13.Execute



I1.2.Anticipation.of.Change_____I1.2.Frequency.of.Change

I1.2.Anticipation.of.Change

UNFORESEEN

0

1

0

FORESEEABLE

6

5

1

DOFORESEEN

0

1

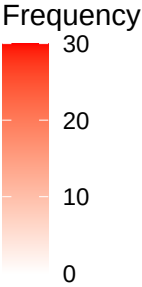
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DOFREQUENT

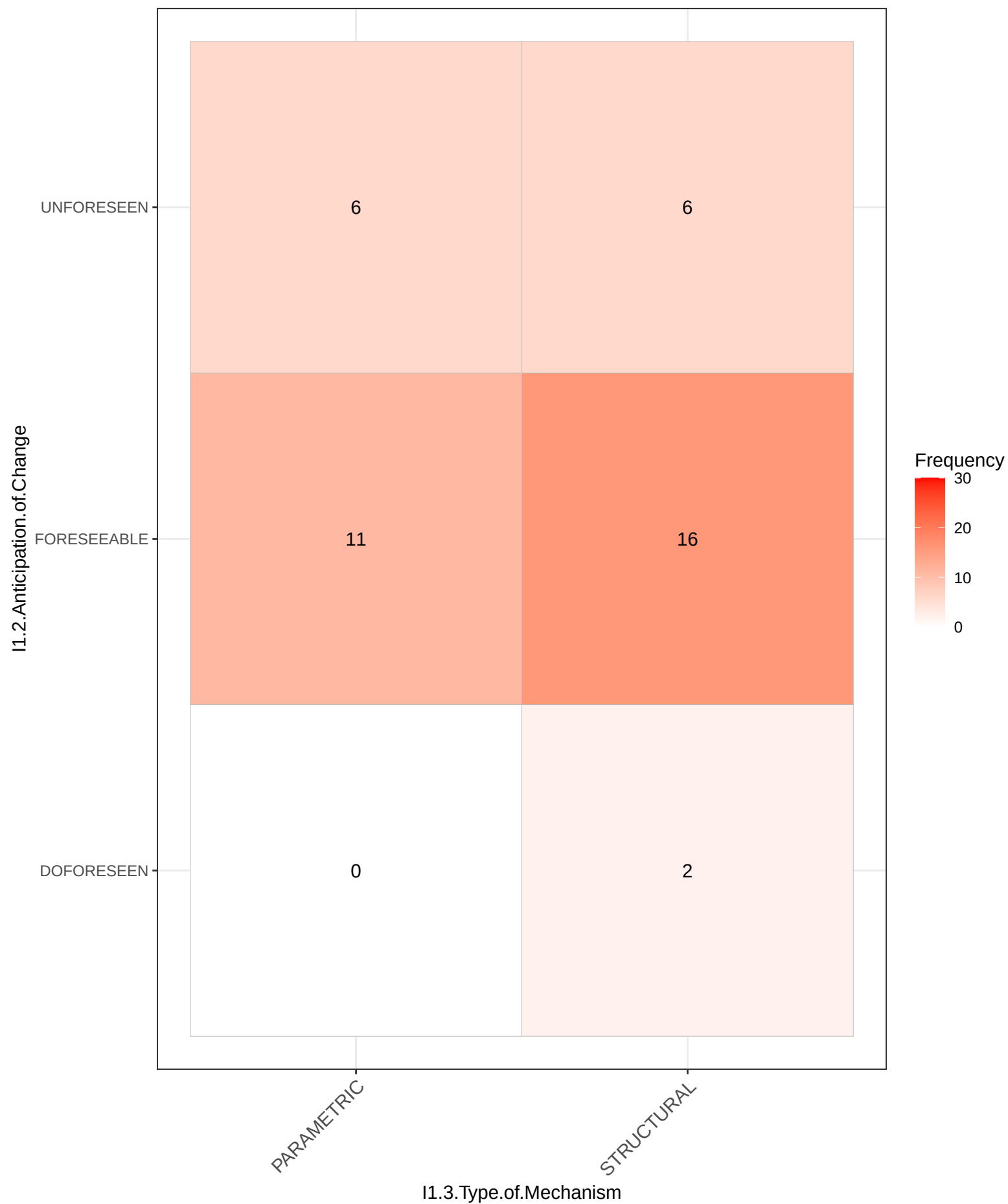
INFREQUENT

RARE

I1.2.Frequency.of.Change



I1.2.Anticipation.of.Change_____I1.3.Type.of.Mechanism



I1.2.Anticipation.of.Change

UNFORESEEN

0

7

FORESEEABLE

4

15

DOFORESEEN

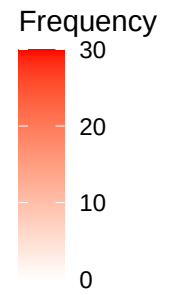
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2

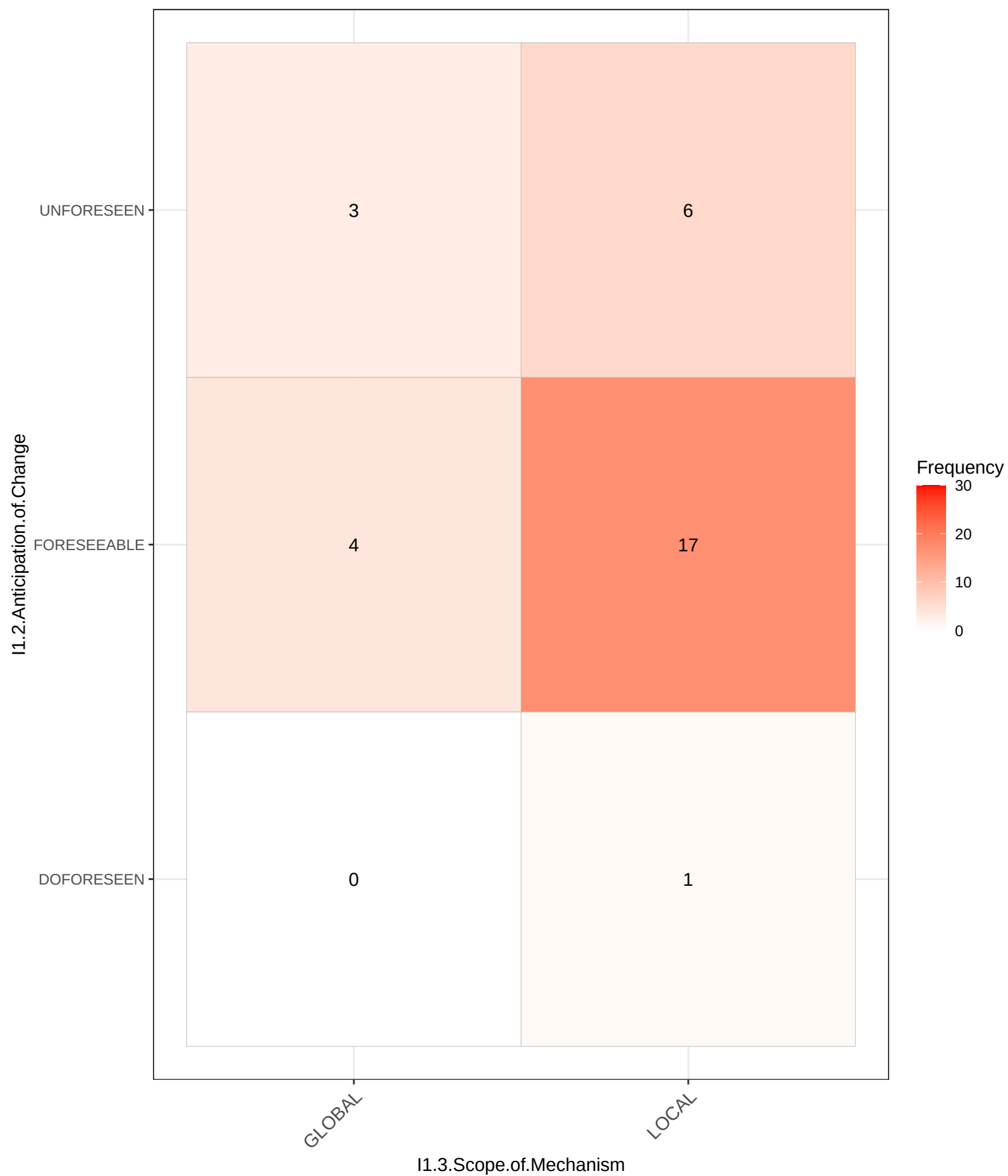
DECENTRALIZED

DOCENTRALIZED

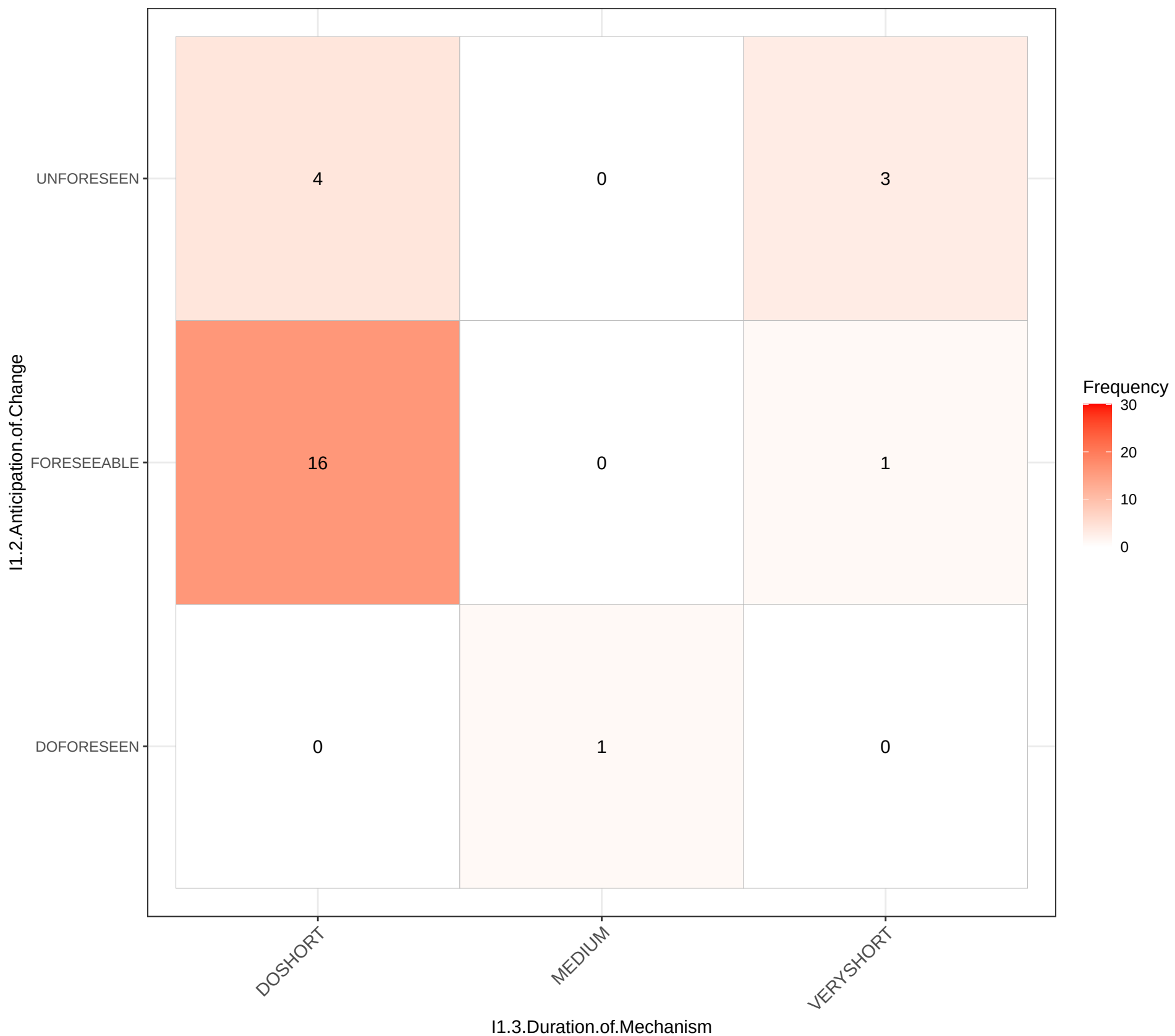
I1.3.Organization.of.Mechanism

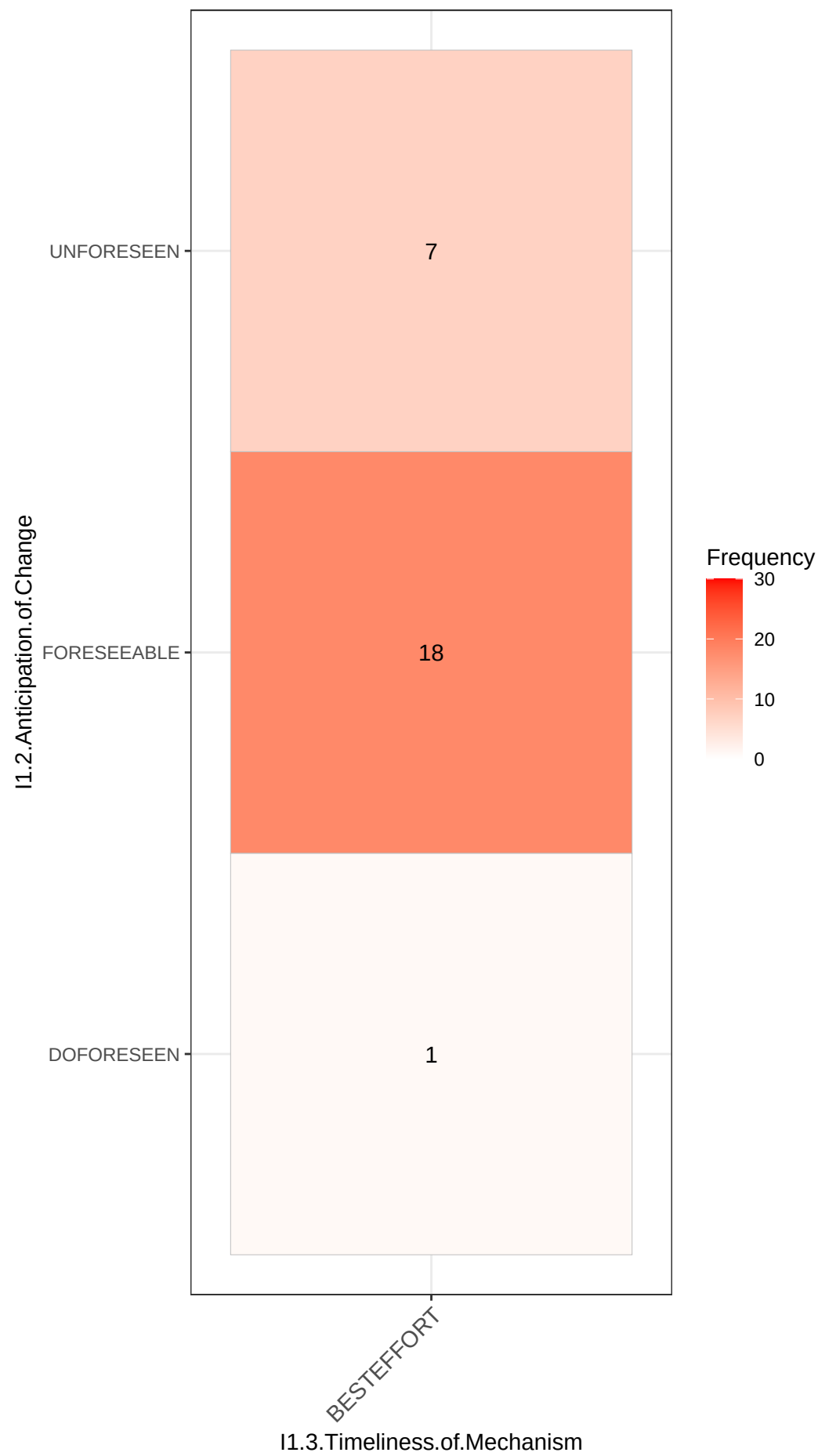


I1.2.Anticipation.of.Change_____I1.3.Scope.of.Mechanism



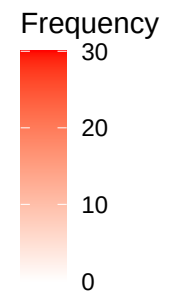
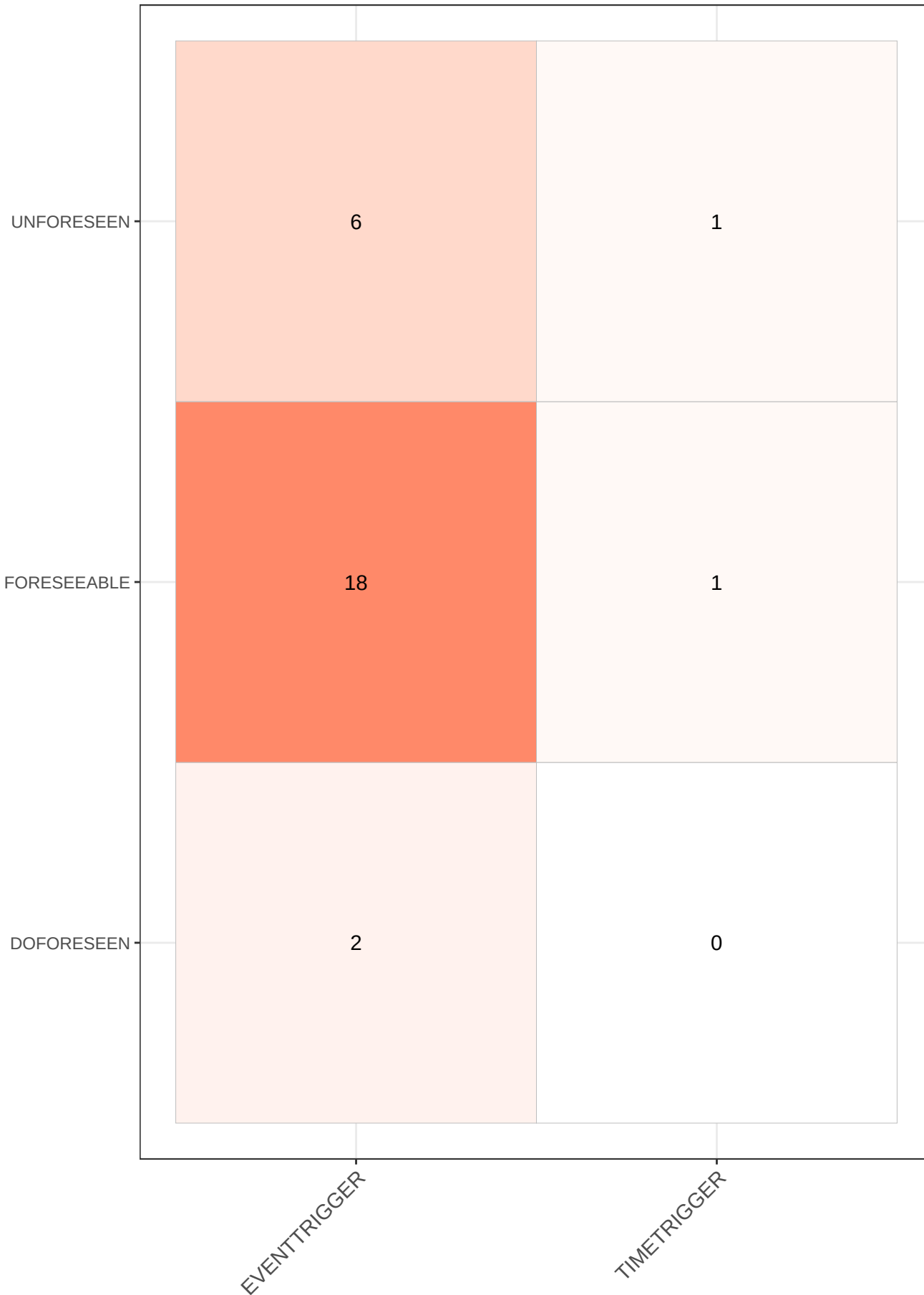
I1.2.Anticipation.of.Change_____I1.3.Duration.of.Mechanism



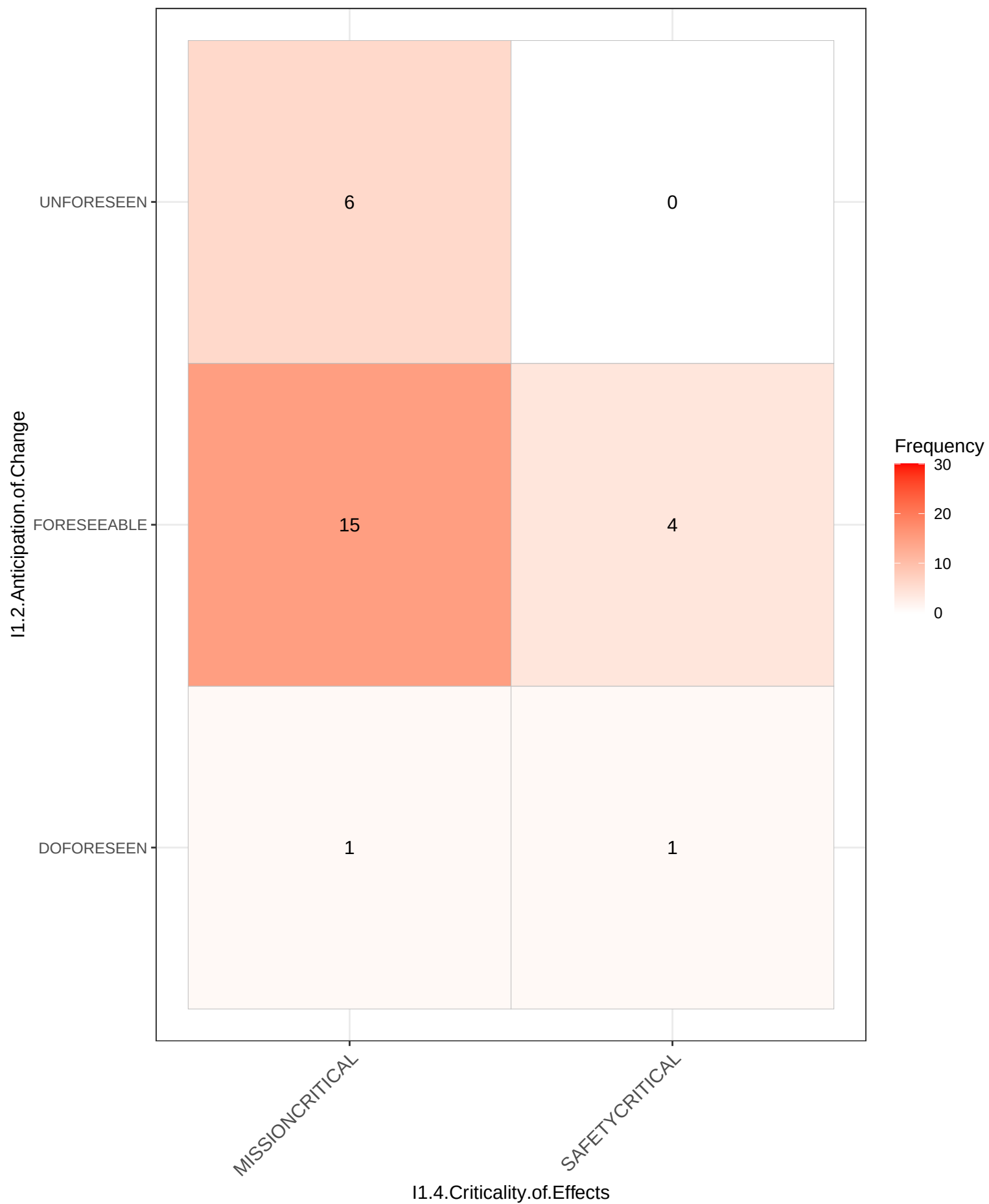


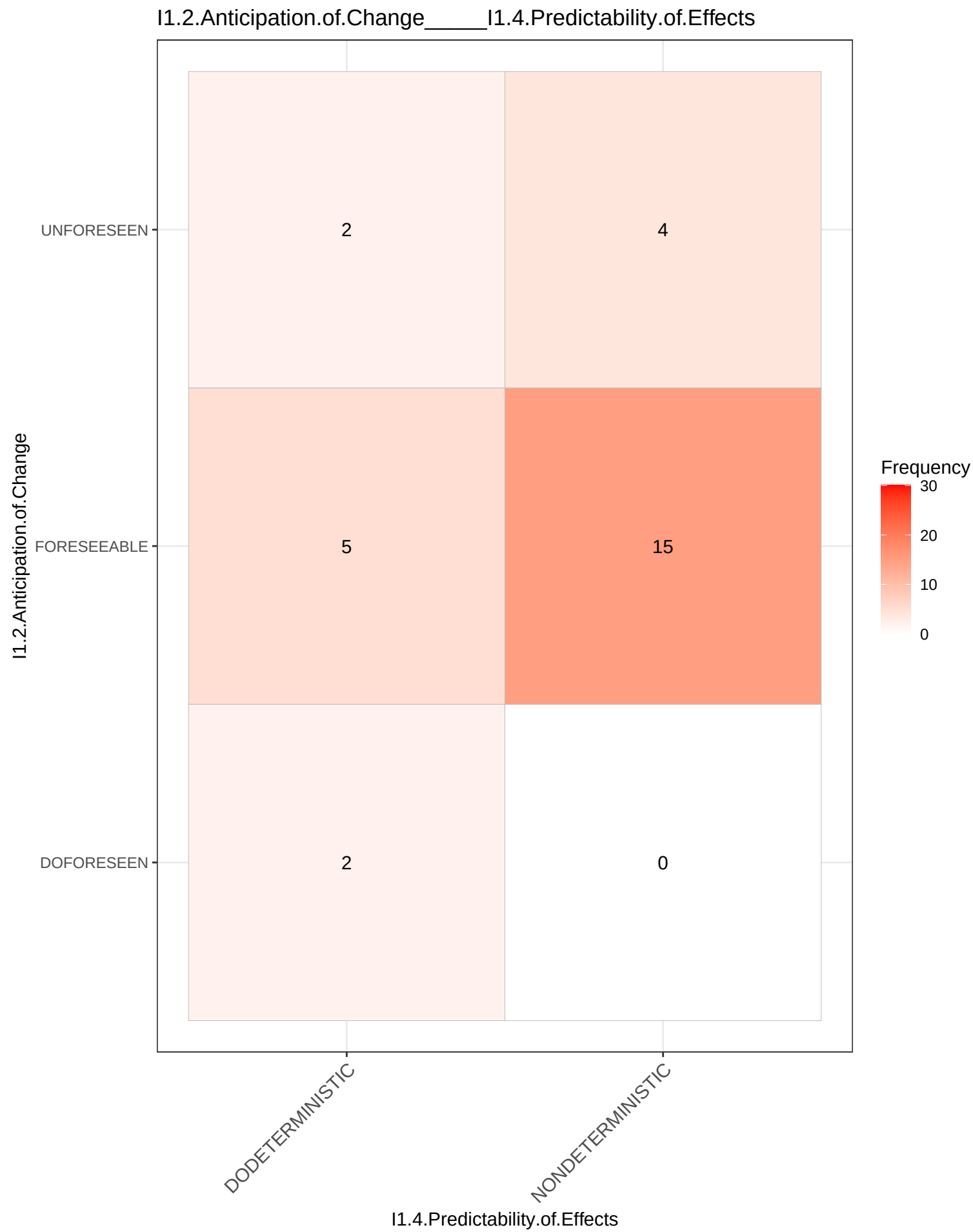
I1.2.Anticipation.of.Change_____I1.3.Trigger.of.Mechanism

I1.2.Anticipation.of.Change



I1.3.Trigger.of.Mechanism





I1.2.Anticipation.of.Change_____I1.4.Overhead.of.Effects

I1.2.Anticipation.of.Change

UNFORESEEN

3

2

1

FORESEEABLE

10

1

4

DOFORESEEN

1

0

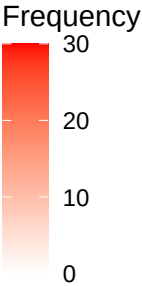
1

DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.2.Anticipation.of.Change_____I1.4.Resilience.of.Effects

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

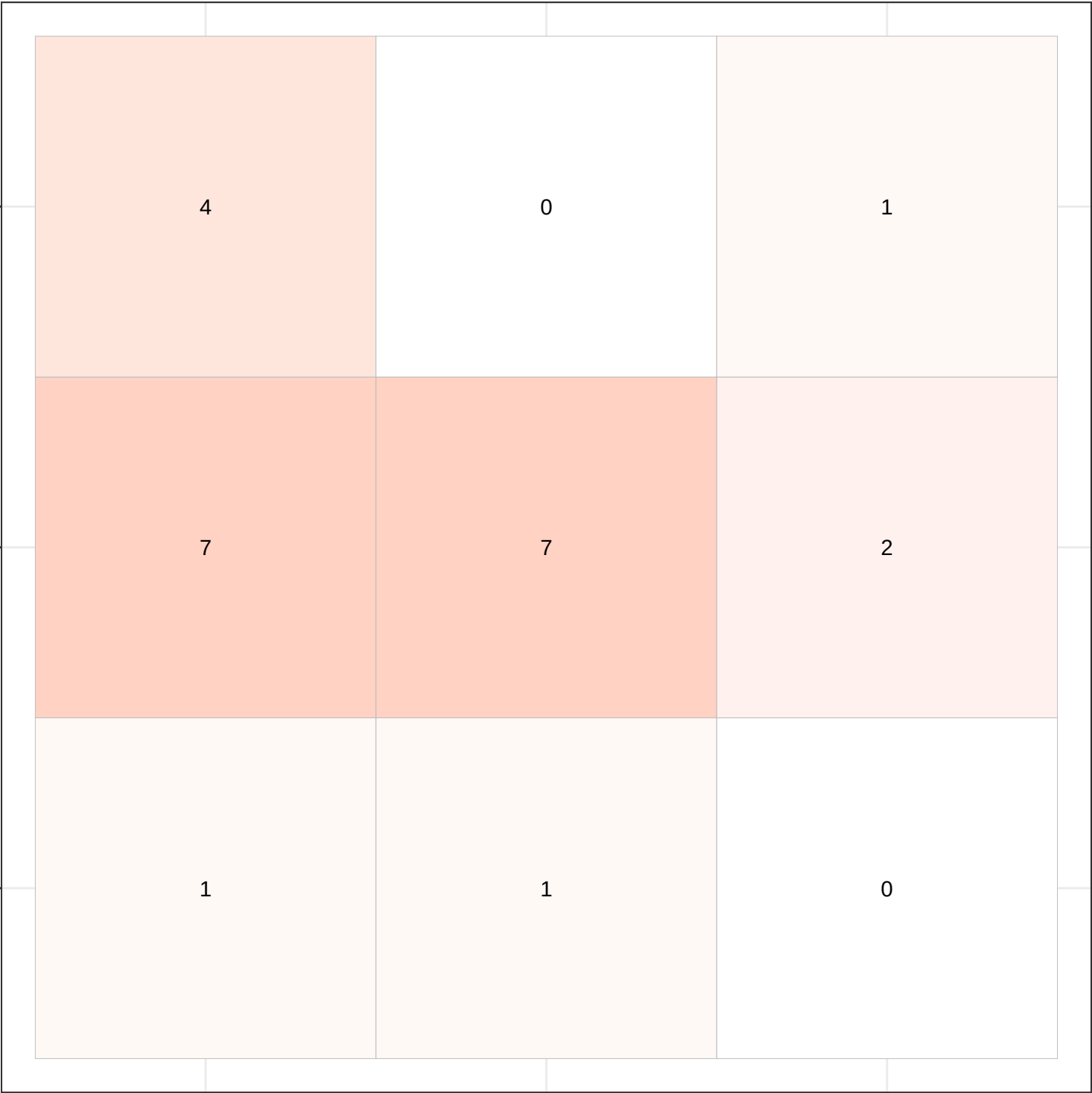
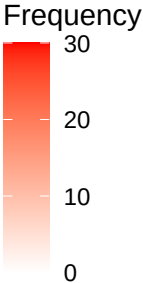
DOFORESEEN

DEPENDENT

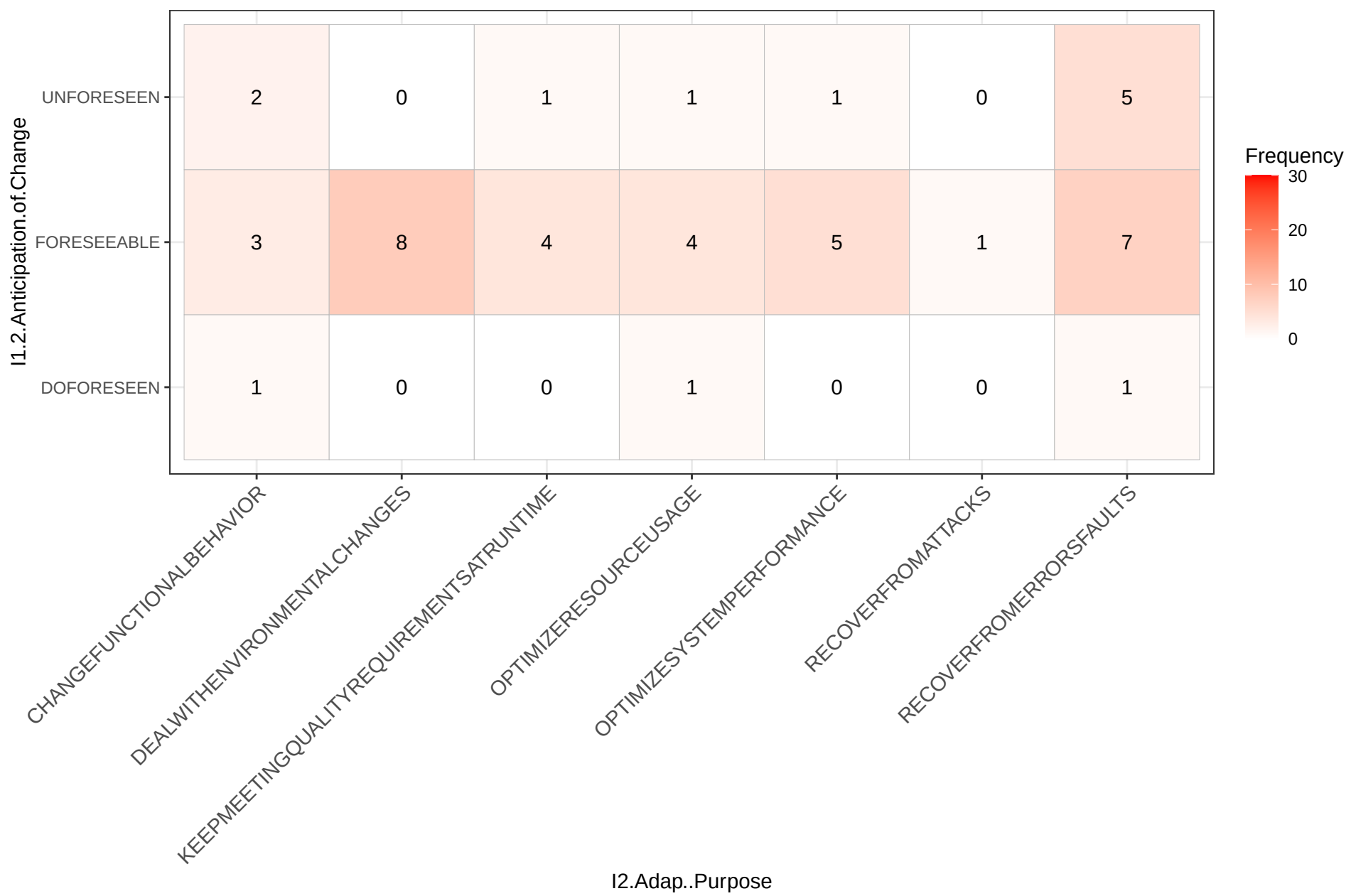
DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects



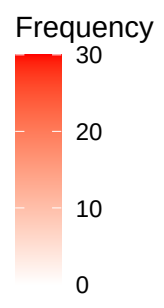
I1.2.Anticipation.of.Change_____I2.Adap..Purpose



I1.2.Anticipation.of.Change

I1.2.Anticipation.of.Change_____I3.Robot.Type

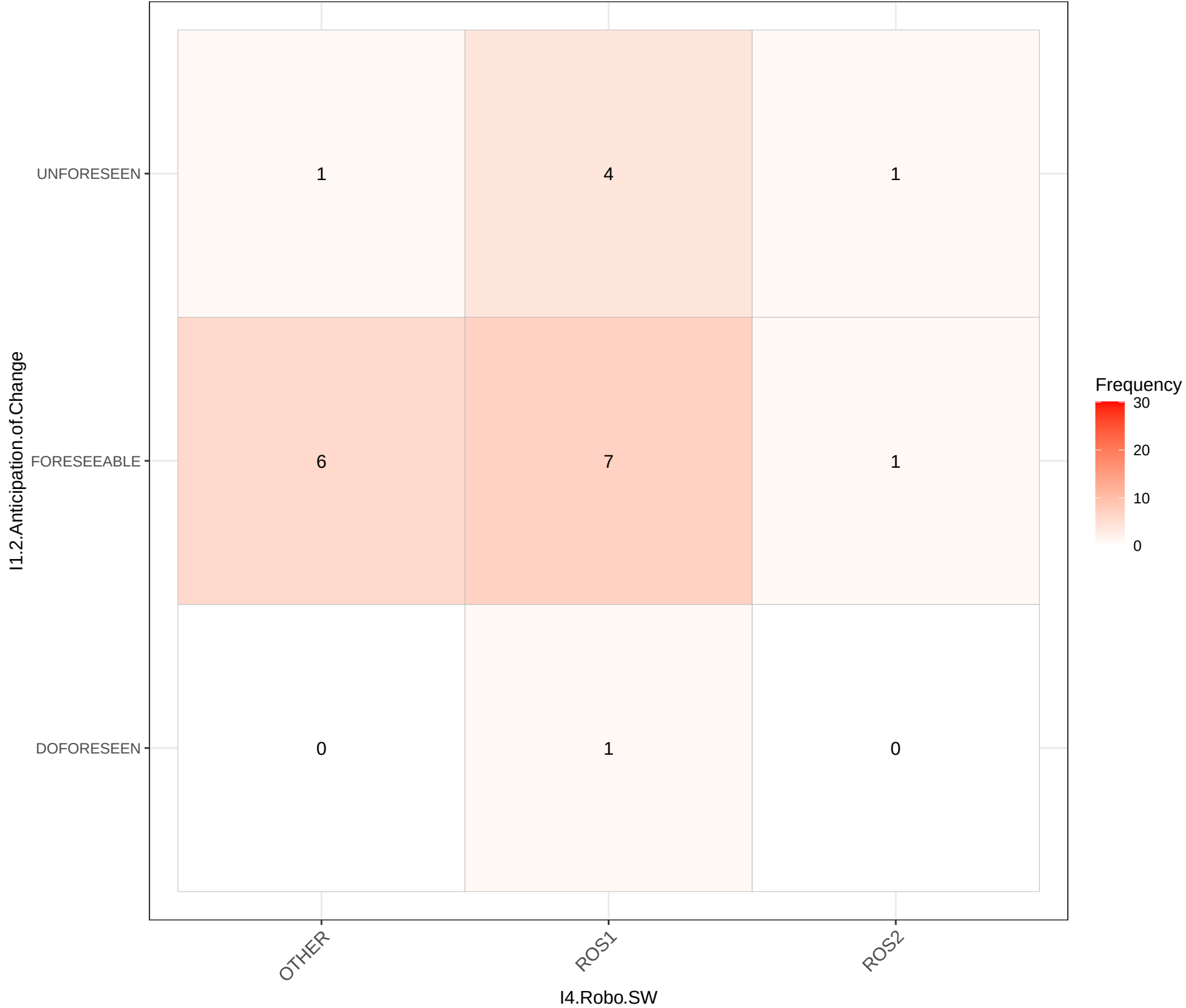
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FORESEEABLE	1	1	0	1	2	1	2	0	1	1	1	1	0	1	1	0	0	0	1	0	1	0	1	2	1	1	0	1
DOFORESEEN	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0



BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOTS
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXAI
INFOTAINMENTROBOT
KUKALIGHTWEIGHTROBOT4LWR4
MOBILESERVICEROBOT
MOBILEMANIPULATOR
MOBILEROBOTS
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLE
QUADROCOPTORFOREVALUATION
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV

I3.Robot.Type

I1.2.Anticipation.of.Change_____I4.Robo.SW



I1.2.Anticipation.of.Change_____I5.QA

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

DOFORESEEN

FUNCTIONALSUITABILITY

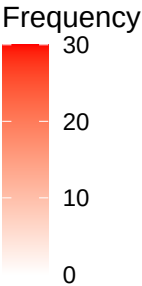
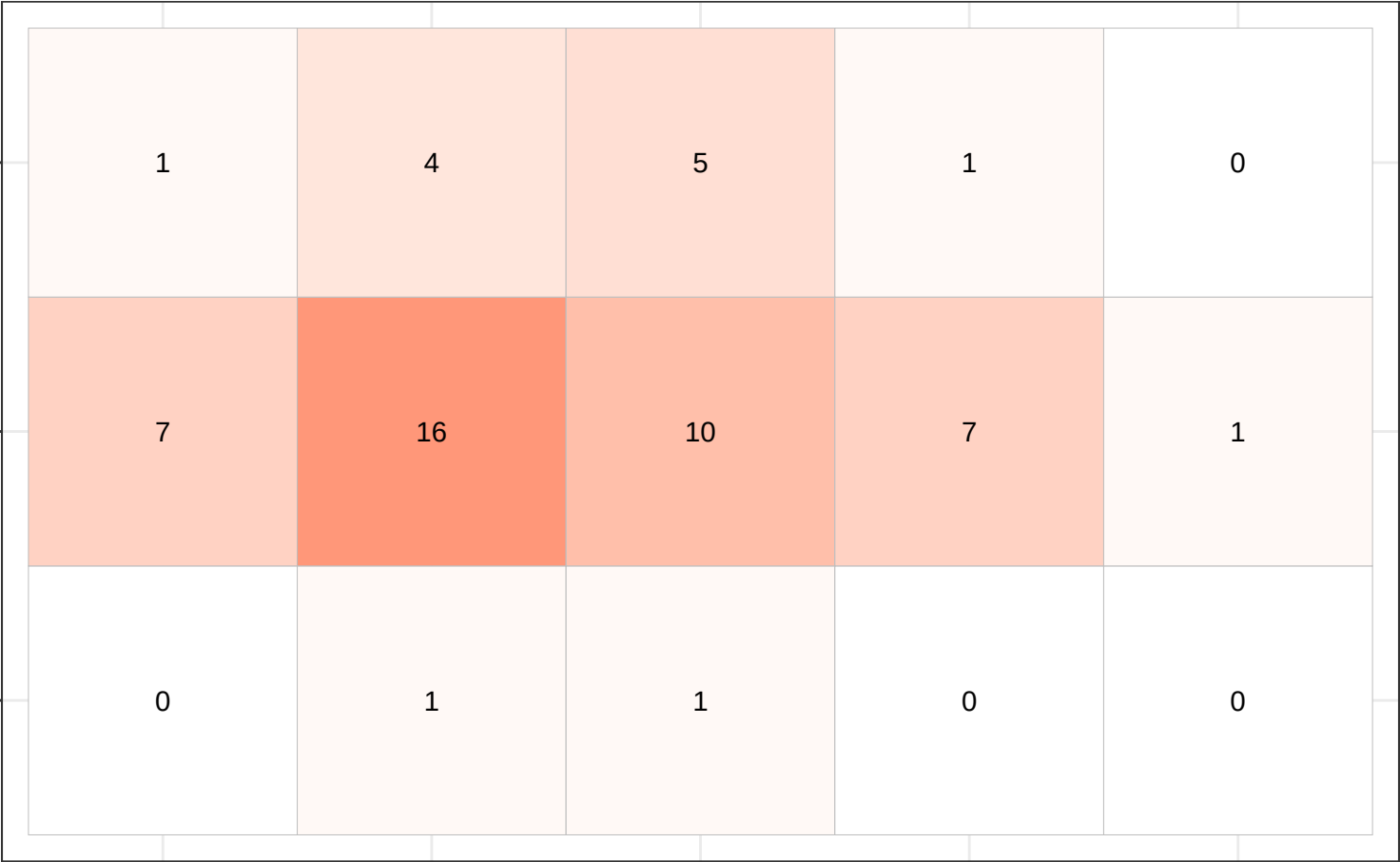
PERFORMANCEEFFICIENCY

RELIABILITY

SAFETY

SECURITY

I5.QA



I1.2.Anticipation.of.Change_____I6.Independence

I1.2.Anticipation.of.Change

UNFORESEEN

FORESEEABLE

DOFORESEEN

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

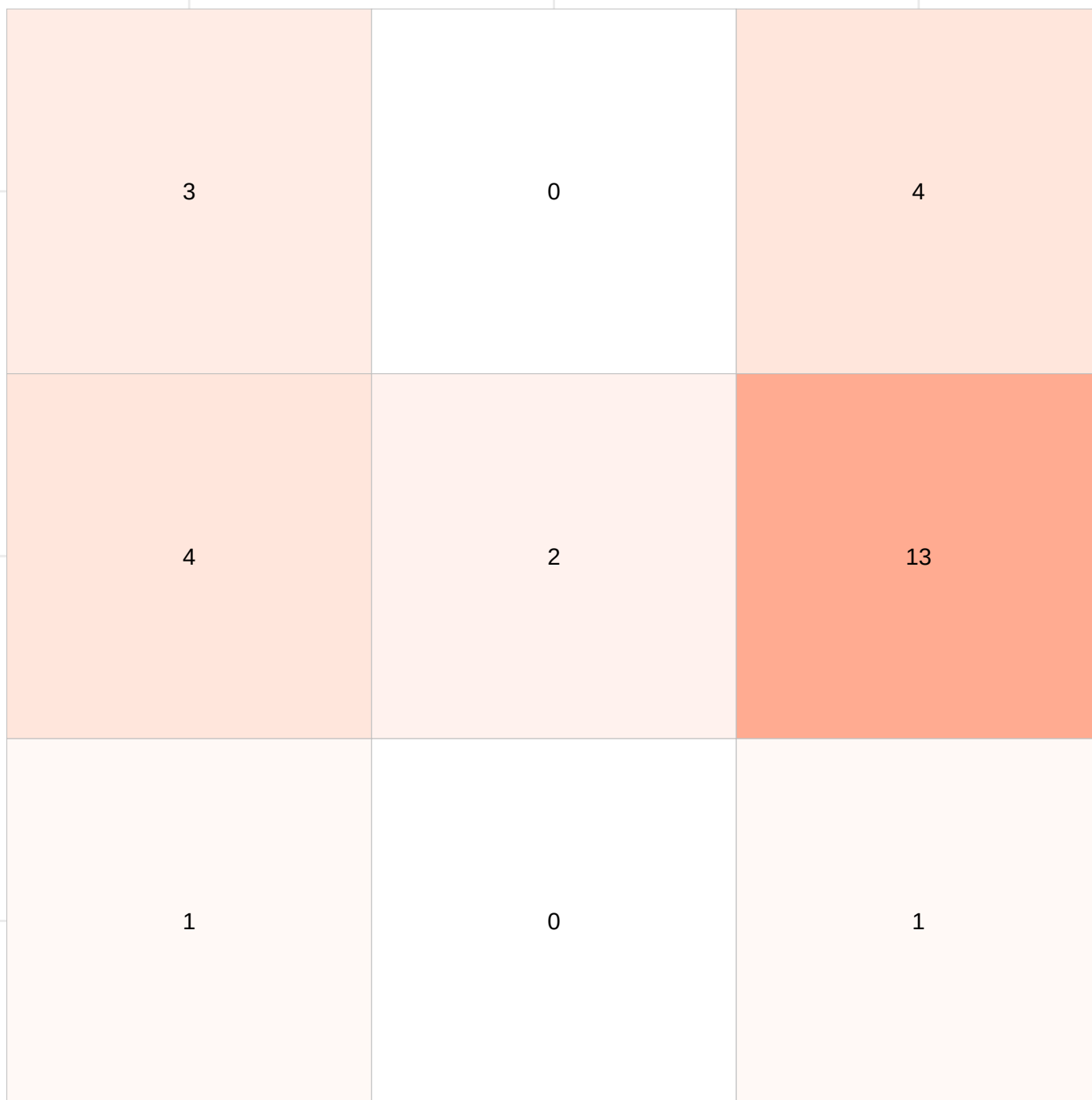
Frequency

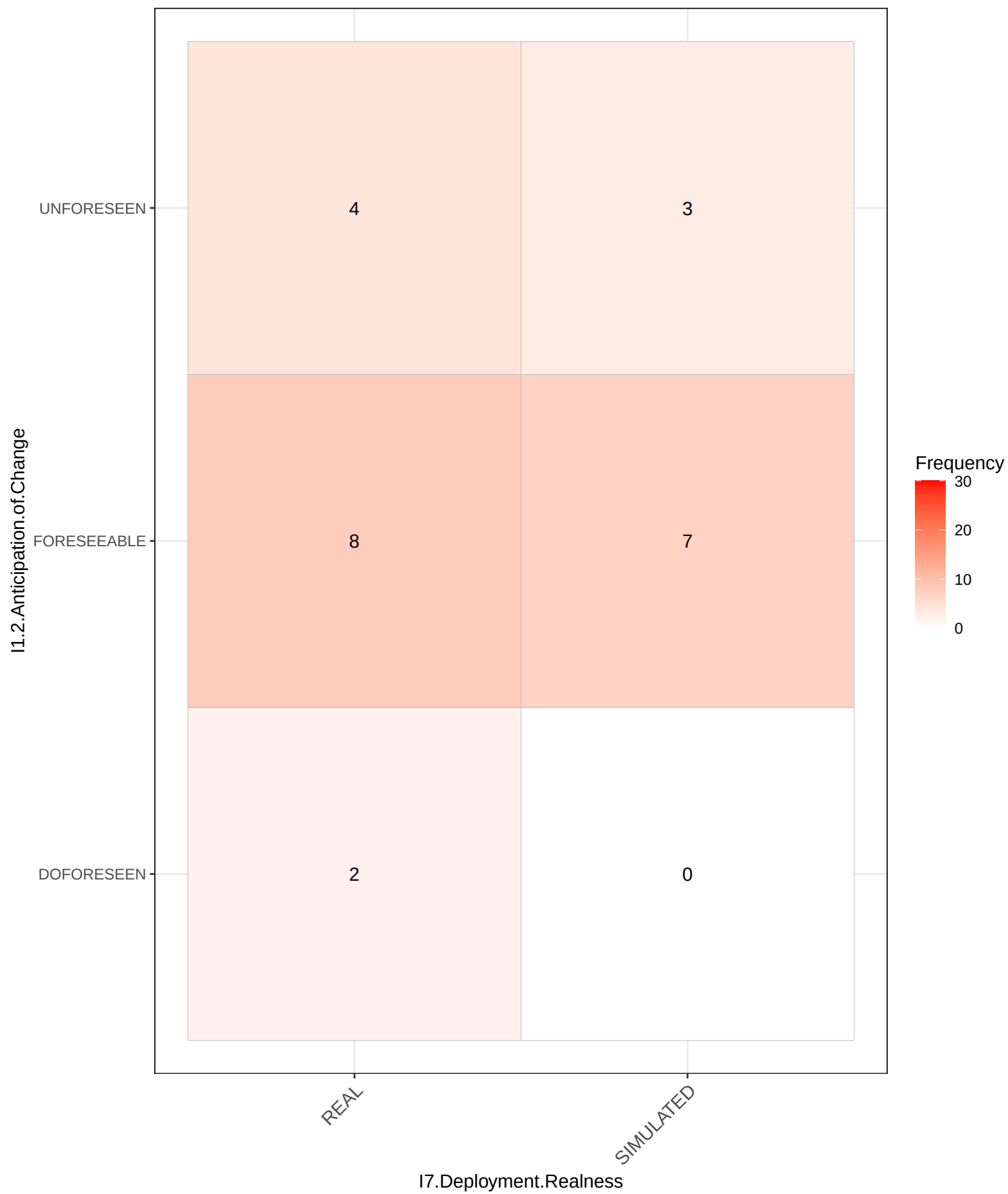
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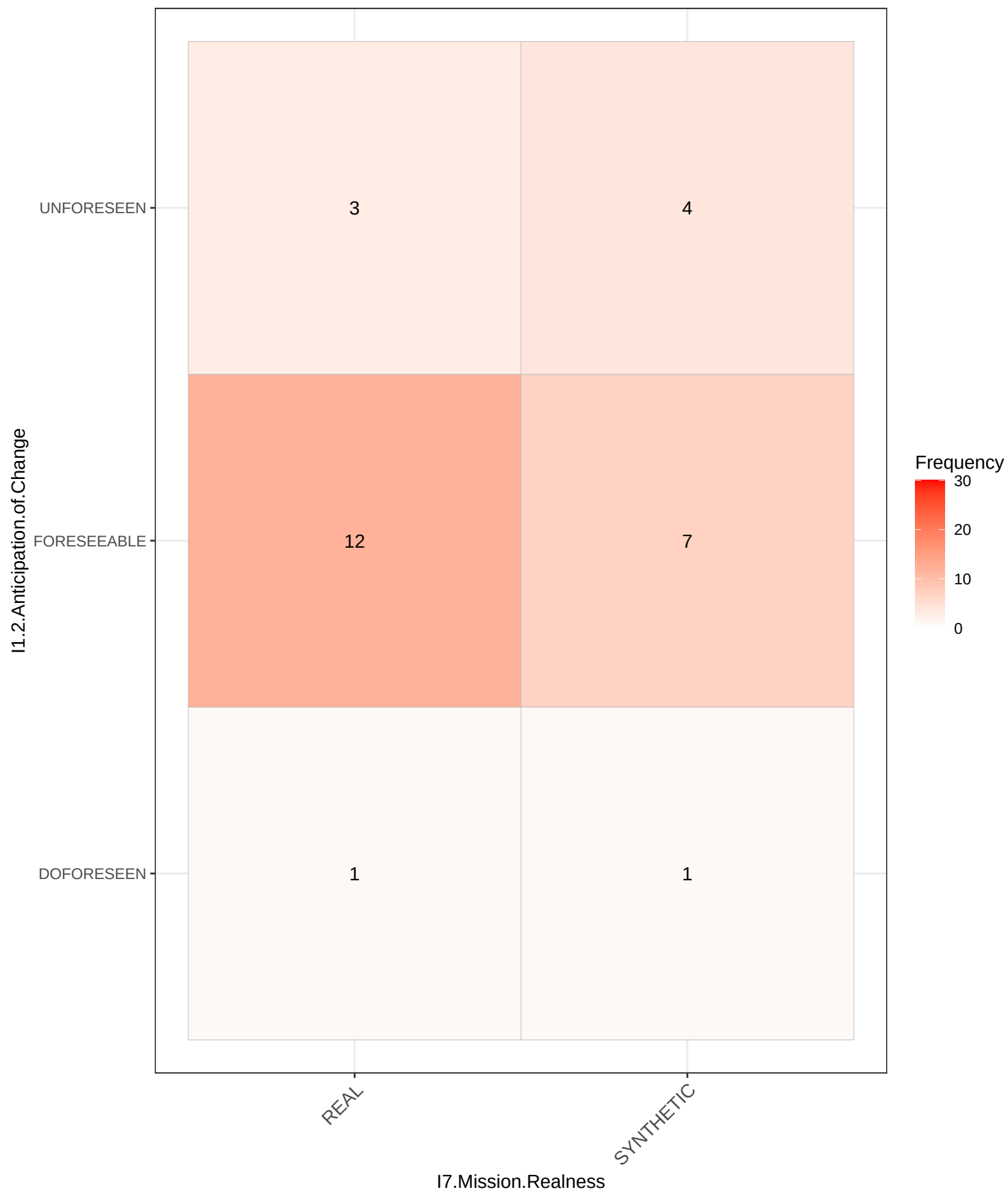
20

10

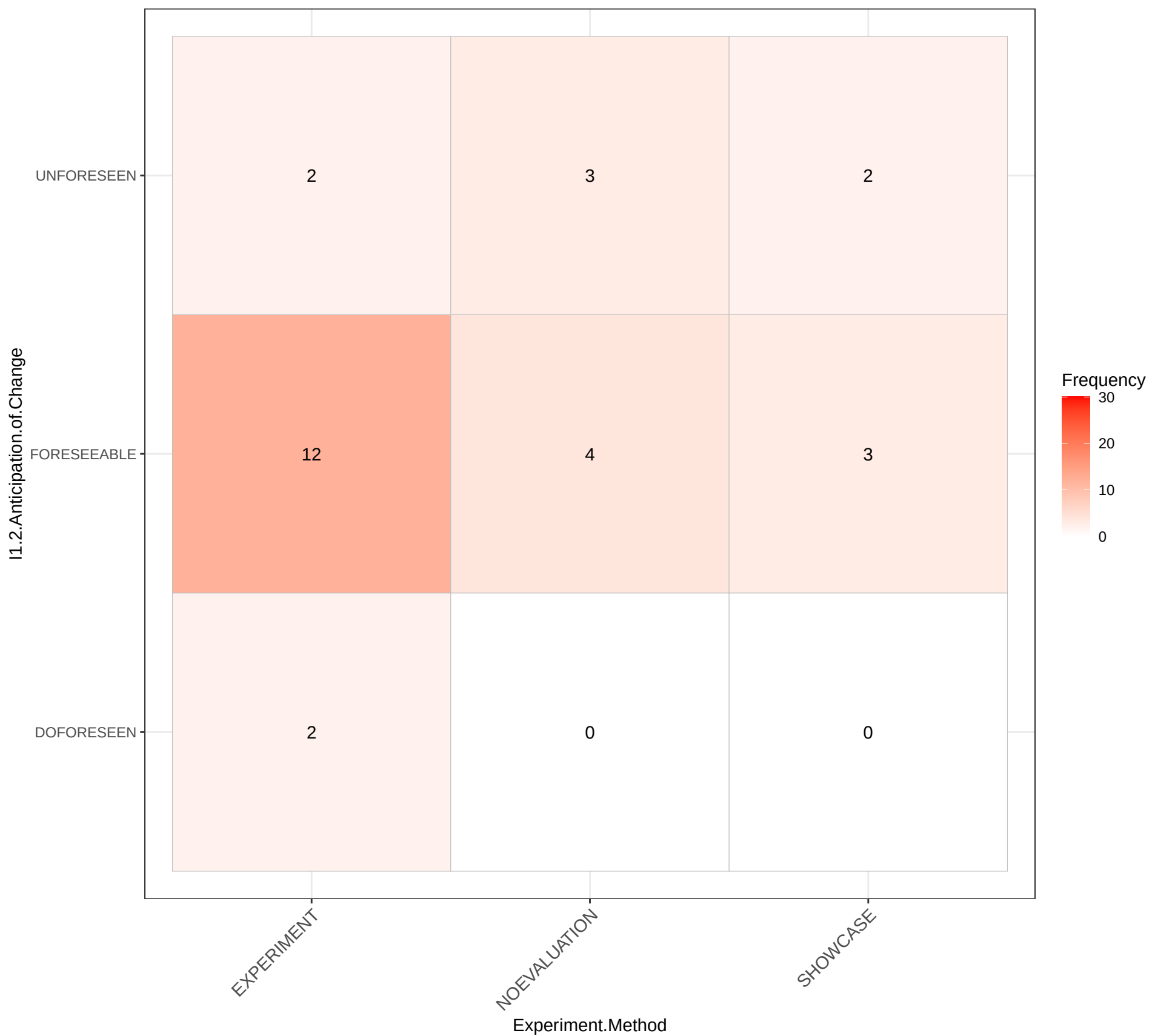
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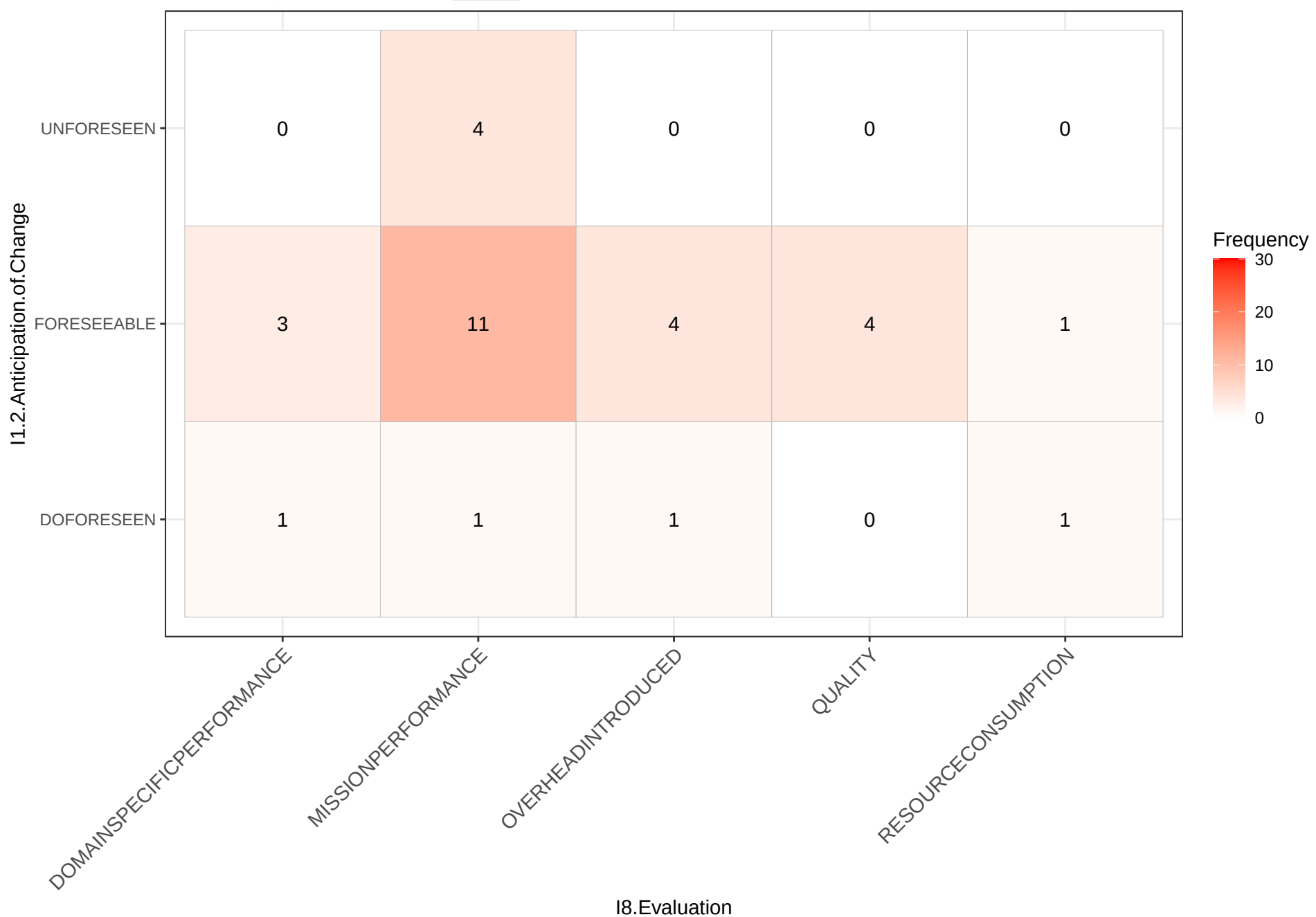




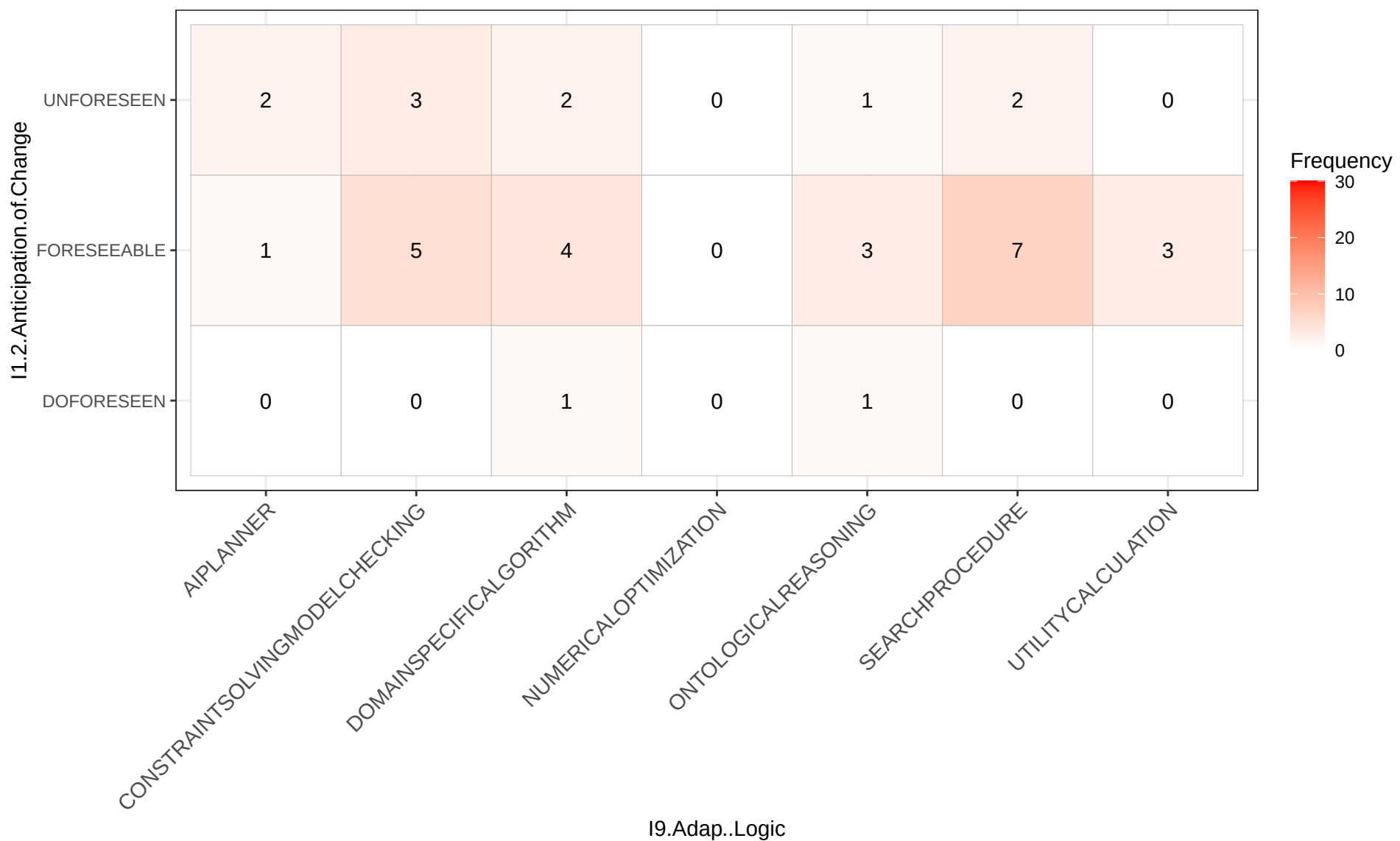
I1.2.Anticipation.of.Change_____Experiment.Method



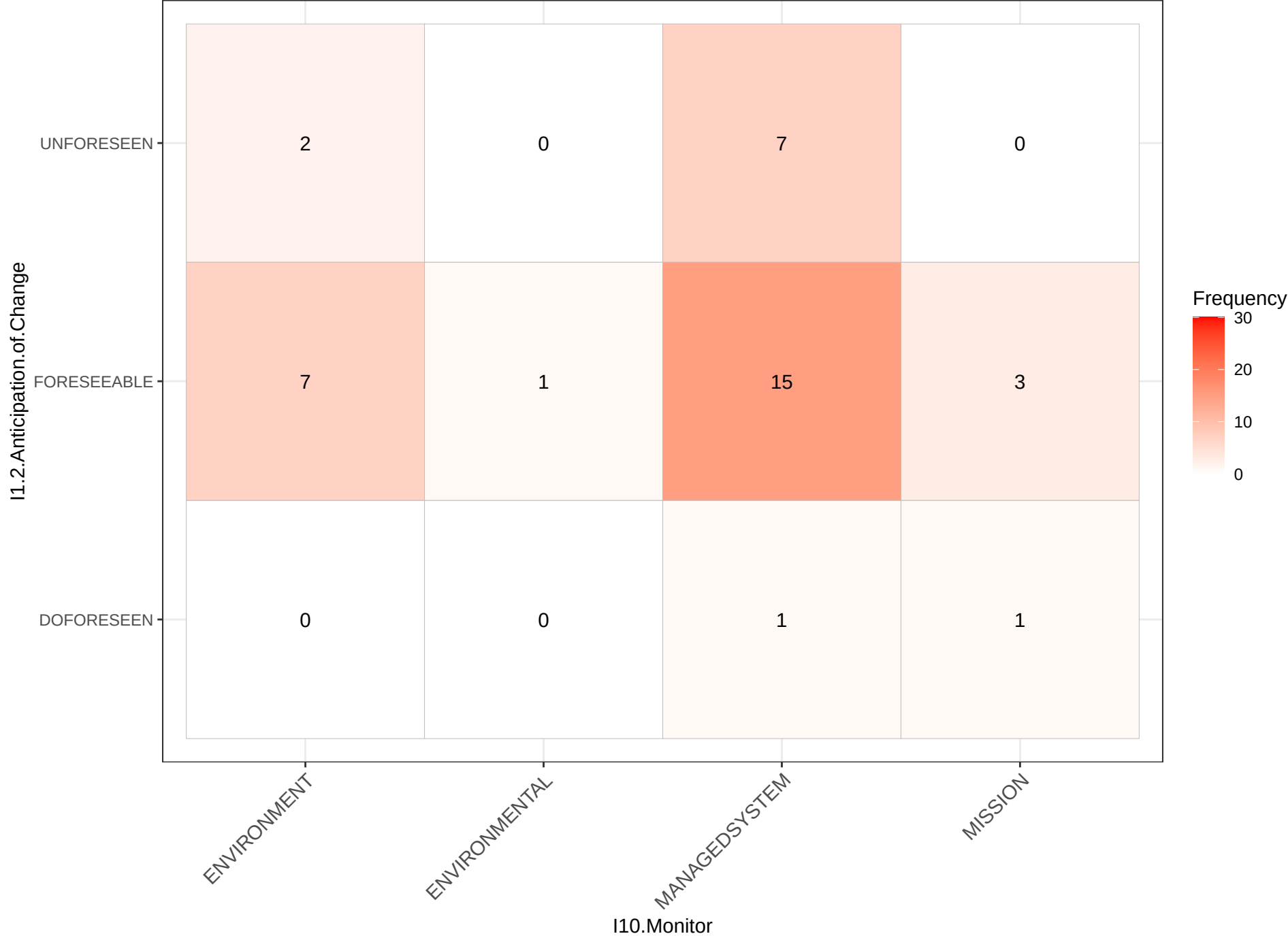
I1.2.Anticipation.of.Change_____I8.Evaluation



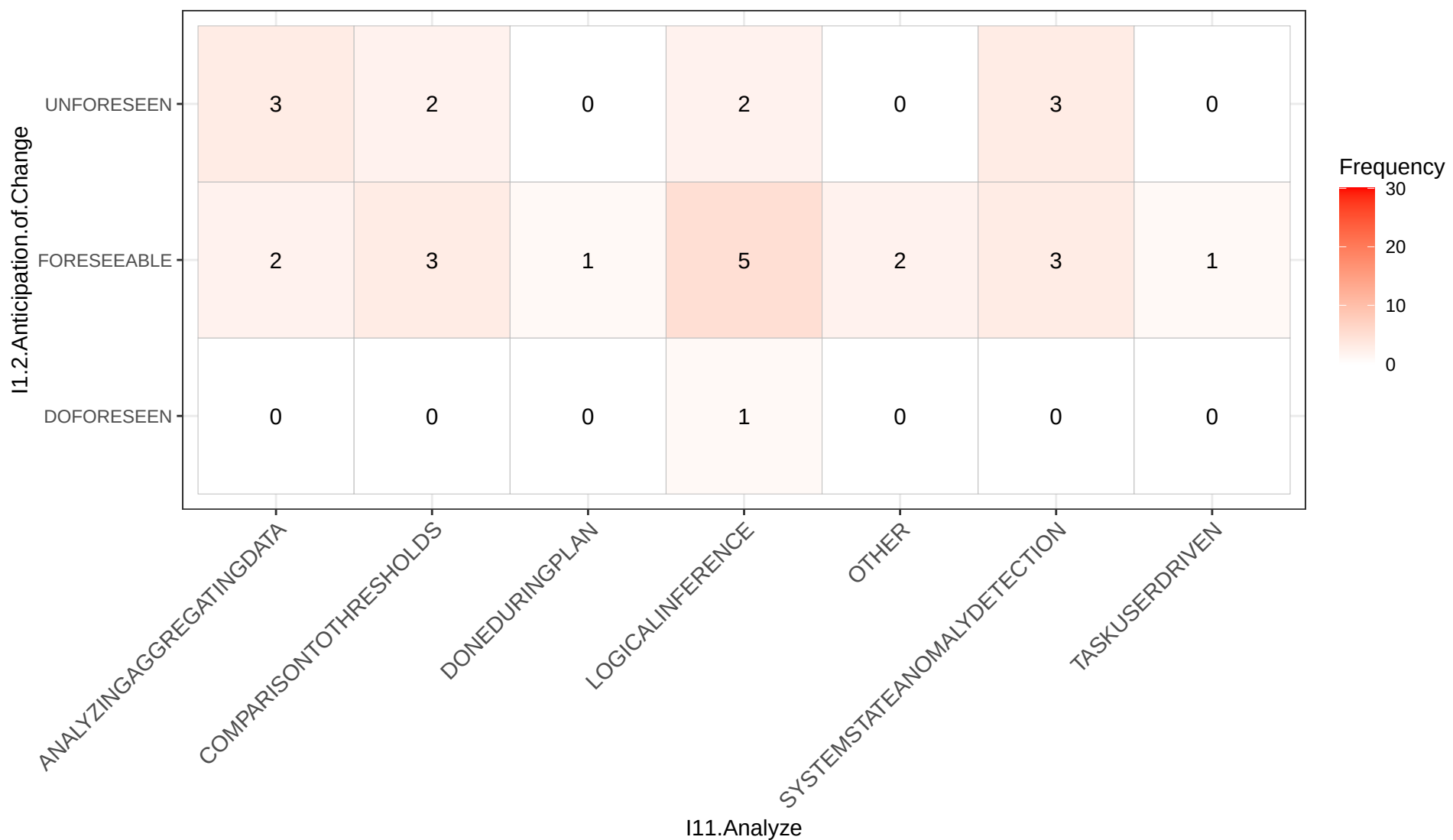
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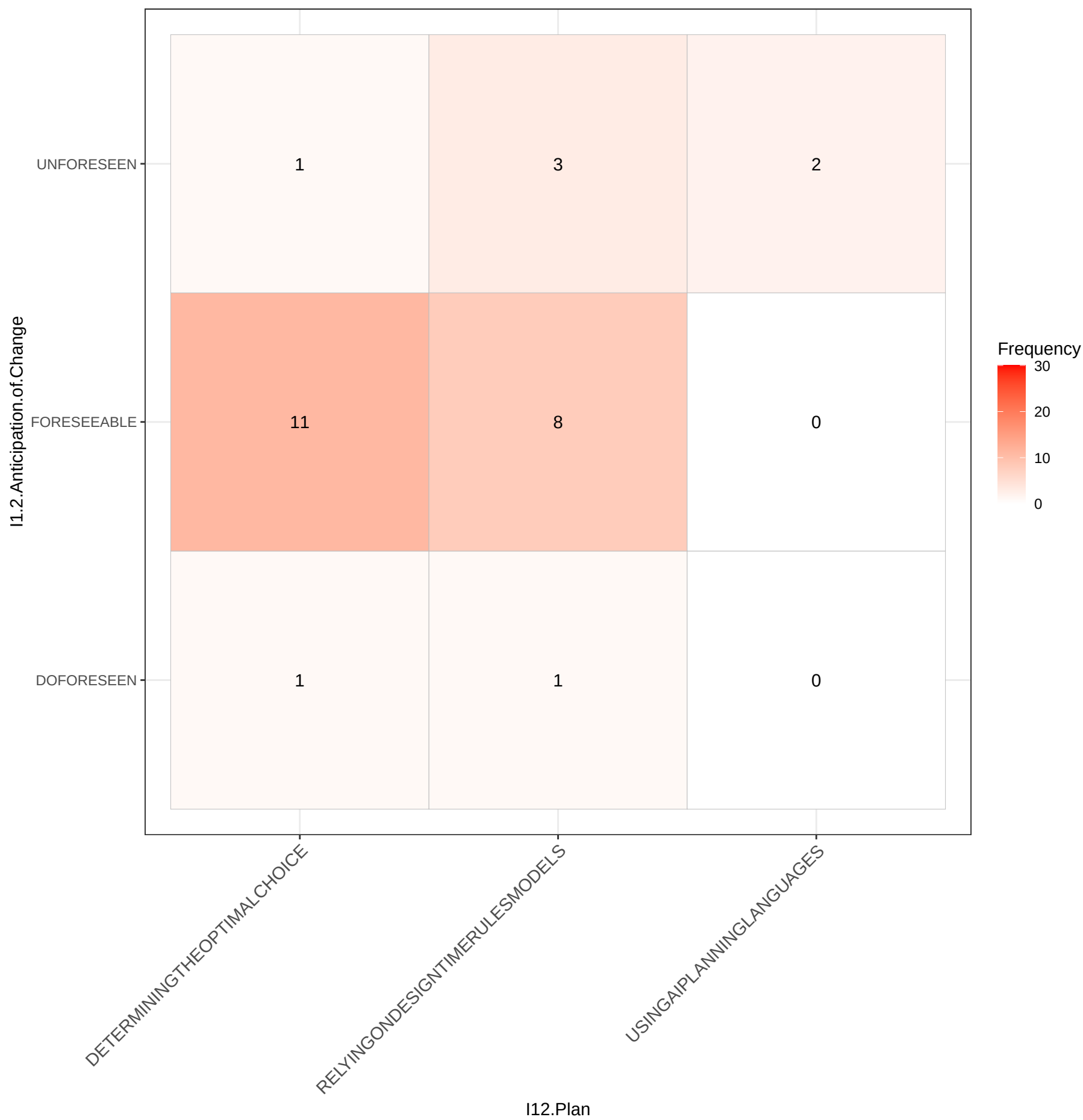
I1.2.Anticipation.of.Change_____I10.Monitor



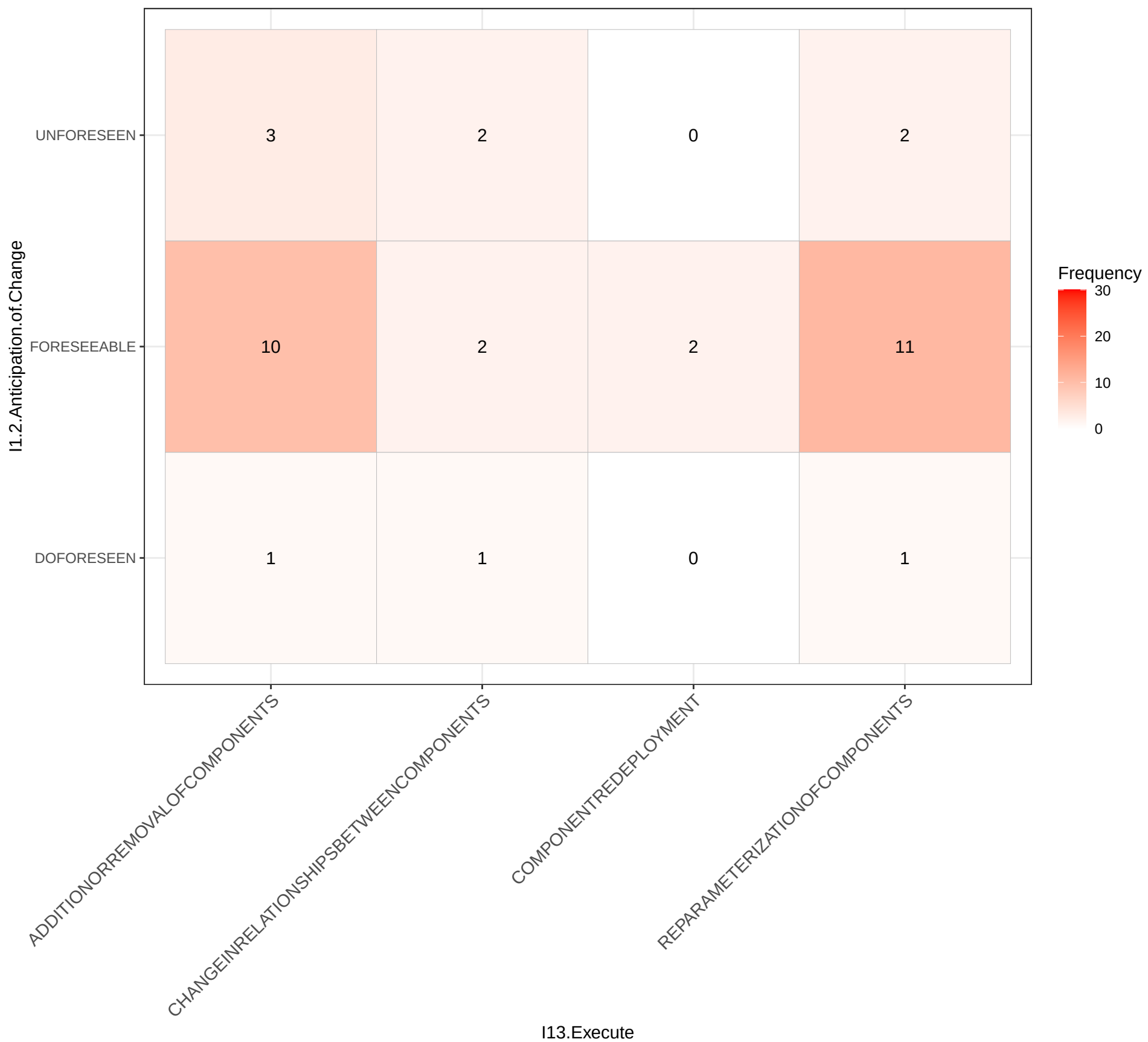
I1.2.Anticipation.of.Change_____I11.Analyze



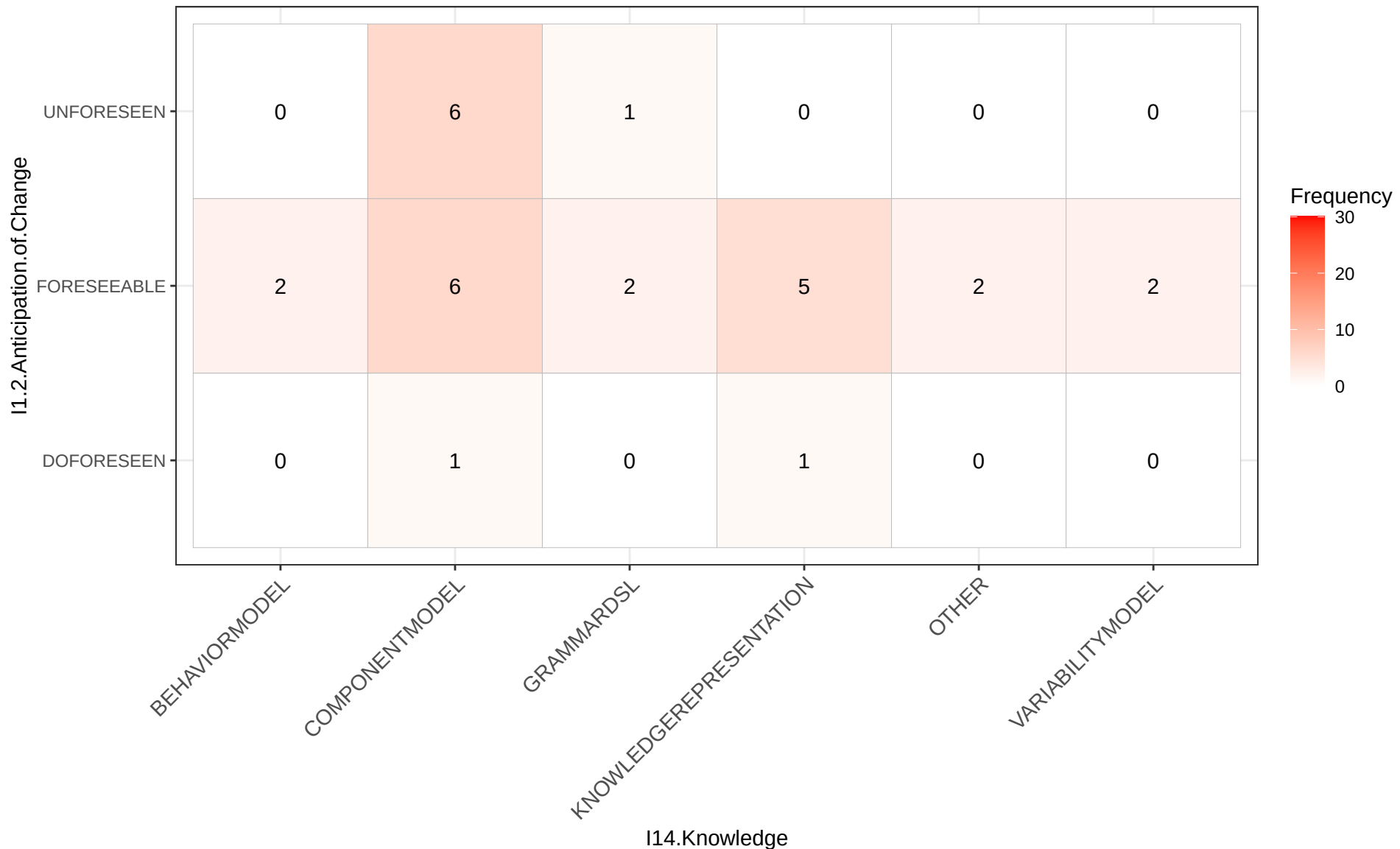
I1.2.Anticipation.of.Change_____I12.Plan

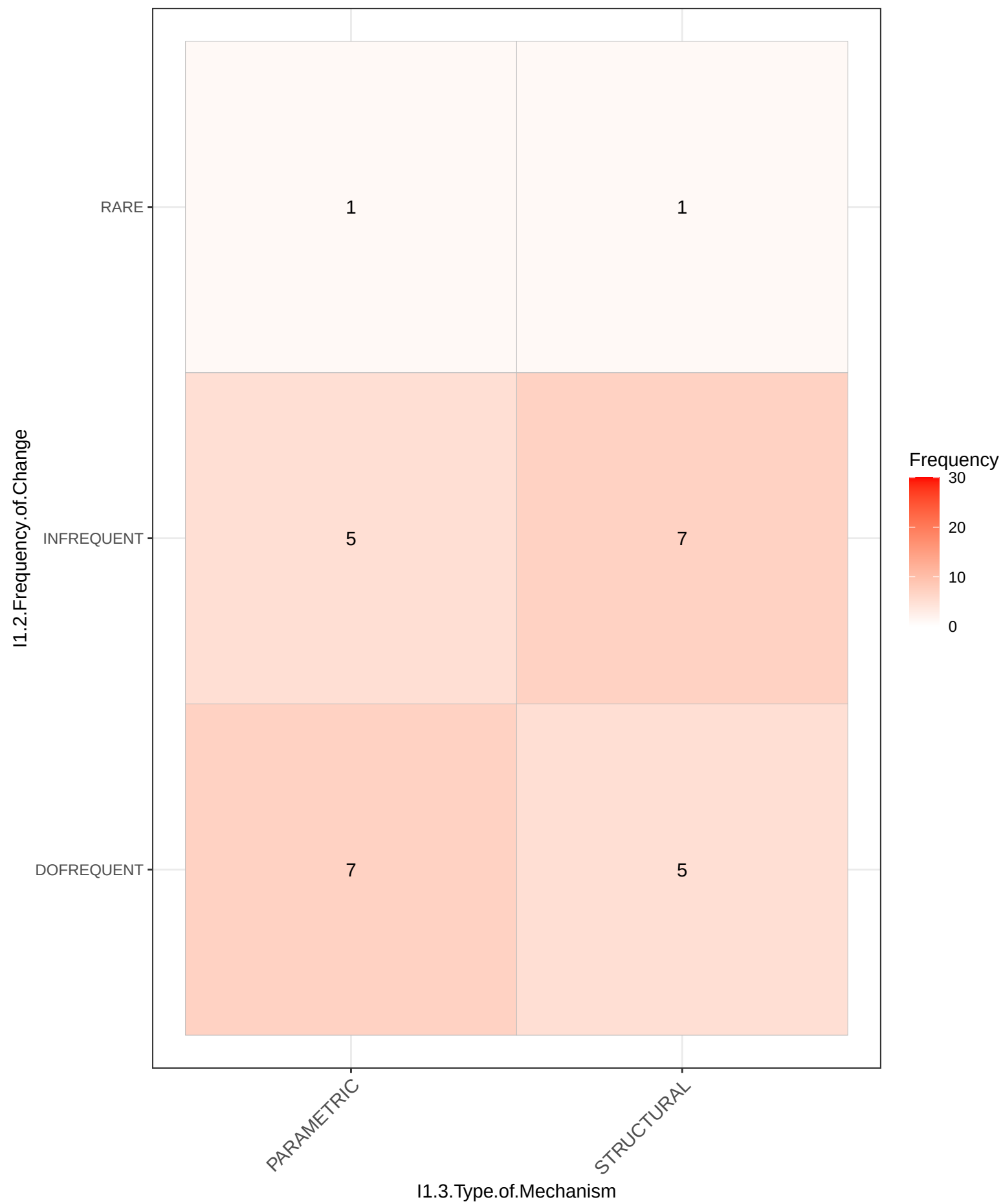


I1.2.Anticipation.of.Change_____I13.Execute

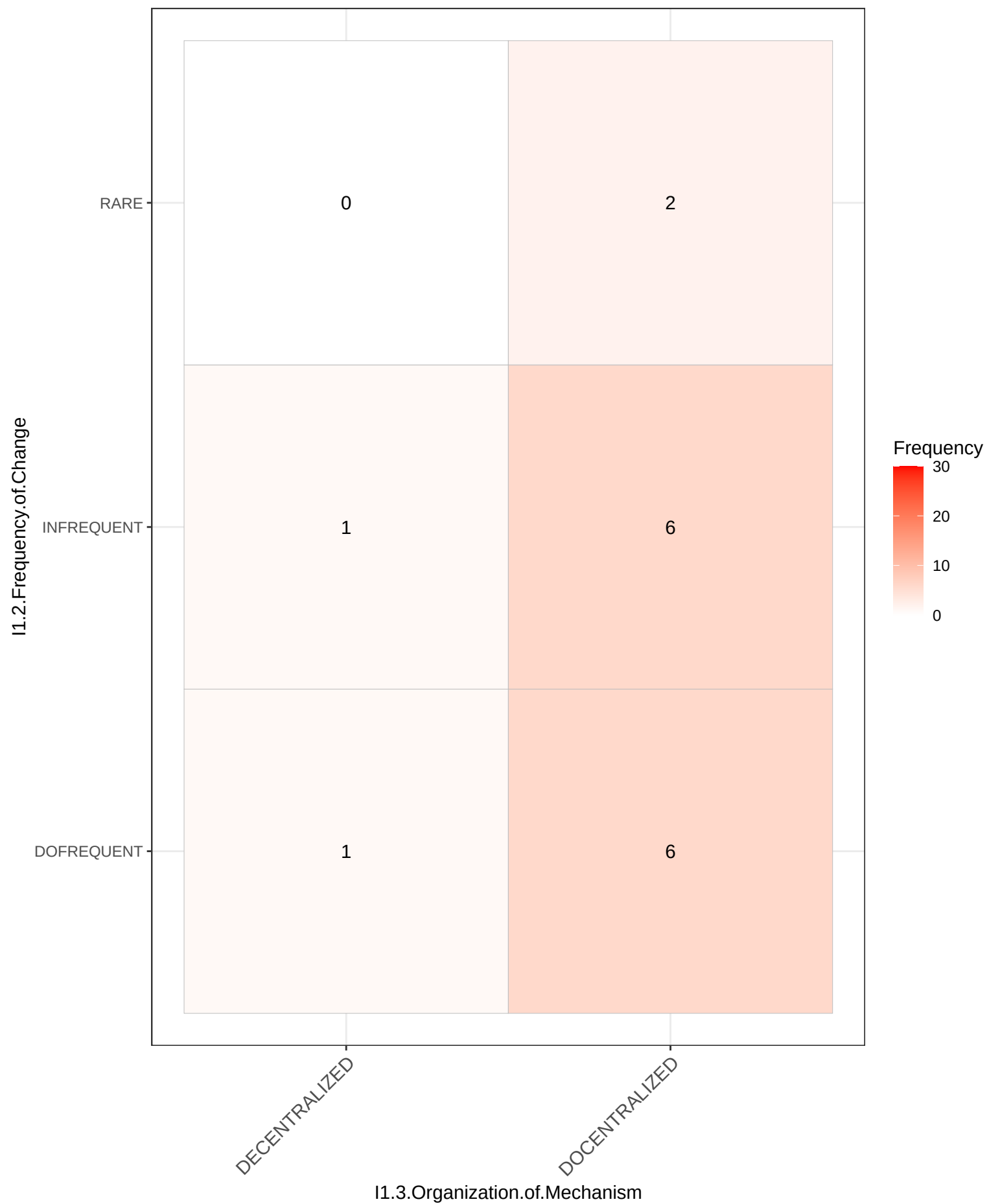


I1.2.Anticipation.of.Change_____I14.Knowledge





I1.2.Frequency.of.Change_____I1.3.Organization.of.Mechanism



I1.2.Frequency.of.Change_____I1.3.Scope.of.Mechanism

I1.2.Frequency.of.Change

RARE

INFREQUENT

DOFREQUENT

0

1

4

2

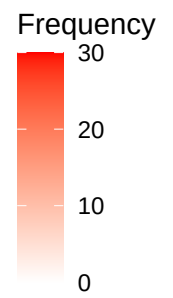
6

6

GLOBAL

LOCAL

I1.3.Scope.of.Mechanism



I1.2.Frequency.of.Change_____I1.3.Duration.of.Mechanism

I1.2.Frequency.of.Change

RARE

1

1

0

INFREQUENT

5

0

0

DOFREQUENT

5

0

1

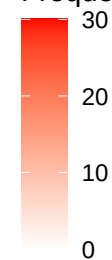
DOSHORT

MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism

Frequency



I1.2.Frequency.of.Change_____I1.3.Timeliness.of.Mechanism

I1.2.Frequency.of.Change

RARE

2

INFREQUENT

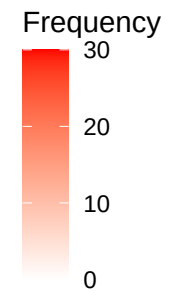
6

DOFREQUENT

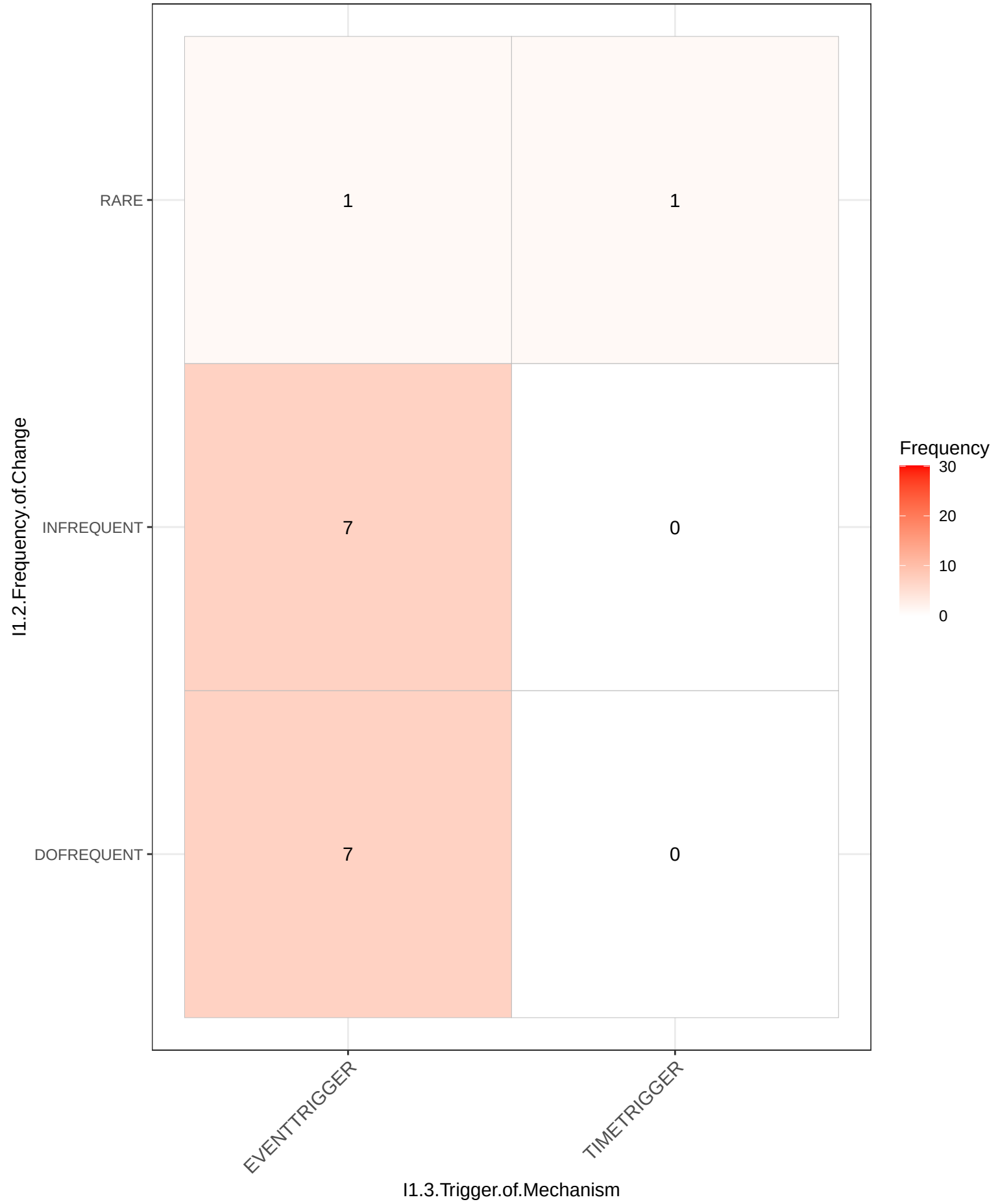
6

BESTEFFECT

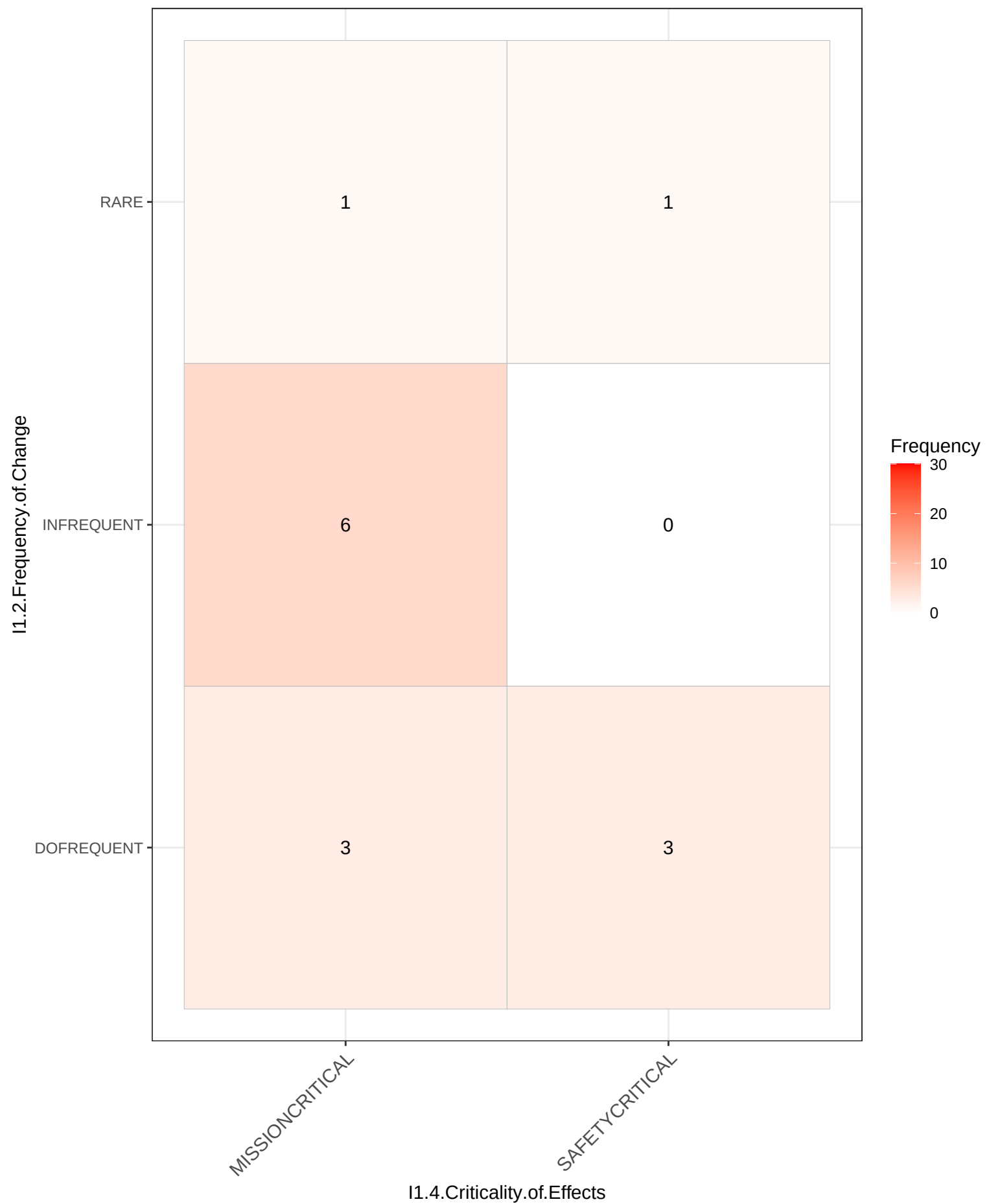
I1.3.Timeliness.of.Mechanism

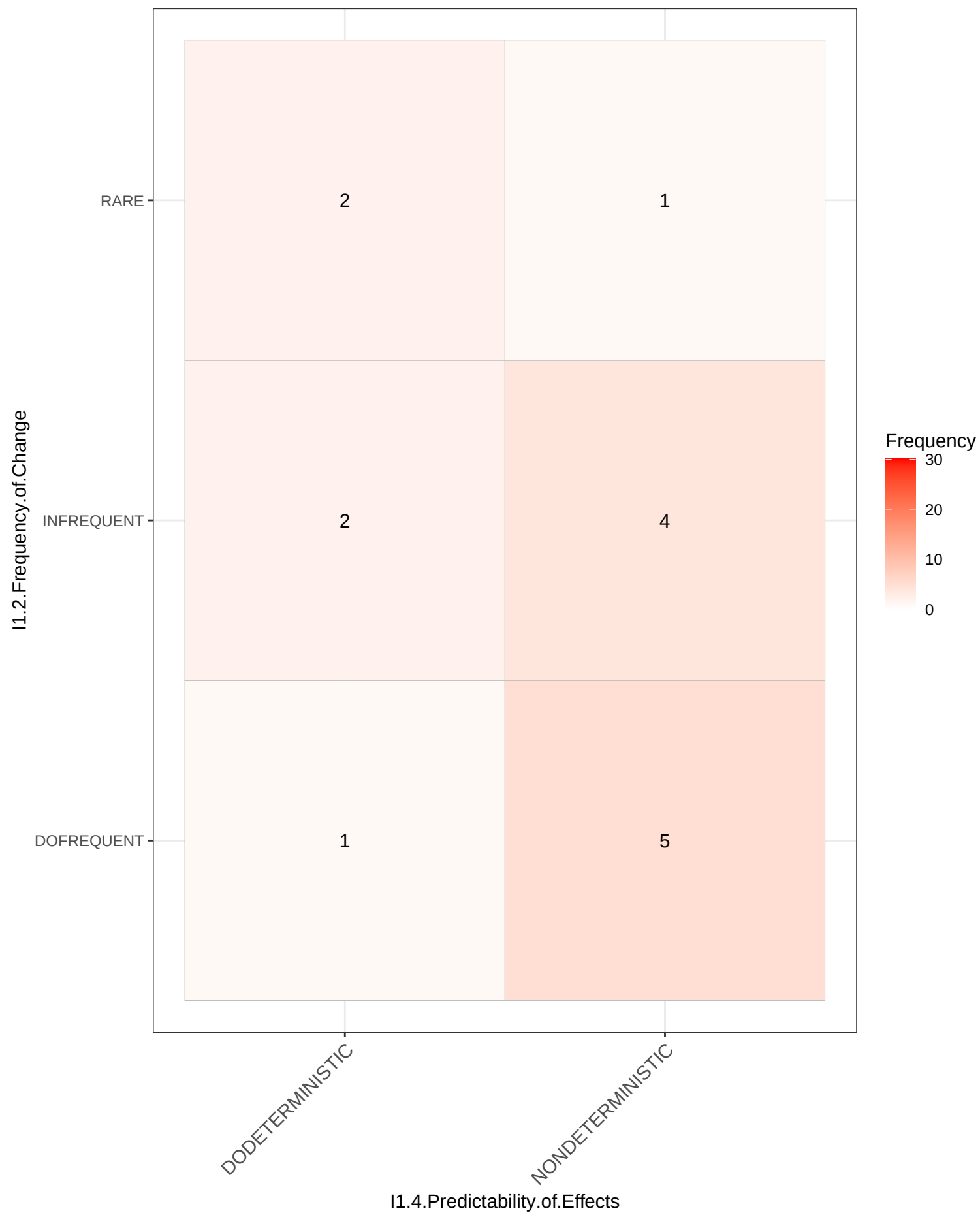


I1.2.Frequency.of.Change_____I1.3.Trigger.of.Mechanism



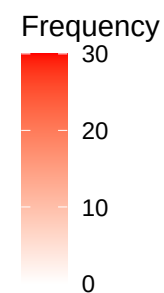
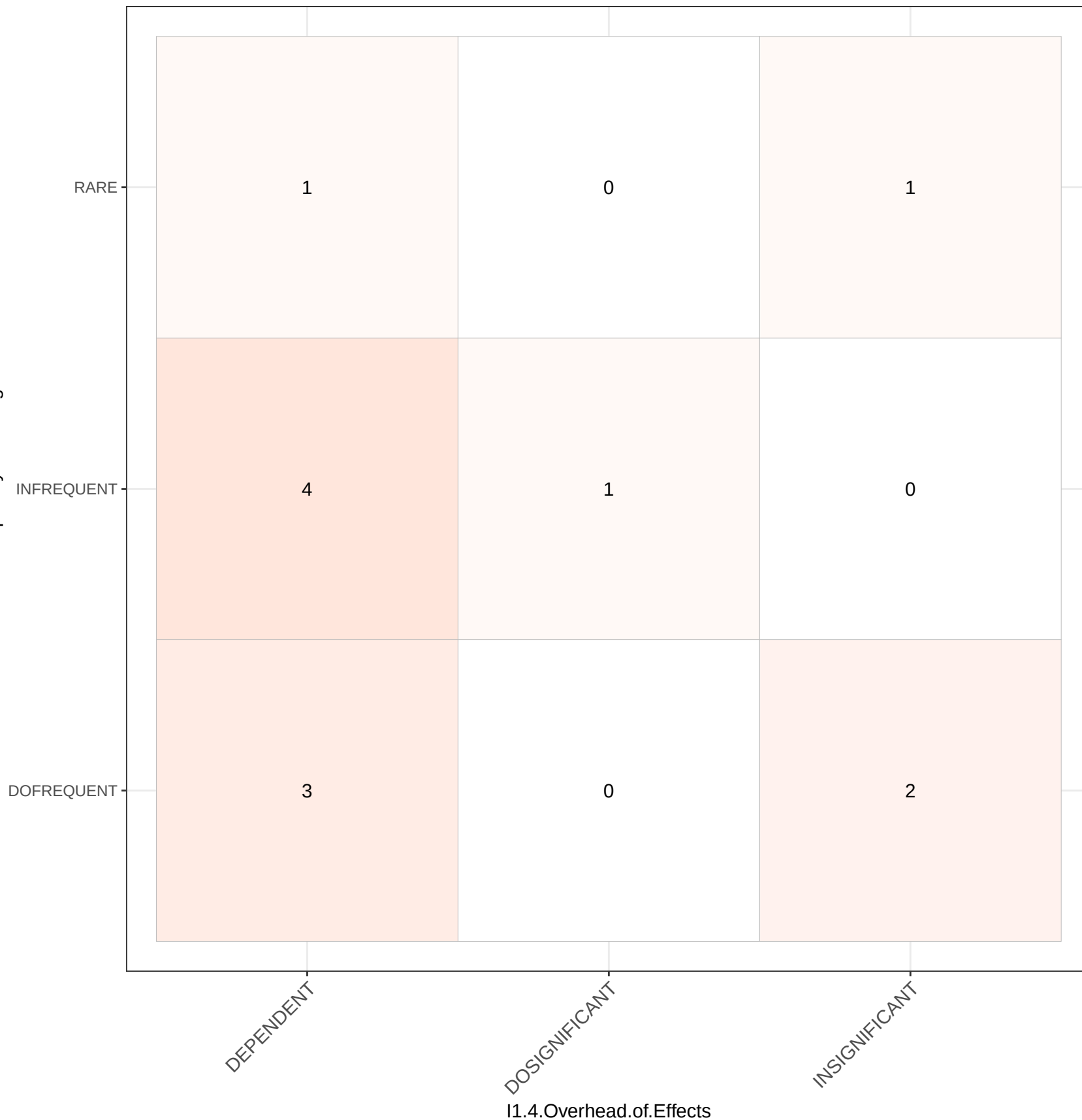
I1.2.Frequency.of.Change_____I1.4.Criticality.of.Effects





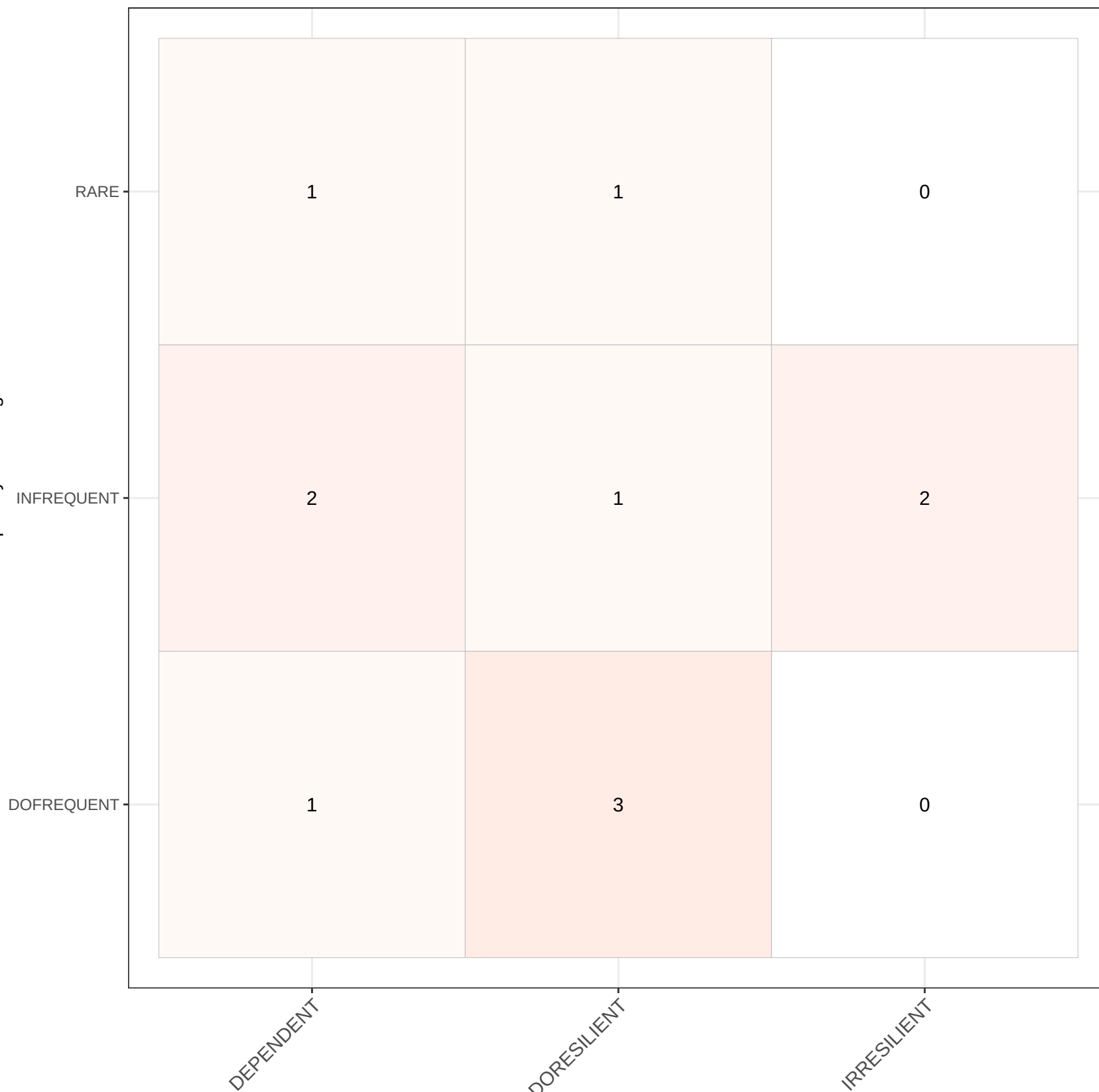
I1.2.Frequency.of.Change_____I1.4.Overhead.of.Effects

I1.2.Frequency.of.Change



I1.2.Frequency.of.Change_____I1.4.Resilience.of.Effects

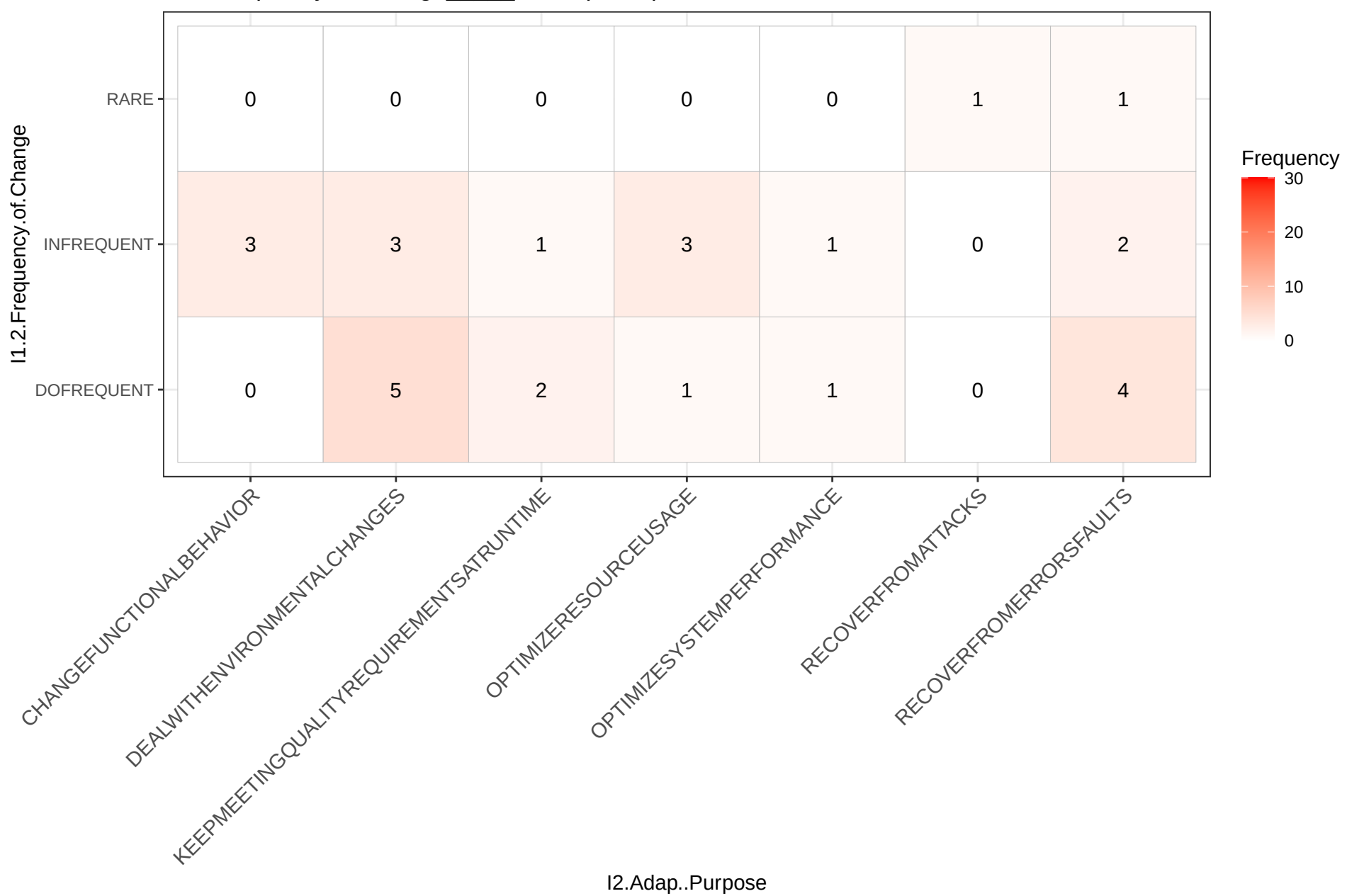
I1.2.Frequency.of.Change



Frequency
30
20
10
0

I1.4.Resilience.of.Effects

I1.2.Frequency.of.Change_____I2.Adap..Purpose



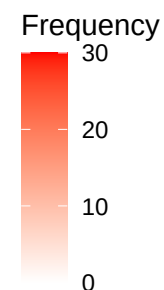
I1.2.Frequency.of.Change

I1.2.Frequency.of.Change I3.Robot.Type

RARE	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
INFREQUENT	0	0	0	0	0	0	1	0	1	0	1	0	1	0	0	1	0	1	0	0	0	0	1	0	0	0
DOFREQUENT	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0	1	0	0	0	0	0	2	0	0	0	1

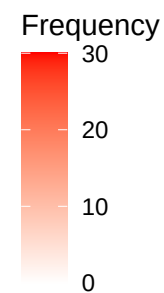
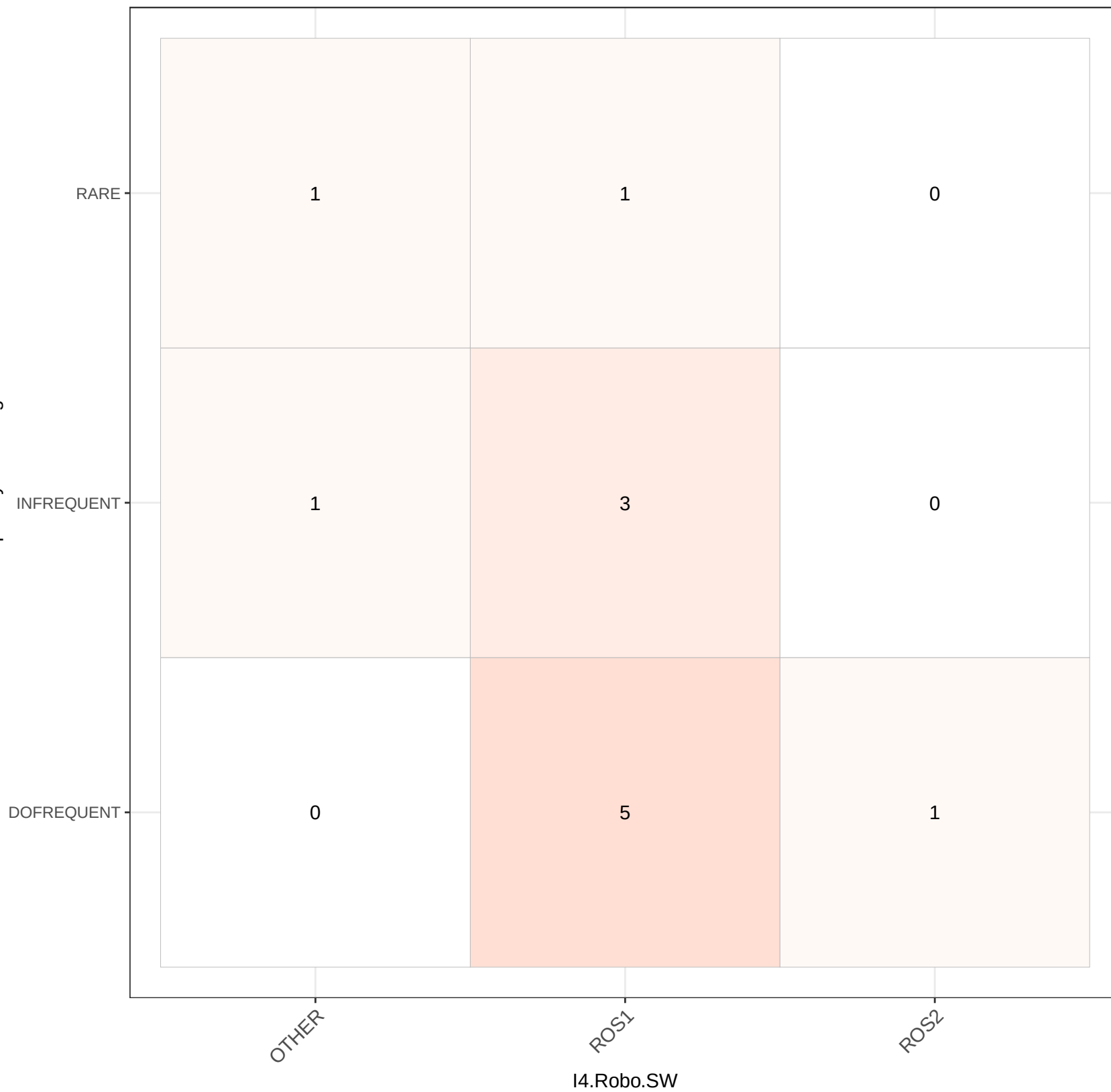
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOT
HETEROGENOUSROBOTS
HEXMANIPULATOR
HEXAII
INFOTAINMENTROBOTMOBILESERVICEROBOT
IROBOTCREATE2
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
RESCUE
SINGLESERVINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

I3.Robot.Type

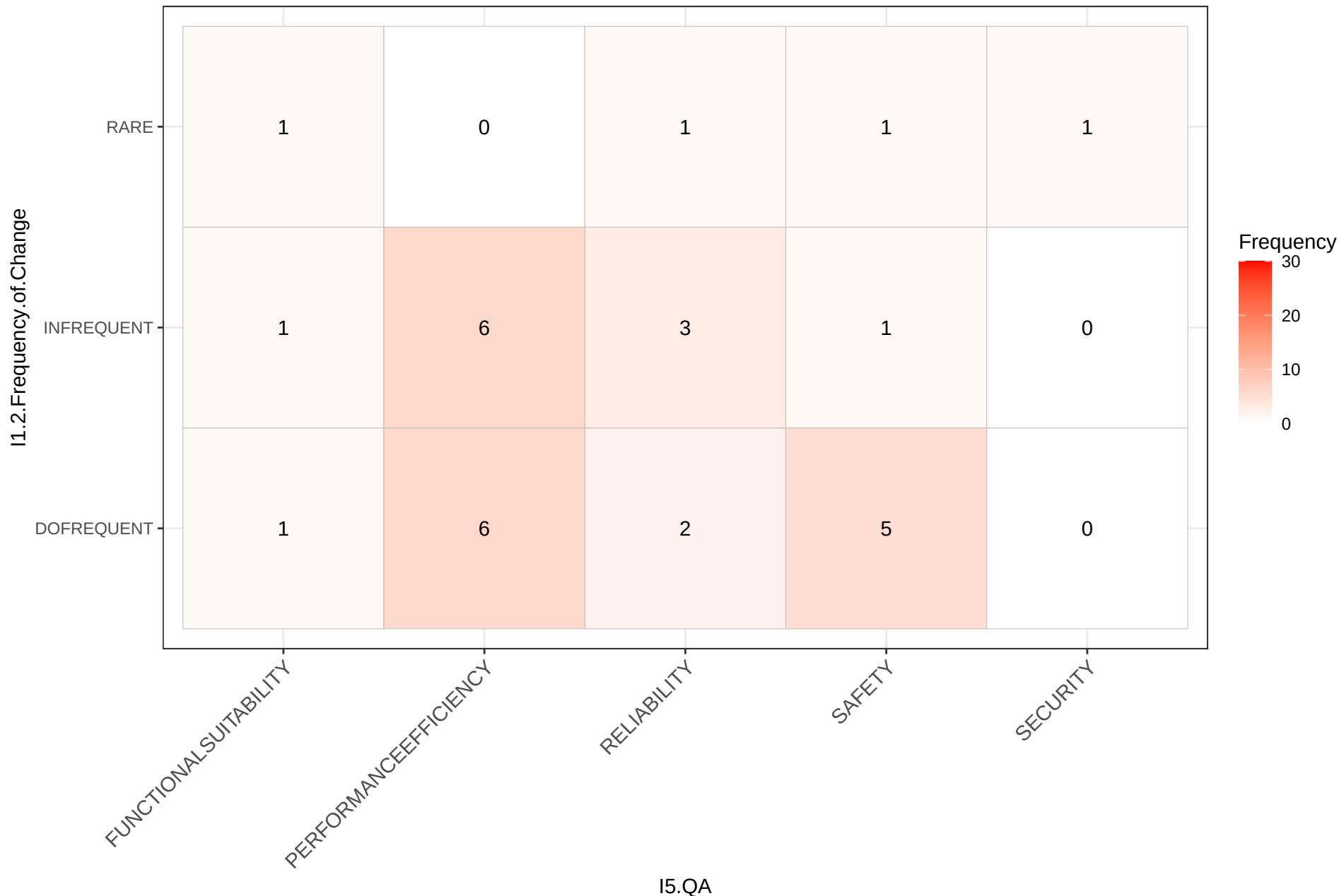


I1.2.Frequency.of.Change_____I4.Robo.SW

I1.2.Frequency.of.Change

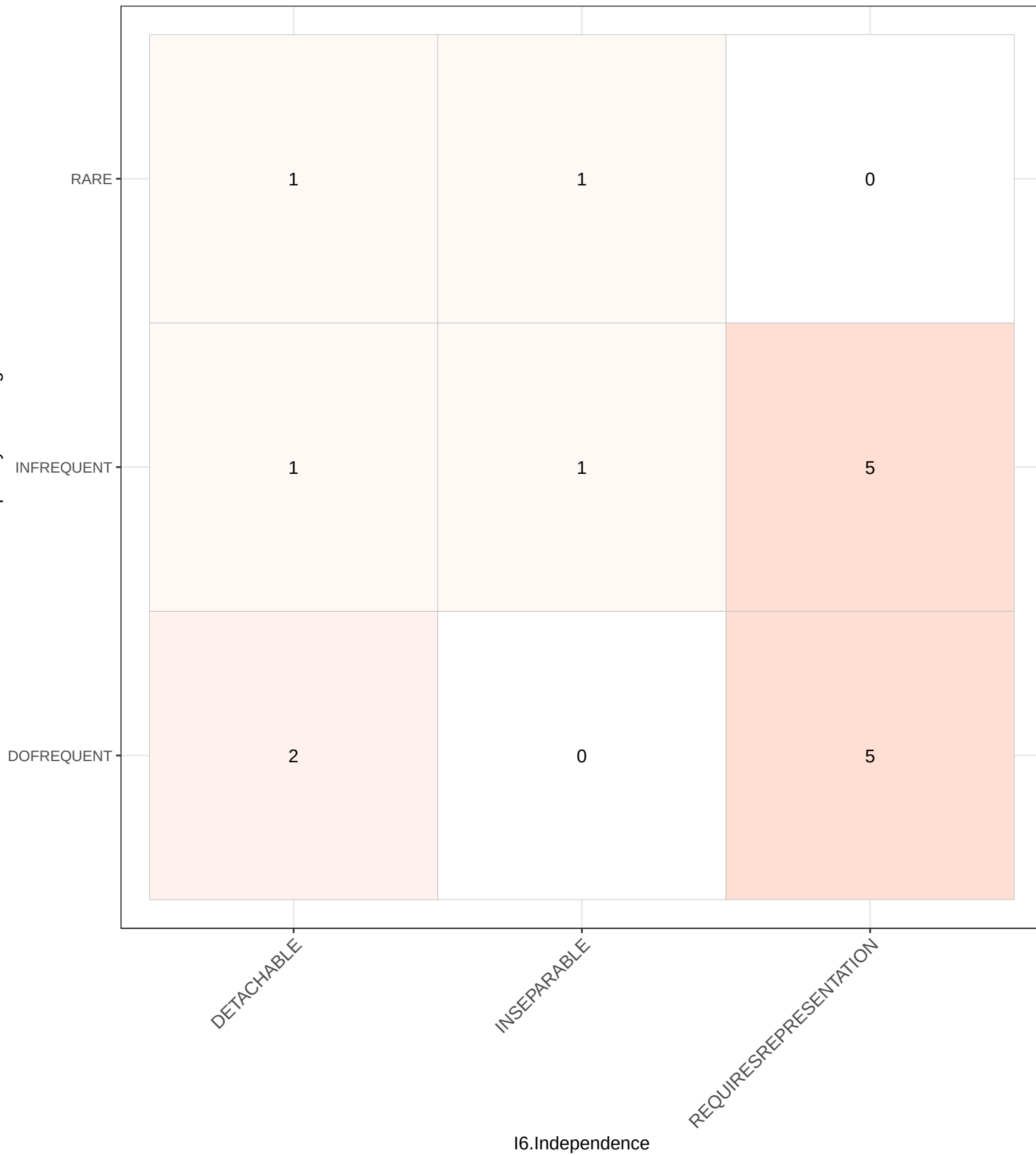


I1.2.Frequency.of.Change_____I5.QA

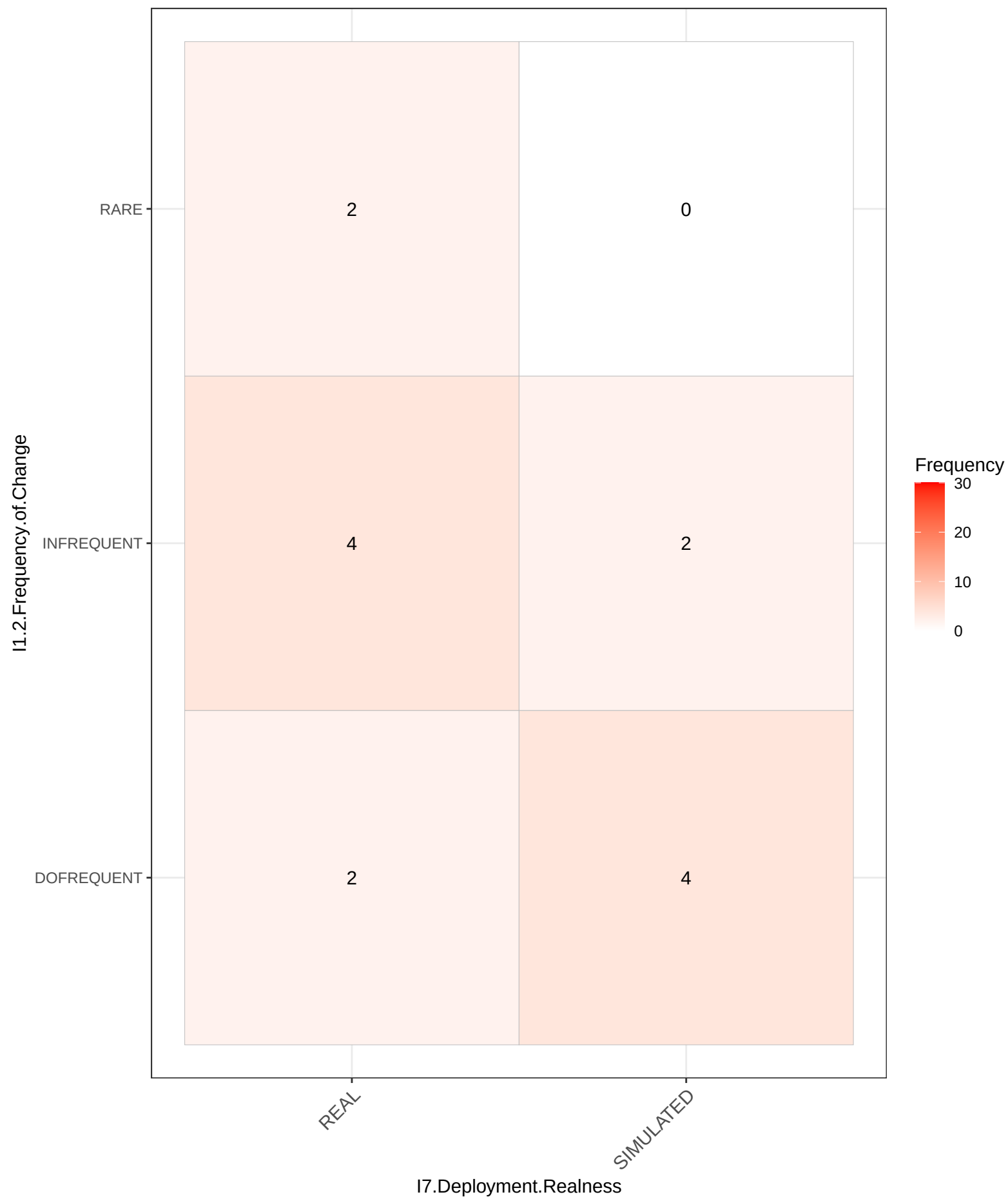


I1.2.Frequency.of.Change_____I6.Independence

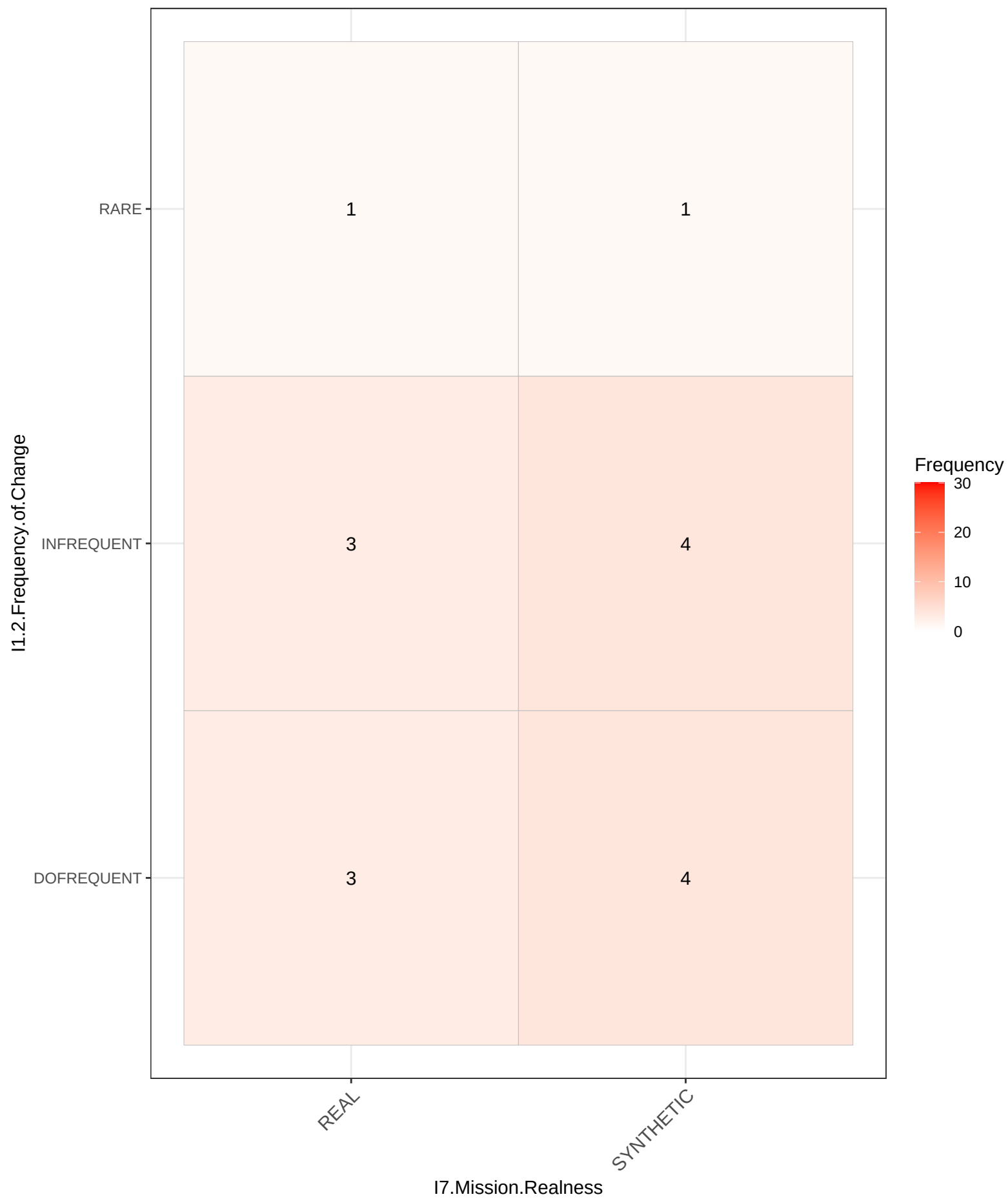
I1.2.Frequency.of.Change



I1.2.Frequency.of.Change_____I7.Deployment.Realness

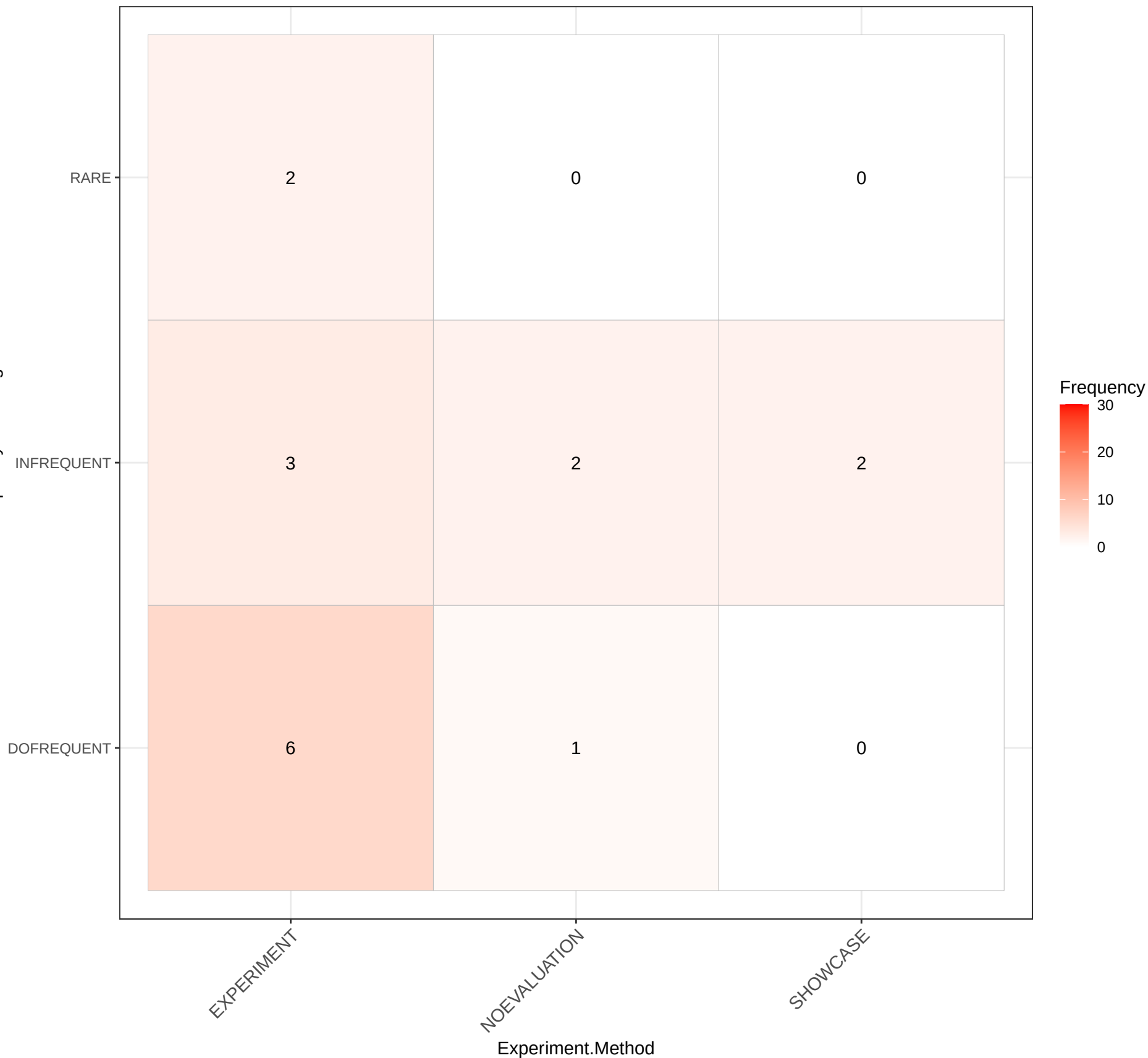


I1.2.Frequency.of.Change_____I7.Mission.Realness

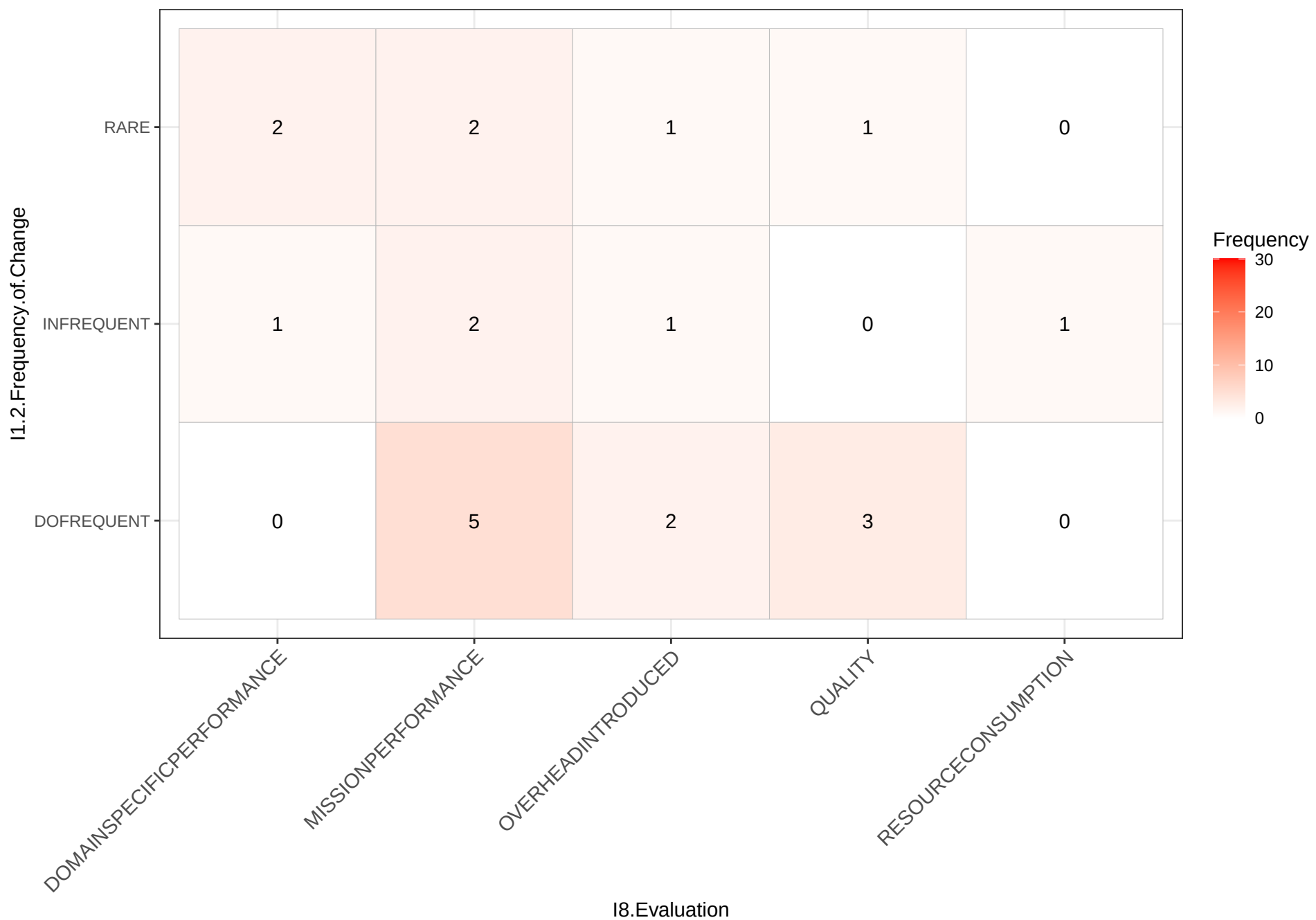


I1.2.Frequency.of.Change_____Experiment.Method

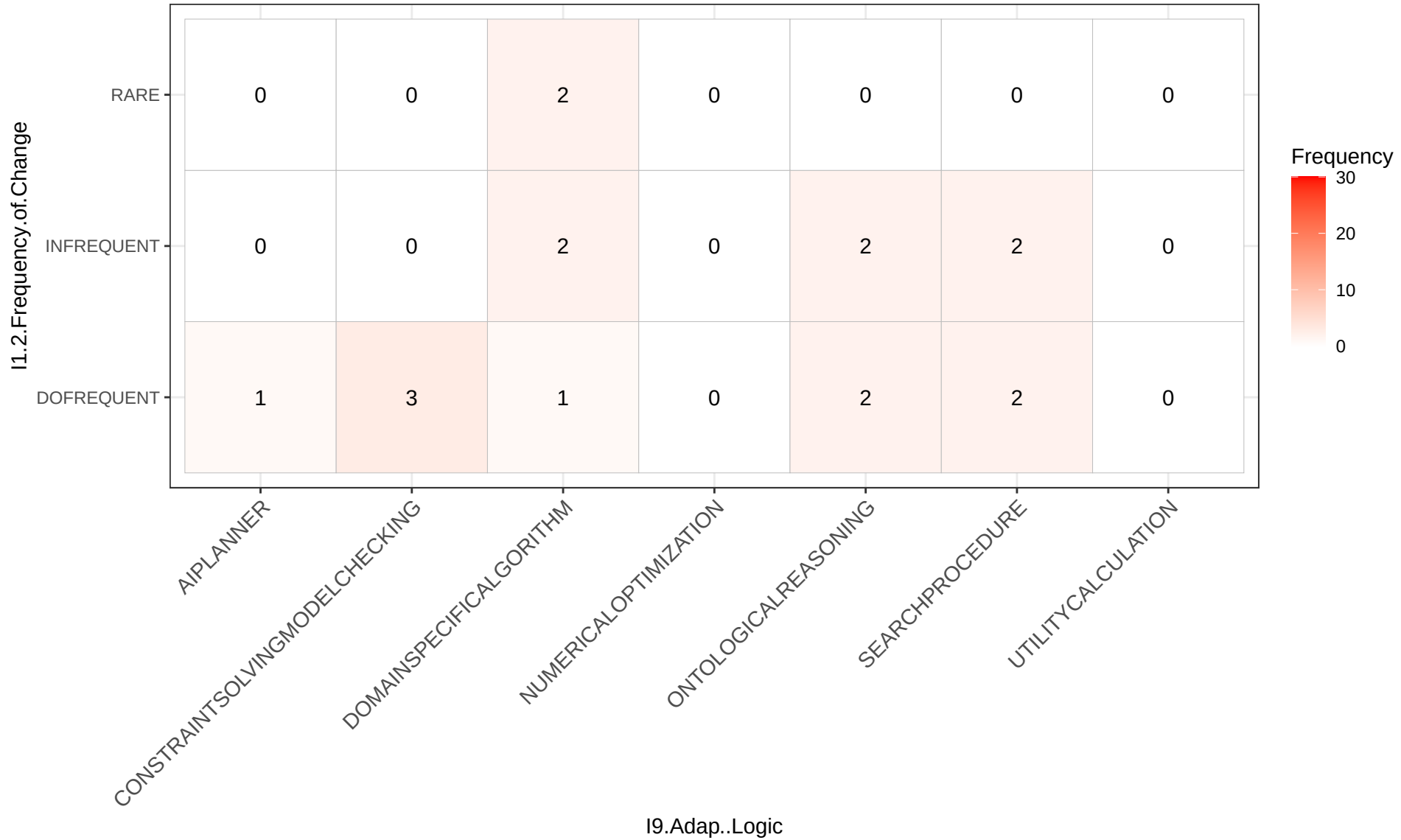
I1.2.Frequency.of.Change



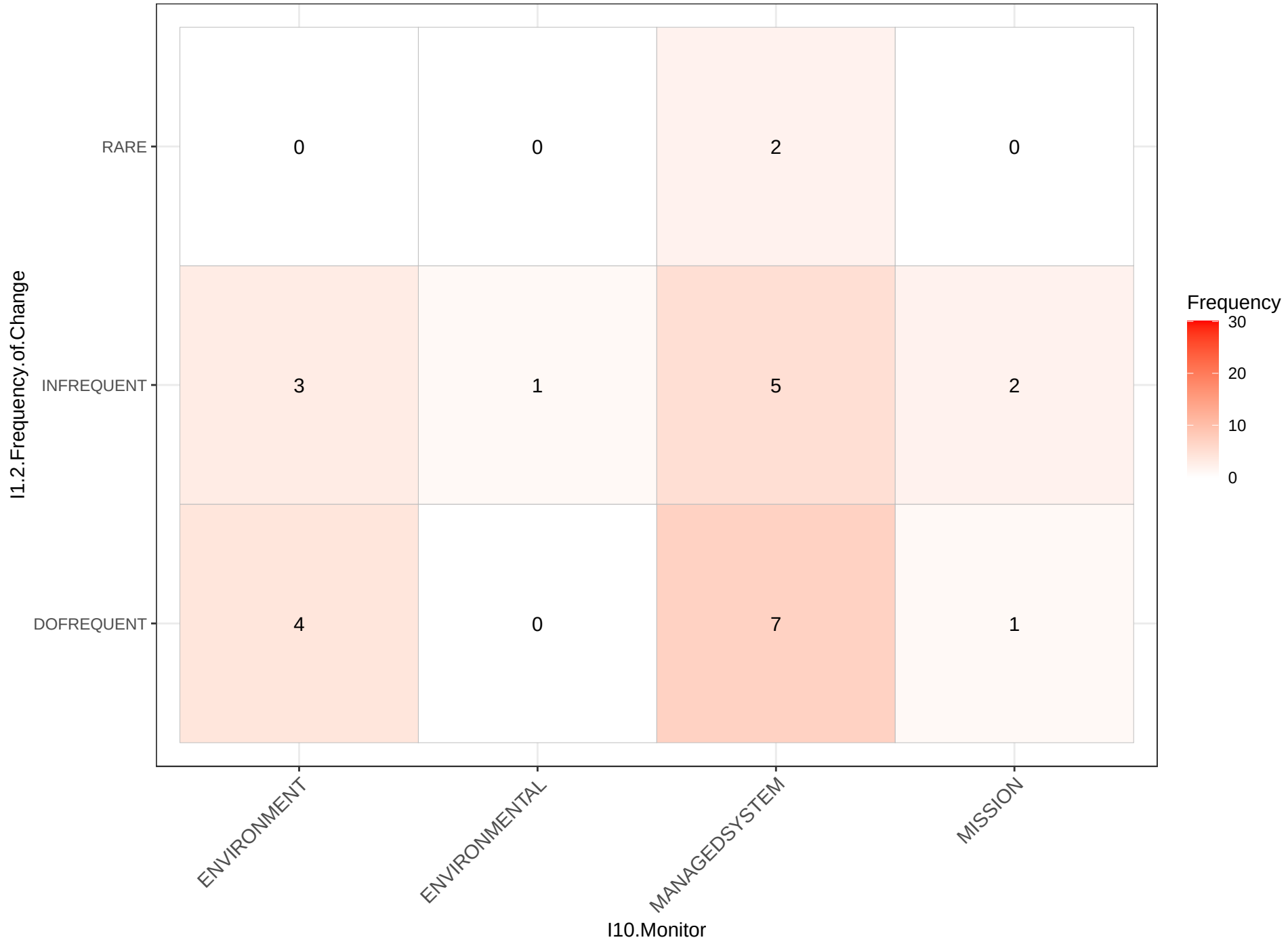
I1.2.Frequency.of.Change_____I8.Evaluation



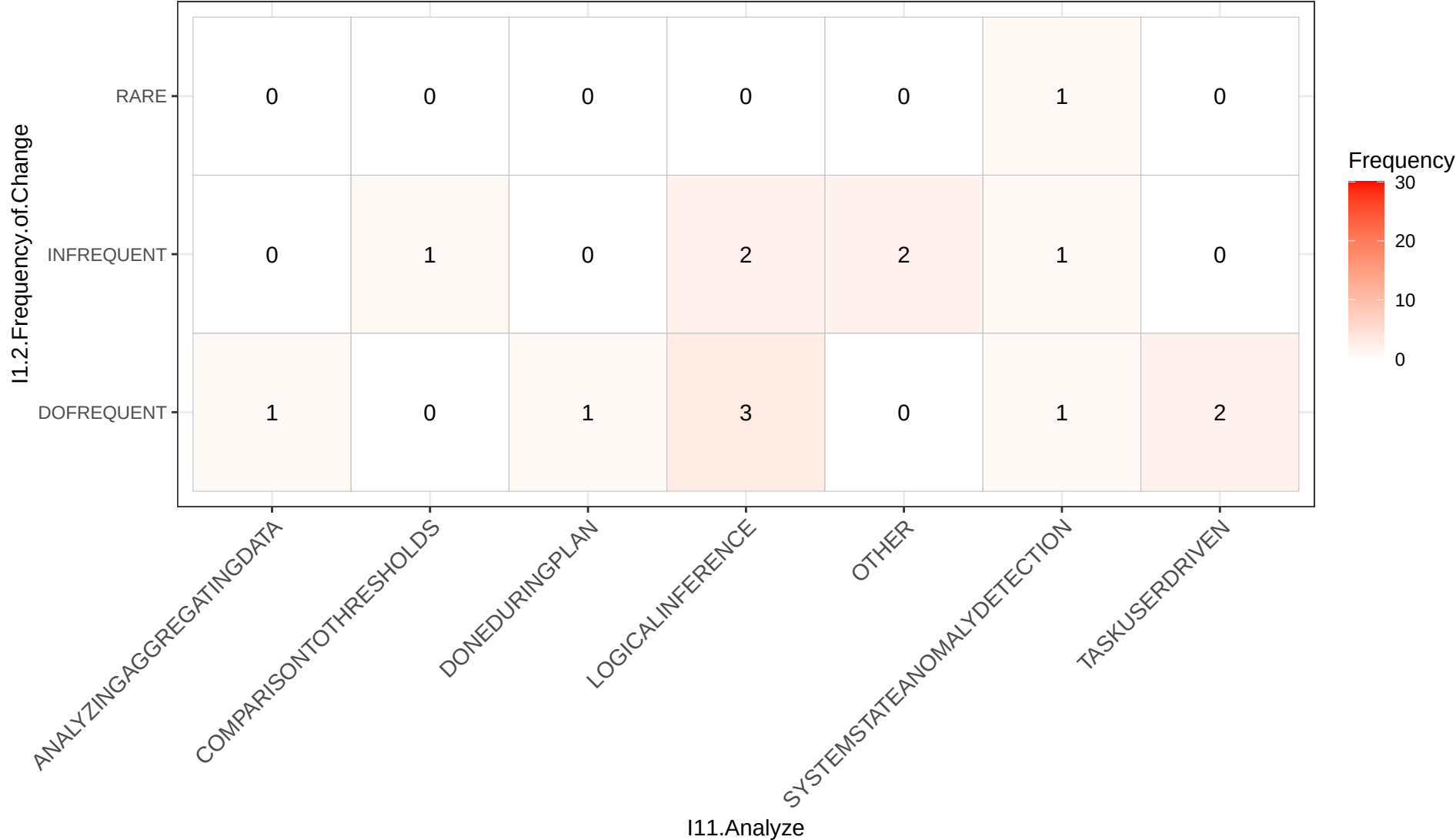
I1.2.Frequency.of.Change_____I9.Adap..Logic



I1.2.Frequency.of.Change_____I10.Monitor



I1.2.Frequency.of.Change_____I11.Analyze



I1.2.Frequency.of.Change_____I12.Plan

I1.2.Frequency.of.Change

RARE

2

0

INFREQUENT

2

5

0

DOFREQUENT

6

0

0

Frequency

30

20

10

0

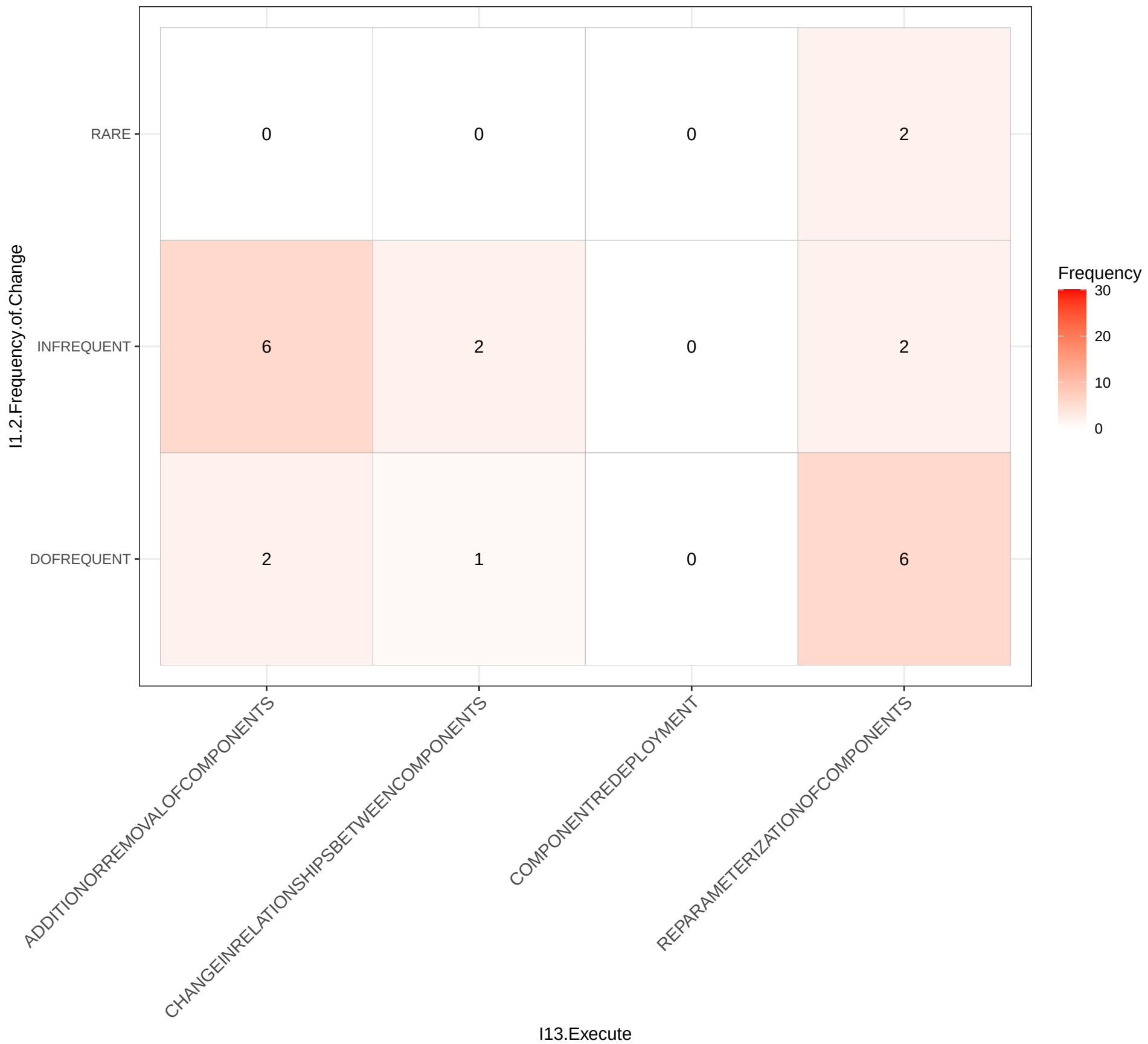
DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

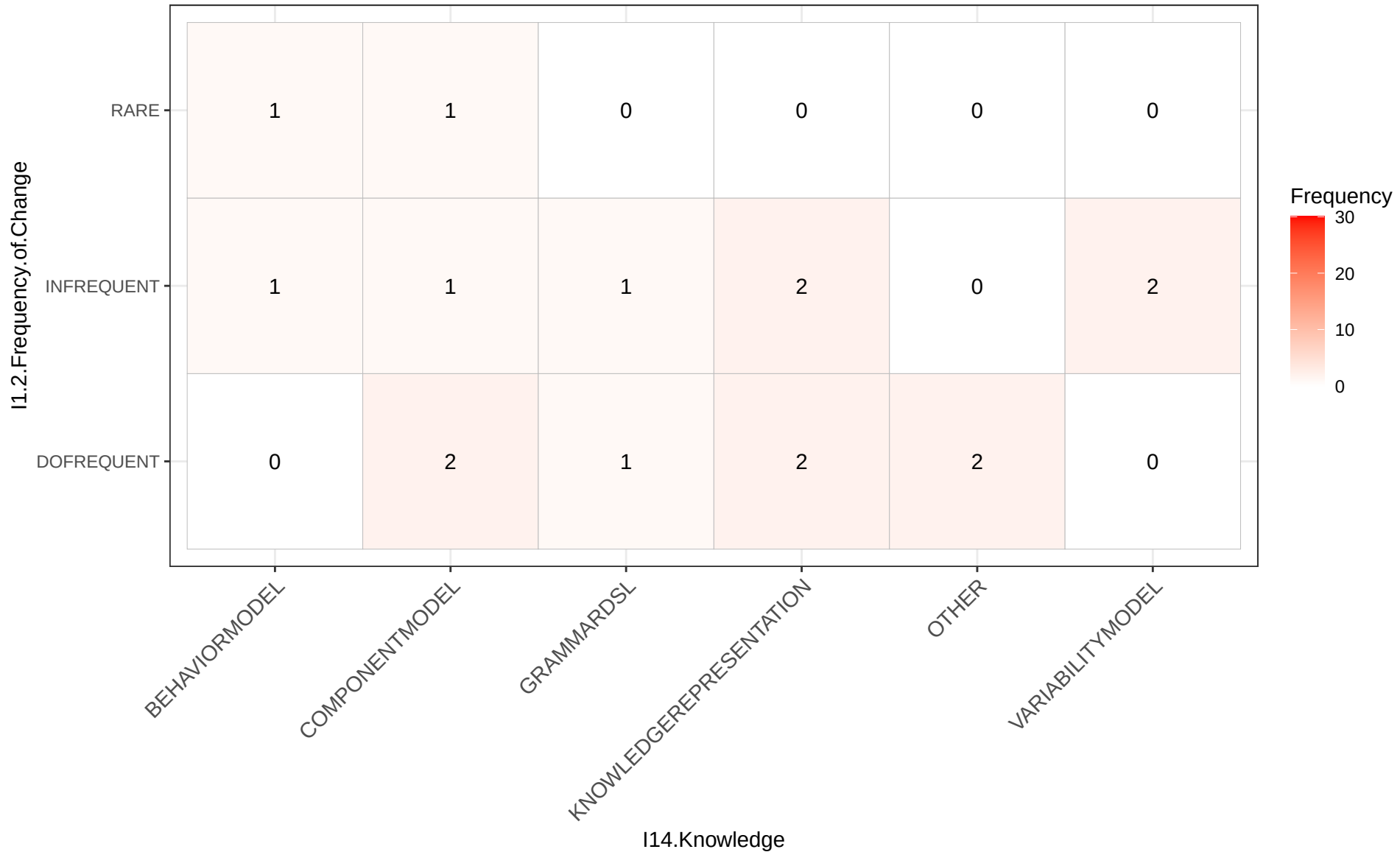
USINGAIPLANNINGLANGUAGES

I12.Plan

I1.2.Frequency.of.Change_____I13.Execute



I1.2.Frequency.of.Change_____I14.Knowledge



I1.3.Type.of.Mechanism_____I1.3.Organization.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

3

22

PARAMETRIC

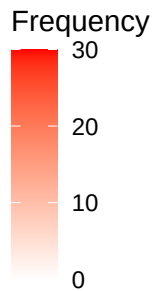
2

17

DECENTRALIZED

DOCENTRALIZED

I1.3.Organization.of.Mechanism



I1.3.Type.of.Mechanism_____I1.3.Scope.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

7

21

PARAMETRIC

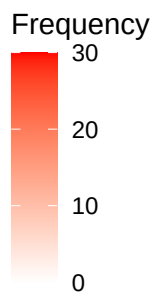
8

17

GLOBAL

LOCAL

I1.3.Scope.of.Mechanism



I1.3.Type.of.Mechanism____I1.3.Duration.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

18

1

3

PARAMETRIC

13

0

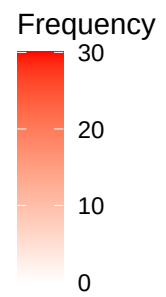
4

DOSHORT

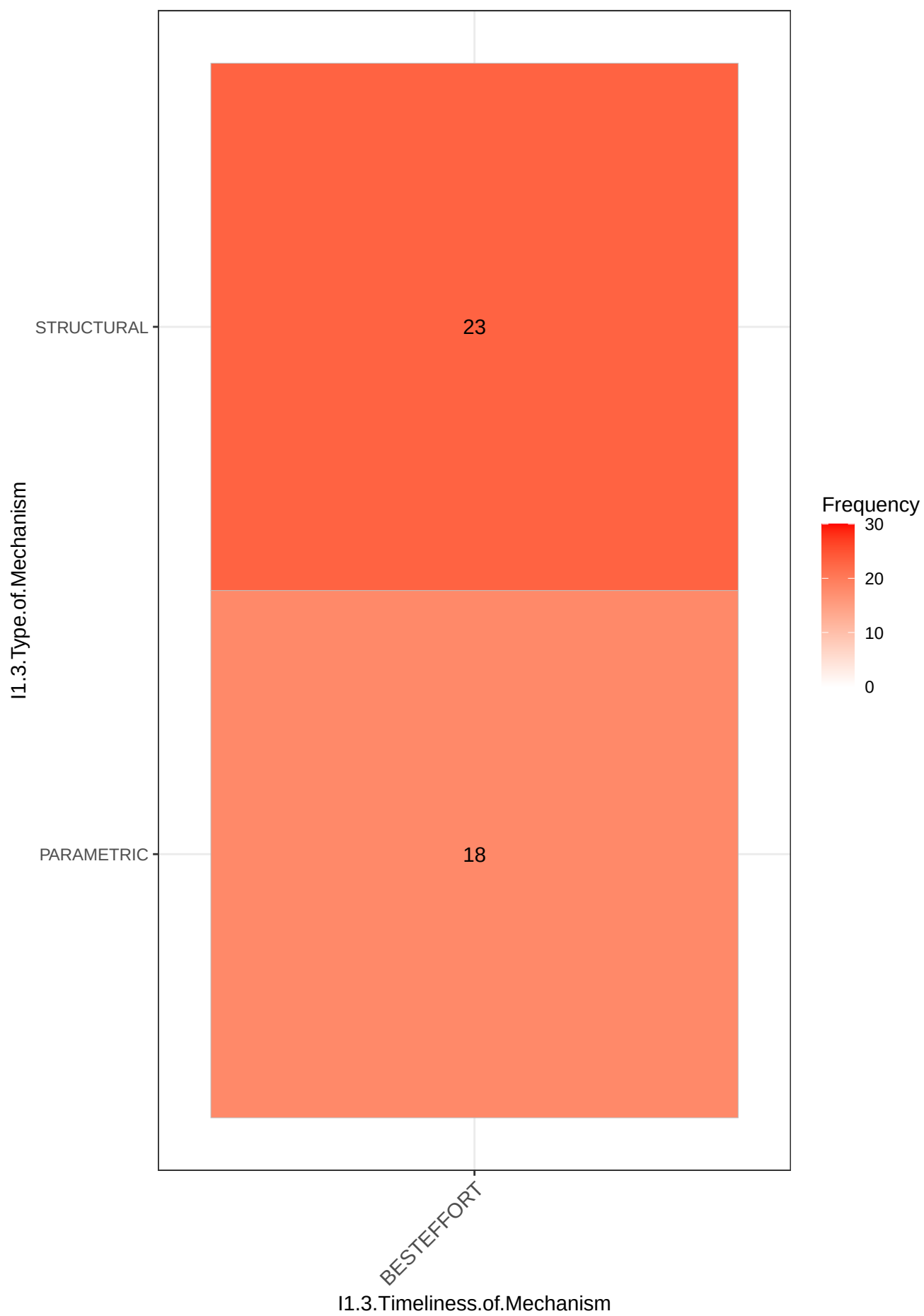
MEDIUM

VERYSHORT

I1.3.Duration.of.Mechanism



I1.3.Type.of.Mechanism_____I1.3.Timeliness.of.Mechanism



I1.3.Type.of.Mechanism_____I1.3.Trigger.of.Mechanism

I1.3.Type.of.Mechanism

STRUCTURAL

24

1

PARAMETRIC

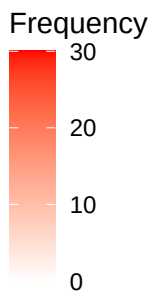
16

2

EVENTTRIGGER

TIMETRIGGER

I1.3.Trigger.of.Mechanism



I1.3.Type.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

18

5

PARAMETRIC

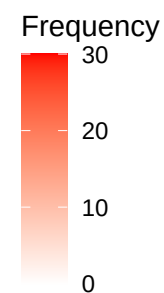
13

4

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects



I1.3.Type.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

8

15

PARAMETRIC

5

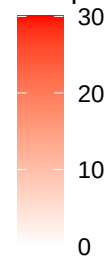
14

DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects

Frequency



I1.3.Type.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

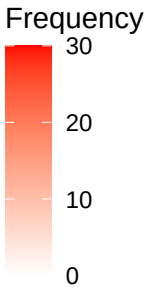
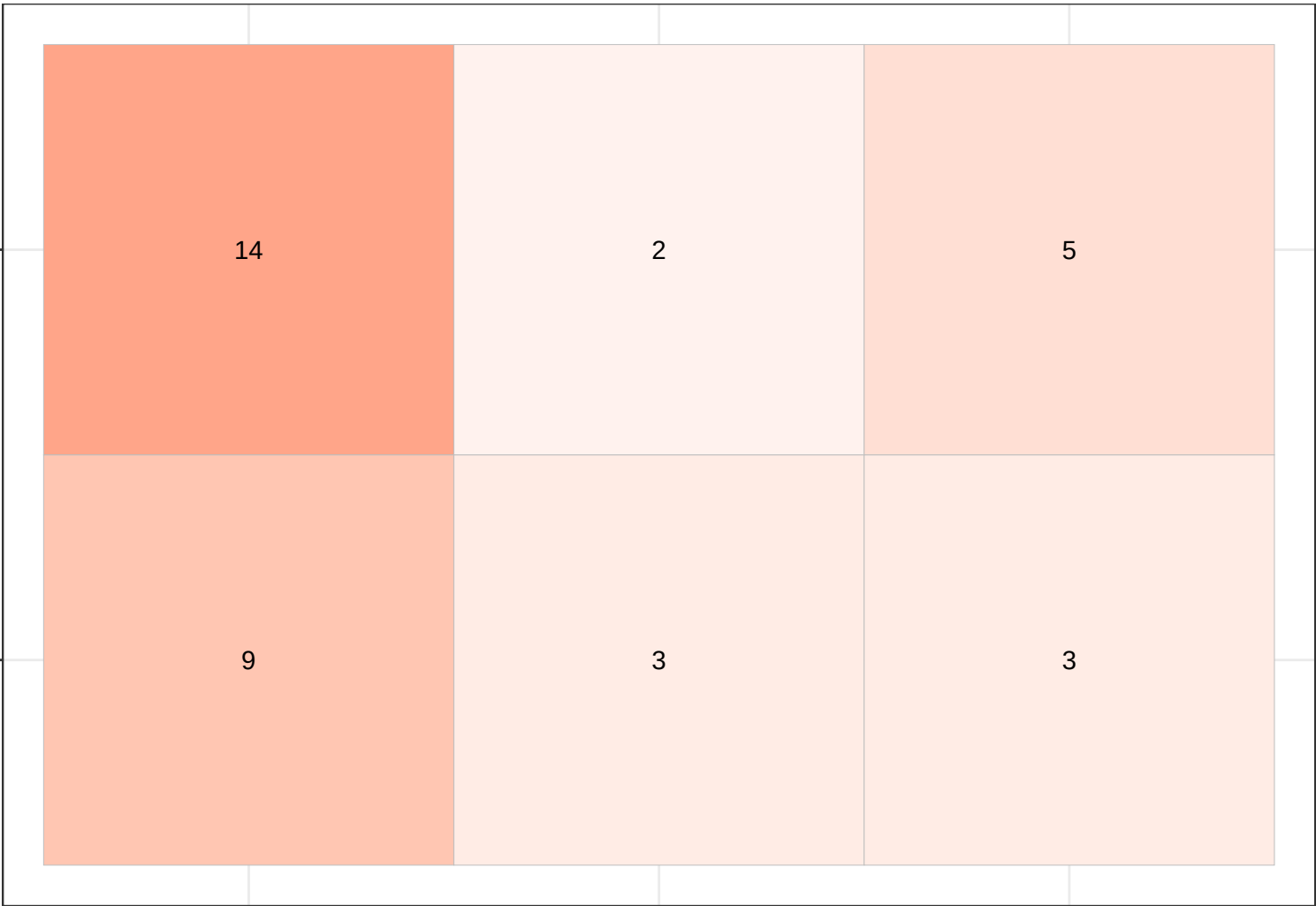
PARAMETRIC

DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.3.Type.of.Mechanism_____I1.4.Resilience.of.Effects

I1.3.Type.of.Mechanism

STRUCTURAL

PARAMETRIC

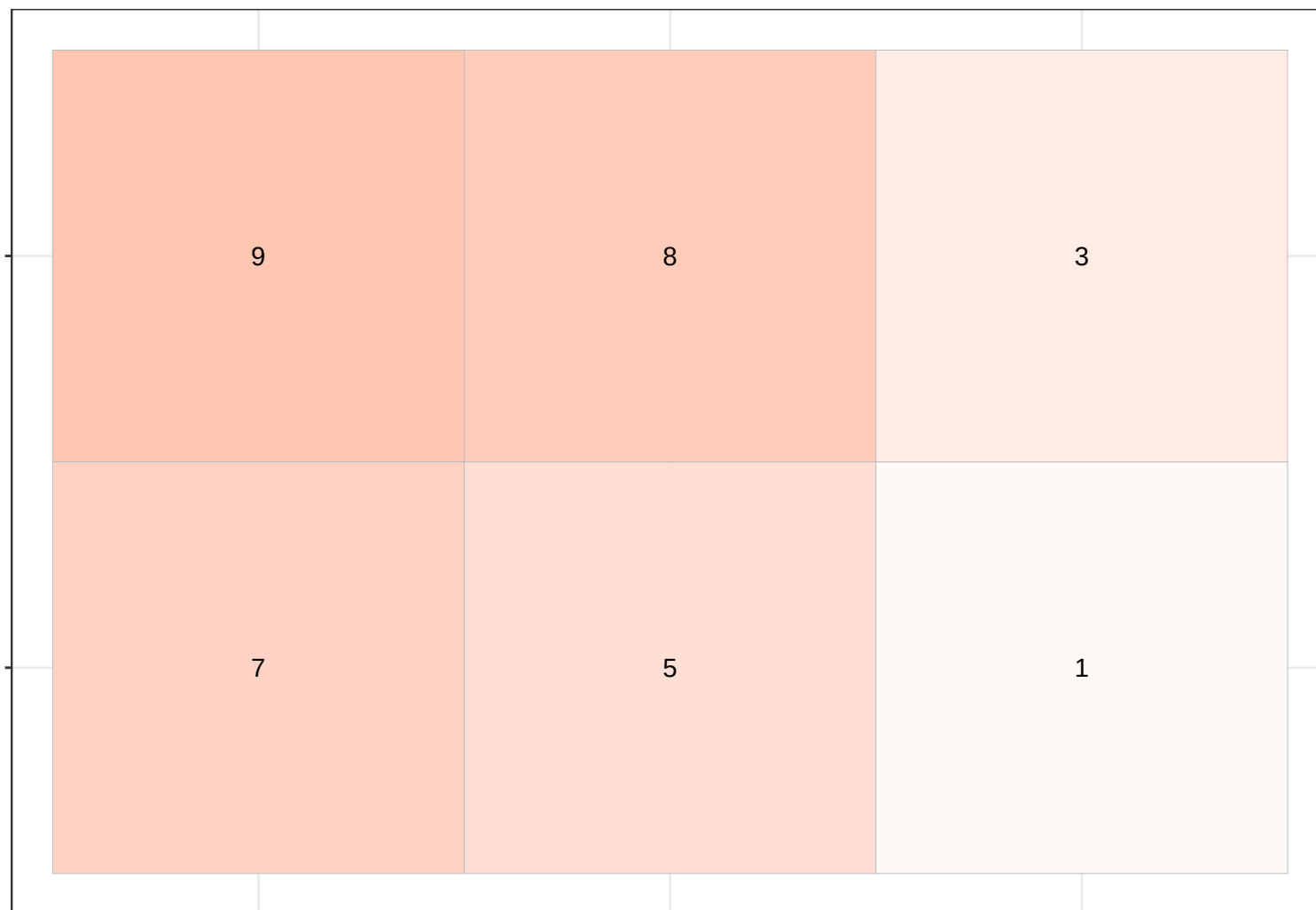
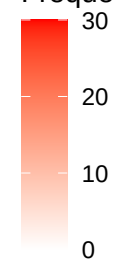
DEPENDENT

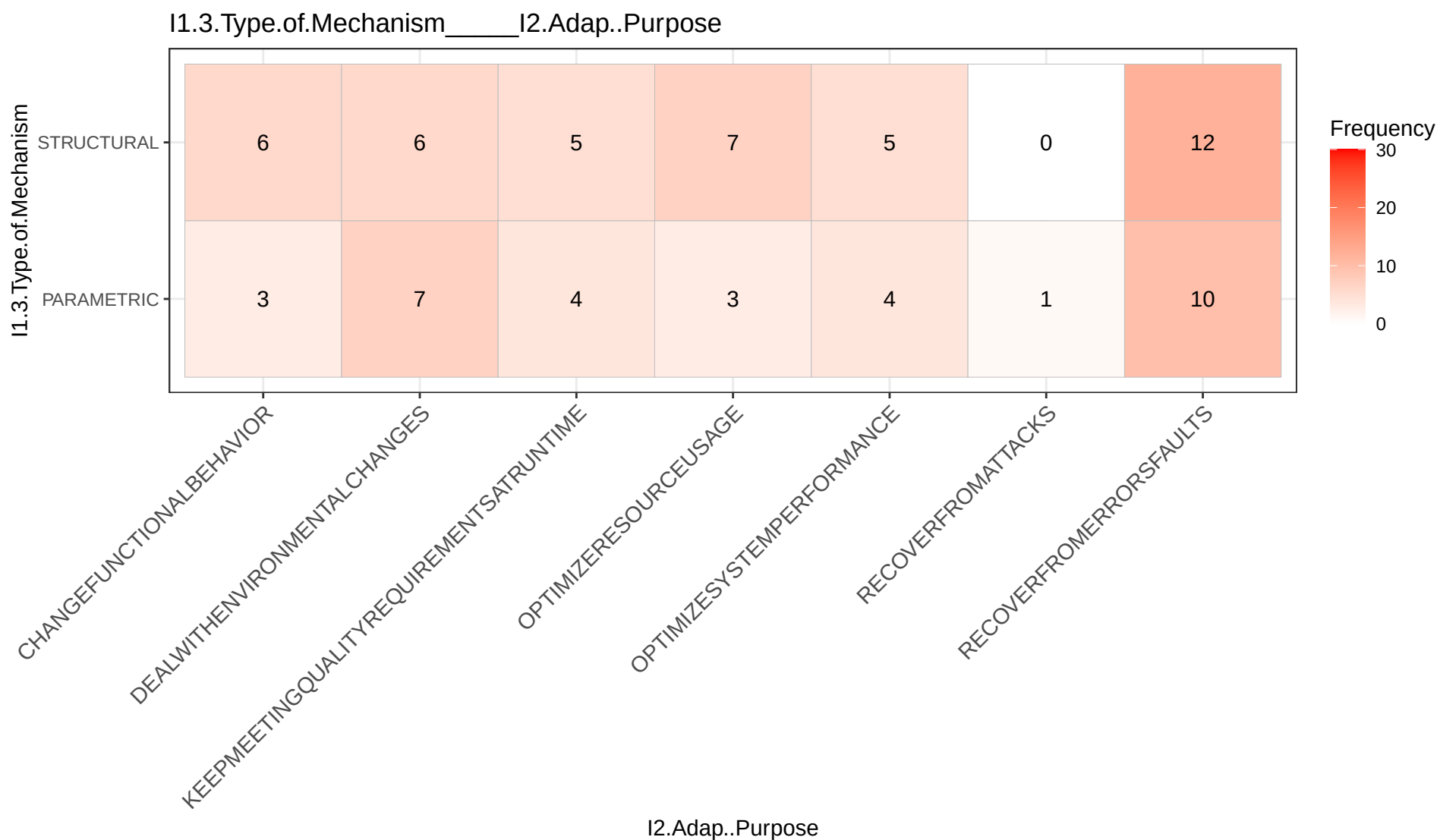
DORESILIENT

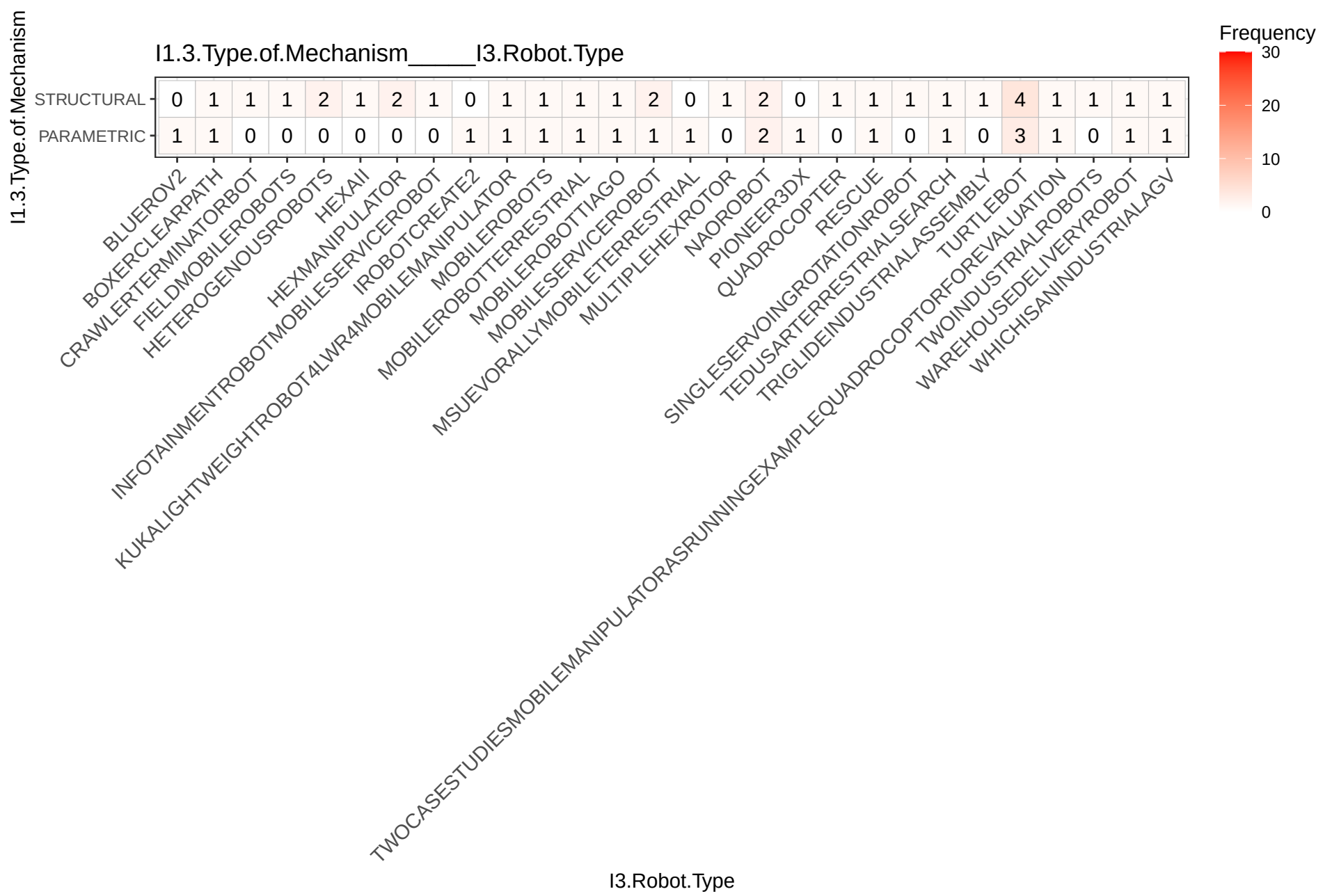
IRRESILIENT

I1.4.Resilience.of.Effects

Frequency

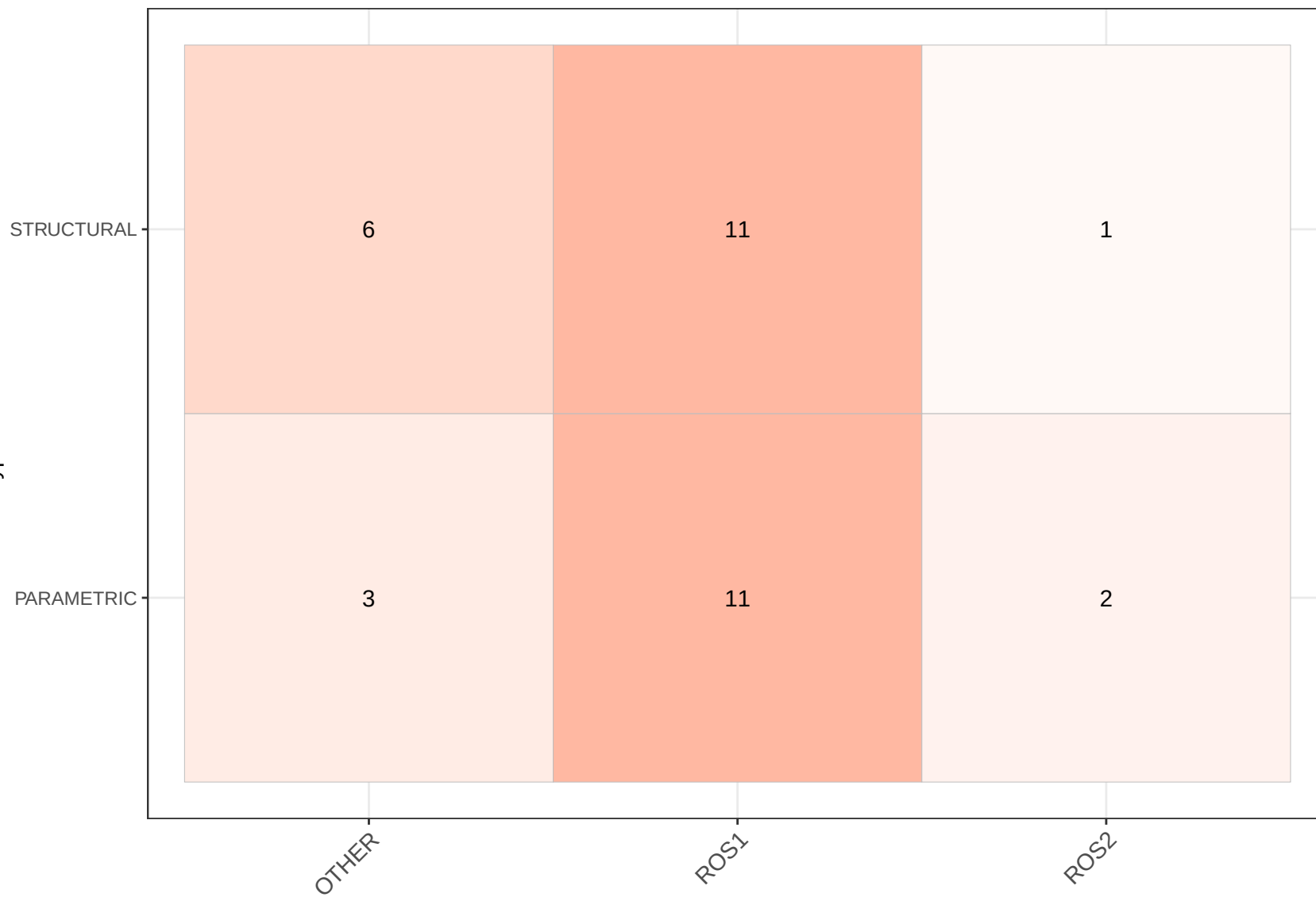






I1.3.Type.of.Mechanism____I4.Robo.SW

I1.3.Type.of.Mechanism



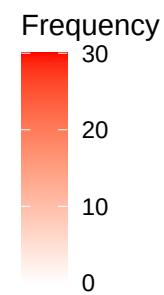
PARAMETRIC

OTHER

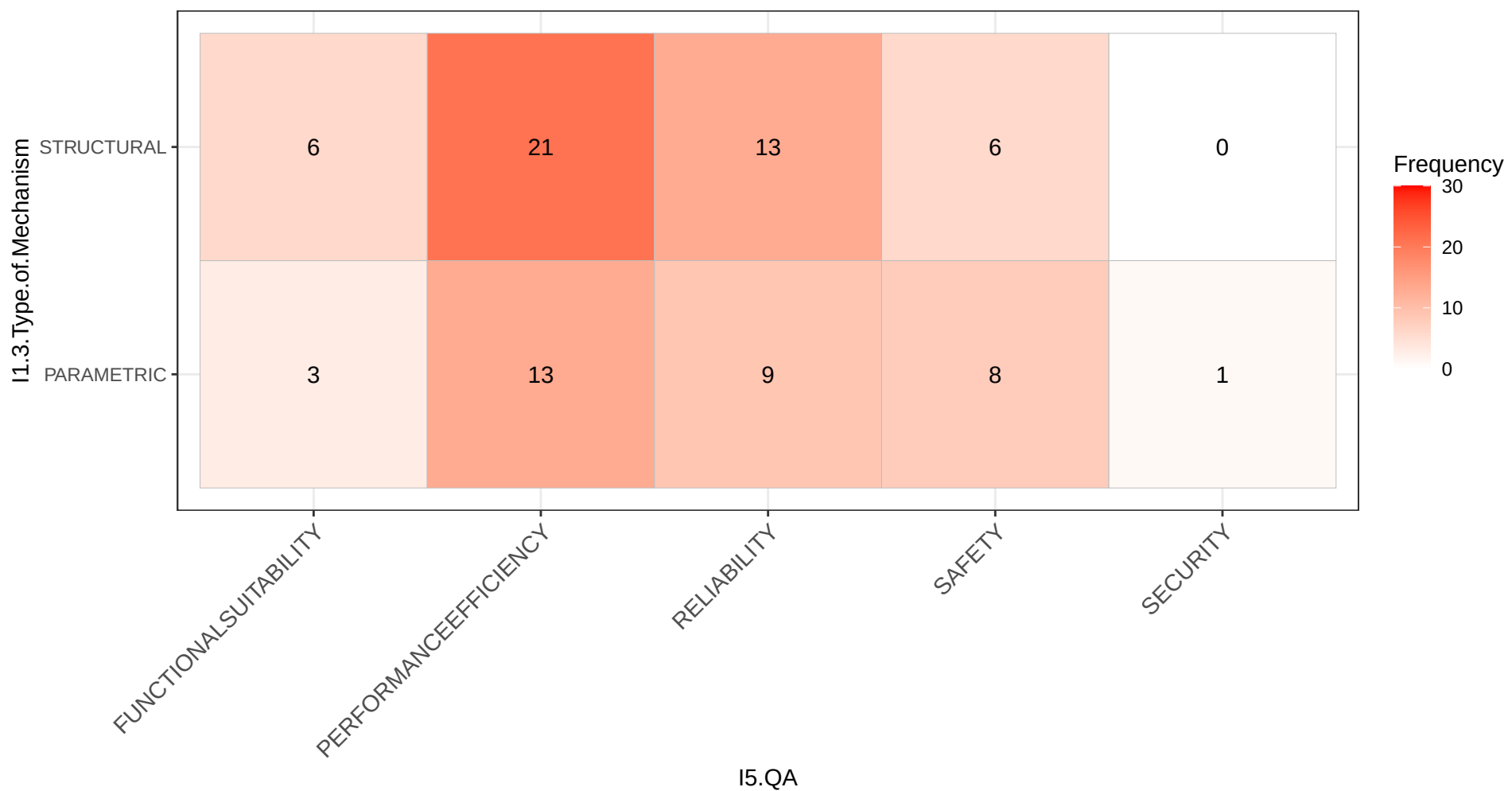
I4.Robo.SW

ROS1

ROS2



I1.3.Type.of.Mechanism_____I5.QA



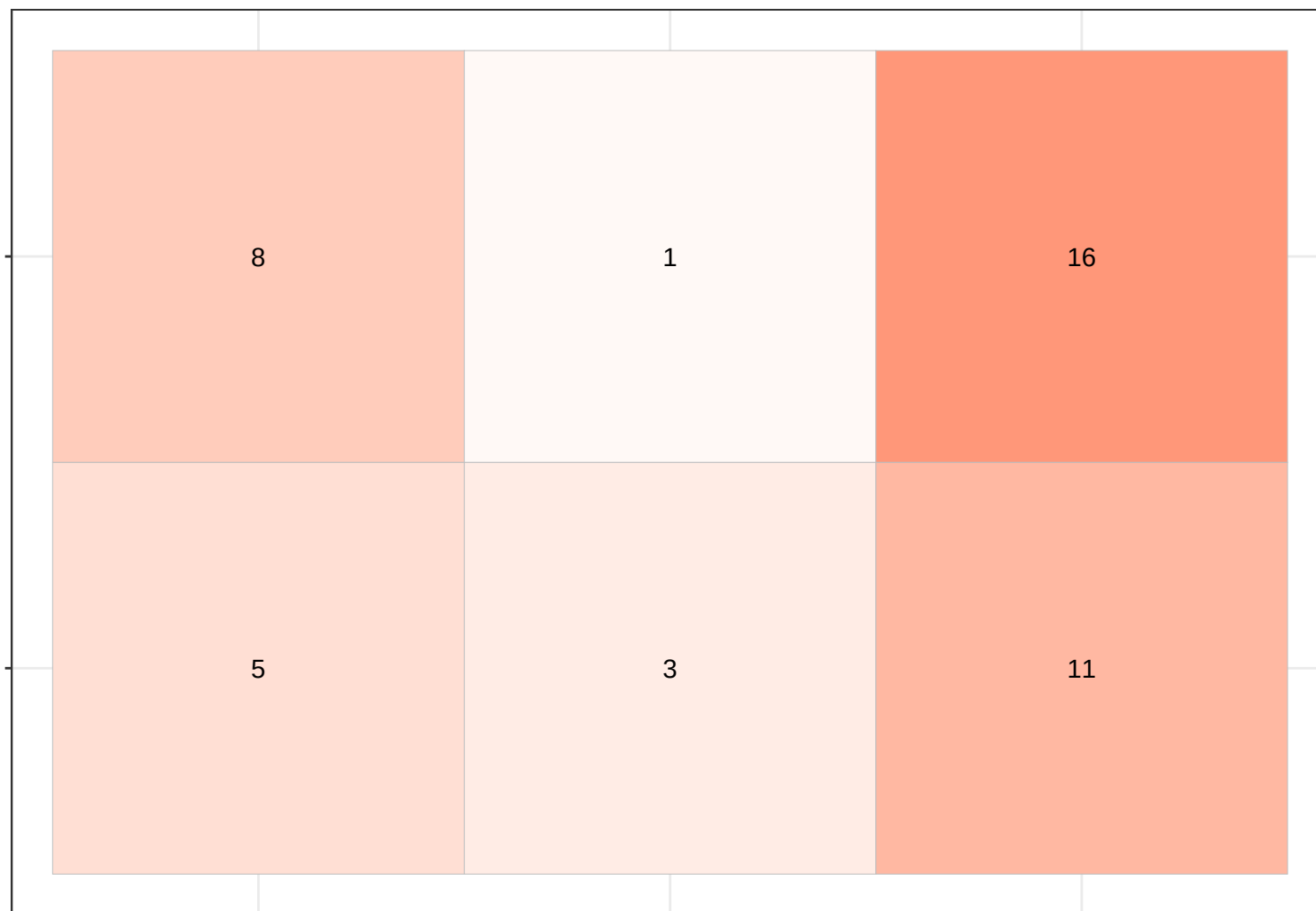
I5.QA

I1.3.Type.of.Mechanism_____I6.Independence

I1.3.Type.of.Mechanism

STRUCTURAL

PARAMETRIC



DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency

30

20

10

0

I1.3.Type.of.Mechanism____I7.Deployment.Realness

I1.3.Type.of.Mechanism

STRUCTURAL

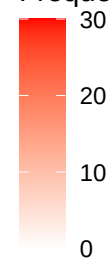
PARAMETRIC

REAL

SIMULATED

I7.Deployment.Realness

Frequency



12

9

9

8

I1.3.Type.of.Mechanism____I7.Mission.Realness

I1.3.Type.of.Mechanism

STRUCTURAL

14

11

PARAMETRIC

8

11

REAL

SYNTHETIC

I7.Mission.Realness

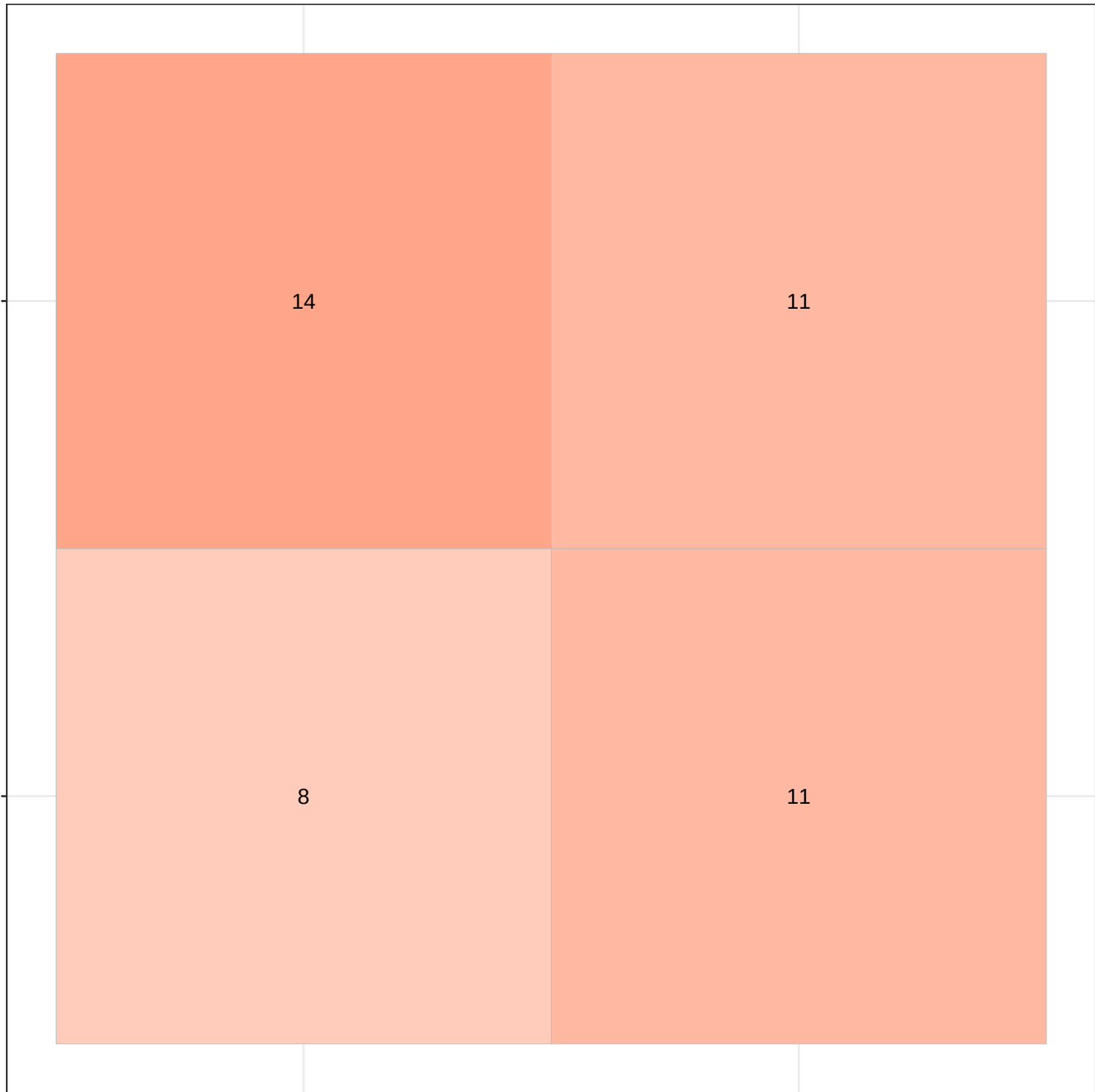
Frequency

30

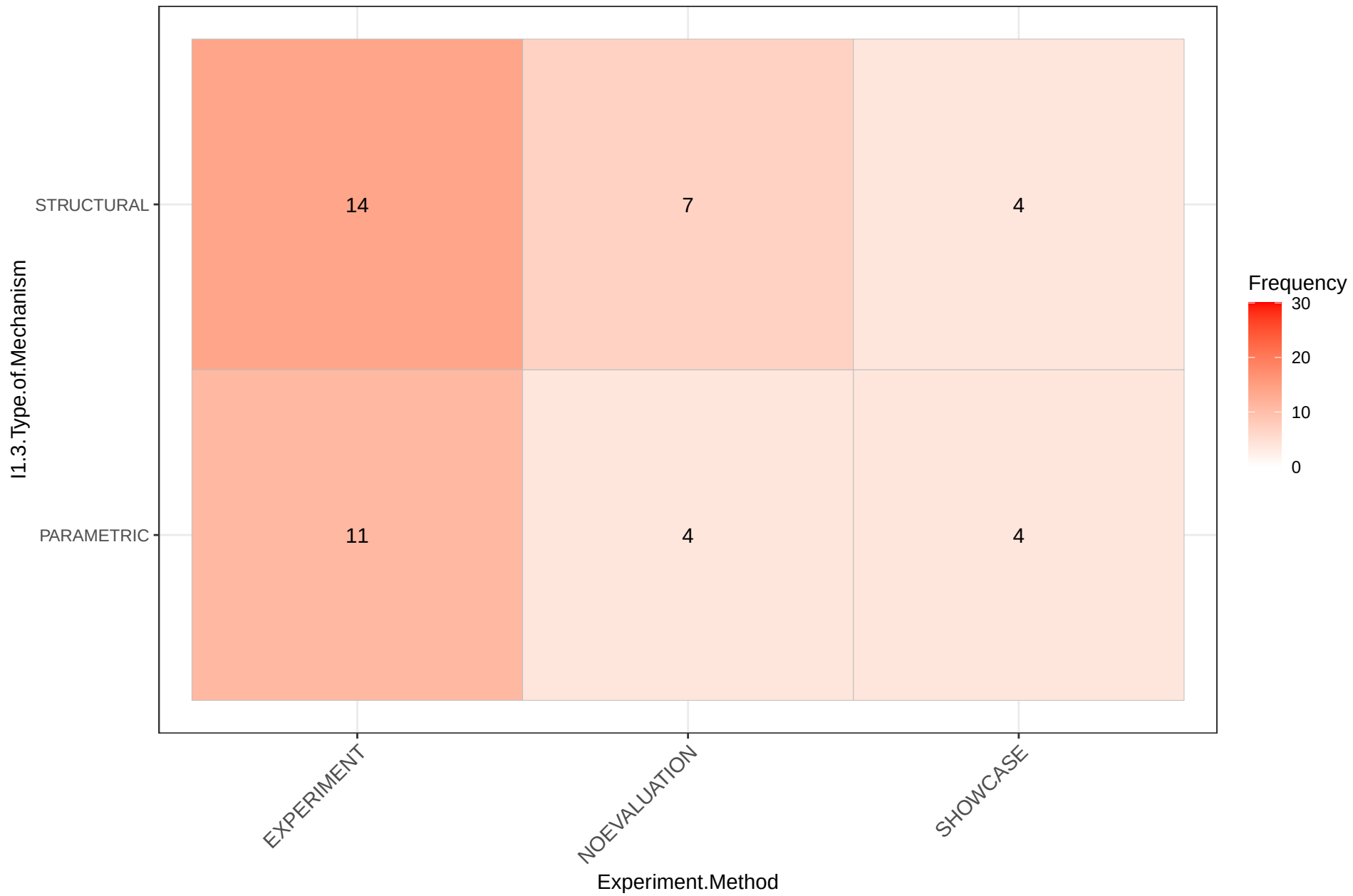
20

10

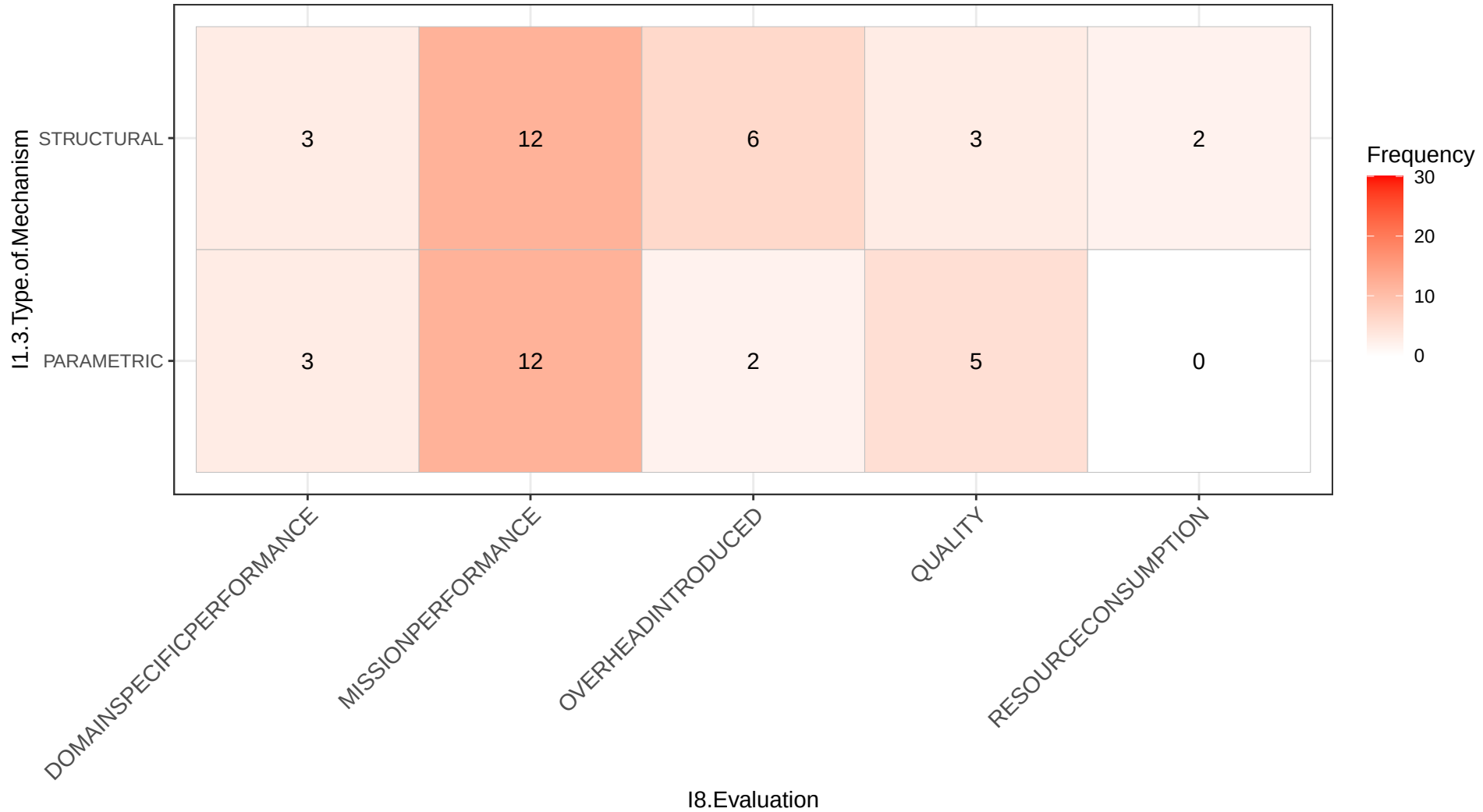
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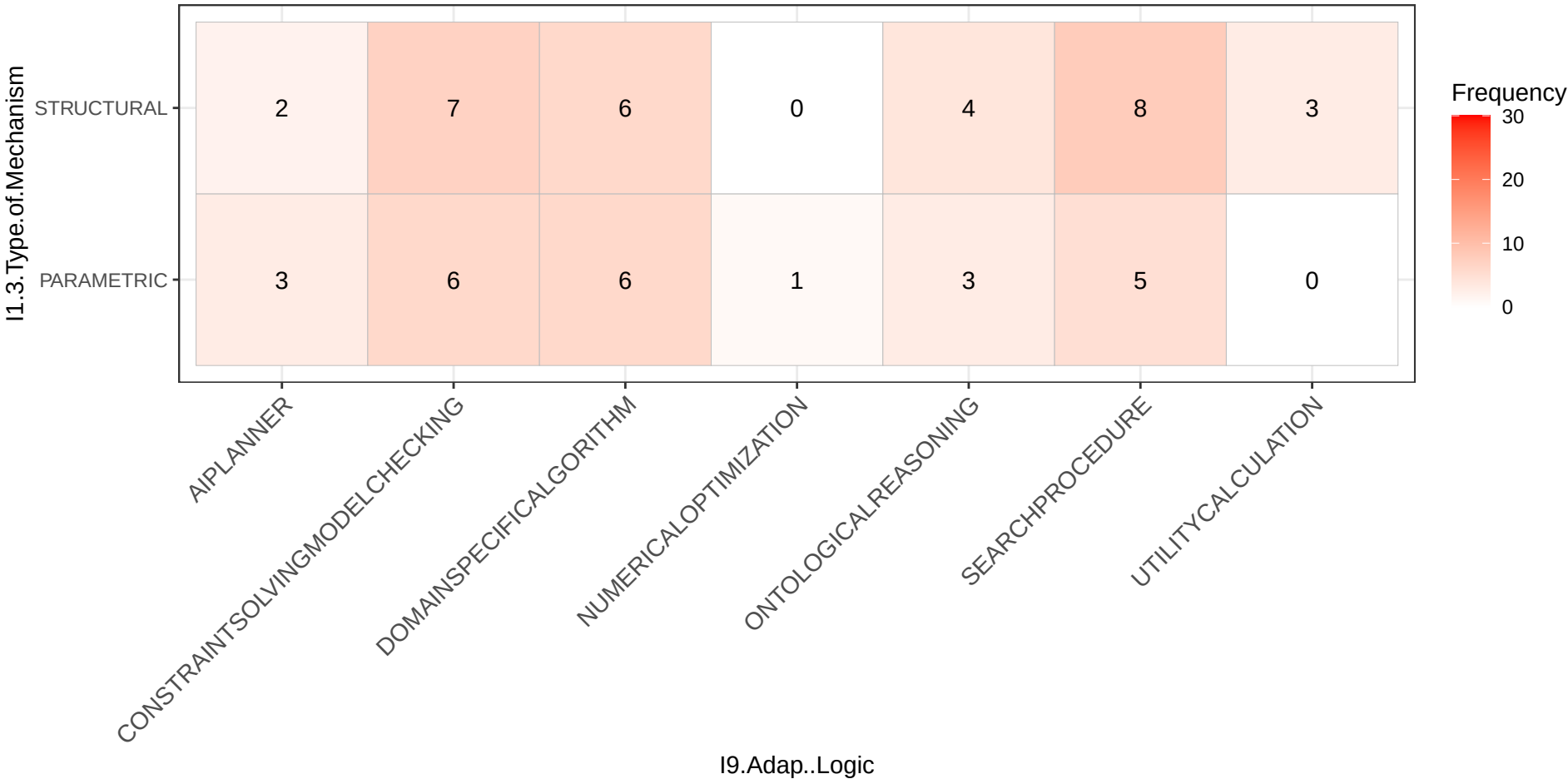
I1.3.Type.of.Mechanism_____Experiment.Method



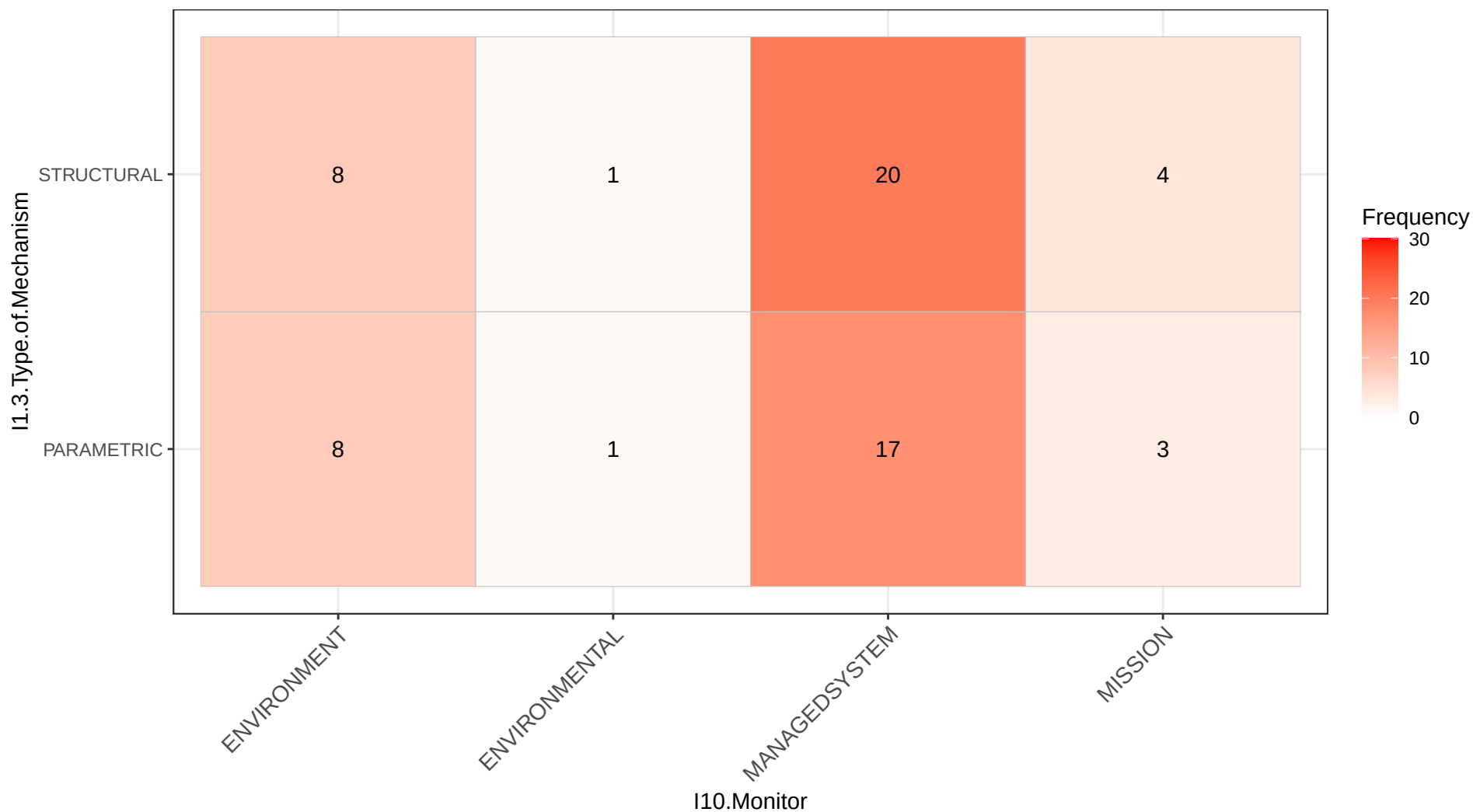
I1.3.Type.of.Mechanism_____I8.Evaluation

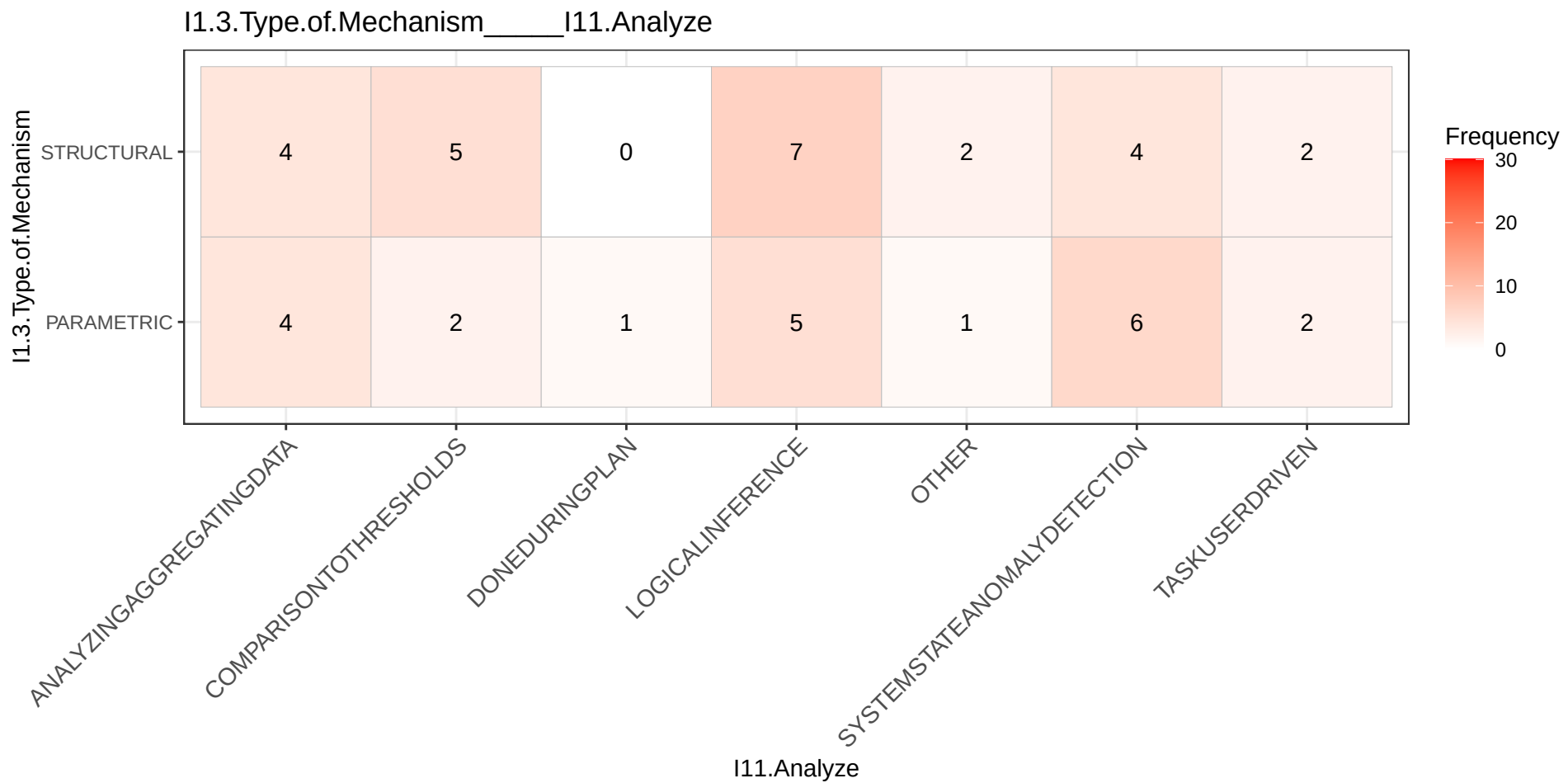


I1.3.Type.of.Mechanism_____I9.Adap..Logic



I1.3.Type.of.Mechanism____I10.Monitor





I1.3.Type.of.Mechanism_____I12.Plan

I1.3.Type.of.Mechanism

STRUCTURAL

11

11

1

PARAMETRIC

8

8

2

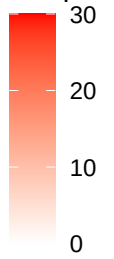
DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

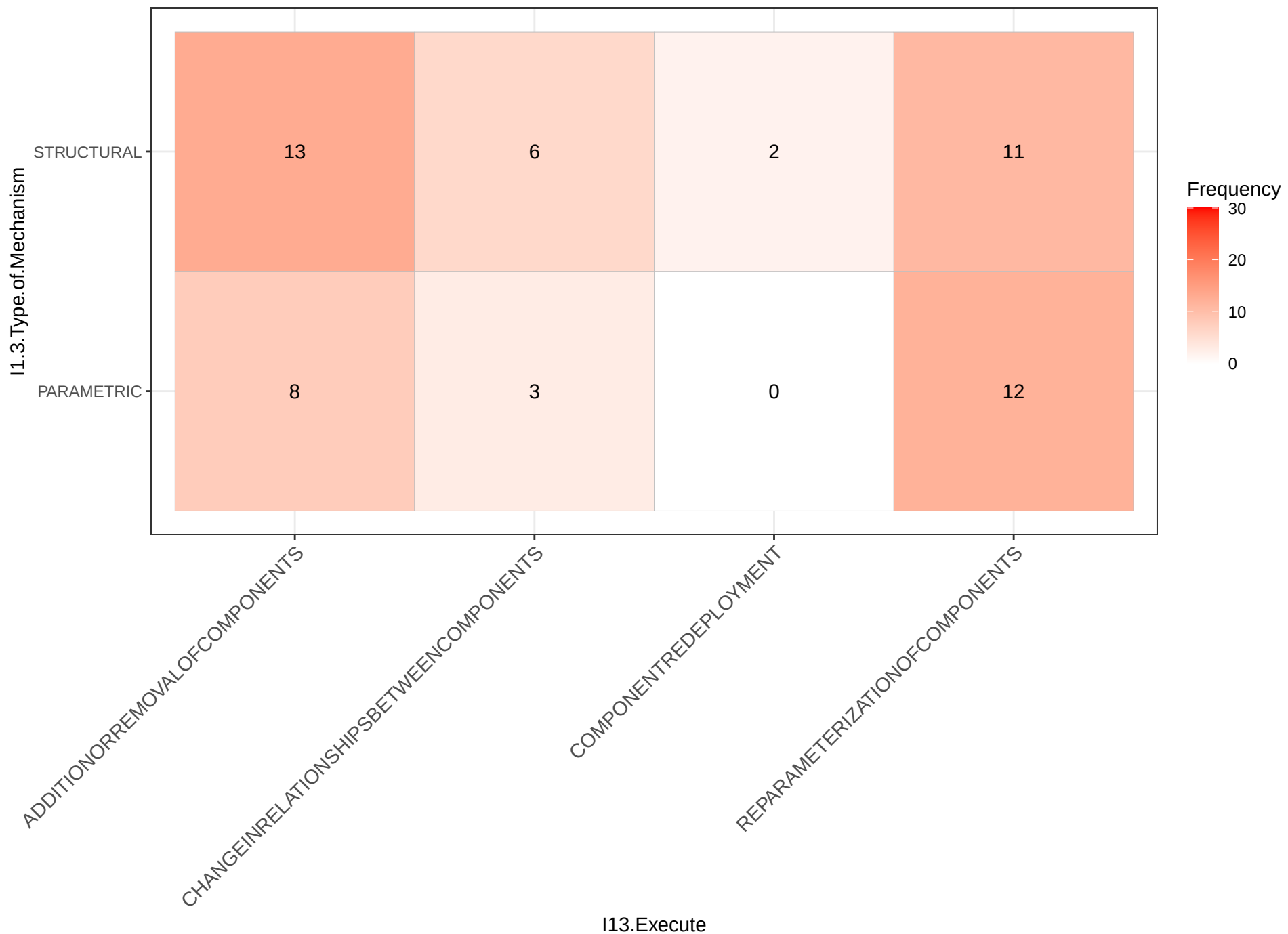
USINGAIPLANNINGLANGUAGES

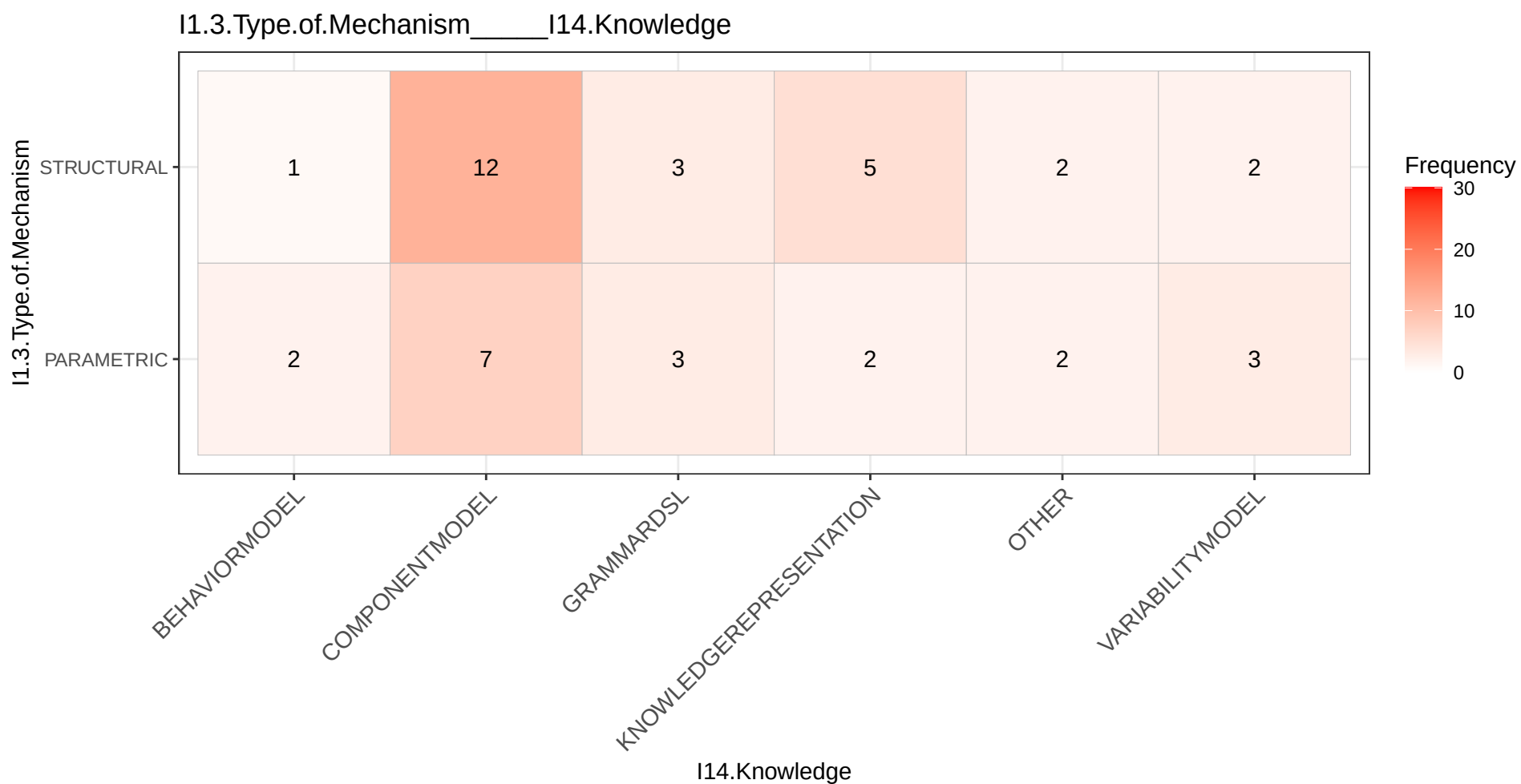
I12.Plan

Frequency



I1.3.Type.of.Mechanism____I13.Execute





I1.3.Organization.of.Mechanism_____I1.3.Scope.of.Mechanism

I1.3.Organization.of.Mechanism

DOCENTRALIZED

8

22

DECENTRALIZED

1

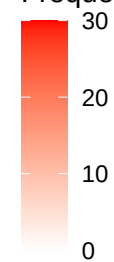
4

GLOBAL

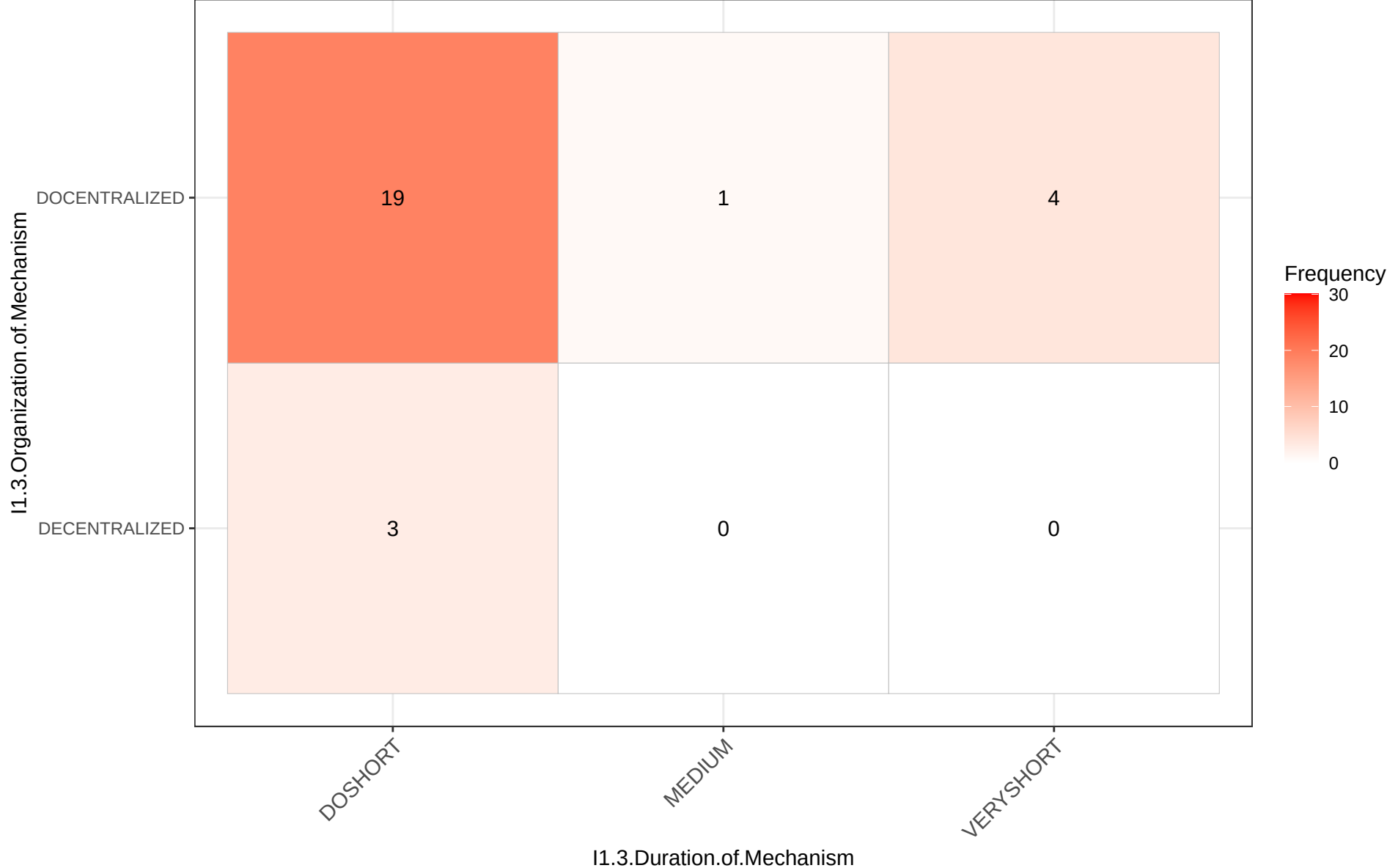
LOCAL

I1.3.Scope.of.Mechanism

Frequency



I1.3.Organization.of.Mechanism_____I1.3.Duration.of.Mechanism



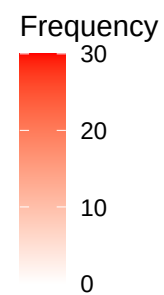
I1.3.Organization.of.Mechanism

DOCENTRALIZED

DECENTRALIZED

BESTEFFECT

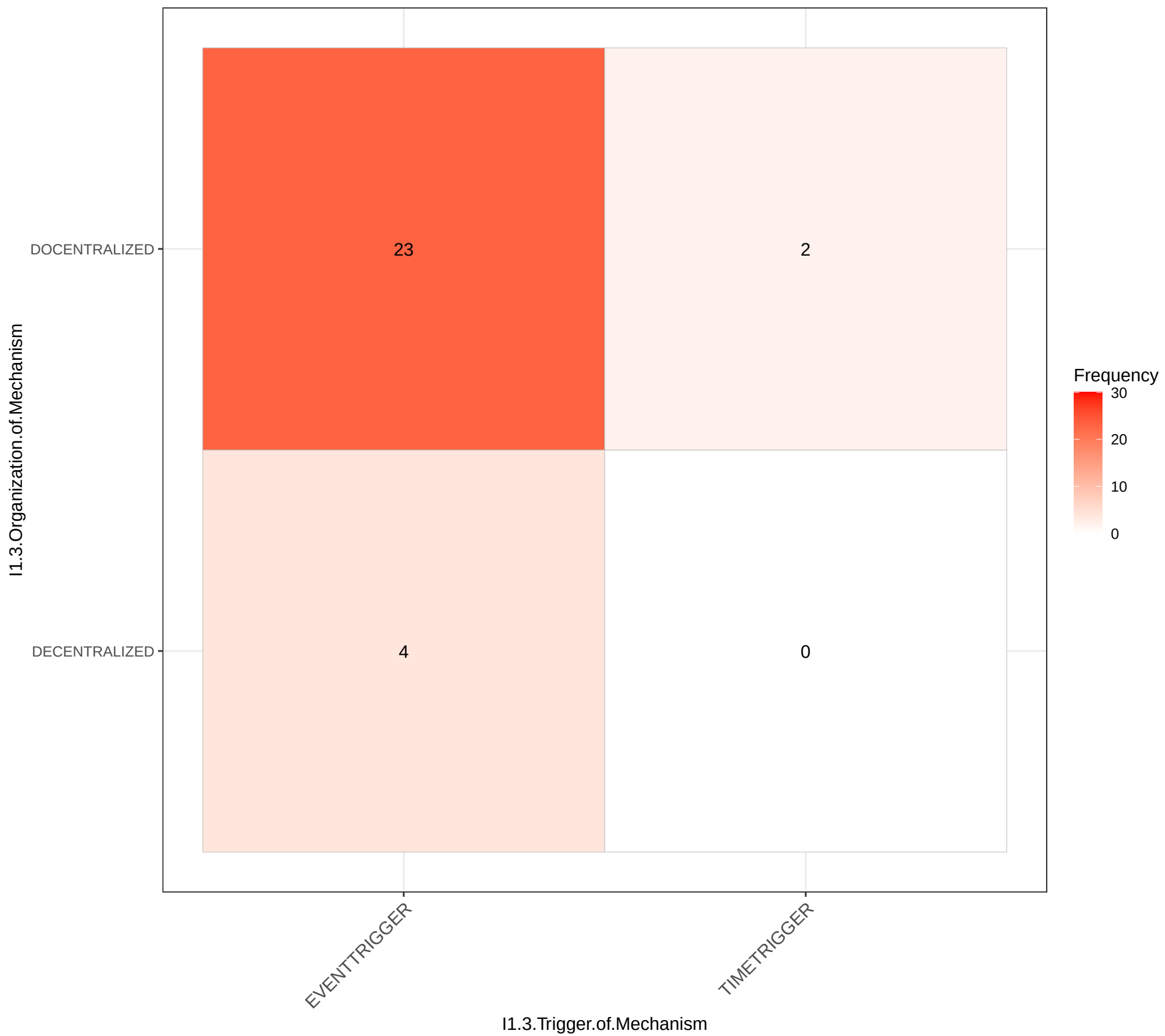
I1.3.Timeliness.of.Mechanism



24

4

I1.3.Organization.of.Mechanism_____I1.3.Trigger.of.Mechanism



I1.3.Organization.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Organization.of.Mechanism

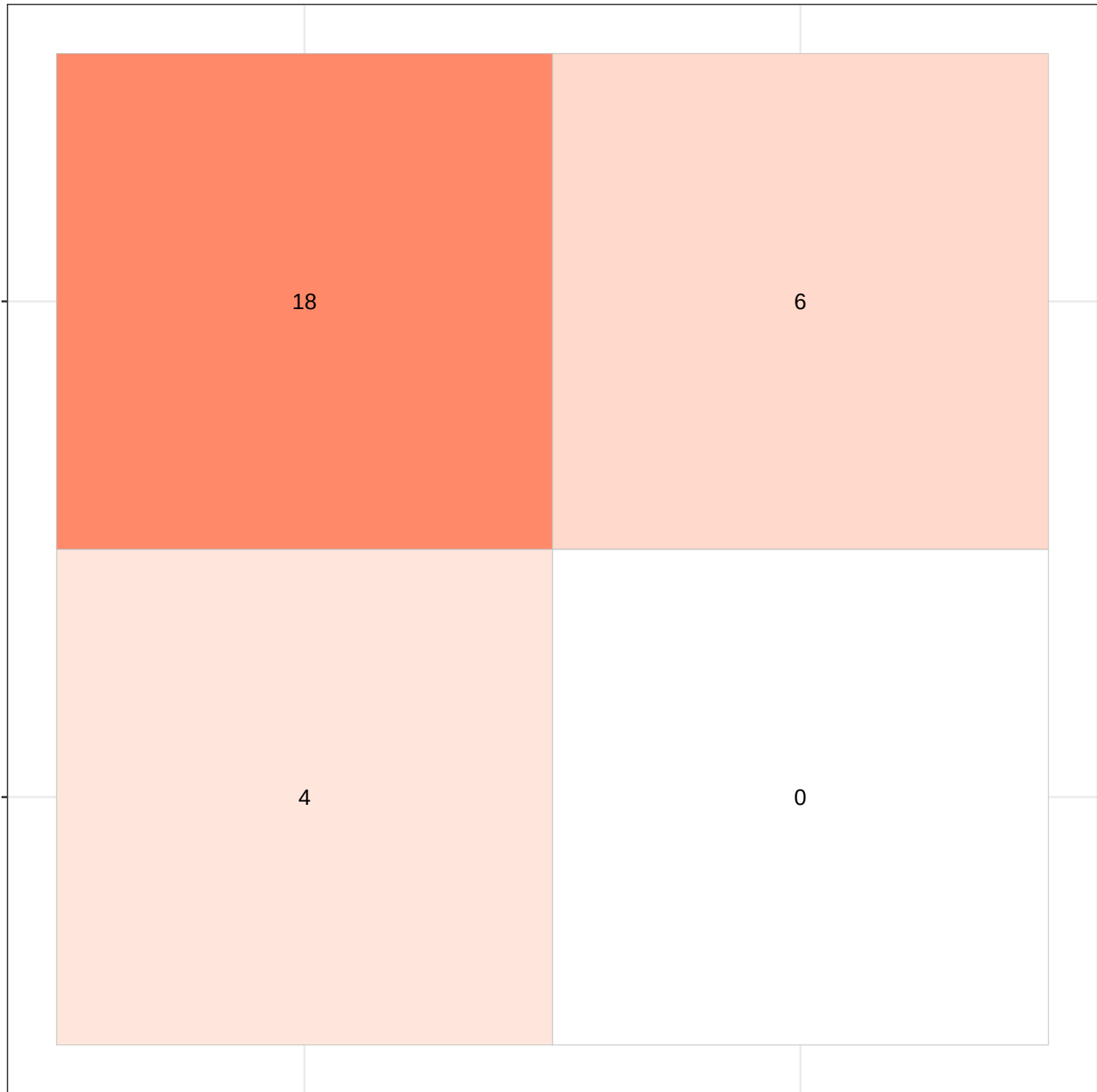
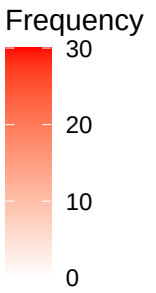
DOCENTRALIZED

DECENTRALIZED

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects



I1.3.Organization.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Organization.of.Mechanism

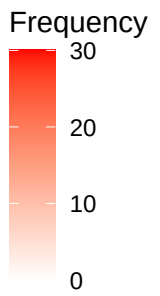
DOCENTRALIZED

DECENTRALIZED

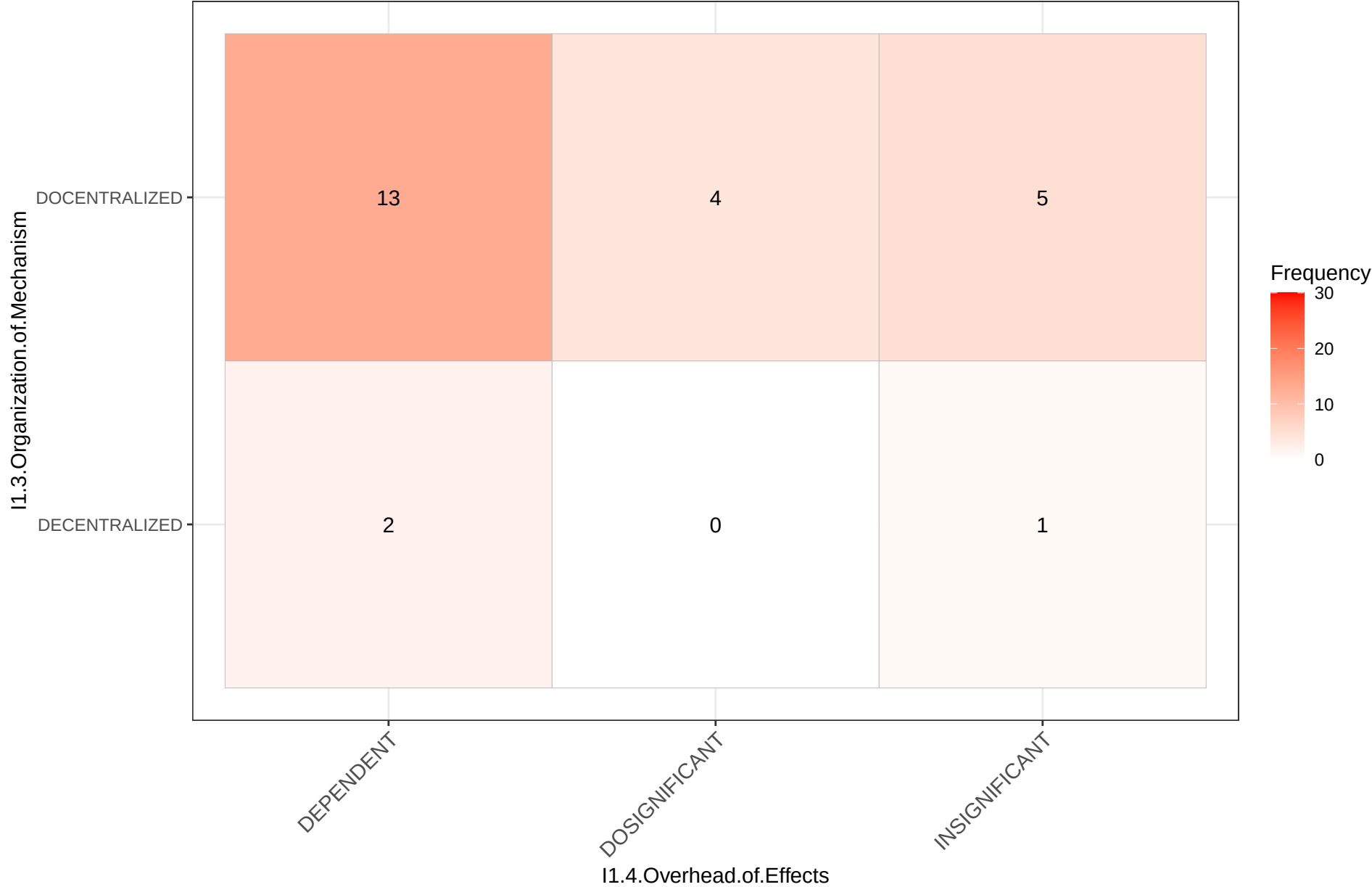
DODETERMINISTIC

NONDETERMINISTIC

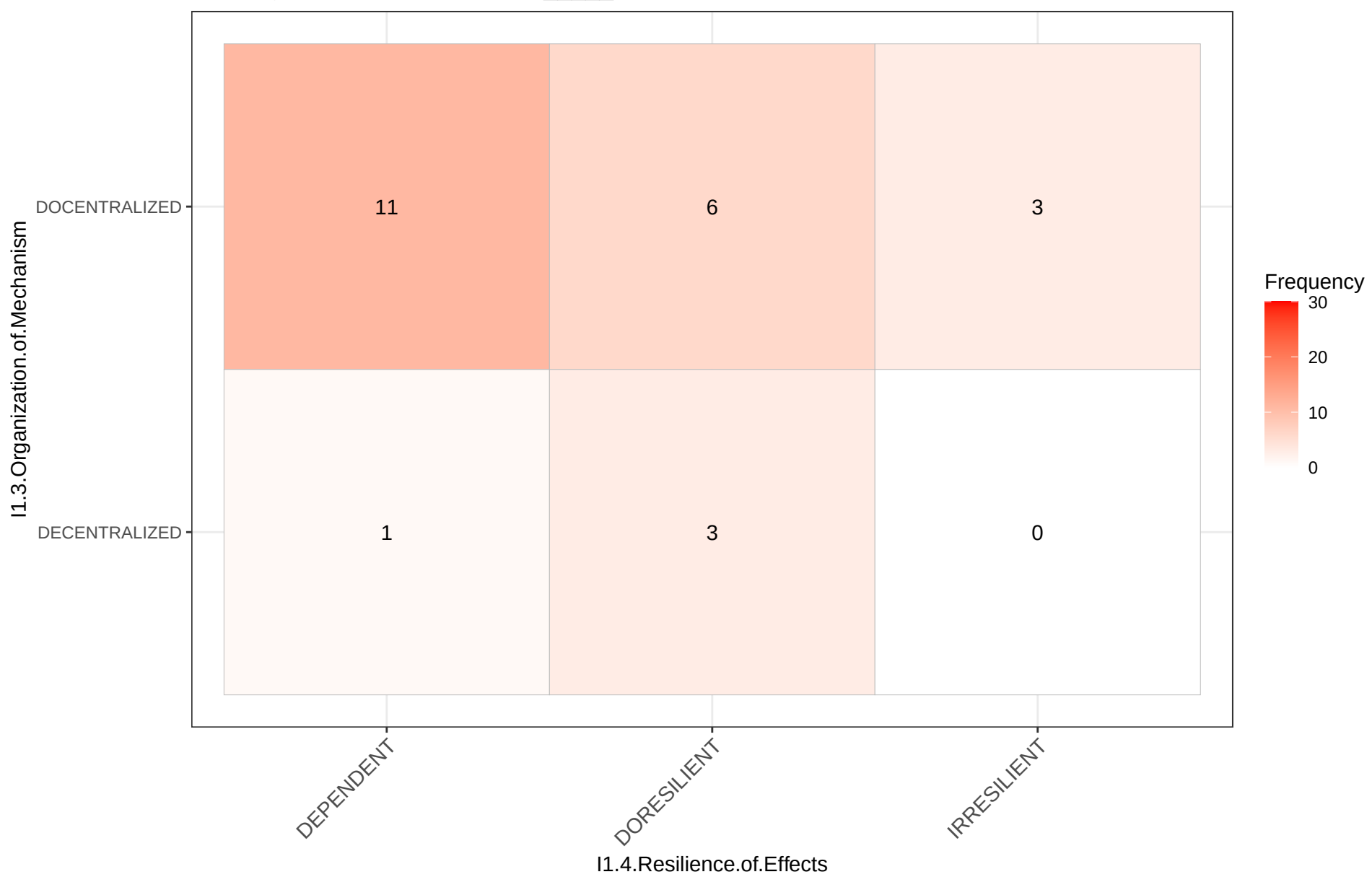
I1.4.Predictability.of.Effects



I1.3.Organization.of.Mechanism_____I1.4.Overhead.of.Effects



I1.3.Organization.of.Mechanism_____I1.4.Resilience.of.Effects



I1.3.Organization.of.Mechanism_____I2.Adap..Purpose

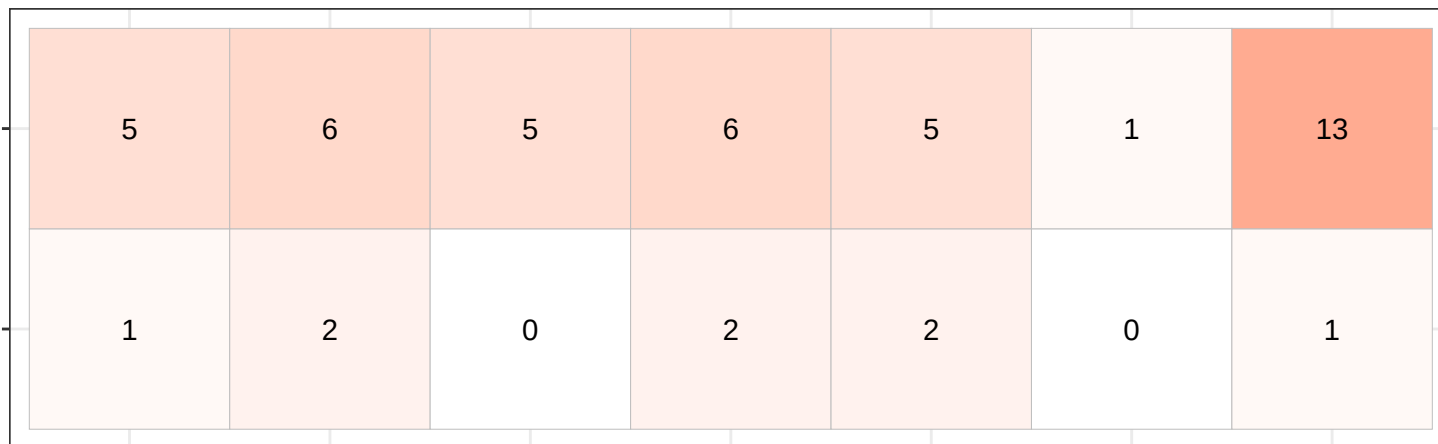
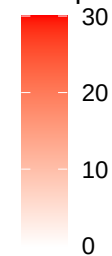
DOCENTRALIZED

DECENTRALIZED

CHANGEFUNCTIONALBEHAVIOR
DEALWITHENVIRONMENTALCHANGES
KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME
OPTIMIZERESOURCEUSAGE
OPTIMIZESYSTEMPERFORMANCE
RECOVERFROMATTACKS
RECOVERFROMERRORSFAULTS

I2.Adap..Purpose

Frequency



I1.3.Organization.of.Mechanism_____I3.Robot.Type

Frequency



30
20
10
0

I3.Robot.Type

I1.3.Organization.of.Mechanism_____I4.Robo.SW

I1.3.Organization.of.Mechanism

DOCENTRALIZED

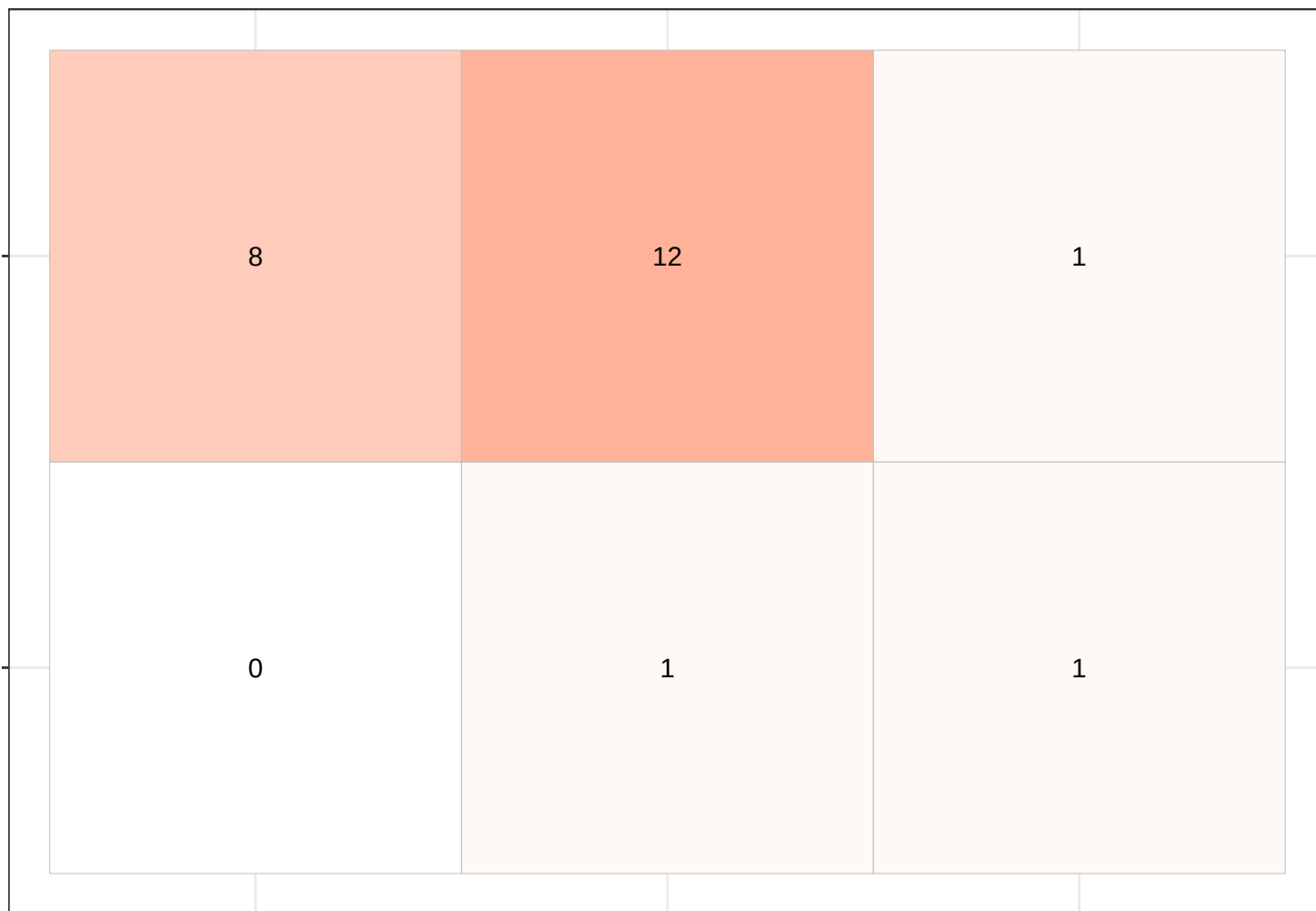
DECENTRALIZED

OTHER

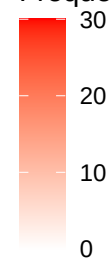
ROS1

ROS2

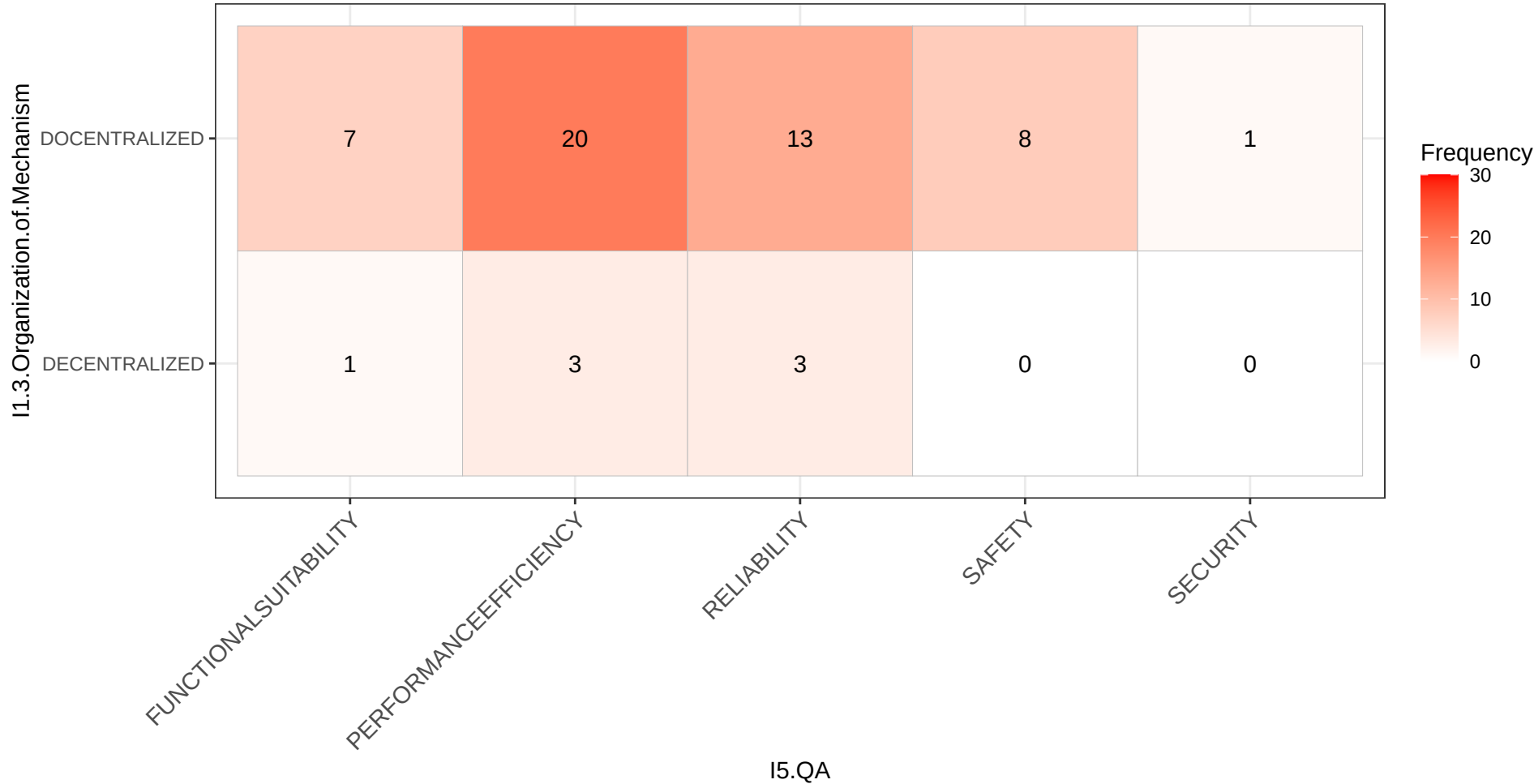
I4.Robo.SW



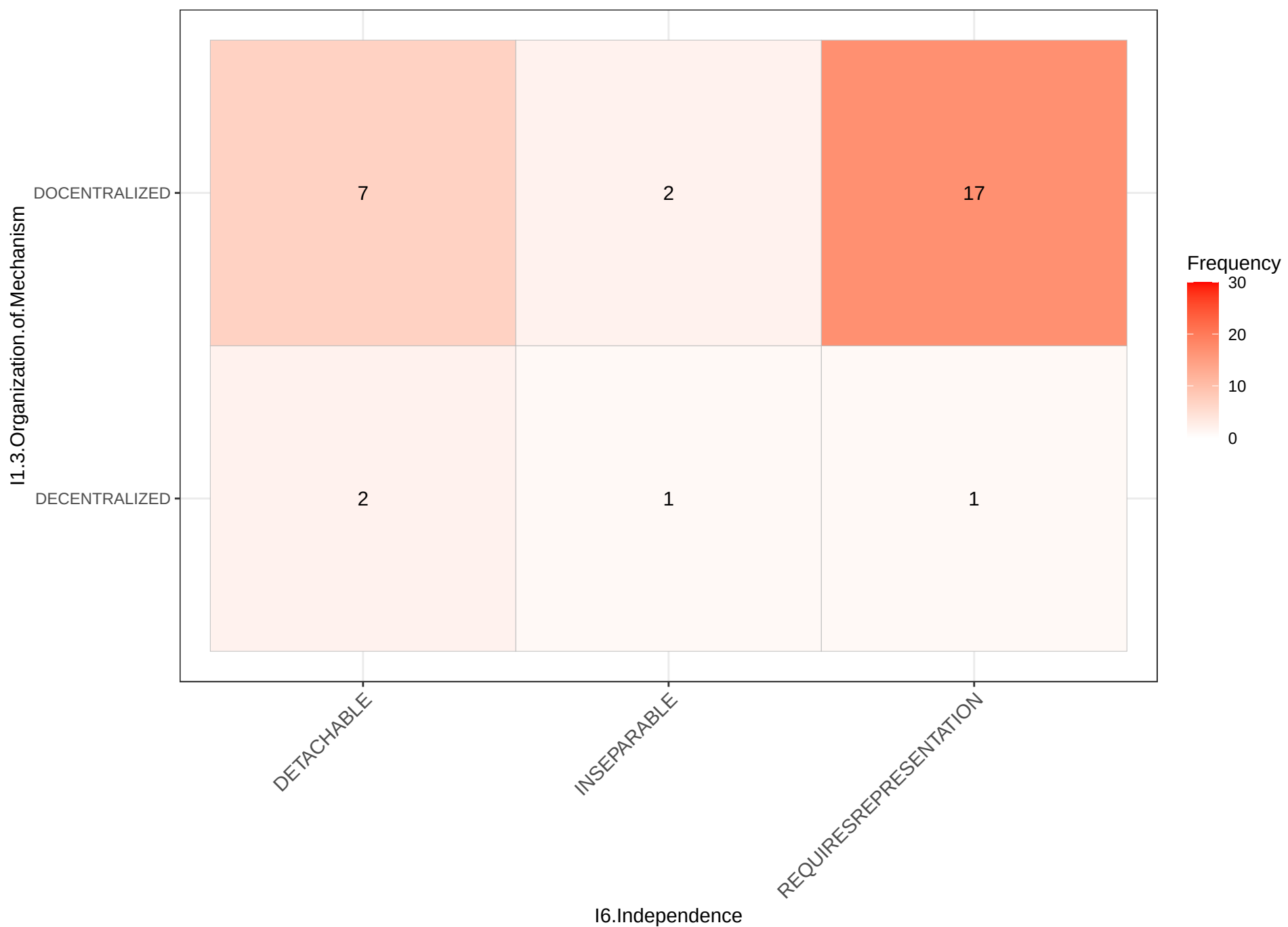
Frequency



I1.3.Organization.of.Mechanism_____I5.QA

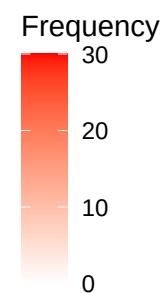
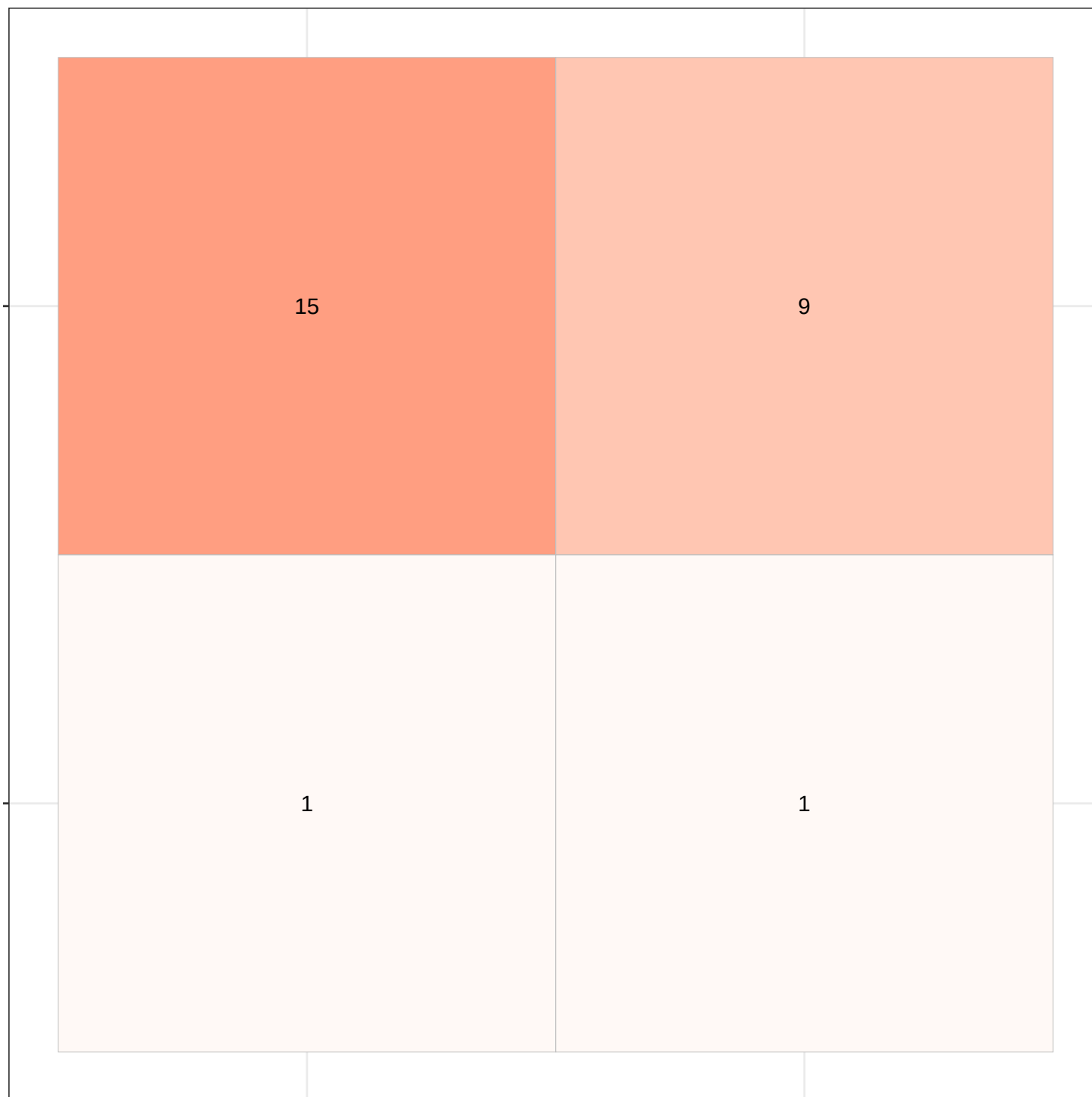


I1.3.Organization.of.Mechanism_____I6.Independence



I1.3.Organization.of.Mechanism_____I7.Deployment.Realness

I1.3.Organization.of.Mechanism



REAL

SIMULATED

I7.Deployment.Realness

I1.3.Organization.of.Mechanism_____I7.Mission.Realness

I1.3.Organization.of.Mechanism

DOCENTRALIZED

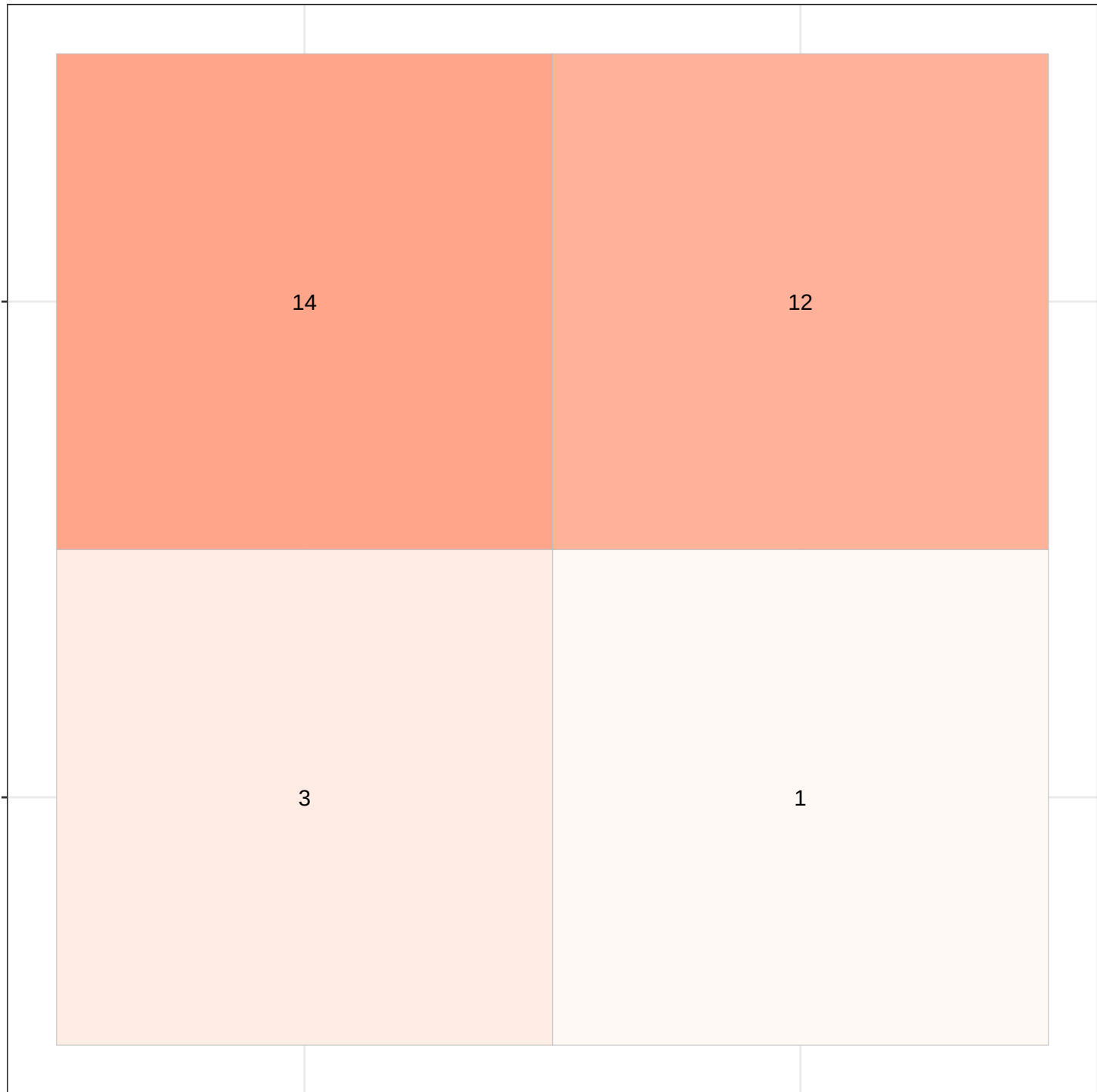
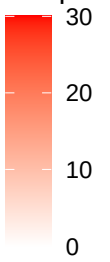
DECENTRALIZED

REAL

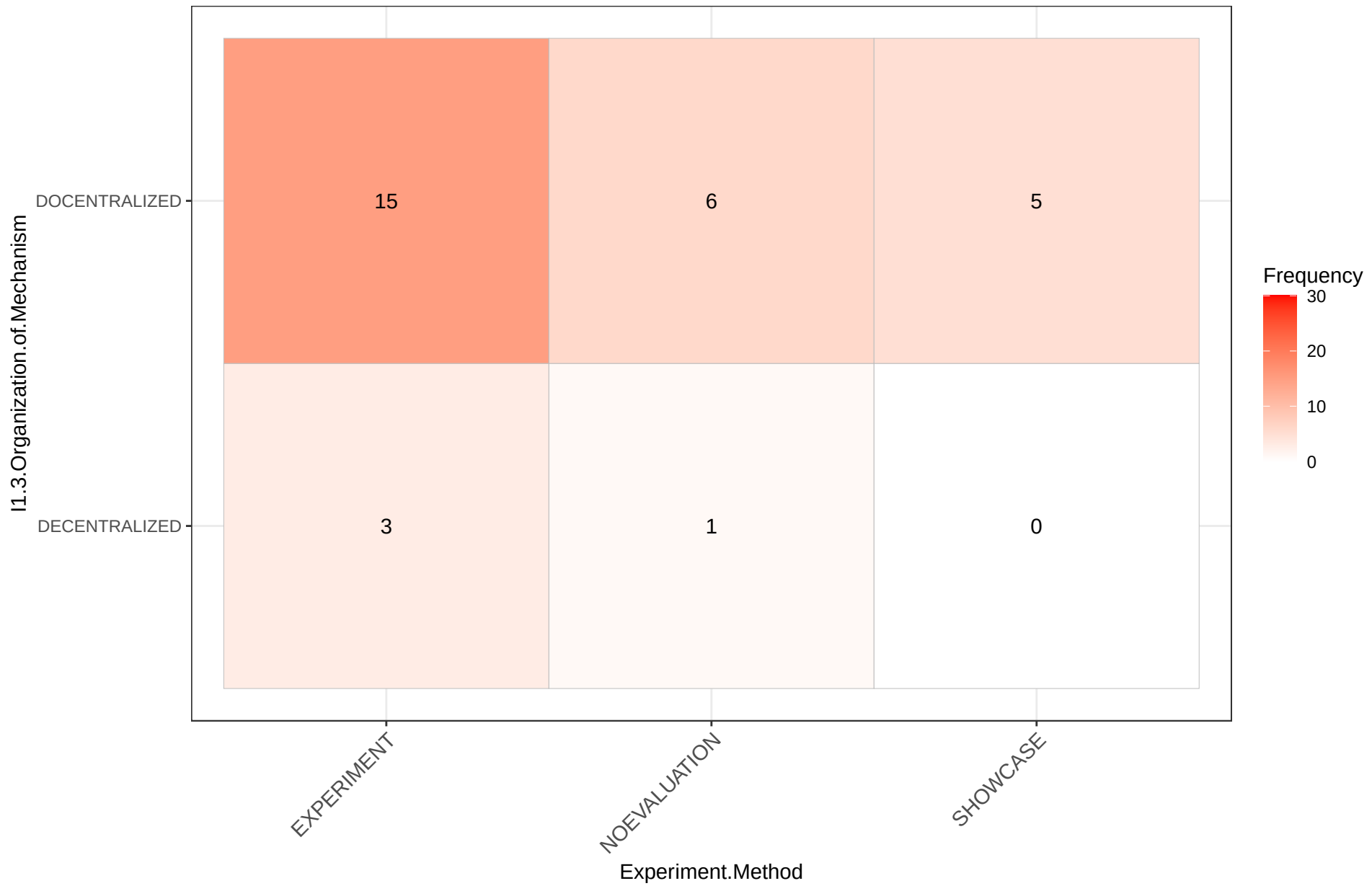
SYNTHETIC

I7.Mission.Realness

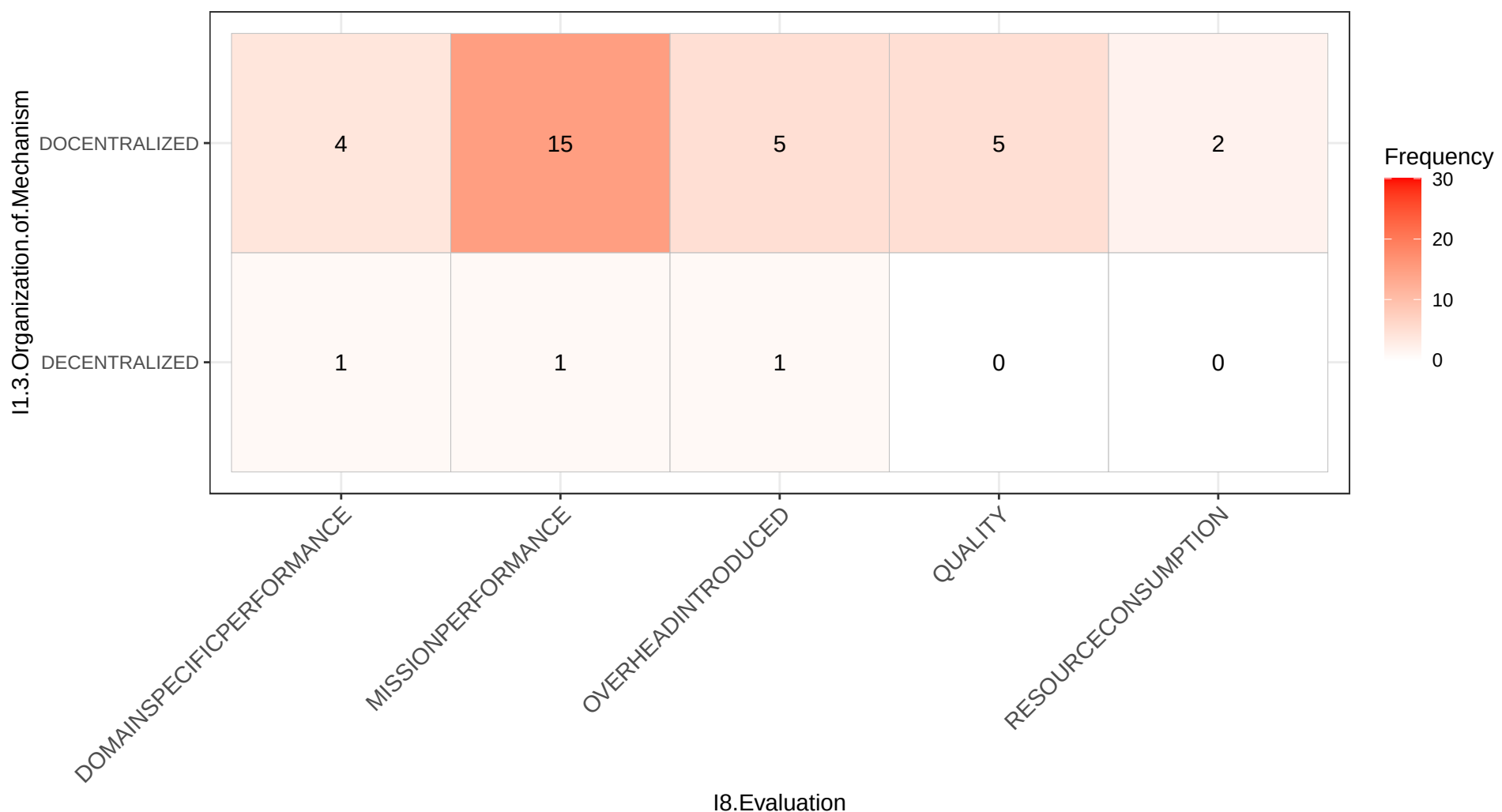
Frequency



I1.3.Organization.of.Mechanism____Experiment.Method

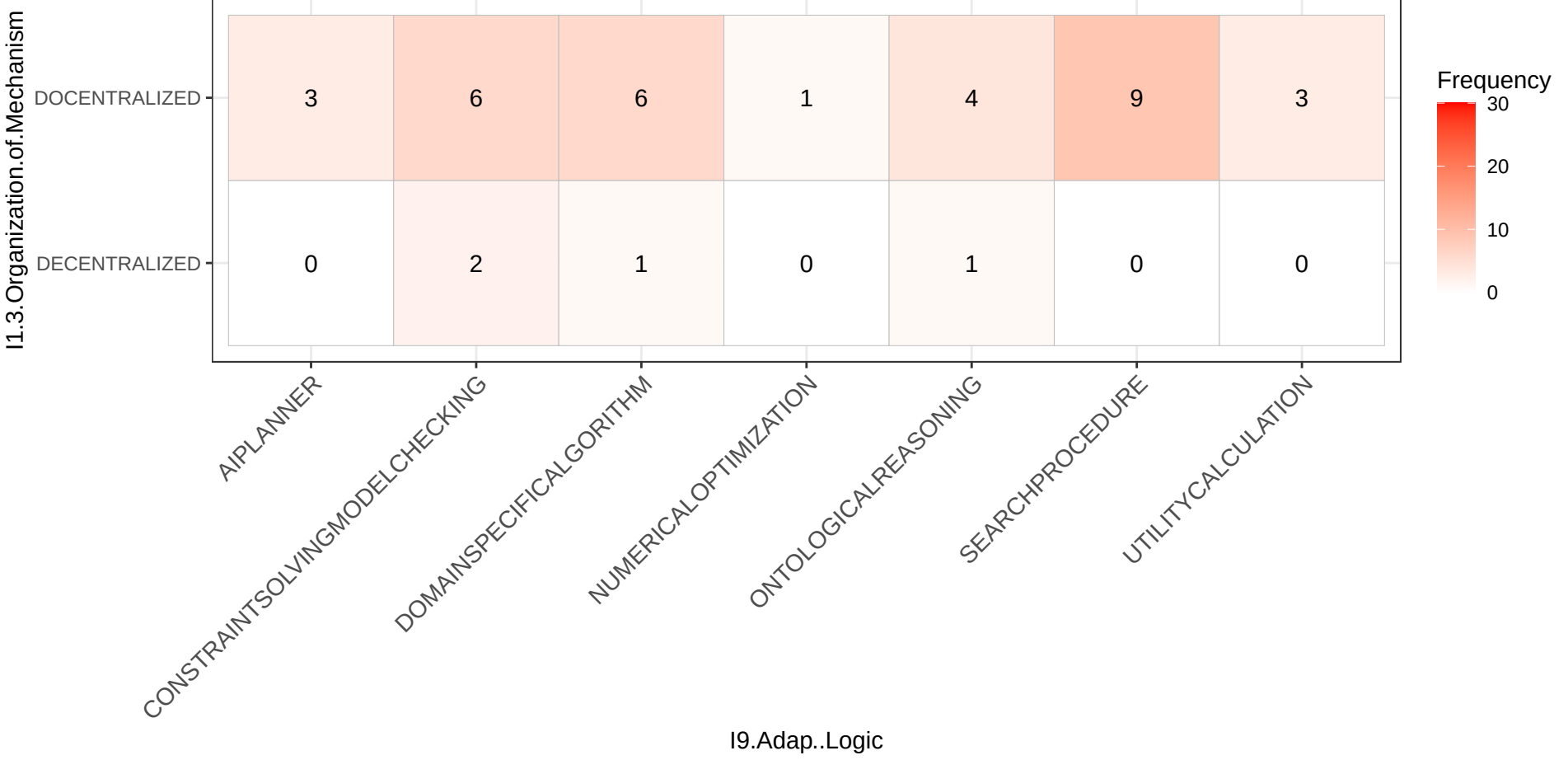


I1.3.Organization.of.Mechanism_____I8.Evaluation

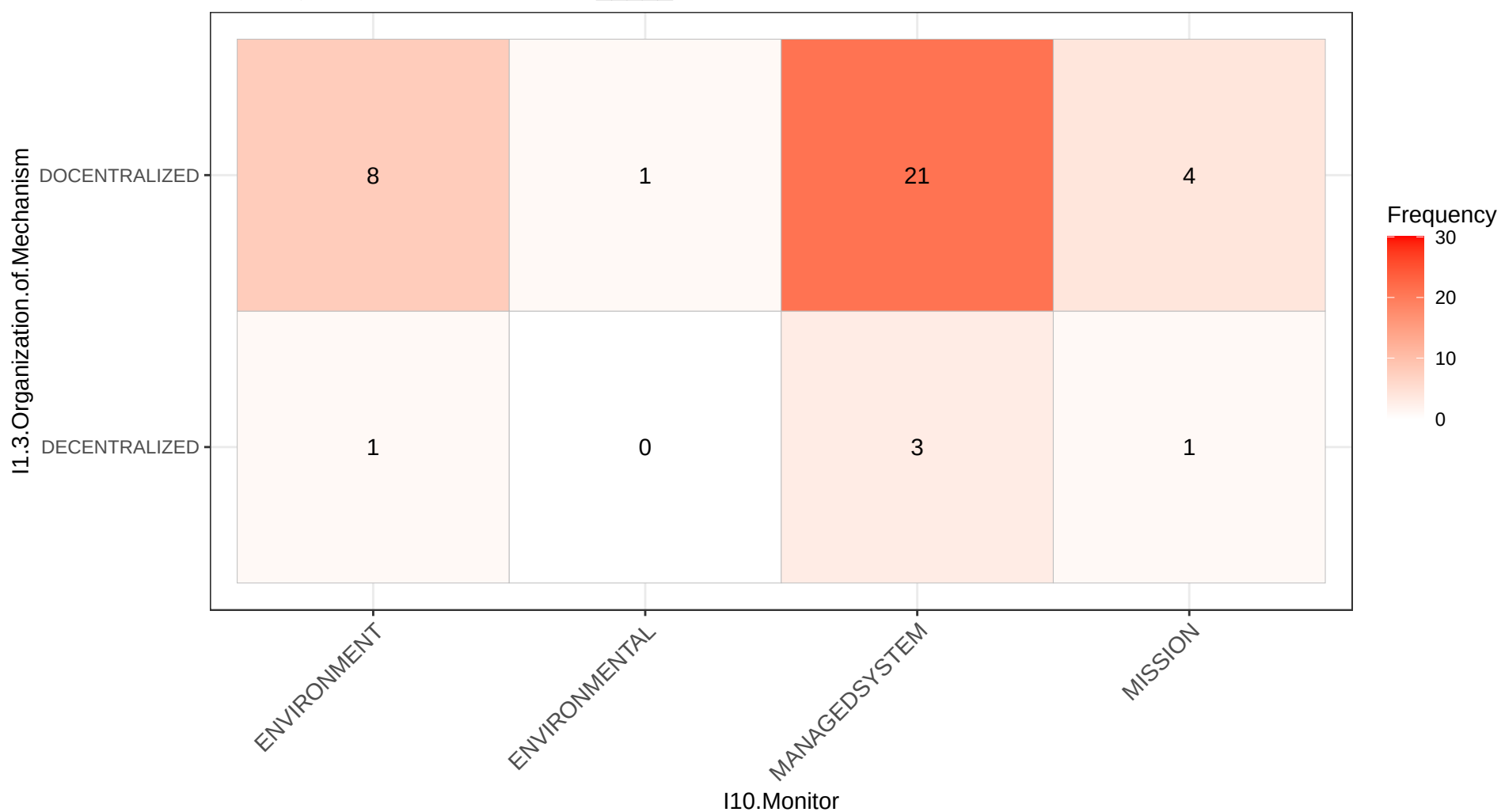


I8.Evaluation

I1.3.Organization.of.Mechanism_____I9.Adap..Logic

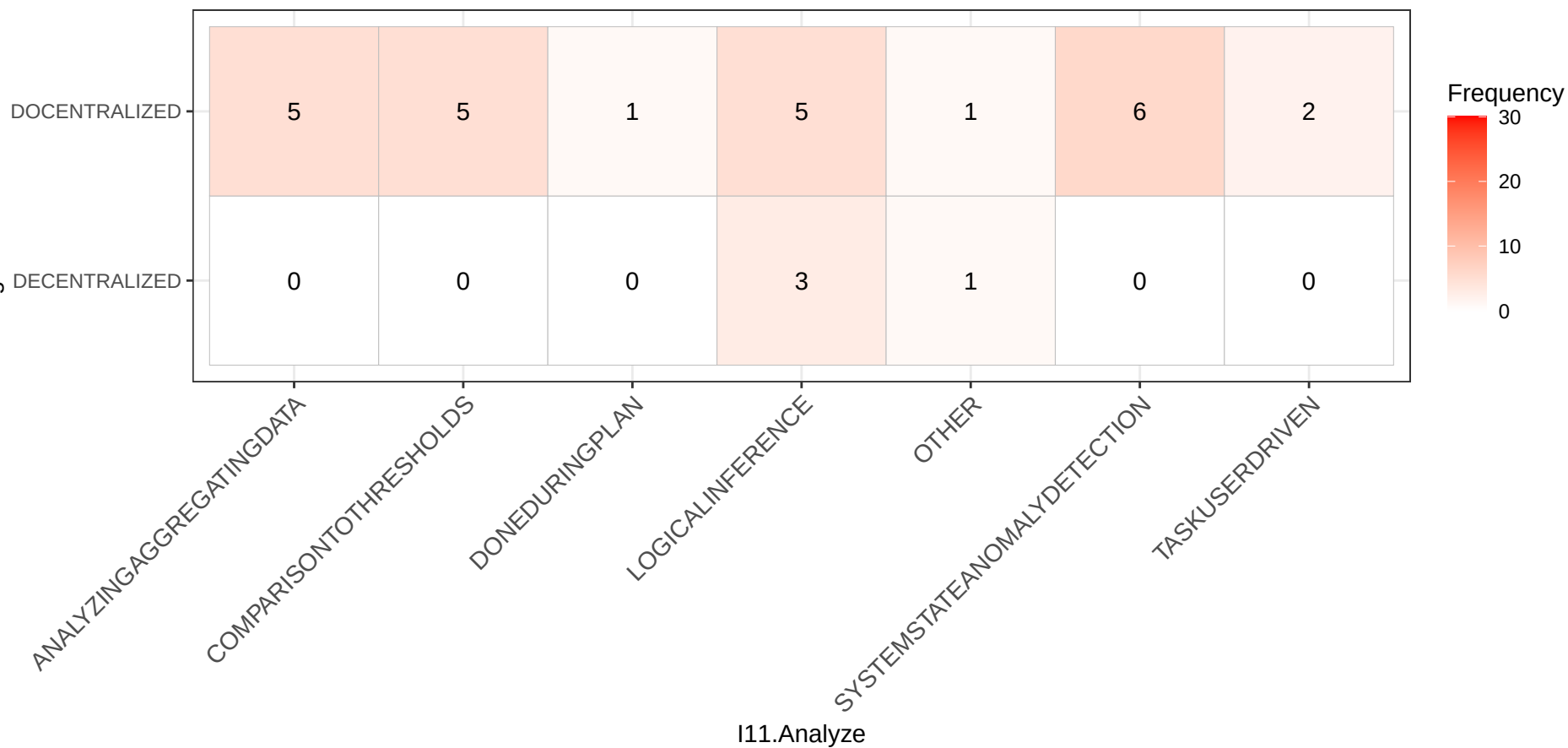


I1.3.Organization.of.Mechanism_____I10.Monitor

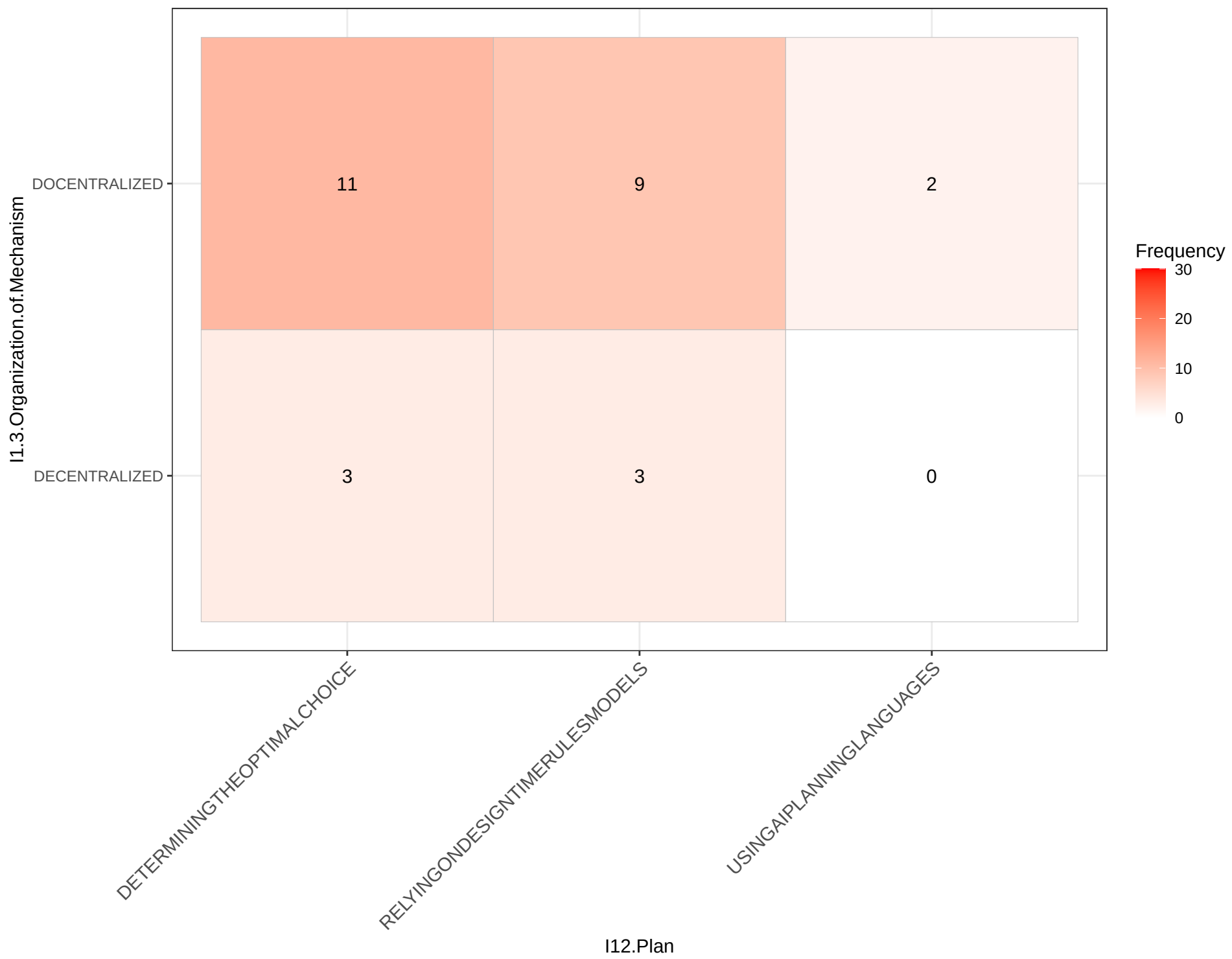


I1.3.Organization.of.Mechanism

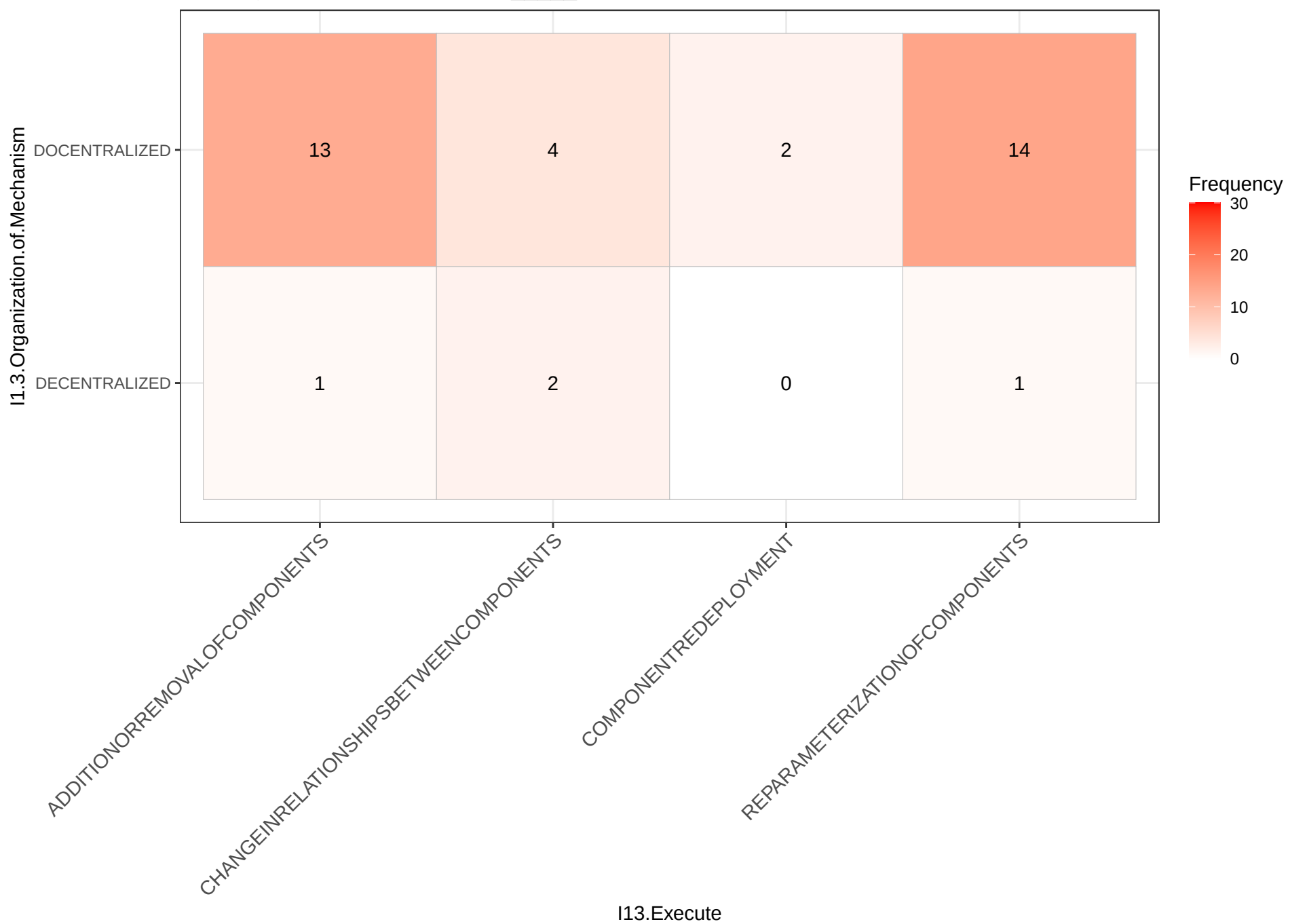
I1.3.Organization.of.Mechanism_____I11.Analyze

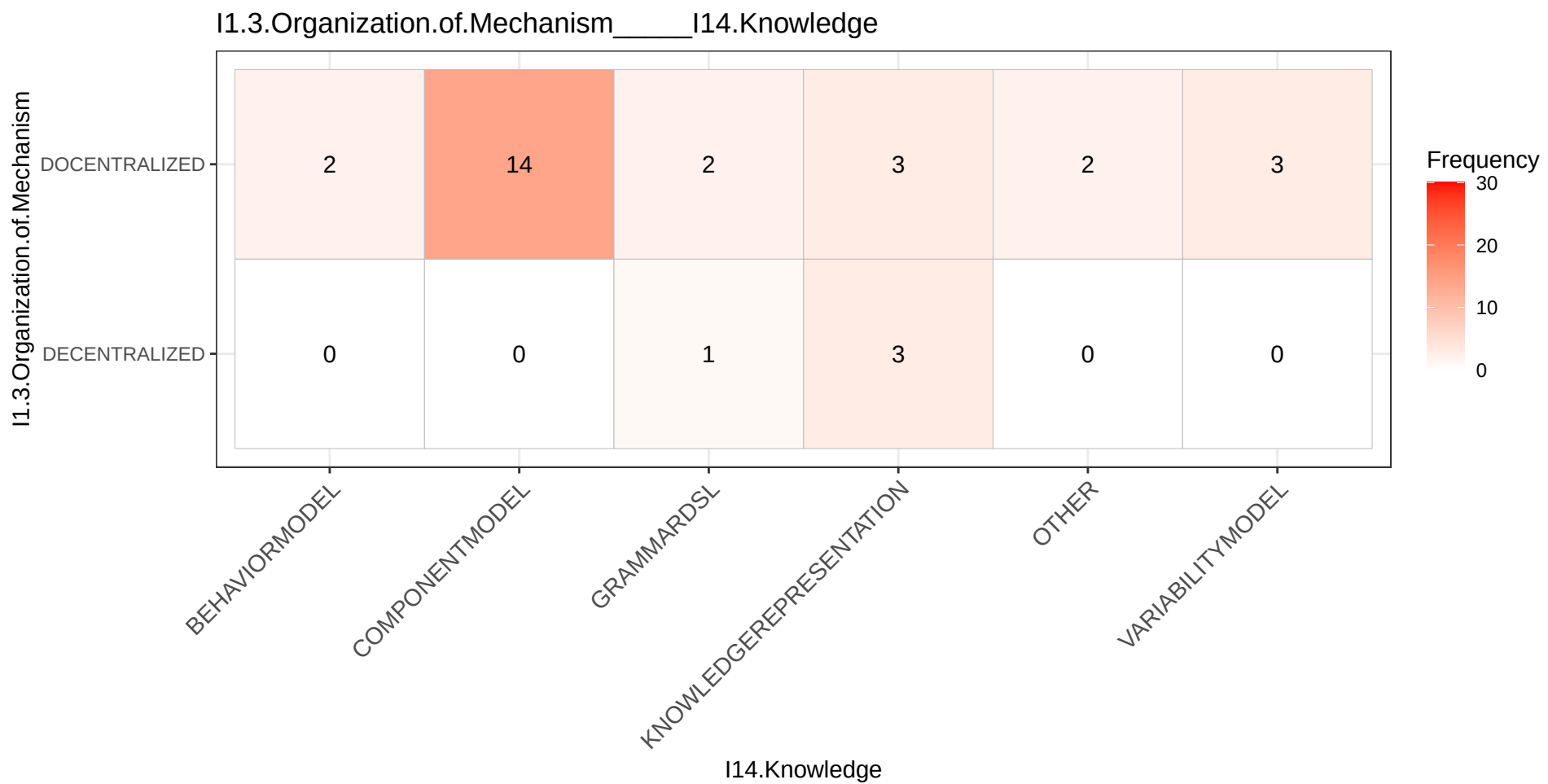


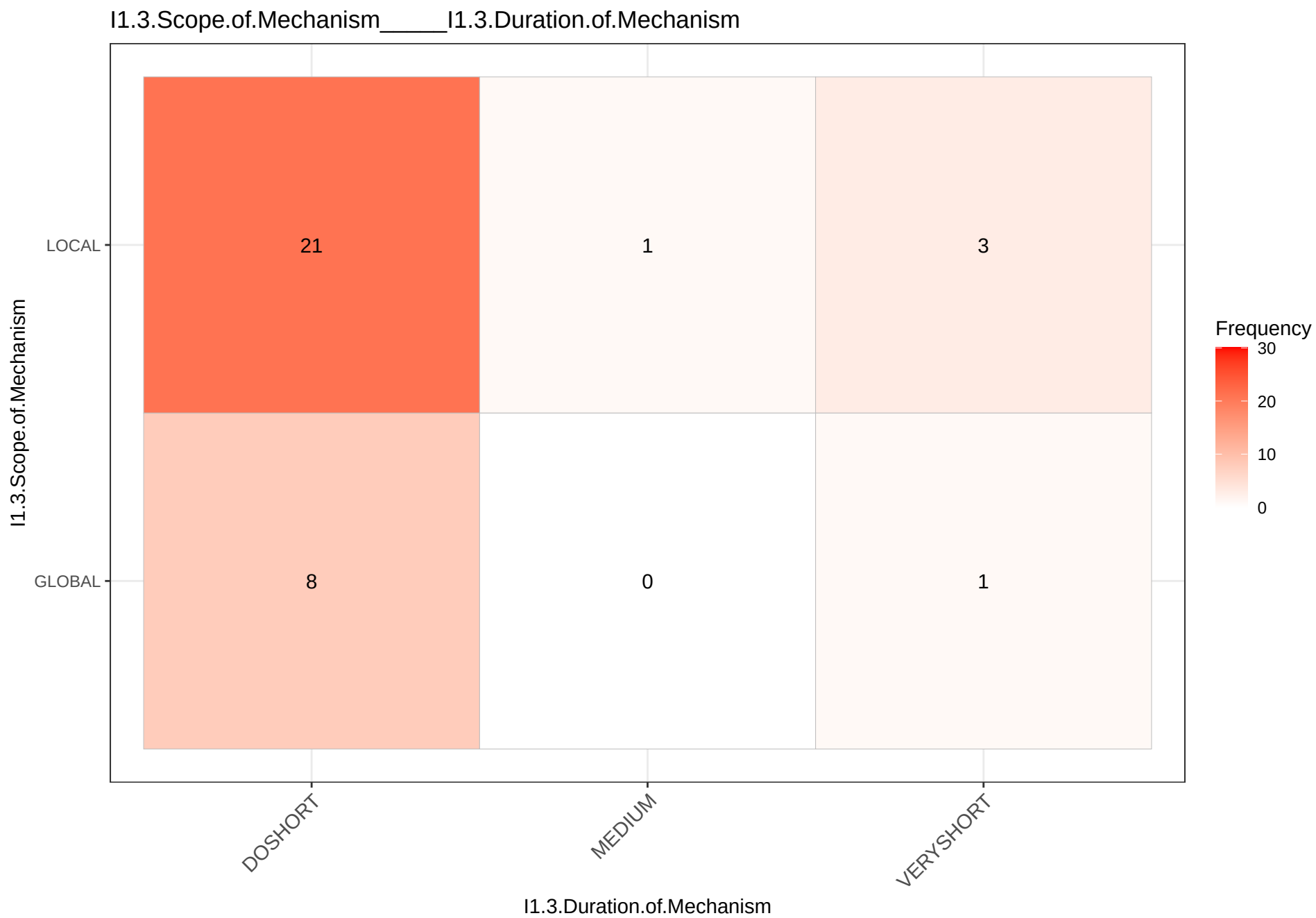
I1.3.Organization.of.Mechanism_____I12.Plan



I1.3.Organization.of.Mechanism_____I13.Execute







I1.3.Scope.of.Mechanism

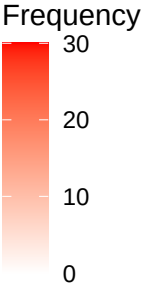
LOCAL

GLOBAL

EVENTTRIGGER

TIMETRIGGER

I1.3.Trigger.of.Mechanism



23

2

7

1

I1.3.Scope.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Scope.of.Mechanism

LOCAL

19

5

GLOBAL

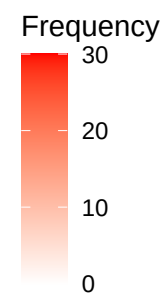
3

4

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects



I1.3.Scope.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Scope.of.Mechanism

LOCAL

5

20

GLOBAL

3

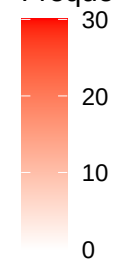
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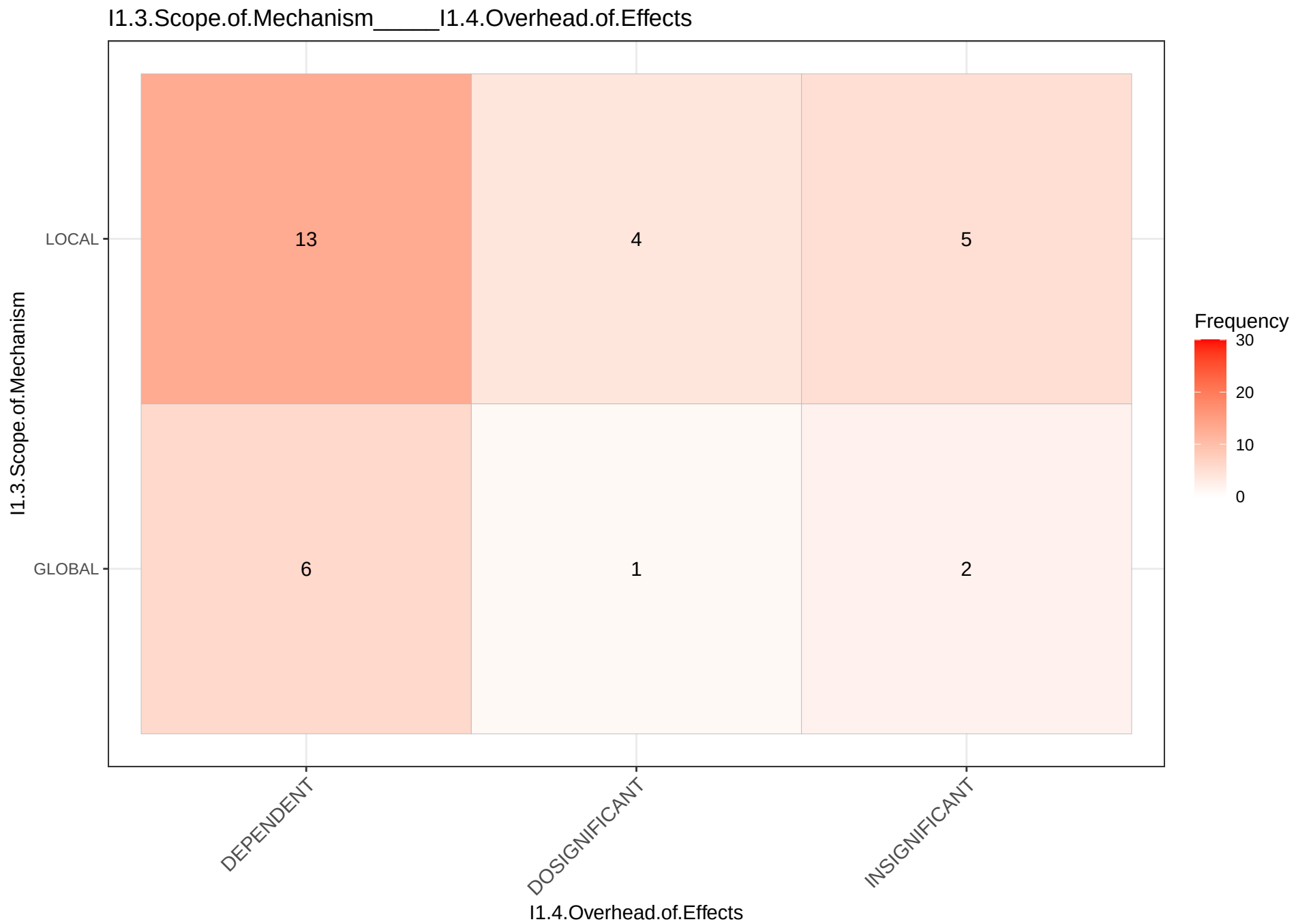
DODETERMINISTIC

NONDETERMINISTIC

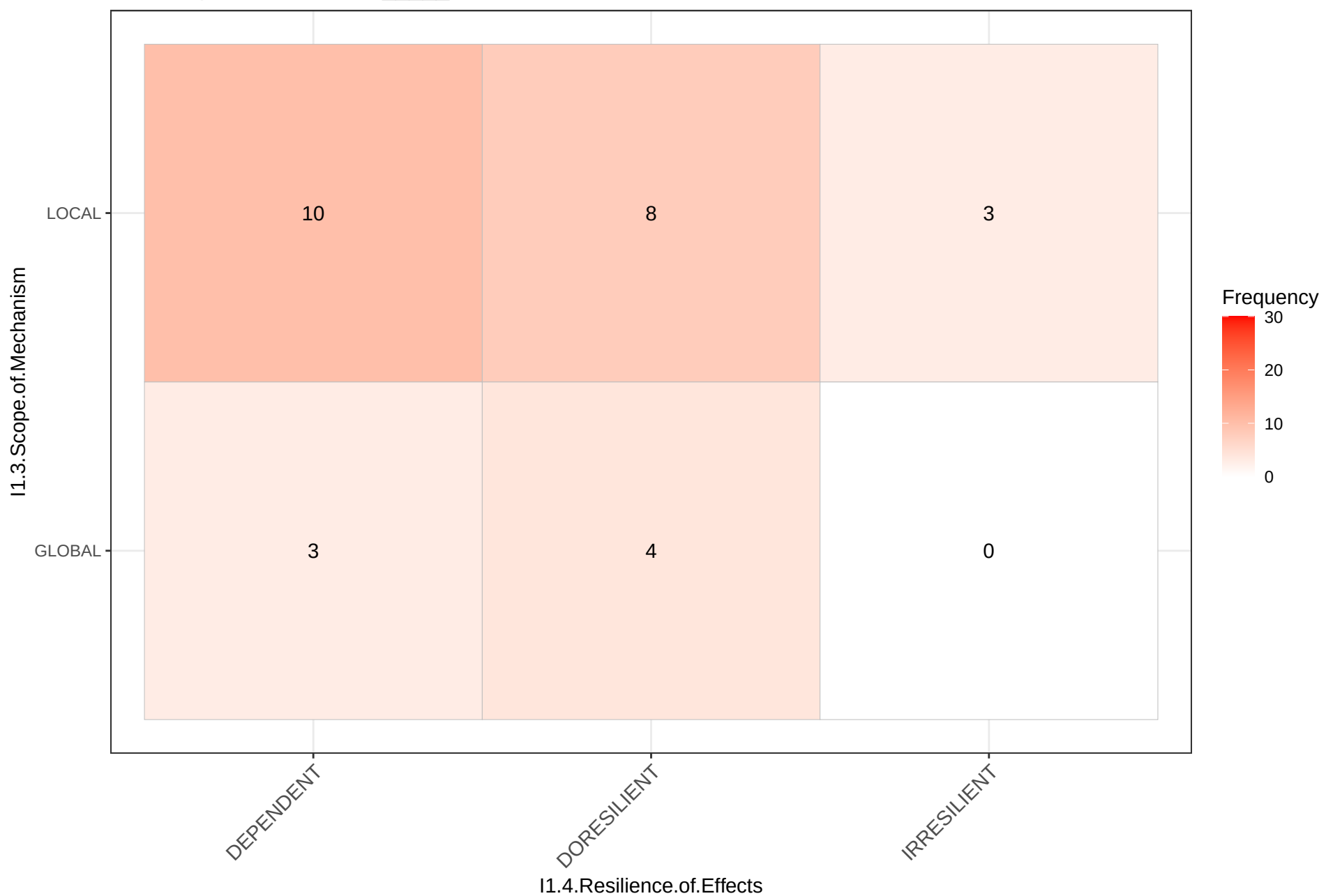
I1.4.Predictability.of.Effects

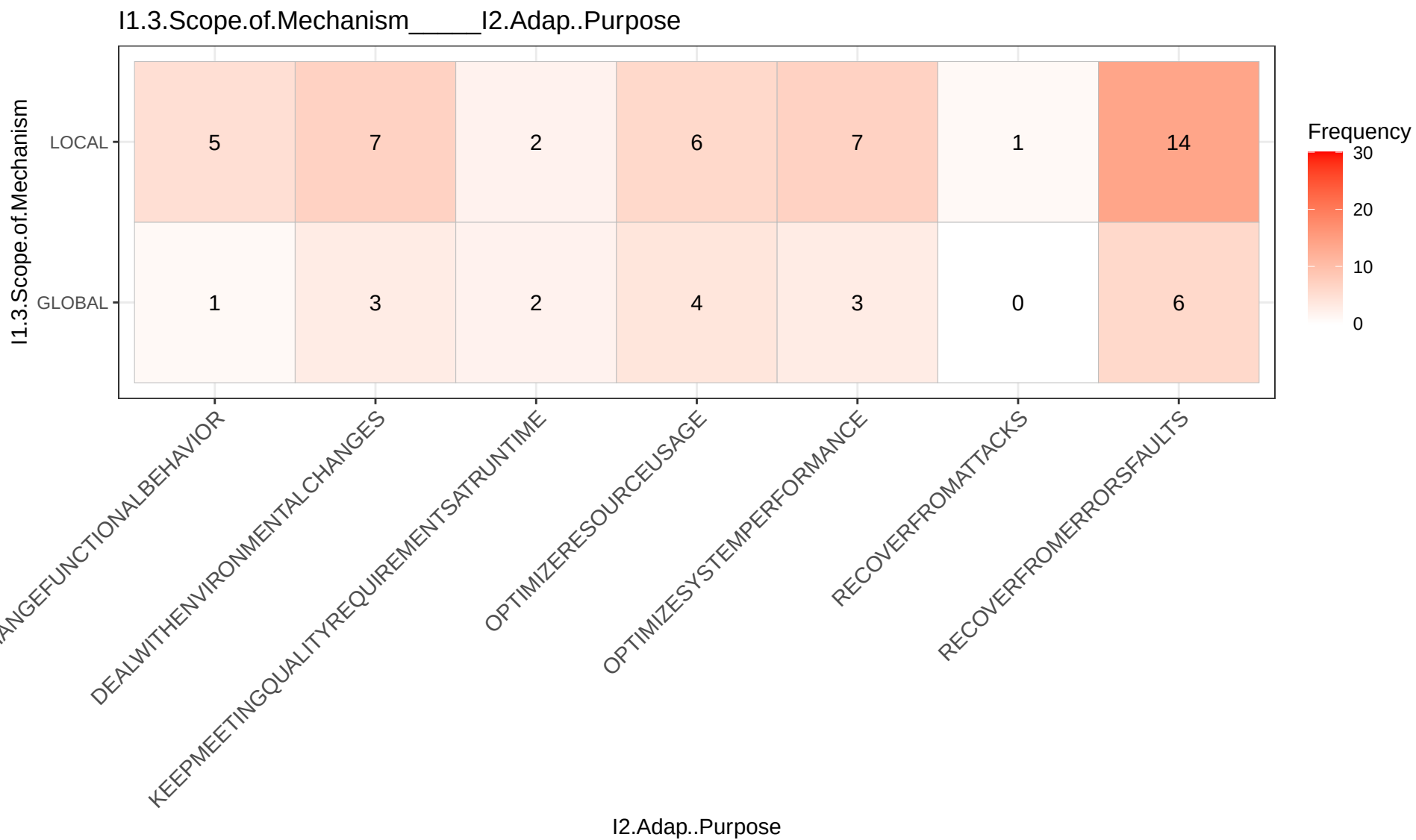
Frequency





I1.3.Scope.of.Mechanism_____I1.4.Resilience.of.Effects



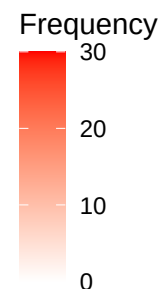


I1.3.Scope.of.Mechanism

I1.3.Scope.of.Mechanism I3.Robot.Type

LOCAL
GLOBAL

1	1	1	1	2	0	2	0	1	1	2	1	0	0	1	1	2	1	1	1	1	1	0	4	1	1	1	1
1	0	0	0	0	1	0	0	0	0	1	0	1	0	0	0	2	0	0	0	0	0	1	3	0	0	0	0



BLUEROV2
BOXERCLEARPATH
CRAWLER TERMINATOR BOT
FIELD MOBILEROBOT
HETEROGENOUS ROBOTS
INFOTAINMENTROBOT
KUKALIGHTWEIGHTROBOT4LWR4
MOBILESERVICEROBOT
HEXMANIPULATOR
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
RESCUE
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
TWO CASE STUDIES MOBILE MANIPULATOR AS RUNNING EXAMPLE QUADRO COPTOR

I3.Robot.Type

I1.3.Scope.of.Mechanism_____I4.Robo.SW

I1.3.Scope.of.Mechanism

LOCAL

8

13

1

GLOBAL

1

4

2

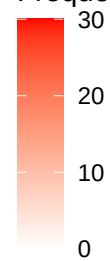
OTHER

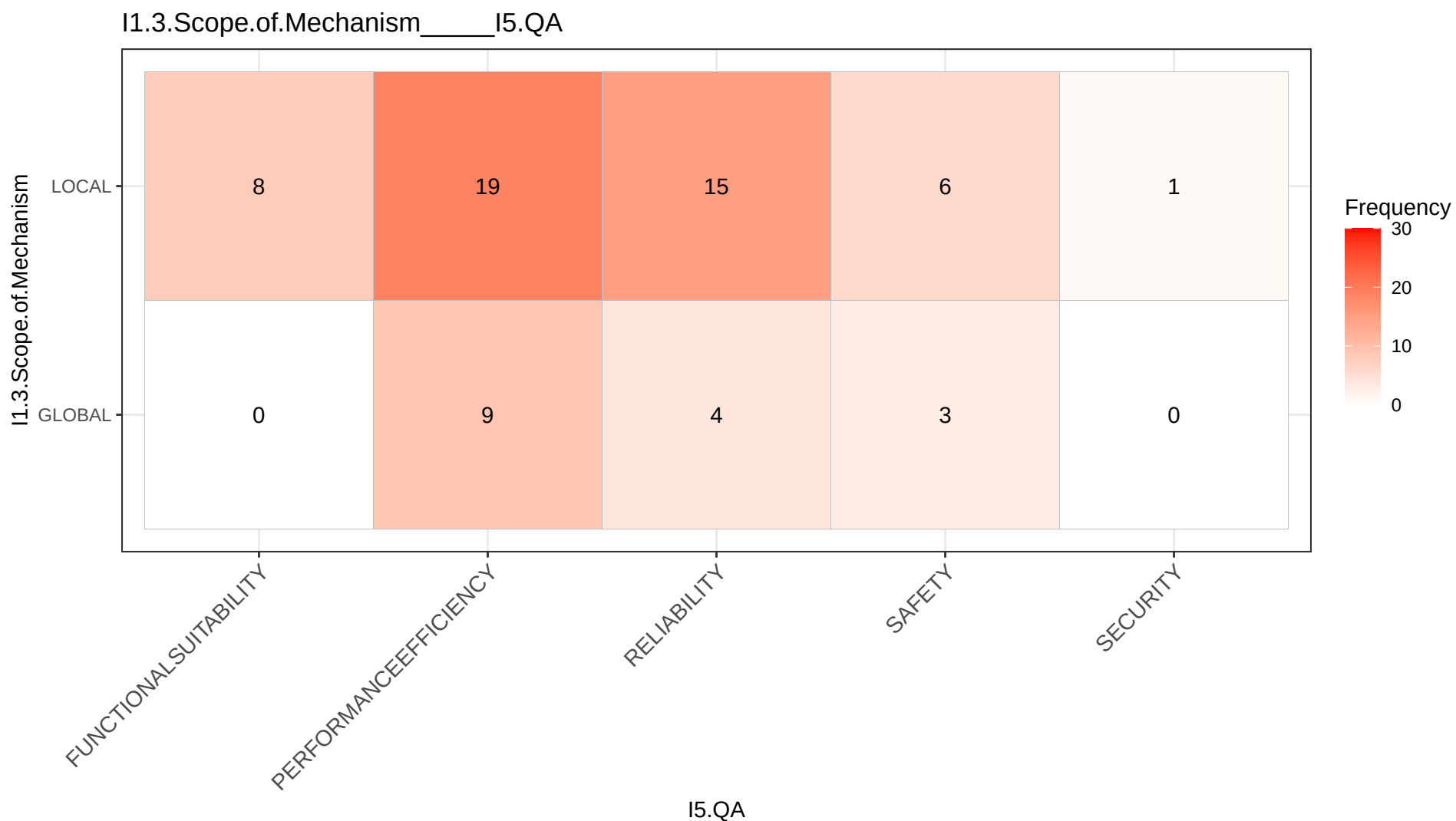
ROS1

ROS2

I4.Robo.SW

Frequency





I1.3.Scope.of.Mechanism_____I6.Independence

I1.3.Scope.of.Mechanism

LOCAL

GLOBAL

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency

30

20

10

0

7

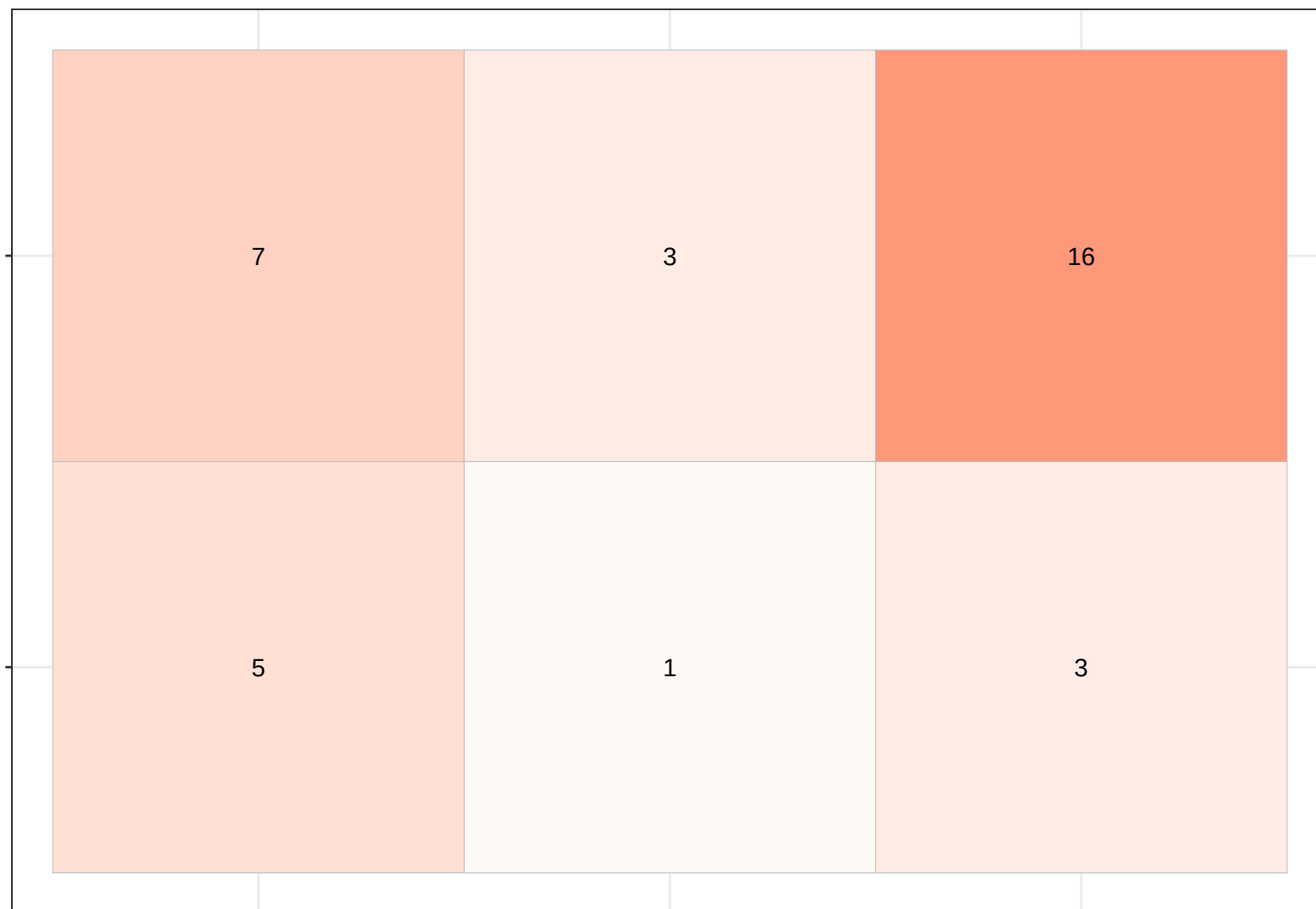
3

16

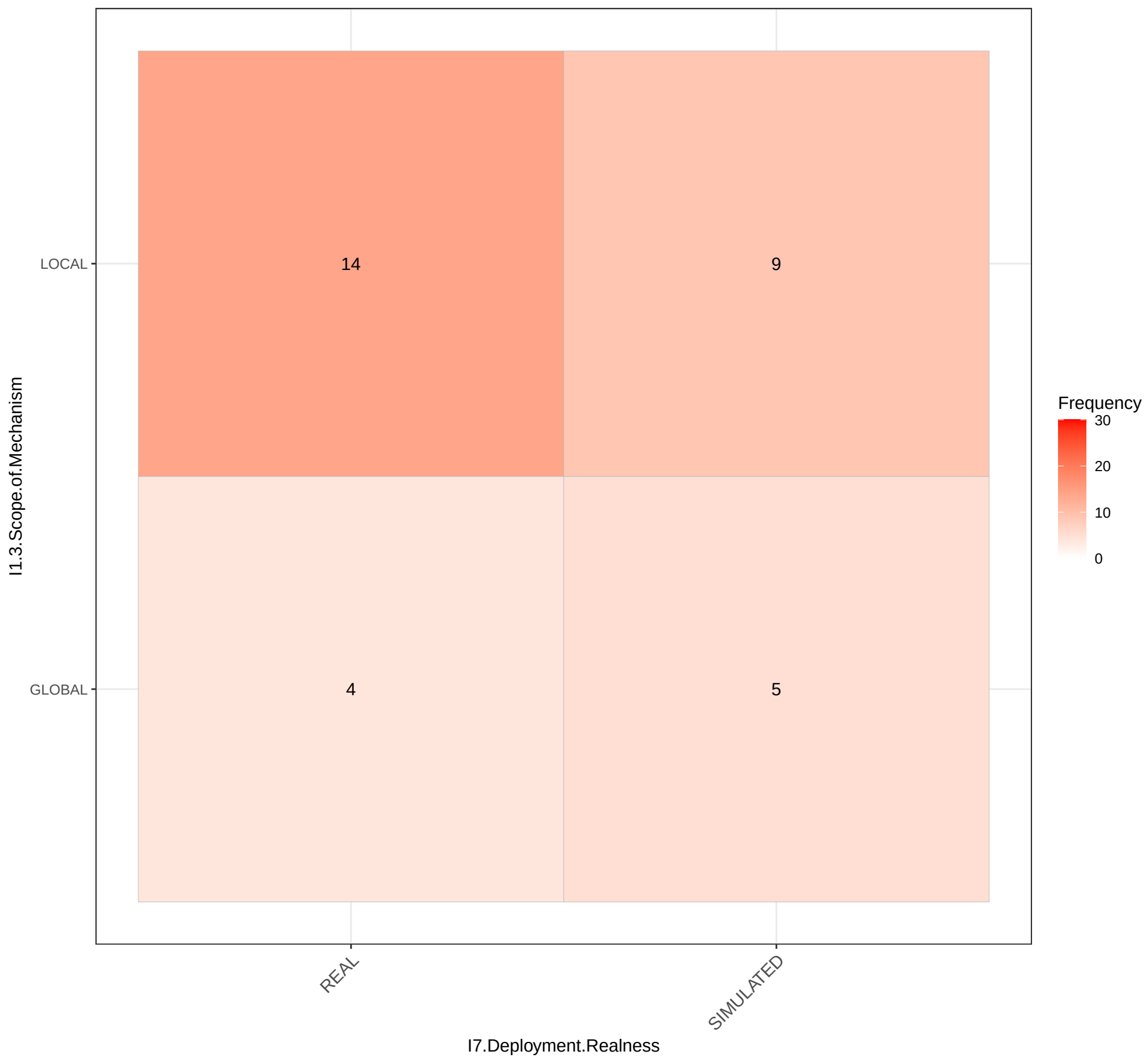
5

1

3



I1.3.Scope.of.Mechanism_____I7.Deployment.Realness



I1.3.Scope.of.Mechanism_____I7.Mission.Realness

I1.3.Scope.of.Mechanism

LOCAL

13

13

GLOBAL

4

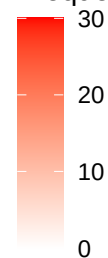
5

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



I1.3.Scope.of.Mechanism _____ Experiment.Method

I1.3.Scope.of.Mechanism

LOCAL

17

5

4

GLOBAL

6

2

1

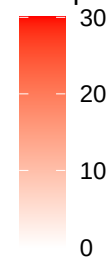
EXPERIMENT

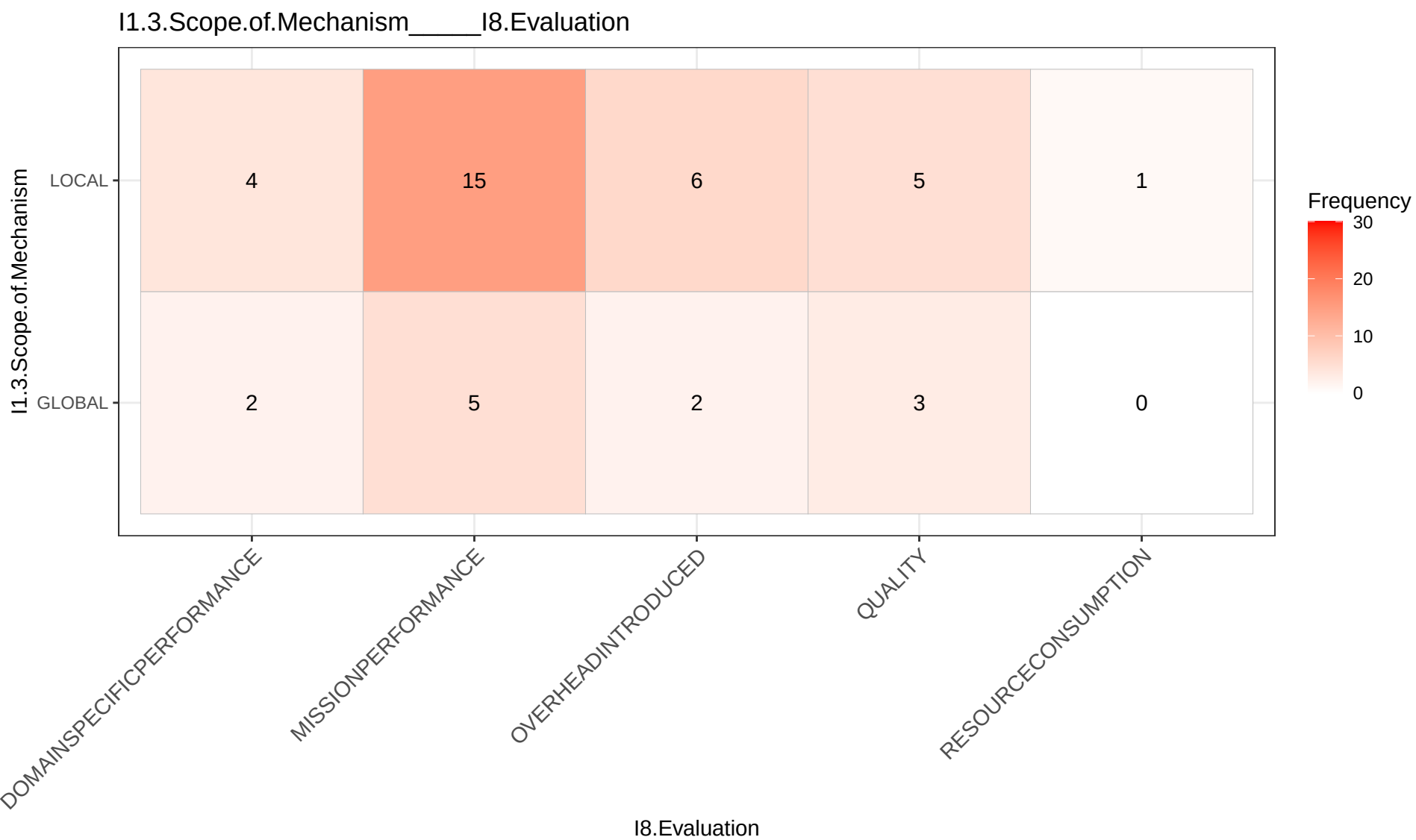
NOEVALUATION

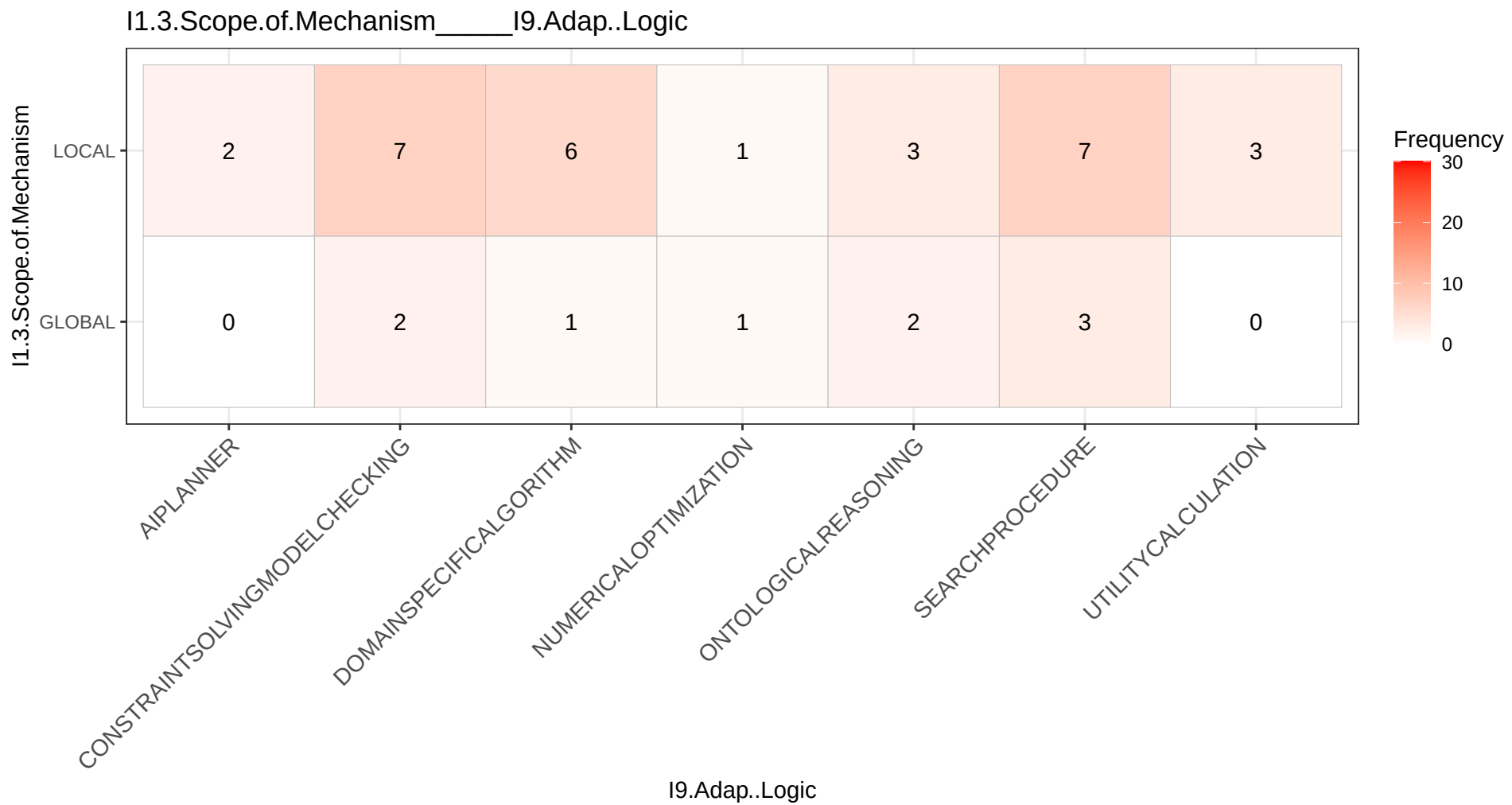
SHOWCASE

Experiment.Method

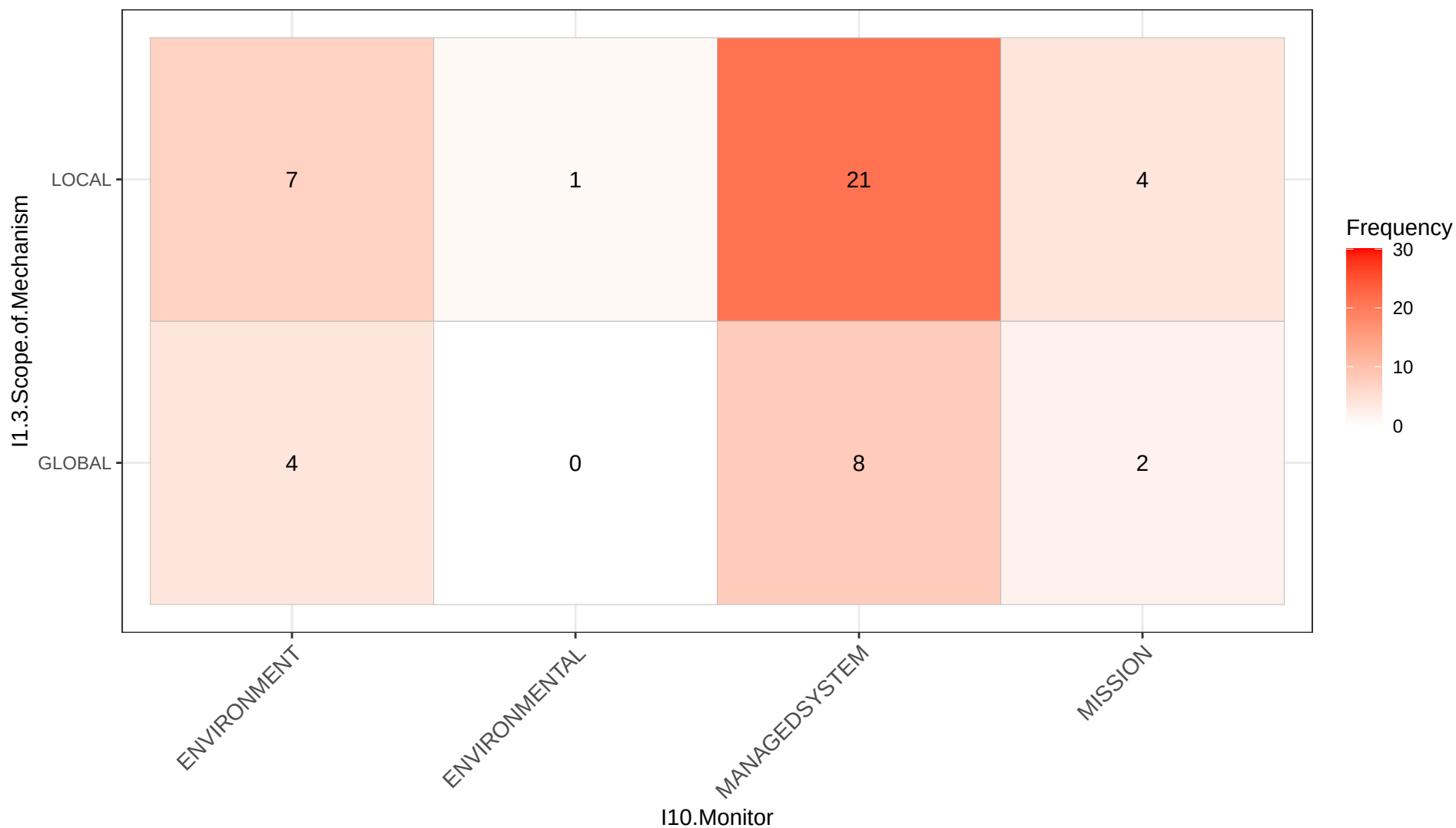
Frequency

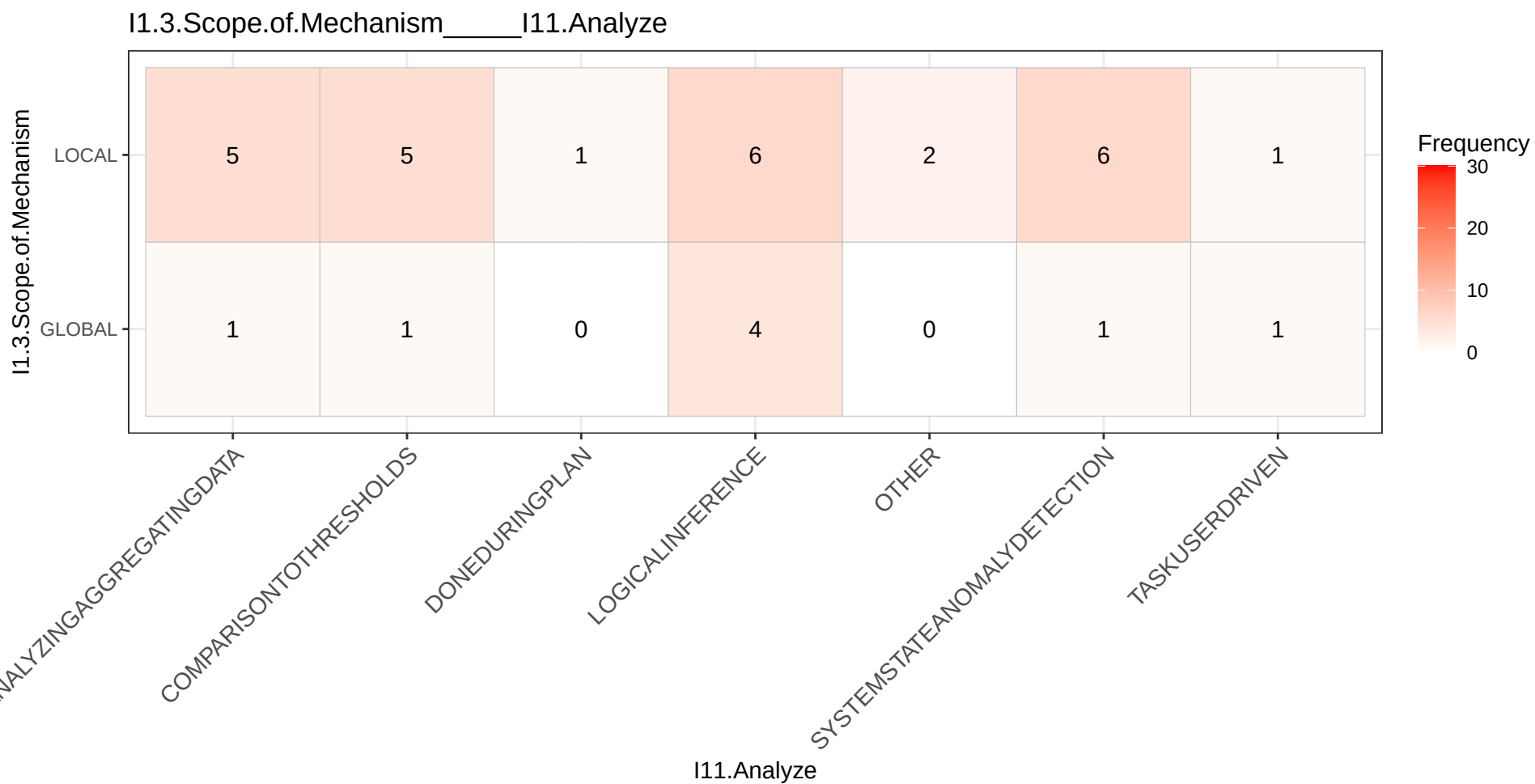




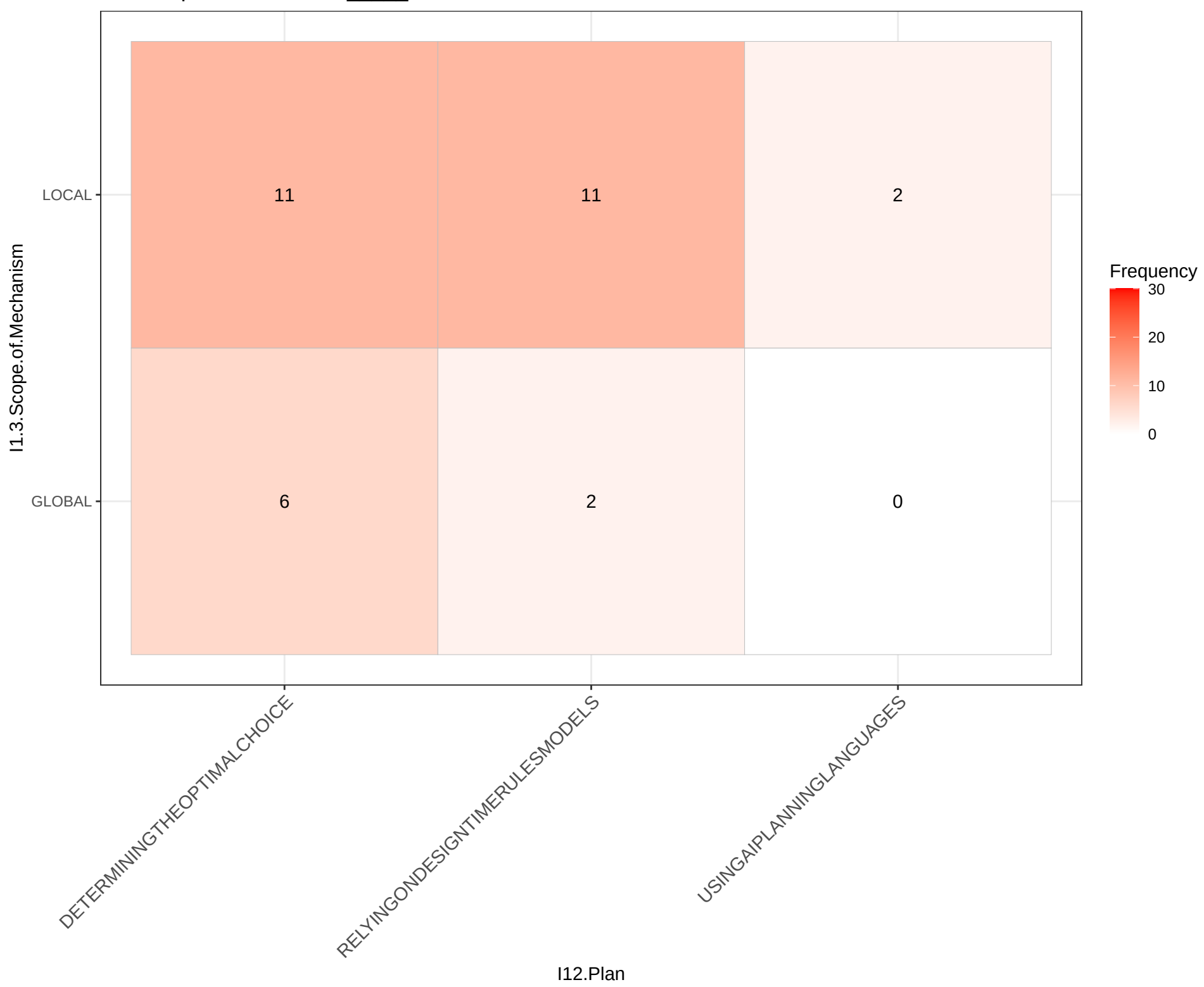


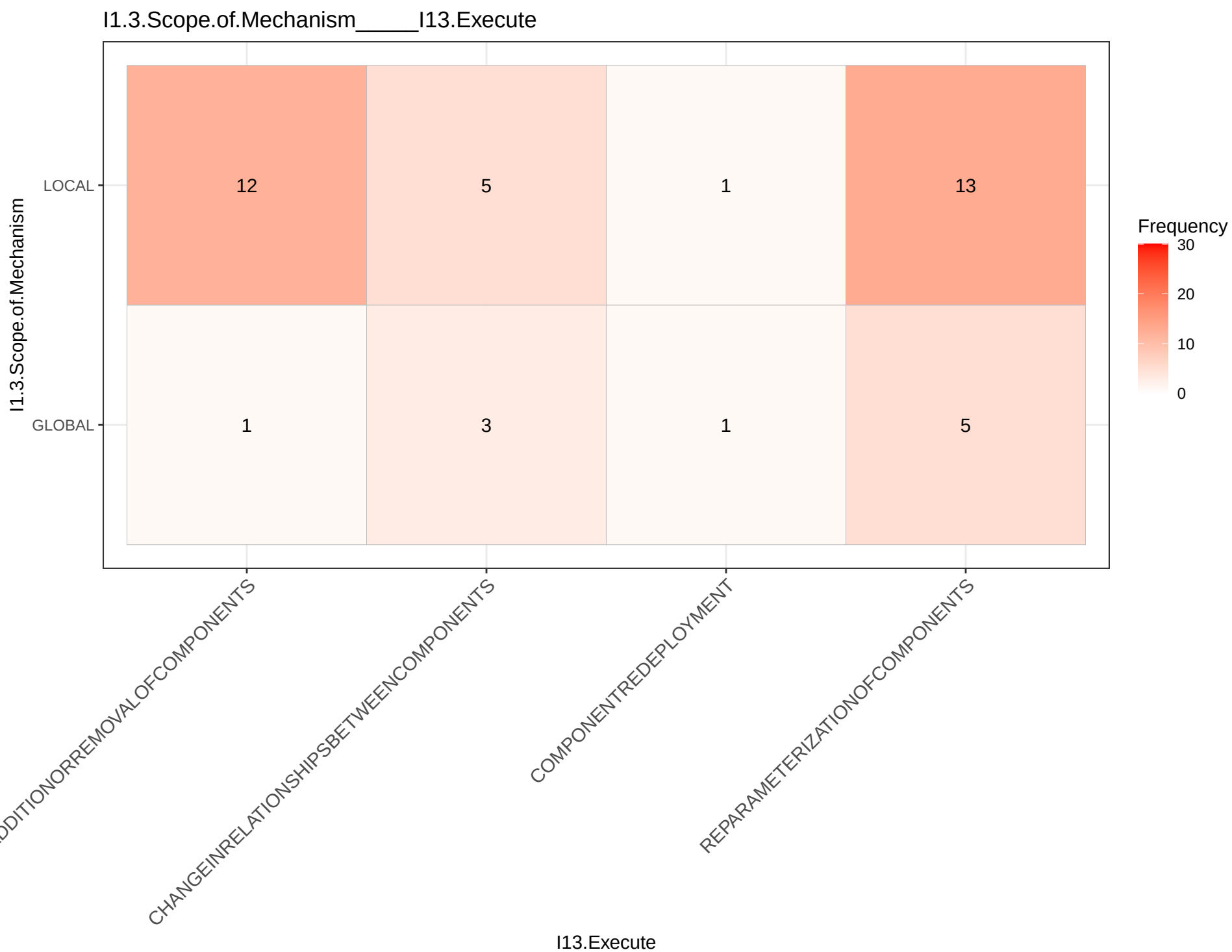
I1.3.Scope.of.Mechanism_____I10.Monitor

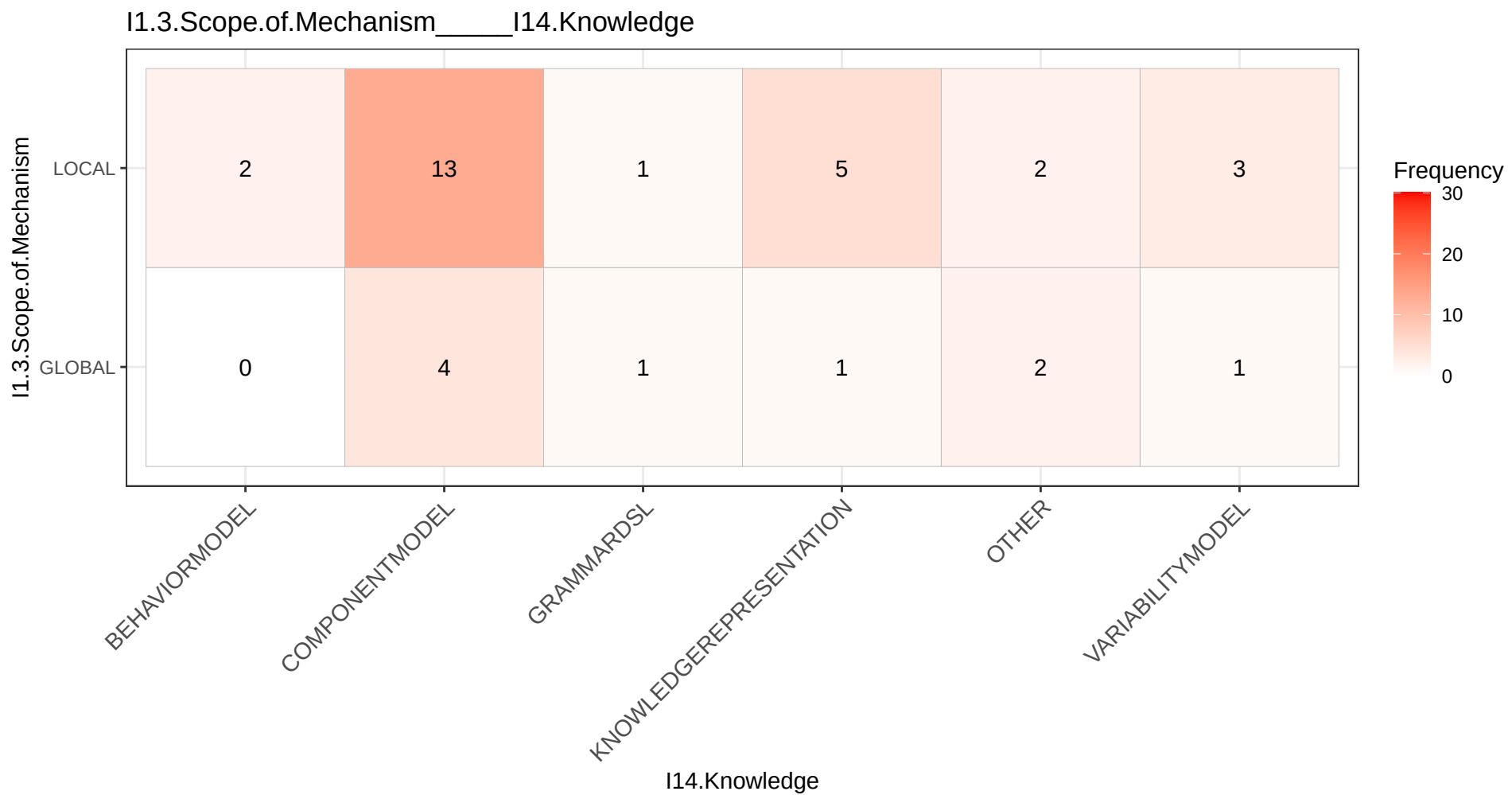


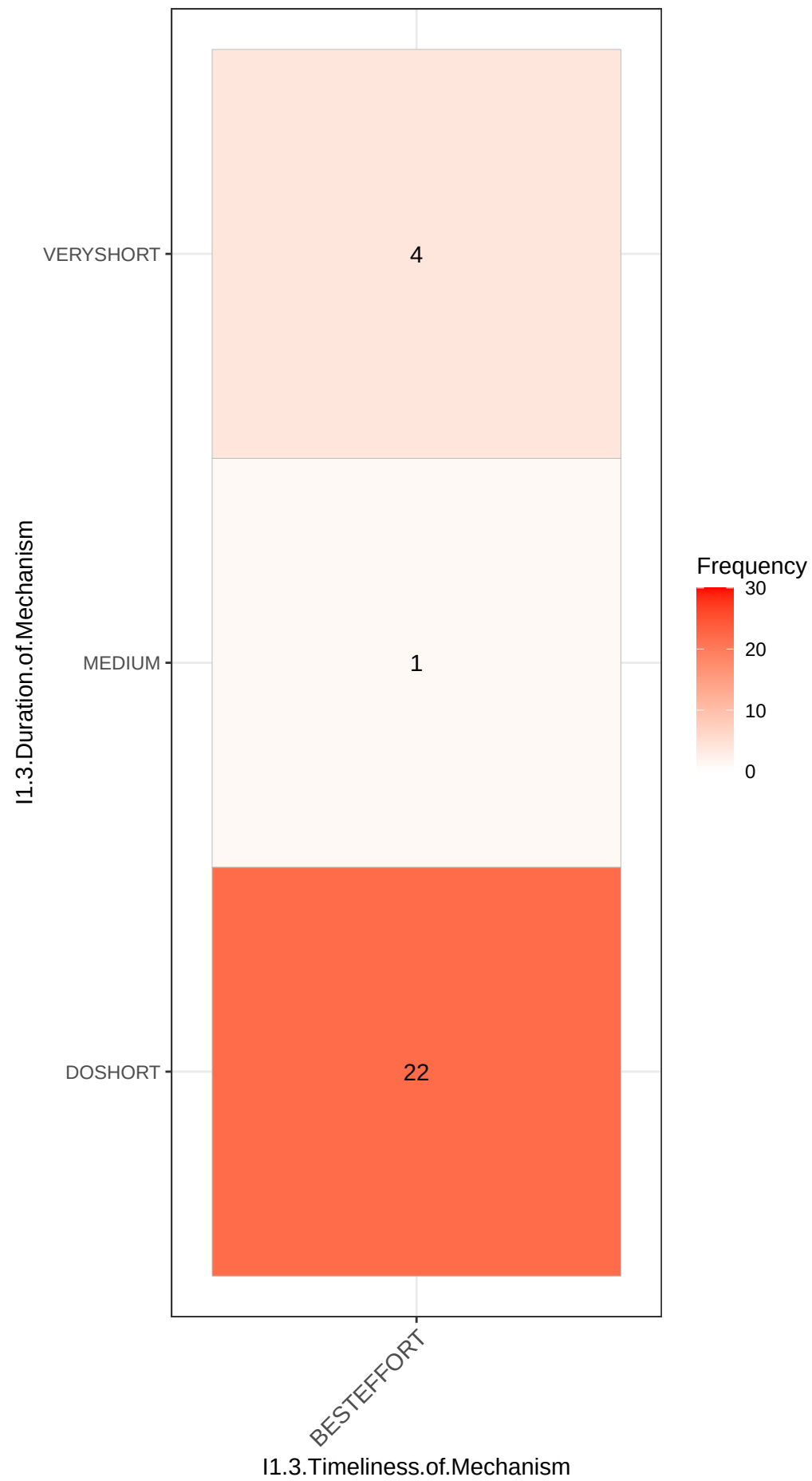


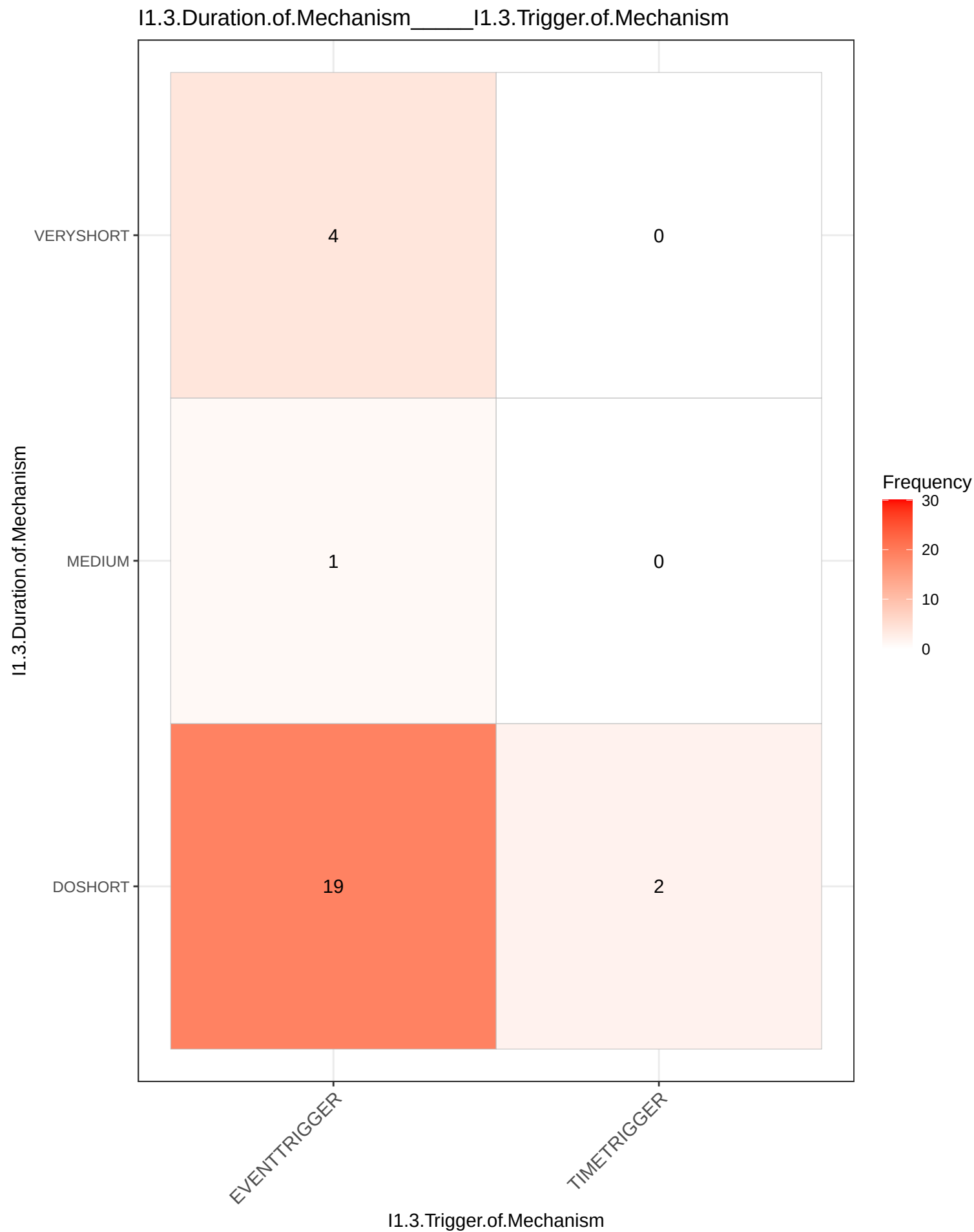
I1.3.Scope.of.Mechanism_____I12.Plan



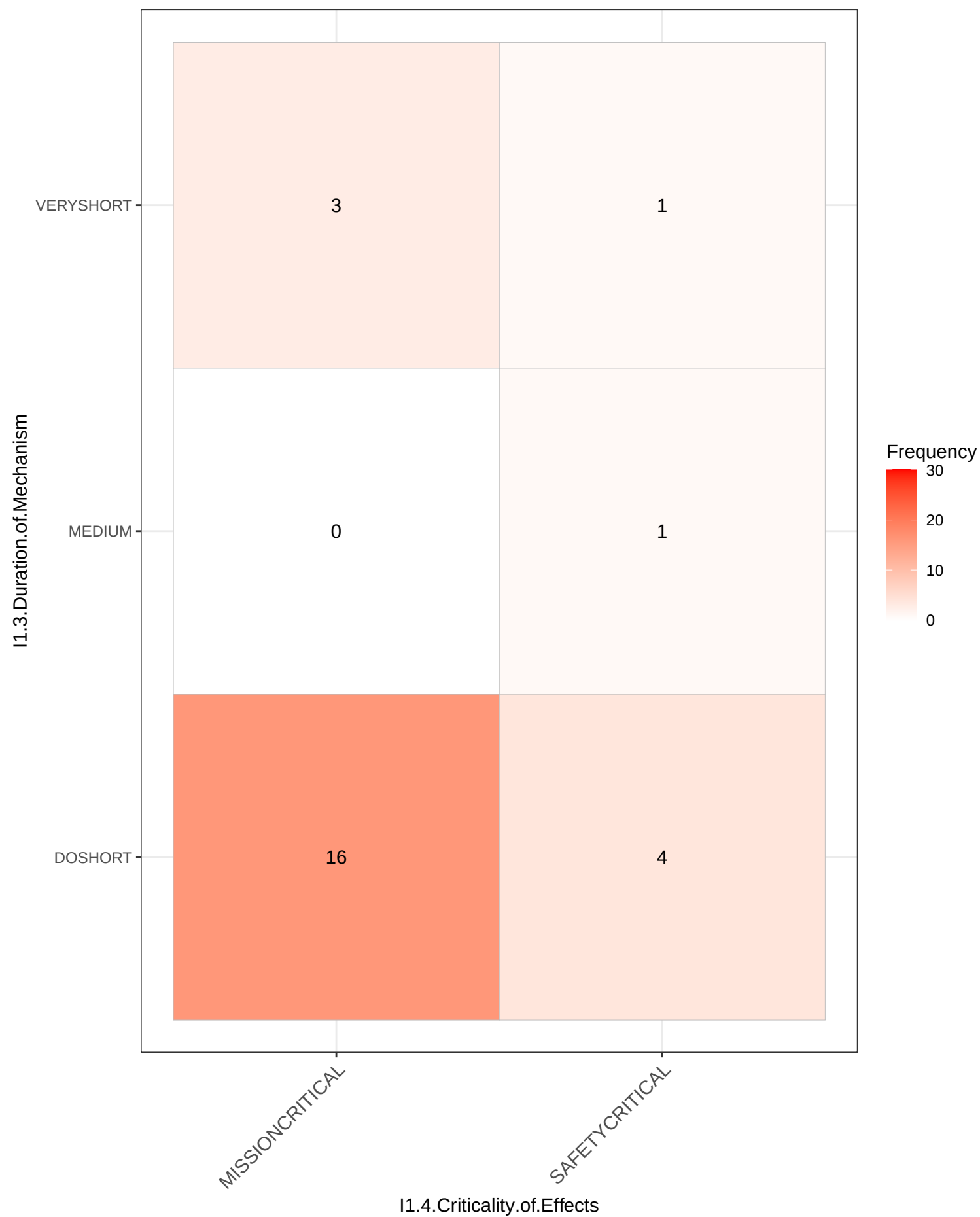


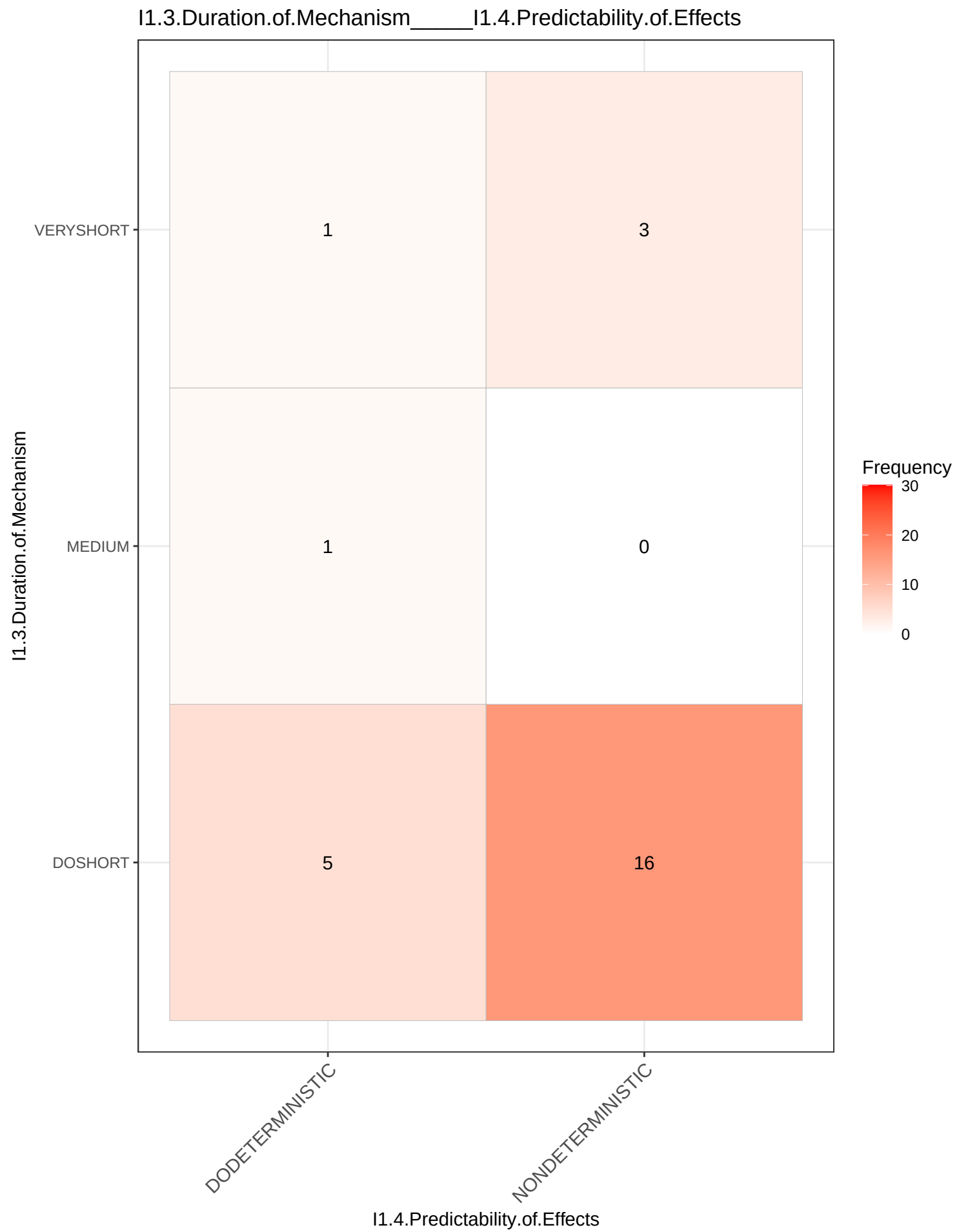






I1.3.Duration.of.Mechanism_____I1.4.Criticality.of.Effects





I1.3.Duration.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Duration.of.Mechanism

VERYSHORT

1

1

2

MEDIUM

0

0

1

DOSHORT

13

3

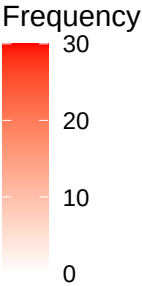
3

DEPENDENT

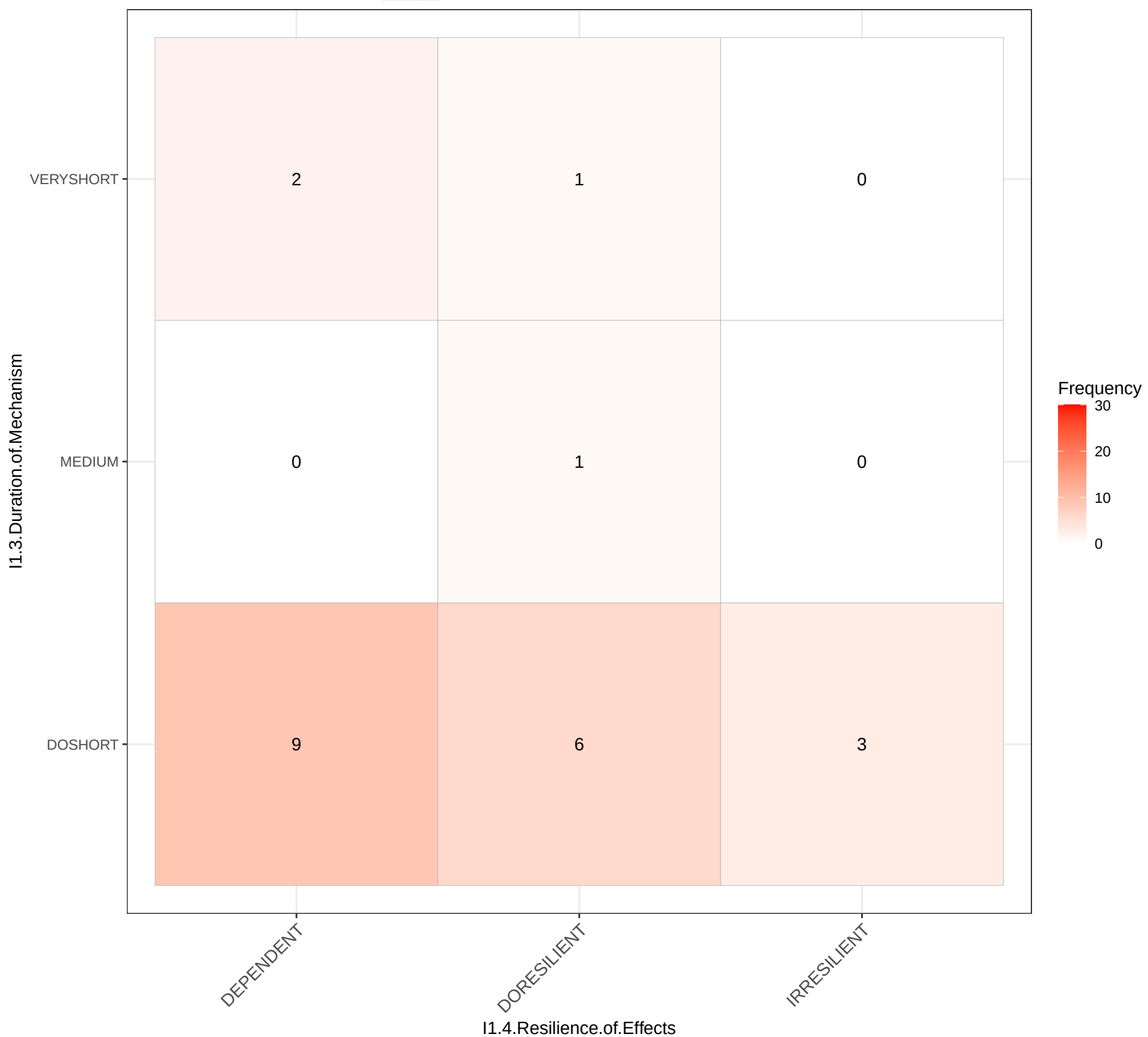
DOSIGNIFICANT

INSIGNIFICANT

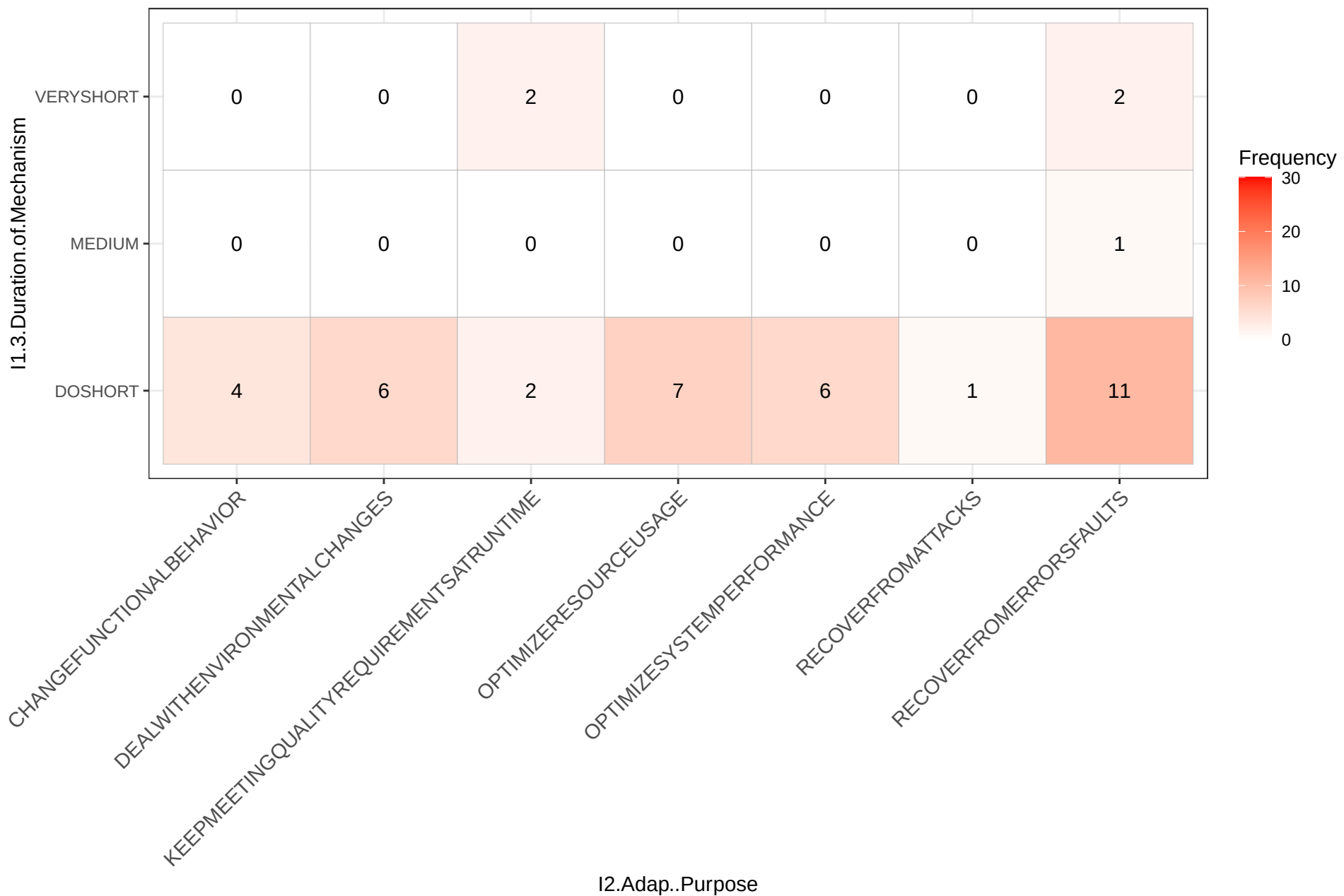
I1.4.Overhead.of.Effects



I1.3.Duration.of.Mechanism_____I1.4.Resilience.of.Effects



I1.3.Duration.of.Mechanism_____I2.Adap..Purpose



I1.3.Duration.of.Mechanism

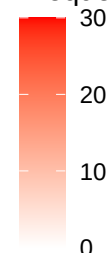
I1.3.Duration.of.Mechanism_____I3.Robot.Type

VERYSHORT	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	1	1	
MEDIUM	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	
DOSHORT	1	0	1	1	2	1	2	0	1	0	2	1	0	0	1	1	2	0	1	1	1	1	1	3	1	1	0

BLUEROV2
BOXERCLEARPATH
CRAWLER TERMINATOR BOT
FIELD MOBILEROBOTS
HETEROGENOUS ROBOTS
HEXAII
HEX MANIPULATOR
INFOTAINMENTROBOT
KUKALIGHTWEIGHTROBOT4LWR4
MOBILESERVICEROBOT
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
RESCUE
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIAL
TRIGLIDEINDUSTRIAL
TWO CASE STUDIES MOBILE MANIPULATOR AS RUNNING EXAMPLE
QUADROCOPTOR FOR ASSEMBLY
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSE DELIVERY ROBOT
WHICH IS AN INDUSTRIAL AGV

I3.Robot.Type

Frequency



I1.3.Duration.of.Mechanism_____I4.Robo.SW

I1.3.Duration.of.Mechanism

VERYSHORT

0

3

1

MEDIUM

0

1

0

DOSHORT

8

9

1

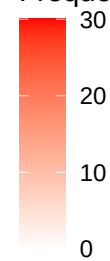
OTHER

ROS1

ROS2

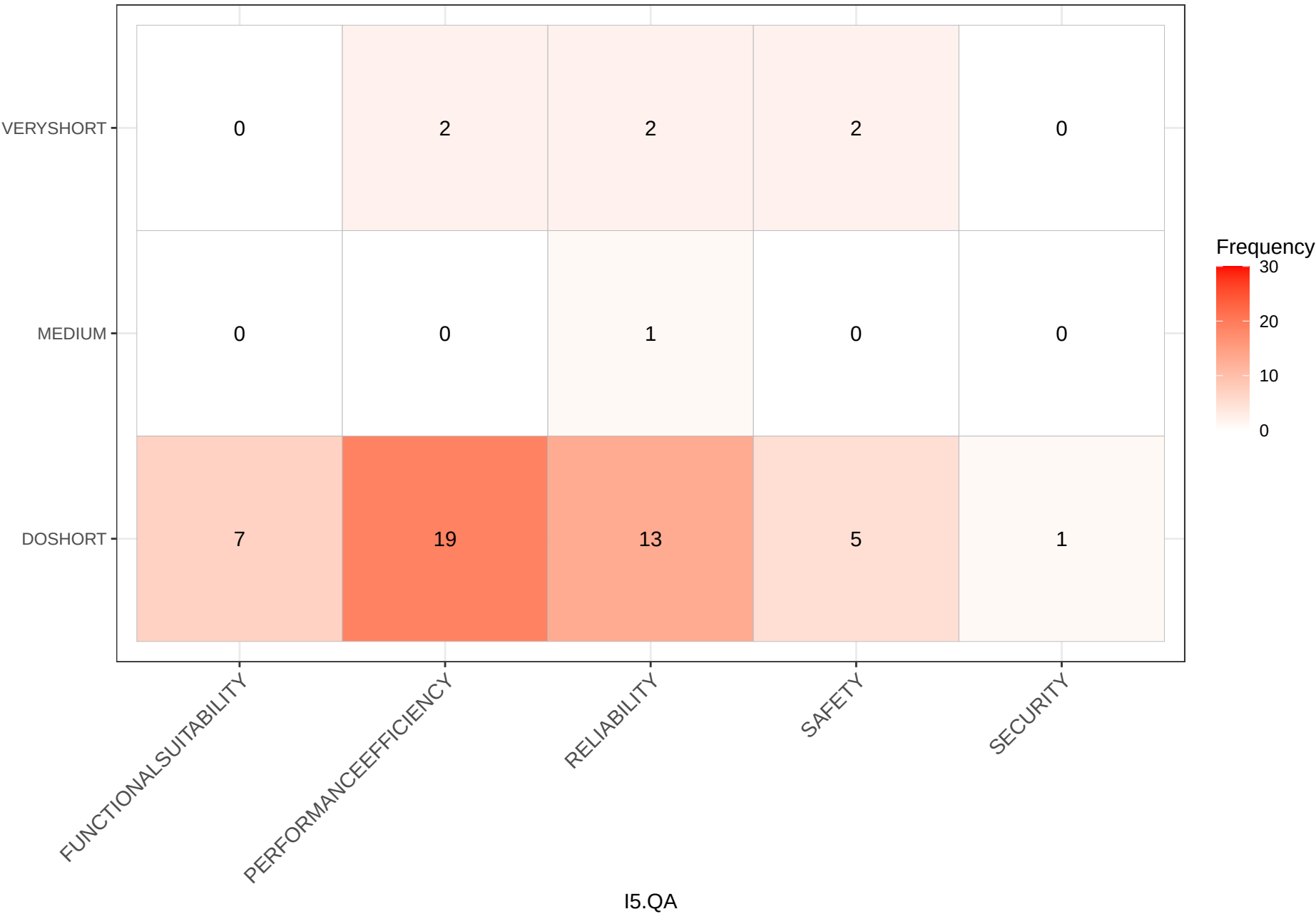
I4.Robo.SW

Frequency

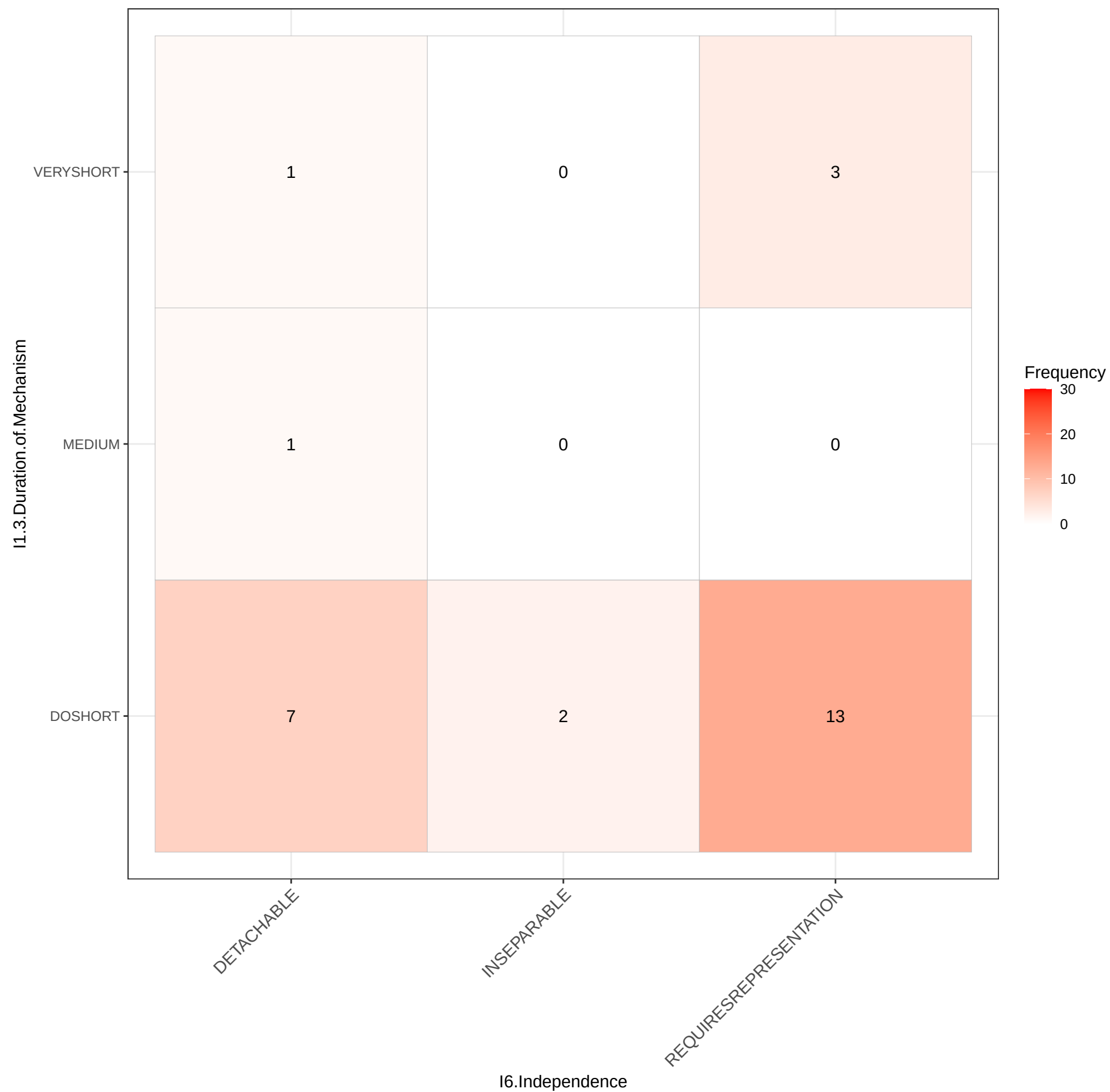


I1.3.Duration.of.Mechanism_____I5.QA

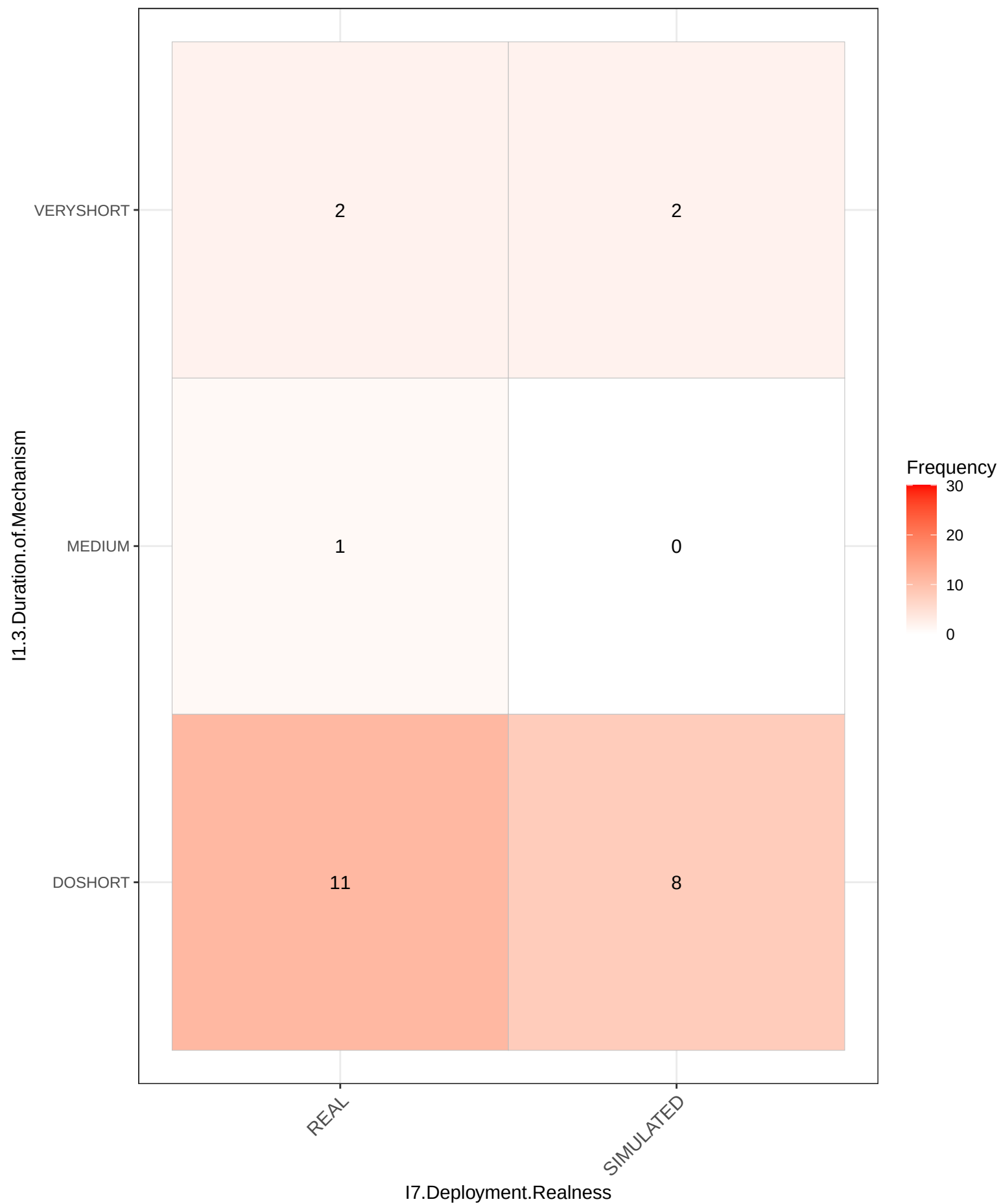
I1.3.Duration.of.Mechanism



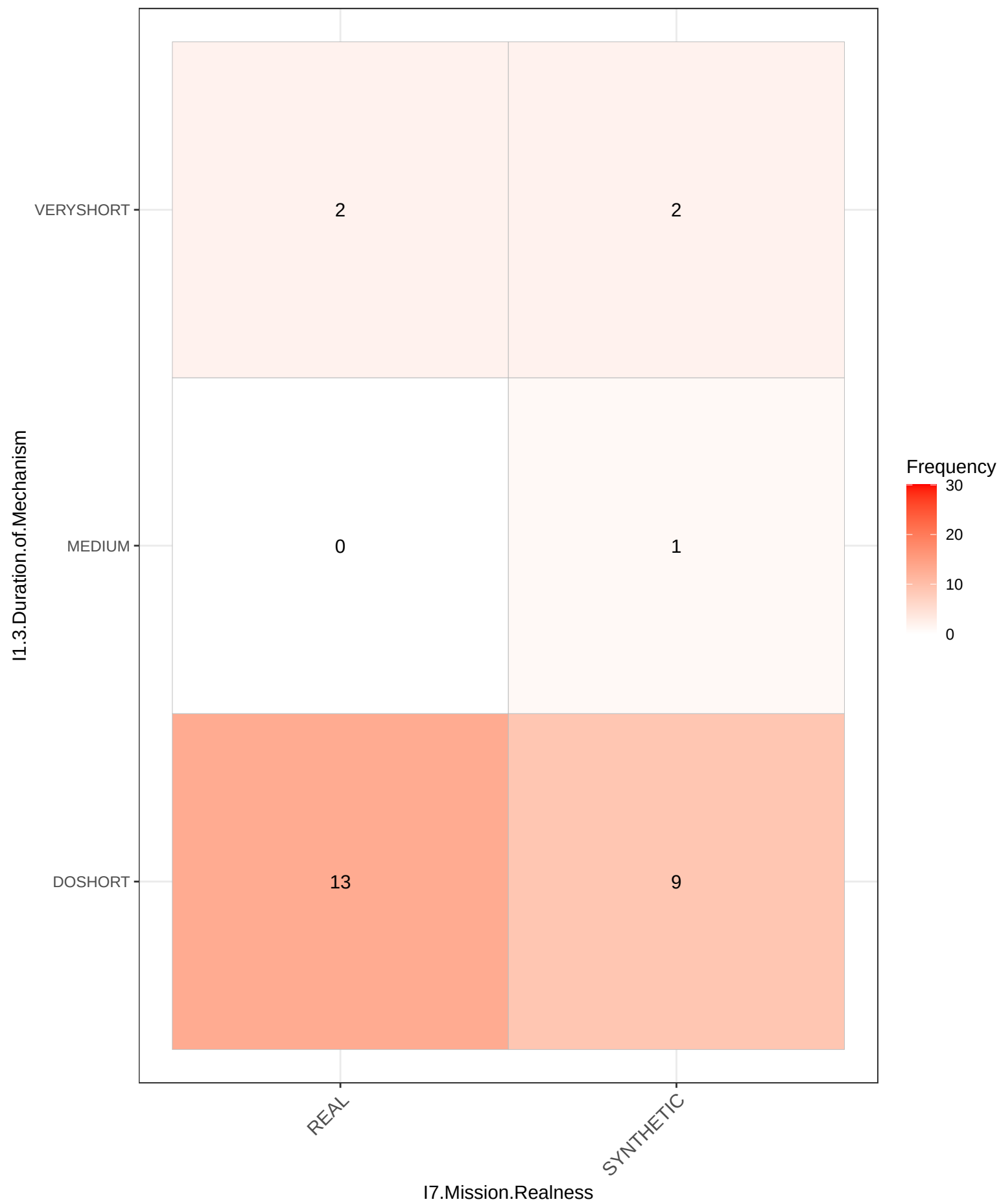
I1.3.Duration.of.Mechanism_____I6.Independence



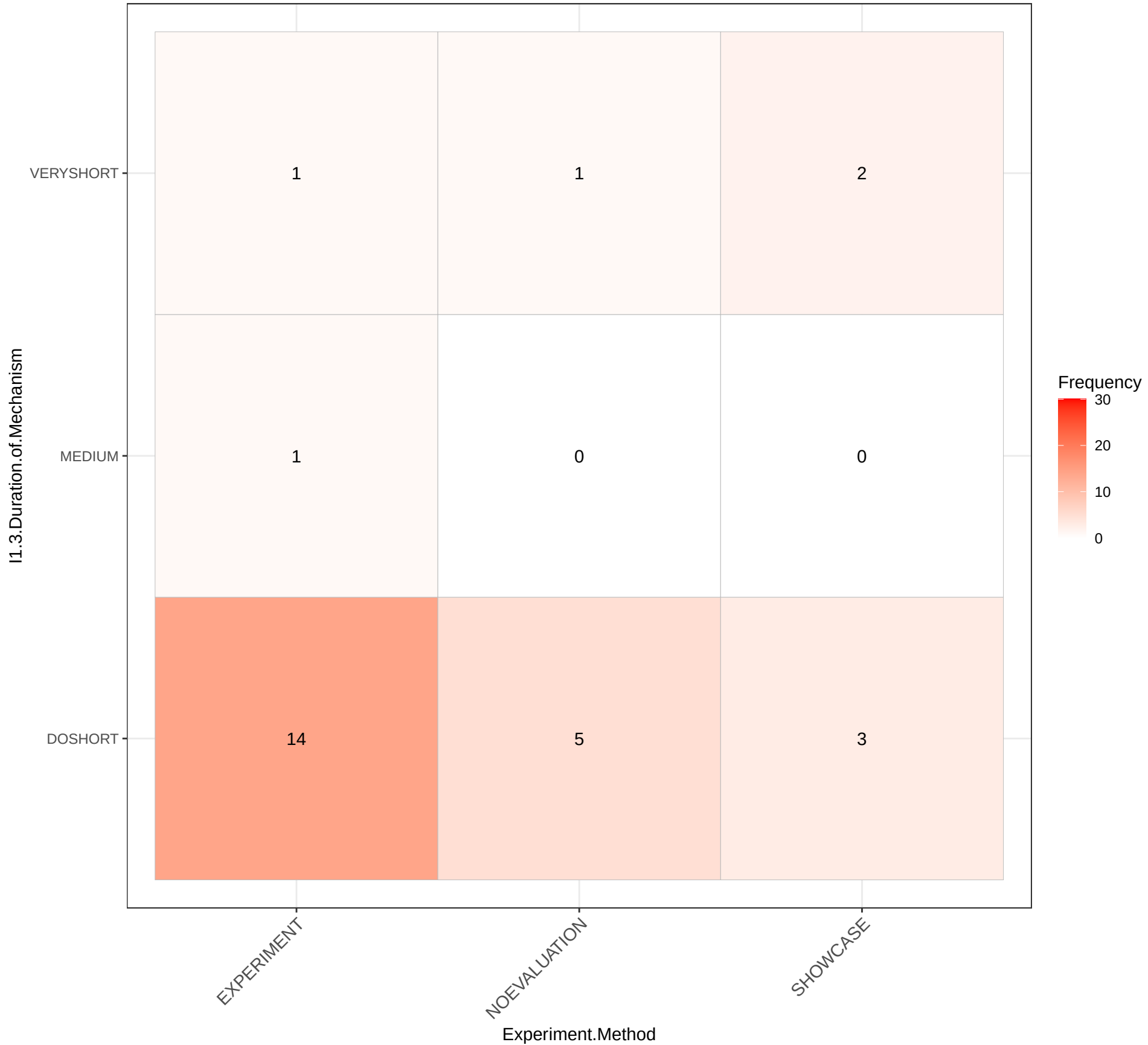
I1.3.Duration.of.Mechanism_____I7.Deployment.Realness



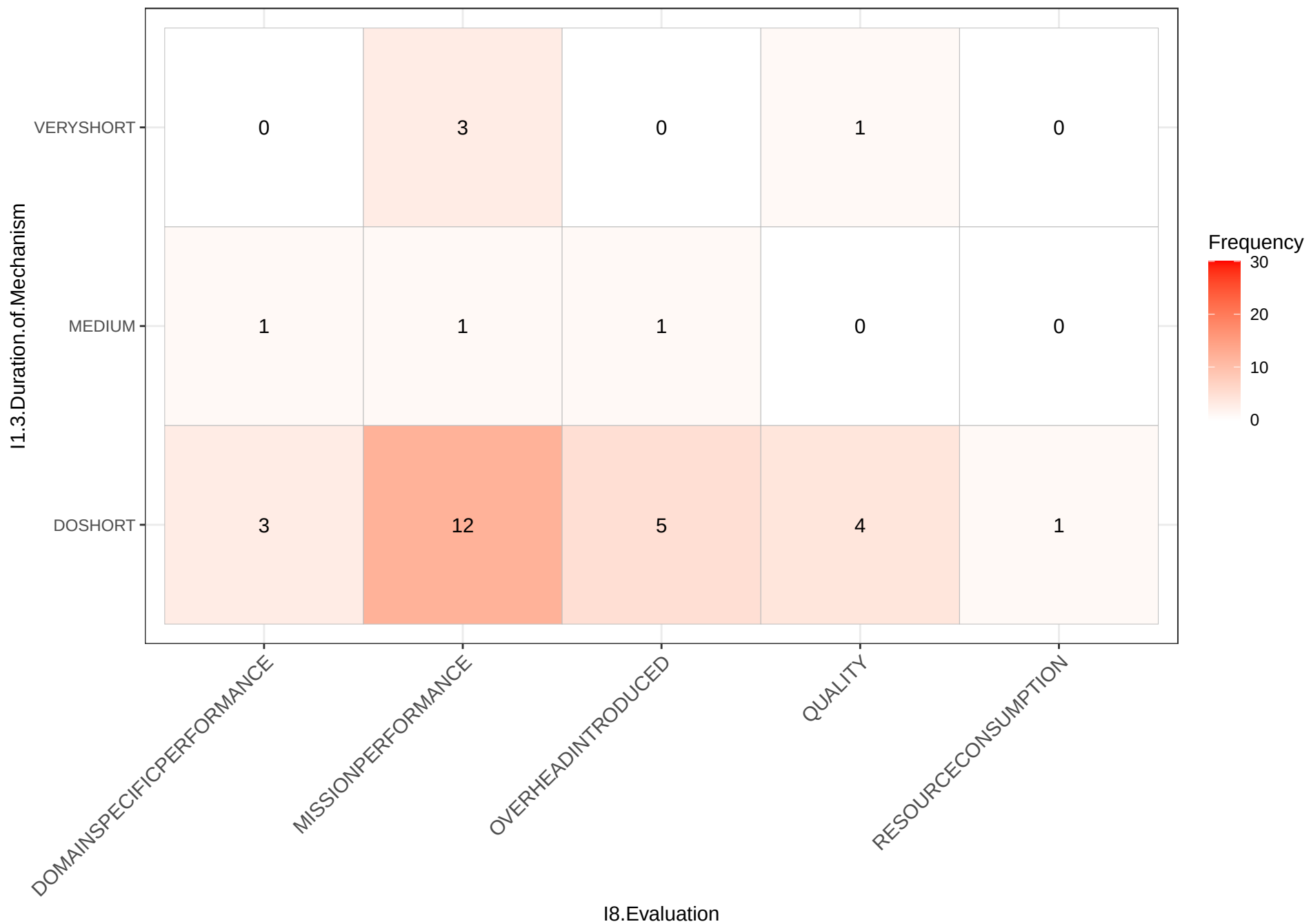
I1.3.Duration.of.Mechanism_____I7.Mission.Realness



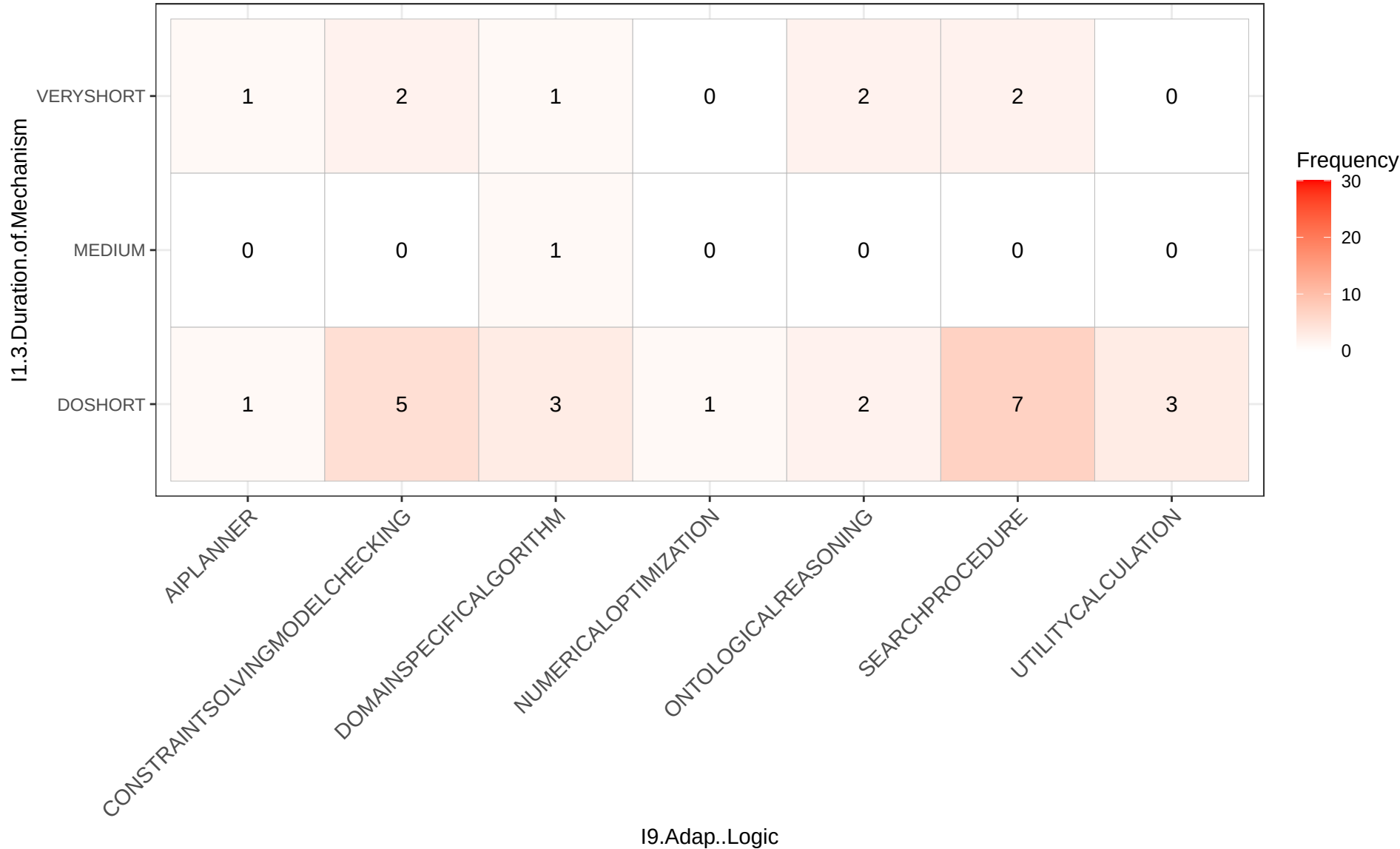
I1.3.Duration.of.Mechanism_____Experiment.Method



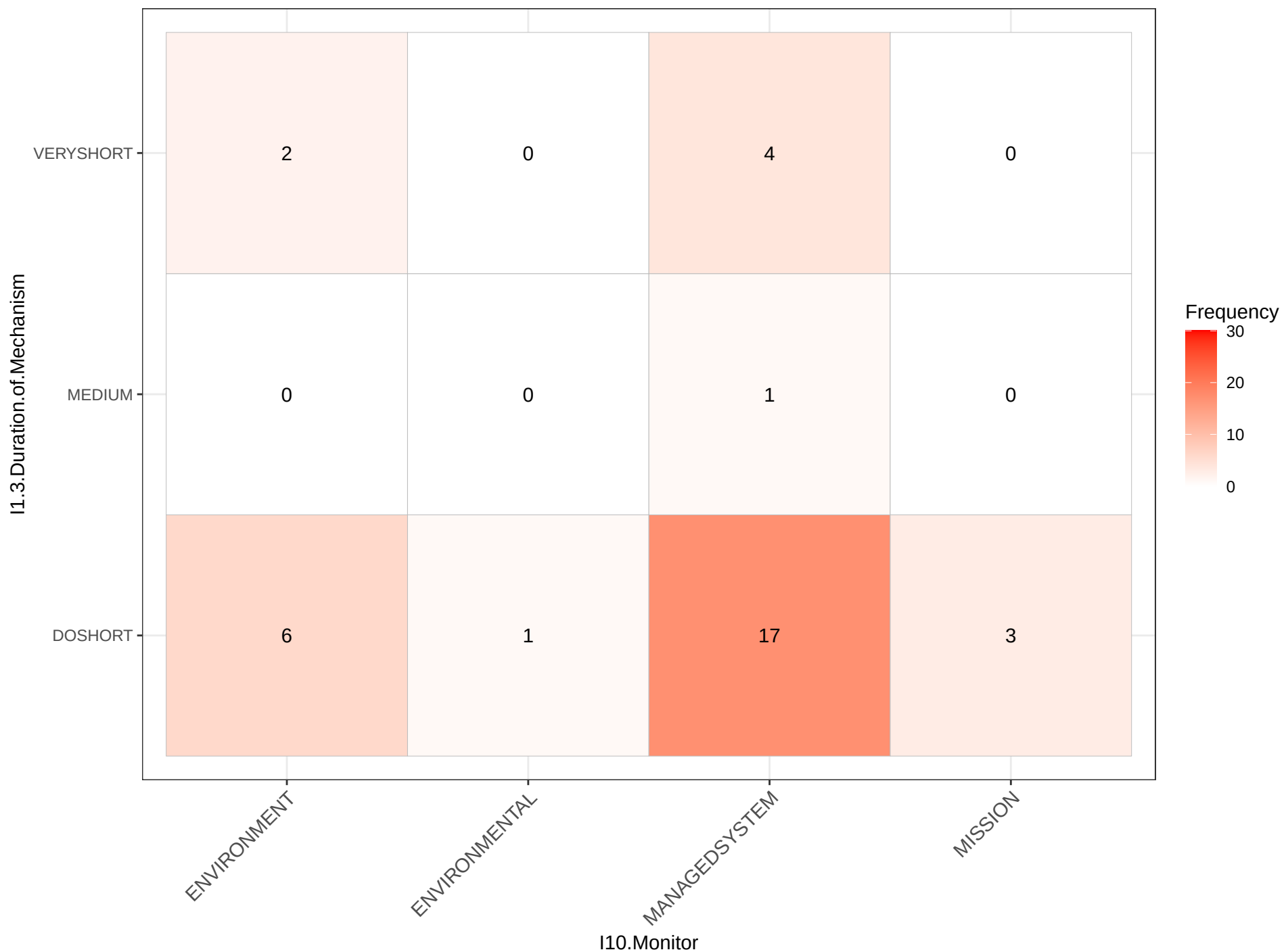
I1.3.Duration.of.Mechanism_____I8.Evaluation



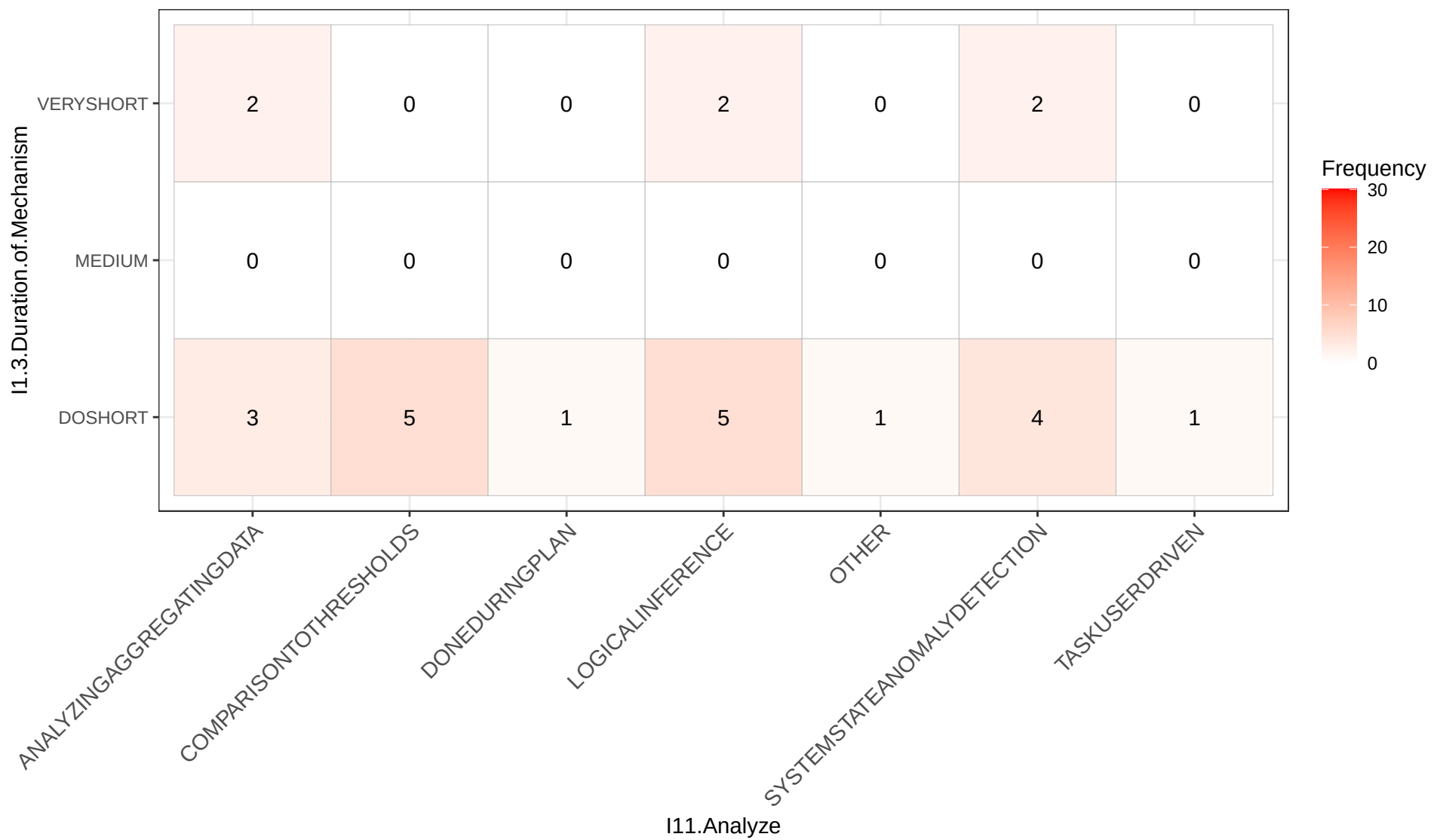
I1.3.Duration.of.Mechanism_____I9.Adap..Logic



I1.3.Duration.of.Mechanism_____I10.Monitor



I1.3.Duration.of.Mechanism_____I11.Analyze



I1.3.Duration.of.Mechanism

VERYSHORT

MEDIUM

DOSHORT

Frequency

30

20

10

0

1

2

1

0

1

0

11

8

1

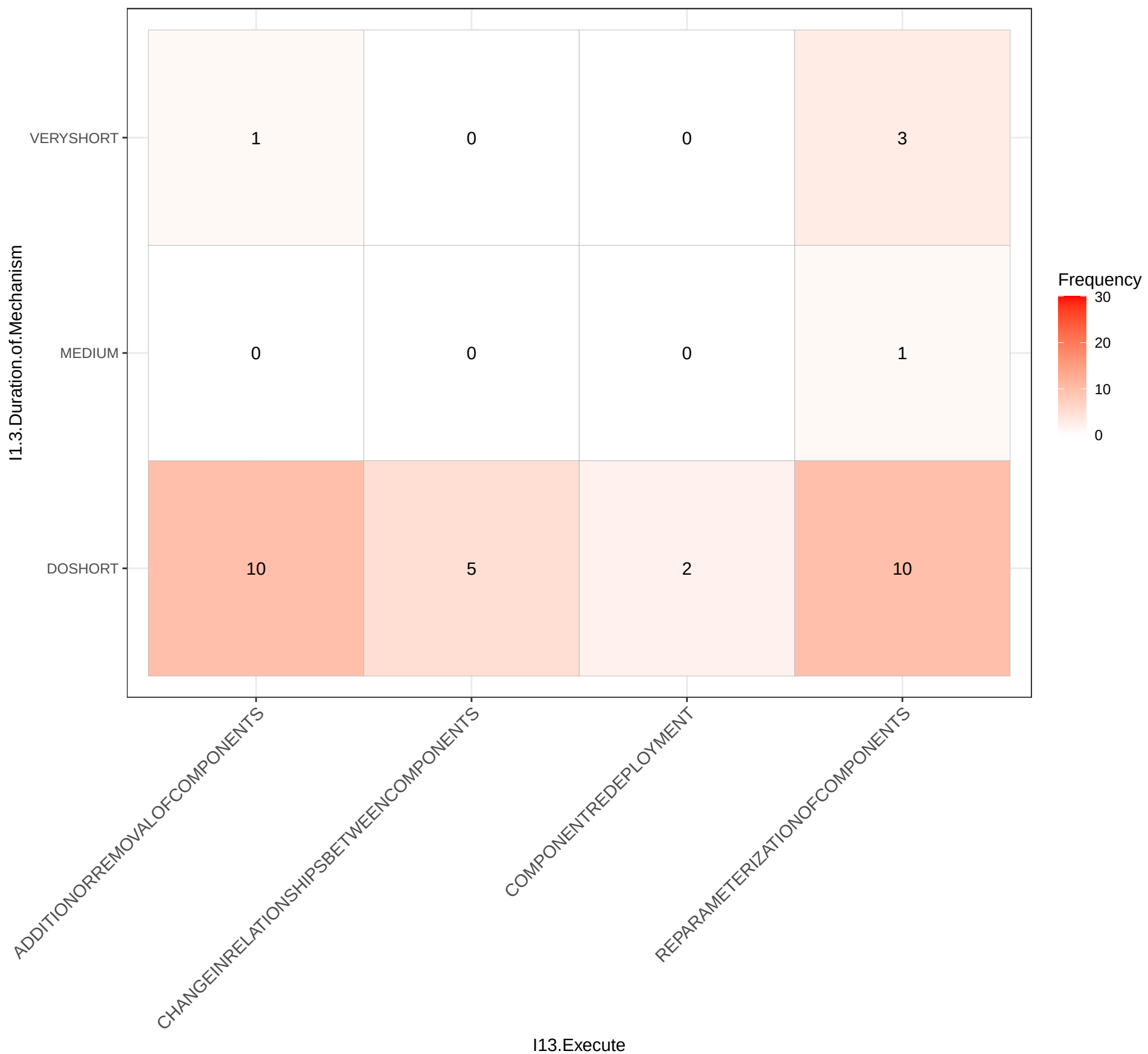
DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

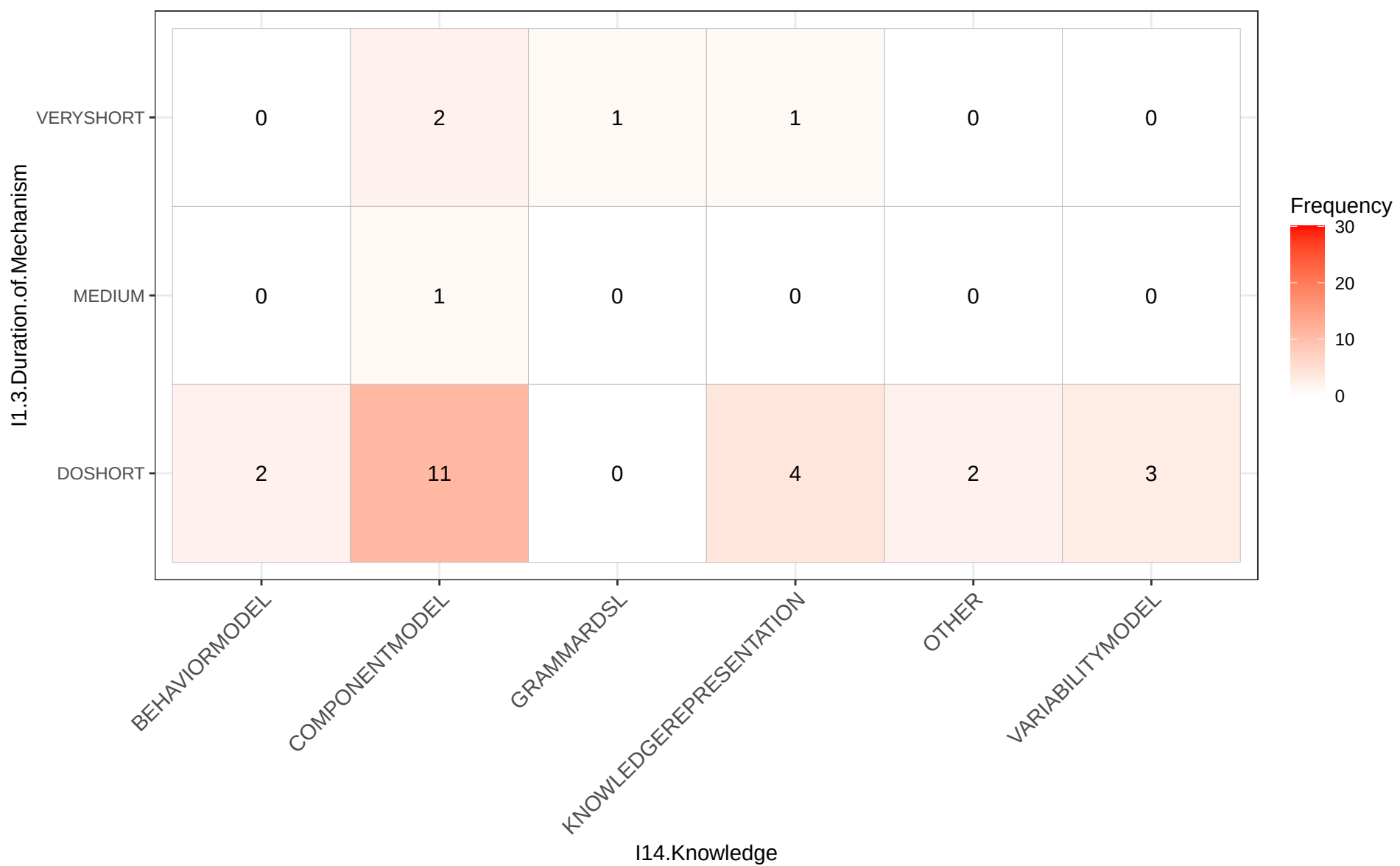
USINGAIPANNINGLANGUAGES

I12.Plan

I1.3.Duration.of.Mechanism_____I13.Execute



I1.3.Duration.of.Mechanism_____I14.Knowledge



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I1.3.Trigger.of.Mechanism

BESTEFFORT

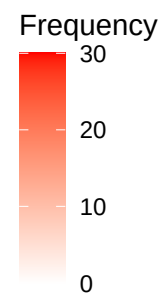
25

2

EVENTTRIGGER

TIMETRIGGER

I1.3.Trigger.of.Mechanism



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I1.4.Criticality.of.Effects

BESTEFFORT

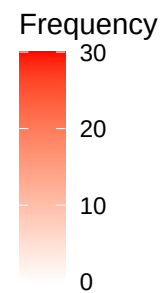
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6

MISSIONCRITICAL

SAFETYCRITICAL

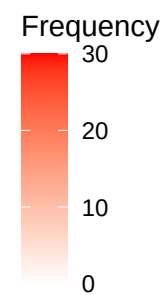
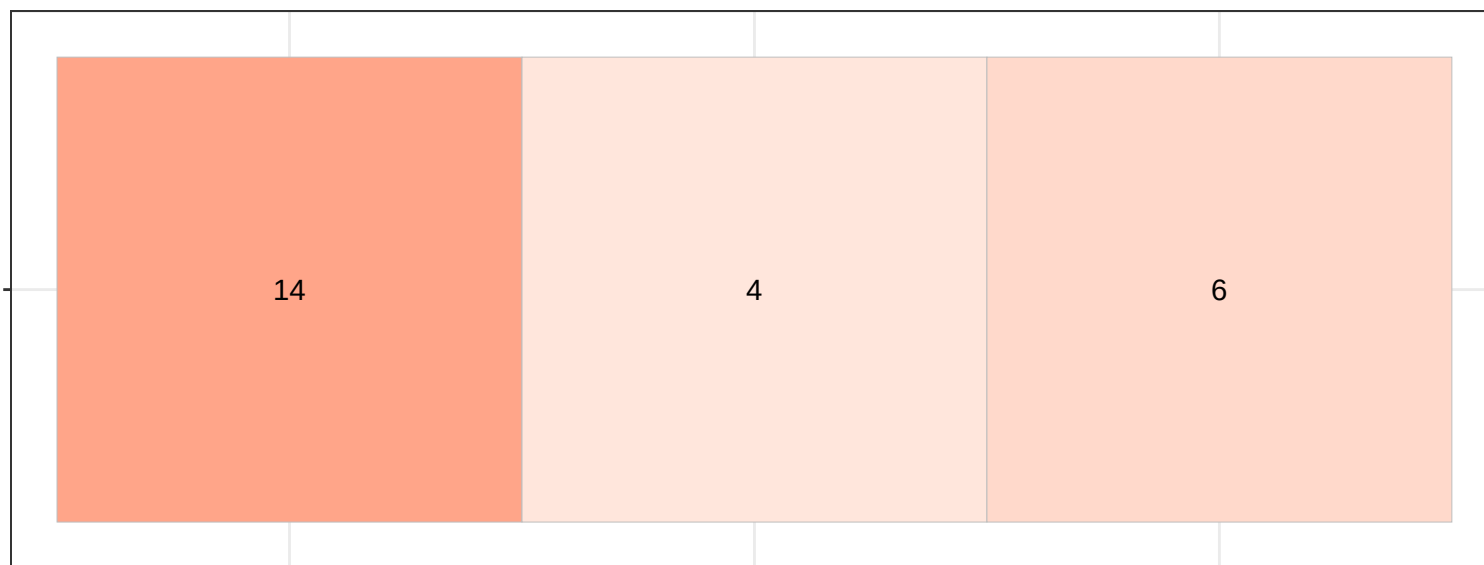
I1.4.Criticality.of.Effects



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I1.4.Overhead.of.Effects

BESTEFFORT

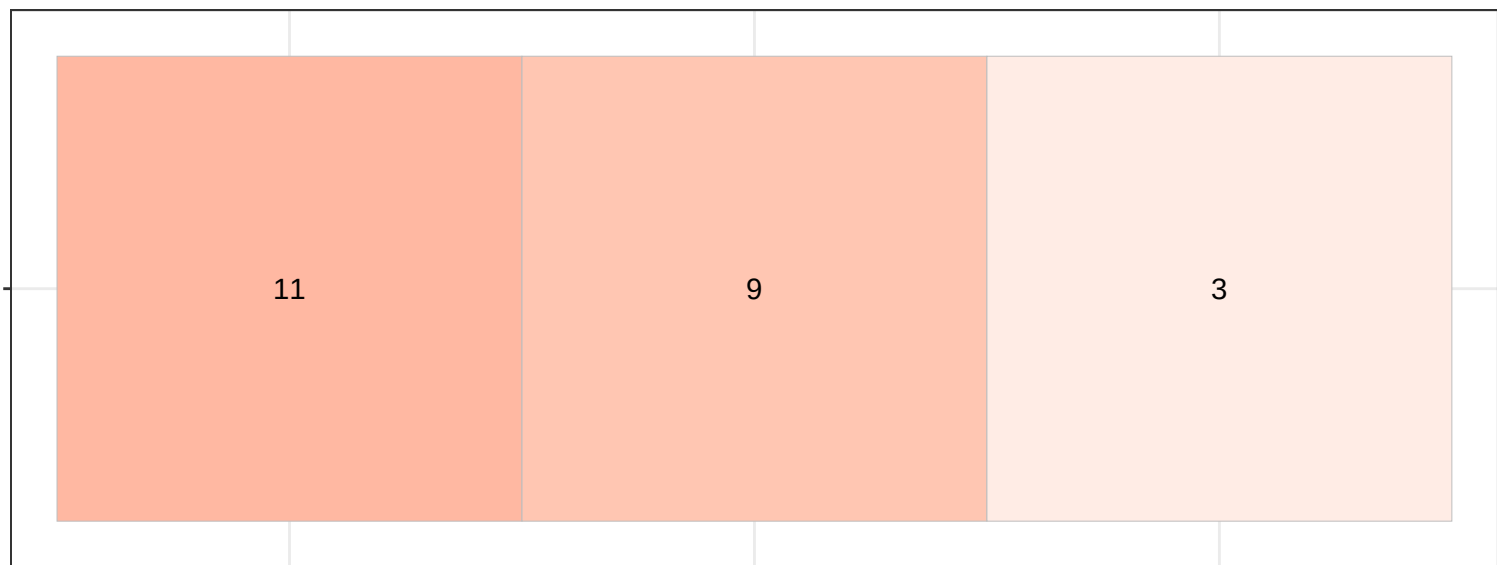


I1.4.Overhead.of.Effects

I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I1.4.Resilience.of.Effects

BESTEFFORT

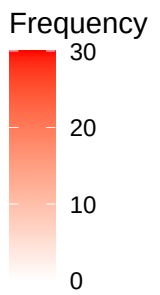


DEPENDENT

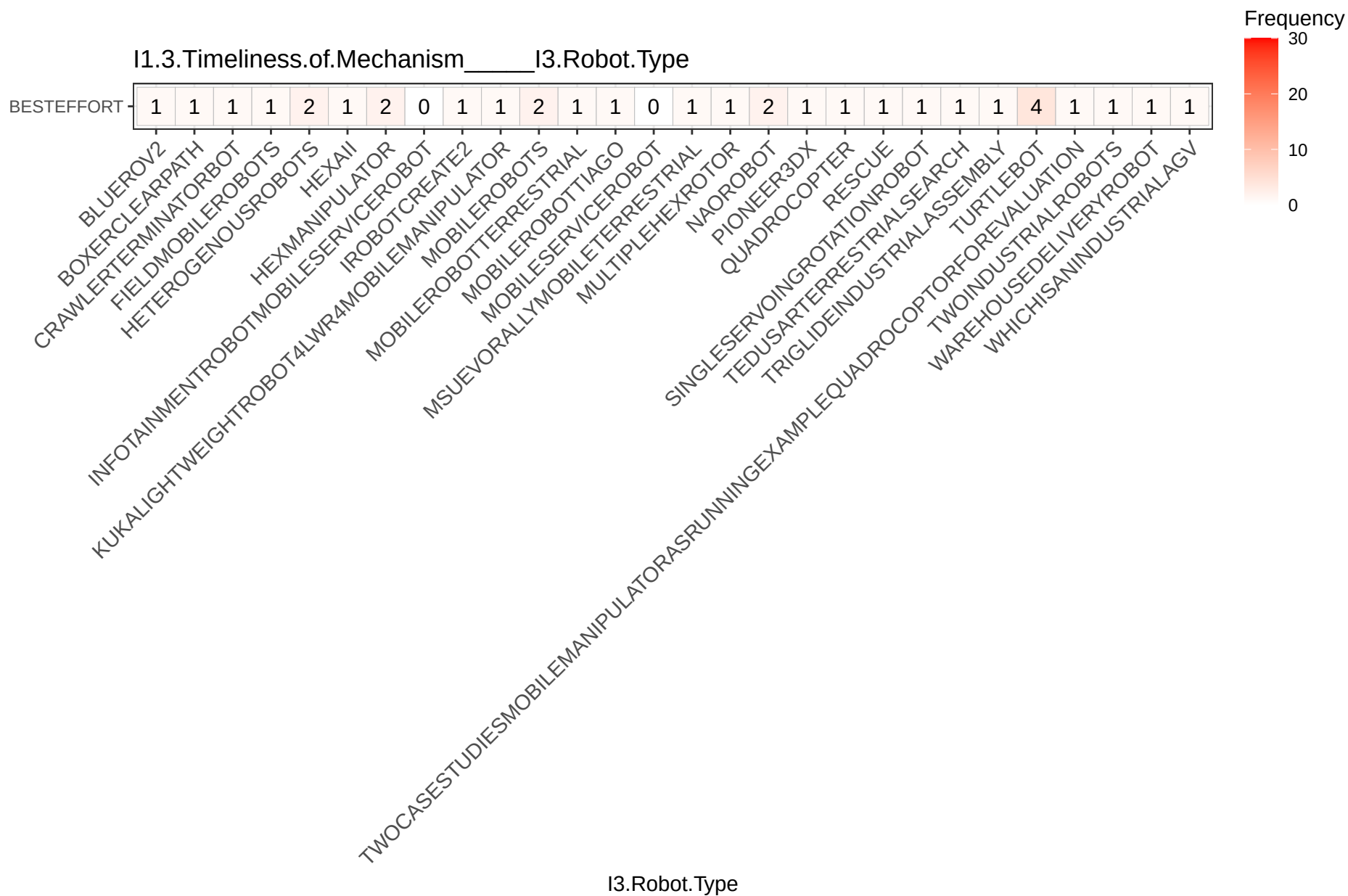
DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects



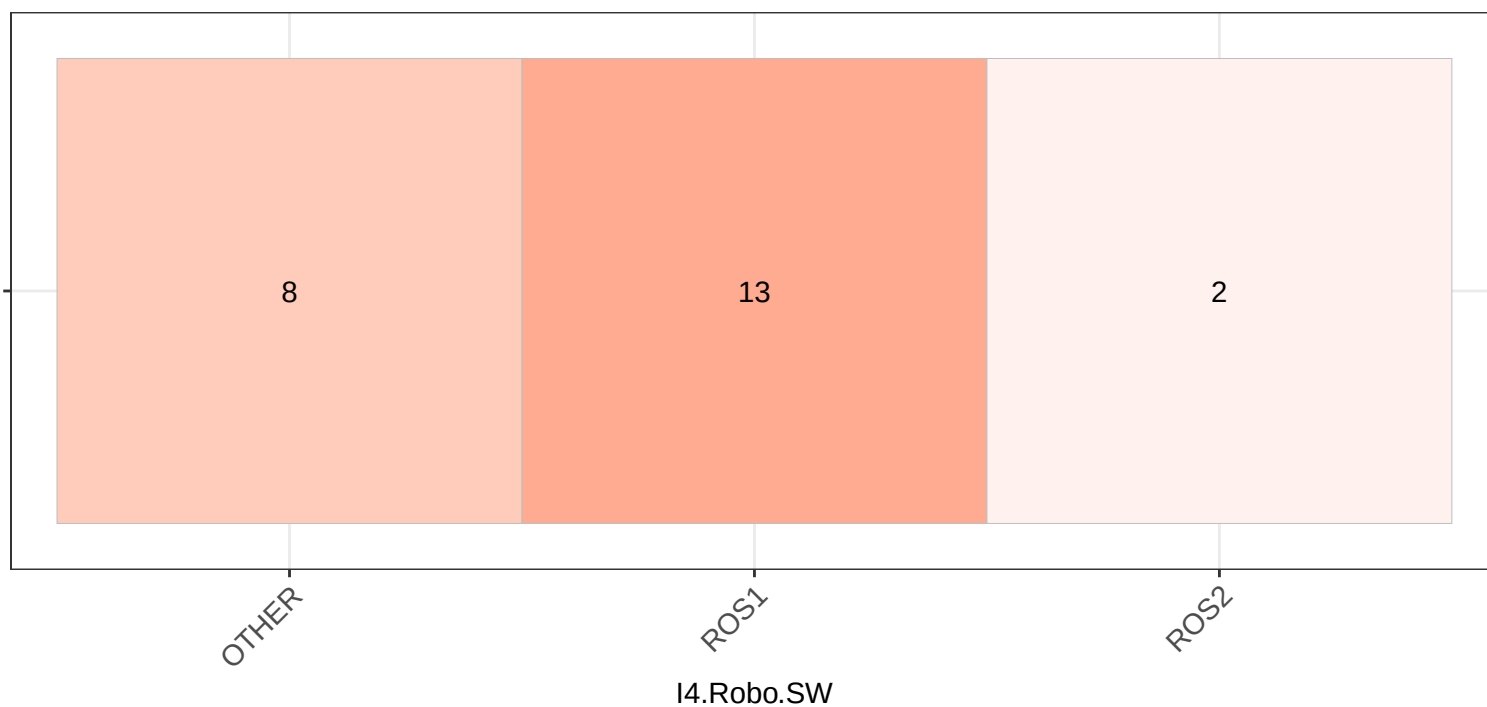
I1.3.Timeliness.of.Mechanism_____I3.Robot.Type



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I4.Robo.SW

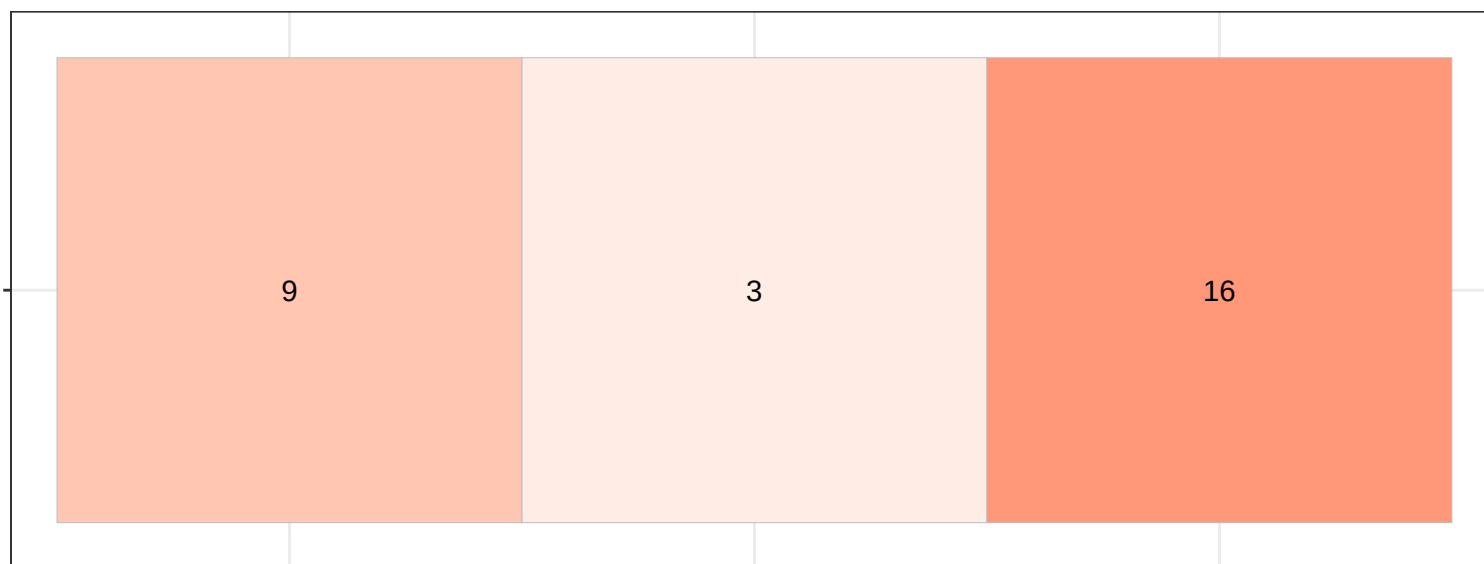
BESTEFFORT



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I6.Independence

BESTEFFORT



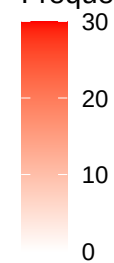
DETACHABLE

INSEPARABLE

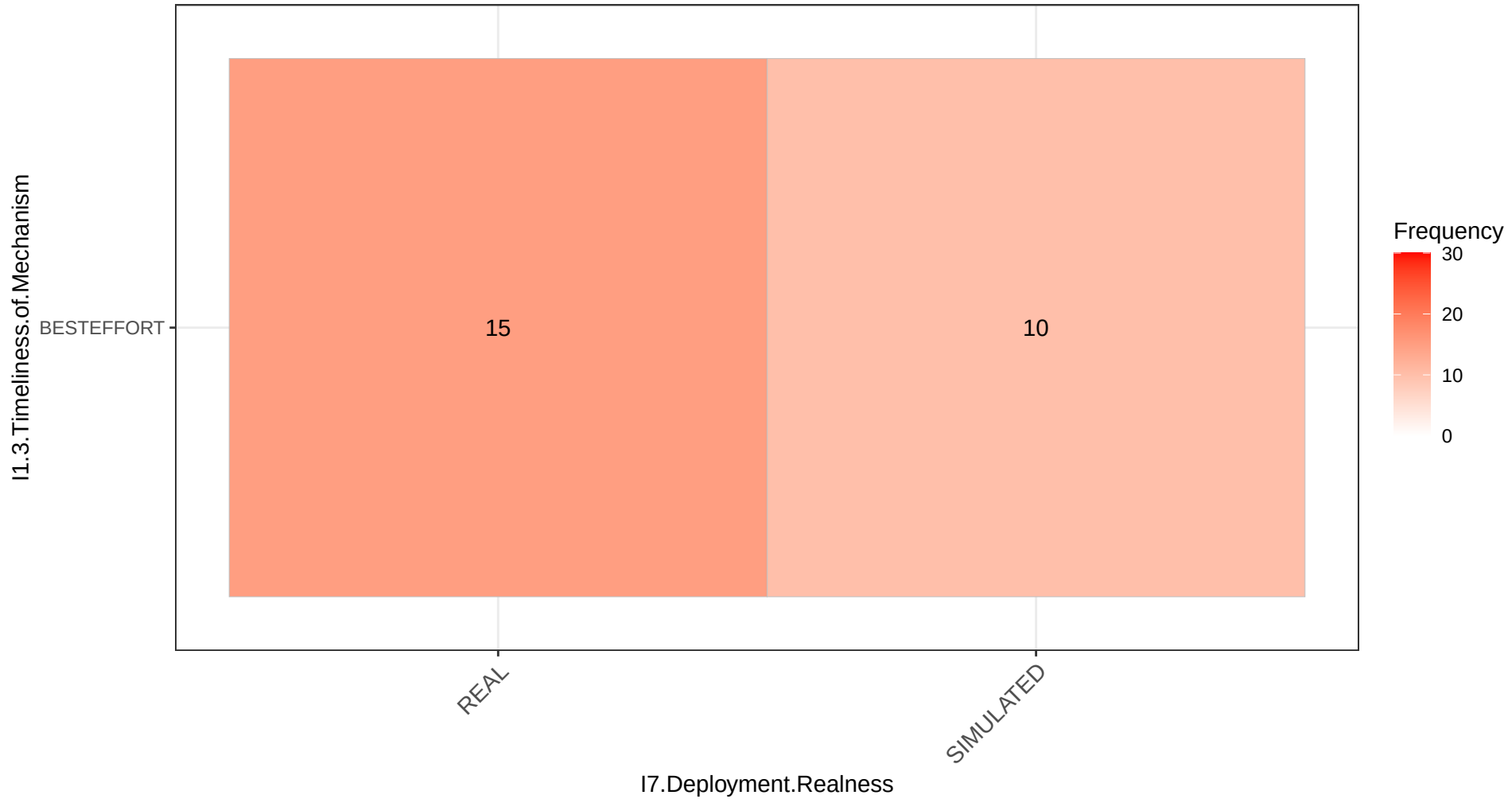
REQUIRESREPRESENTATION

I6.Independence

Frequency



I1.3.Timeliness.of.Mechanism_____I7.Deployment.Realness



I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I7.Mission.Realness

BESTEFFORT

REAL

SYNTHETIC

I7.Mission.Realness

15

13

Frequency

30

20

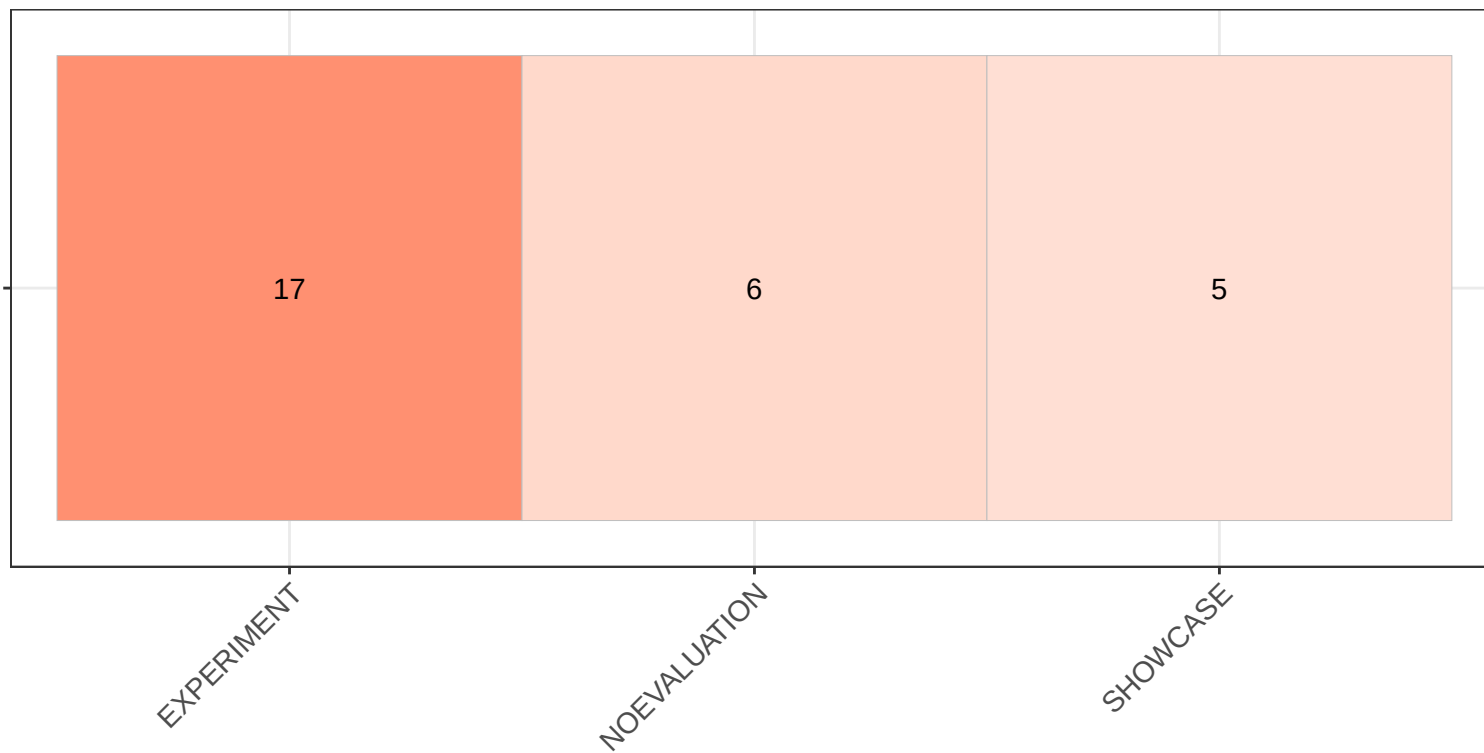
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I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____Experiment.Method

BESTEFFORT

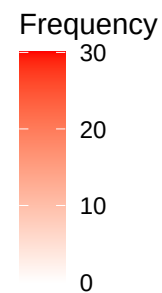
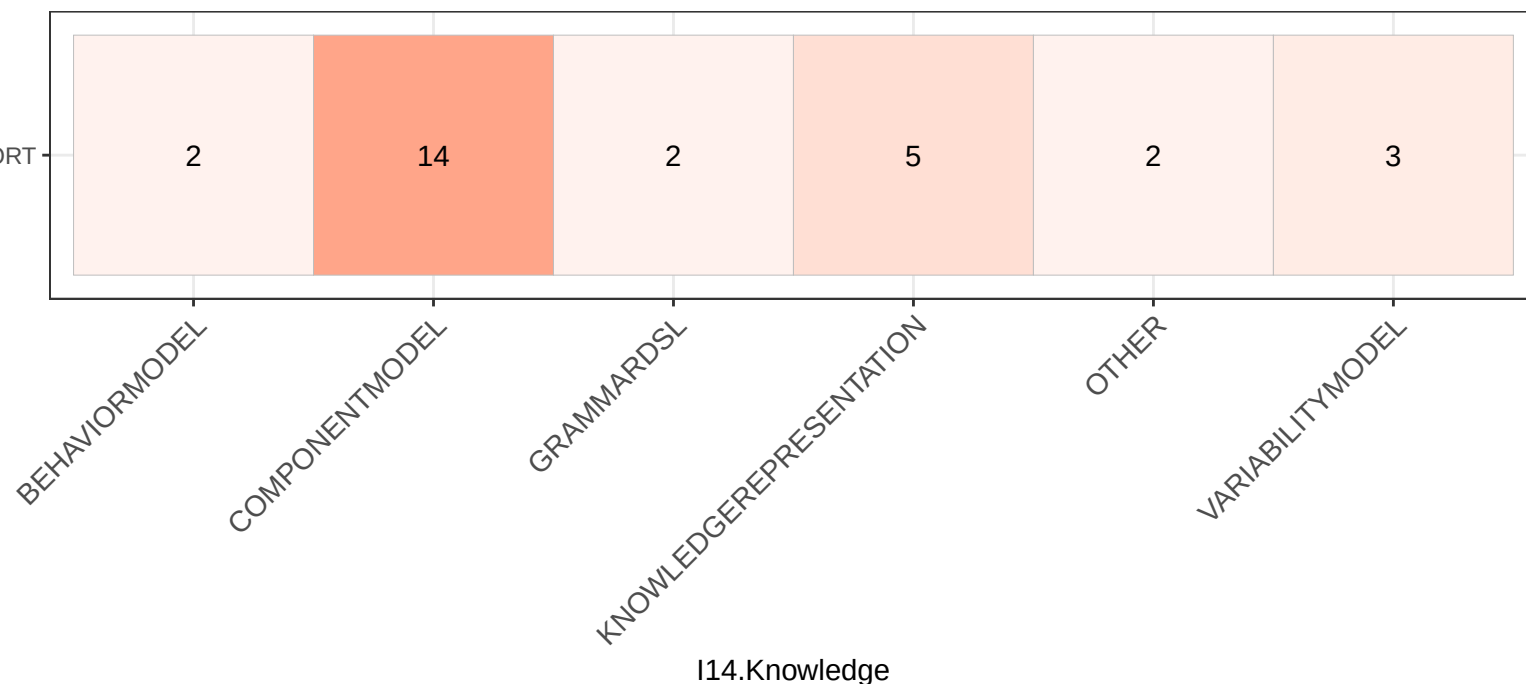


Experiment.Method

I1.3.Timeliness.of.Mechanism

I1.3.Timeliness.of.Mechanism_____I14.Knowledge

BESTEFFORT



I1.3.Trigger.of.Mechanism_____I1.4.Criticality.of.Effects

I1.3.Trigger.of.Mechanism

TIMETRIGGER

2

0

EVENTTRIGGER

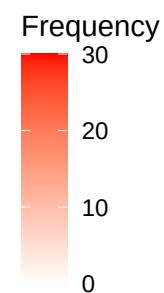
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5

MISSIONCRITICAL

SAFETYCRITICAL

I1.4.Criticality.of.Effects



I1.3.Trigger.of.Mechanism_____I1.4.Predictability.of.Effects

I1.3.Trigger.of.Mechanism

TIMETRIGGER

2

2

EVENTTRIGGER

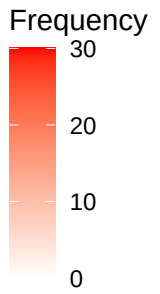
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17

DODETERMINISTIC

NONDETERMINISTIC

I1.4.Predictability.of.Effects



I1.3.Trigger.of.Mechanism_____I1.4.Overhead.of.Effects

I1.3.Trigger.of.Mechanism

TIMETRIGGER

2

0

0

EVENTTRIGGER

13

3

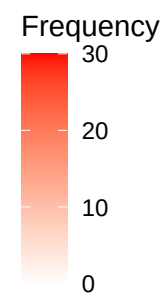
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DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.3.Trigger.of.Mechanism_____I1.4.Resilience.of.Effects

I1.3.Trigger.of.Mechanism

TIMETRIGGER

2

0

0

EVENTTRIGGER

10

8

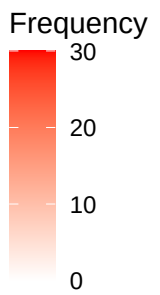
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DEPENDENT

DORESILIENT

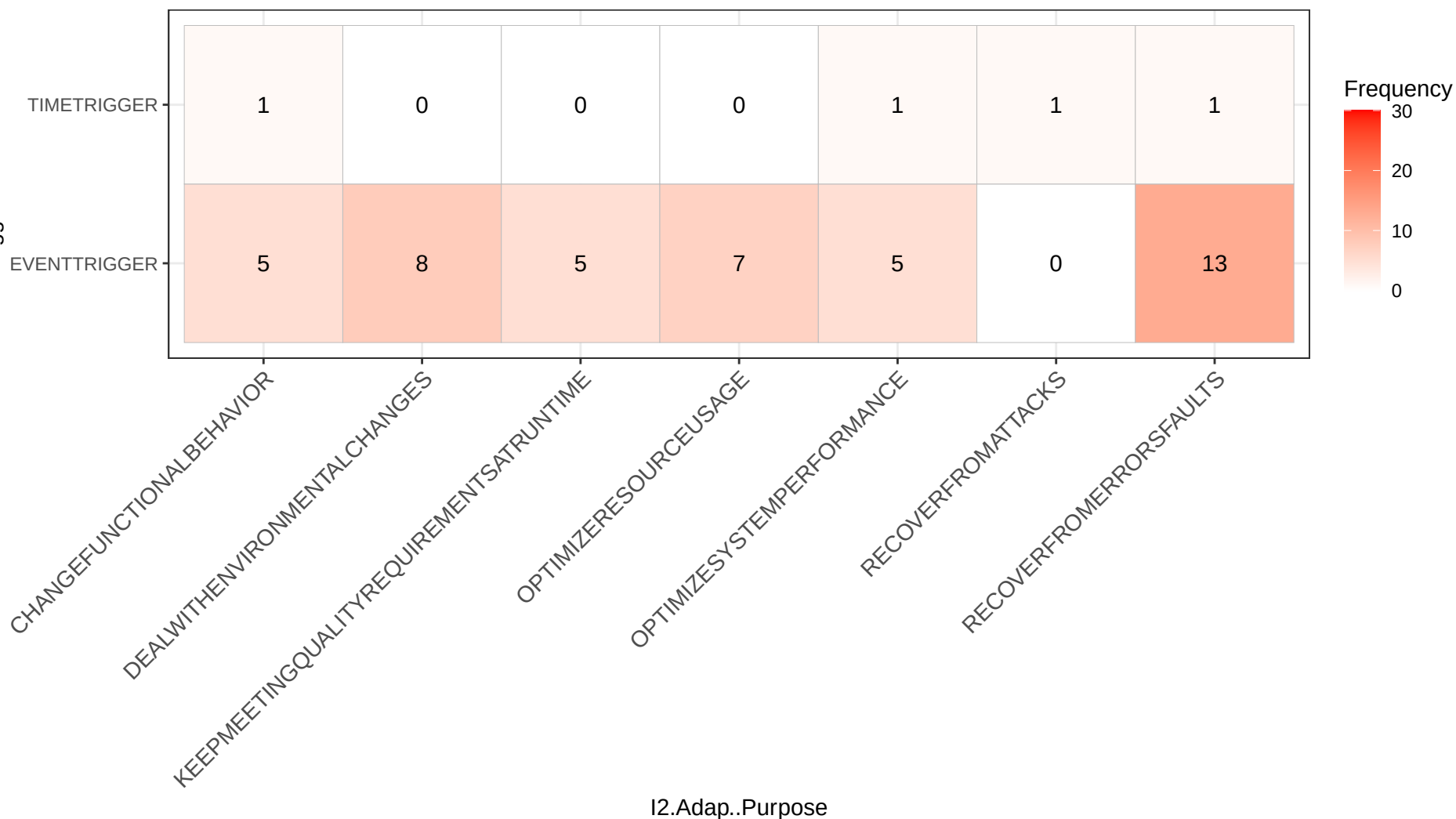
IRRESILIENT

I1.4.Resilience.of.Effects



I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I2.Adap..Purpose



I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I3.Robot.Type

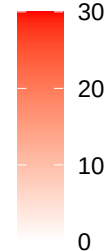
TIMETRIGGER
EVENTTRIGGER

0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
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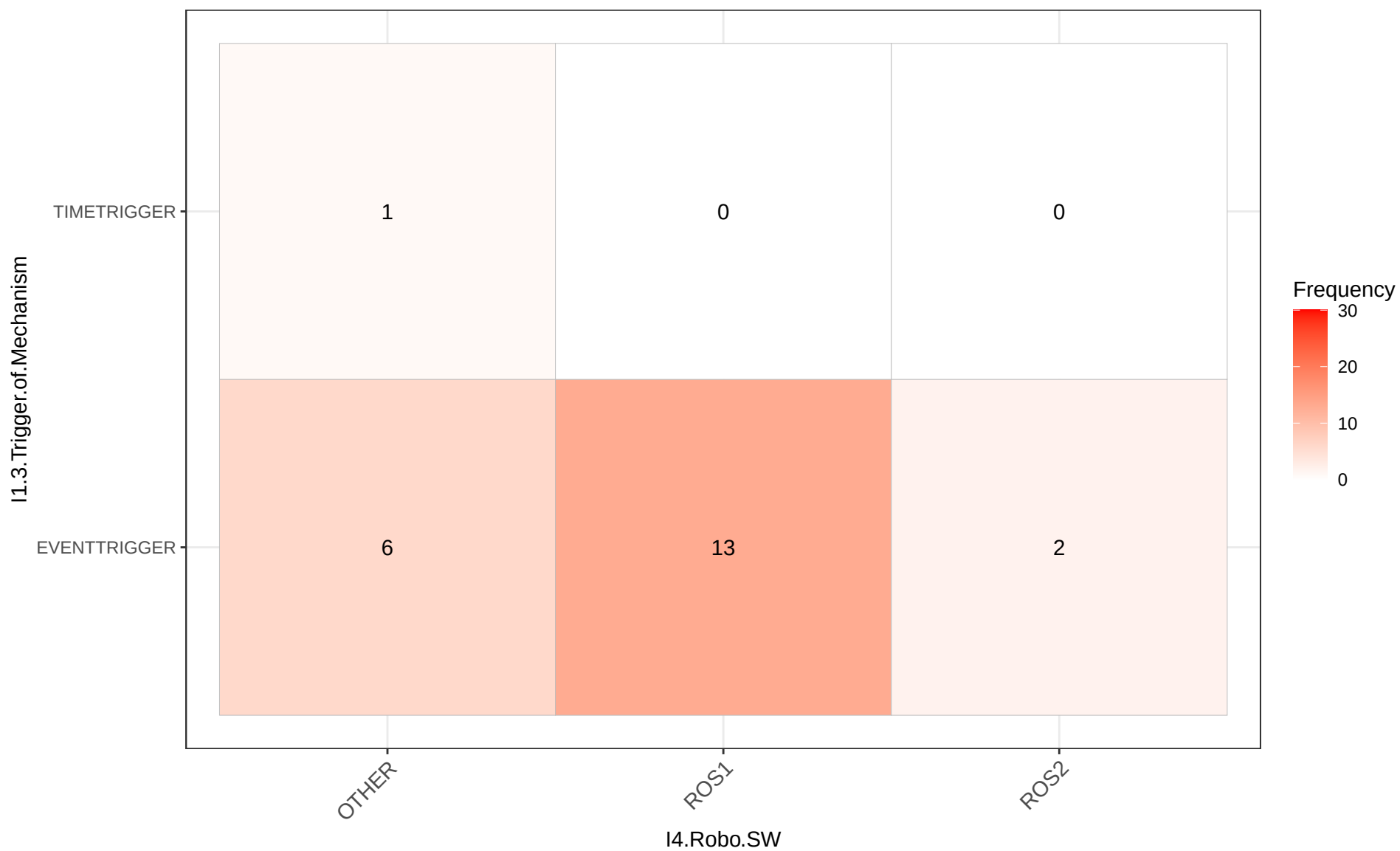
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOTS
HETEROGENOUSROBOTS
HEXAII
HEXMANIPULATOR
IROBOTSERVICEROBOT
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MULTIPLEHEXROTOR
NAOROBOT
PIONEER3DX
QUADROCOPTER
RESCUE
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TURTLEBOT
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

I3.Robot.Type

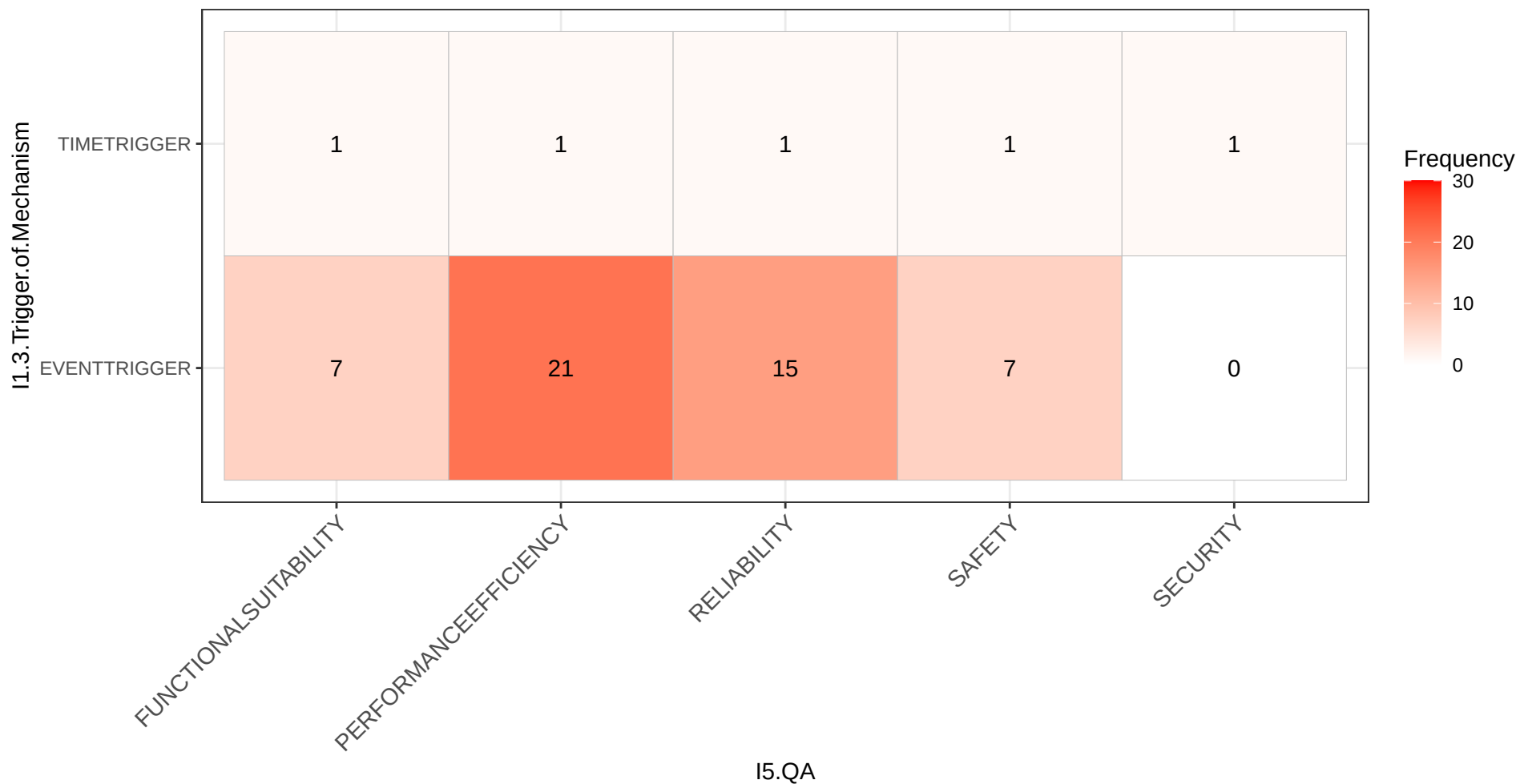
Frequency



I1.3.Trigger.of.Mechanism_____I4.Robo.SW



I1.3.Trigger.of.Mechanism_____I5.QA



I5.QA

I1.3.Trigger.of.Mechanism_____I6.Independence

I1.3.Trigger.of.Mechanism

TIMETRIGGER

1

1

0

EVENTTRIGGER

8

1

18

DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency

30

20

10

0

I1.3.Trigger.of.Mechanism_____I7.Deployment.Realness

I1.3.Trigger.of.Mechanism

TIMETRIGGER

EVENTTRIGGER

REAL

SIMULATED

I7.Deployment.Realness

Frequency

30

20

10

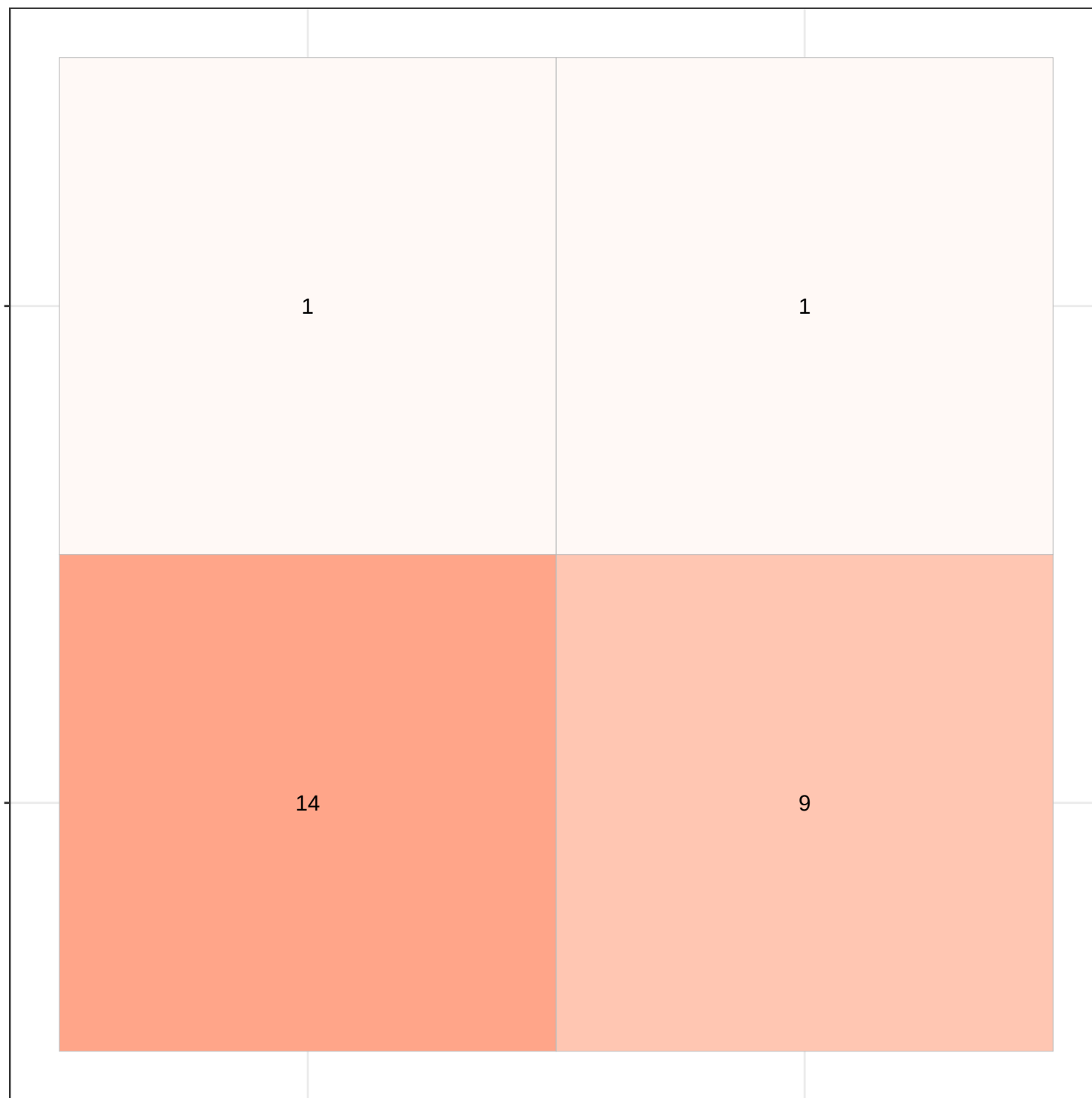
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1

1

14

9



I1.3.Trigger.of.Mechanism_____I7.Mission.Realness

I1.3.Trigger.of.Mechanism

TIMETRIGGER

EVENTTRIGGER

REAL

SYNTHETIC

I7.Mission.Realness

Frequency

30

20

10

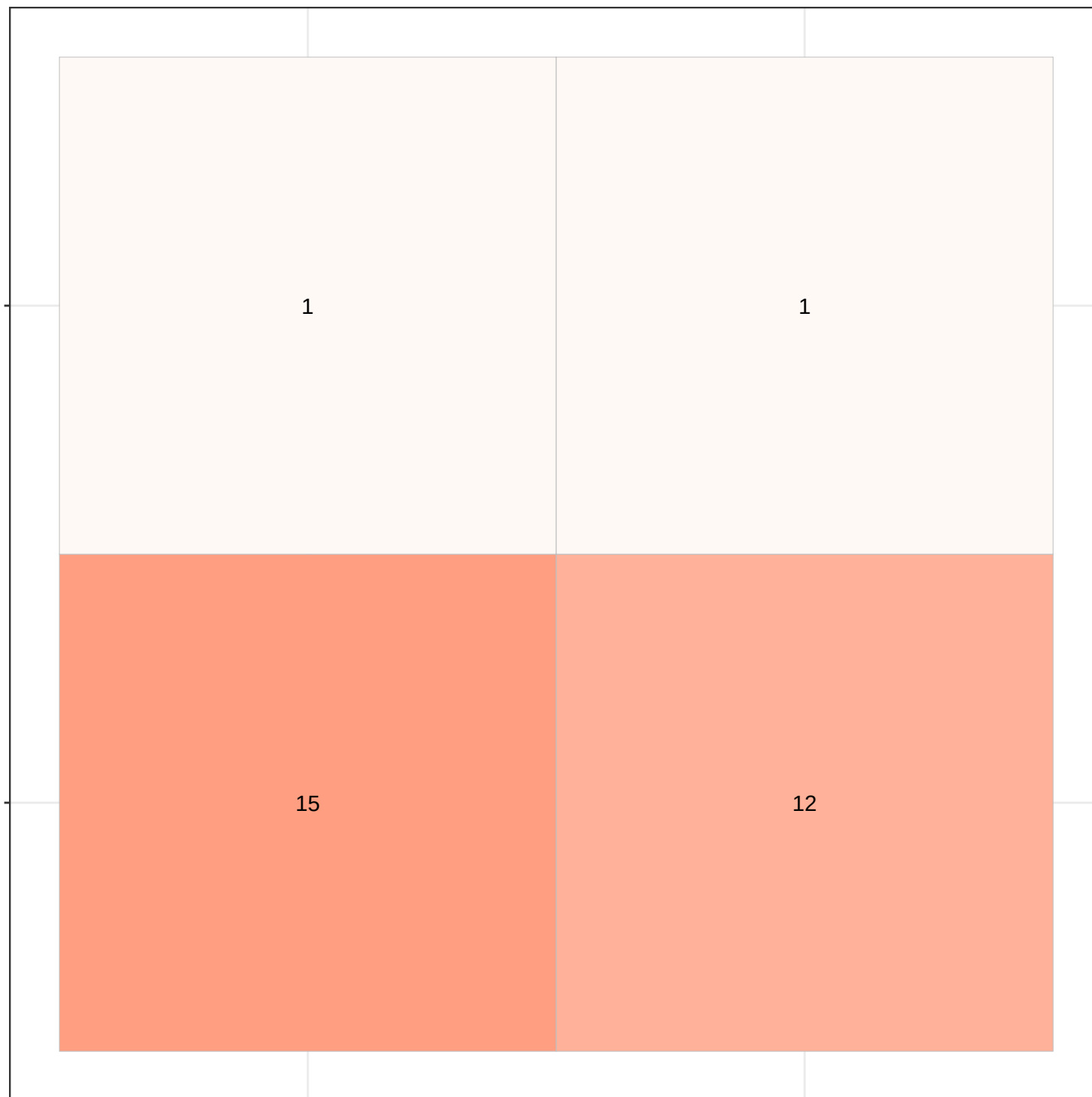
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1

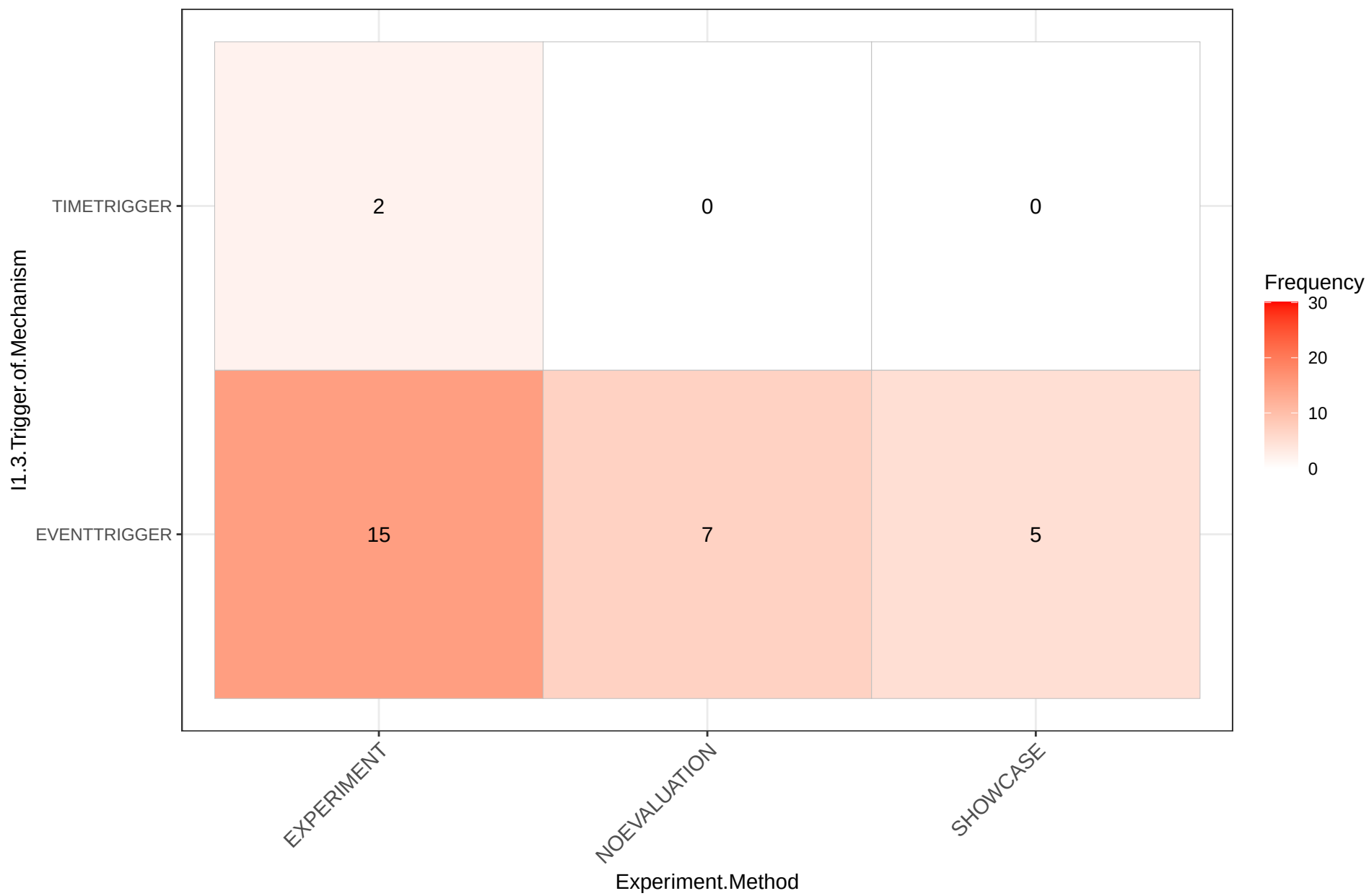
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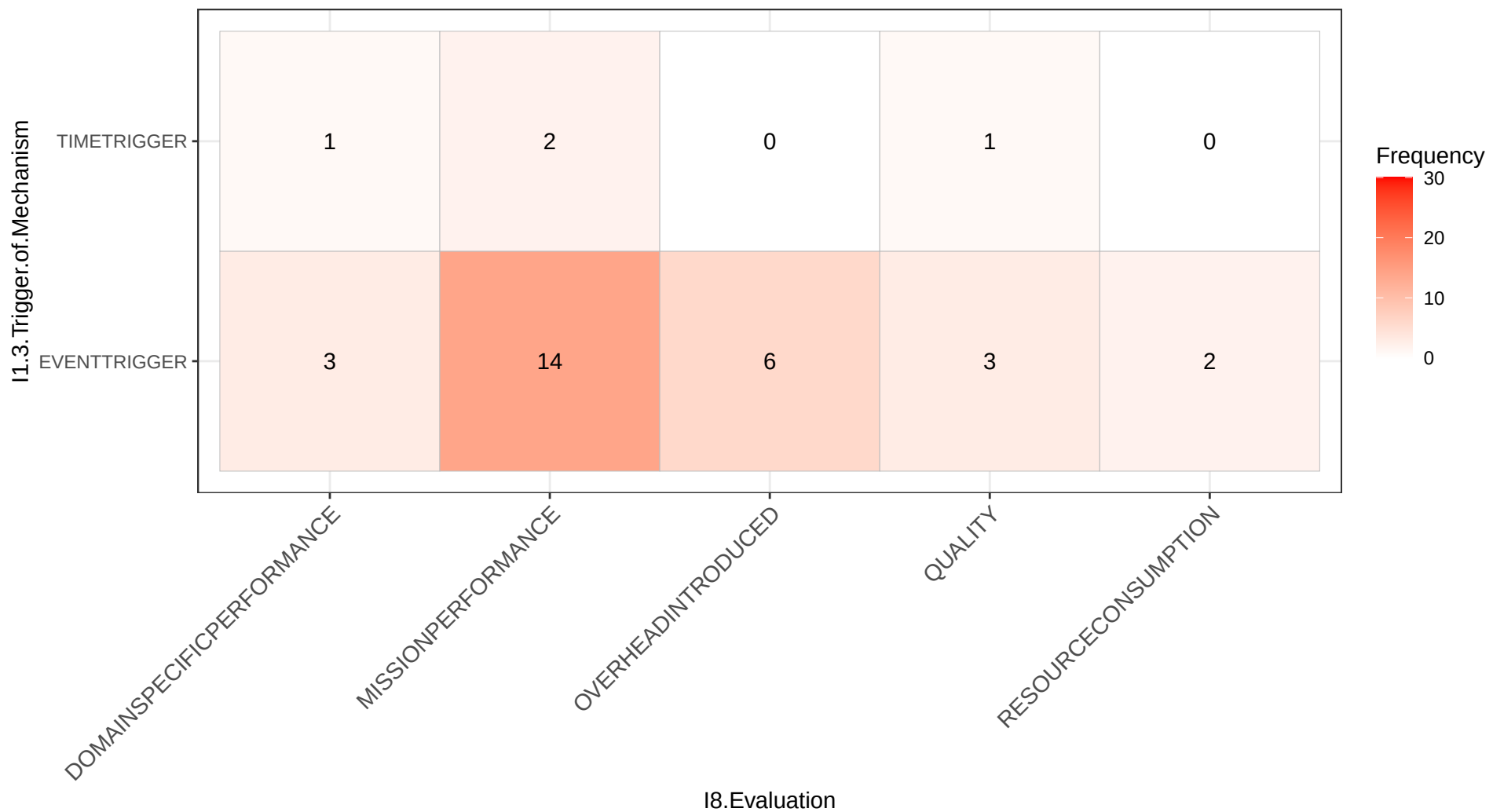
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I1.3.Trigger.of.Mechanism_____Experiment.Method

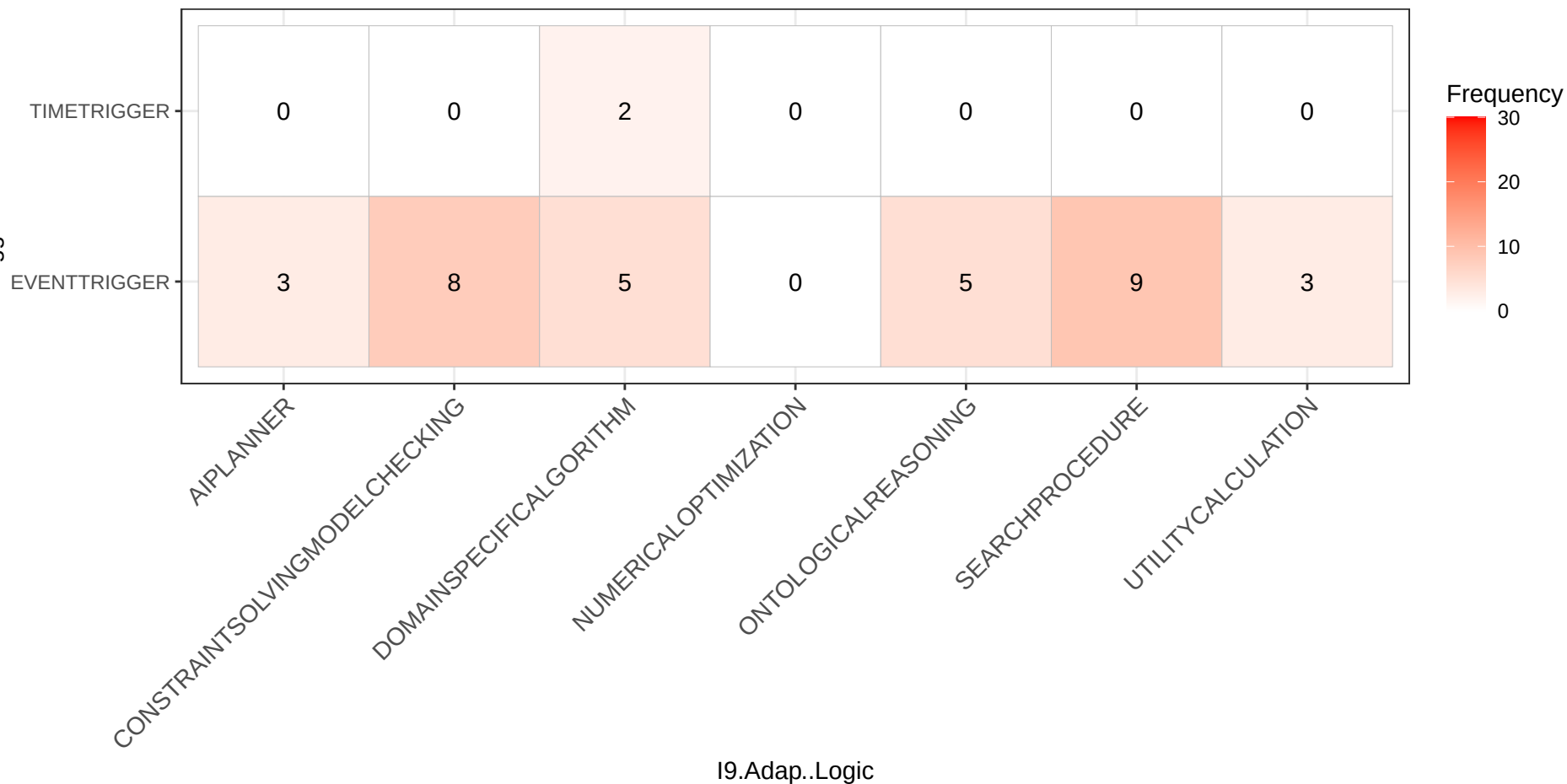


I1.3.Trigger.of.Mechanism_____I8.Evaluation

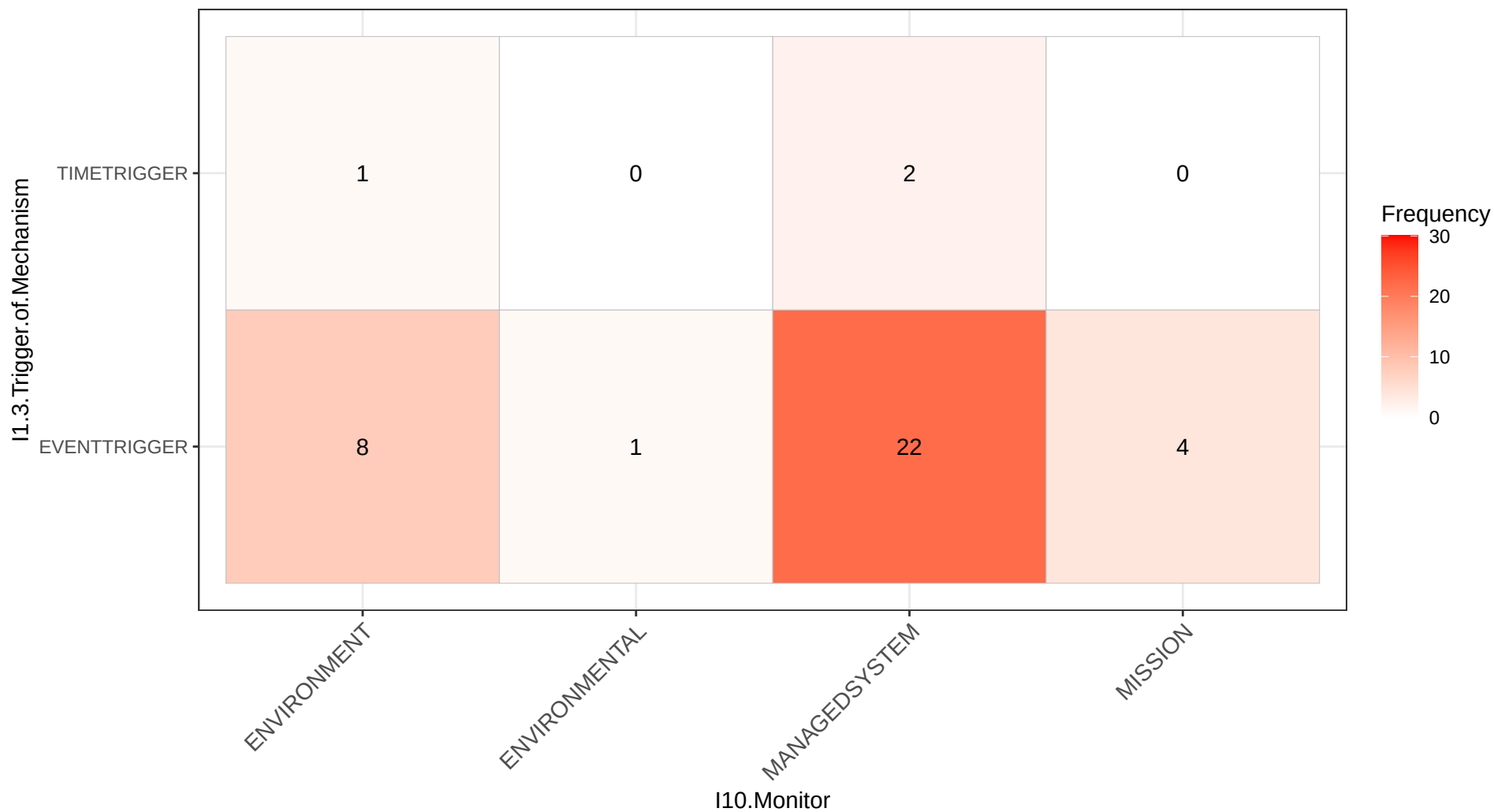


I1.3. Trigger.of.Mechanism

I1.3. Trigger.of.Mechanism_____I9.Adap..Logic

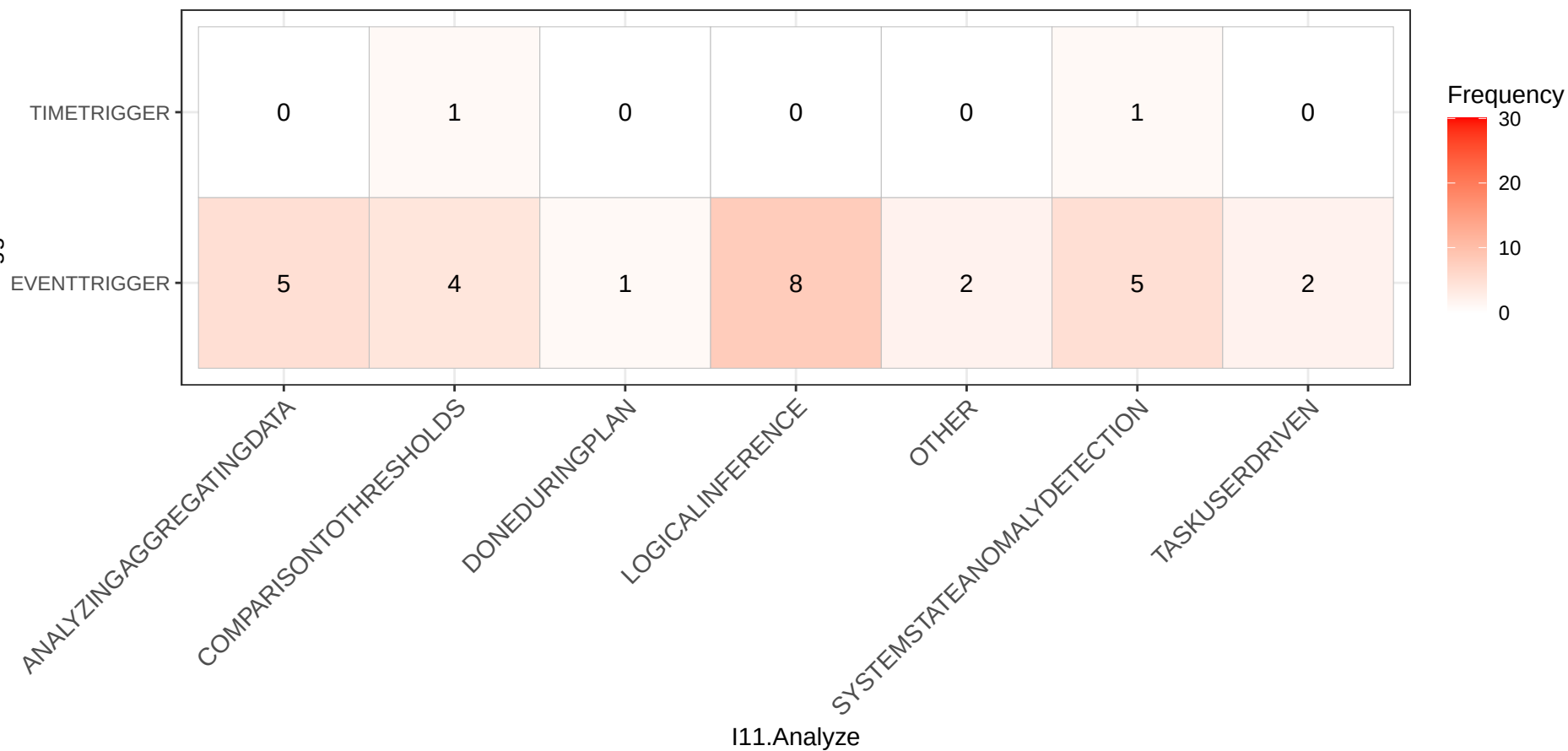


I1.3.Trigger.of.Mechanism_____I10.Monitor



I1.3.Trigger.of.Mechanism

I1.3.Trigger.of.Mechanism_____I11.Analyze



I1.3.Trigger.of.Mechanism_____I12.Plan

I1.3.Trigger.of.Mechanism

TIMETRIGGER

0

2

0

EVENTTRIGGER

13

10

2

DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency

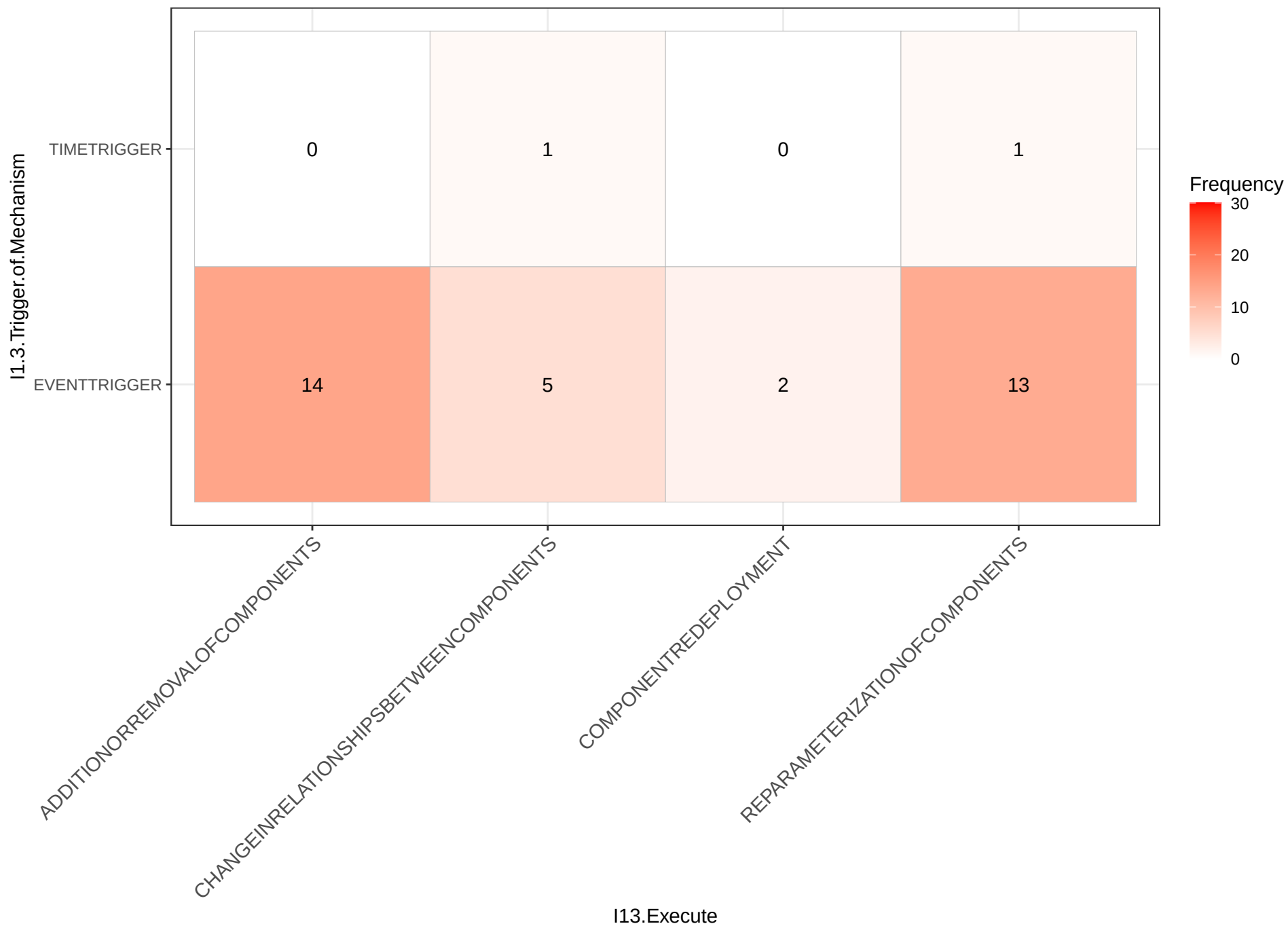
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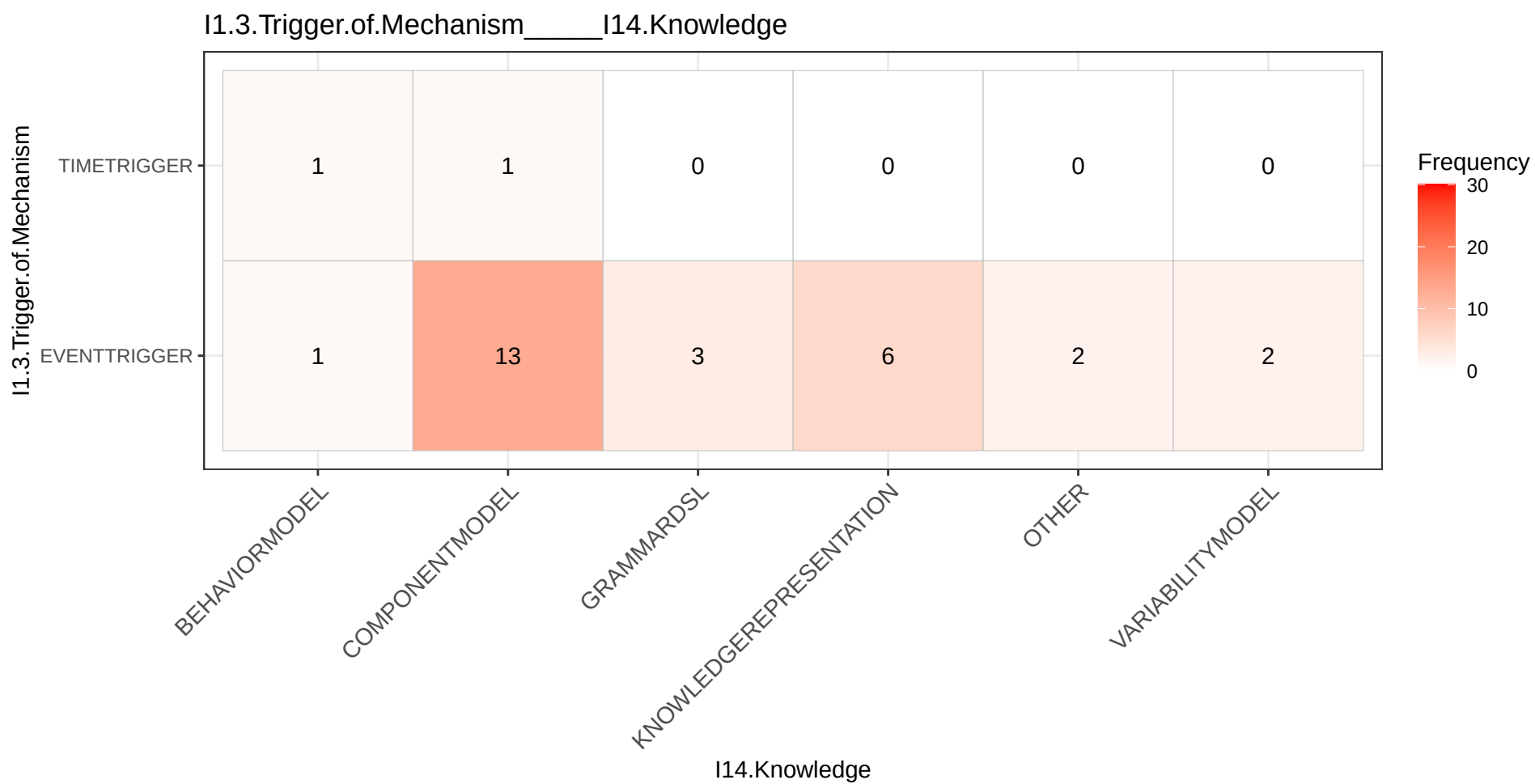
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10

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I1.3.Trigger.of.Mechanism____I13.Execute





I1.4.Criticality.of.Effects_____I1.4.Predictability.of.Effects

I1.4.Criticality.of.Effects

SAFETYCRITICAL

2

4

MISSIONCRITICAL

7

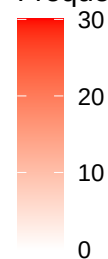
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DO
DETERMINISTIC

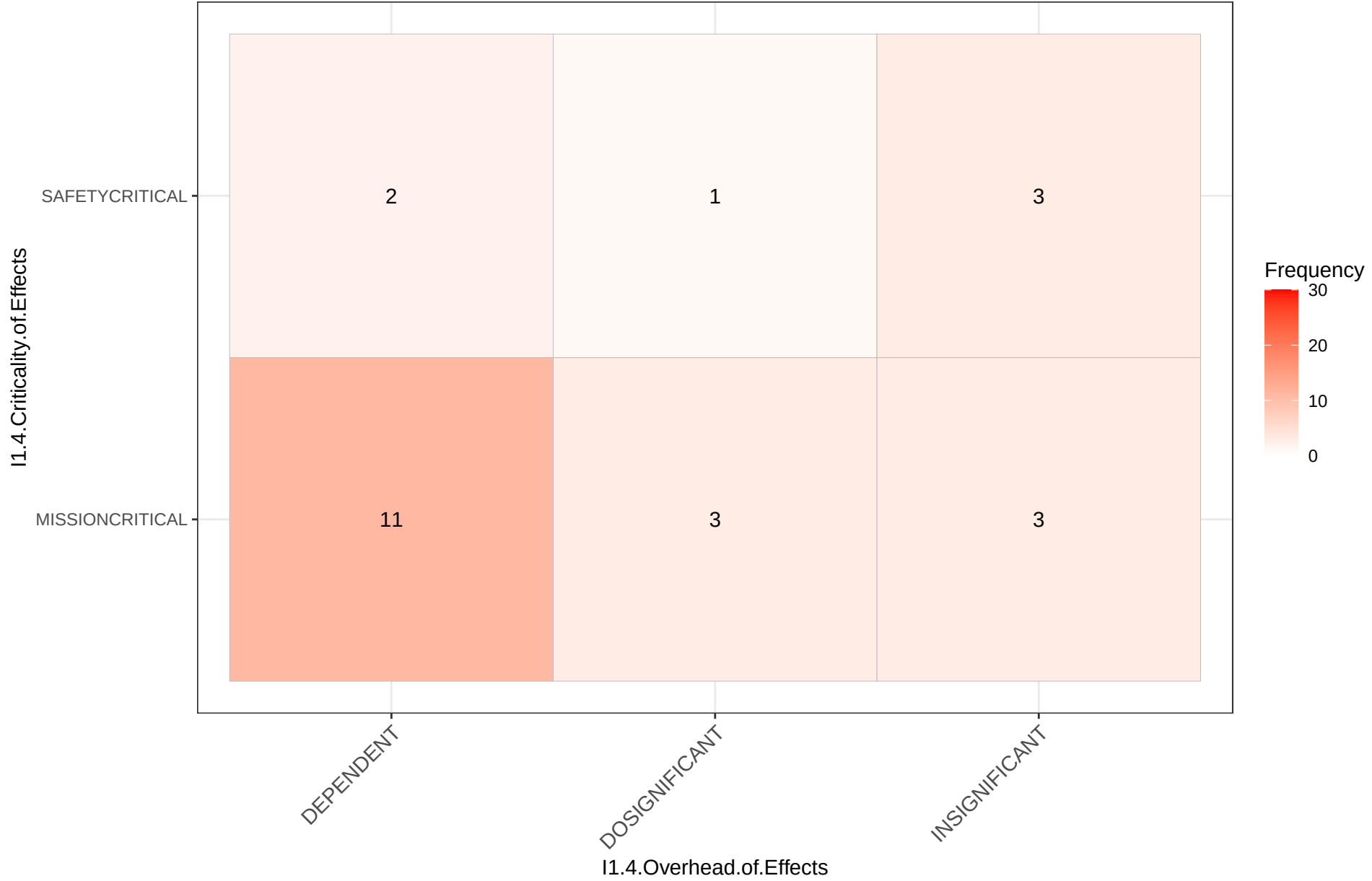
NON
DETERMINISTIC

I1.4.Predictability.of.Effects

Frequency



I1.4.Criticality.of.Effects_____I1.4.Overhead.of.Effects



I1.4.Criticality.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Criticality.of.Effects

SAFETYCRITICAL

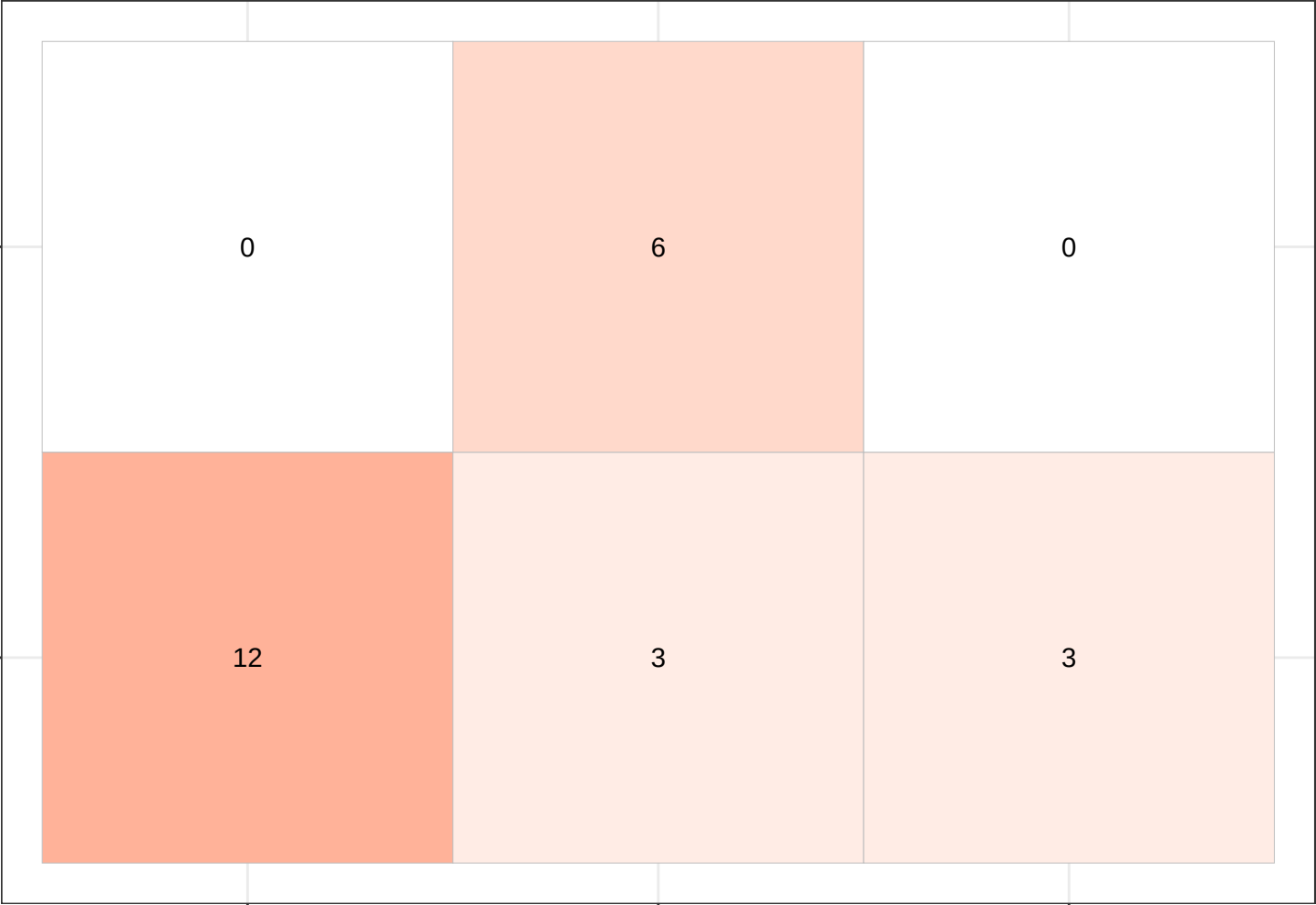
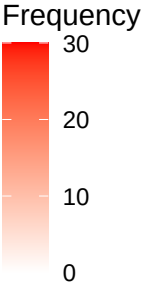
MISSIONCRITICAL

DEPENDENT

DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects



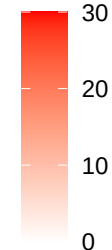
I1.4.Criticality.of.Effects

I1.4.Criticality.of.Effects_____I2.Adap..Purpose

SAFETYCRITICAL

MISSIONCRITICAL

Frequency



CHANGEFUNCTIONALBEHAVIOR	DEALWITHENVIRONMENTALCHANGES	KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME	OPTIMIZERESOURCEUSAGE	OPTIMIZESYSTEMPERFORMANCE	RECOVERFROMATTACKS	RECOVERFROMERRORSFAULTS
0	2	2	2	1	0	3
6	6	3	4	6	1	9

I2.Adap..Purpose

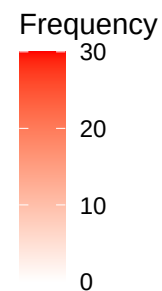
I1.4.Criticality.of.Effects

I1.4.Criticality.of.Effects_____I3.Robot.Type

SAFETYCRITICAL
MISSIONCRITICAL

0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	3	0	0	0	1
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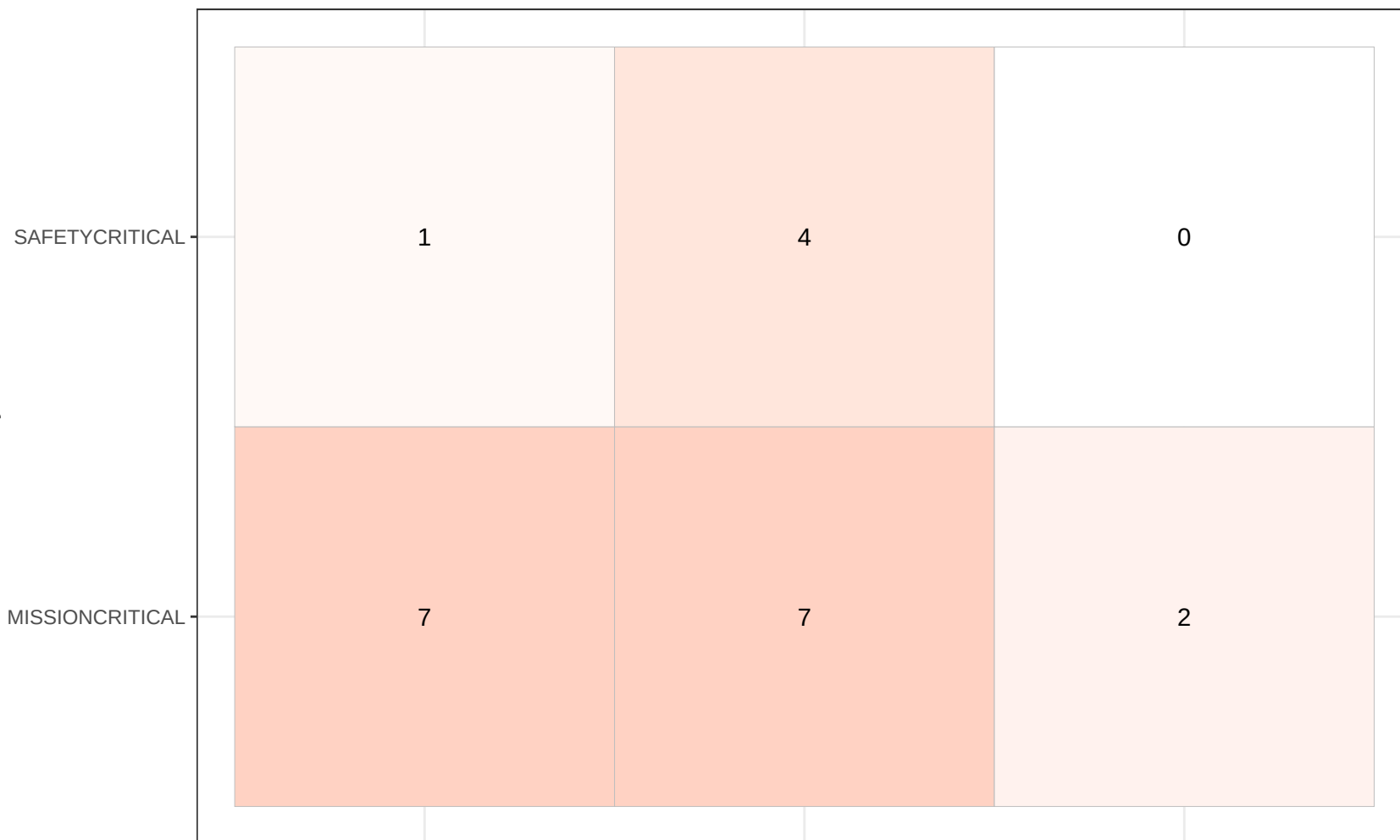
BLUEROV2
BOXERCLEARPATH
CRAWLERTERMINATORBOT
FIELDMOBILEROBOT
HETEROGENOUSROBOTS
INFOTAINMENTROBOT
KUKALIGHTWEIGHTROBOT4LWR4
HEXMANIPULATOR
HEXMANIPULATOR
IROBOTCREATE2
MOBILEROBOT
MOBILEROBOTTERRESTRIAL
MOBILEROBOTTIAGO
MOBILESERVICEROBOT
MSUEVORALLYMOBILETERRESTRIAL
MULTIPLEHEXROTAR
NAOROBOT
PIONEER3DX
QUADROCOPTER
SINGLESERVOINGROTATIONROBOT
TEDUSARTERRESTRIALSEARCH
TRIGLIDEINDUSTRIALASSEMBLY
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION
TWOINDUSTRIALROBOTS
WAREHOUSEDELIVERYROBOT
WHICHISANINDUSTRIALAGV
TURTLEBOT



I3.Robot.Type

I1.4.Criticality.of.Effects _____ I4.Robo.SW

I1.4.Criticality.of.Effects



MISSIONCRITICAL

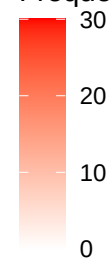
OTHER

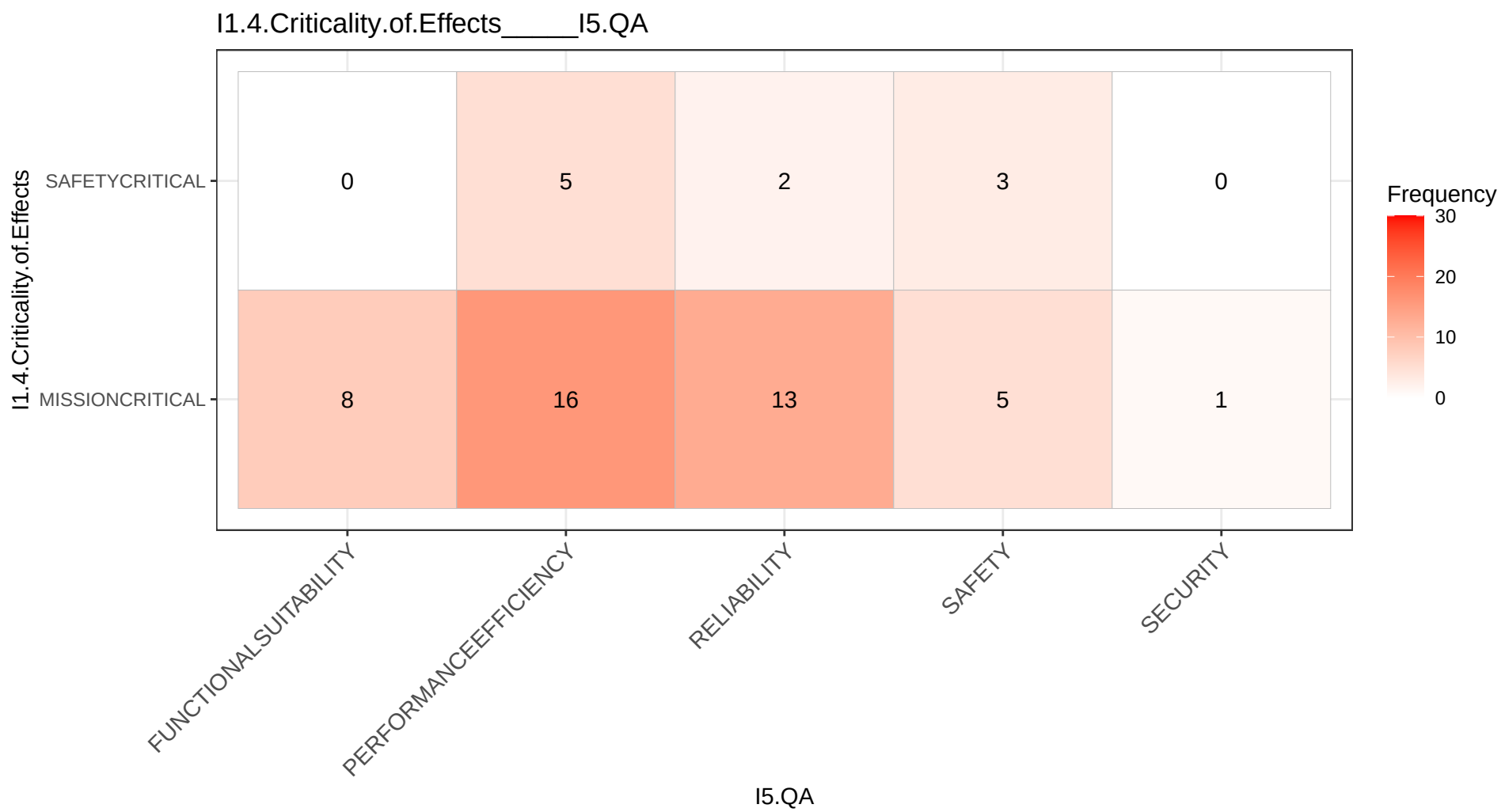
ROS1

ROS2

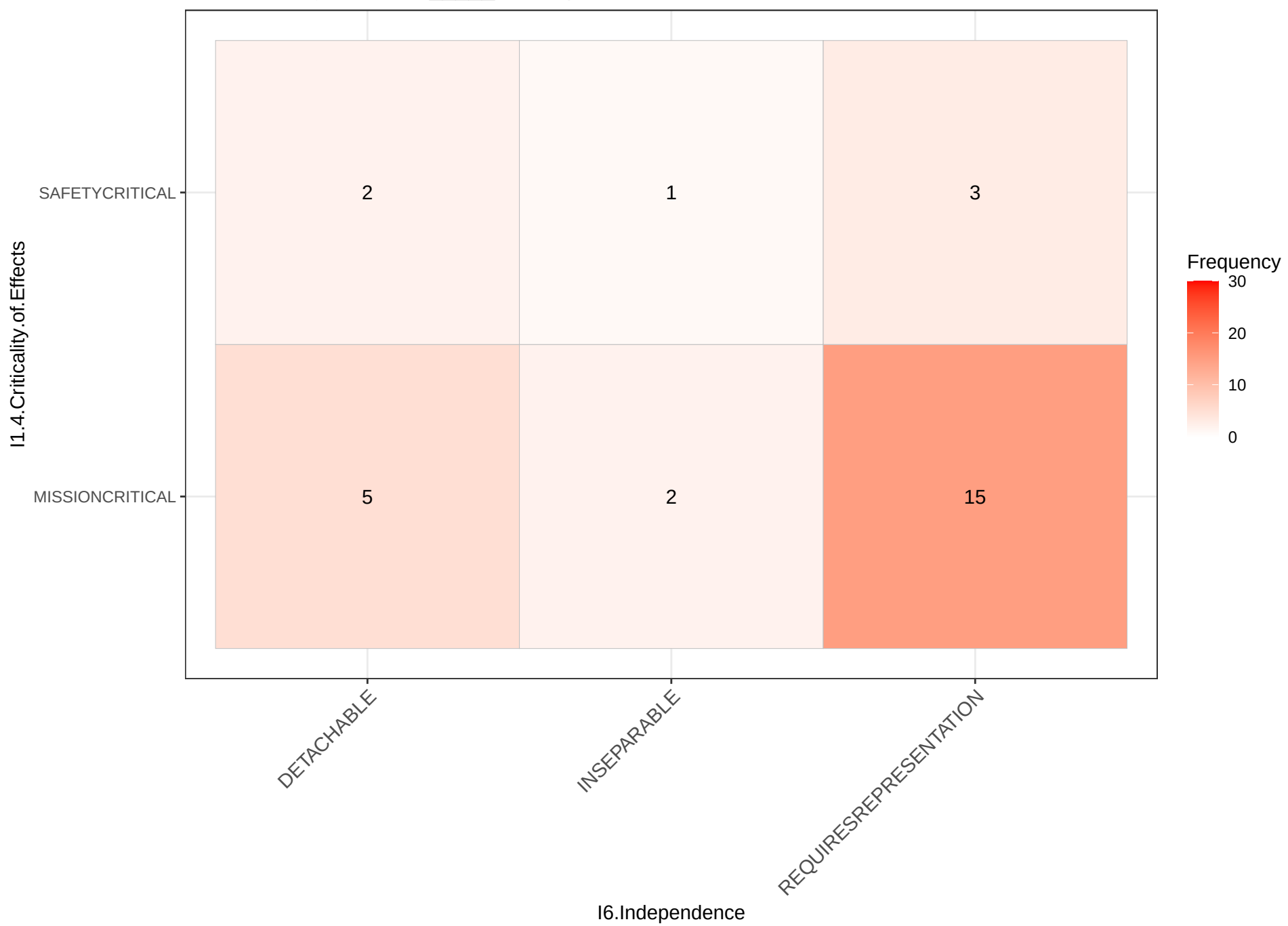
I4.Robo.SW

Frequency





I1.4.Criticality.of.Effects_____I6.Independence



I1.4.Criticality.of.Effects_____I7.Deployment.Realness

I1.4.Criticality.of.Effects

SAFETYCRITICAL

3

3

MISSIONCRITICAL

11

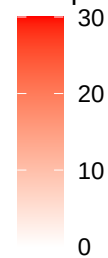
7

REAL

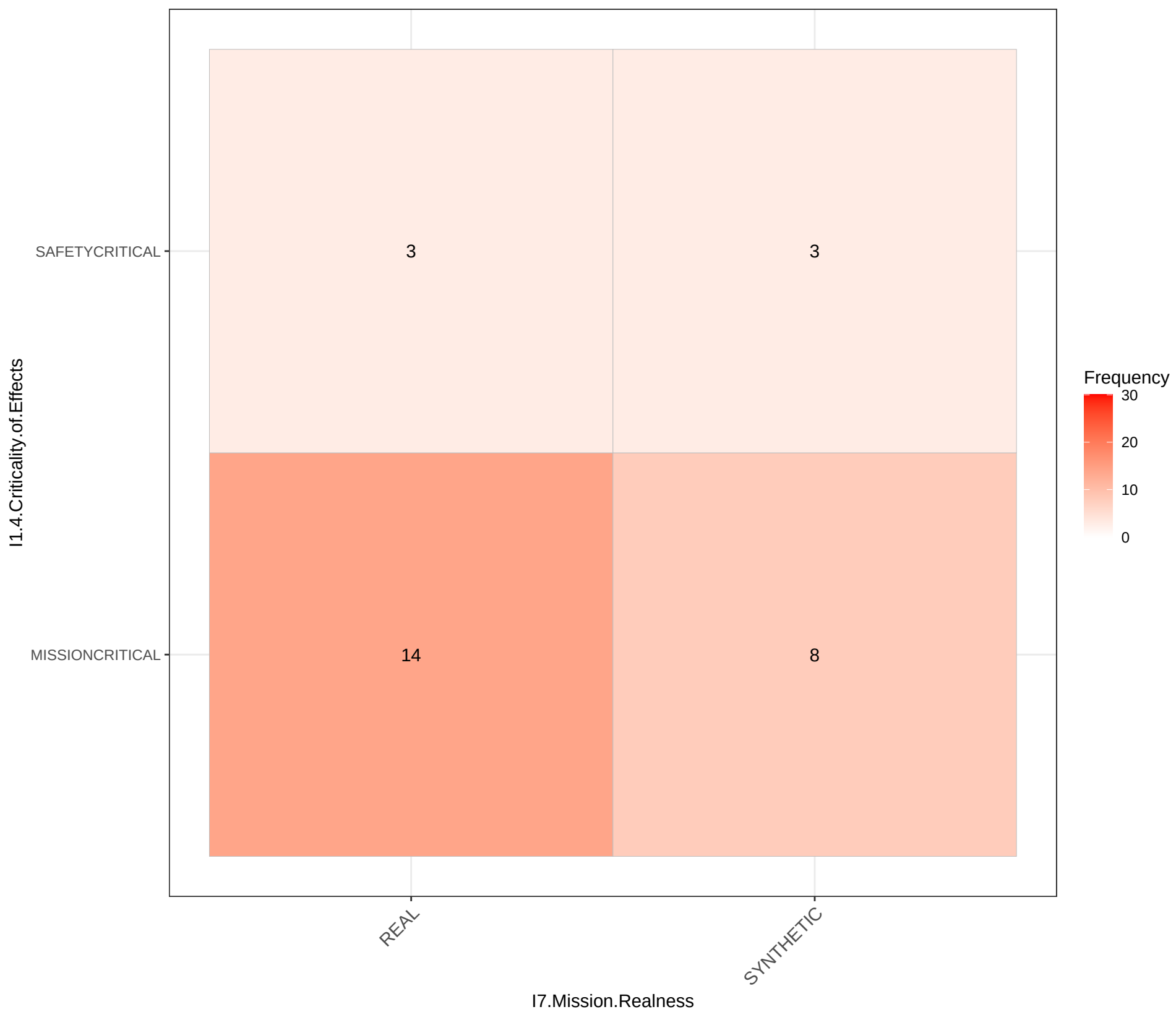
SIMULATED

I7.Deployment.Realness

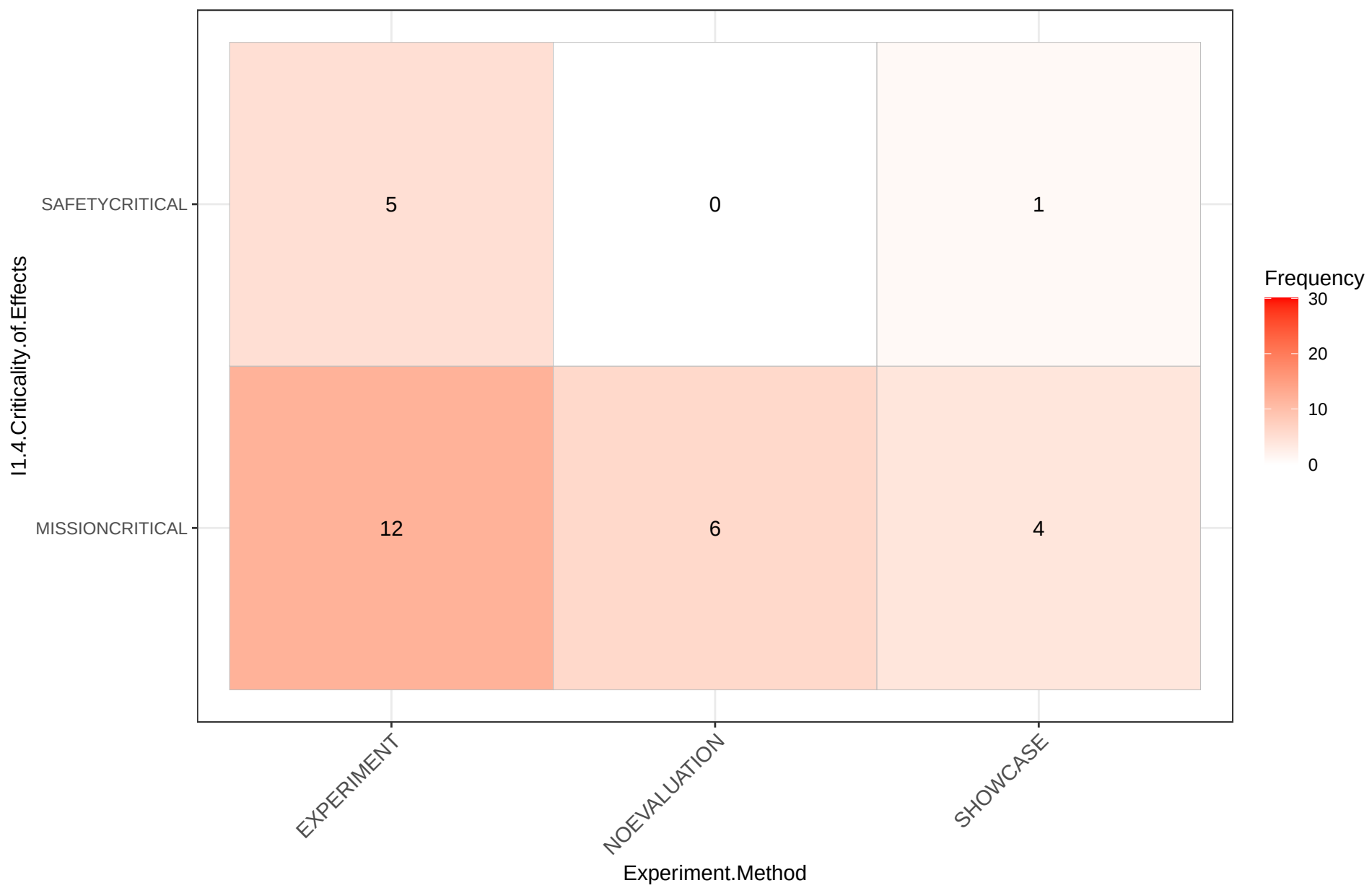
Frequency



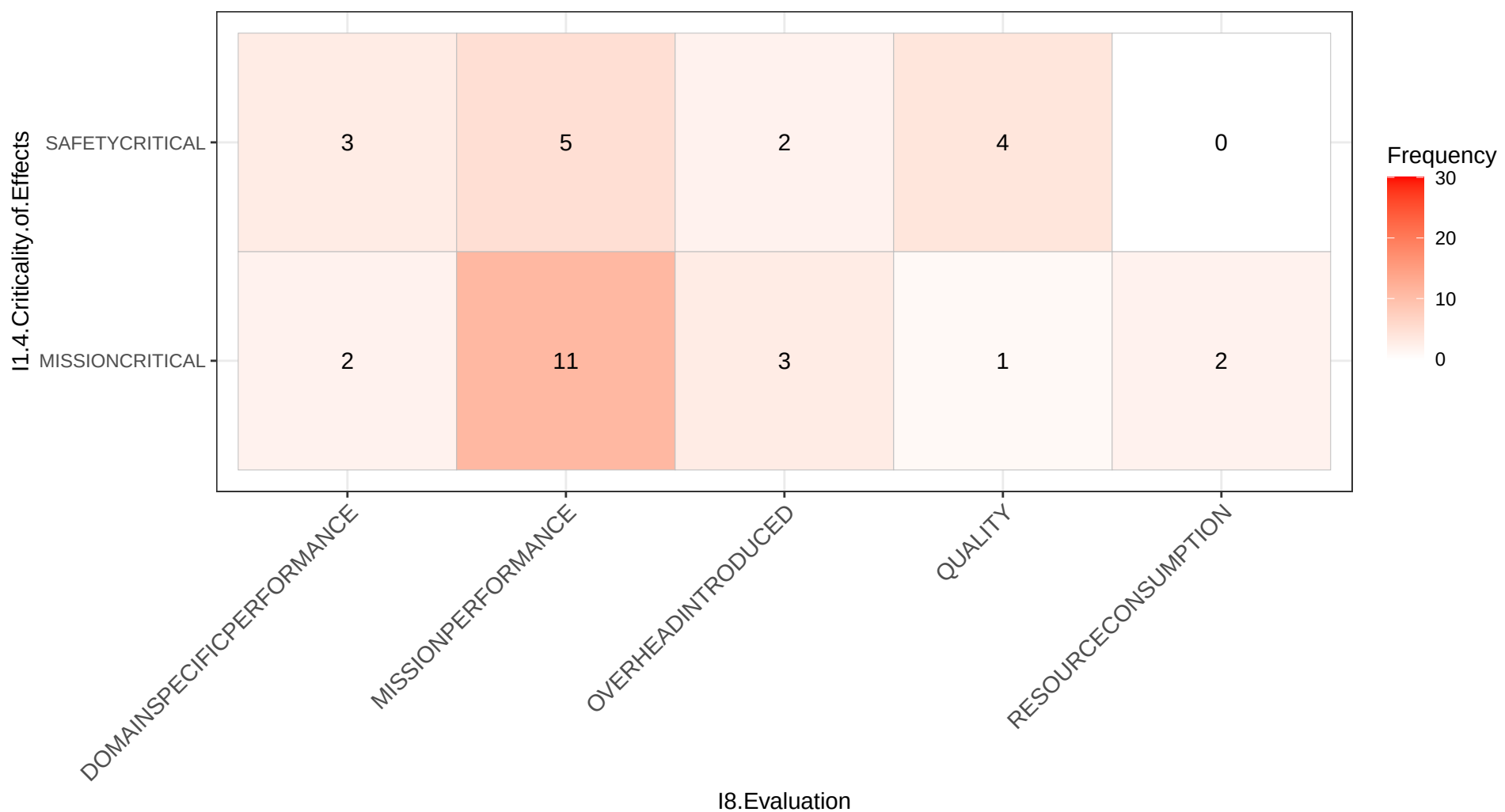
I1.4.Criticality.of.Effects_____I7.Mission.Realness

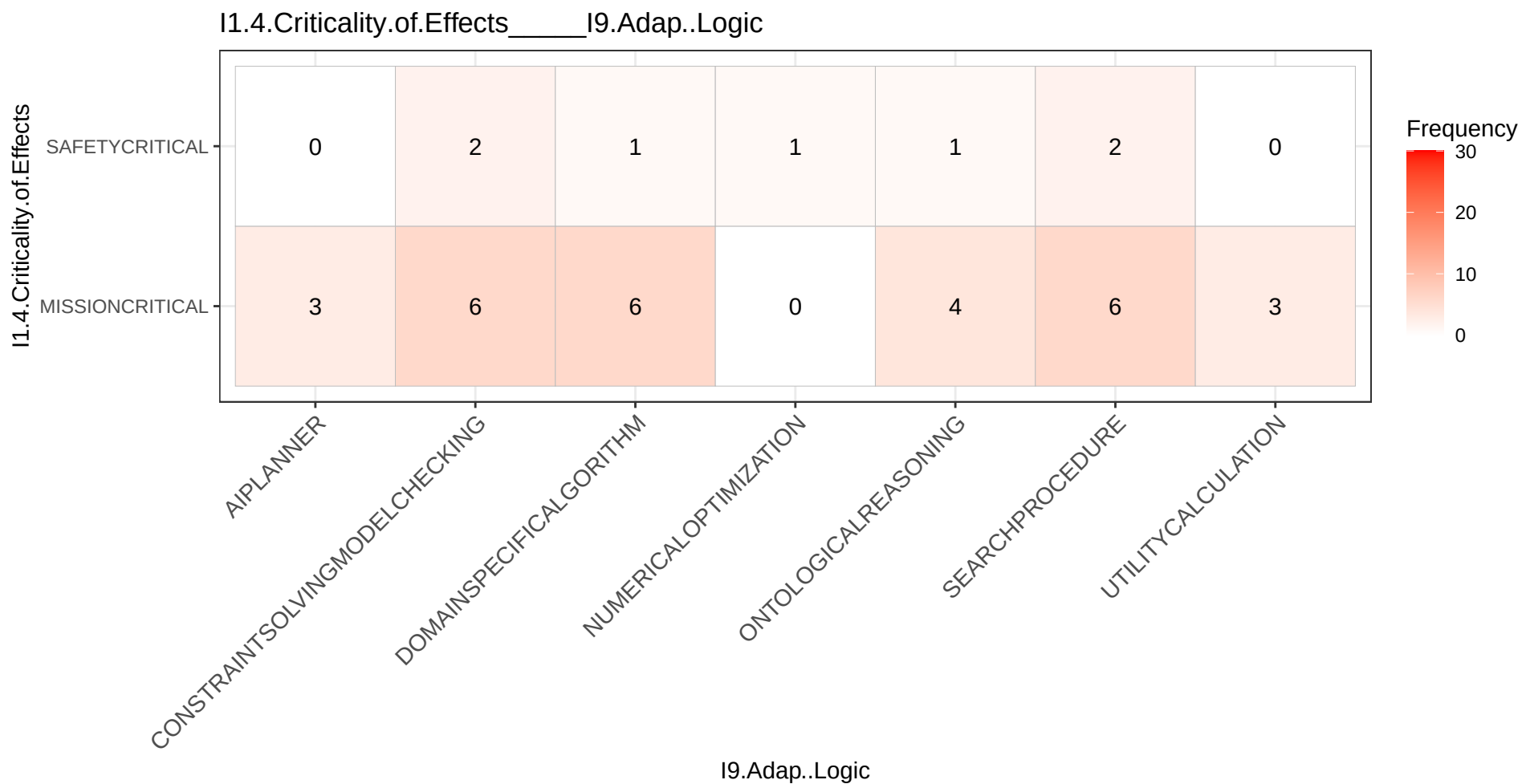


I1.4.Criticality.of.Effects_____Experiment.Method

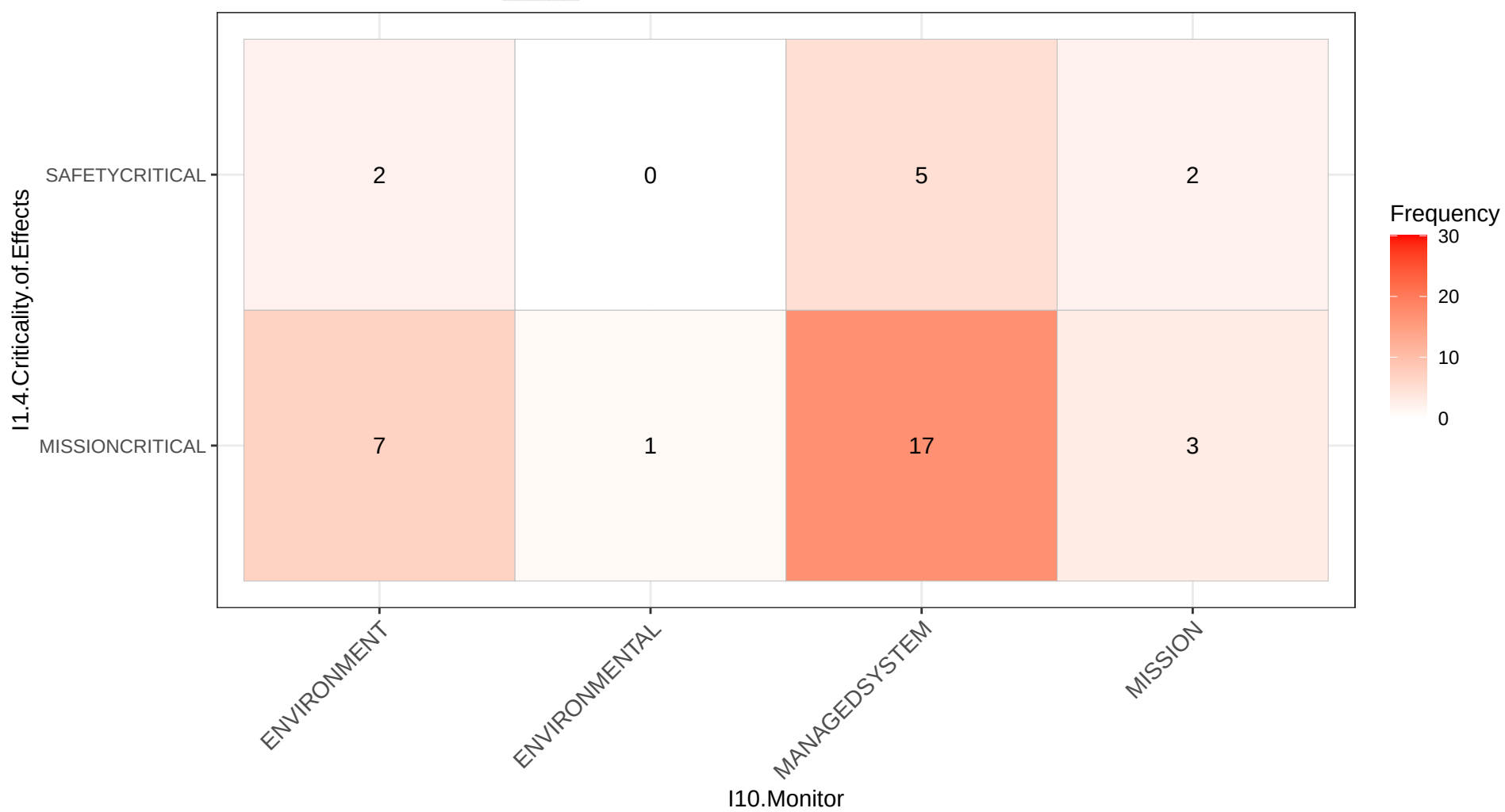


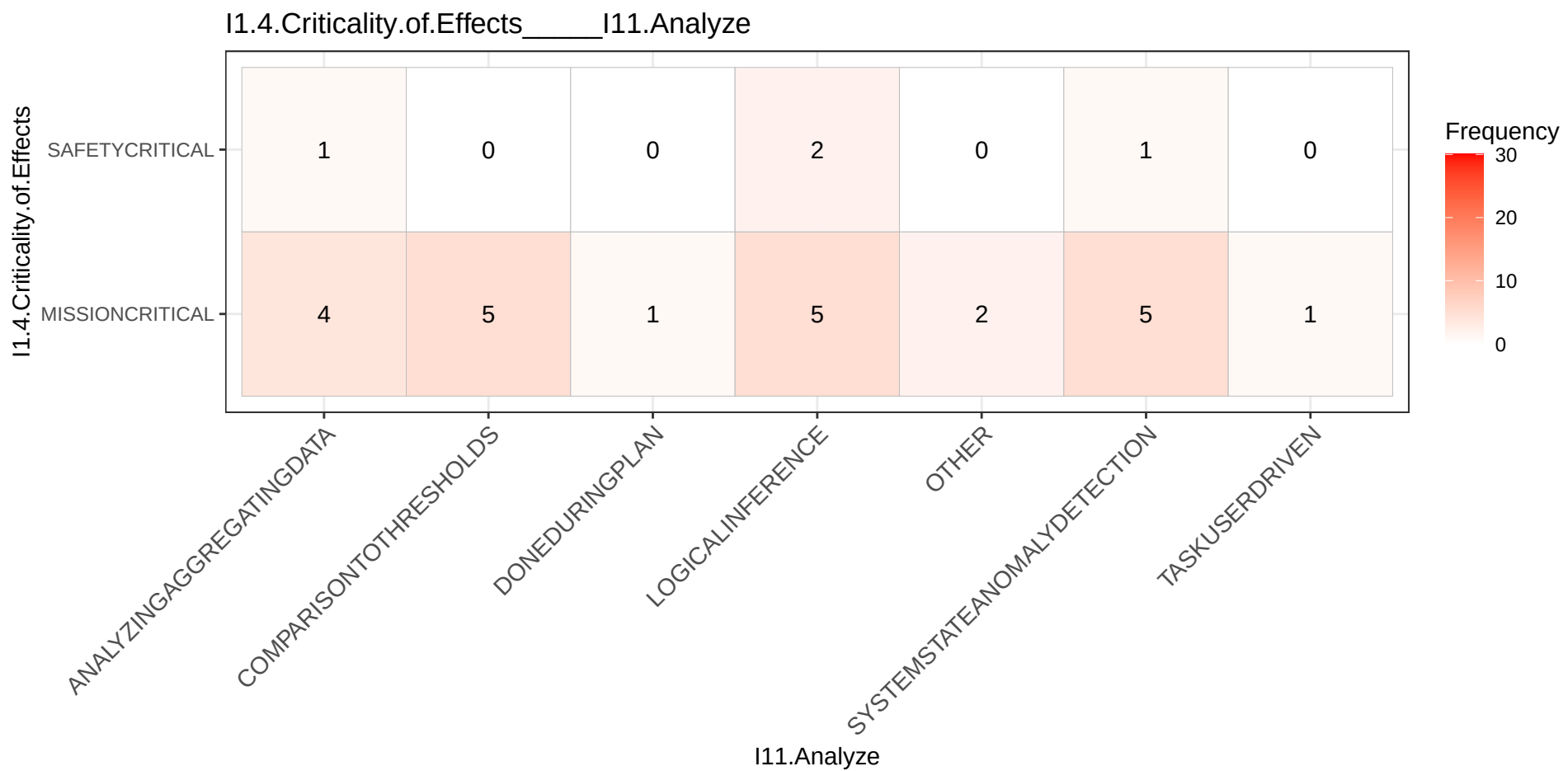
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I1.4.Criticality.of.Effects_____I10.Monitor



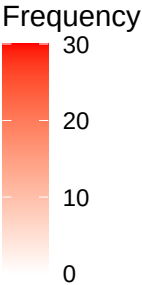


I1.4.Criticality.of.Effects_____I12.Plan

I1.4.Criticality.of.Effects

SAFETYCRITICAL

MISSIONCRITICAL

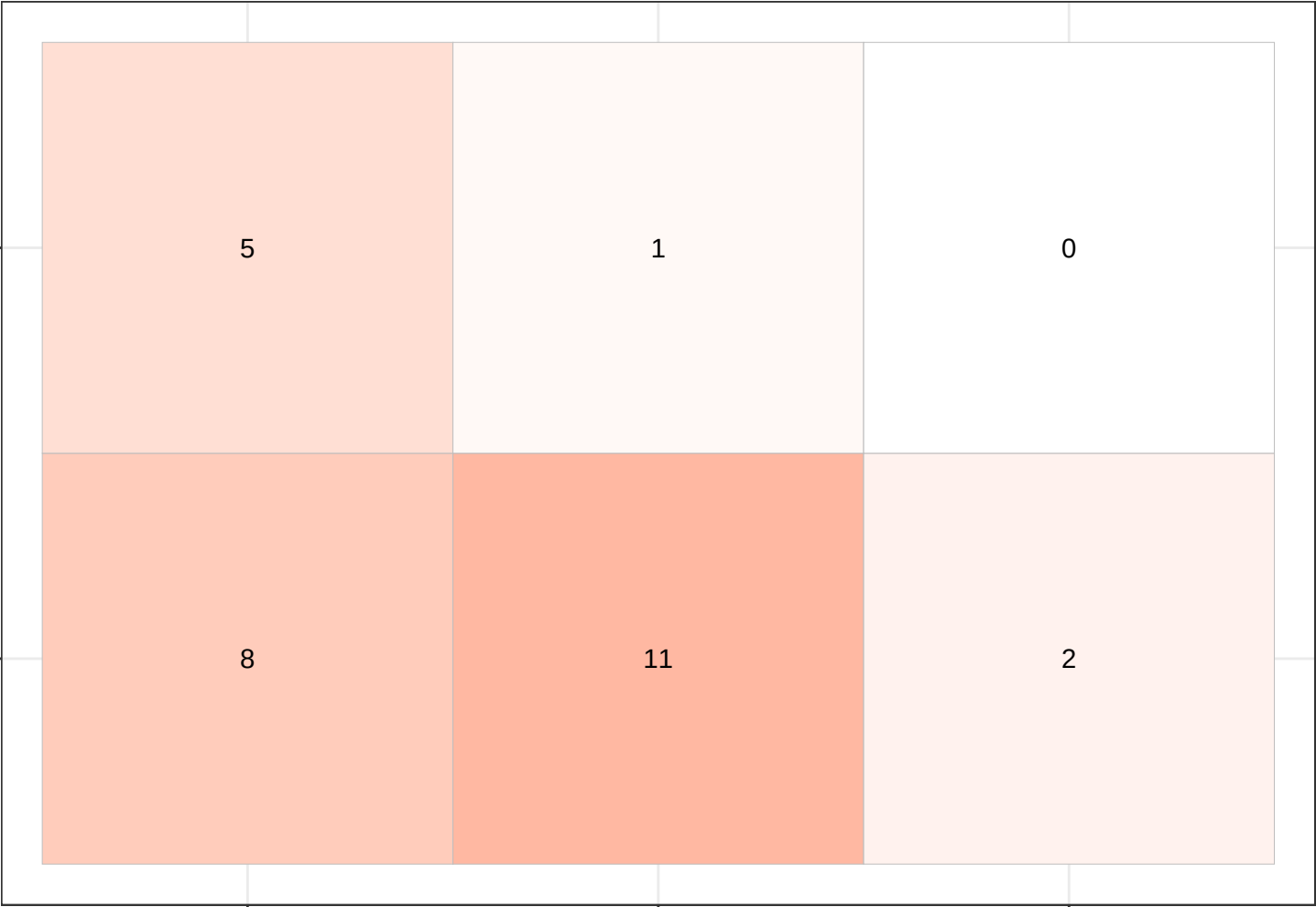


DETERMININGTHEOPTIMALCHOICE

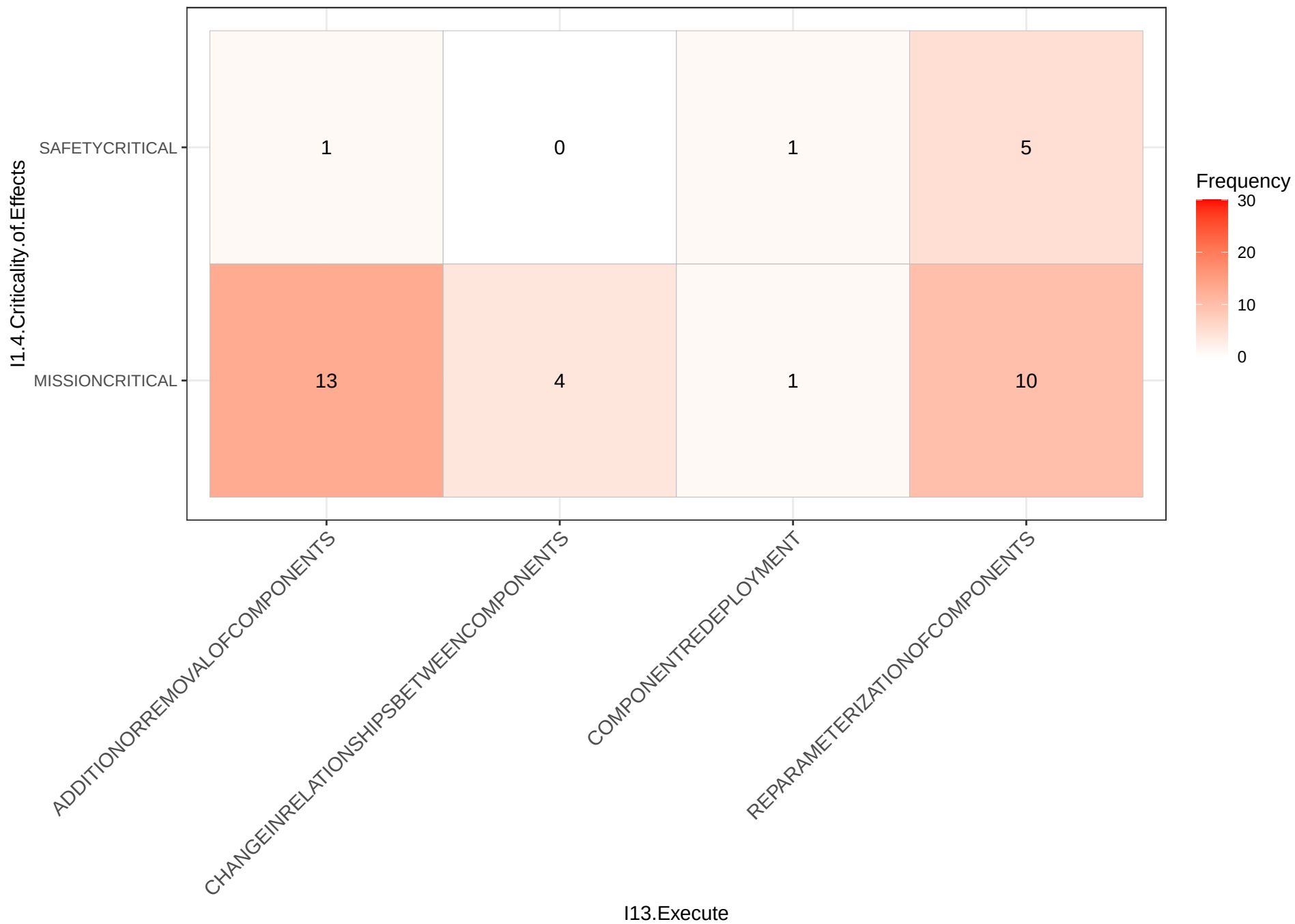
RELYINGONDESIGNTIMERULESMODELS

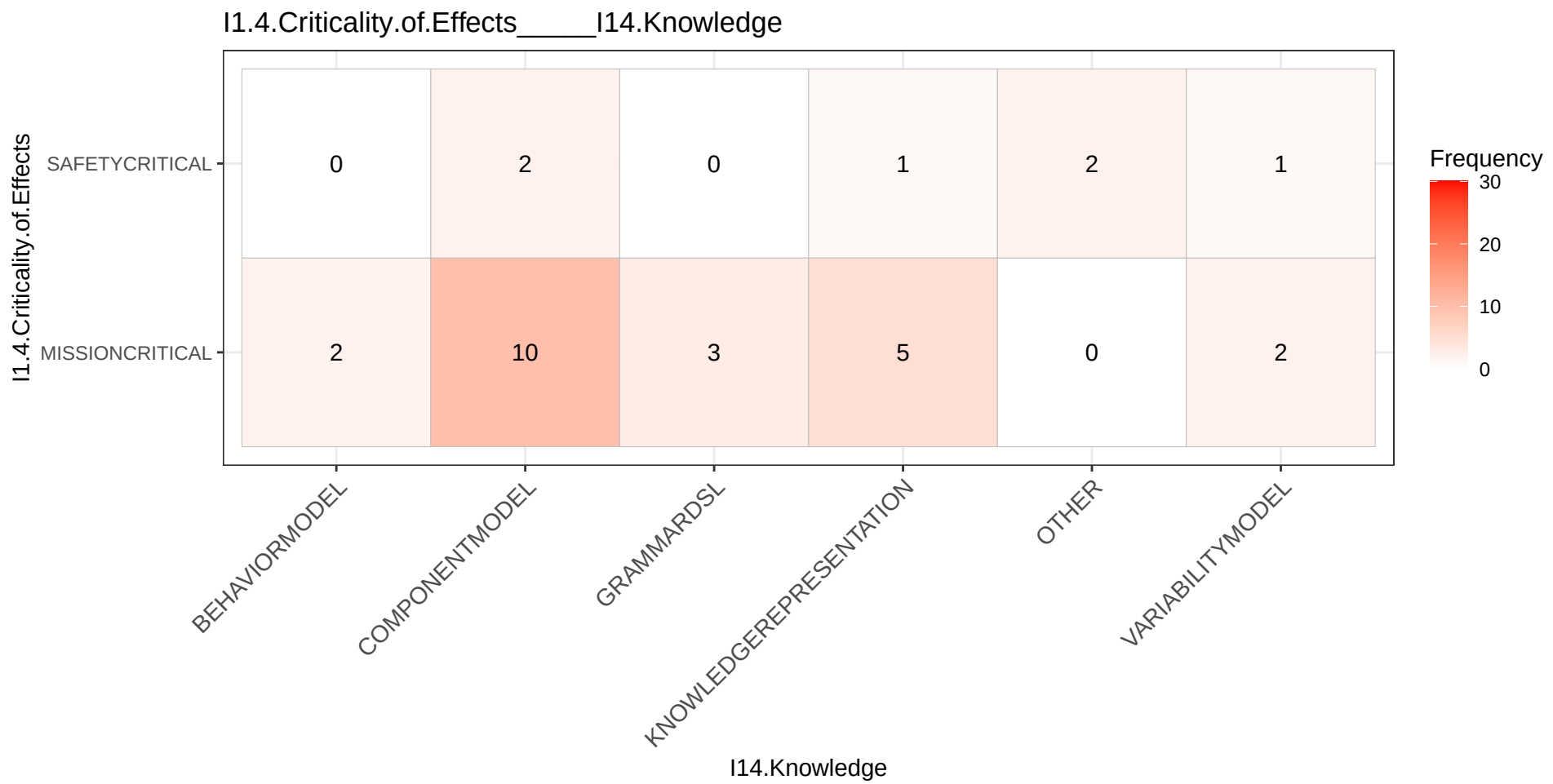
USINGAIPANNINGLANGUAGES

I12.Plan



I1.4.Criticality.of.Effects_____I13.Execute





I1.4.Predictability.of.Effects_____I1.4.Overhead.of.Effects

I1.4.Predictability.of.Effects

NONDETERMINISTIC

9

4

4

DODETERMINISTIC

6

0

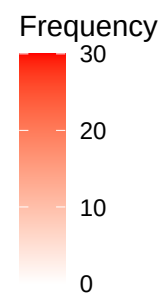
2

DEPENDENT

DOSIGNIFICANT

INSIGNIFICANT

I1.4.Overhead.of.Effects



I1.4.Predictability.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Predictability.of.Effects

NONDETERMINISTIC

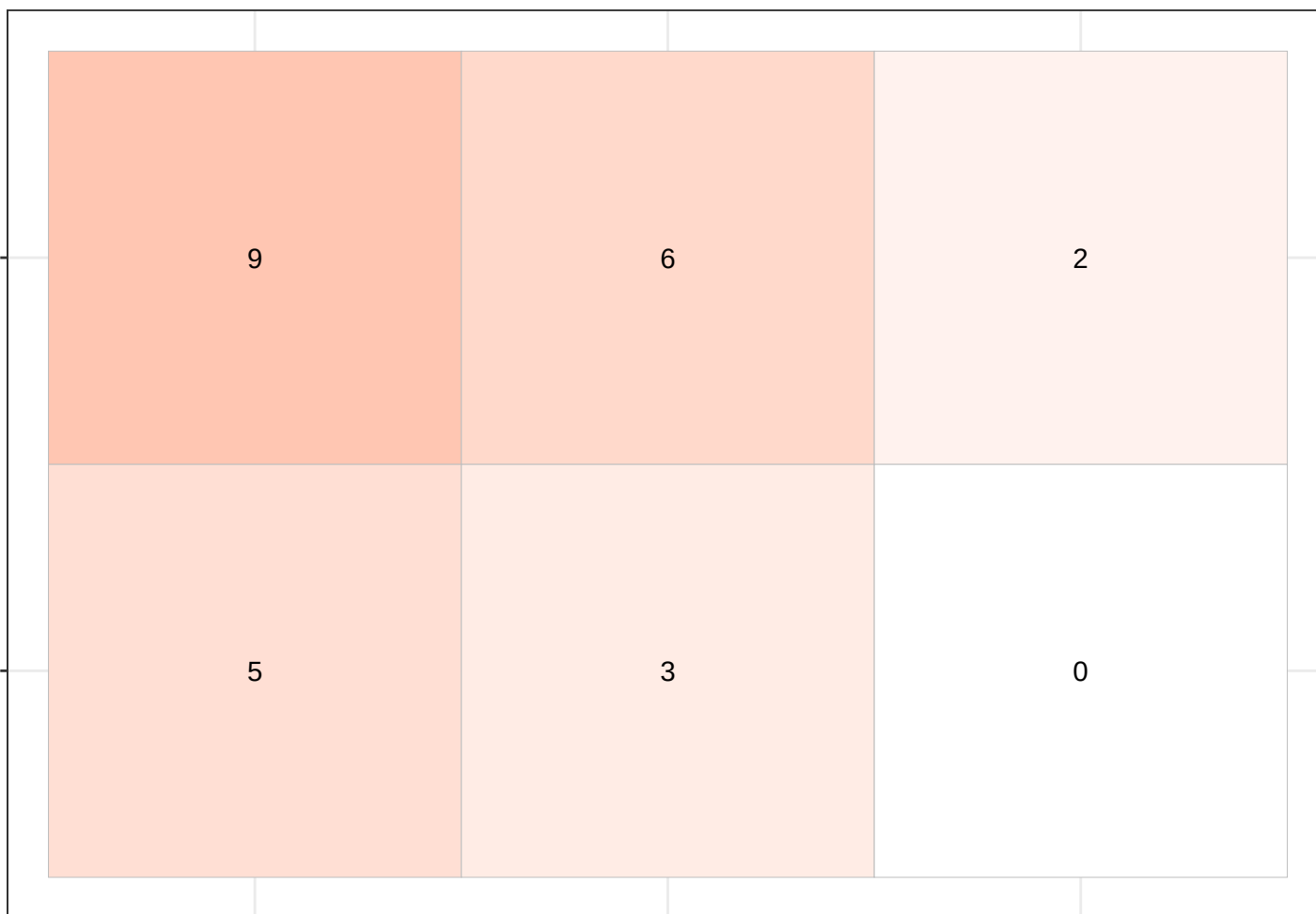
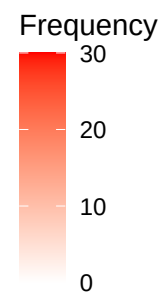
DODETERMINISTIC

DEPENDENT

DORESILIENT

IRRESILIENT

I1.4.Resilience.of.Effects



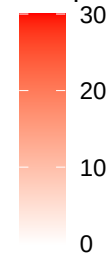
I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I2.Adap..Purpose

NONDETERMINISTIC

DODETERMINISTIC

Frequency



CHANGEFUNCTIONALBEHAVIOR

DEALWITHENVIRONMENTALCHANGES

KEEPMEETINGQUALITYREQUIREMENTSATRUNTIME

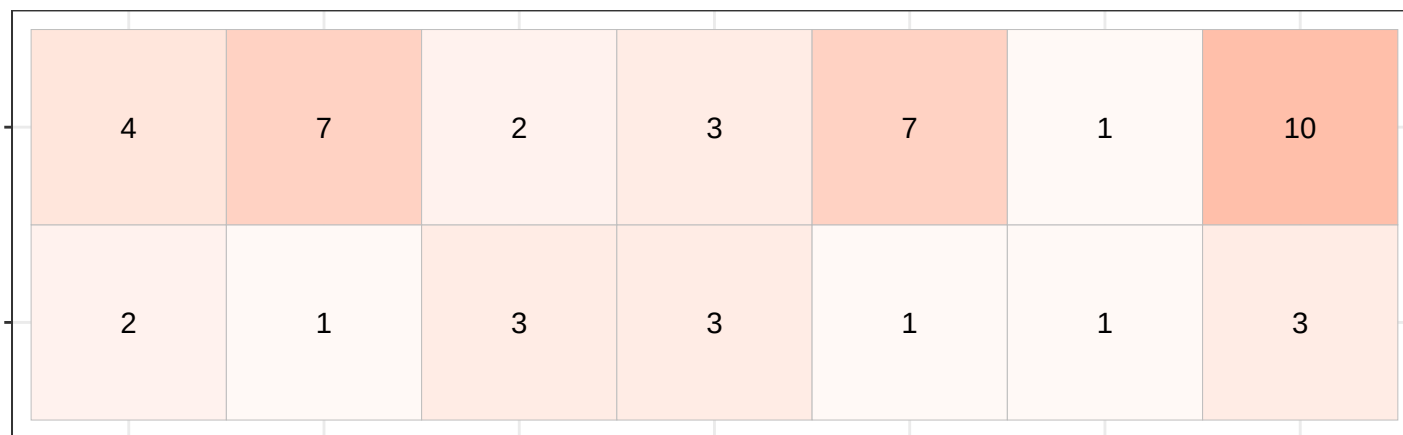
OPTIMIZERESOURCEUSAGE

OPTIMIZESEYSTEMPERFORMANCE

RECOVERFROMATTACKS

RECOVERFROMERRORSFAULTS

I2.Adap..Purpose

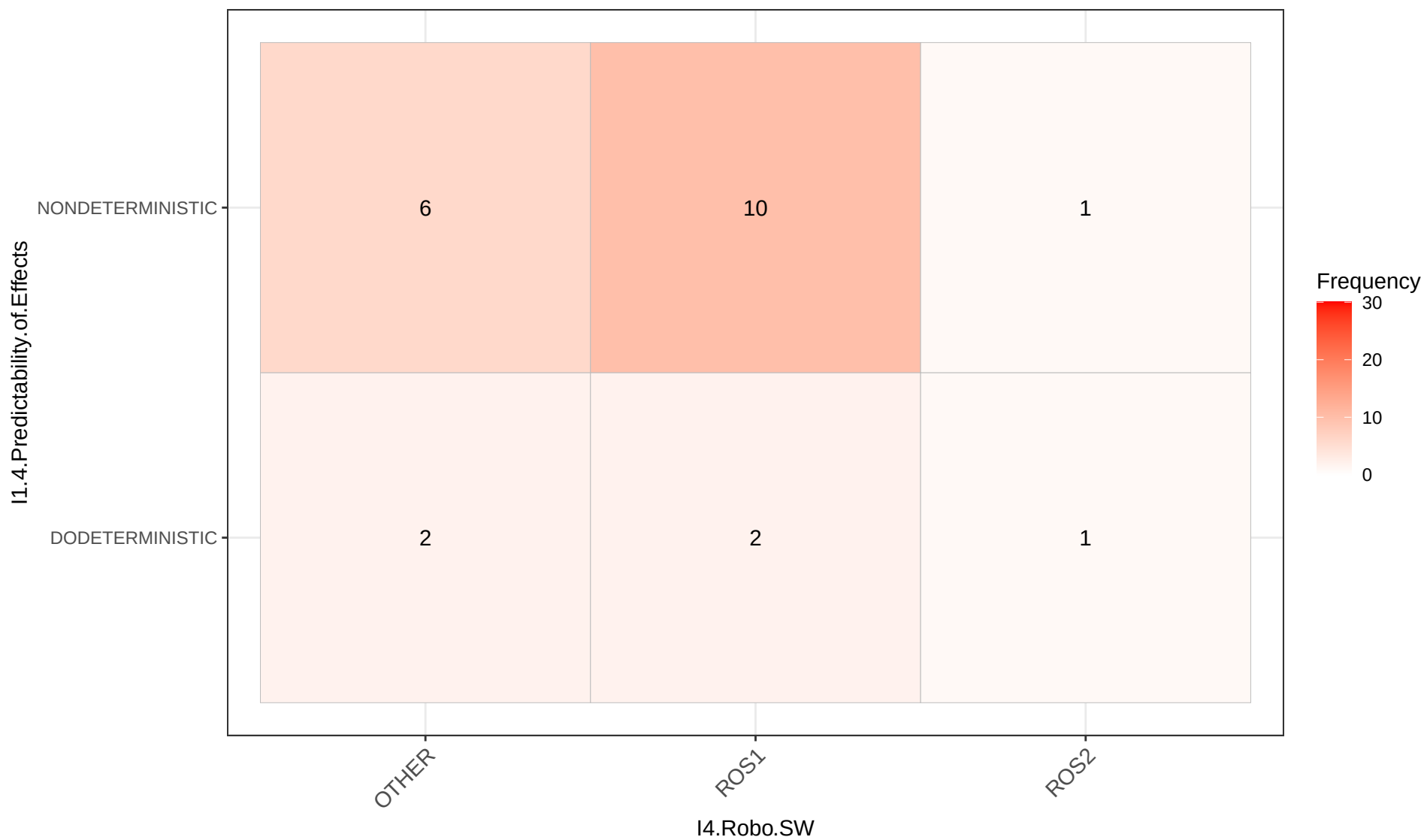


I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I3.Robot.Type



I1.4.Predictability.of.Effects_____I4.Robo.SW



I1.4. Predictability of Effects

I1.4. Predictability of Effects _____ I5. QA

NONDETERMINISTIC

DODETERMINISTIC

FUNCTIONALSUITABILITY

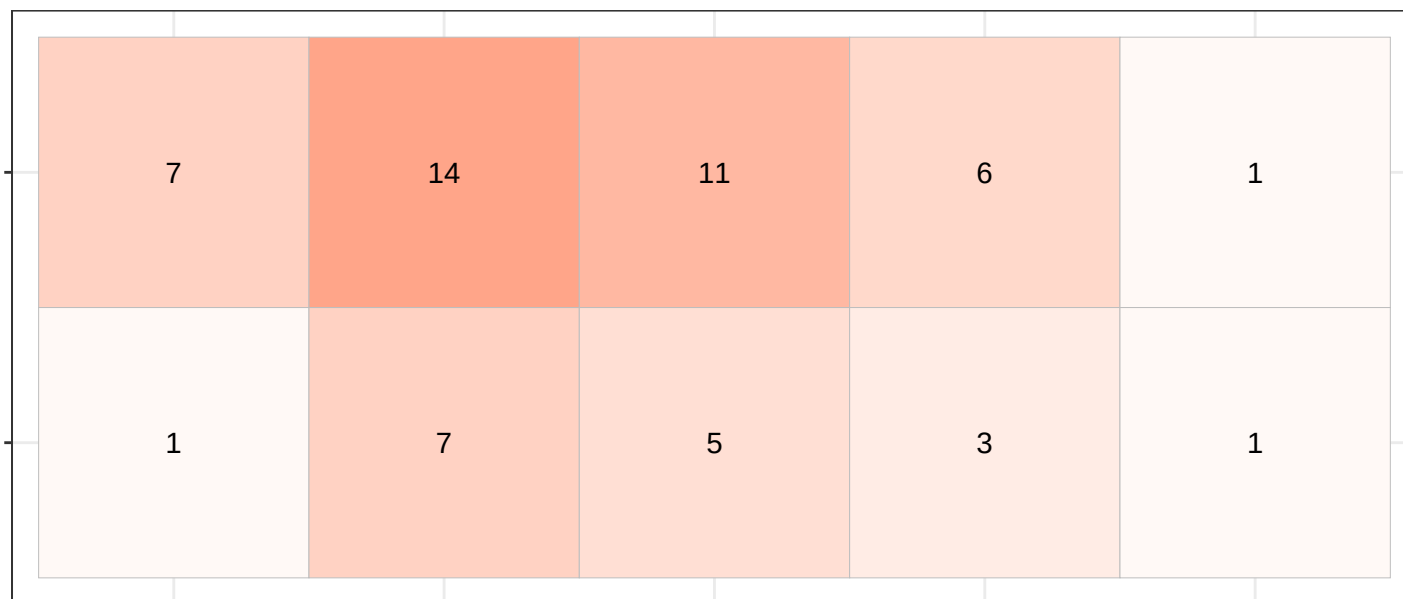
PERFORMANCEEFFICIENCY

RELIABILITY

SAFETY

SECURITY

I5. QA



Frequency

30

20

10

0

I1.4.Predictability.of.Effects_____I6.Independence

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

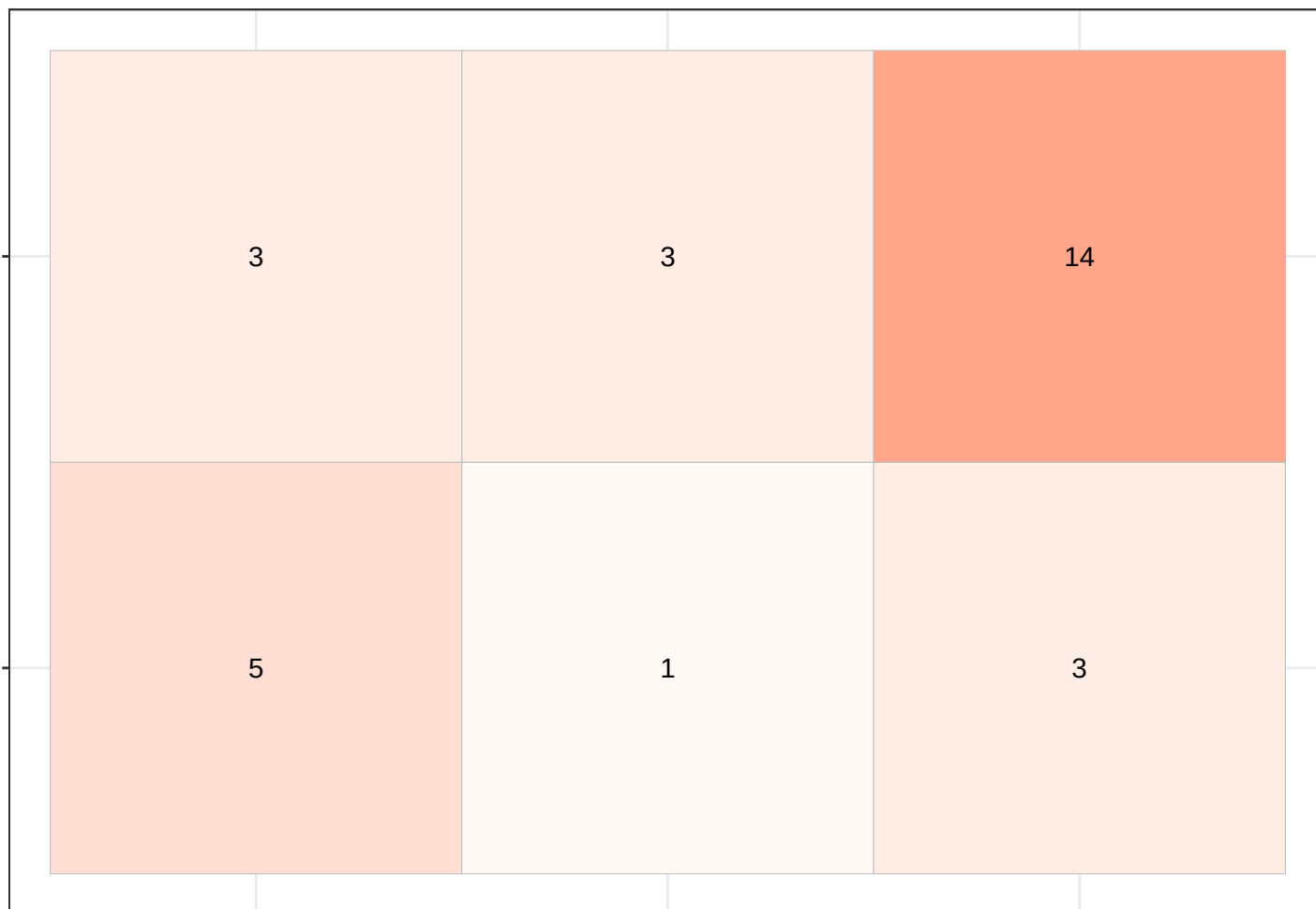
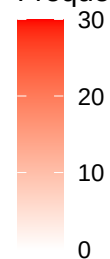
DETACHABLE

INSEPARABLE

REQUIRESREPRESENTATION

I6.Independence

Frequency



I1.4.Predictability.of.Effects_____I7.Deployment.Realness

I1.4.Predictability.of.Effects

NONDETERMINISTIC

11

8

DODETERMINISTIC

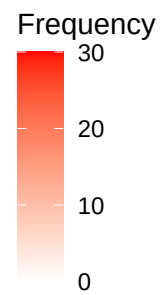
4

2

REAL

SIMULATED

I7.Deployment.Realness



I1.4.Predictability.of.Effects_____I7.Mission.Realness

I1.4.Predictability.of.Effects

NONDETERMINISTIC

11

9

DODETERMINISTIC

6

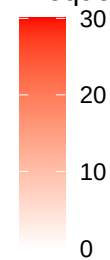
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REAL

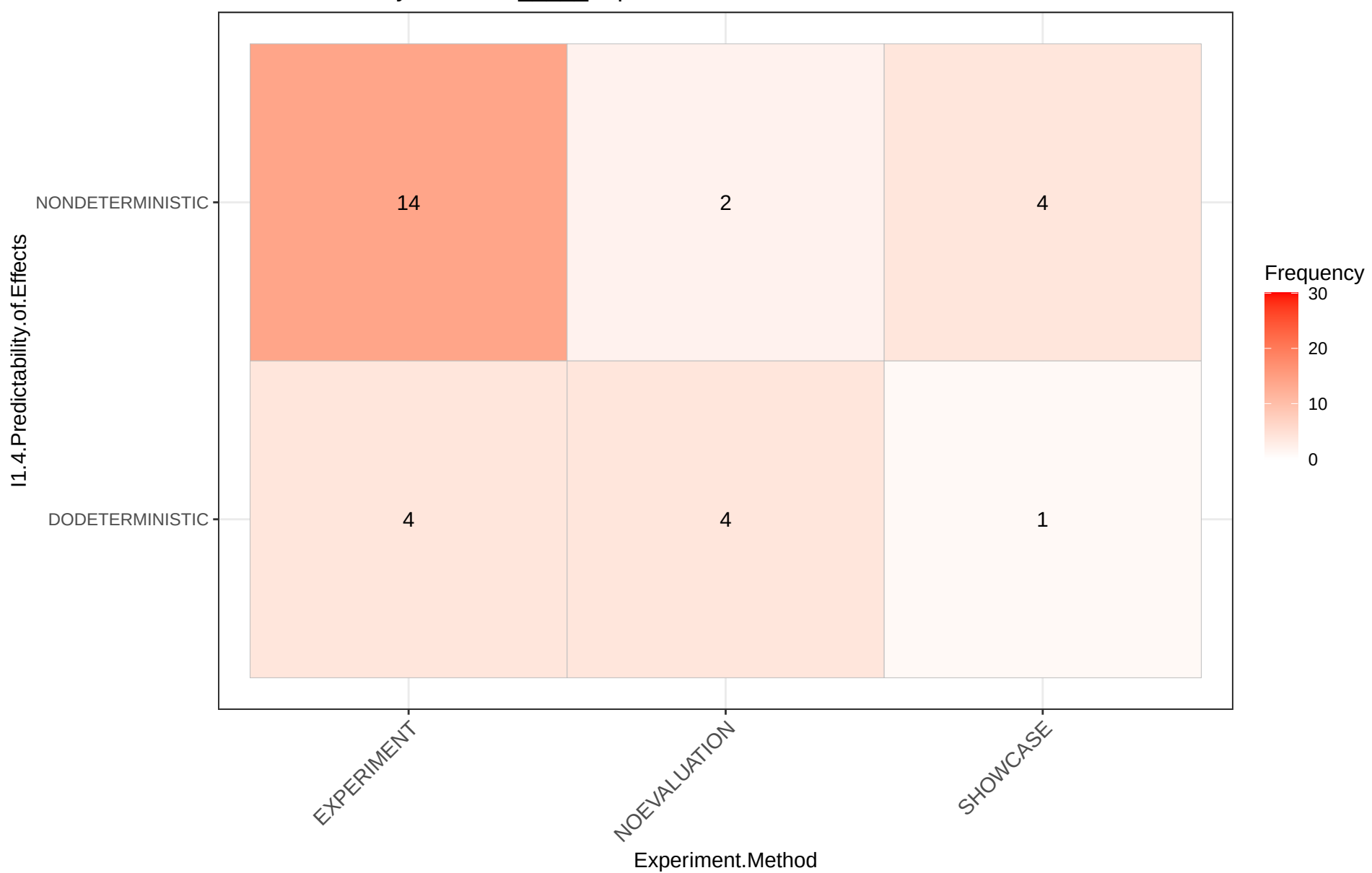
SYNTHETIC

I7.Mission.Realness

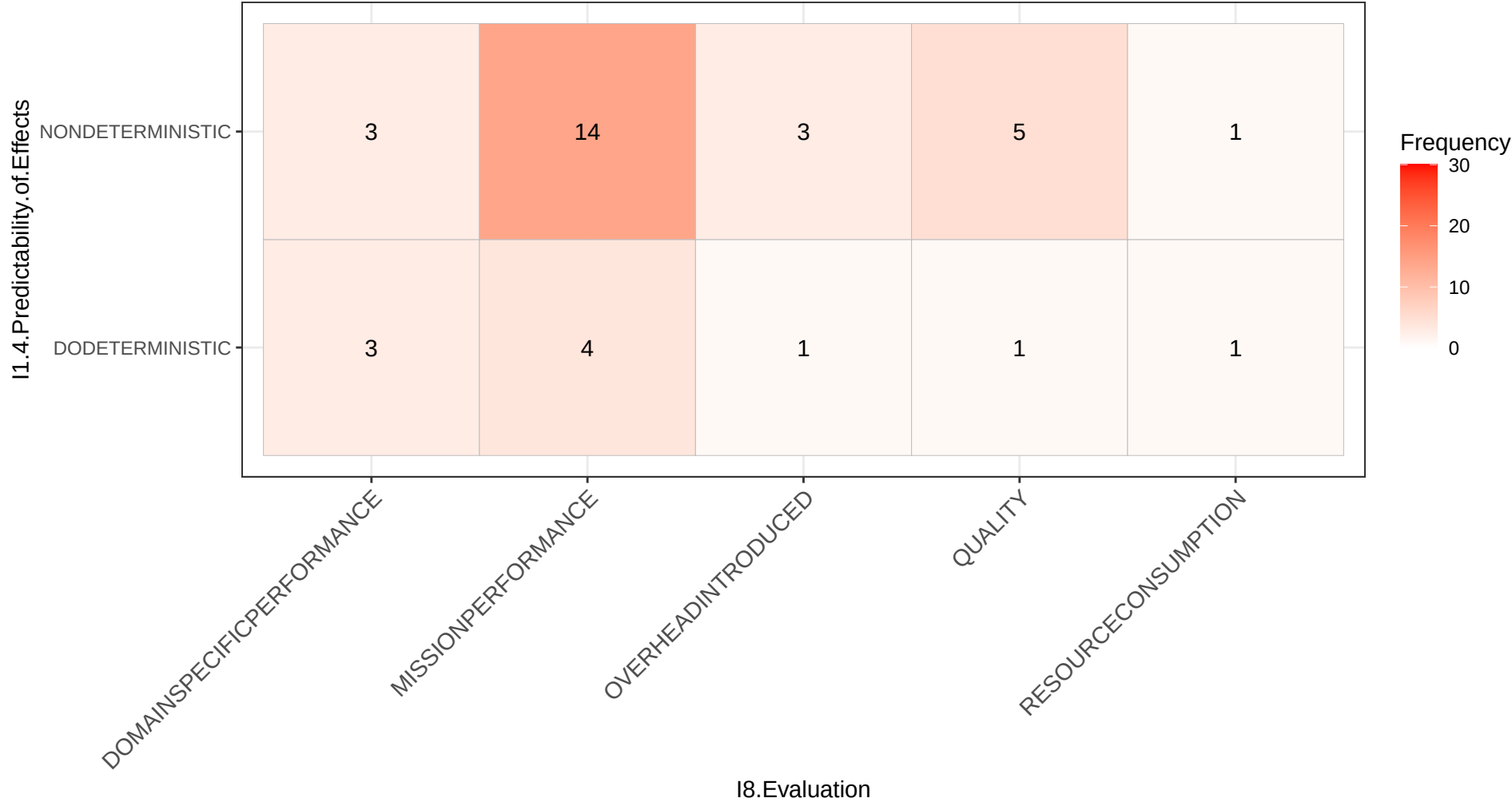
Frequency



I1.4.Predictability.of.Effects_____Experiment.Method

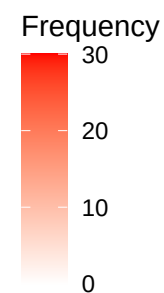
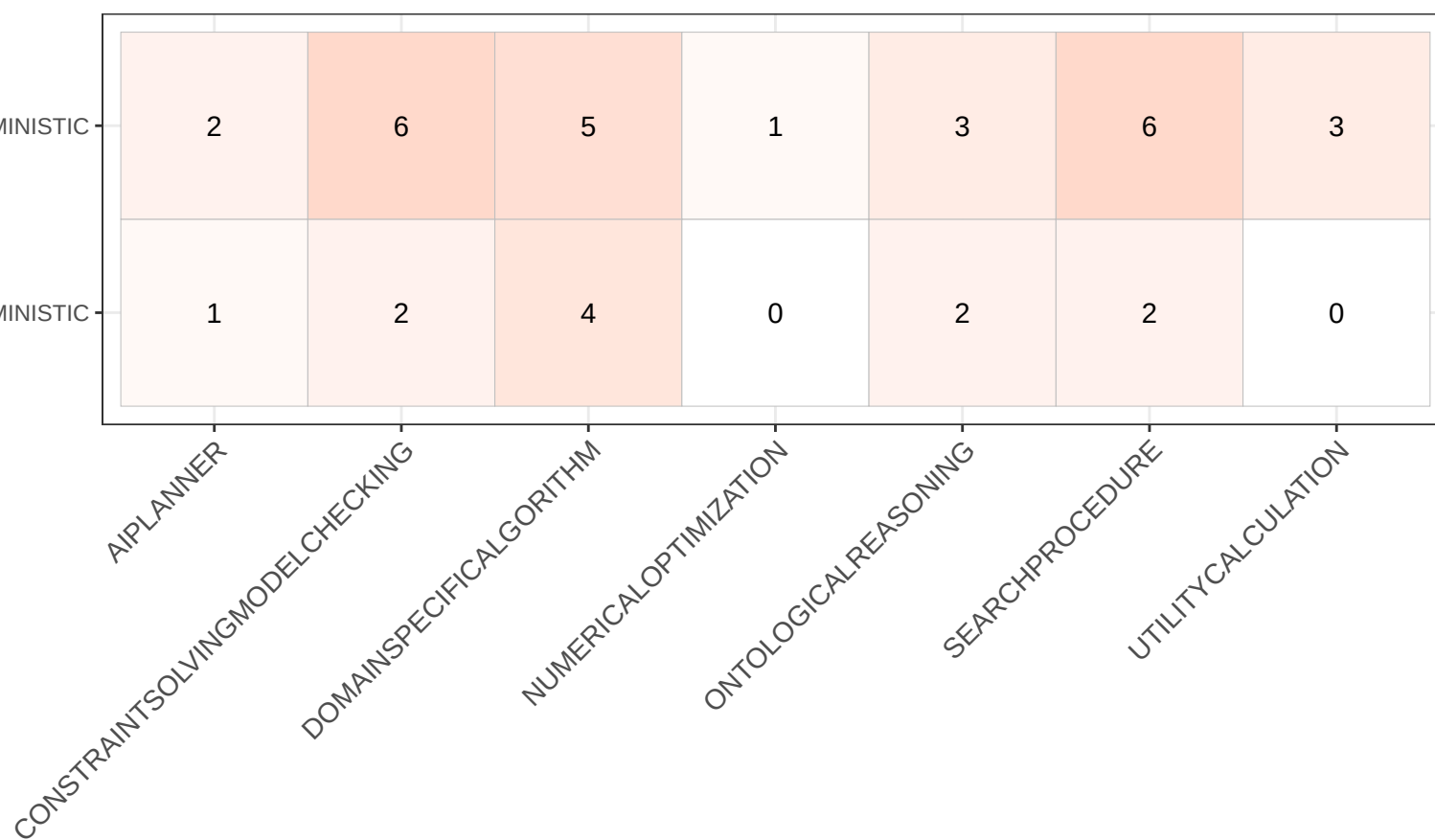


I1.4.Predictability.of.Effects_____I8.Evaluation



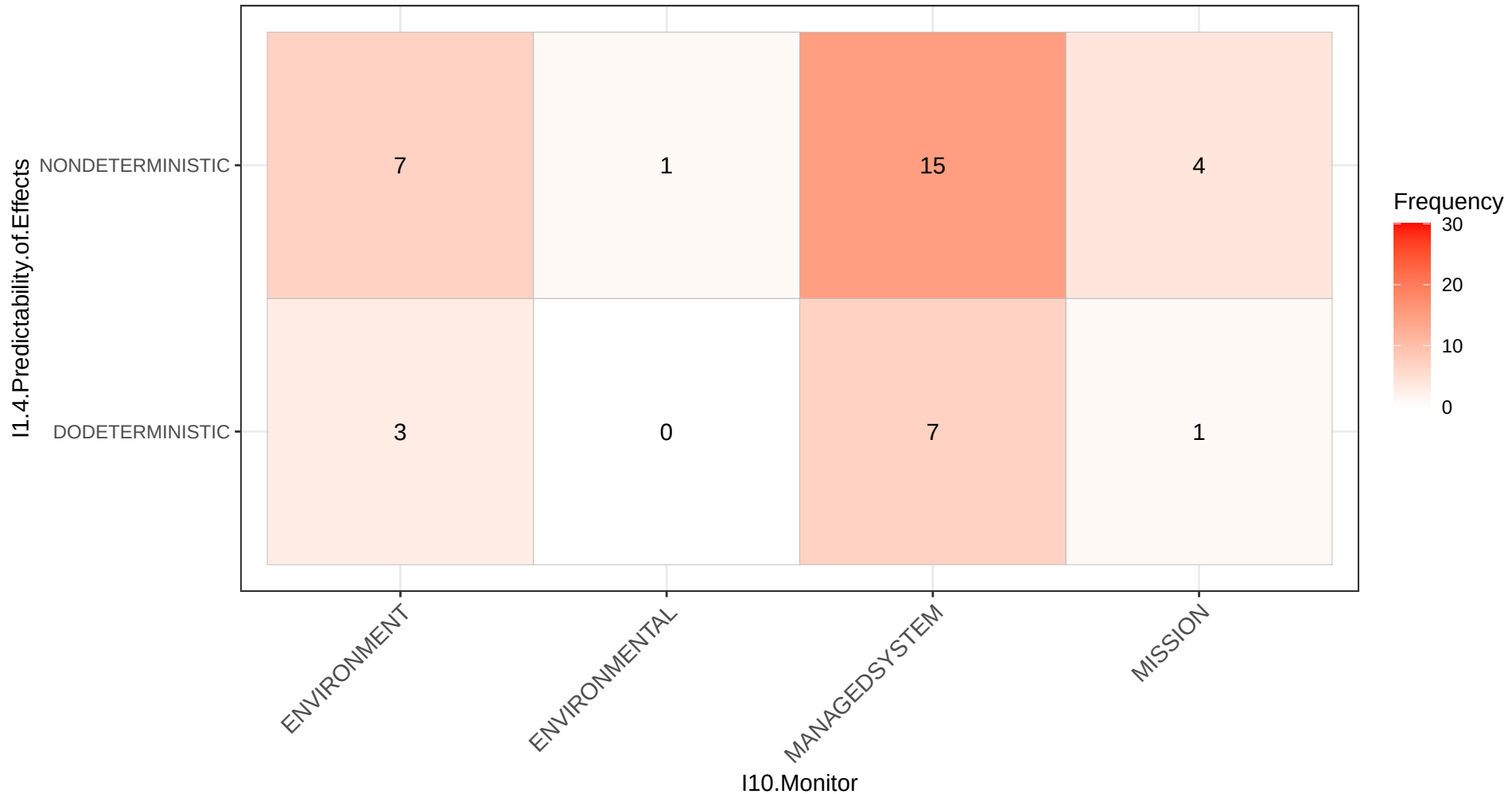
I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects _____ I9.Adap..Logic



I9.Adap..Logic

I1.4.Predictability.of.Effects_____I10.Monitor



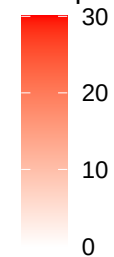
I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I11.Analyze

NONDETERMINISTIC

DODETERMINISTIC

Frequency



ANALYZINGAGGREGATINGDATA

COMPARISONTOHRESHOLDS

DONEDURINGPLAN

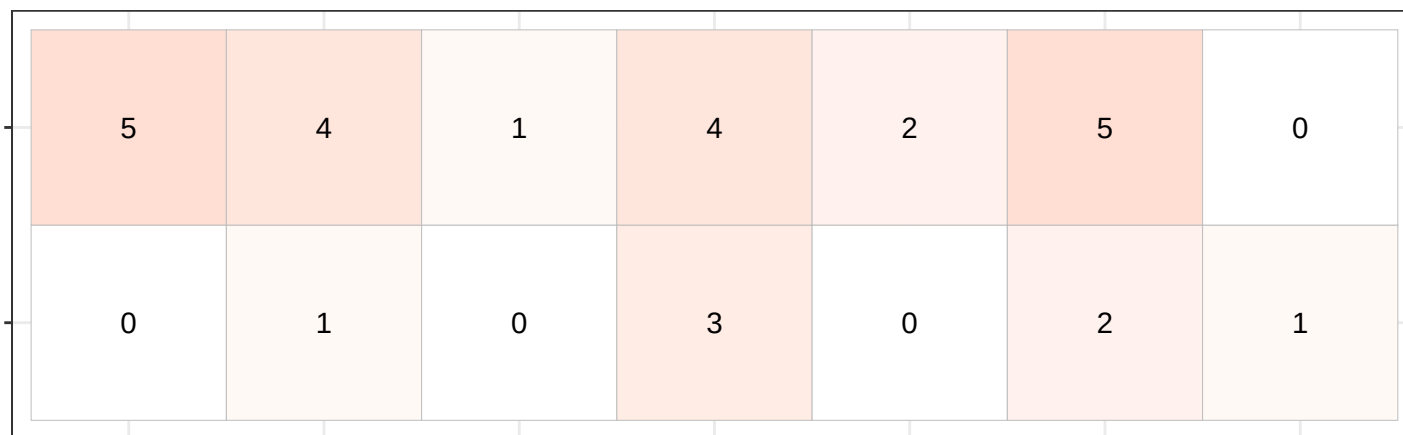
LOGICALINFERENCE

OTHER

SYSTEMSTATEANOMALYDETECTION

TASKUSERDRIVEN

I11.Analyze



I1.4.Predictability.of.Effects_____I12.Plan

I1.4.Predictability.of.Effects

NONDETERMINISTIC

9

8

2

DODETERMINISTIC

4

6

0

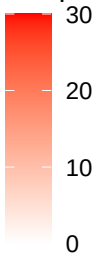
DETERMININGTHEOPTIMALCHOICE

RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

I12.Plan

Frequency



I1.4.Predictability.of.Effects_____I13.Execute

I1.4.Predictability.of.Effects

NONDETERMINISTIC

DODETERMINISTIC

Frequency

30

20

10

0

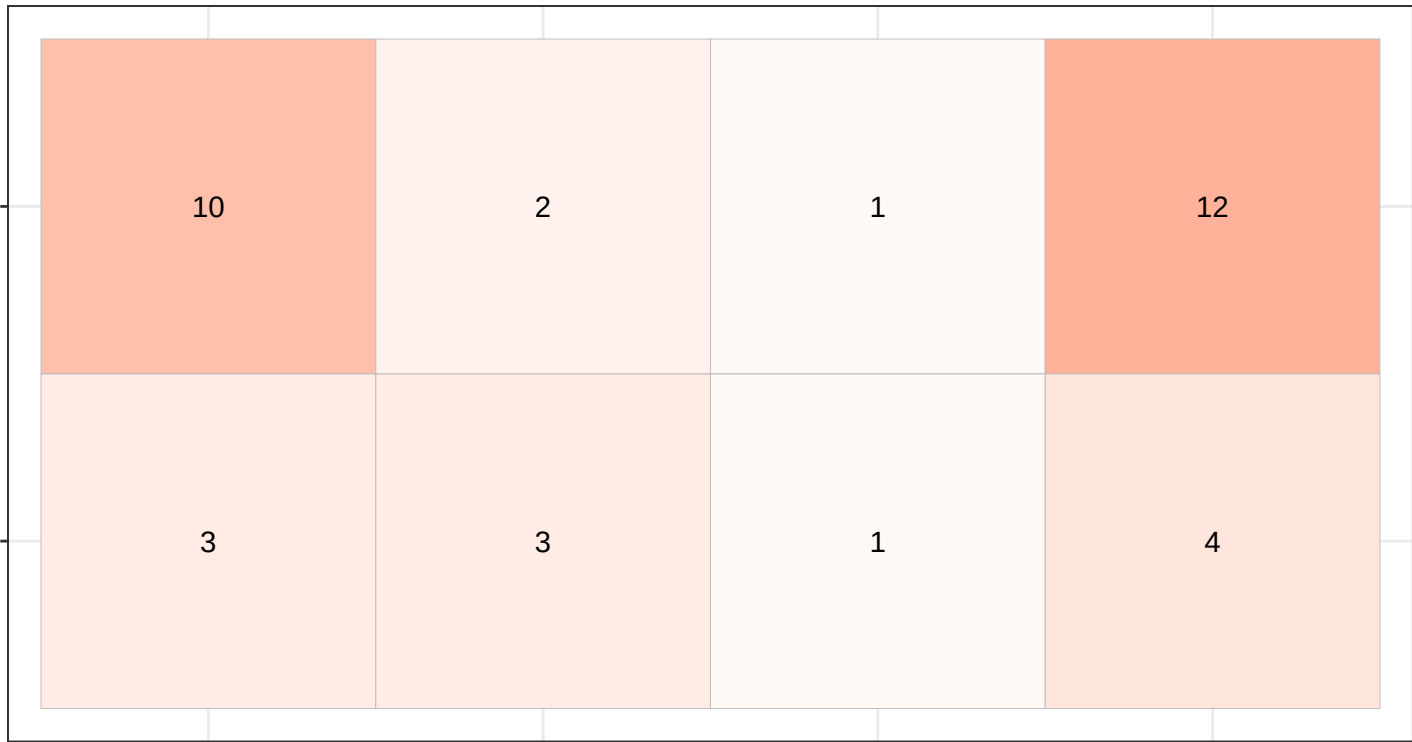
ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPS BETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

I13.Execute



I1.4.Predictability.of.Effects

I1.4.Predictability.of.Effects_____I14.Knowledge

NONDETERMINISTIC

DODETERMINISTIC

BEHAVIORMODEL

COMPONENTMODEL

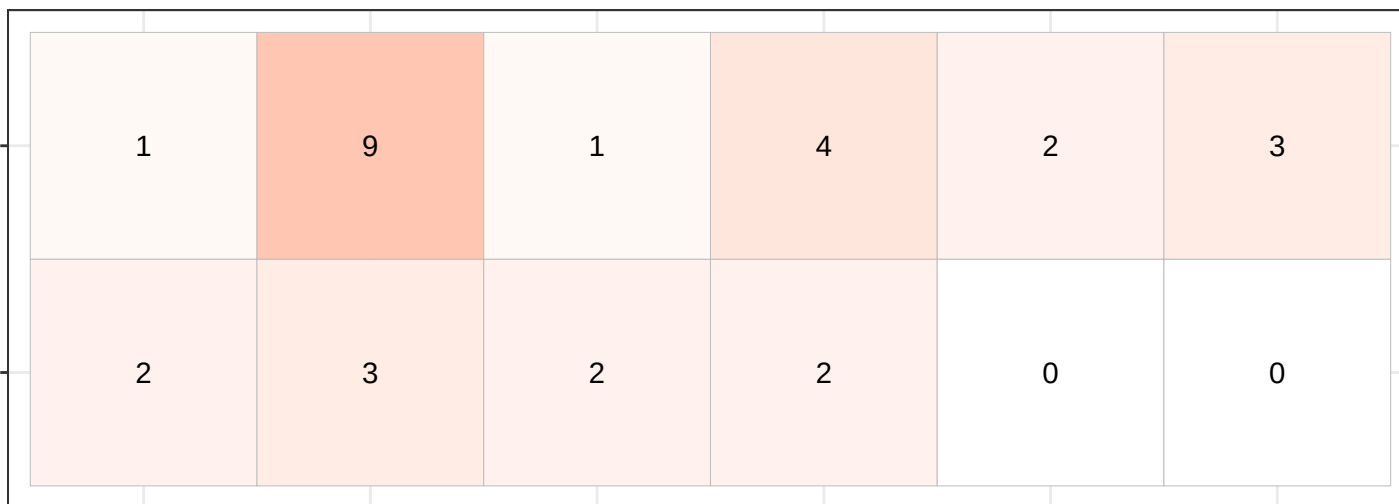
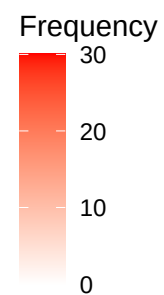
GRAMMARDSL

KNOWLEDGEREPRESENTATION

OTHER

VARIABILITYMODEL

I14.Knowledge



I1.4.Overhead.of.Effects_____I1.4.Resilience.of.Effects

I1.4.Overhead.of.Effects

INSIGNIFICANT

2

3

0

DOSIGNIFICANT

2

1

1

DEPENDENT

8

4

1

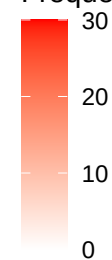
DEPENDENT

DORESILIENT

IRRESILIENT

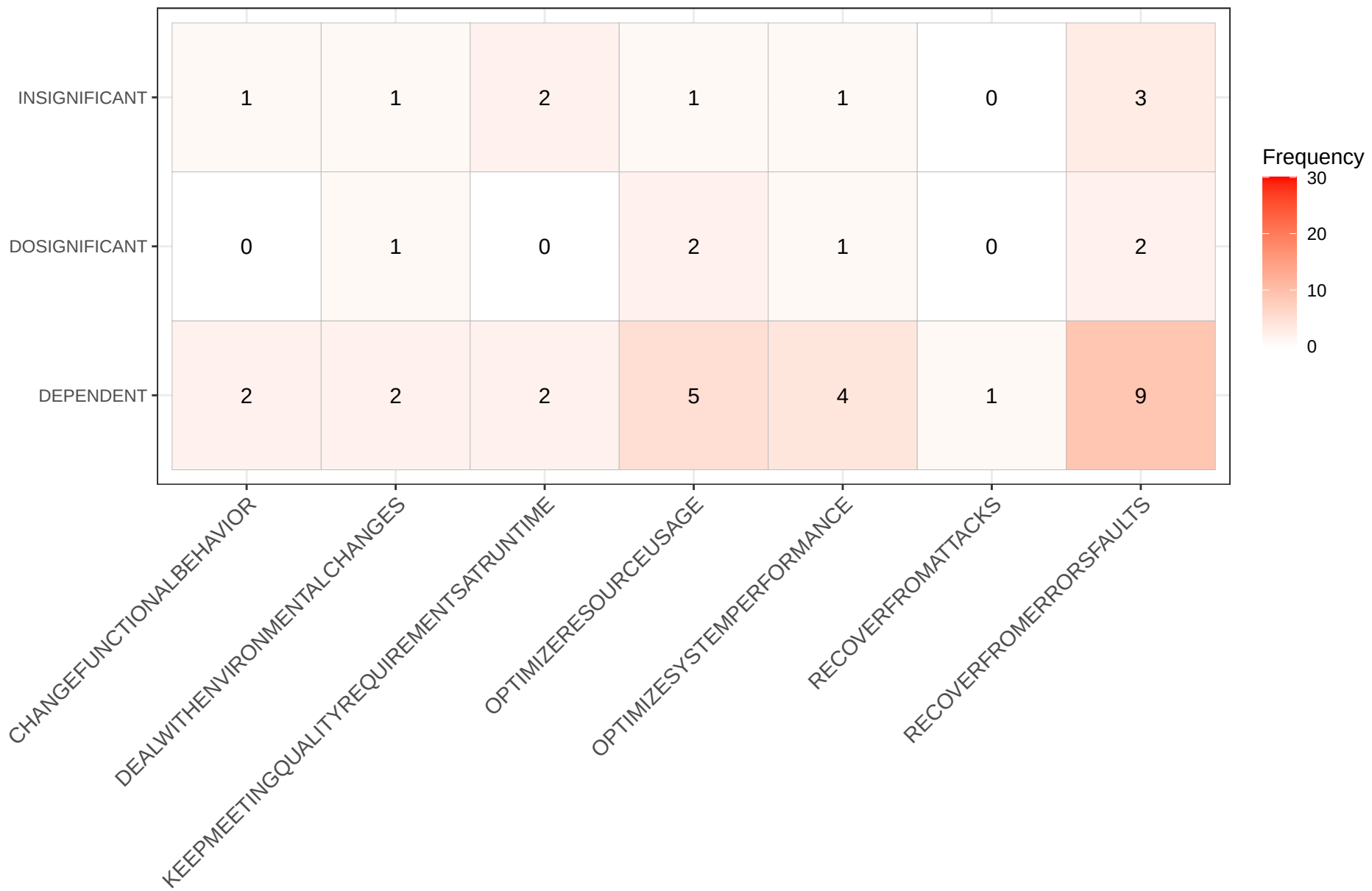
I1.4.Resilience.of.Effects

Frequency



I1.4.Overhead.of.Effects_____I2.Adap..Purpose

I1.4.Overhead.of.Effects

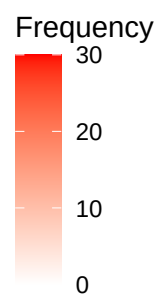


I2.Adap..Purpose

I1.4.Overhead.of.Effects

I1.4.Overhead.of.Effects_____I3.Robot.Type

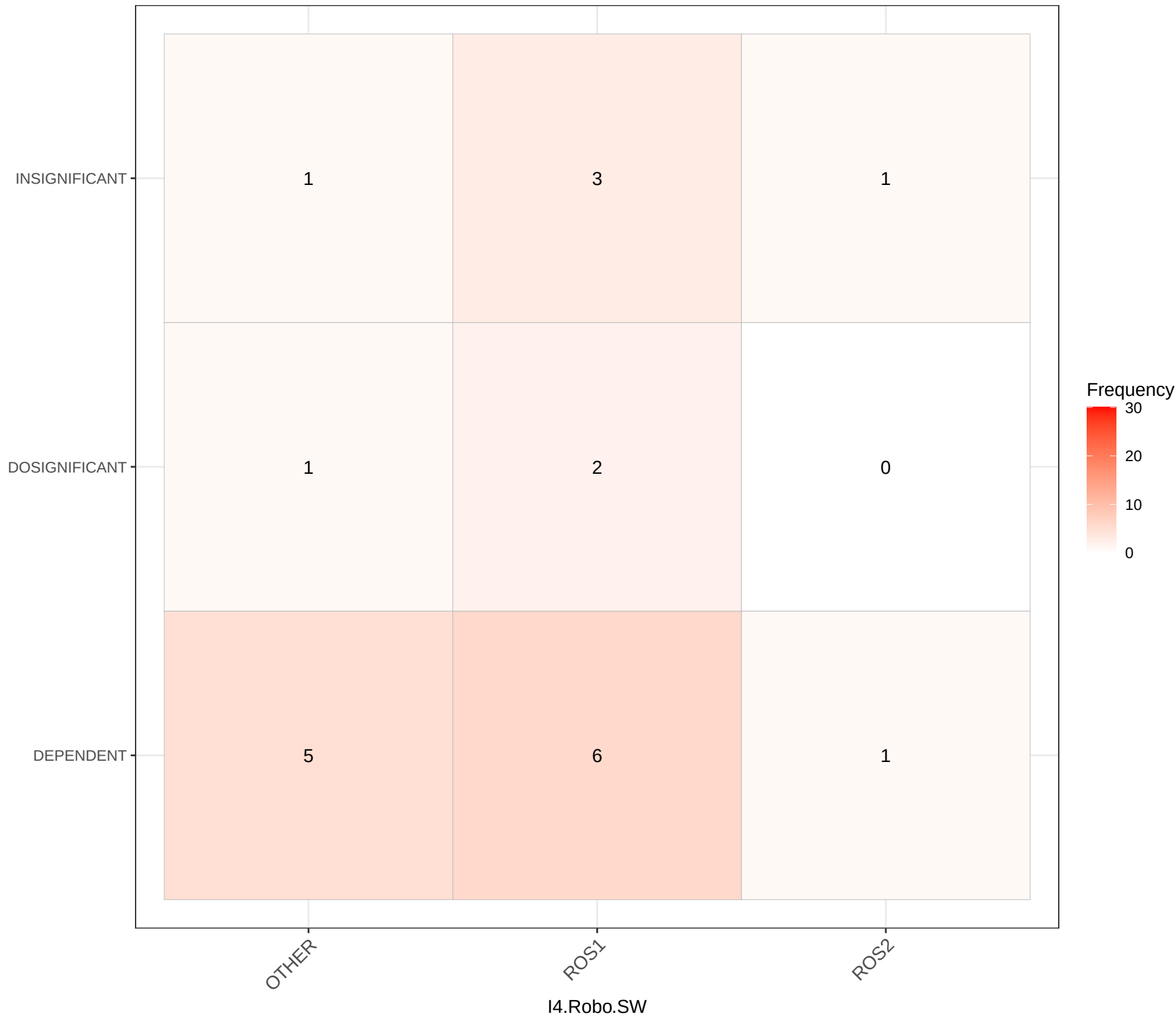
INSIGNIFICANT	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	1	1	1
DOSIGNIFICANT	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	1	0	1	0	0	0	0	0	
DEPENDENT	0	0	0	1	2	0	2	1	1	0	1	1	1	1	0	0	2	0	0	0	1	0	0	3	0	0	0
	BLUEOV2	BOXERCLEARPATH	CRAWLERTERMINATORBOT	FIELDMOBILEROBOTS	HETEROGENOUSROBOTS	HEXAII	HEXMANIPULATOR	IROBOTCREATE2	MOBILEROBOTMANIPULATOR	MOBILEROBOTS	MOBILERTERRESTRIAL	MOBILEROBOTTIAGO	MOBILESERVICECERBOT	MULTIPLEHXRTRIAL	NAOROBOT	PIONEER3DX	QUADROCOPTER	RESCUE	SINGLESERVOINGROTATIONROBOT	TEDUSARTERRESTRIALSEARCH	TRIGLIDEINDUSTRIALASSEMBLY	TURTLEBOT	TWOINDUSTRIALROBOTS	WAREHOUSEDELIVERYROBOT	WHICHISANINDUSTRIALAGV	TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	



I3.Robot.Type

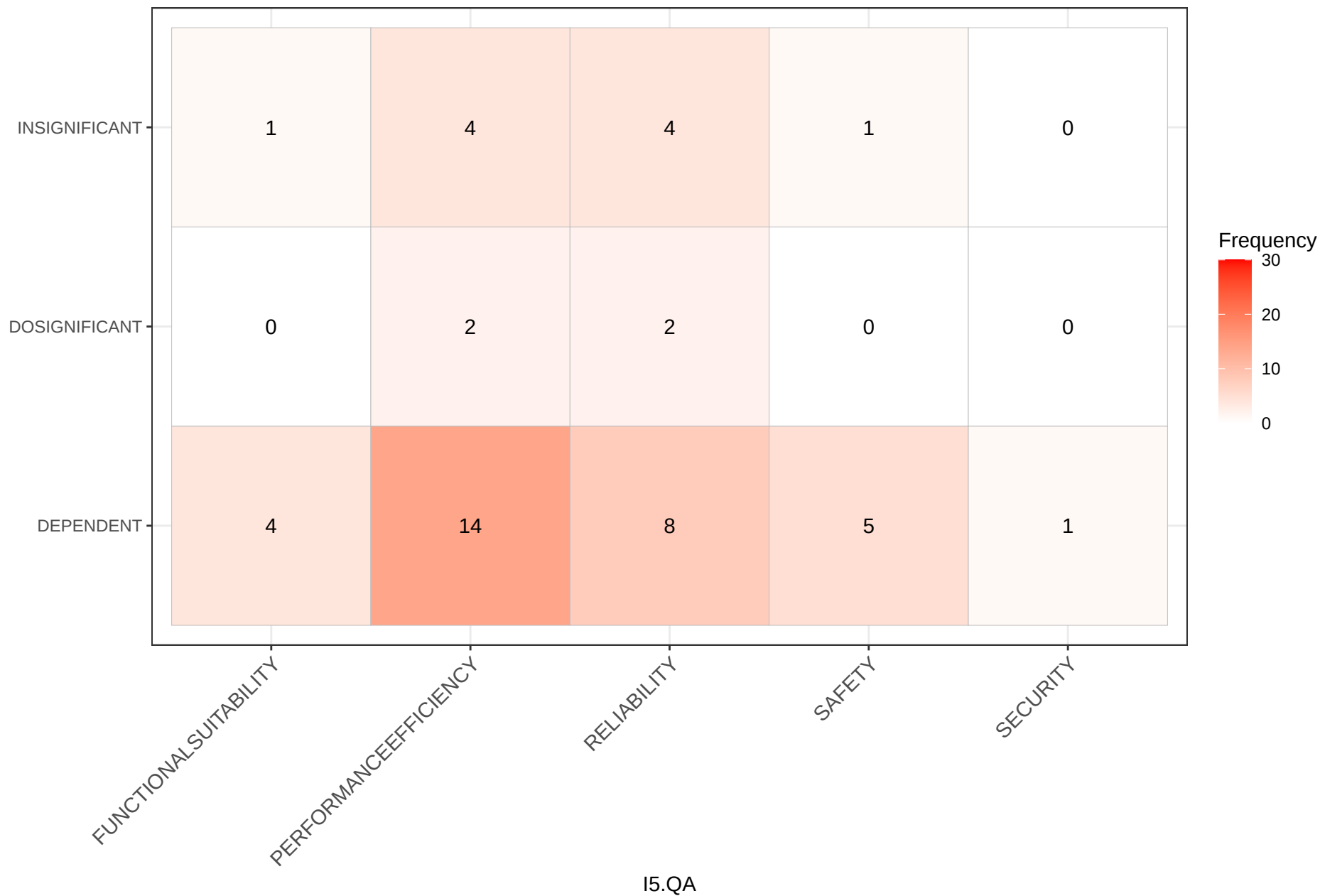
I1.4.Overhead.of.Effects_____I4.Robo.SW

I1.4.Overhead.of.Effects

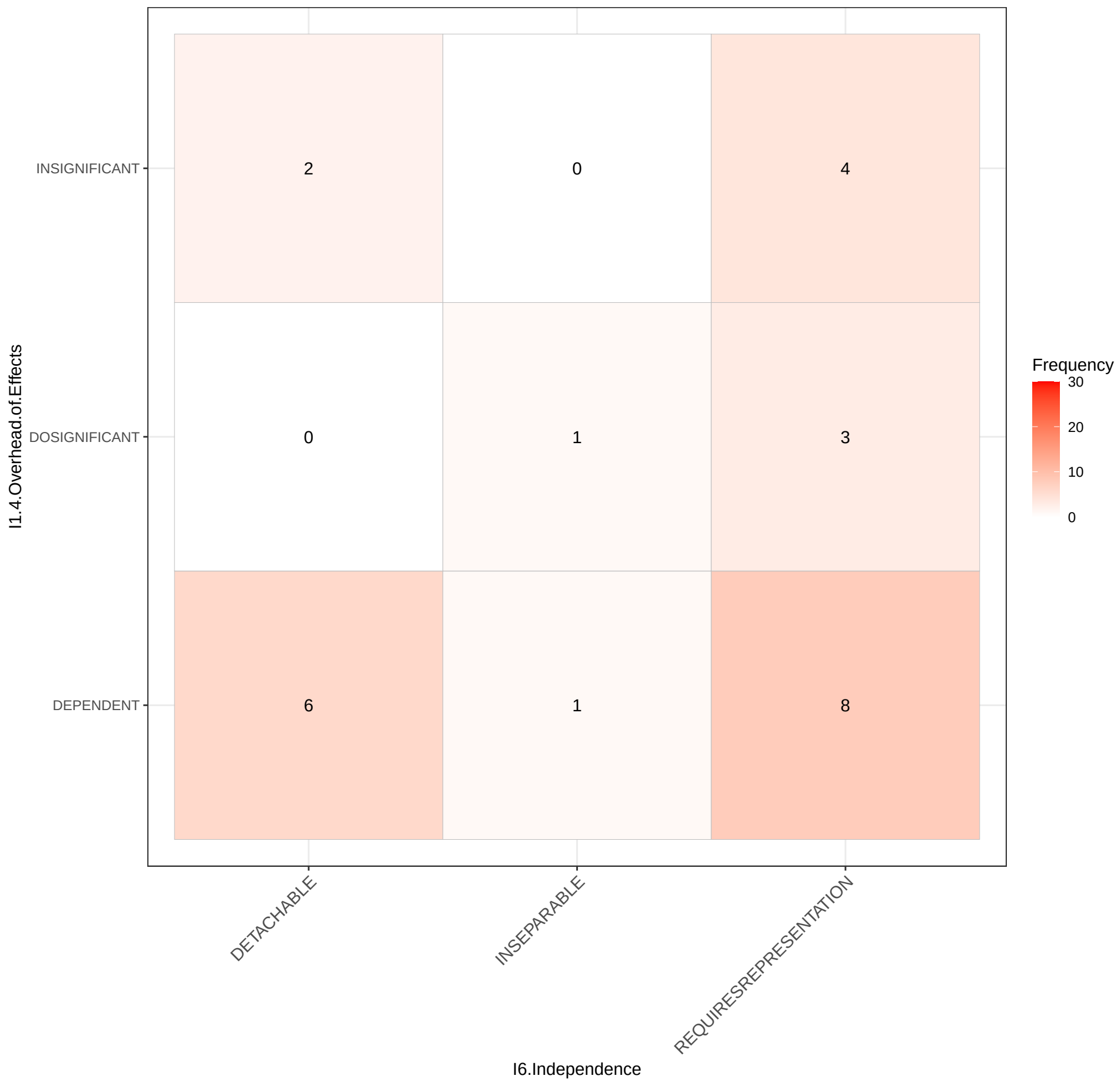


I1.4.Overhead.of.Effects_____I5.QA

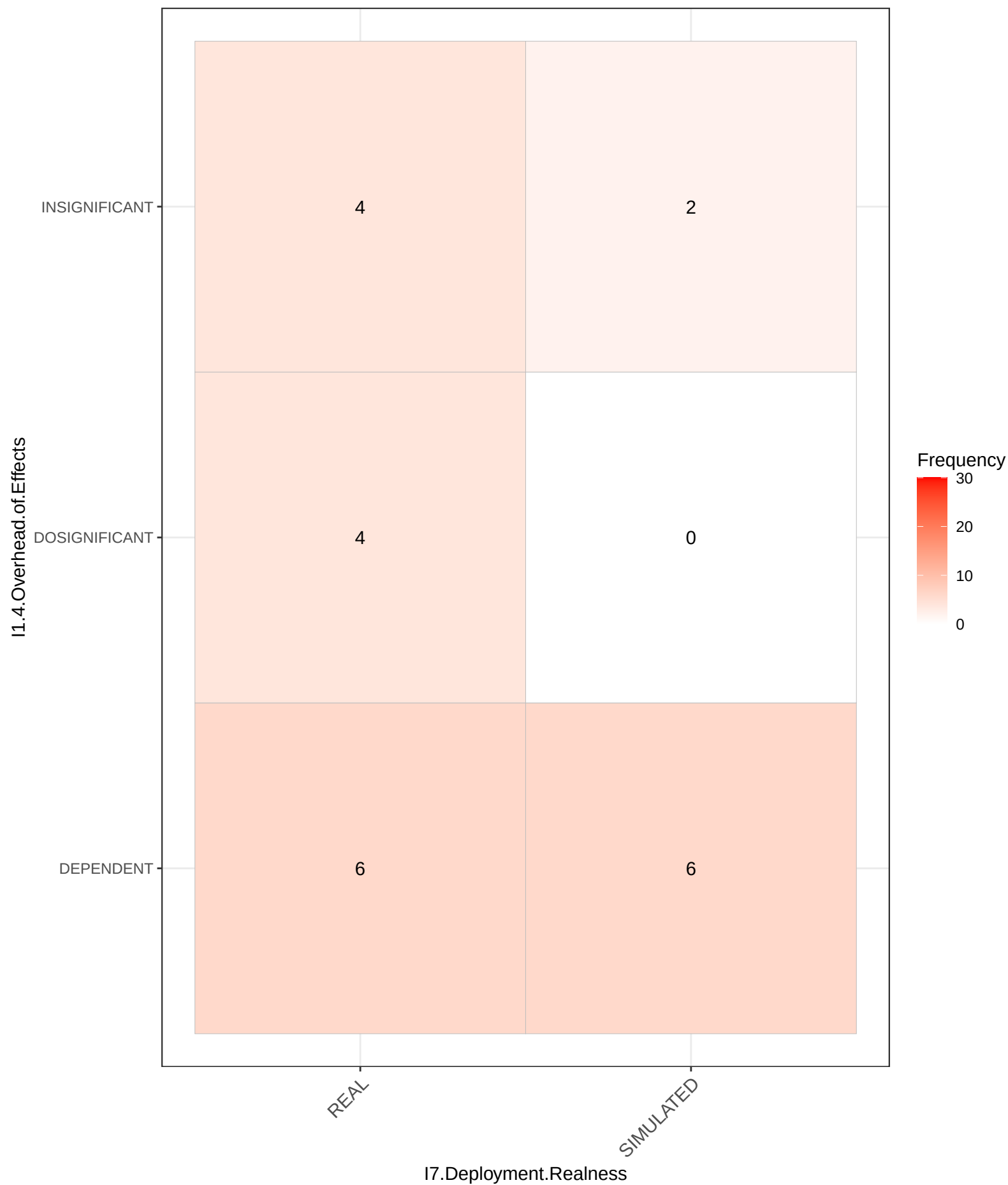
I1.4.Overhead.of.Effects



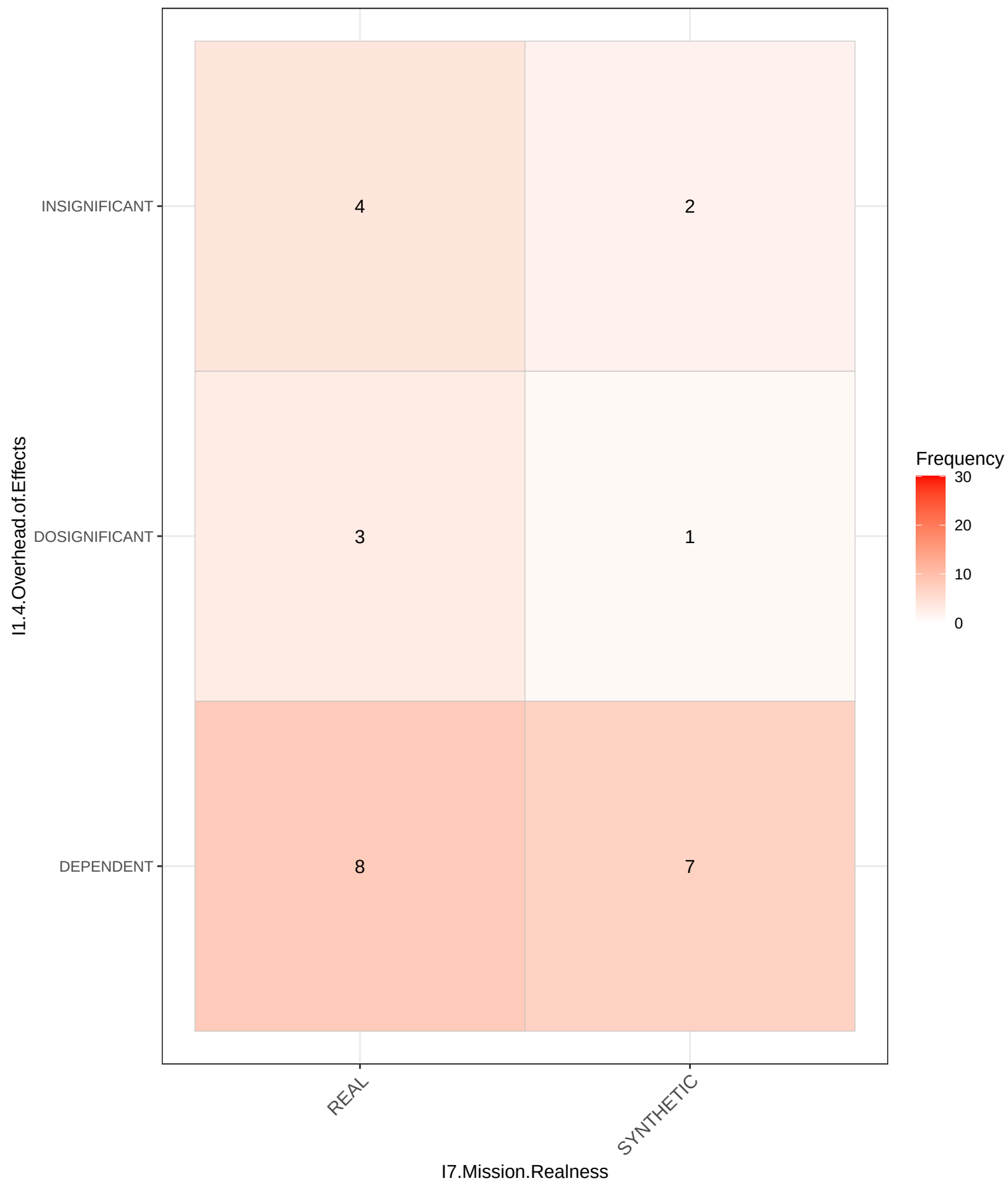
I1.4.Overhead.of.Effects_____I6.Independence



I1.4.Overhead.of.Effects_____I7.Deployment.Realness

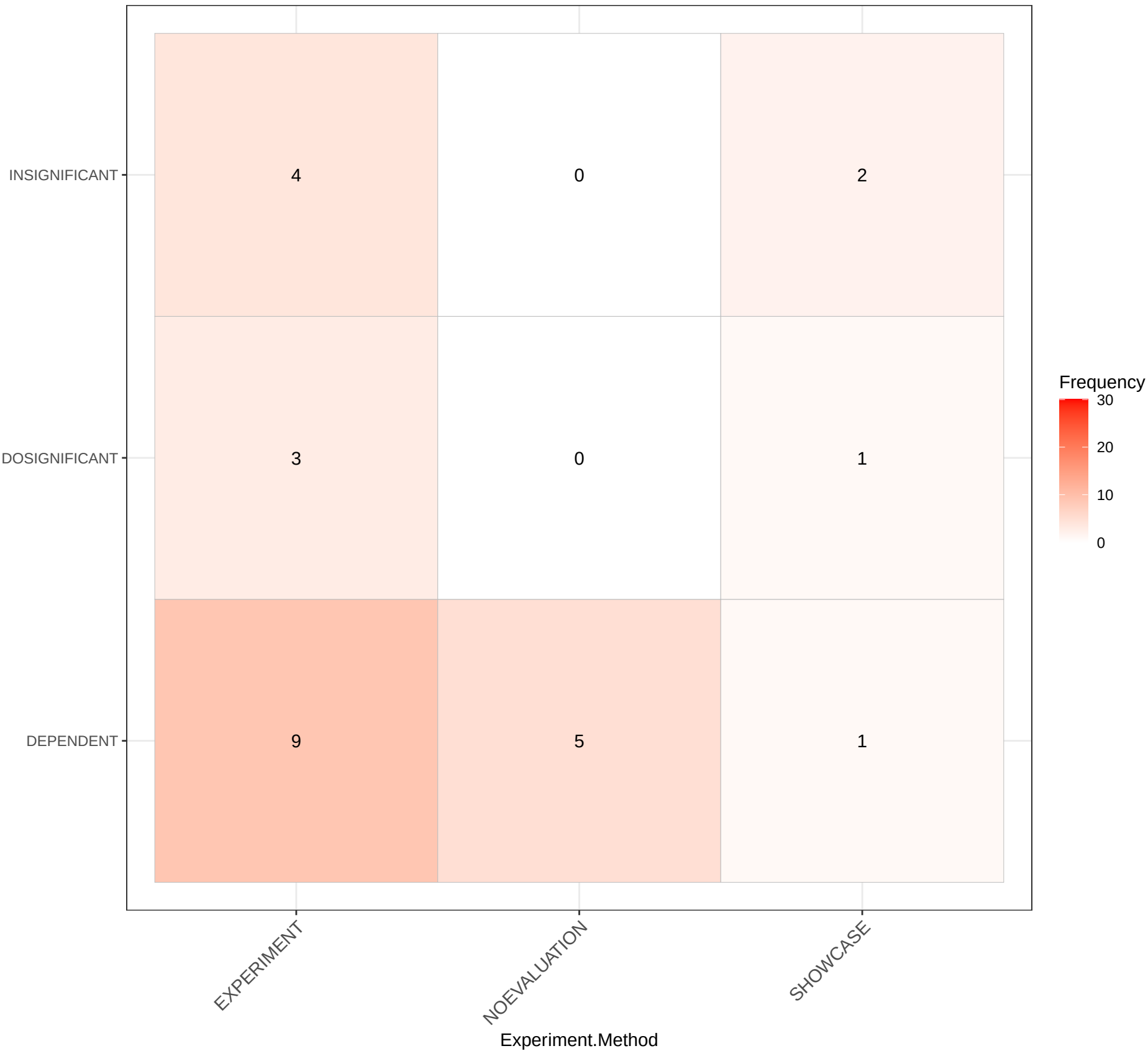


I1.4.Overhead.of.Effects_____I7.Mission.Realness



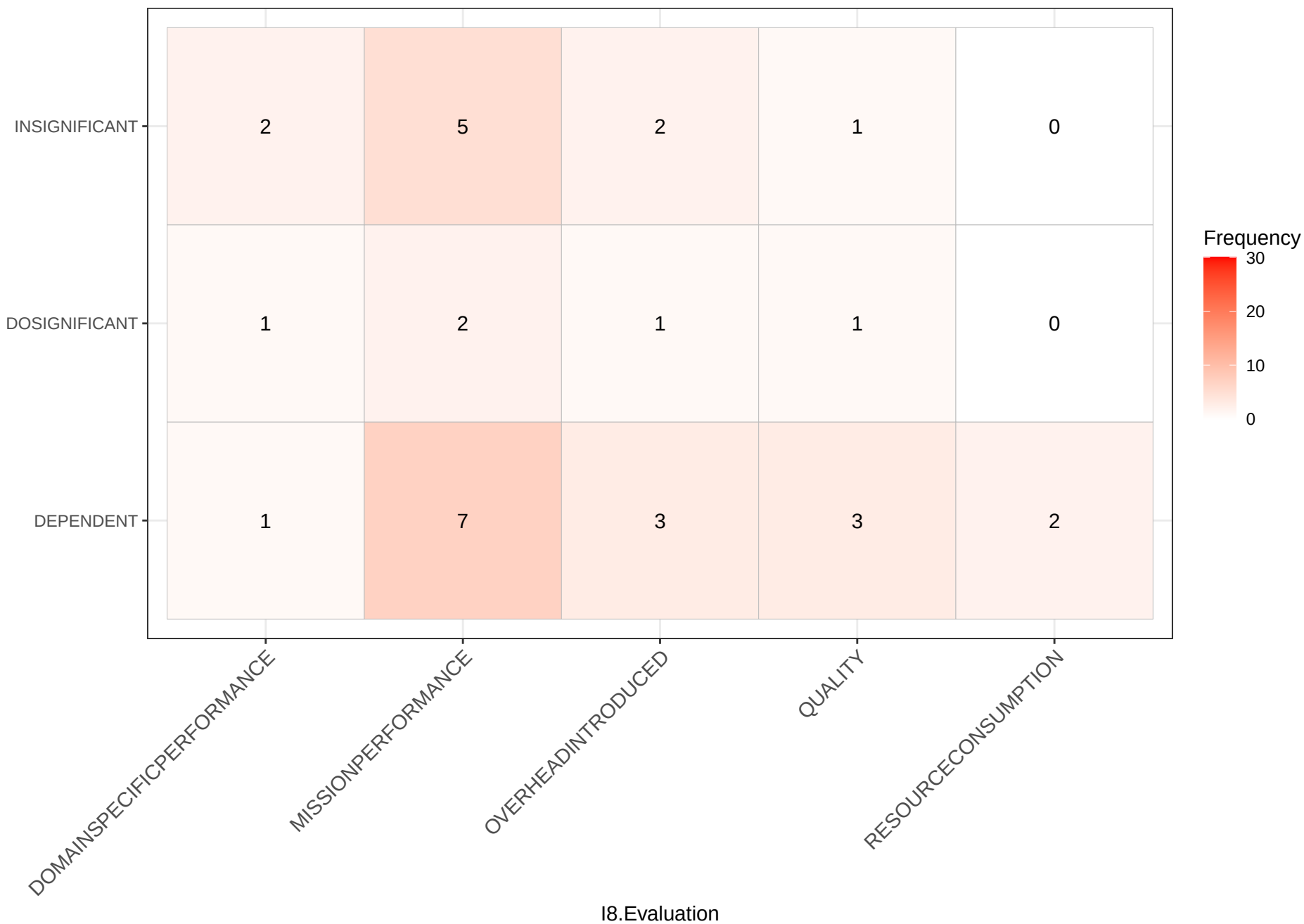
I1.4.Overhead.of.Effects_____Experiment.Method

I1.4.Overhead.of.Effects

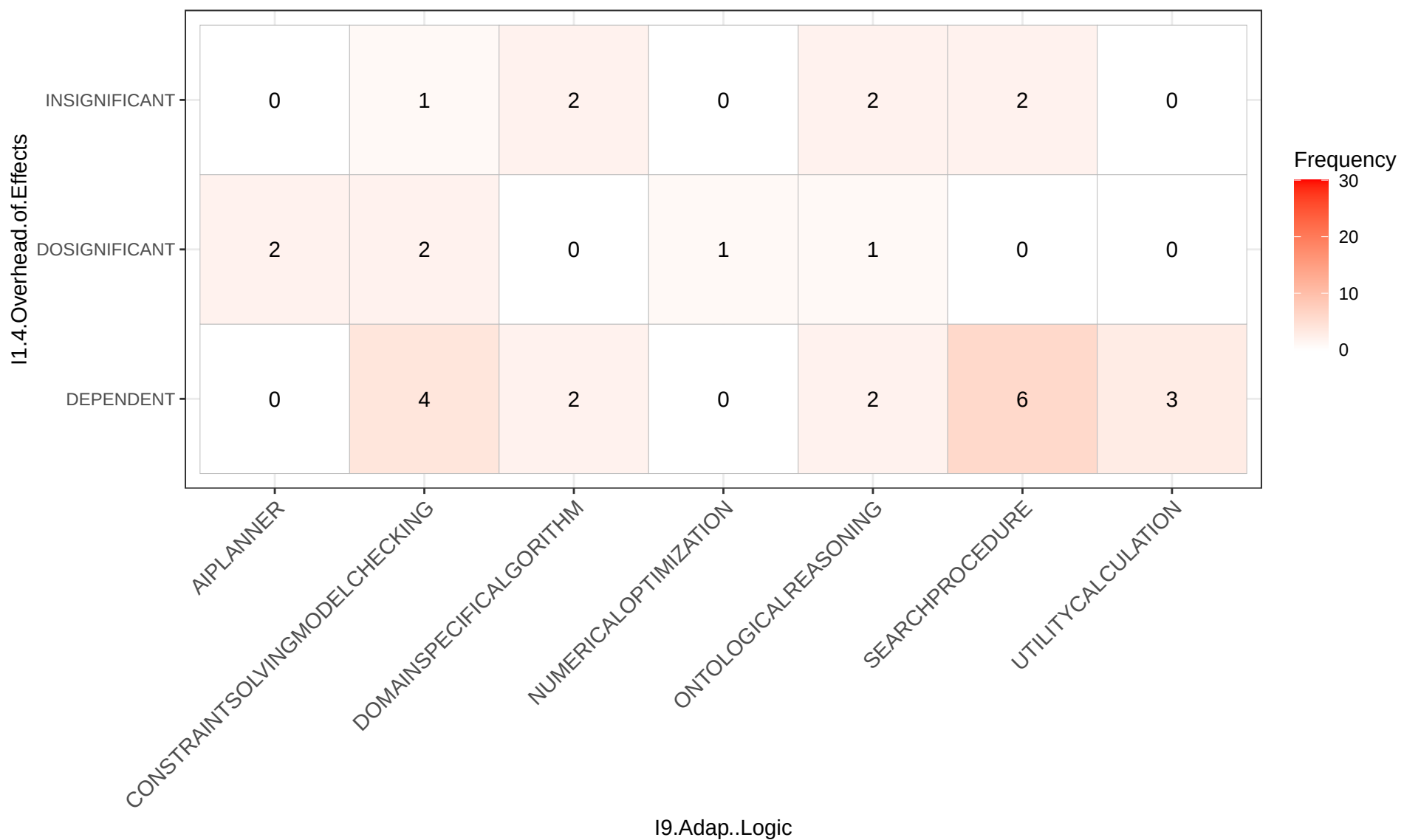


I1.4.Overhead.of.Effects_____I8.Evaluation

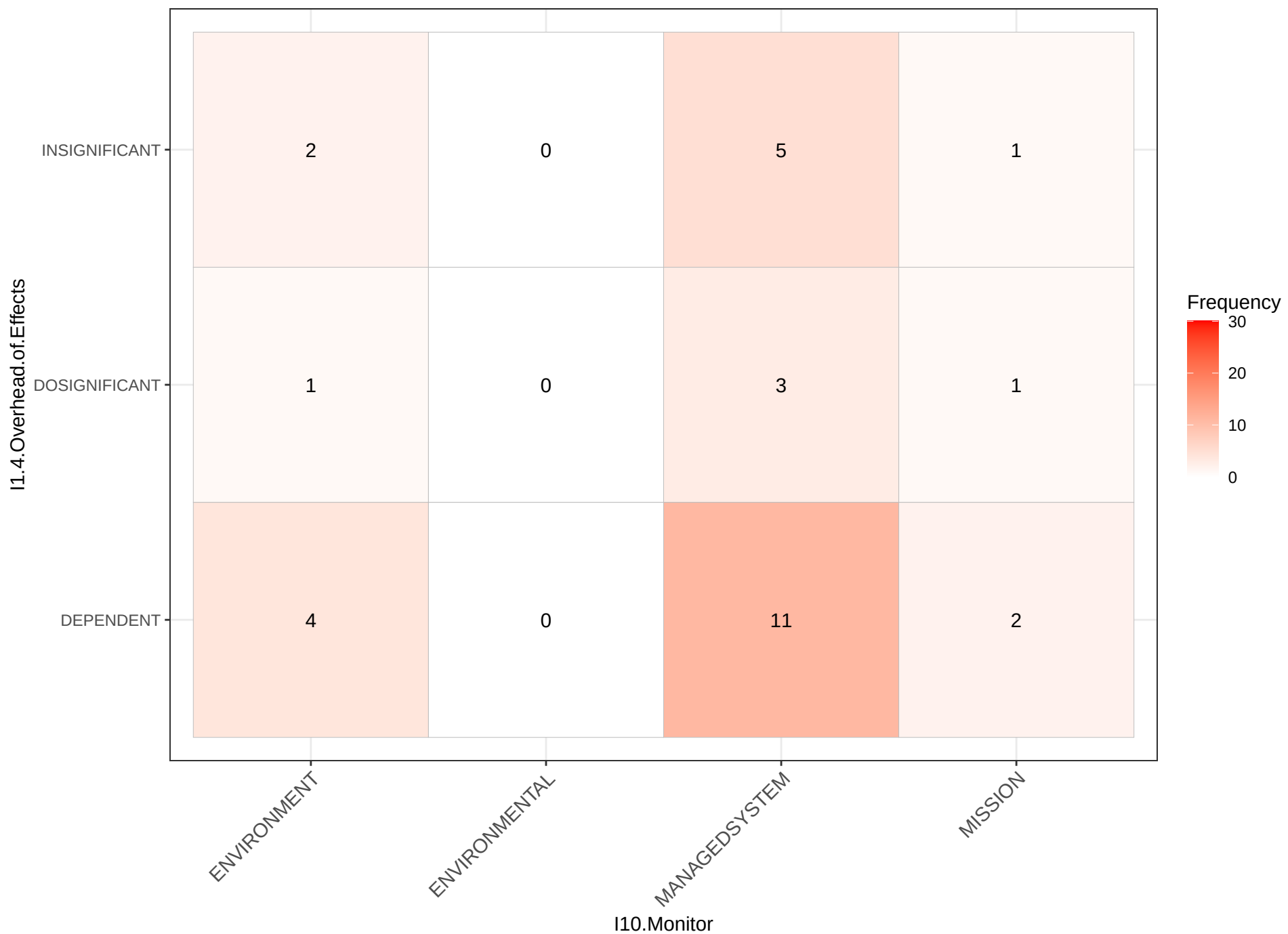
I1.4.Overhead.of.Effects



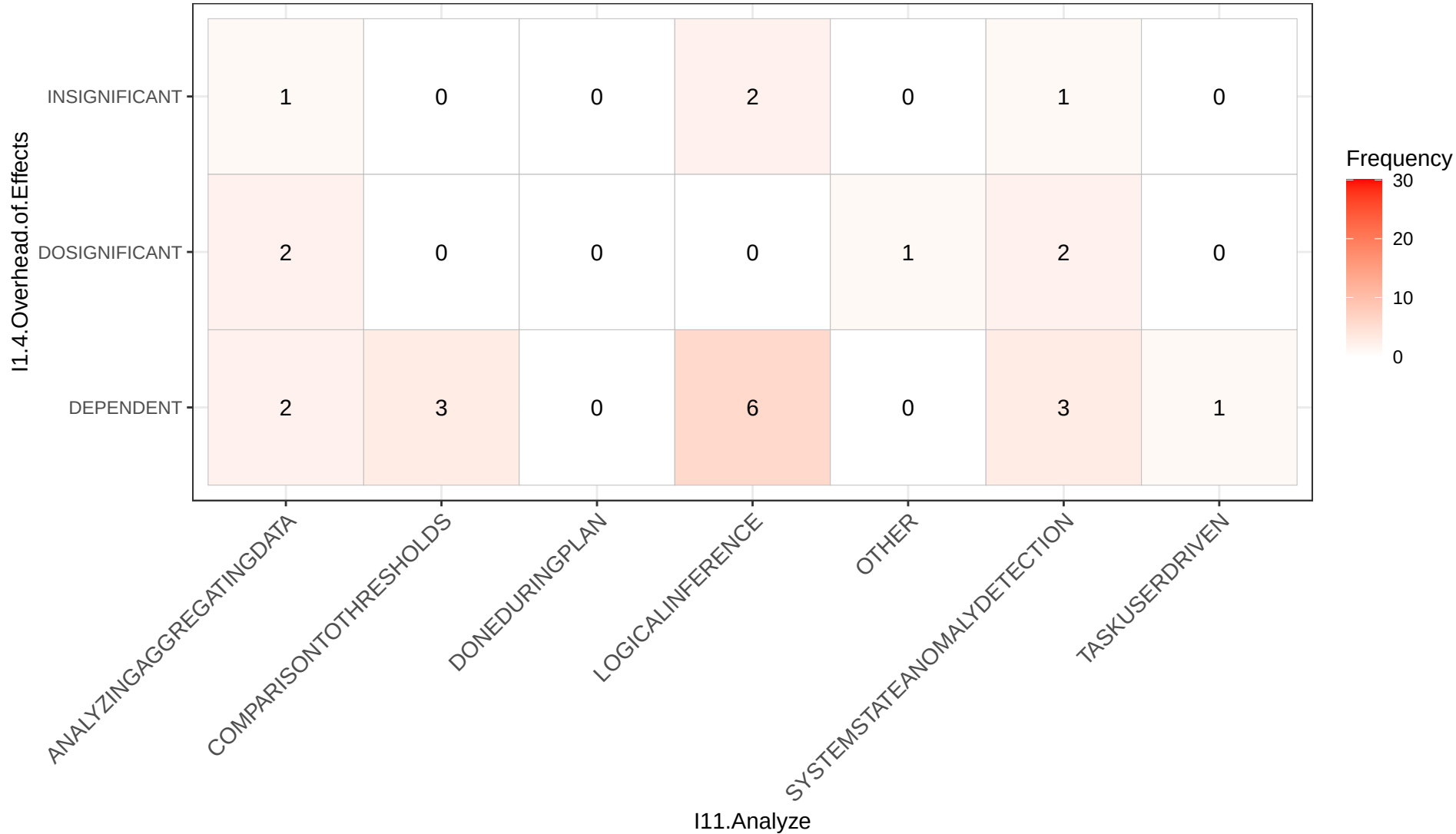
I1.4.Overhead.of.Effects_____I9.Adap..Logic



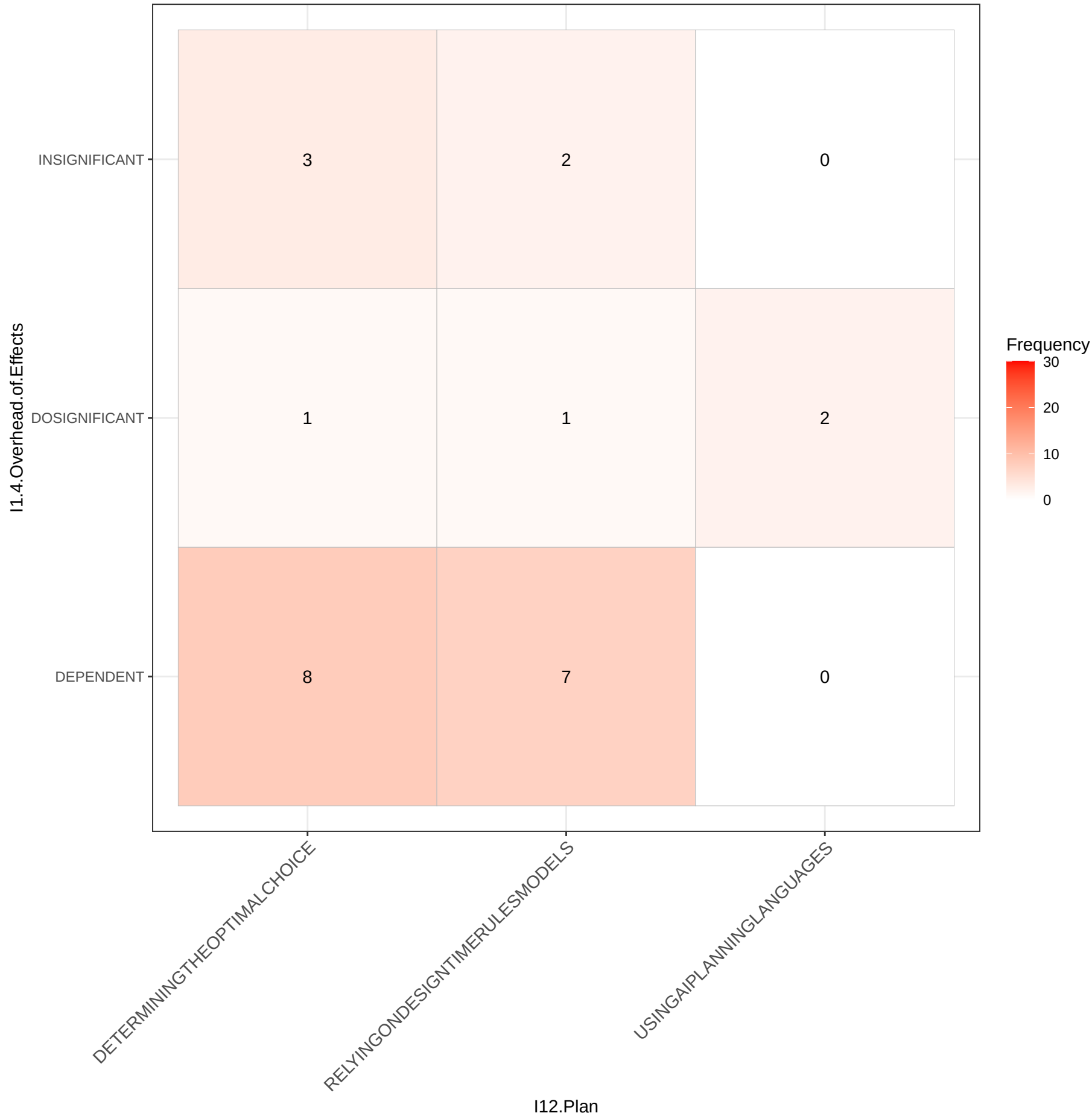
I1.4.Overhead.of.Effects_____I10.Monitor



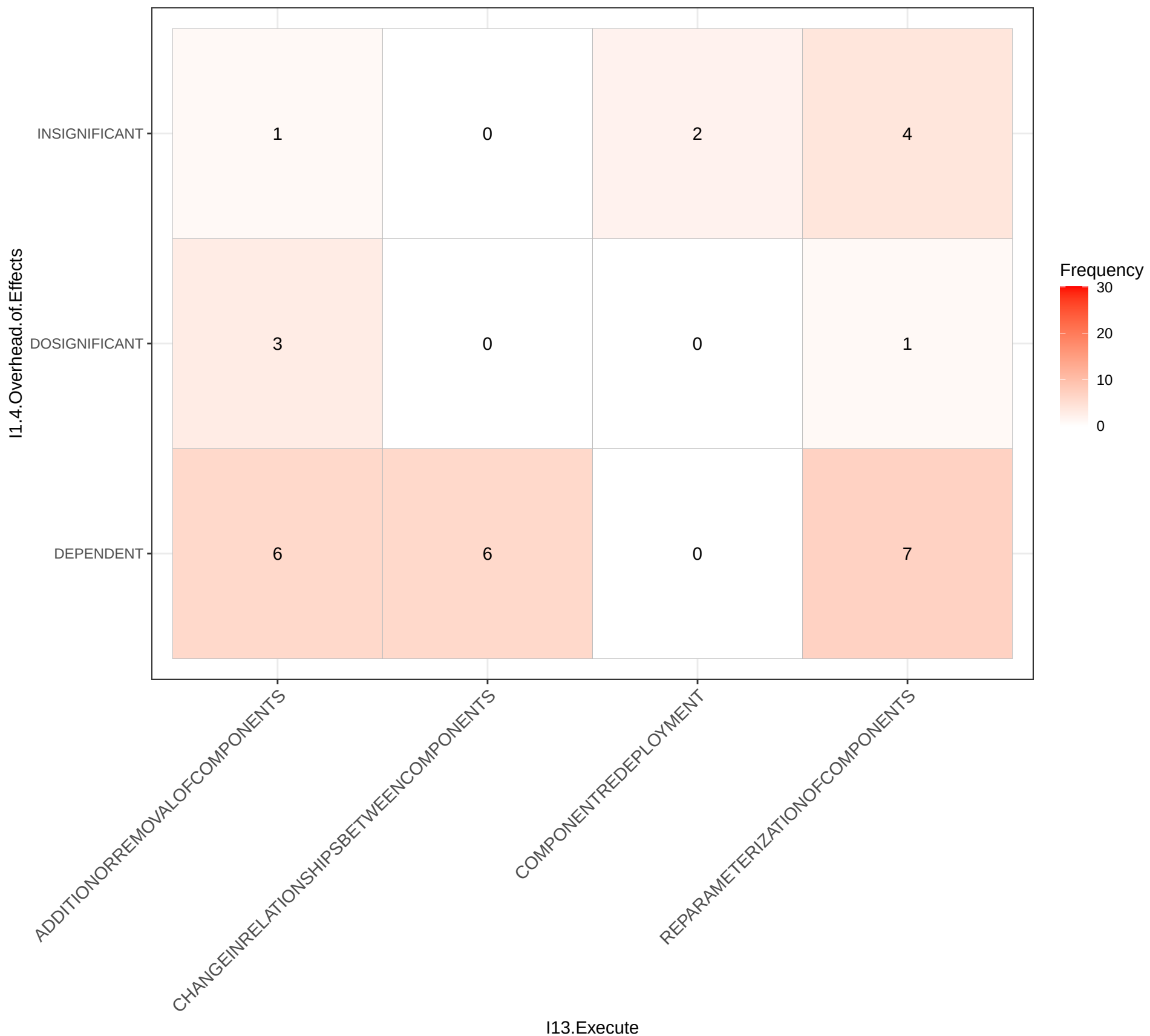
I1.4.Overhead.of.Effects_____I11.Analyze



I1.4.Overhead.of.Effects_____I12.Plan

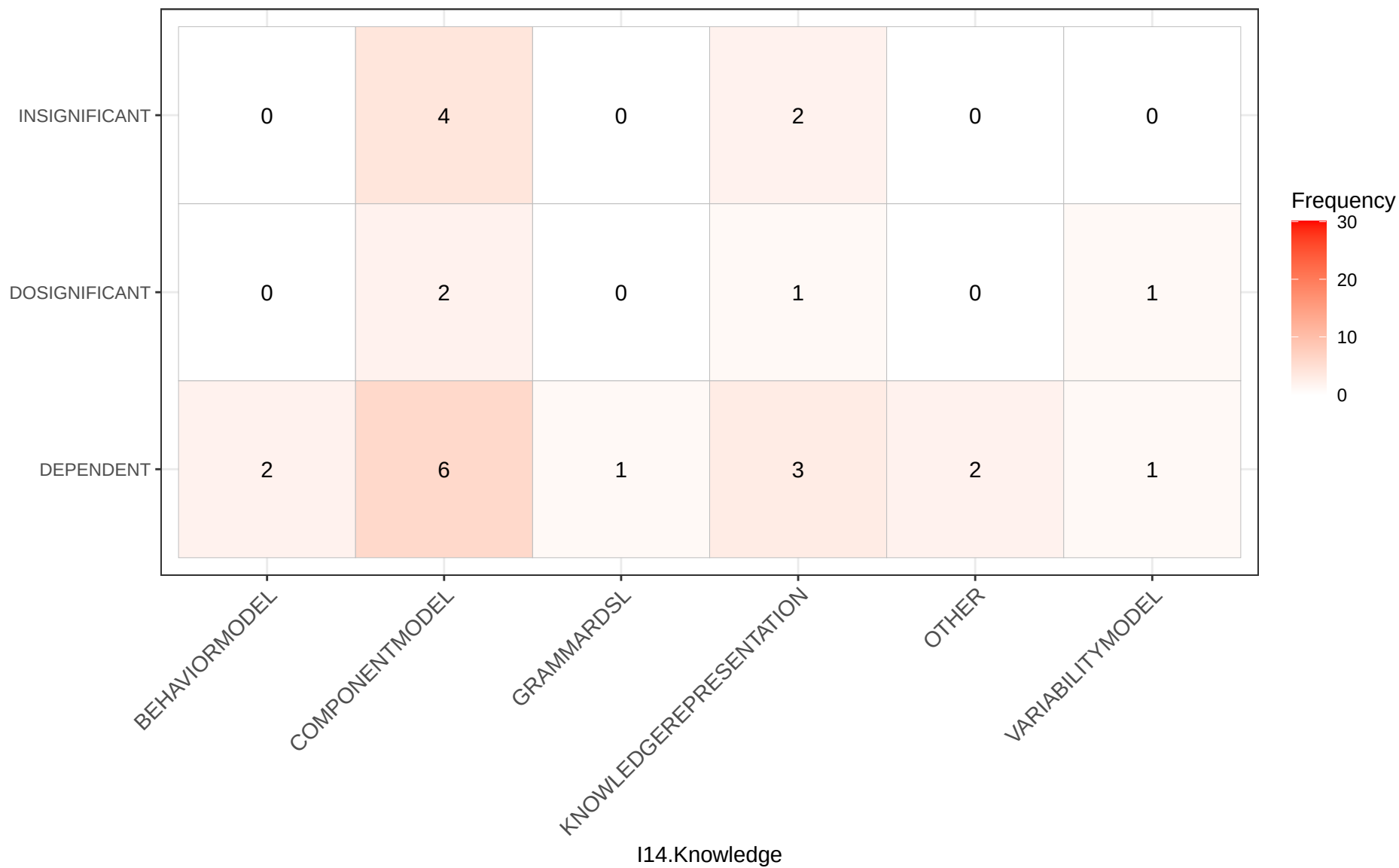


I1.4.Overhead.of.Effects_____I13.Execute

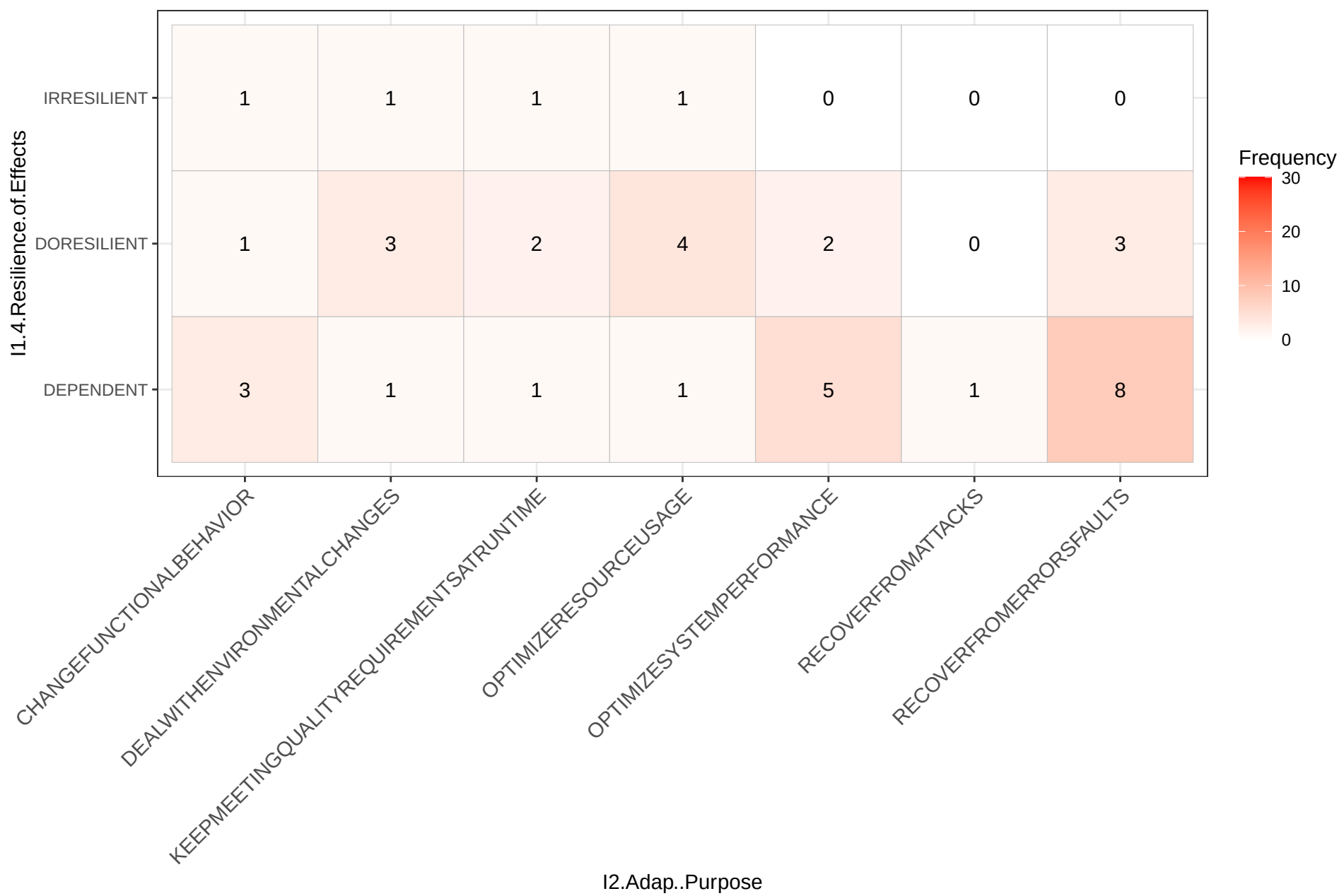


I1.4.Overhead.of.Effects

I1.4.Overhead.of.Effects_____I14.Knowledge



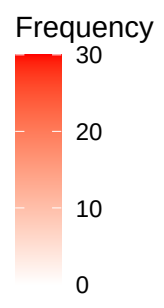
I1.4.Resilience.of.Effects_____I2.Adap..Purpose



I1.4.Resilience.of.Effects

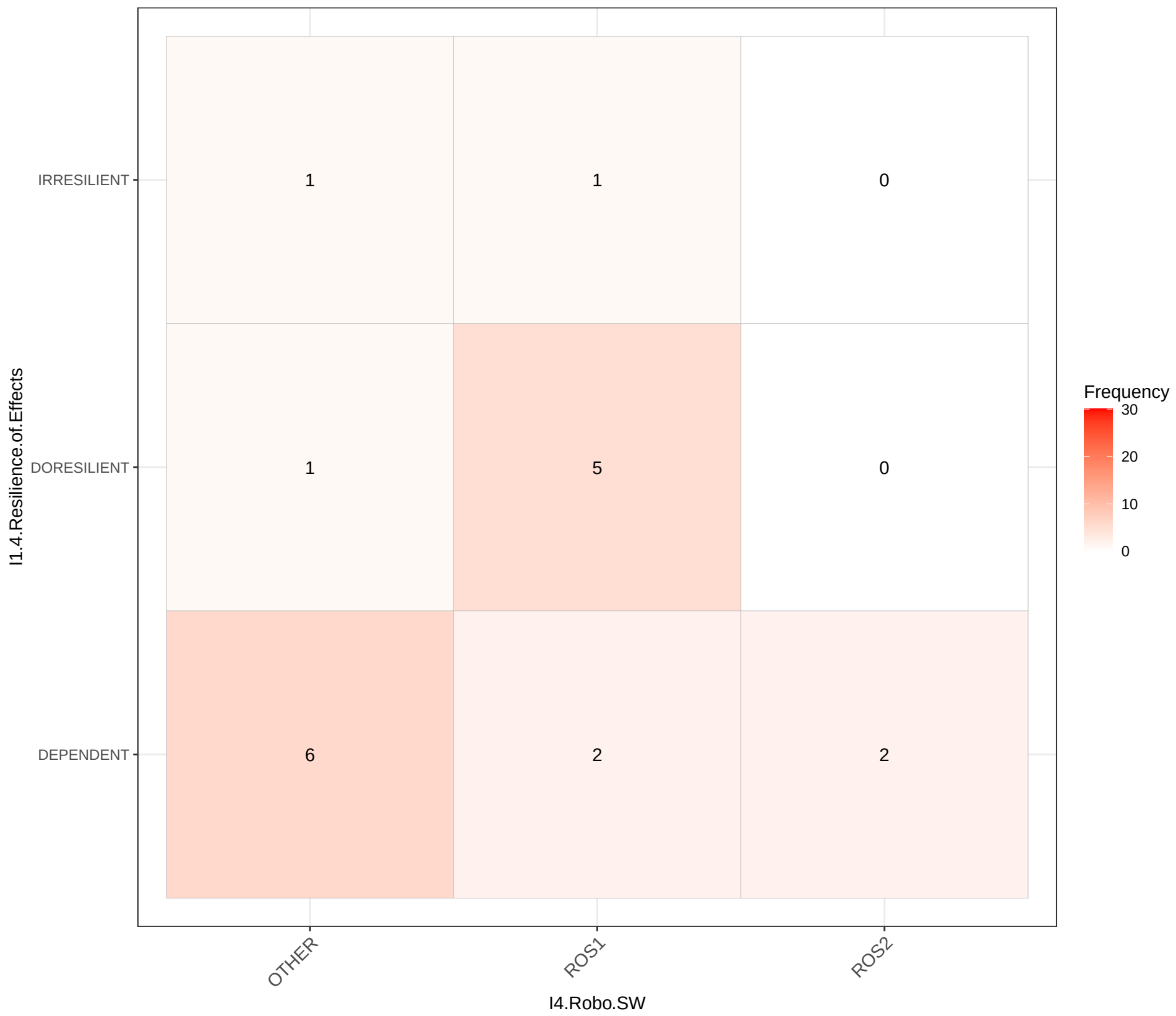
I1.4.Resilience.of.Effects_____I3.Robot.Type

IRRESILIENT	0	0	1	0	0	0	0	0	0	0	1	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0
DORESILIENT	0	1	0	0	2	1	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1	3	0	0	0	1
DEPENDENT	1	0	0	1	0	0	2	1	1	0	1	0	1	1	0	0	1	0	1	1	1	1	0	1	0	1	0
BLUE ROV2																											
BOXER CLEARPATH																											
CRAWLER TERMINATOR BOT																											
FIELD MOBILEROBOTS																											
HETEROGENOUS ROBOTS																											
INFOTAINMENT ROBOT																											
KUKA LIGHTWEIGHT ROBOT 4LWR4																											
MOBILE SERVICEROBOT																											
HEX MANIPULATOR																											
IROBOT CREATE2																											
MOBILE ROBOT MANIPULATOR																											
MOBILEROBOTTERRESTRIAL																											
MOBILE SERVICE ROBOT																											
MOBILEROBOT TIAGO																											
MSUEVORALLY MOBILE TERRESTRIAL																											
MULTIPLE HEX ROTOR																											
NAO ROBOT																											
PIONEER 3-DX																											
QUADRO COPTER																											
SINGLE SERVING ROTATION ROBOT																											
TEDUS ARTERRESTRIAL SEARCH																											
TRIGLIDE INDUSTRIAL ASSEMBLY																											
TWO CASE STUDIES MOBILE MANIPULATOR AS RUNNING EXAMPLE																											
QUADRO COPTOR FOR EVALUATION																											
TURTLEBOT																											
TWO INDUSTRIAL ROBOTS																											
WAREHOUSE DELIVERY ROBOT																											
WHICH IS AN INDUSTRIAL AGV																											

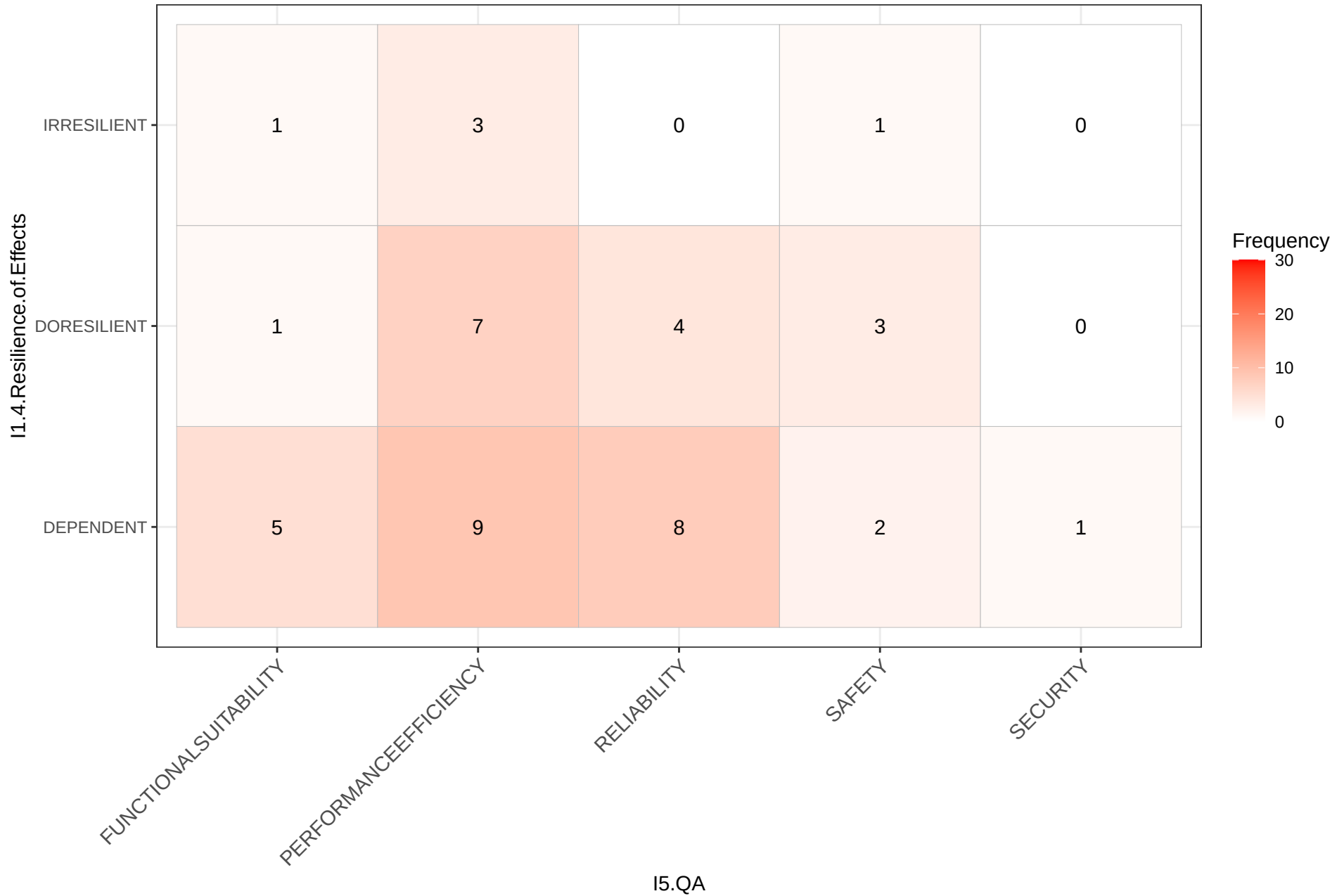


I3.Robot.Type

I1.4.Resilience.of.Effects_____I4.Robo.SW

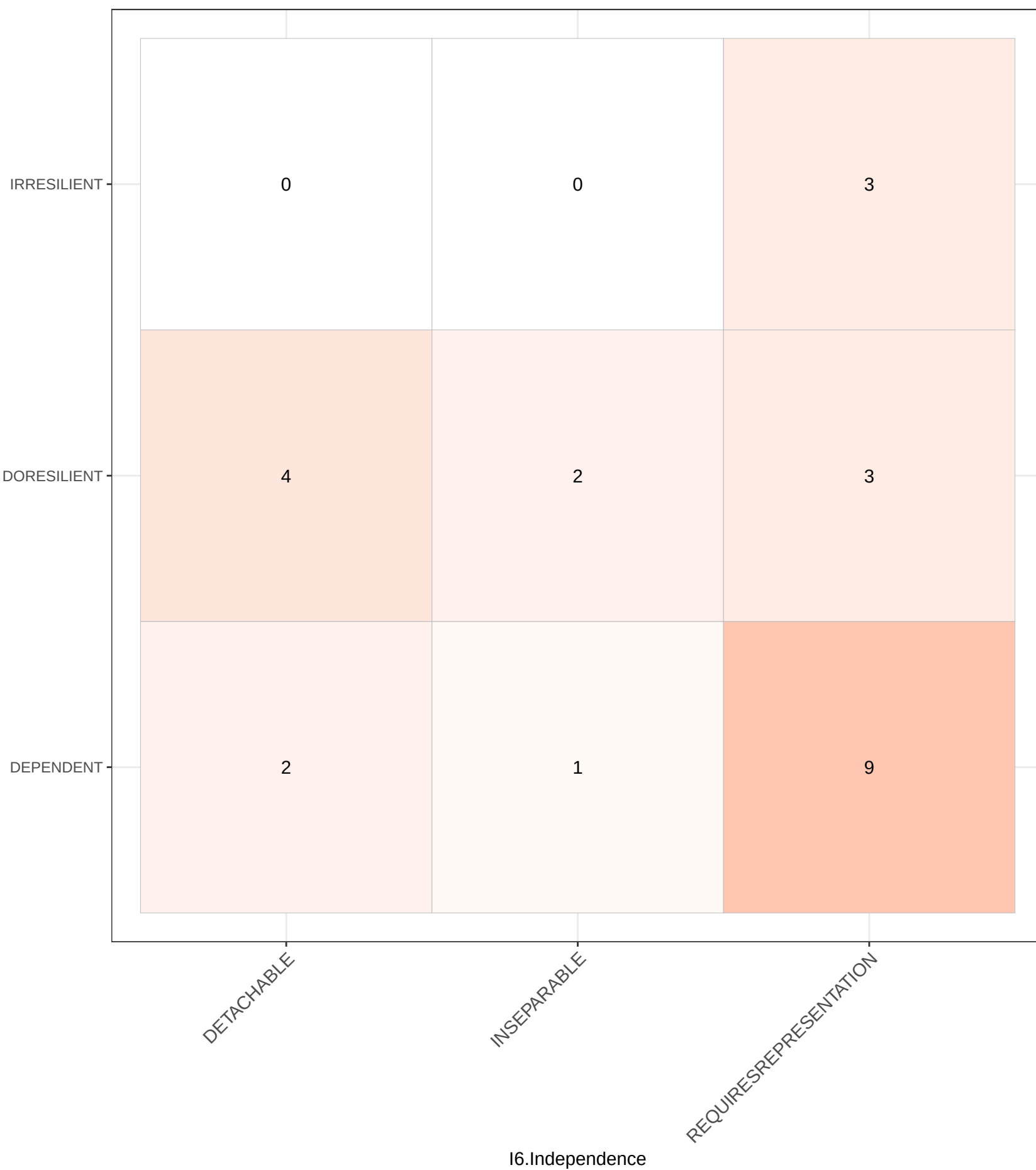


I1.4.Resilience.of.Effects_____I5.QA

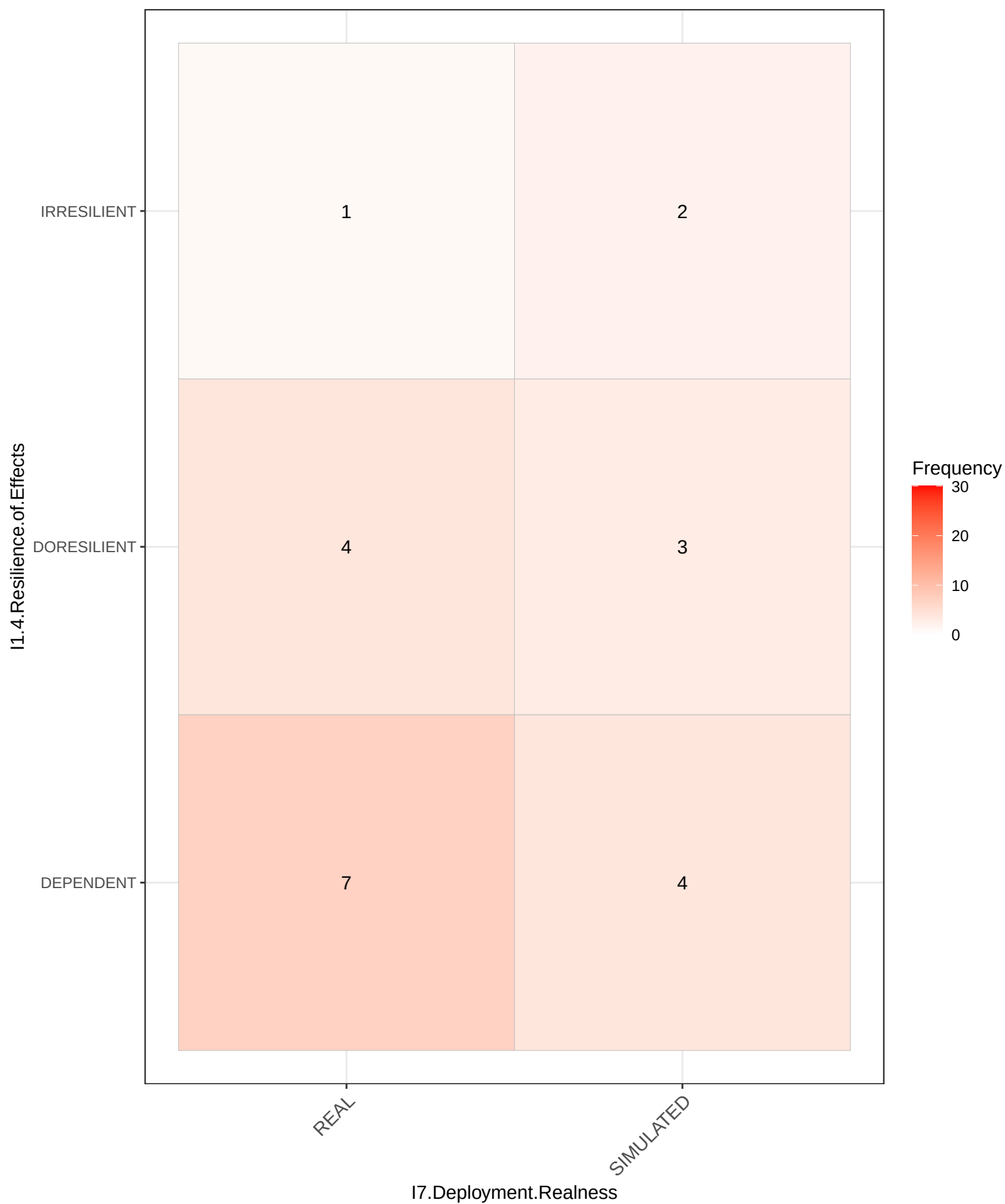


I1.4.Resilience.of.Effects_____I6.Independence

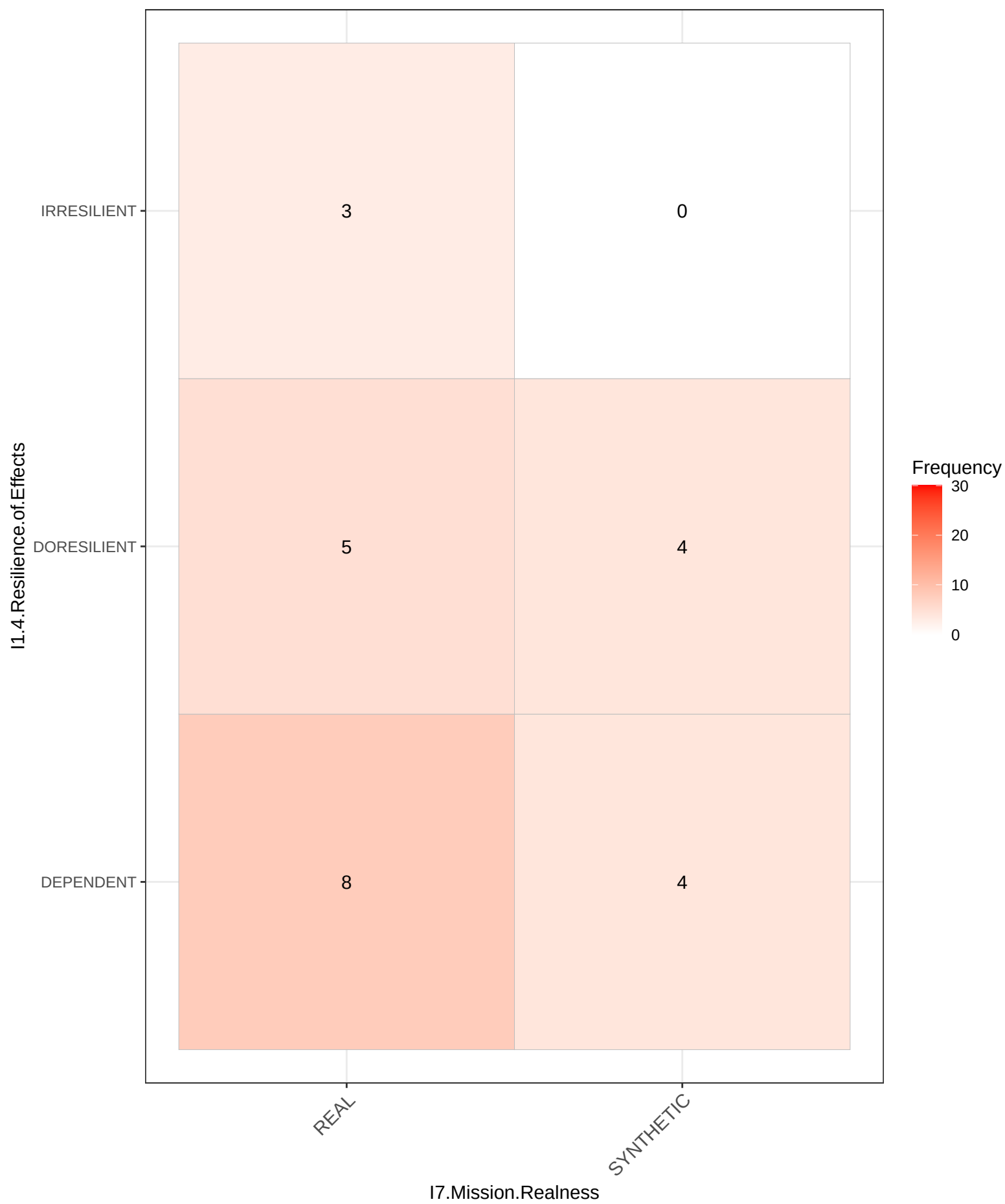
I1.4.Resilience.of.Effects



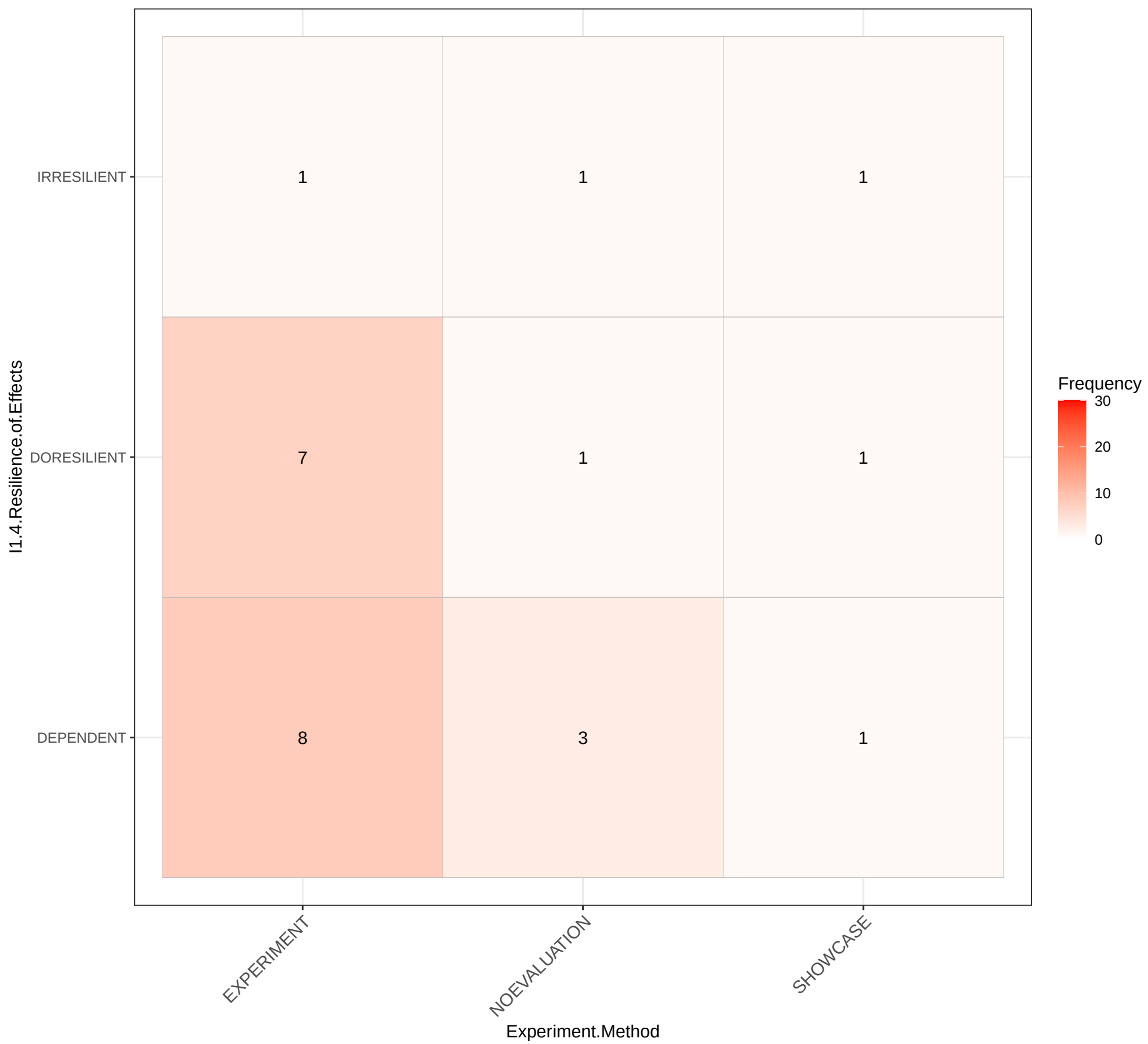
I1.4.Resilience.of.Effects_____I7.Deployment.Realness



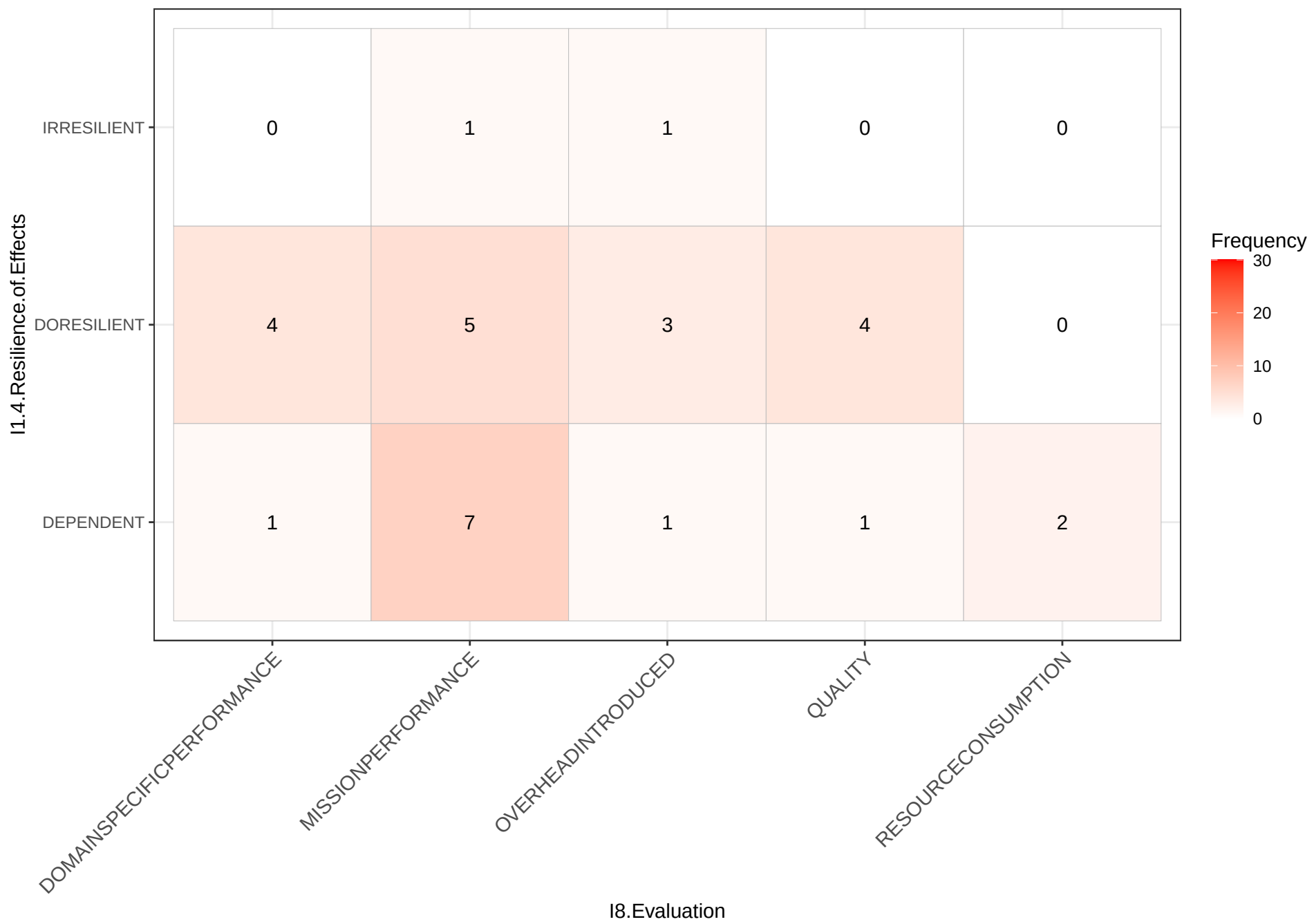
I1.4.Resilience.of.Effects_____I7.Mission.Realness



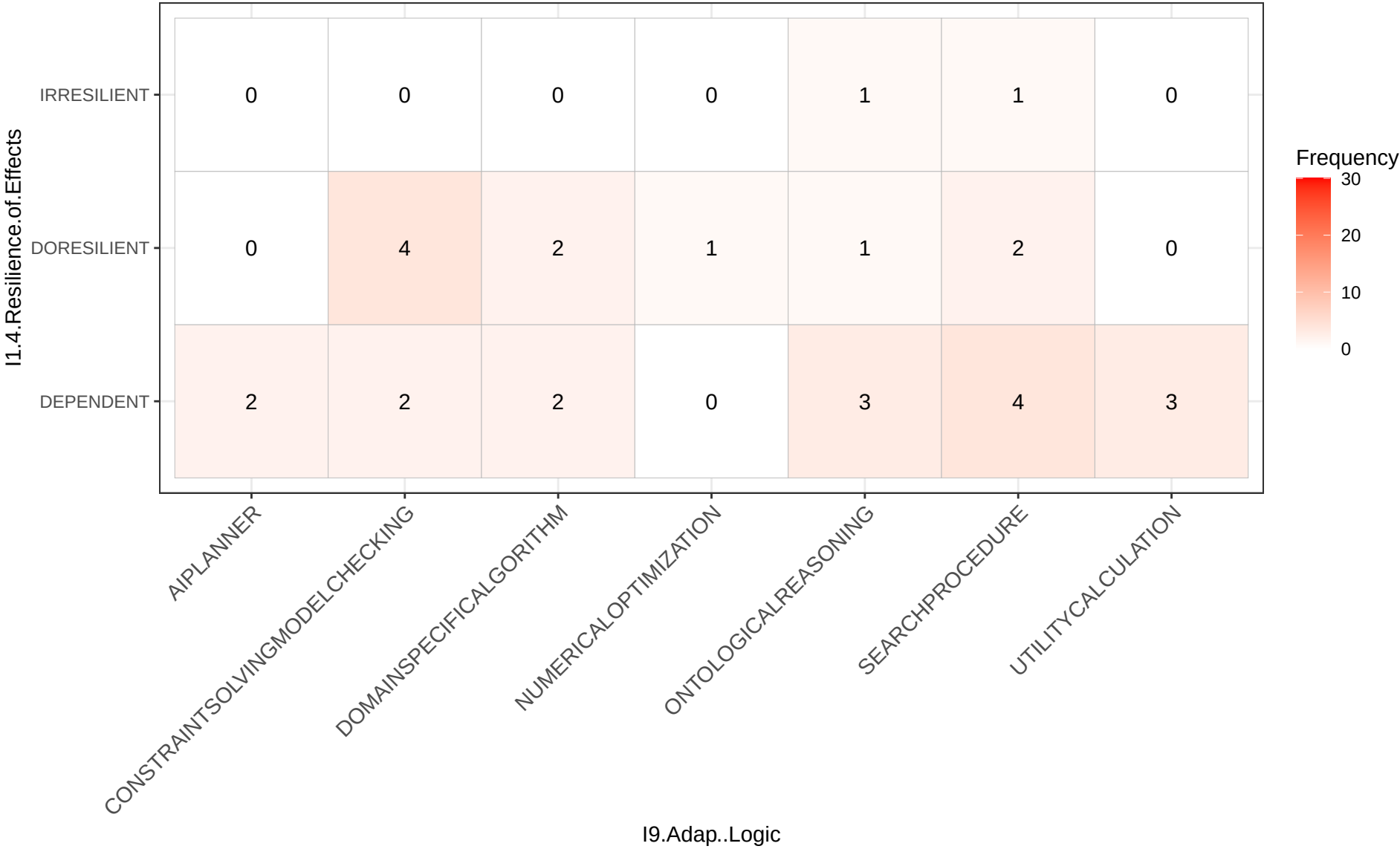
I1.4.Resilience.of.Effects_____Experiment.Method



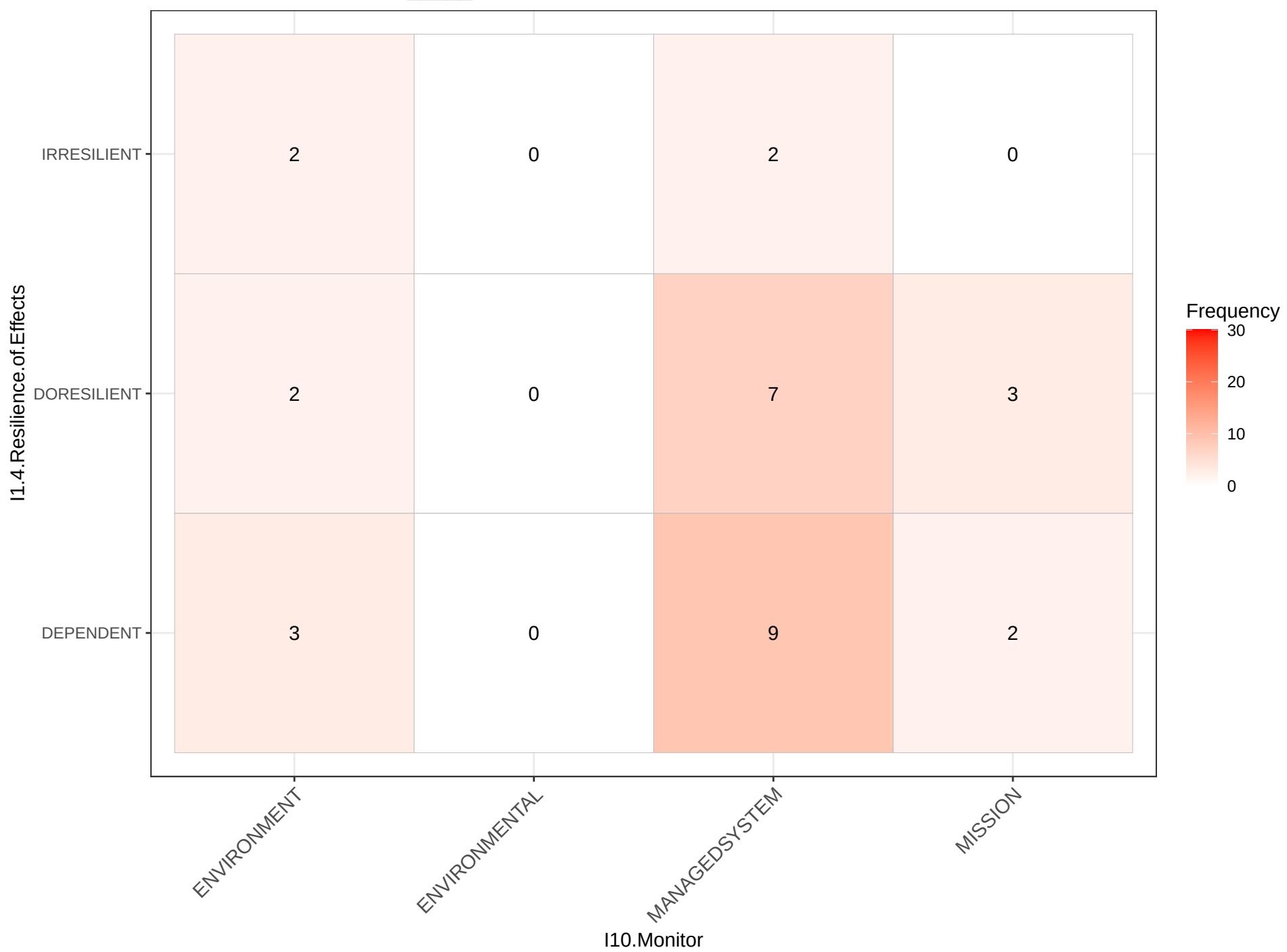
I1.4.Resilience.of.Effects_____I8.Evaluation



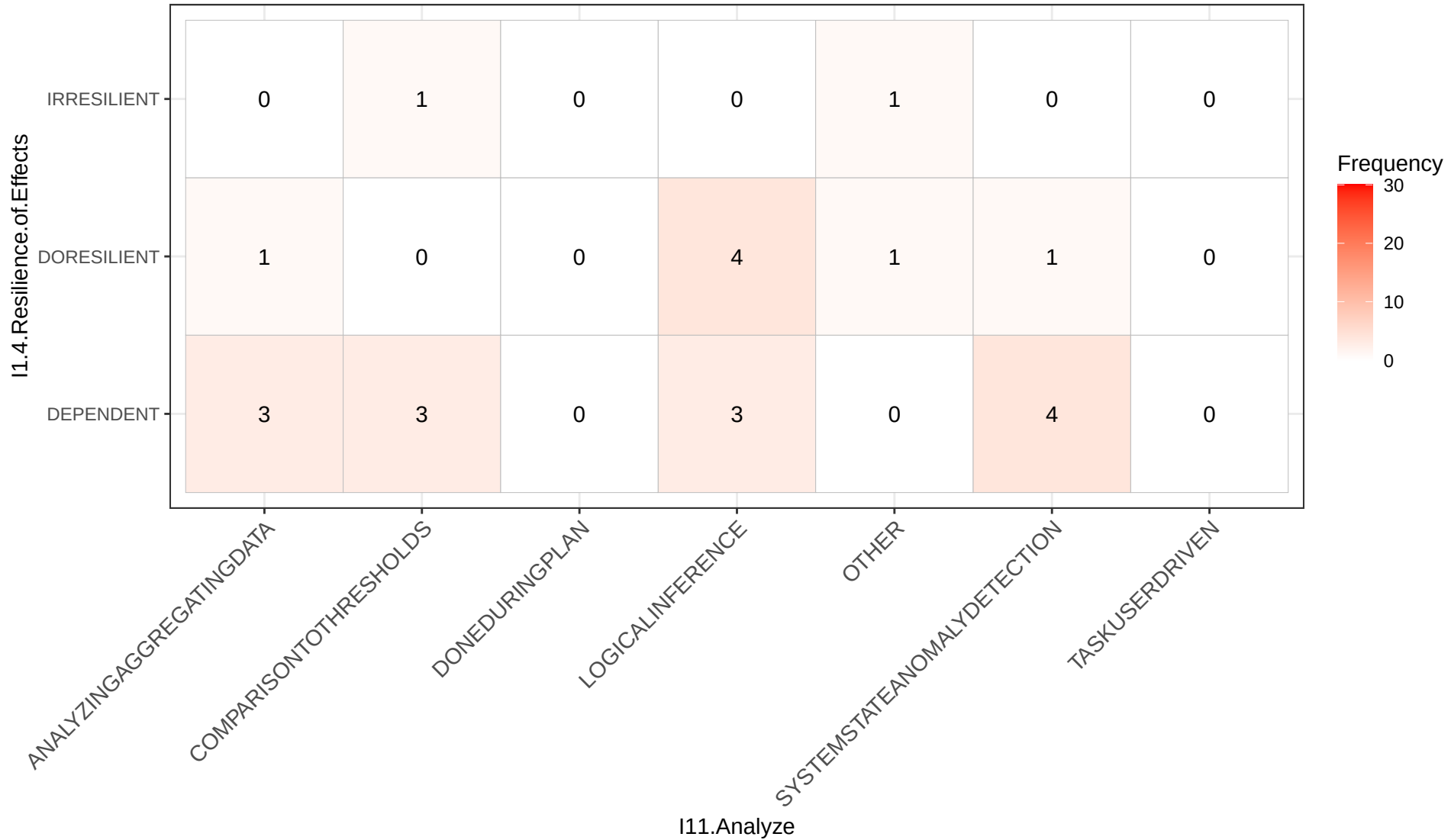
I1.4.Resilience.of.Effects_____I9.Adap..Logic

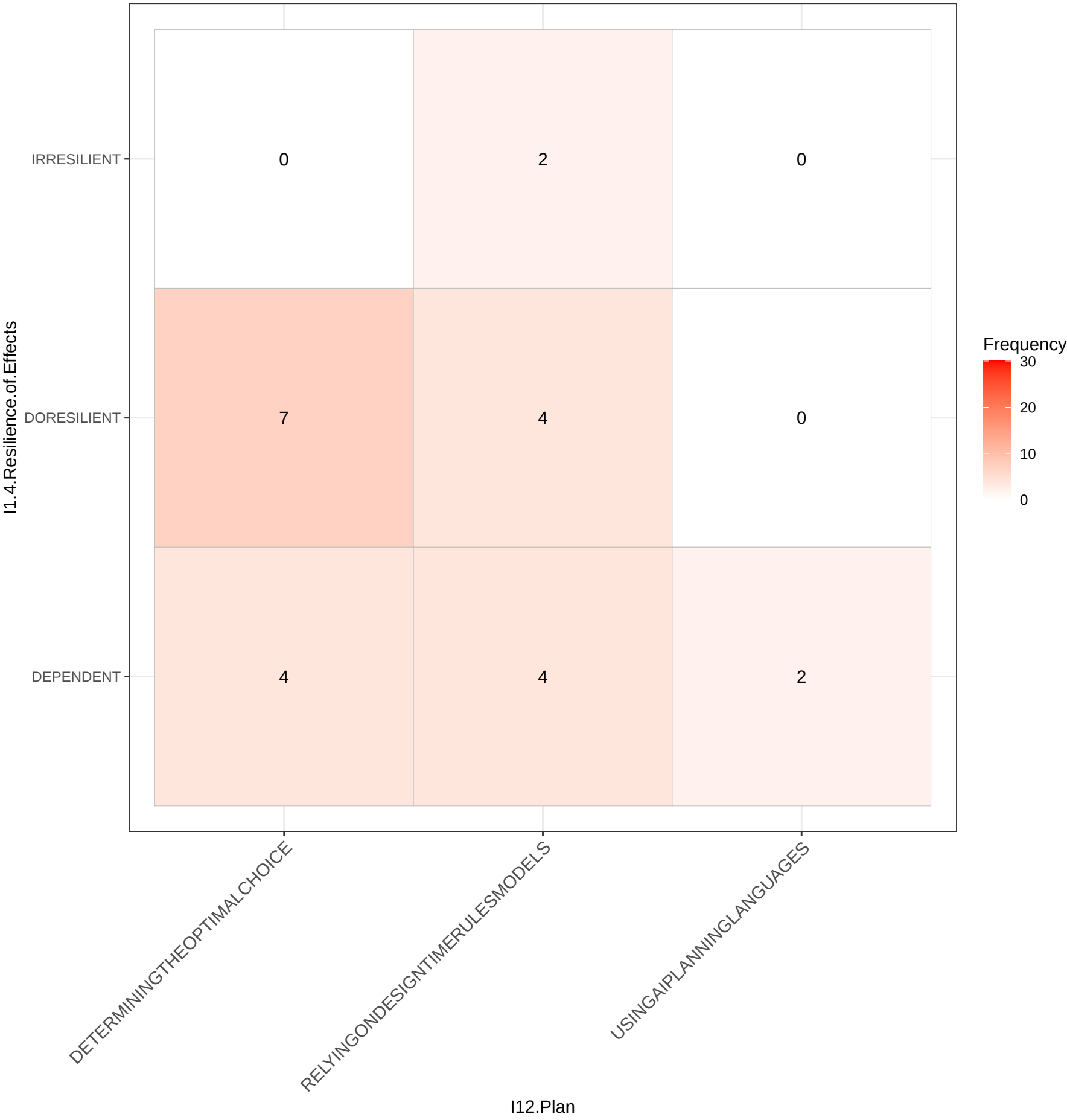


I1.4.Resilience.of.Effects_____I10.Monitor

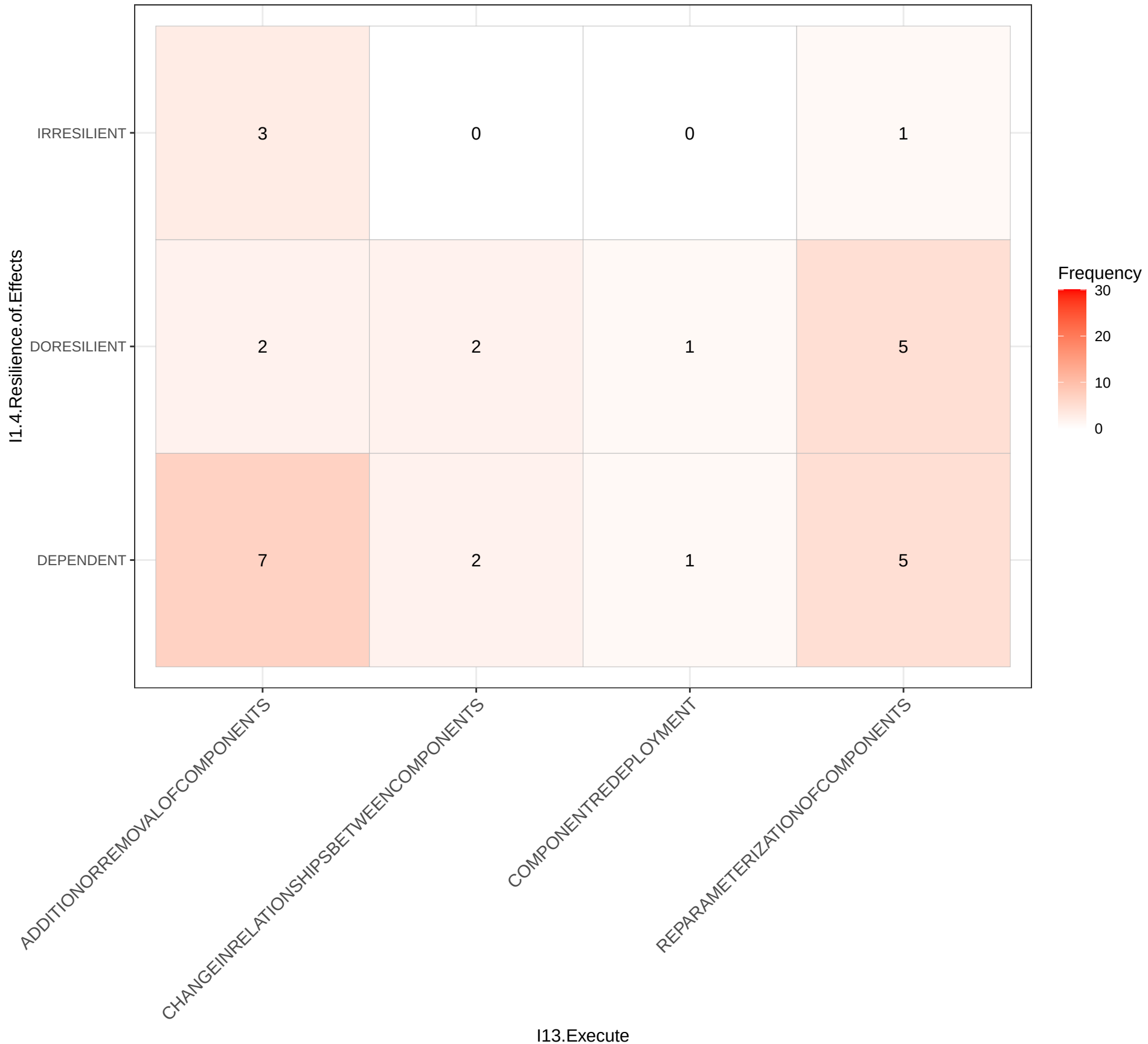


I1.4.Resilience.of.Effects_____I11.Analyze

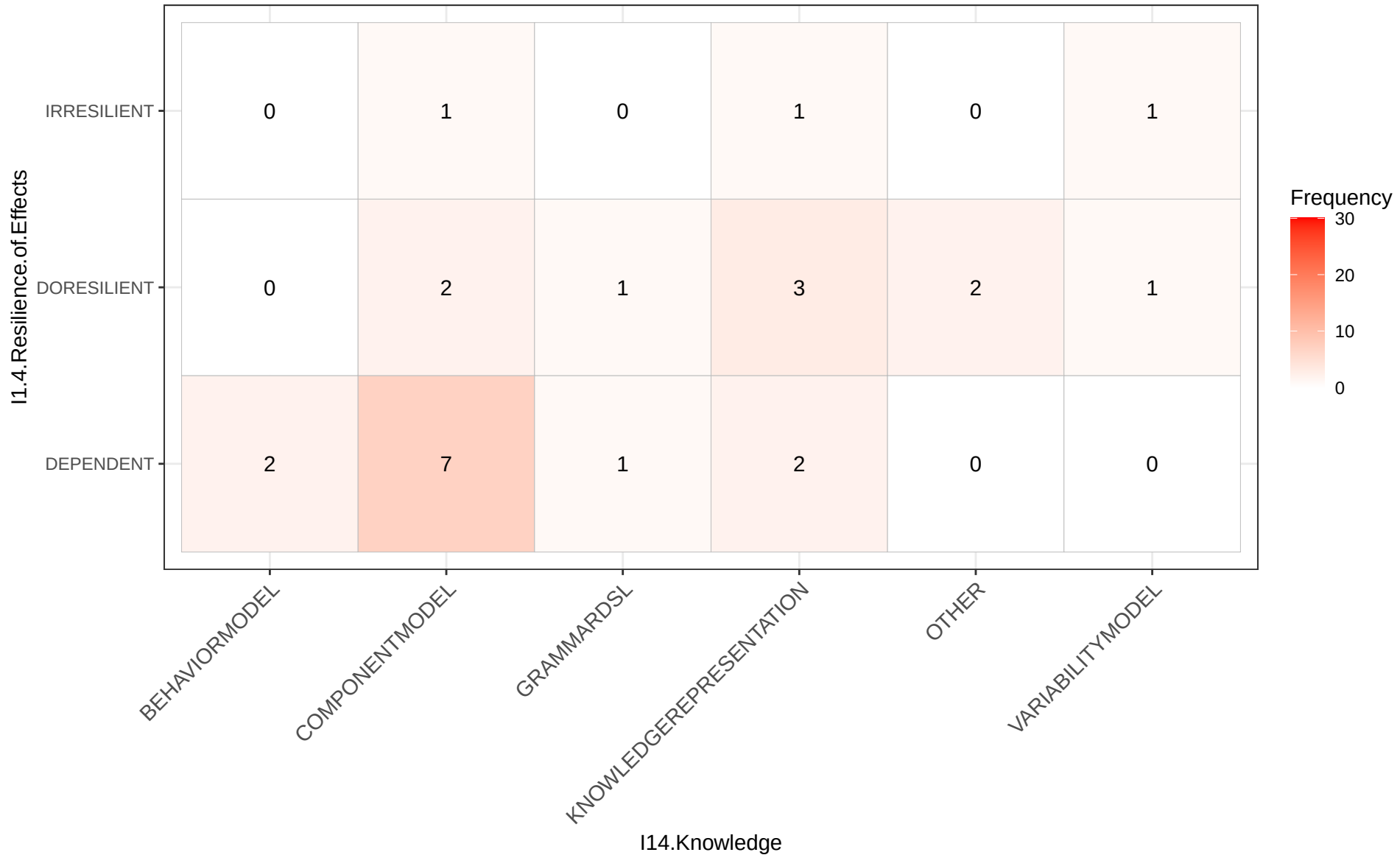




I1.4.Resilience.of.Effects_____I13.Execute

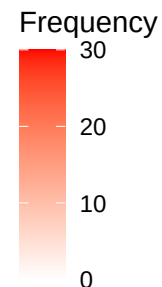
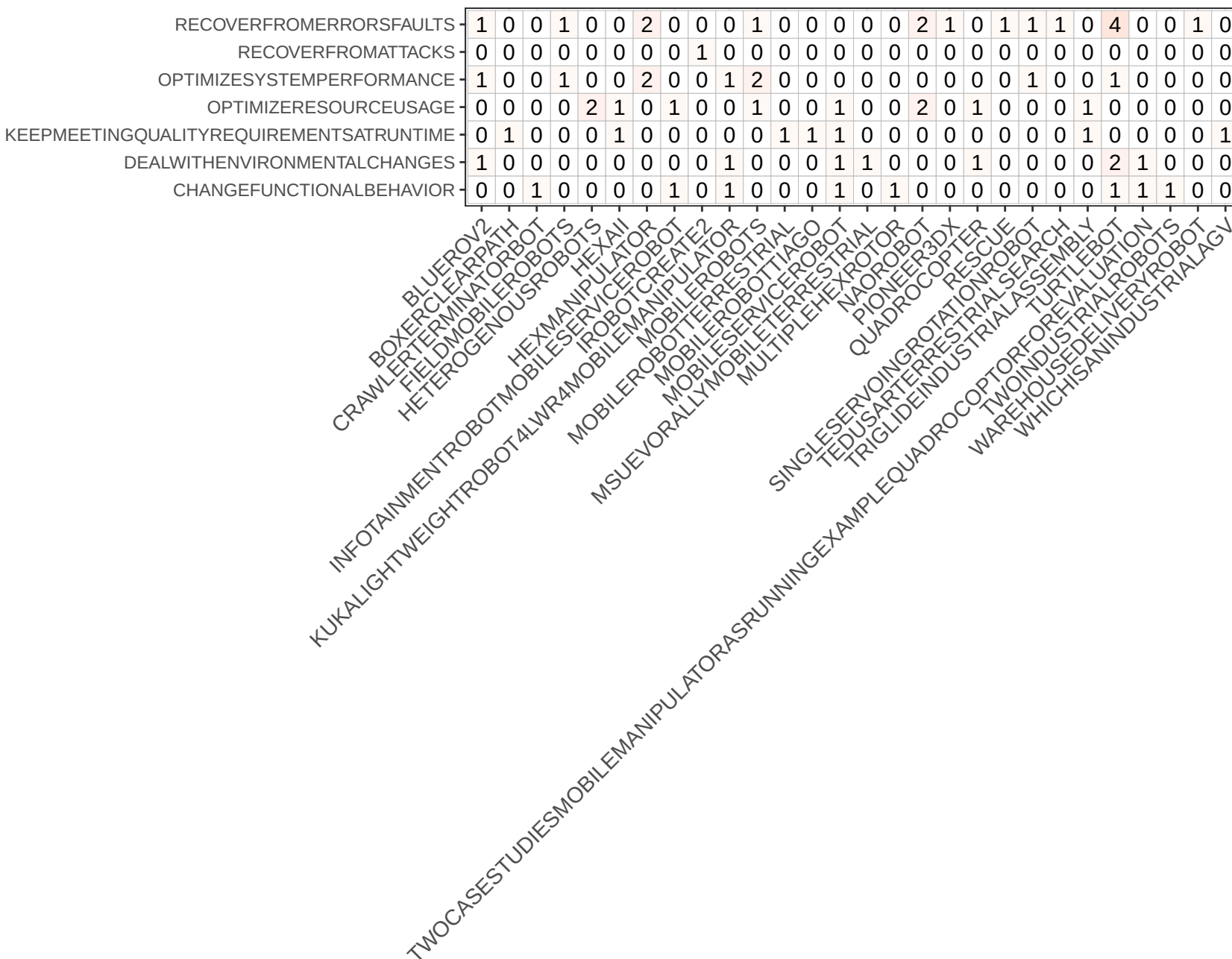


I1.4.Resilience.of.Effects_____I14.Knowledge



I2.Adap..Purpose

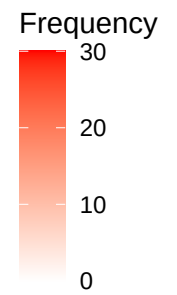
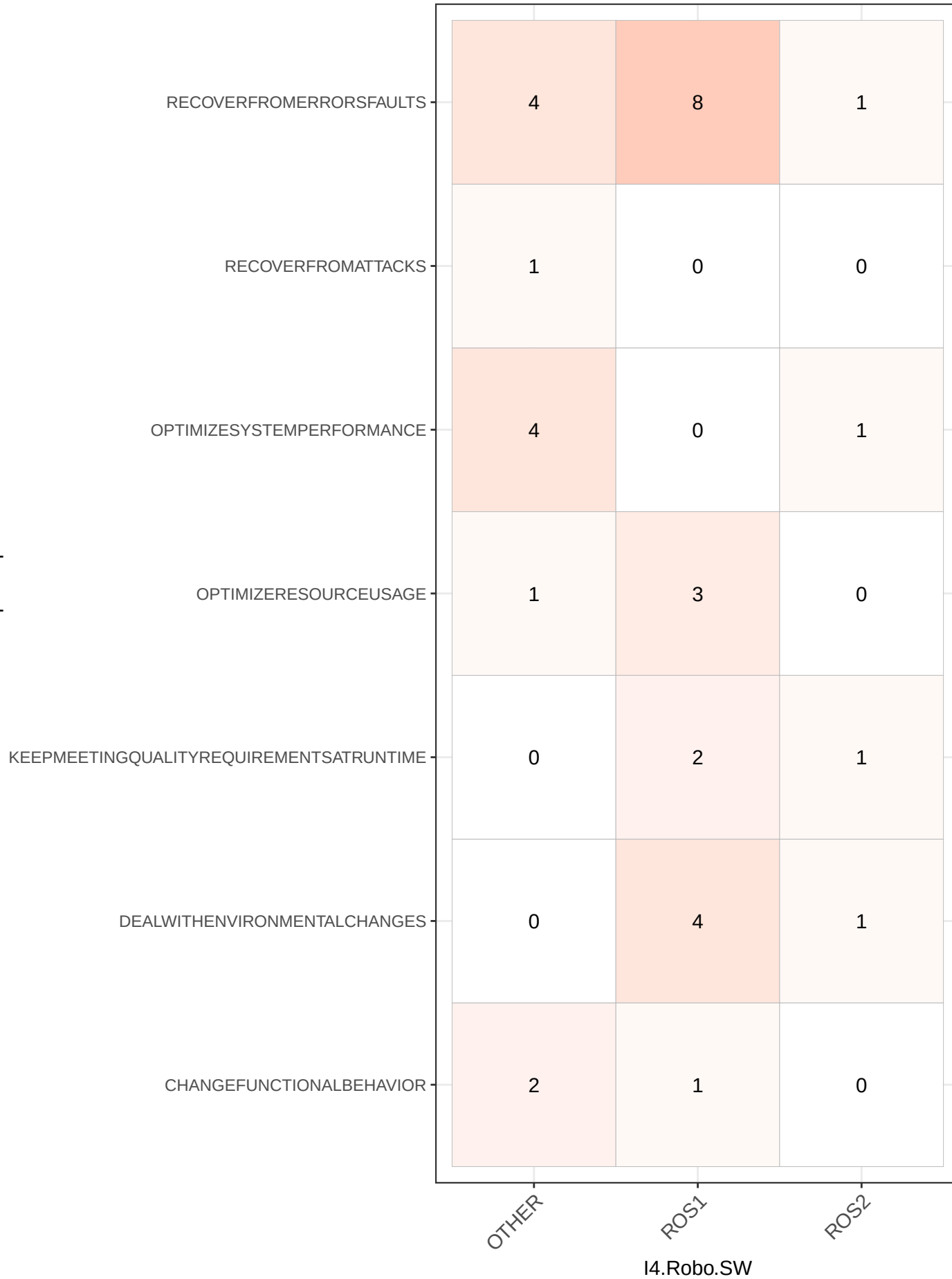
I2.Adap..Purpose _____ I3.Robot.Type



I3.Robot.Type

I2.Adap..Purpose_____I4.Robo.SW

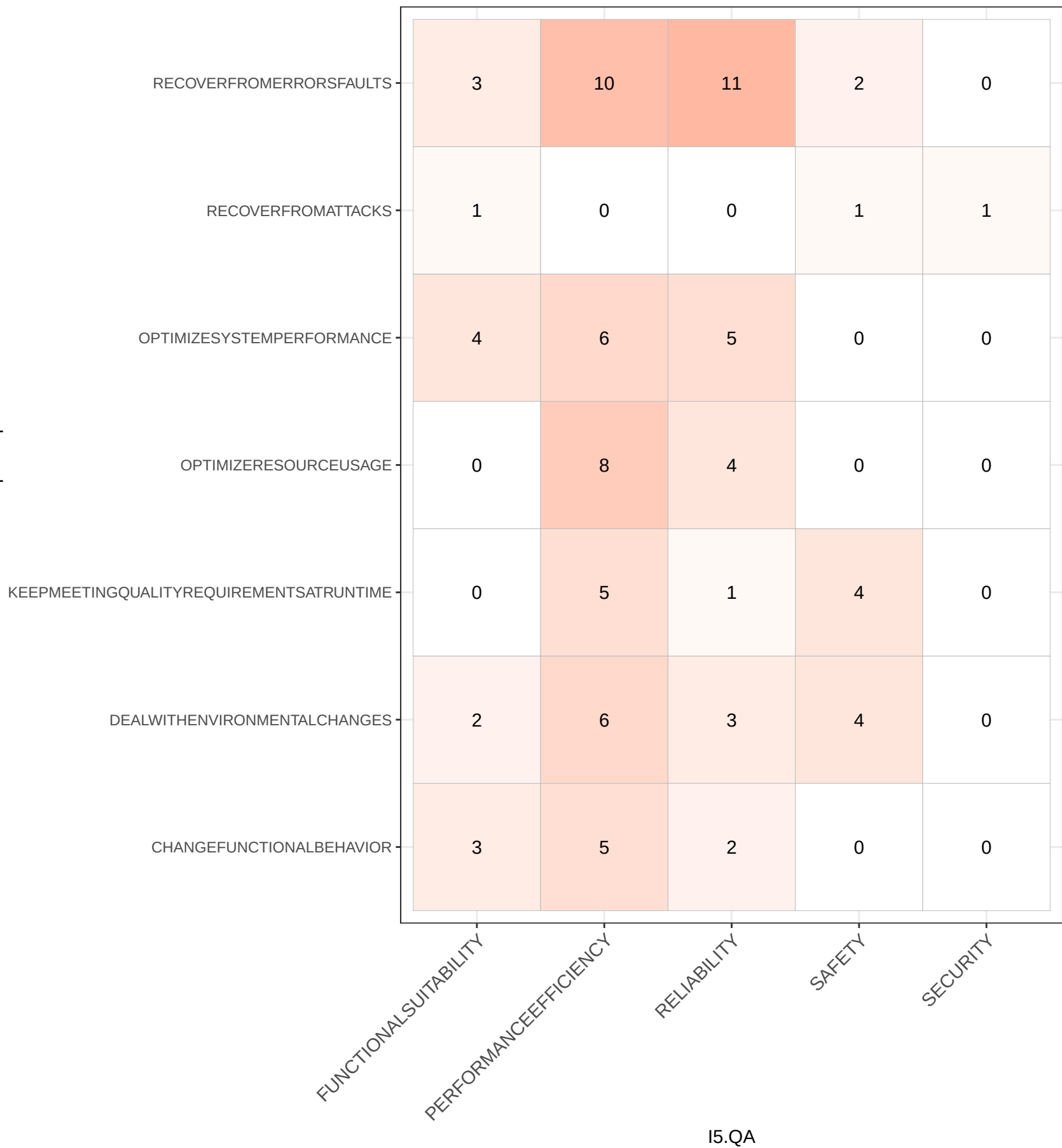
I2.Adap..Purpose

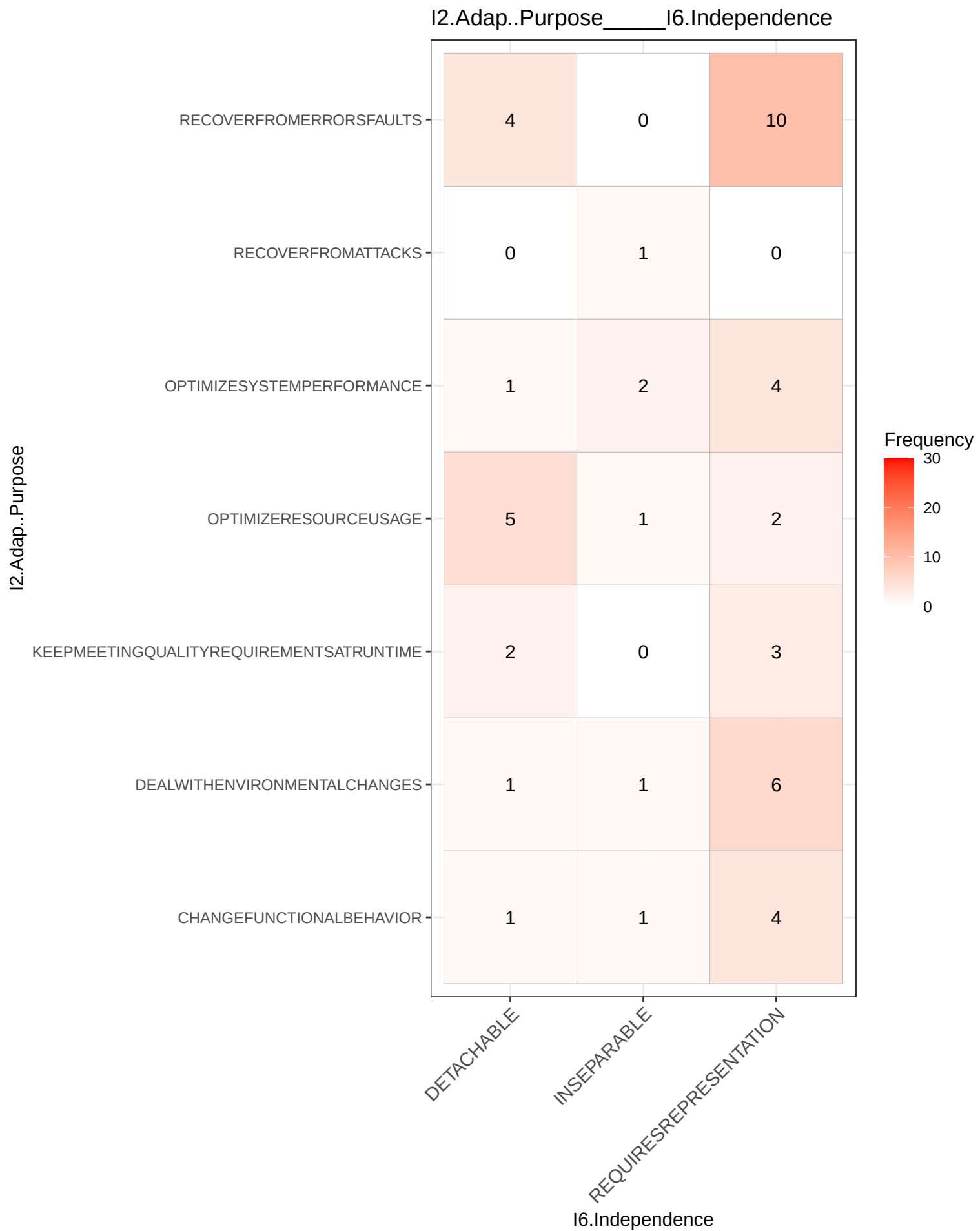


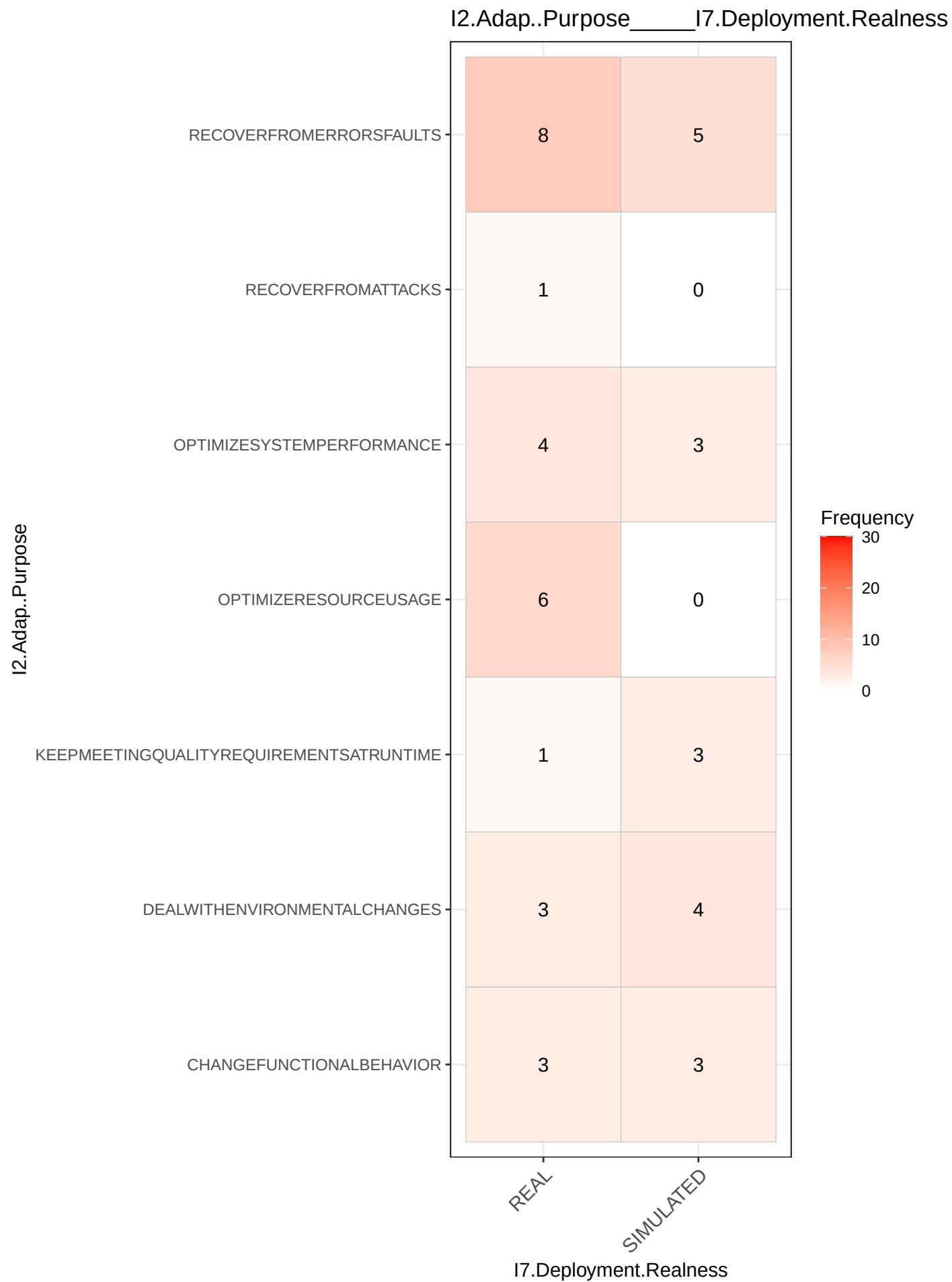
I4.Robo.SW

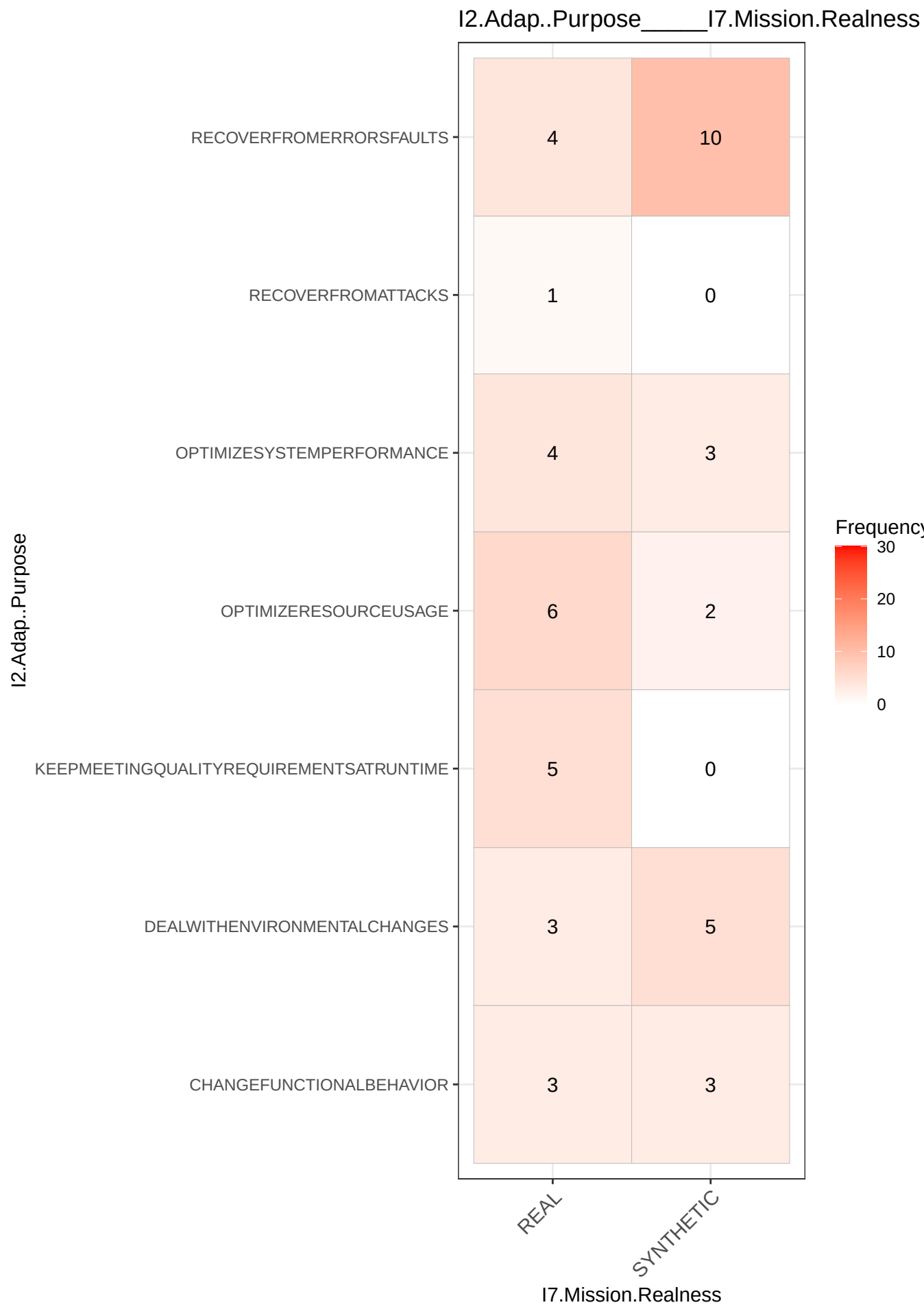
I2.Adap..Purpose_____I5.QA

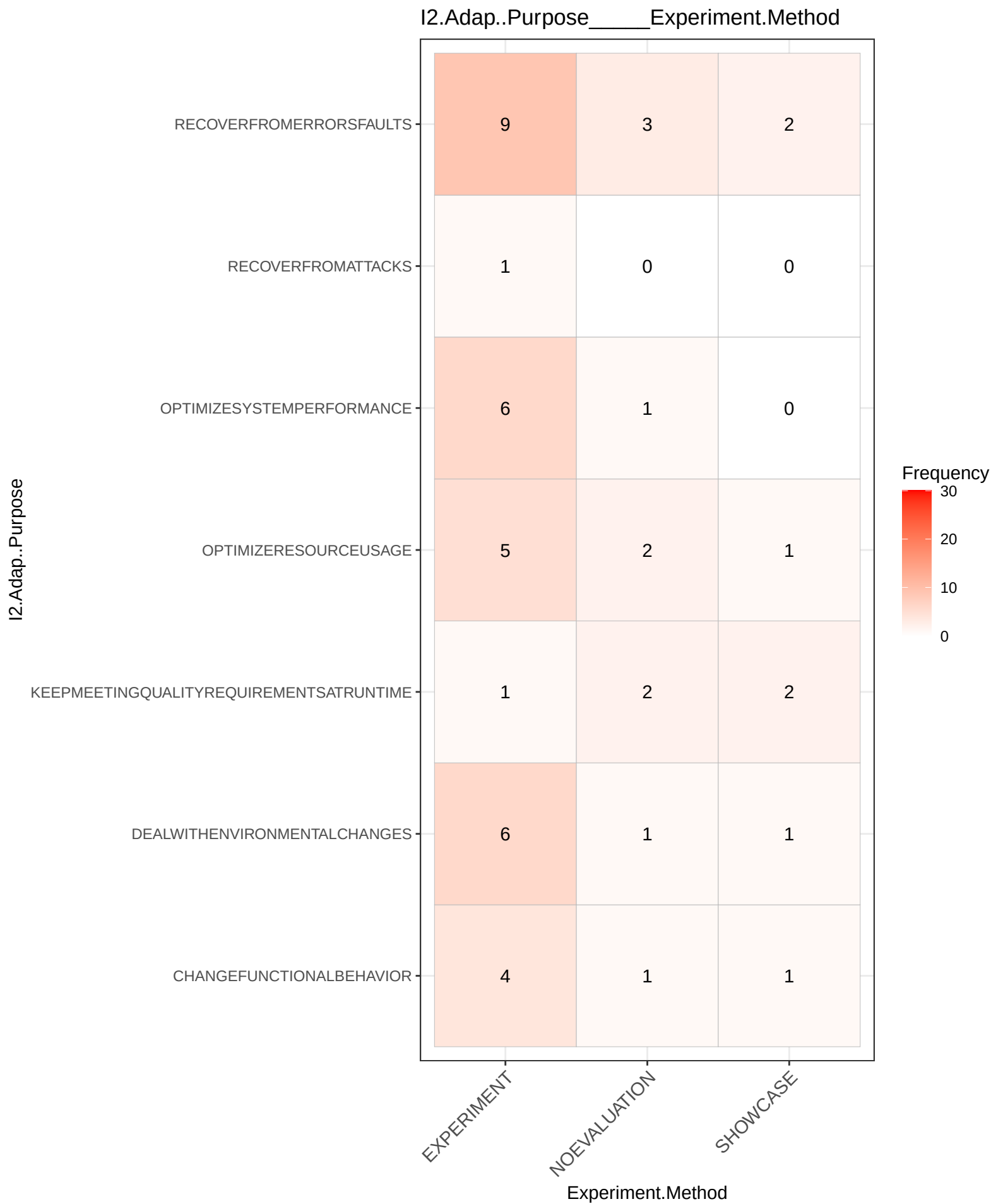
I2.Adap..Purpose

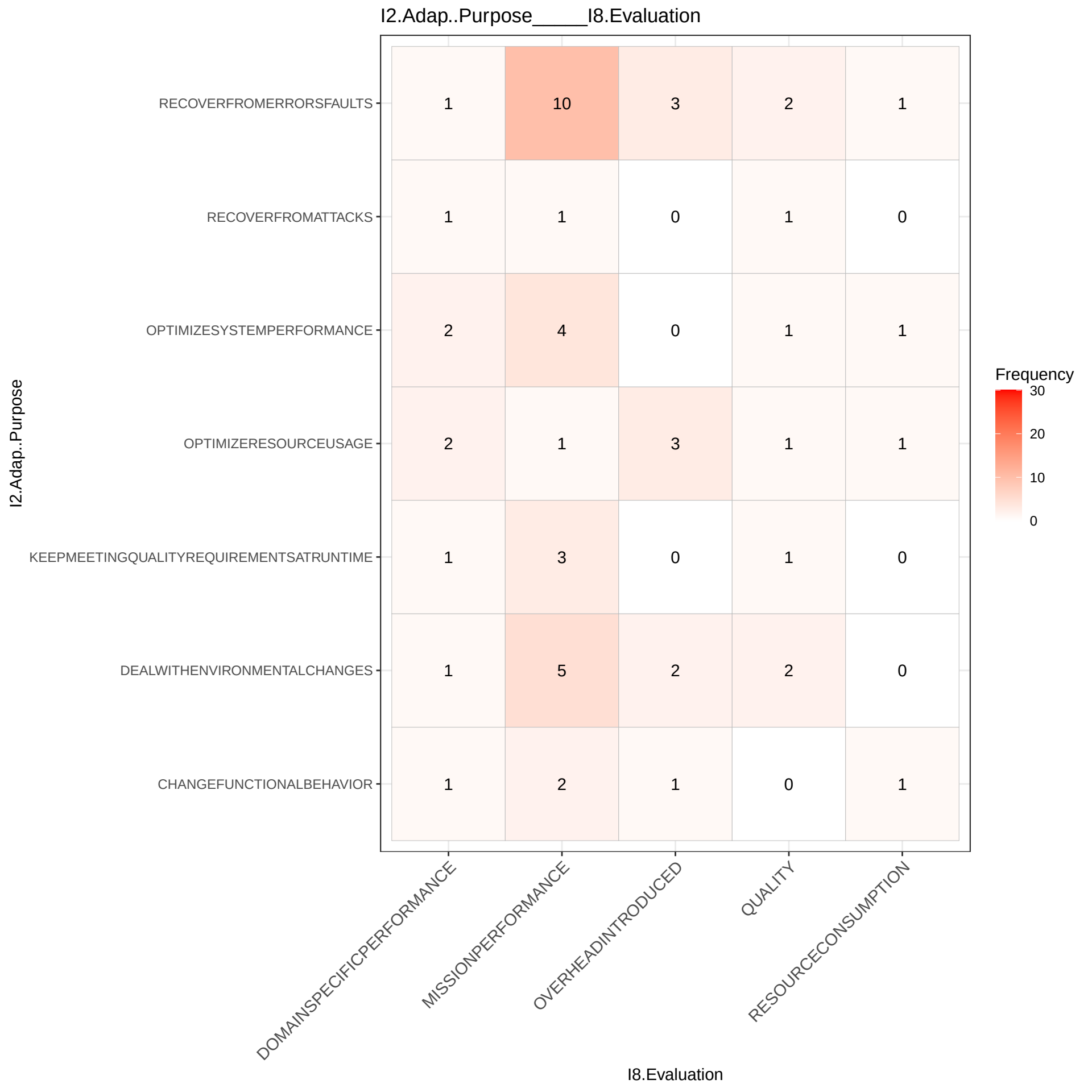


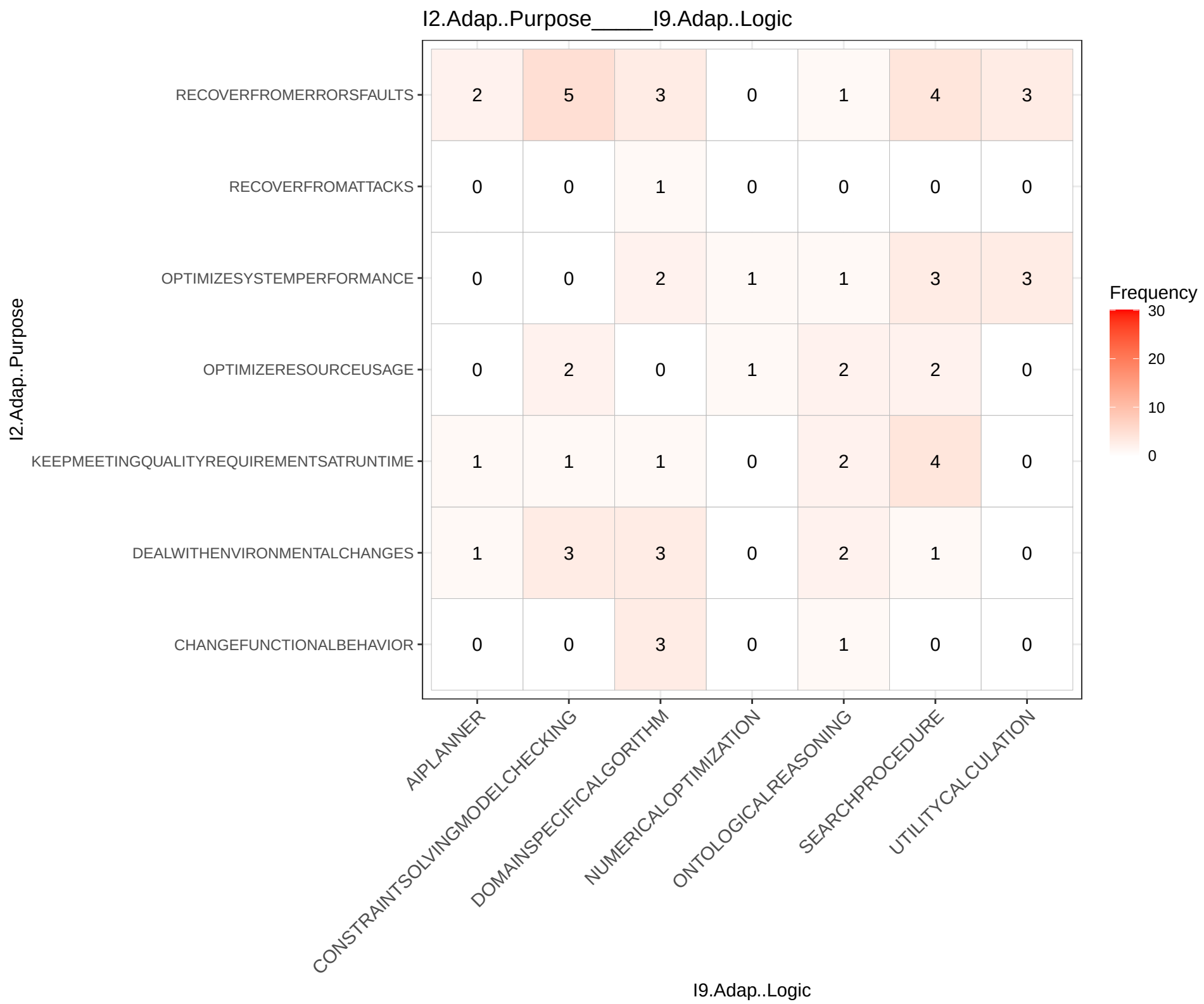






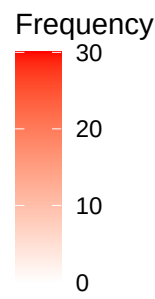
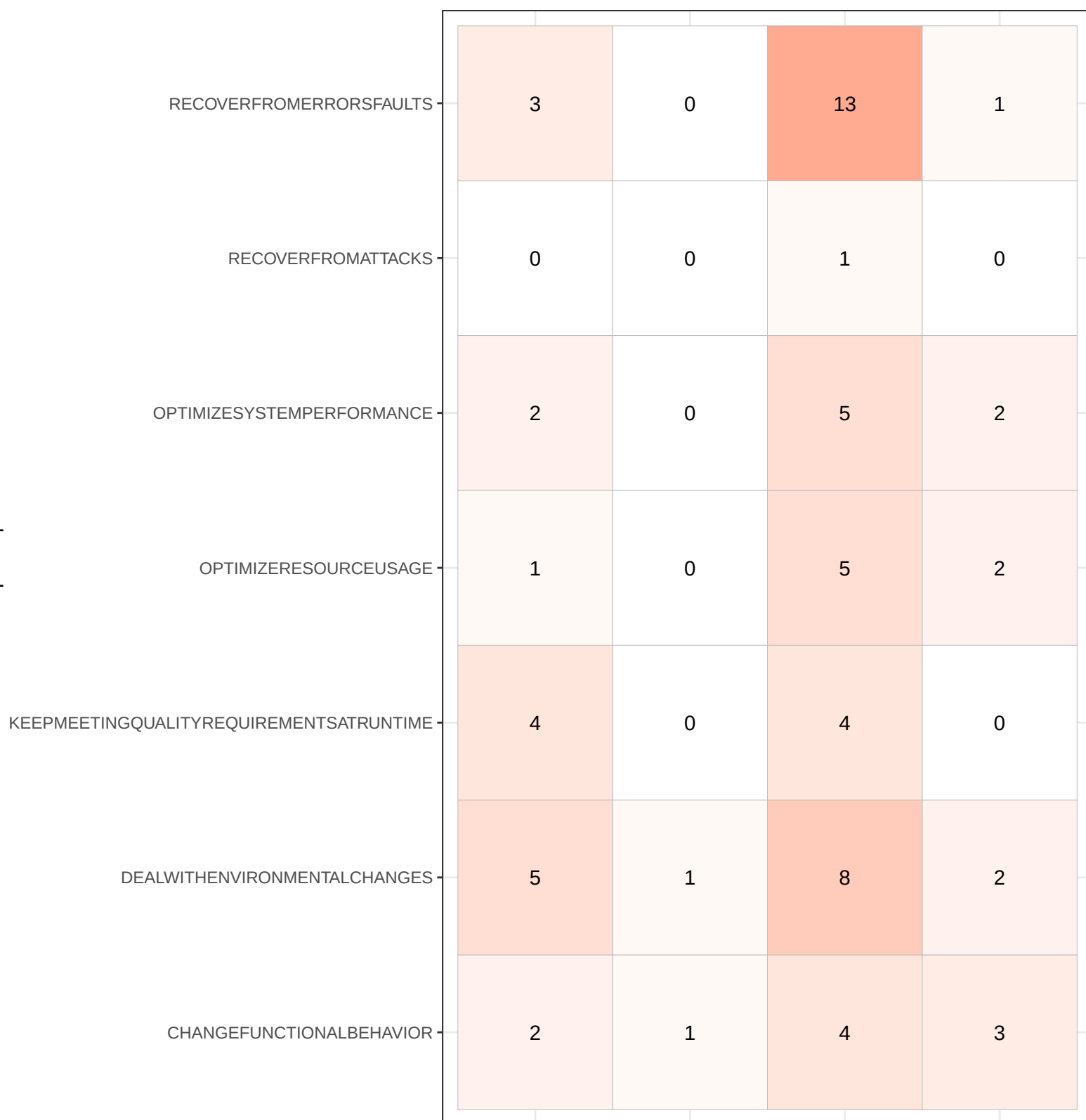






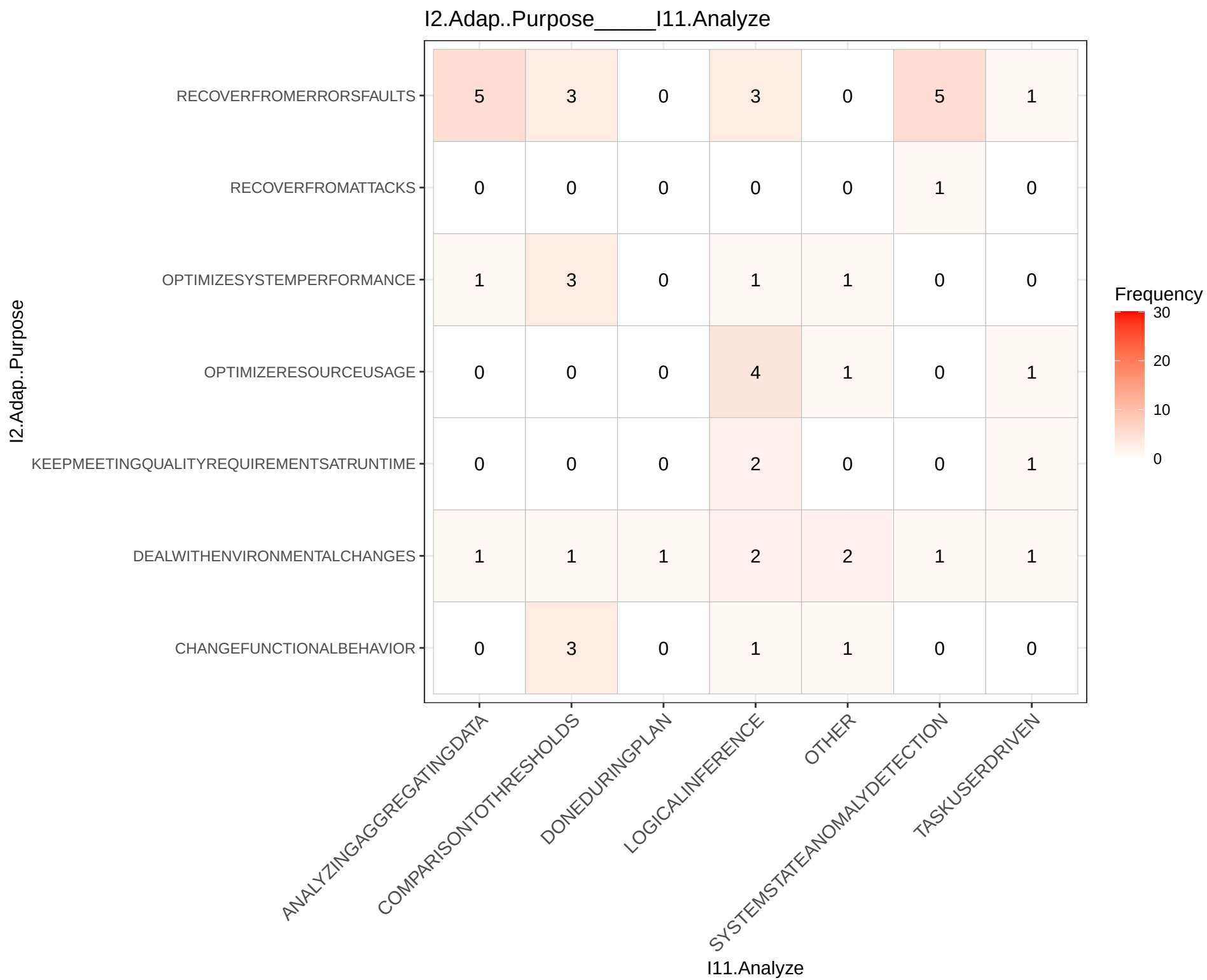
I2.Adap..Purpose_____I10.Monitor

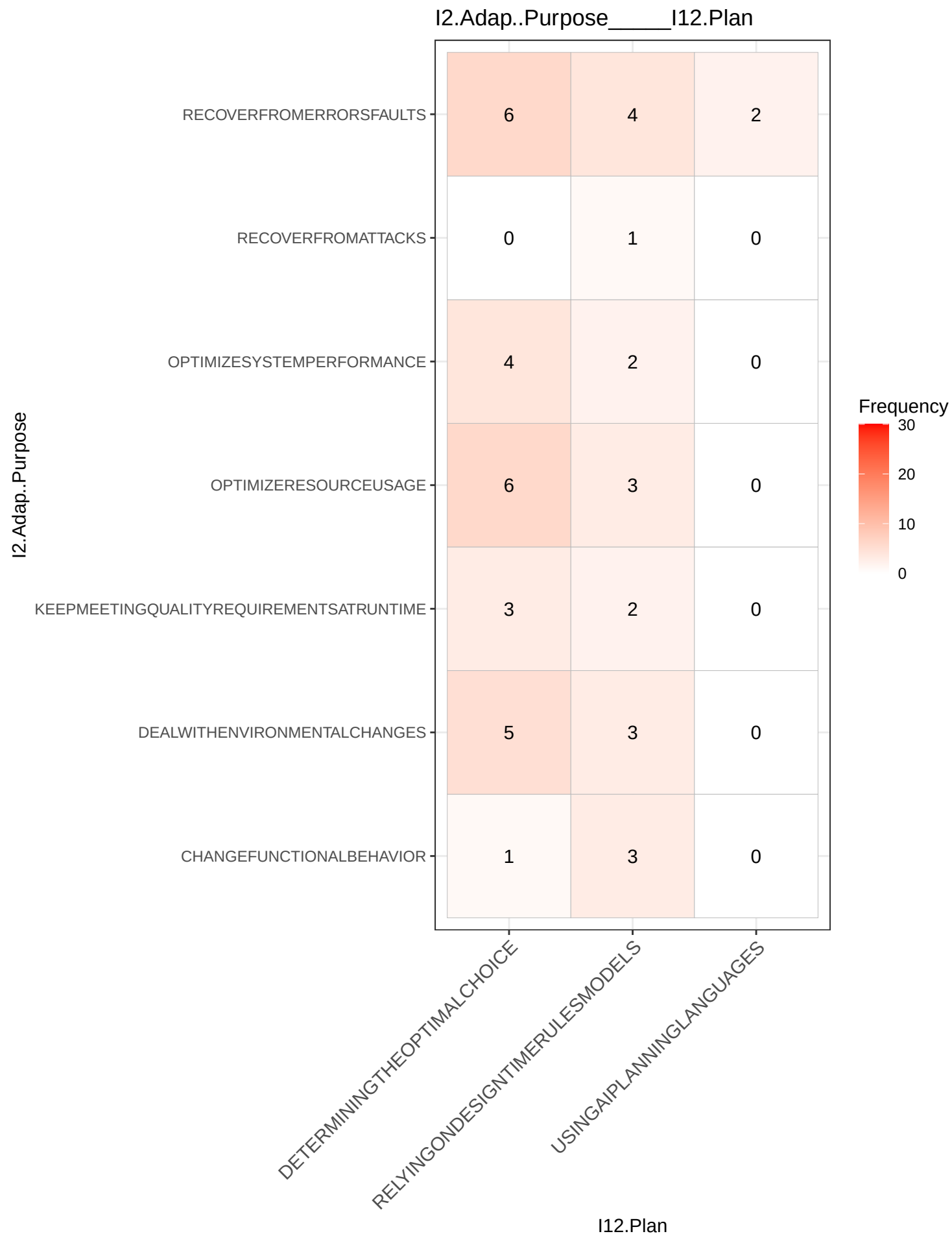
I2.Adap..Purpose



I10.Monitor

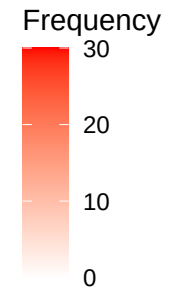
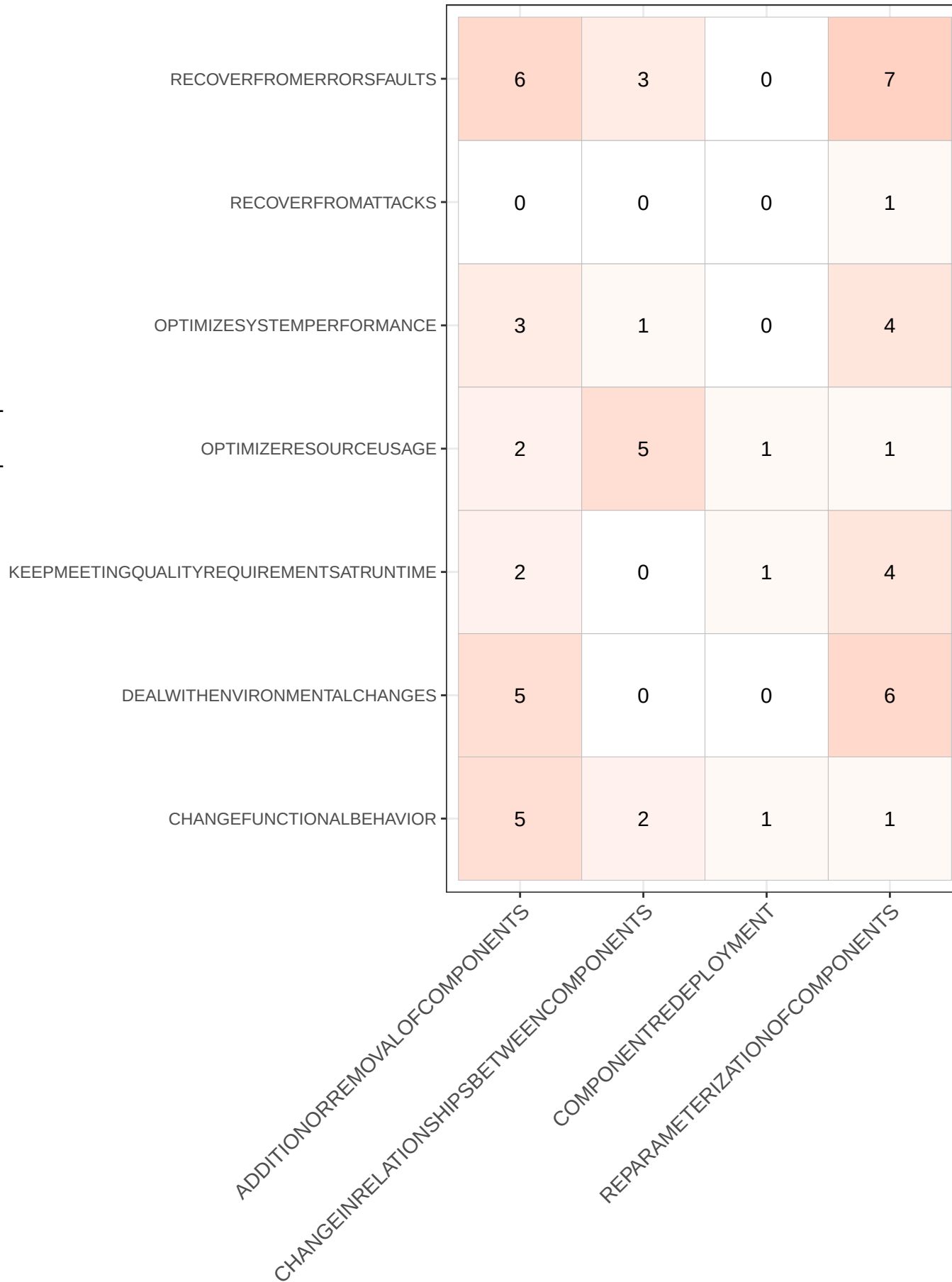
ENVIRONMENT
ENVIRONMENTAL
MANAGEDSYSTEM
MISSION





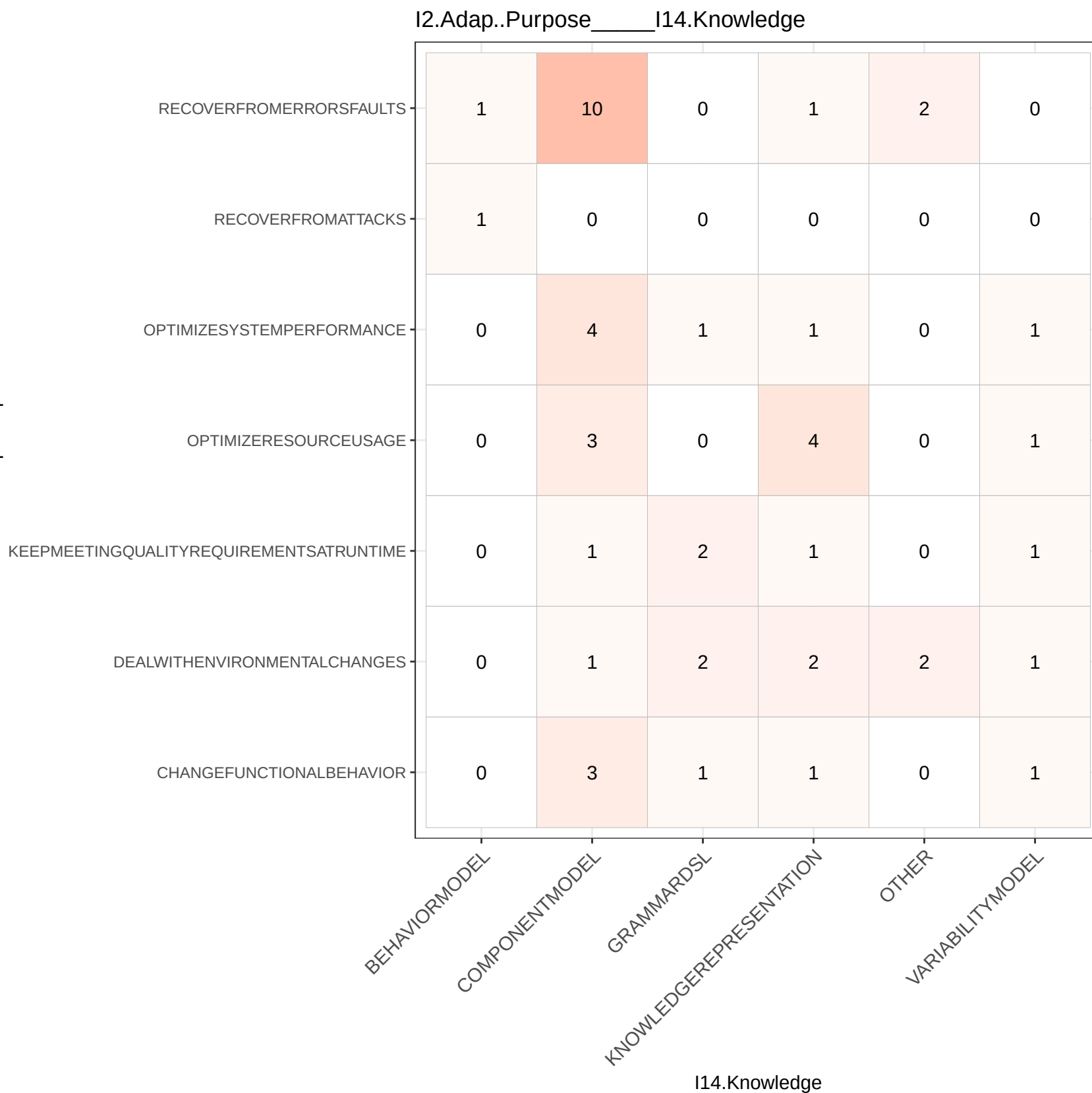
I2.Adap..Purpose_____I13.Execute

I2.Adap..Purpose



I13.Execute

I2.Adap..Purpose



I3.Robot.Type



I3.Robot.Type_____I5.QA

I3.Robot.Type

WHICHISANINDUSTRIALAGV

WAREHOUSEDELIVERYROBOT

TWOINDUSTRIALROBOTS

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

TURTLEBOT

TRIGLIDEINDUSTRIALASSEMBLY

TEDUSARTERRESTRIALSEARCH

SINGLESERVOINGROTATIONROBOT

RESCUE

QUADROCOPTER

PIONEER3DX

NAOROBOT

MULTIPLEHEXROTOR

MSUEVORALLYMOBILETERRESTRIAL

MOBILESERVICEROBOT

MOBILEROBOTTIAGO

MOBILEROBOTTERRESTRIAL

MOBILEROBOTS

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

IROBOTCREATE2

INFOTAINMENTROBOTMOBILESERVICEROBOT

HEXMANIPULATOR

HEXAII

HETEROGENOUSROBOTS

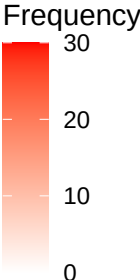
FIELDMOBILEROBOTS

CRAWLERTERMINATORBOT

BOXERCLEARPATH

BLUEROV2

0	1	0	1	0
0	0	1	0	0
1	1	0	0	0
0	1	1	0	0
0	3	2	2	0
0	1	1	0	0
0	0	1	0	0
1	1	1	0	0
0	0	1	0	0
0	1	0	0	0
0	0	1	0	0
0	2	1	0	0
1	1	0	0	0
1	0	1	1	0
0	2	0	1	0
0	1	0	1	0
0	1	0	1	0
1	2	1	0	0
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1	0	0	1	1
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0	1	1	0	0
0	2	2	0	0
1	1	1	0	0
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0	1	1	0	0



FUNCTIONALSUITABILITY

PERFORMANCEEFFICIENCY

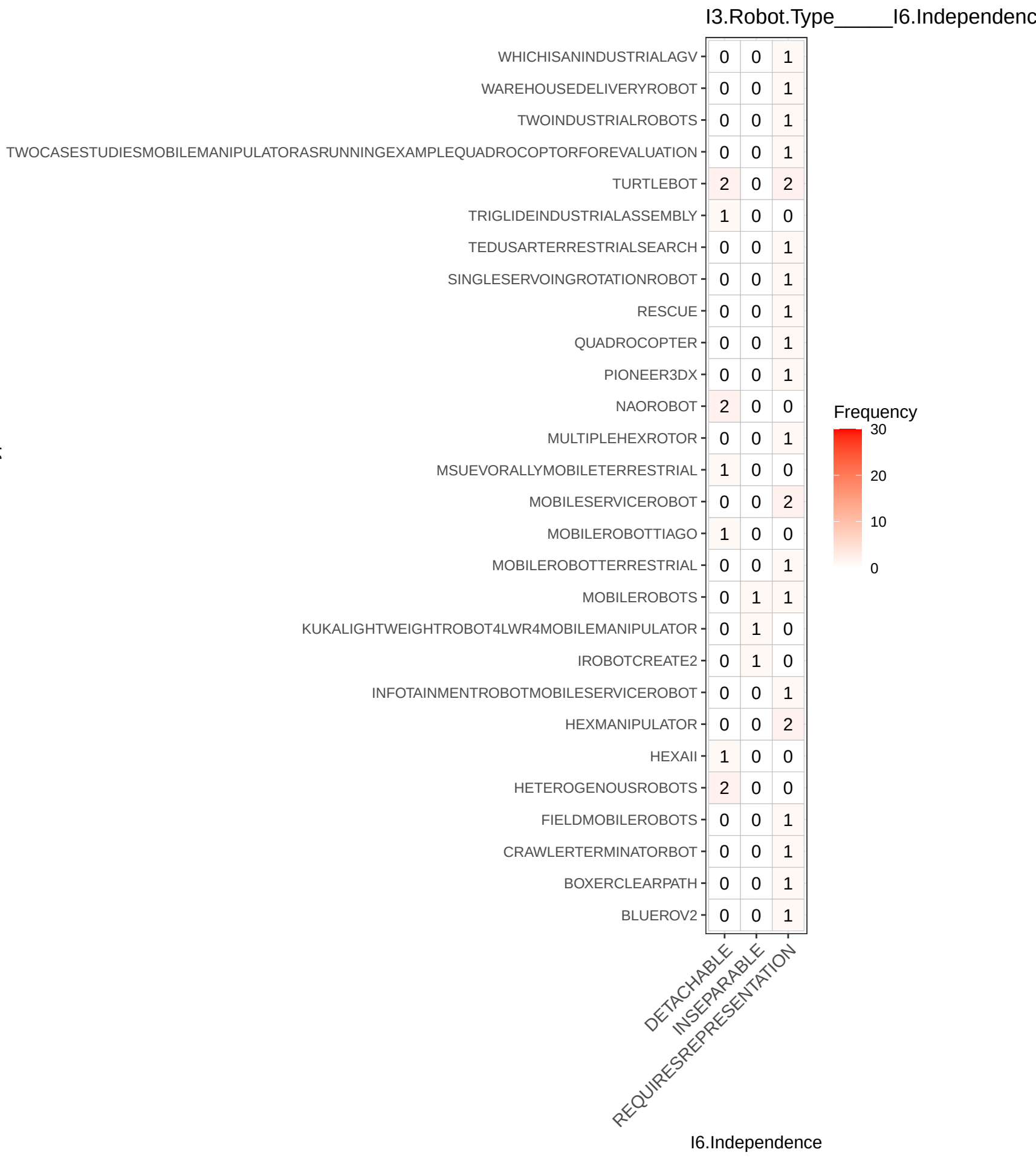
RELIABILITY

SAFETY

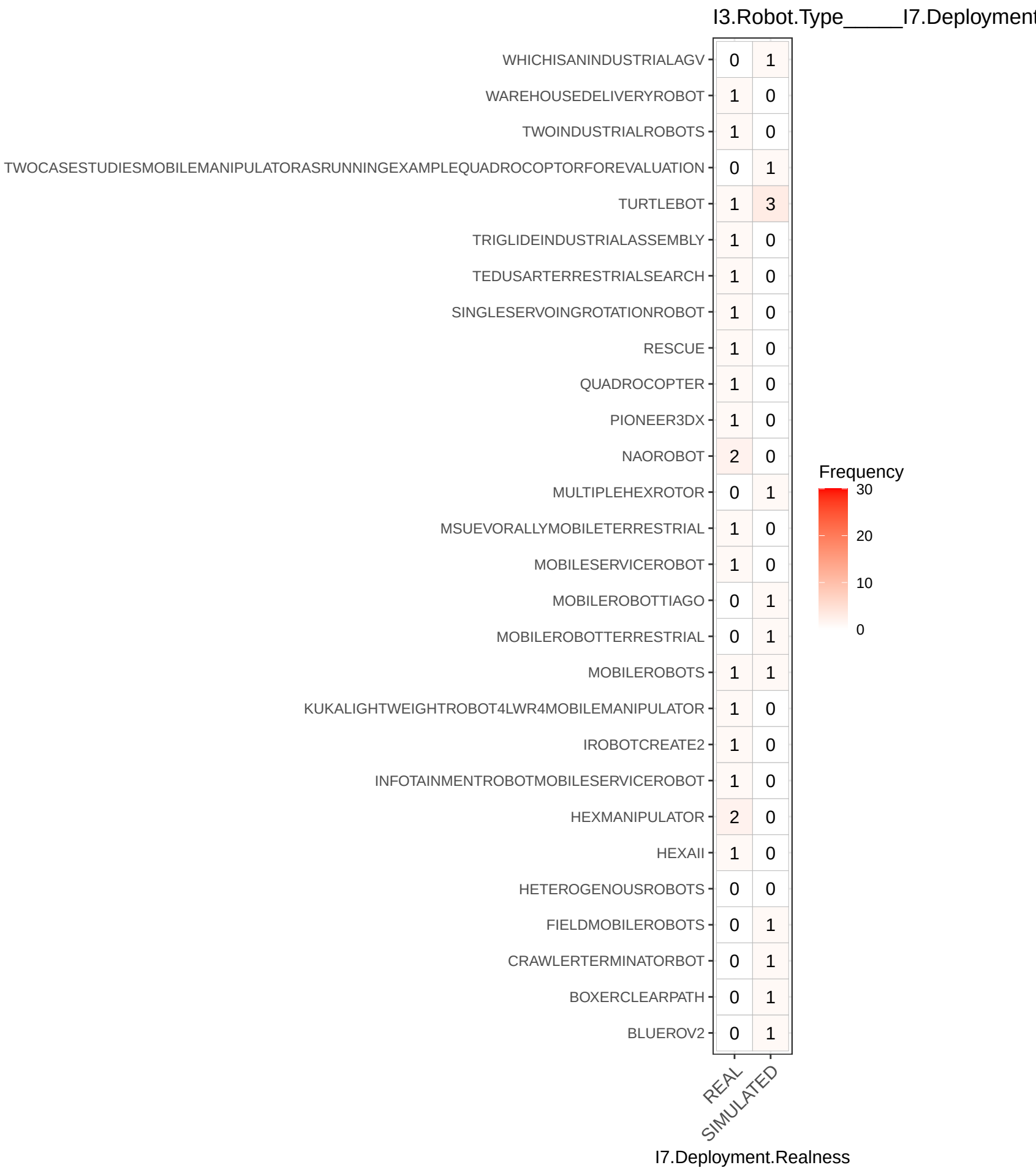
SECURITY

I5.QA

I3.Robot.Type



I3.Robot.Type



I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

INFOTAINMENTROBOTMOBILESERVICEROBOT

SINGLESERVOINGROTATIONROBOT

MOBILEROBOTTERRESTRIAL

MOBILEROBOTTIAGO

MOBILESERVICEROBOT

MSUEVORALLYMOBILETERRESTRIAL

MULTIPLEHEXROTOR

NAOROBOT

PIONEER3DX

QUADROCOPTER

RESCUE

SINGLESERVOINGROTATIONROBOT

TEDUSARTERRESTRIALSEARCH

TRIGLIDEINDUSTRIALASSEMBLY

TURTLEBOT

WAREHOUSEDELIVERYROBOT

WHICHISANINDUSTRIALAGV

SINGLESERVOINGROTATIONROBOT

MOBILEROBOTTERRESTRIAL

MOBILEROBOTTIAGO

MOBILESERVICEROBOT

MSUEVORALLYMOBILETERRESTRIAL

MULTIPLEHEXROTOR

NAOROBOT

PIONEER3DX

QUADROCOPTER

RESCUE

SINGLESERVOINGROTATIONROBOT

TEDUSARTERRESTRIALSEARCH

TRIGLIDEINDUSTRIALASSEMBLY

TURTLEBOT

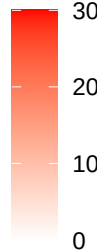
WAREHOUSEDELIVERYROBOT

WHICHISANINDUSTRIALAGV

I7.Mission.Realness

REAL
SYNTHETIC

Frequency

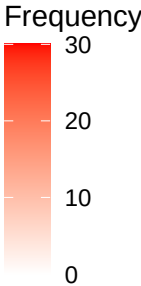


1	0
0	1
1	0
0	1
0	4
1	0
1	0
1	0
1	0
1	0
0	1
0	2
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2	0
1	0
1	0
2	0
0	1
1	0
1	0
1	1
1	0
2	0
1	0
1	0
1	0
1	0

I3.Robot.Type

I3.Robot.TypeExperiment.Method

WHICHISANINDUSTRIALAGV	1	0	0
WAREHOUSEDELIVERYROBOT	0	0	1
TWOINDUSTRIALROBOTS	1	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	0	1
TURTLEBOT	4	0	0
TRIGLIDEINDUSTRIALASSEMBLY	0	0	1
TEDUSARTERRESTRIALSEARCH	1	0	0
SINGLESERVOINGROTATIONROBOT	1	0	0
RESCUE	1	0	0
QUADROCOPTER	1	0	0
PIONEER3DX	0	0	1
NAOROBOT	1	1	0
MULTIPLEHEXROTOR	0	1	0
MSUEVORALLYMOBILETERRESTRIAL	1	0	0
MOBILESERVICEROBOT	1	1	0
MOBILEROBOTTIAGO	0	1	0
MOBILEROBOTTERRESTRIAL	0	0	1
MOBILEROBOTS	2	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0
IROBOTCREATE2	1	0	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	1	0	0
HEXMANIPULATOR	1	1	0
HEXAII	0	0	1
HETEROGENOUSROBOTS	1	1	0
FIELDMOBILEROBOTS	1	0	0
CRAWLERTERMINATORBOT	0	1	0
BOXERCLEARPATH	1	0	0
BLUEROV2	1	0	0

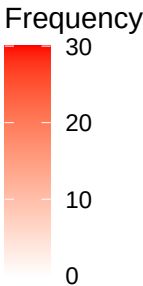


EXPERIMENT
NOEVALUATION
SHOWCASE

Experiment.Method

I3.Robot.Type

WHICHISANINDUSTRIALAGV	0	1	0	1	0
WAREHOUSEDELIVERYROBOT	0	1	0	0	0
TWOINDUSTRIALROBOTS	0	0	1	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	0	0	0
TURTLEBOT	1	4	2	2	0
TRIGLIDEINDUSTRIALASSEMBLY	1	1	0	0	0
TEDUSARTERRESTRIALSEARCH	0	1	0	0	0
SINGLESERVOINGROTATIONROBOT	0	1	0	0	1
RESCUE	0	1	0	0	0
QUADROCOPTER	0	0	1	0	0
PIONEER3DX	0	1	0	0	0
NAOROBOT	0	0	1	0	0
MULTIPLEHEXROTOR	0	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	1	0	0	0
MOBILESERVICEROBOT	0	0	0	0	1
MOBILEROBOTTIAGO	0	0	0	0	0
MOBILEROBOTTERRESTRIAL	0	1	0	0	0
MOBILEROBOTS	1	1	0	1	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0	0	0
IROBOTCREATE2	1	1	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	0	1
HEXMANIPULATOR	0	1	0	0	1
HEXAII	1	1	0	0	0
HETEROGENOUSROBOTS	0	0	1	0	0
FIELDMOBILEROBOTS	0	1	0	0	0
CRAWLERTERMINATORBOT	0	0	0	0	0
BOXERCLEARPATH	0	1	0	1	0
BLUEROV2	0	1	0	0	0

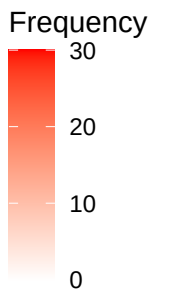


DOMAINSPECIFICPERFORMANCE
MISSIONPERFORMANCE
OVERHEADINTRODUCED
RESOURCECONSUMPTION

I3.Robot.Type

I3.Robot.Type_____I9.Adap..Logic

WHICHISANINDUSTRIALAGV	0	0	0	0	1	1	0
WAREHOUSEDELIVERYROBOT	0	1	1	0	0	0	0
TWOINDUSTRIALROBOTS	0	0	0	0	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	0	1	0	0	0	0
TURTLEBOT	0	2	2	0	0	0	0
TRIGLIDEINDUSTRIALASSEMBLY	0	0	0	0	0	1	0
TEDUSARTERRESTRIALSEARCH	1	1	0	0	0	0	0
SINGLESERVOINGROTATIONROBOT	0	0	0	0	0	1	1
RESCUE	1	1	0	0	0	0	0
QUADROCOPTER	0	0	0	0	1	0	0
PIONEER3DX	1	1	0	0	0	0	0
NAOROBOT	0	0	0	0	0	1	0
MULTIPLEHEXROTOR	0	0	0	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	0	0	0	1	0
MOBILESERVICEROBOT	1	1	1	0	1	0	0
MOBILEROBOTTIAGO	0	0	0	0	1	1	0
MOBILEROBOTTERRESTRIAL	0	0	0	0	0	1	0
MOBILEROBOTS	0	0	0	1	0	1	1
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	1	0	0	0	0
IROBOTCREATE2	0	0	1	0	0	0	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	0	1	0	0
HEXMANIPULATOR	0	0	0	0	0	2	2
HEXAII	0	0	0	0	0	1	0
HETEROGENOUSROBOTS	0	2	0	0	0	0	0
FIELDMOBILEROBOTS	0	0	0	0	0	1	1
CRAWLERTERMINATORBOT	0	0	0	0	0	0	0
BOXERCLEARPATH	0	0	0	0	1	1	0
BLUEROV2	0	0	0	0	1	0	0



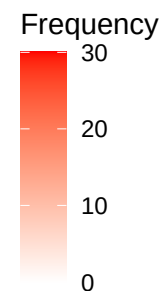
AIPLANNER
CONSTRAINTSOLVINGMODELCHECKING
DOMAINSPECIFICALGORITHM
NUMERICALOPTIMIZATION
ONTOLOGICALREASONING
SEARCHPROCEDURE
UTILITYCALCULATION

I9.Adap..Logic

I3.Robot.Type

I3.Robot.TypeI10.Monitor

WHICHISANINDUSTRIALAGV	1	0	1	0
WAREHOUSEDELIVERYROBOT	0	0	1	0
TWOINDUSTRIALROBOTS	0	0	0	1
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	1	1	1	0
TURTLEBOT	2	0	4	1
TRIGLIDEINDUSTRIALASSEMBLY	0	0	1	0
TEDUSARTERRESTRIALSEARCH	0	0	1	0
SINGLESERVOINGROTATIONROBOT	0	0	1	0
RESCUE	0	0	1	0
QUADROCOPTER	1	0	1	0
PIONEER3DX	0	0	1	0
NAOROBOT	0	0	2	0
MULTIPLEHEXROTOR	0	0	1	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	1	0
MOBILESERVICEROBOT	1	0	1	1
MOBILEROBOTTIAGO	1	0	1	0
MOBILEROBOTTERRESTRIAL	1	0	0	0
MOBILEROBOTS	0	0	1	1
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	1	1
IROBOTCREATE2	0	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	1
HEXMANIPULATOR	0	0	1	0
HEXAII	0	0	1	0
HETEROGENOUSROBOTS	0	0	1	0
FIELDMOBILEROBOTS	0	0	1	0
CRAWLERTERMINATORBOT	0	0	1	0
BOXERCLEARPATH	1	0	1	0
BLUEROV2	1	0	1	0



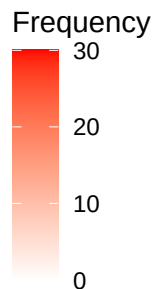
ENVIRONMENT
ENVIRONMENTAL
MANAGEDSYSTEM
MISSION

I10.Monitor

I3.Robot. Type

I3.Robot.Type_____I11.Analyze

WHICHISANINDUSTRIALAGV	0	0	0	1	0	0	0
WAREHOUSEDELIVERYROBOT	1	0	0	0	0	1	0
TWOINDUSTRIALROBOTS	0	0	0	0	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	0	0	0	0	0
TURTLEBOT	1	1	0	1	0	1	0
TRIGLIDEINDUSTRIALASSEMBLY	0	0	0	0	0	0	0
TEDUSARTERRESTRIALSEARCH	1	0	0	0	0	1	0
SINGLESERVOINGROTATIONROBOT	1	1	0	0	0	0	0
RESCUE	1	0	0	0	0	1	0
QUADROCOPTER	0	0	0	0	1	0	0
PIONEER3DX	1	0	0	0	0	1	0
NAOROBOT	0	0	0	1	0	0	1
MULTIPLEHEXROTOR	0	1	0	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	1	0	0	0	0
MOBILESERVICEROBOT	0	0	0	1	0	0	1
MOBILEROBOTTIAGO	0	0	0	1	0	0	0
MOBILEROBOTTERRESTRIAL	0	0	0	0	0	0	0
MOBILEROBOTS	0	1	0	0	0	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	0	0	1	0	0
IROBOTCREATE2	0	0	0	0	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	1	0	0	0
HEXMANIPULATOR	1	1	0	0	0	0	0
HEXAII	0	0	0	0	0	0	0
HETEROGENOUSROBOTS	0	0	0	2	0	0	0
FIELDMOBILEROBOTS	0	1	0	0	0	0	0
CRAWLERTERMINATORBOT	0	1	0	0	0	0	0
BOXERCLEARPATH	0	0	0	1	0	0	0
BLUEROV2	0	0	0	1	0	0	0



ANALYZINGAGGREGATINGDATA
COMPARISONTOHRESHOLDS
DONDURINGPLAN
LOGICALINFERENCE
SYSTEMSTATEANOMALYDETECTION
TASKUSERDRIVEN

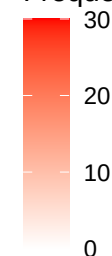
I11.Analyze

I3.Robot.Type

I3.Robot.Type_____I12.Plan

WHICHISANINDUSTRIALAGV	1	0	0
WAREHOUSEDELIVERYROBOT	0	1	0
TWOINDUSTRIALROBOTS	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	1	0
TURTLEBOT	2	2	0
TRIGLIDEINDUSTRIALASSEMBLY	1	0	0
TEDUSARTERRESTRIALSEARCH	0	0	1
SINGLESERVOINGROTATIONROBOT	1	0	0
RESCUE	0	0	1
QUADROCOPTER	0	1	0
PIONEER3DX	0	0	1
NAOROBOT	1	0	0
MULTIPLEHEXROTOR	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	1	0	0
MOBILESERVICEROBOT	2	0	0
MOBILEROBOTTIAGO	0	1	0
MOBILEROBOTTERRESTRIAL	0	1	0
MOBILEROBOTS	2	0	0
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	1	0
IROBOTCREATE2	0	1	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	1	0	0
HEXMANIPULATOR	1	0	0
HEXAII	1	0	0
HETEROGENOUSROBOTS	2	2	0
FIELDMOBILEROBOTS	1	0	0
CRAWLERTERMINATORBOT	0	0	0
BOXERCLEARPATH	1	0	0
BLUEROV2	1	0	0

Frequency



DETERMININGTHEOPTIMALCHOICE
RELYINGONDESIGNTIMERULESMODELS
USINGAIPANNINGLANGUAGES

I12.Plan

I3.Robot.Type

TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION

KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR

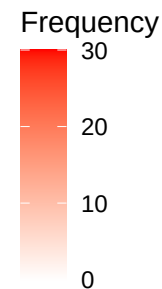
INFOTAINMENTROBOTMOBILESERVICEROBOT

WHICHISANINDUSTRIALAGV	0	0	0	1
WAREHOUSEDELIVERYROBOT	0	0	0	1
TWOINDUSTRIALROBOTS	1	0	1	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	1	0	0	1
TURTLEBOT	1	1	0	3
TRIGLIDEINDUSTRIALASSEMBLY	0	0	1	0
TEDUSARTERRESTRIALSEARCH	1	0	0	0
SINGLESERVOINGROTATIONROBOT	1	0	0	1
RESCUE	1	0	0	0
QUADROCOPTER	1	0	0	0
PIONEER3DX	1	0	0	0
NAOROBOT	0	2	0	0
MULTIPLEHEXROTOR	1	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	0	0	1
MOBILESERVICEROBOT	2	1	0	1
MOBILEROBOTTIAGO	0	0	0	1
MOBILEROBOTTERRESTRIAL	1	0	0	1
MOBILEROBOTS	1	0	0	2
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	1	0	0	0
IROBOTCREATE2	0	0	0	1
INFOTAINMENTROBOTMOBILESERVICEROBOT	1	1	0	0
HEXMANIPULATOR	1	0	0	1
HEXAII	0	0	1	0
HETEROGENOUSROBOTS	0	2	0	0
FIELDMOBILEROBOTS	1	0	0	1
CRAWLERTERMINATORBOT	1	0	0	0
BOXERCLEARPATH	0	0	0	1
BLUEROV2	0	0	0	1

CHANGEINRELATIONSHIPSBETWEENCOMPONENTS
ADDITIONORREMOVALOFCOMPONENTS
REPARAMETERIZATIONOFCOMPONENTS
COMPONENTREDEPLOYMENT

I3.Robot.TypeI3.Execute

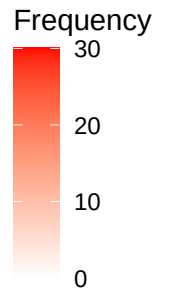
I3.Execute



I3.Robot.Type

I3.Robot.Type_I14.Knowledge

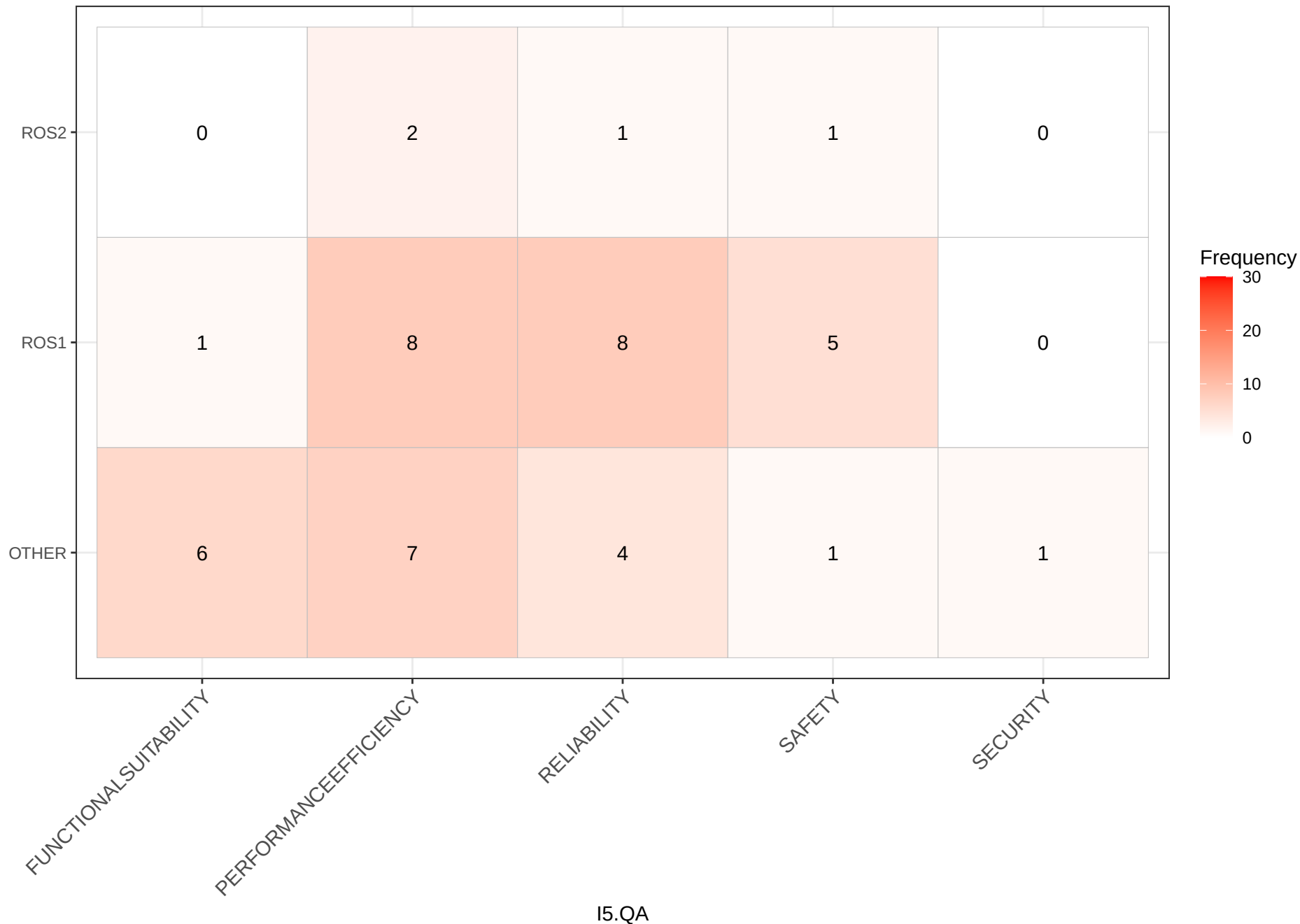
WHICHISANINDUSTRIALAGV	0	0	0	1	0	0
WAREHOUSEDELIVERYROBOT	0	1	0	0	0	0
TWOINDUSTRIALROBOTS	0	1	0	0	0	0
TWOCASESTUDIESMOBILEMANIPULATORASRUNNINGEXAMPLEQUADROCOPTORFOREVALUATION	0	0	0	0	0	1
TURTLEBOT	0	2	0	0	2	0
TRIGLIDEINDUSTRIALASSEMBLY	0	1	0	0	0	0
TEDUSARTERRESTRIALSEARCH	0	1	0	0	0	0
SINGLESERVOINGROTATIONROBOT	0	1	0	0	0	0
RESCUE	0	1	0	0	0	0
QUADROCOPTER	0	0	0	1	0	0
PIONEER3DX	0	1	0	0	0	0
NAOROBOT	0	2	0	0	0	0
MULTIPLEHEXROTOR	0	1	0	0	0	0
MSUEVORALLYMOBILETERRESTRIAL	0	1	0	0	0	0
MOBILESERVICEROBOT	0	0	1	1	0	0
MOBILEROBOTTIAGO	0	0	1	0	0	0
MOBILEROBOTTERRESTRIAL	0	0	0	0	0	1
MOBILEROBOTS	0	1	0	0	0	1
KUKALIGHTWEIGHTROBOT4LWR4MOBILEMANIPULATOR	0	0	1	0	0	0
IROBOTCREATE2	1	0	0	0	0	0
INFOTAINMENTROBOTMOBILESERVICEROBOT	0	0	0	1	0	0
HEXMANIPULATOR	0	2	0	0	0	0
HEXAII	0	1	0	0	0	0
HETEROGENOUSROBOTS	0	0	0	2	0	0
FIELDMOBILEROBOTS	0	1	0	0	0	0
CRAWLERTERMINATORBOT	0	1	0	0	0	0
BOXERCLEARPATH	0	0	0	1	0	0
BLUEROV2	0	0	0	1	0	0



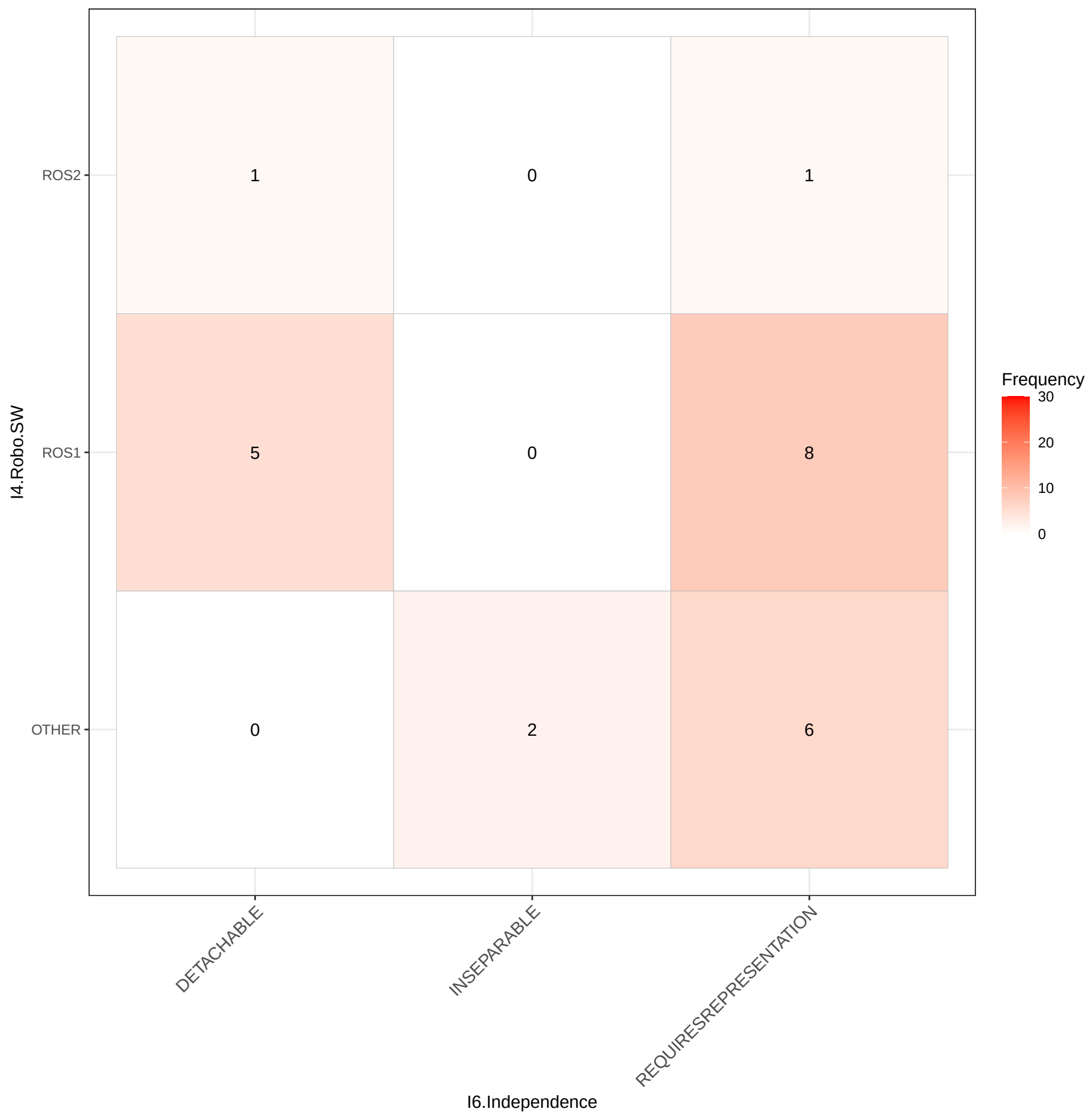
BEHAVIORMODEL
COMPONENTMODEL
GRAMMARDSL
KNOWLEDGEREPRESENTATION
OTHER
VARIABILITYMODEL

I14.Knowledge

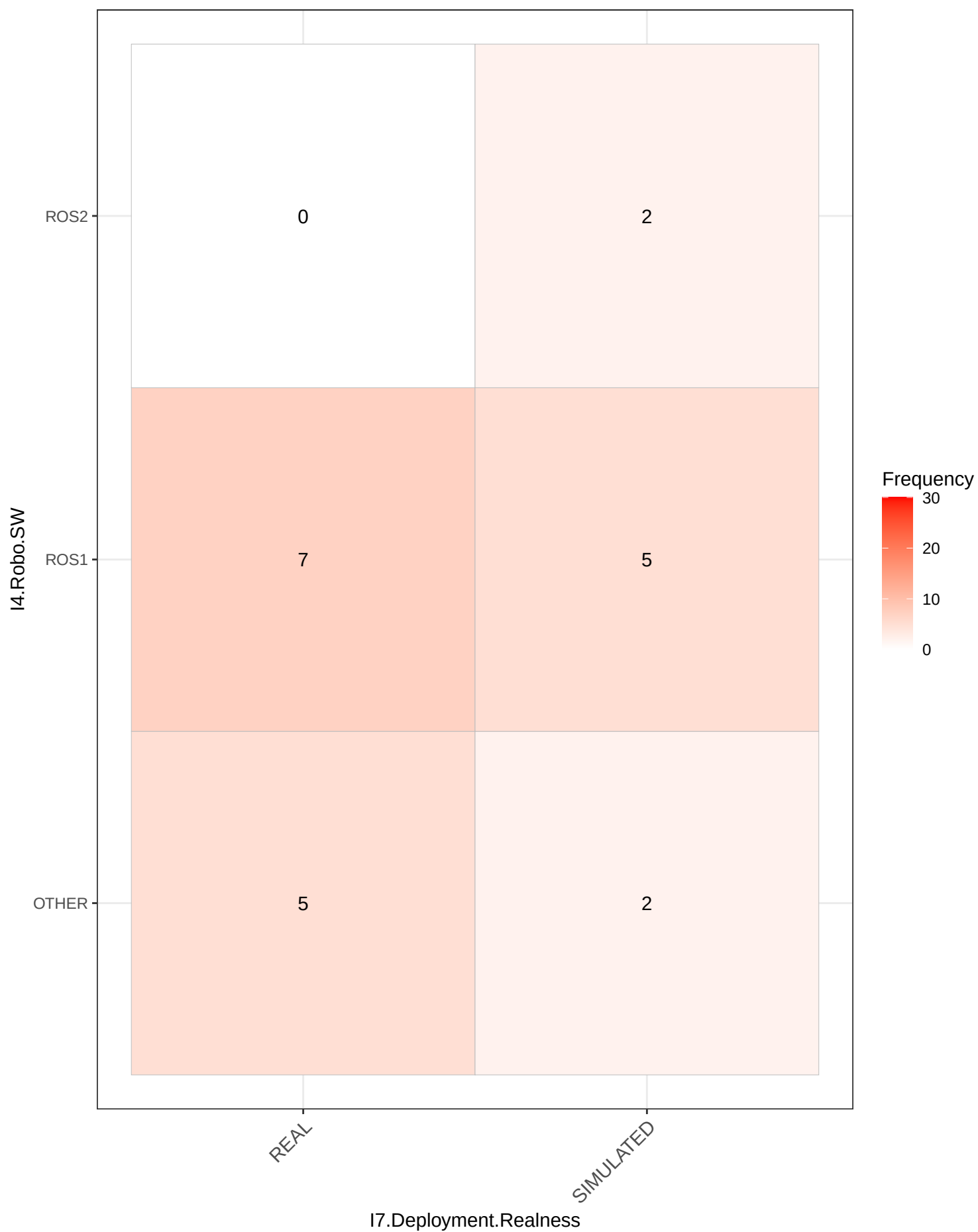
I4.Robo.SW _____ I5.QA



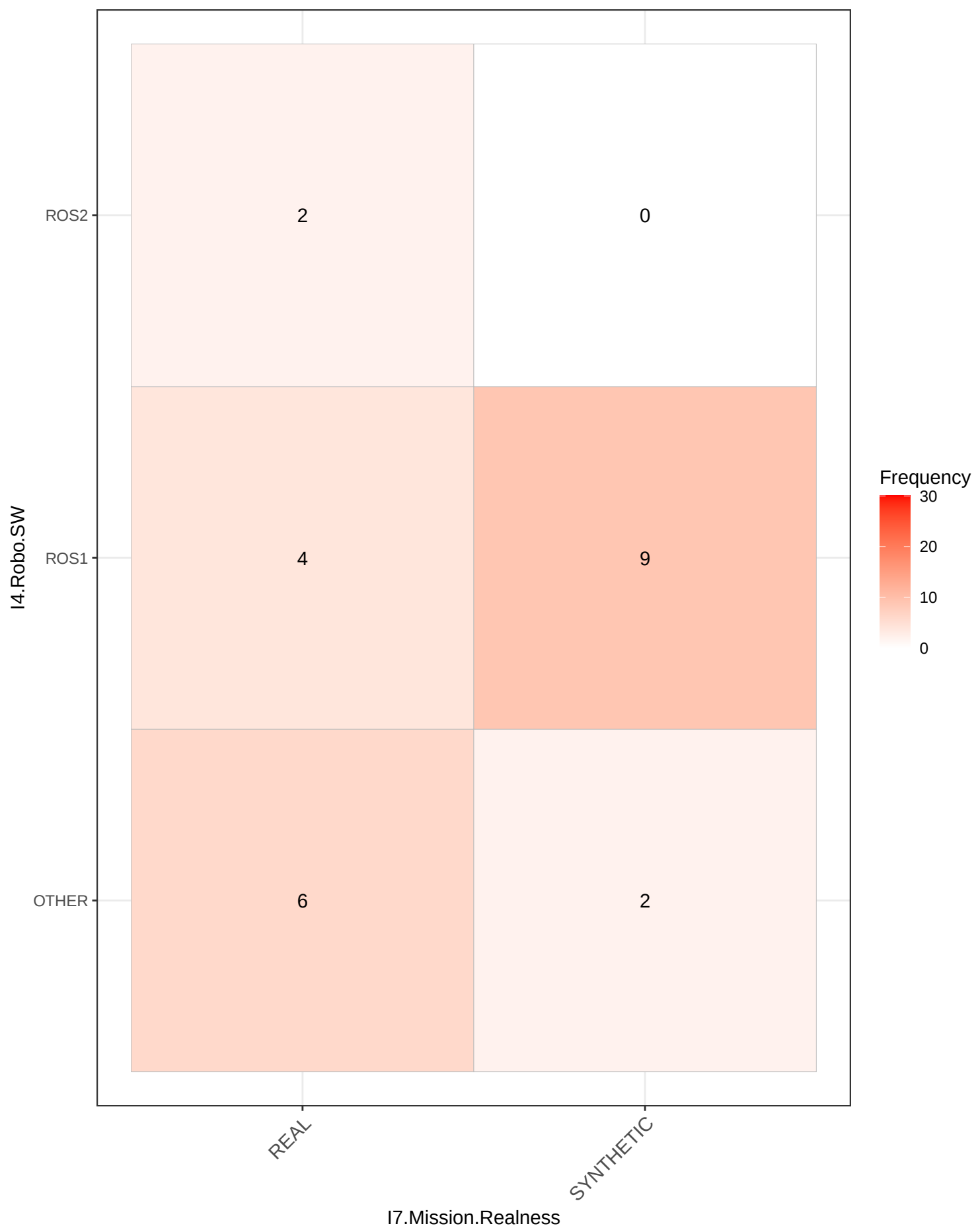
I4.Robo.SW_____I6.Independence



I4.Robo.SW_____I7.Deployment.Realness



I4.Robo.SW _____ I7.Mission.Realness



I4.Robo.SW_____Experiment.Method

I4.Robo.SW

ROS2

ROS1

OTHER

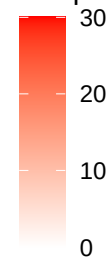
EXPERIMENT

NOEVALUATION

SHOWCASE

Experiment.Method

Frequency



1

1

0

7

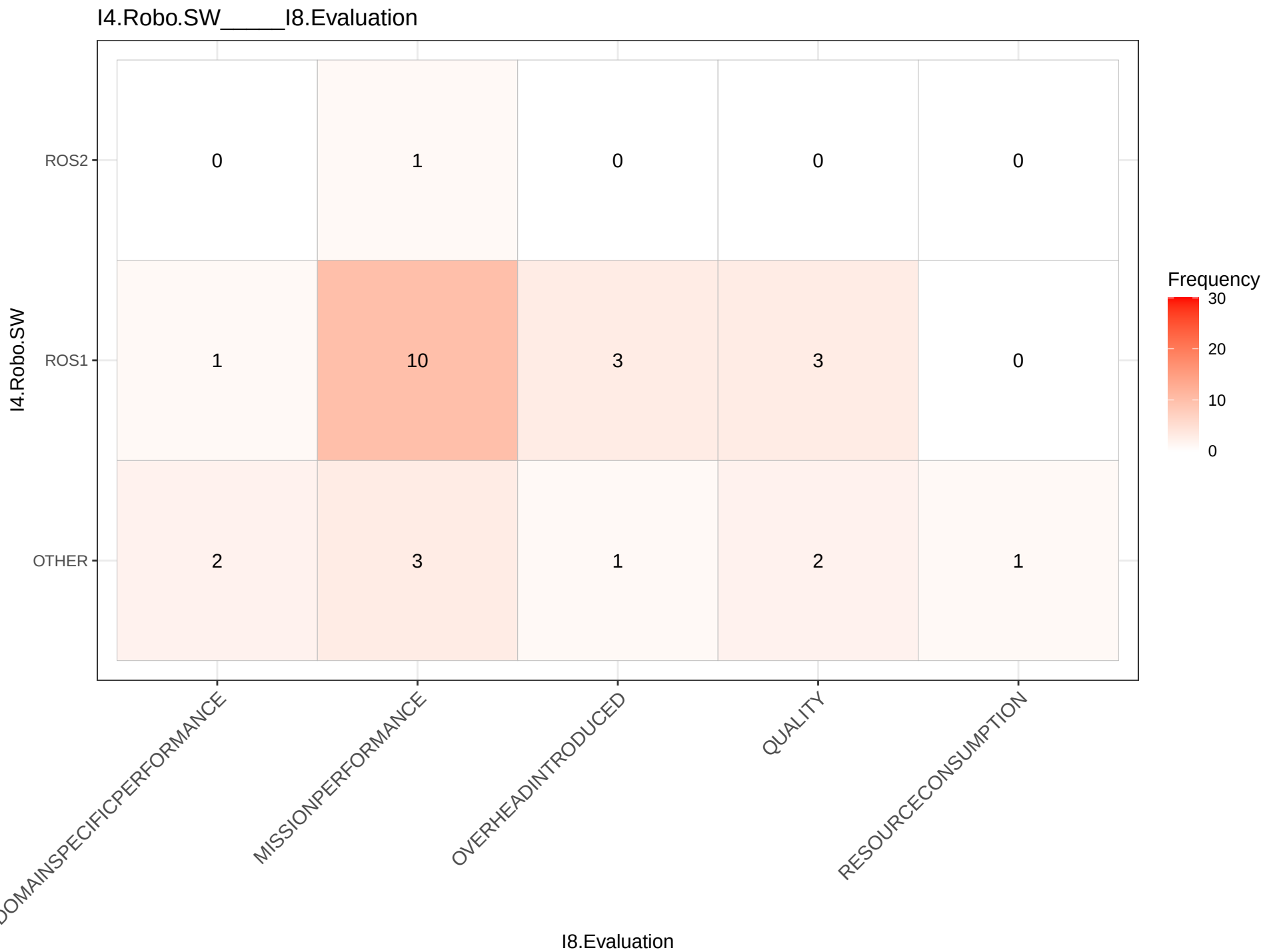
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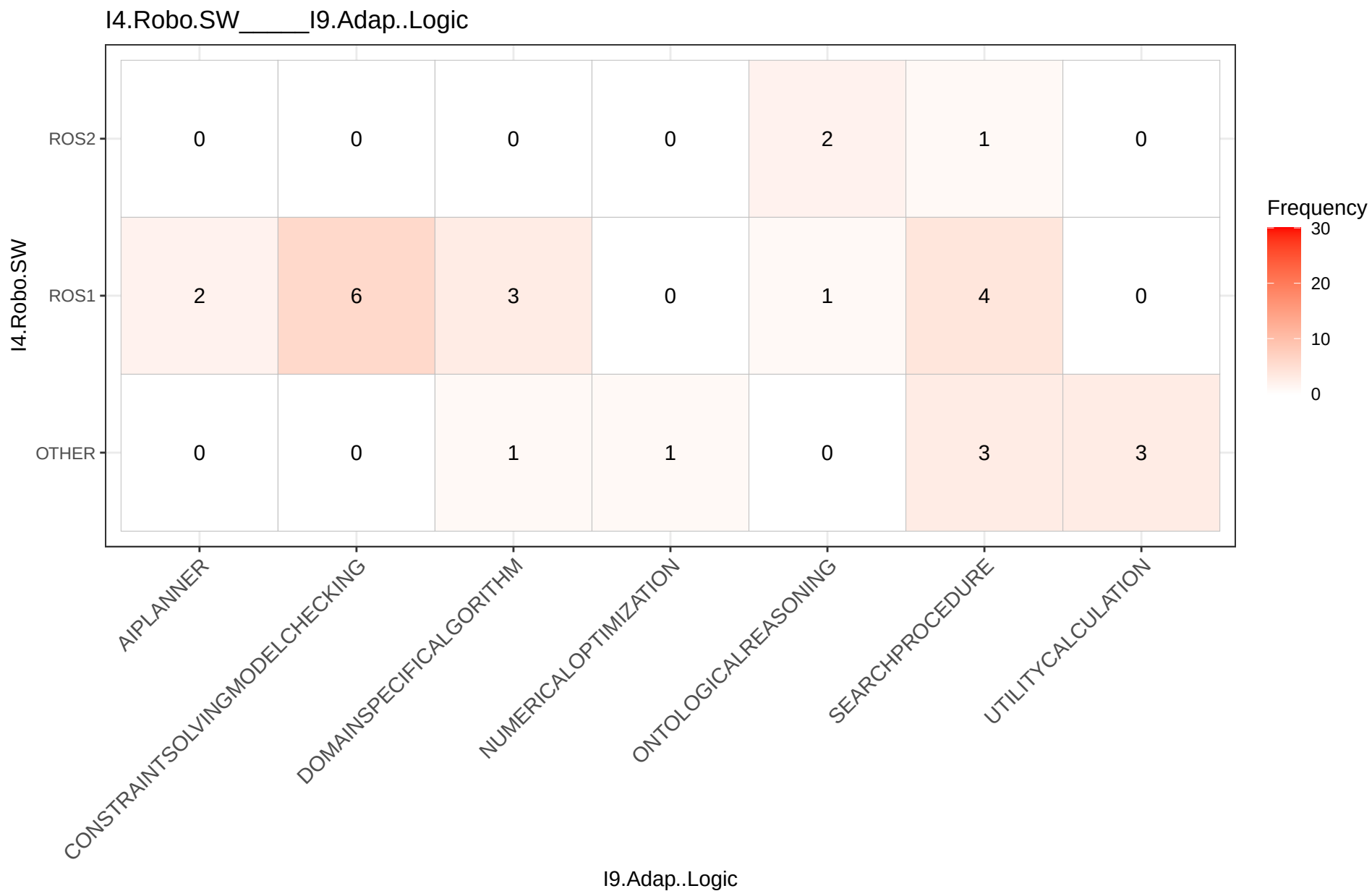
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5

3

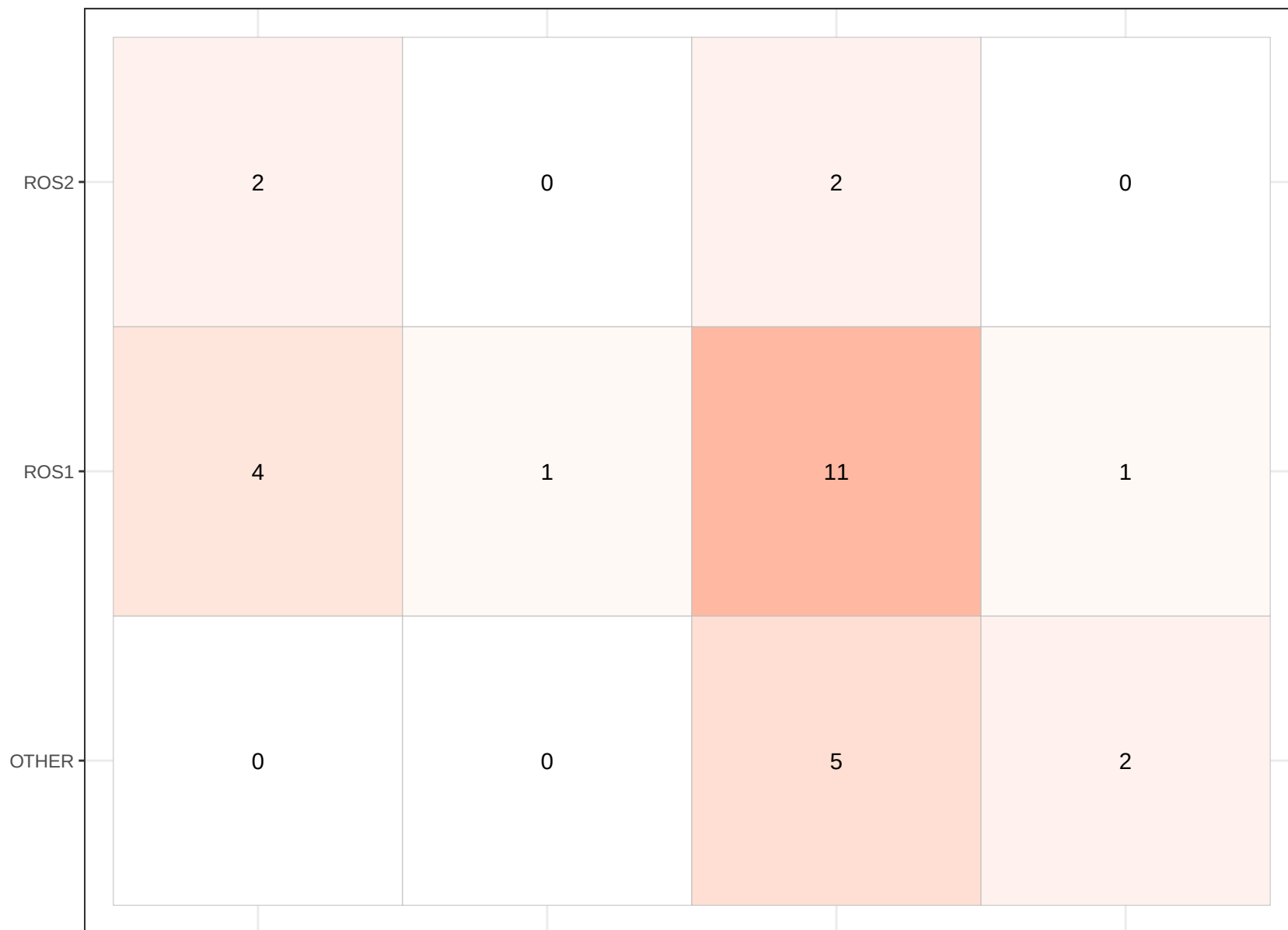
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I4.Robo.SW _____ I10.Monitor

I4.Robo.SW



OTHER

ROS2

ROS1

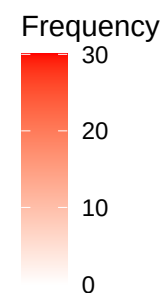
ENVIRONMENTAL

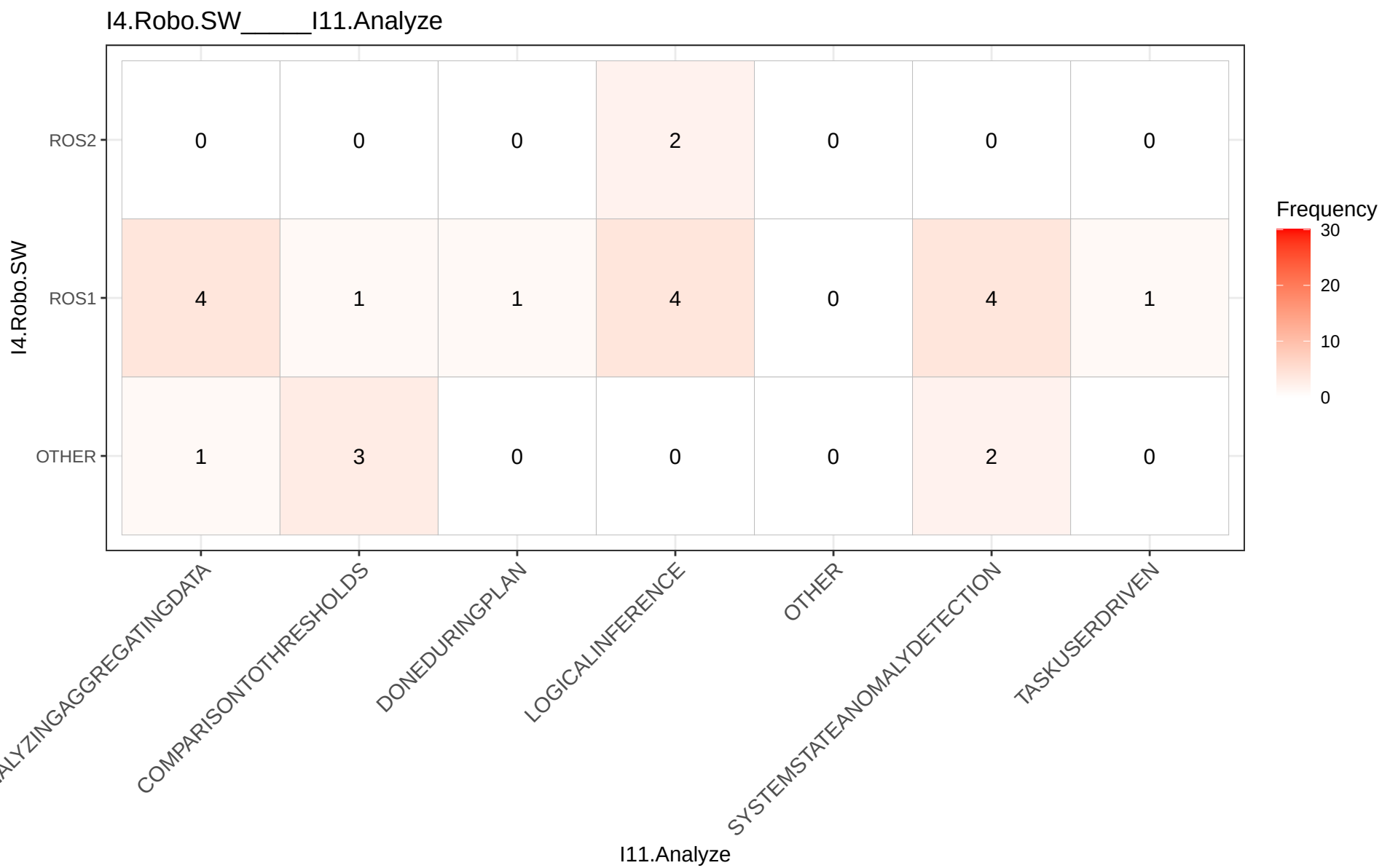
ENVIRONMENT

I10.Monitor

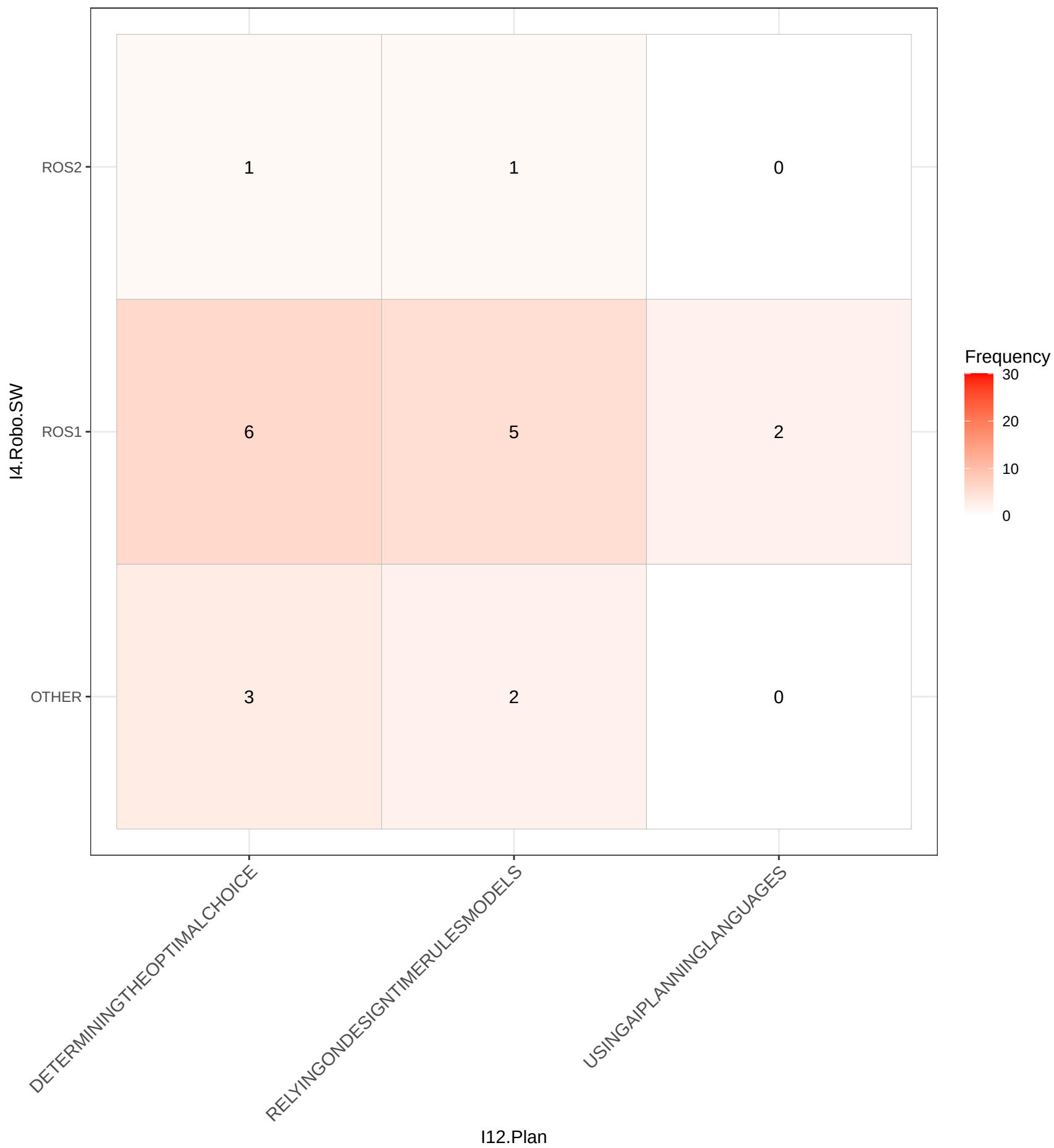
MANAGEDSYSTEM

MISSION

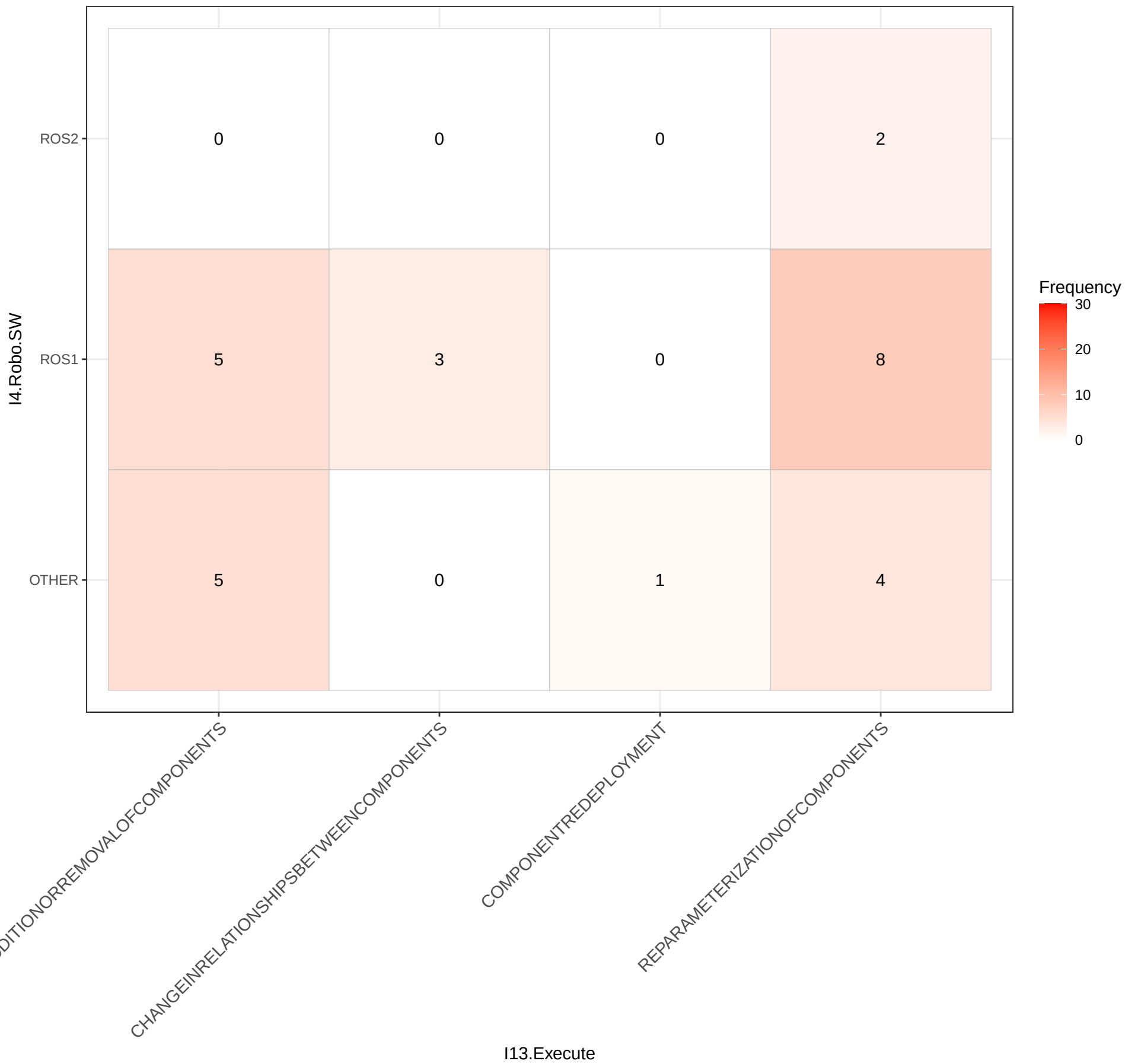




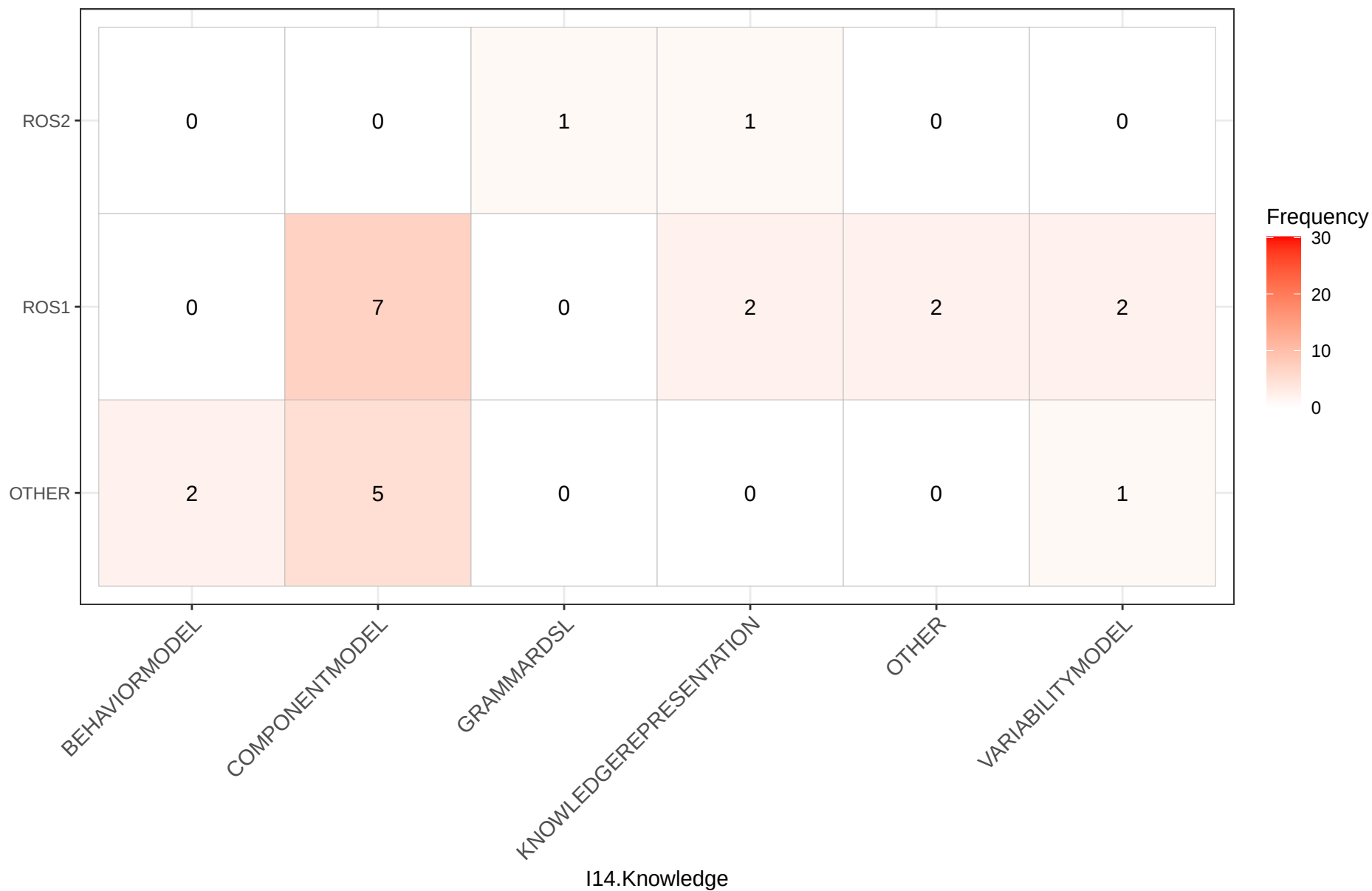
I4.Robo.SW_____I12.Plan



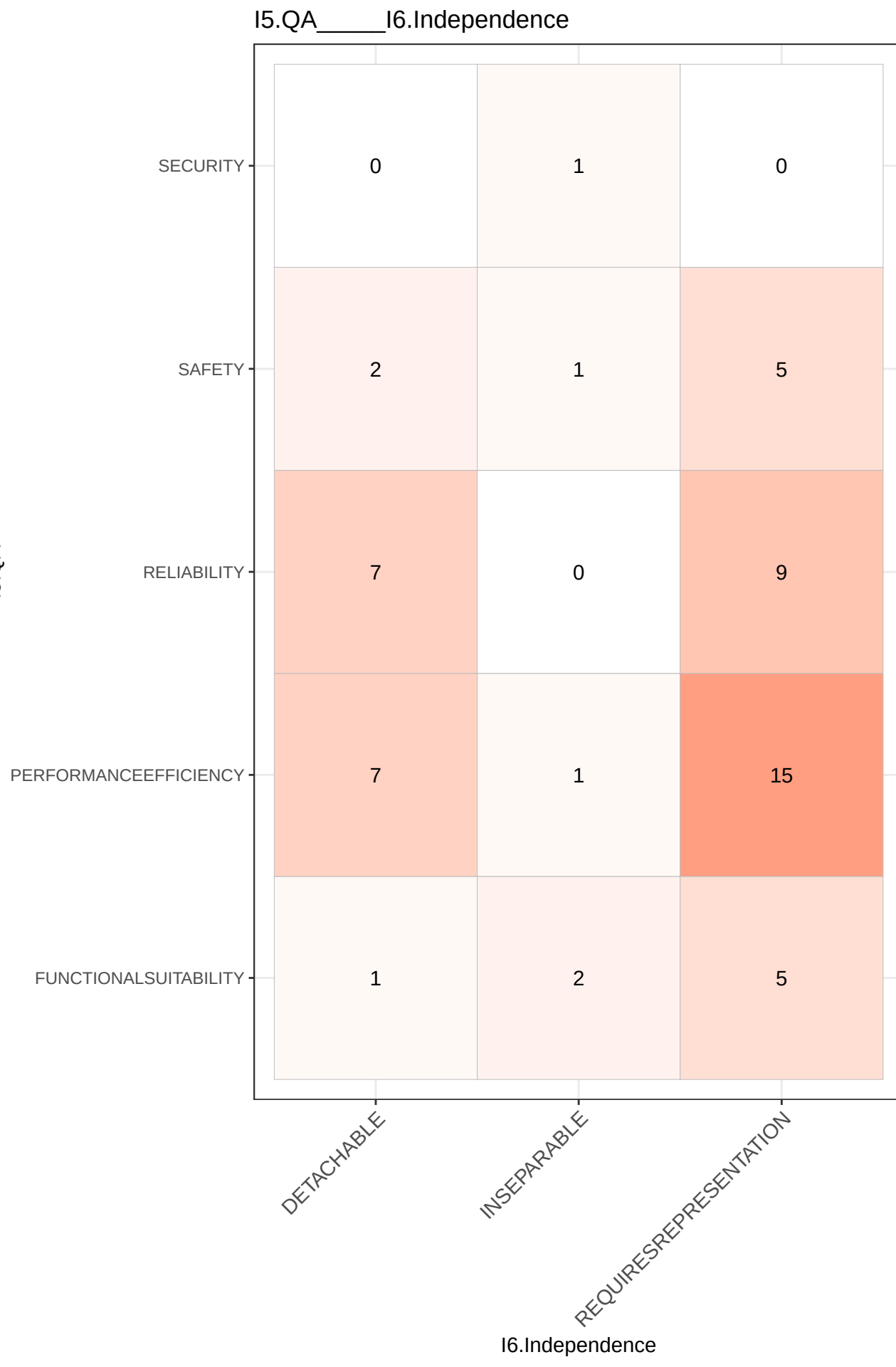
I4.Robo.SW_____I13.Execute

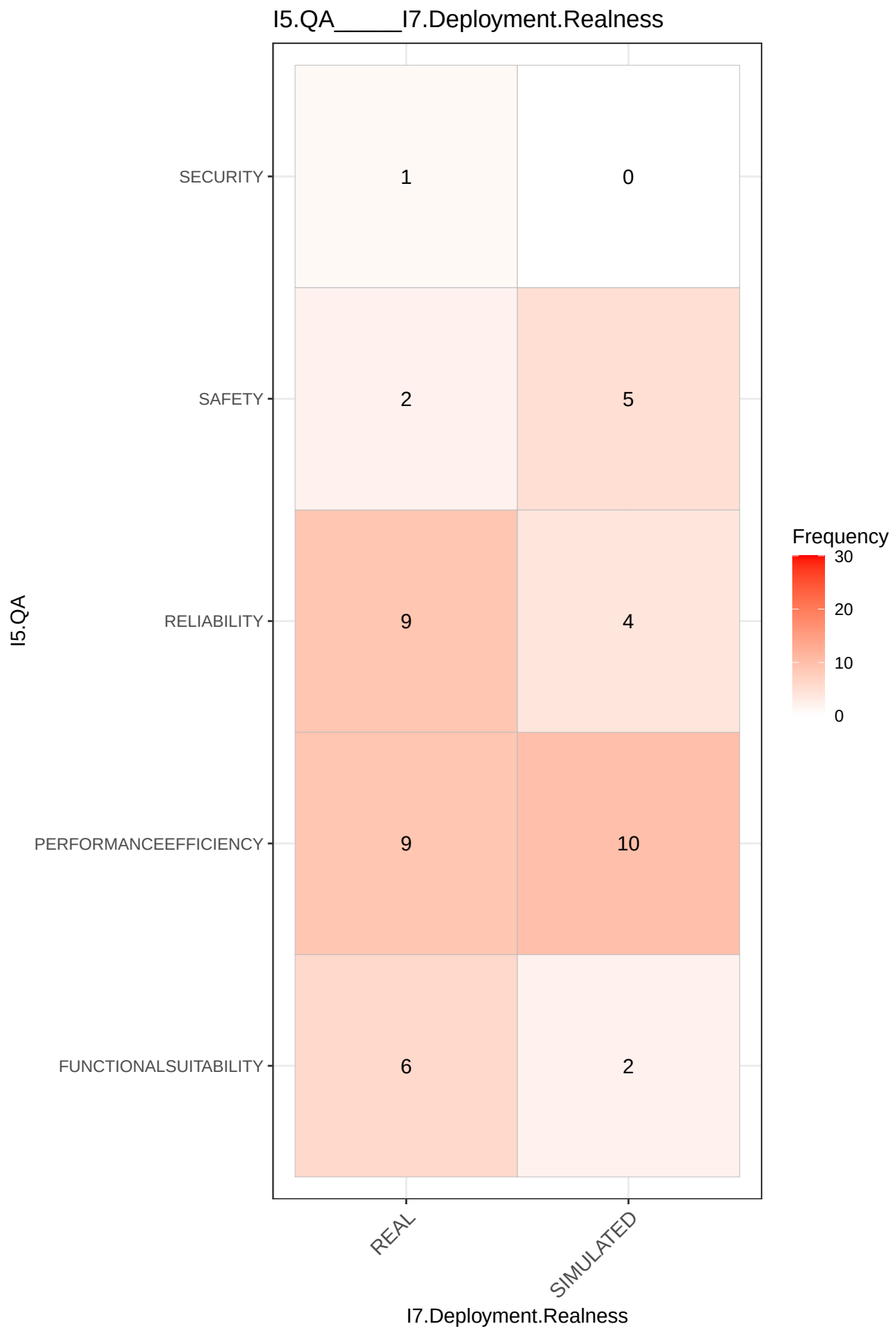


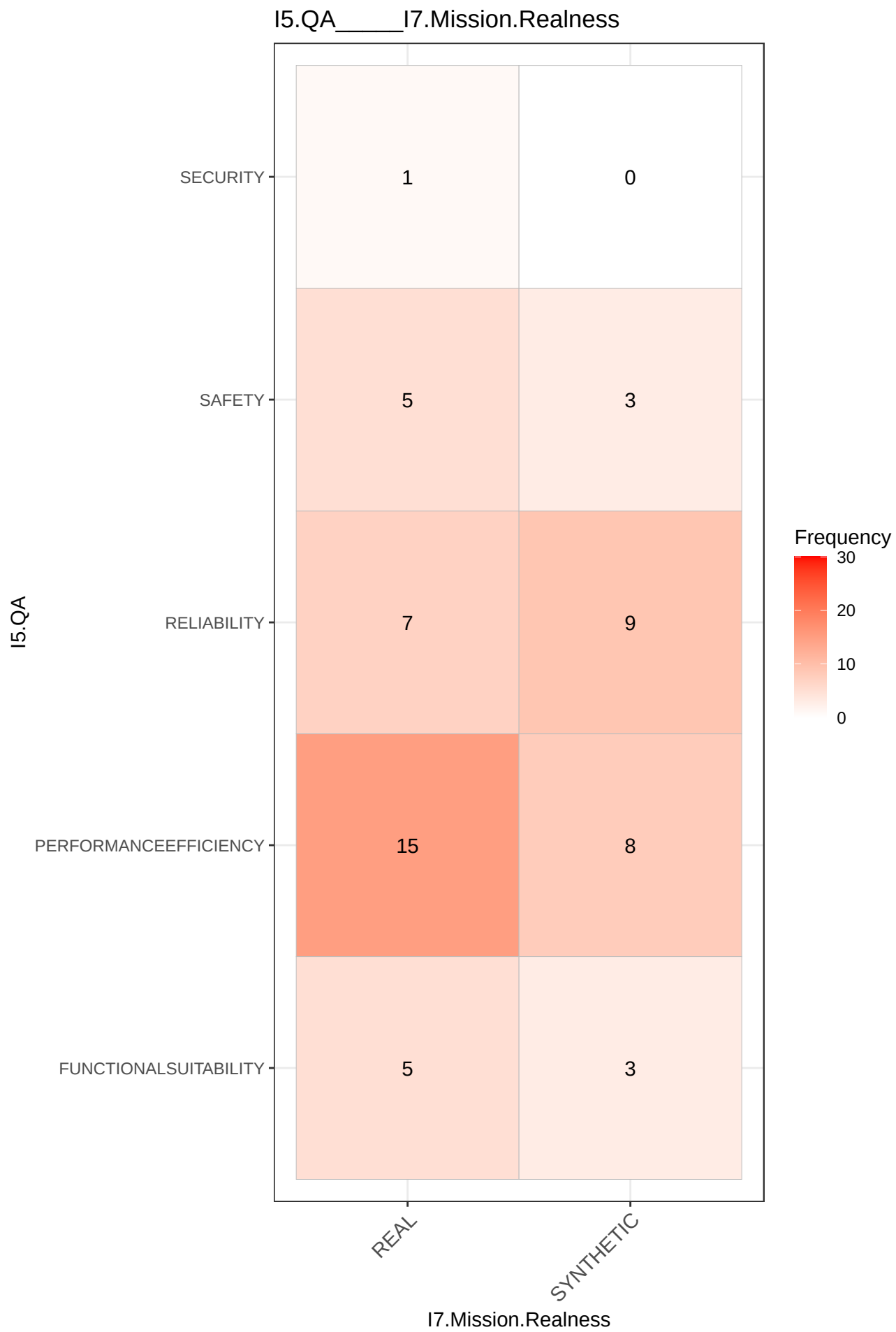
I4.Robo.SW ____ I14.Knowledge

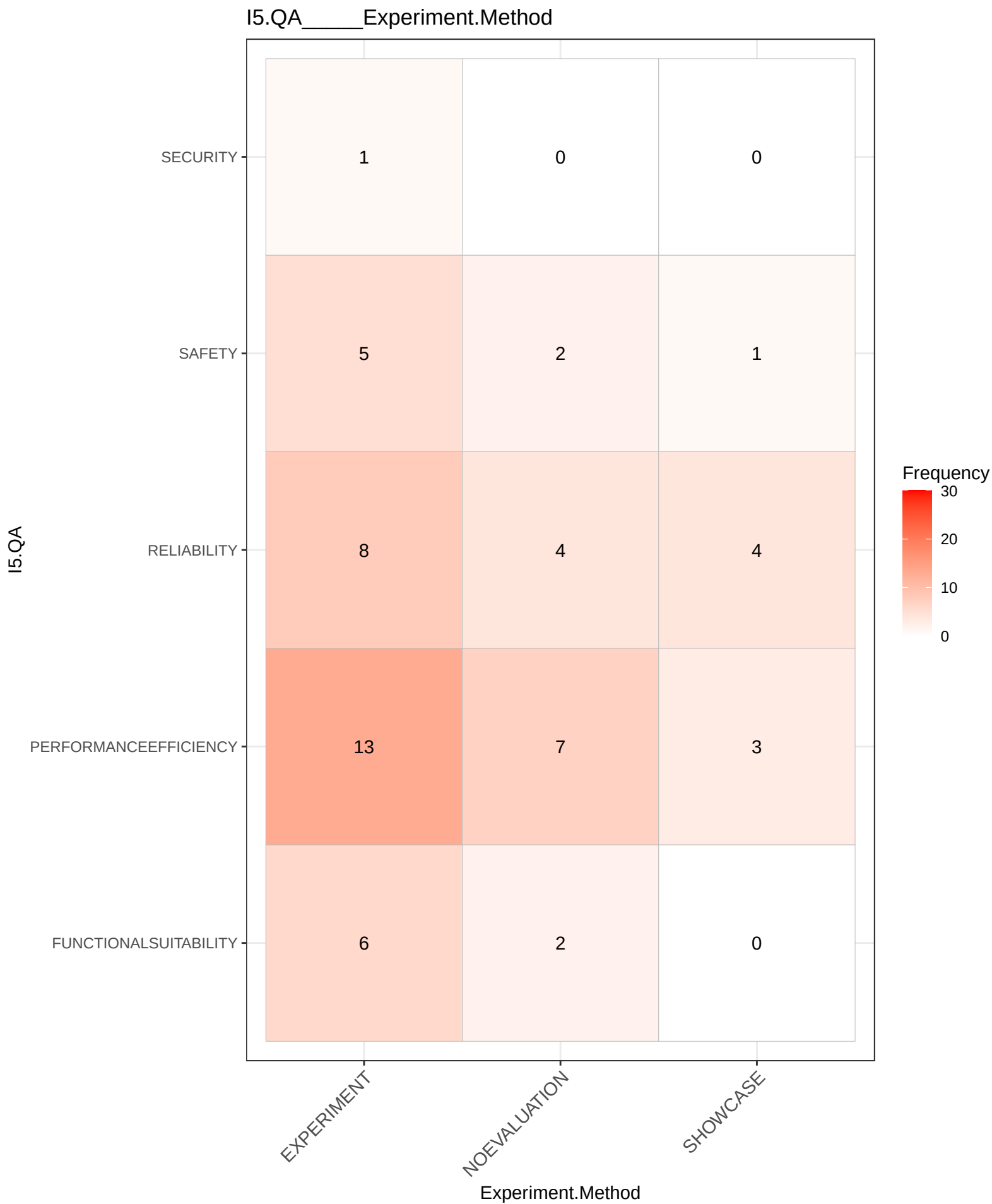


I5.QA









I5.QA

I5.QA_____I8.Evaluation

SECURITY

1

1

0

1

0

SAFETY

1

6

1

4

0

RELIABILITY

2

11

2

0

1

PERFORMANCEEFFICIENCY

2

10

5

4

2

FUNCTIONALSUITABILITY

2

4

1

1

1

DOMAINSPECIFICPERFORMANCE

MISSIONPERFORMANCE

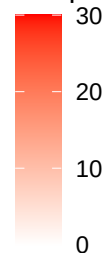
OVERHEADINTRODUCED

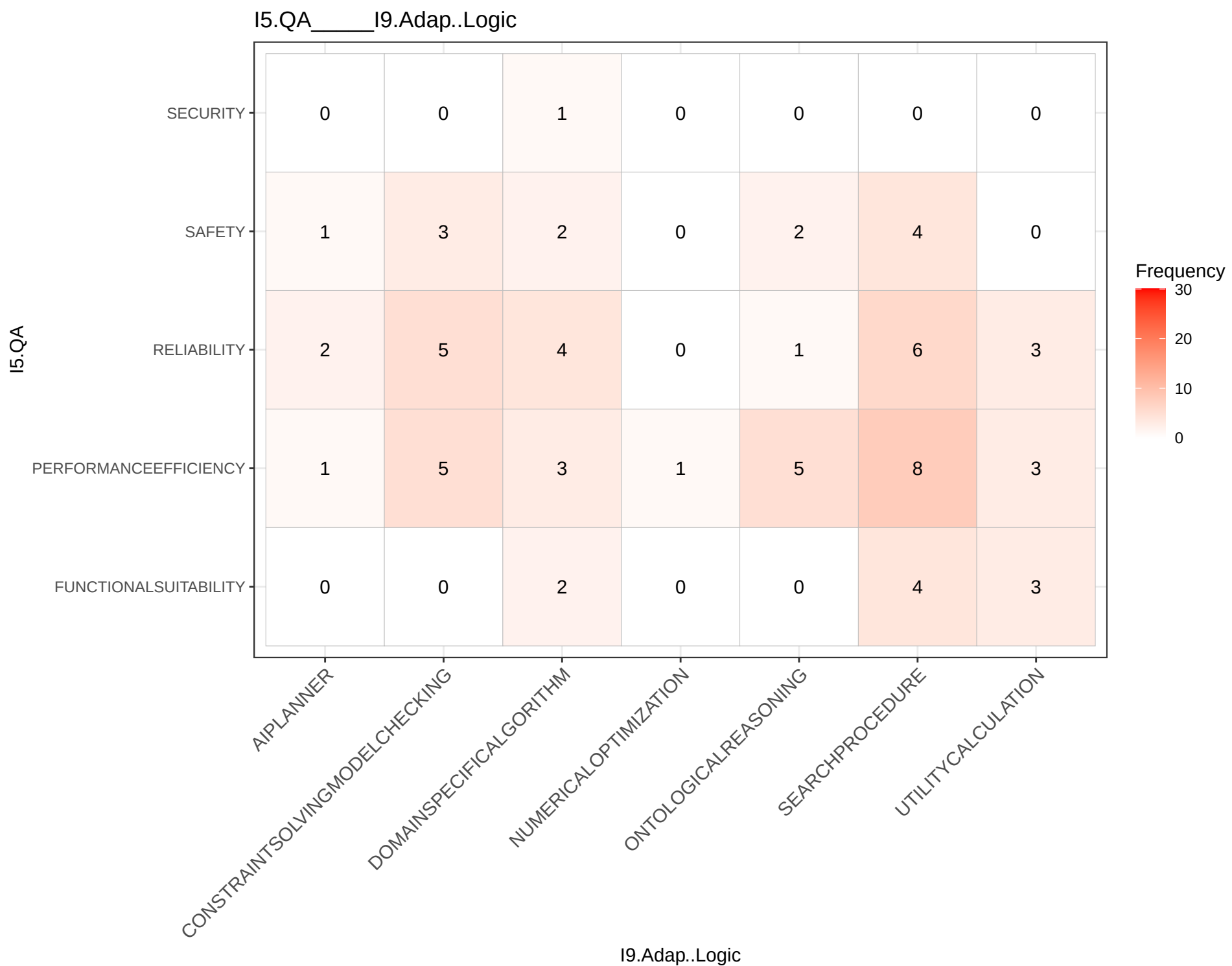
QUALITY

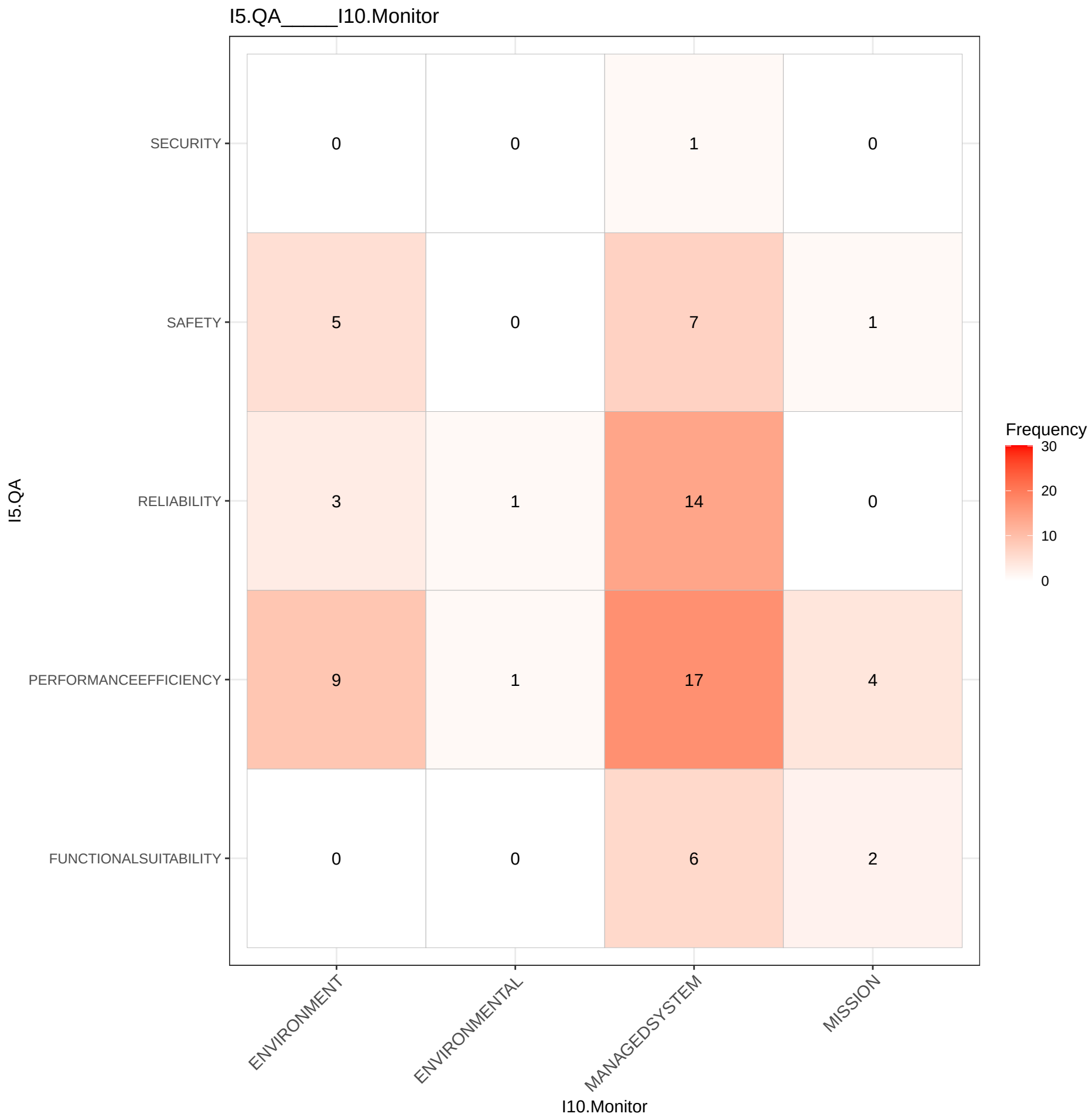
RESOURCECONSUMPTION

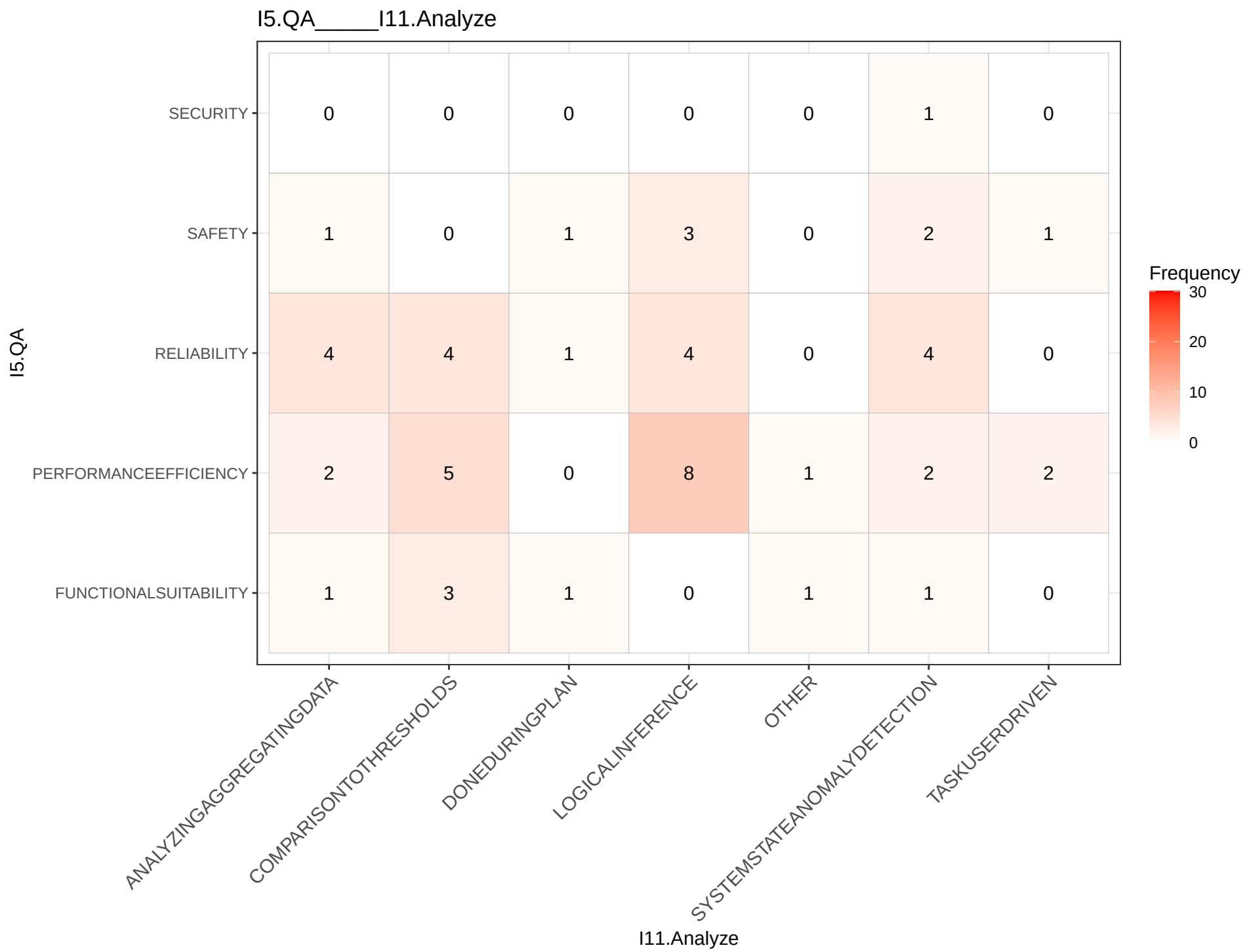
I8.Evaluation

Frequency









I5.QA

SECURITY

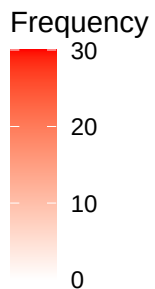
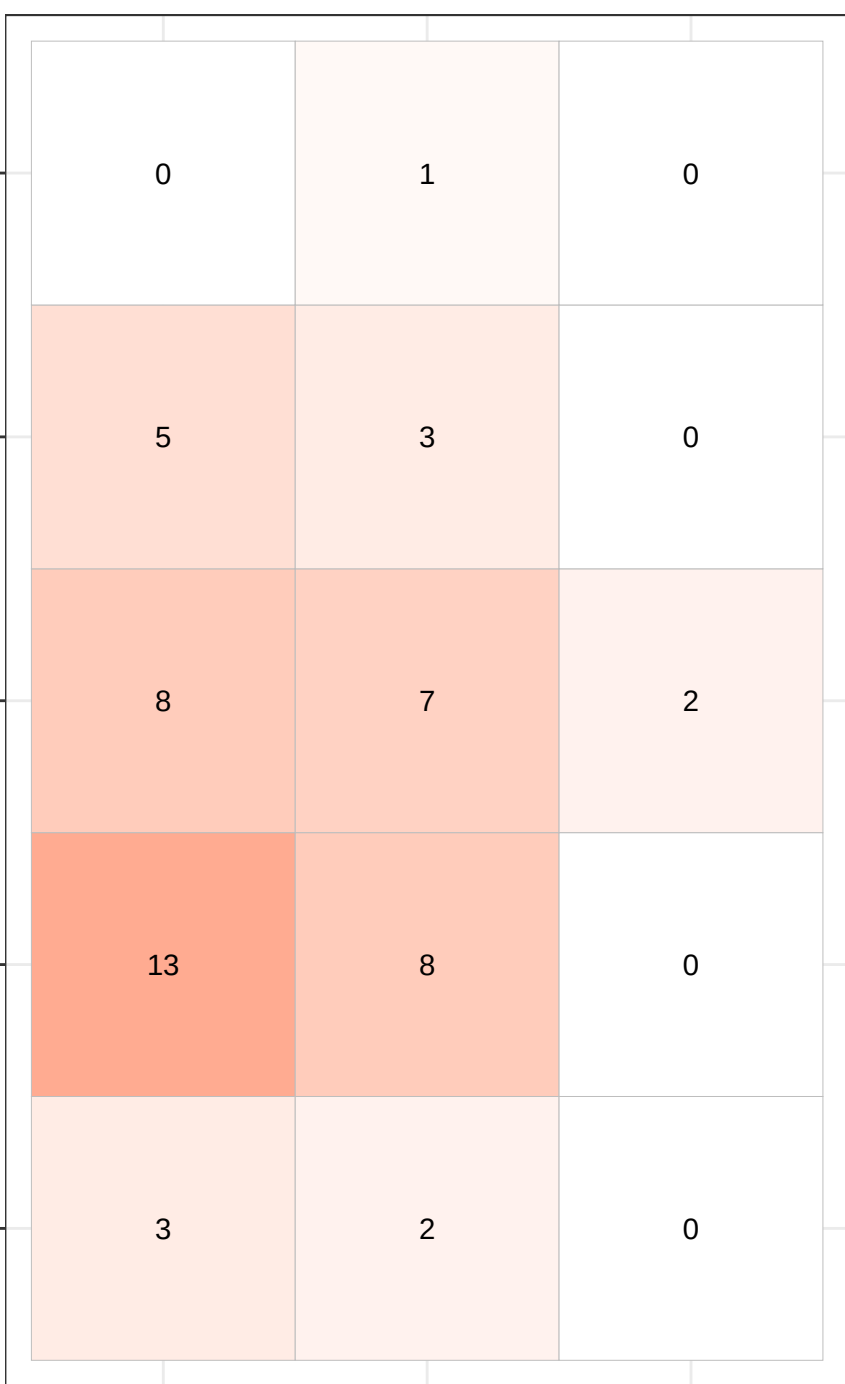
SAFETY

RELIABILITY

PERFORMANCEEFFICIENCY

FUNCTIONALSUITABILITY

I5.QA_____I12.Plan



DETERMININGTHEOPTIMALCHOICE

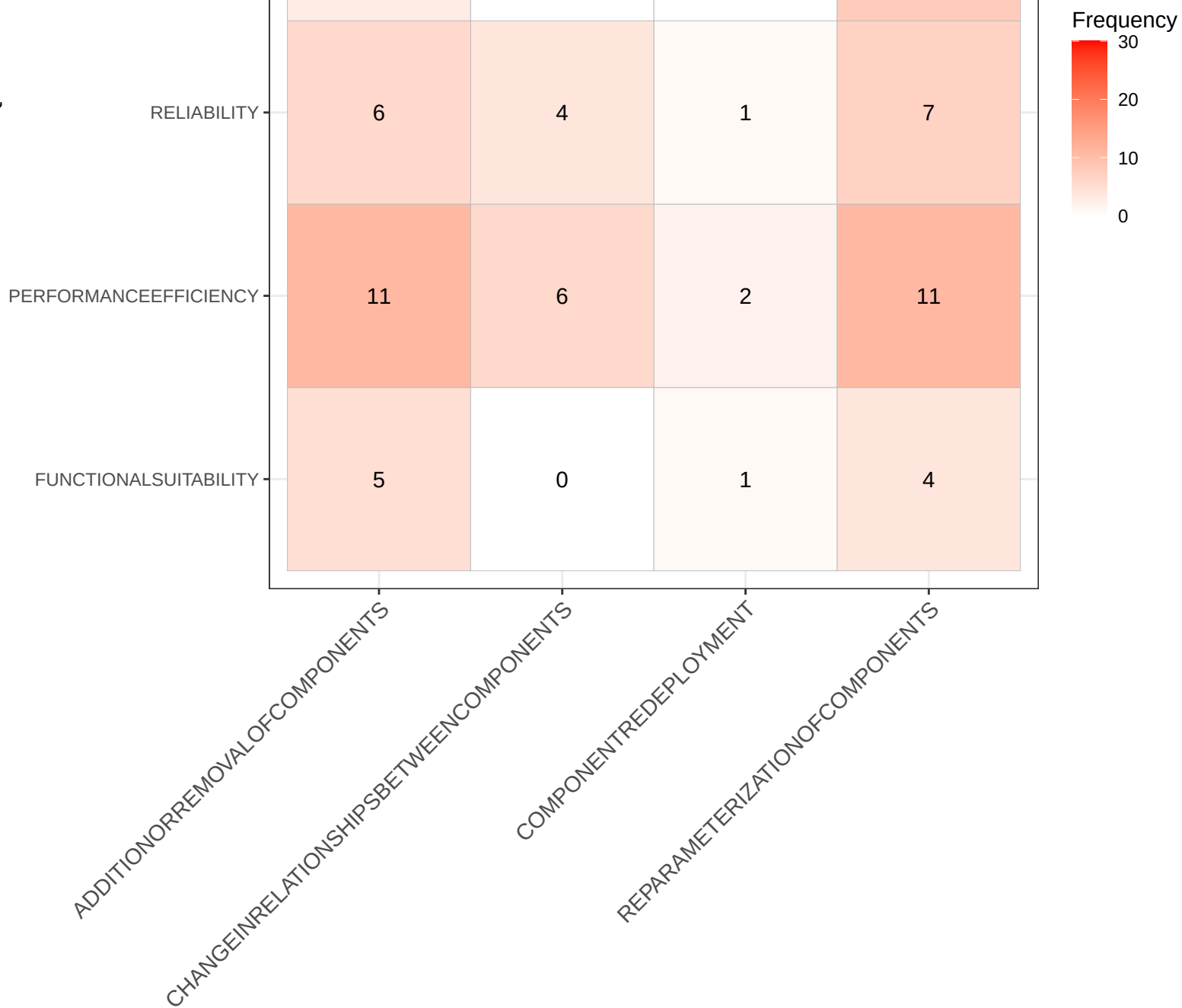
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

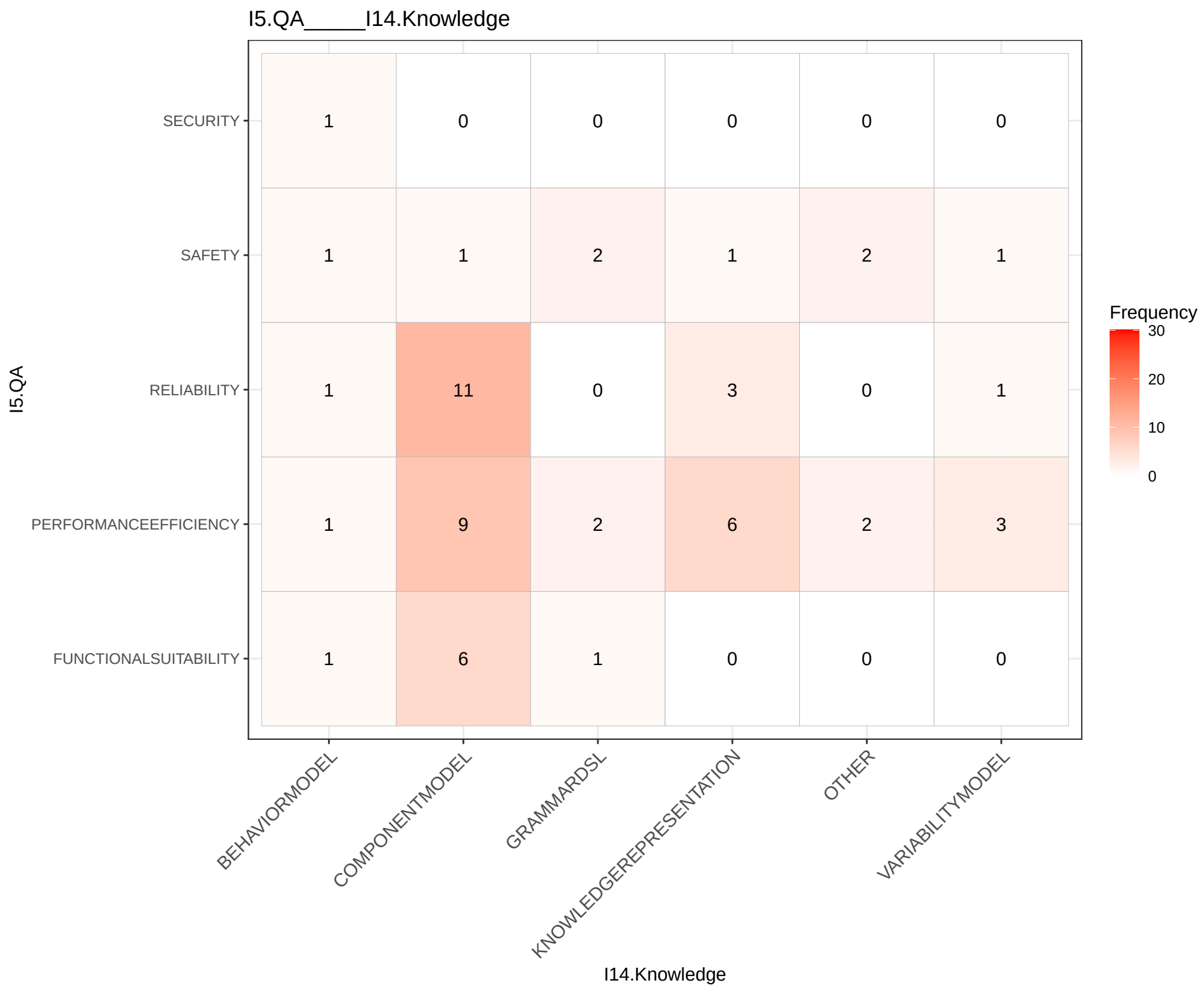
I12.Plan

I5.QA_____I13.Execute

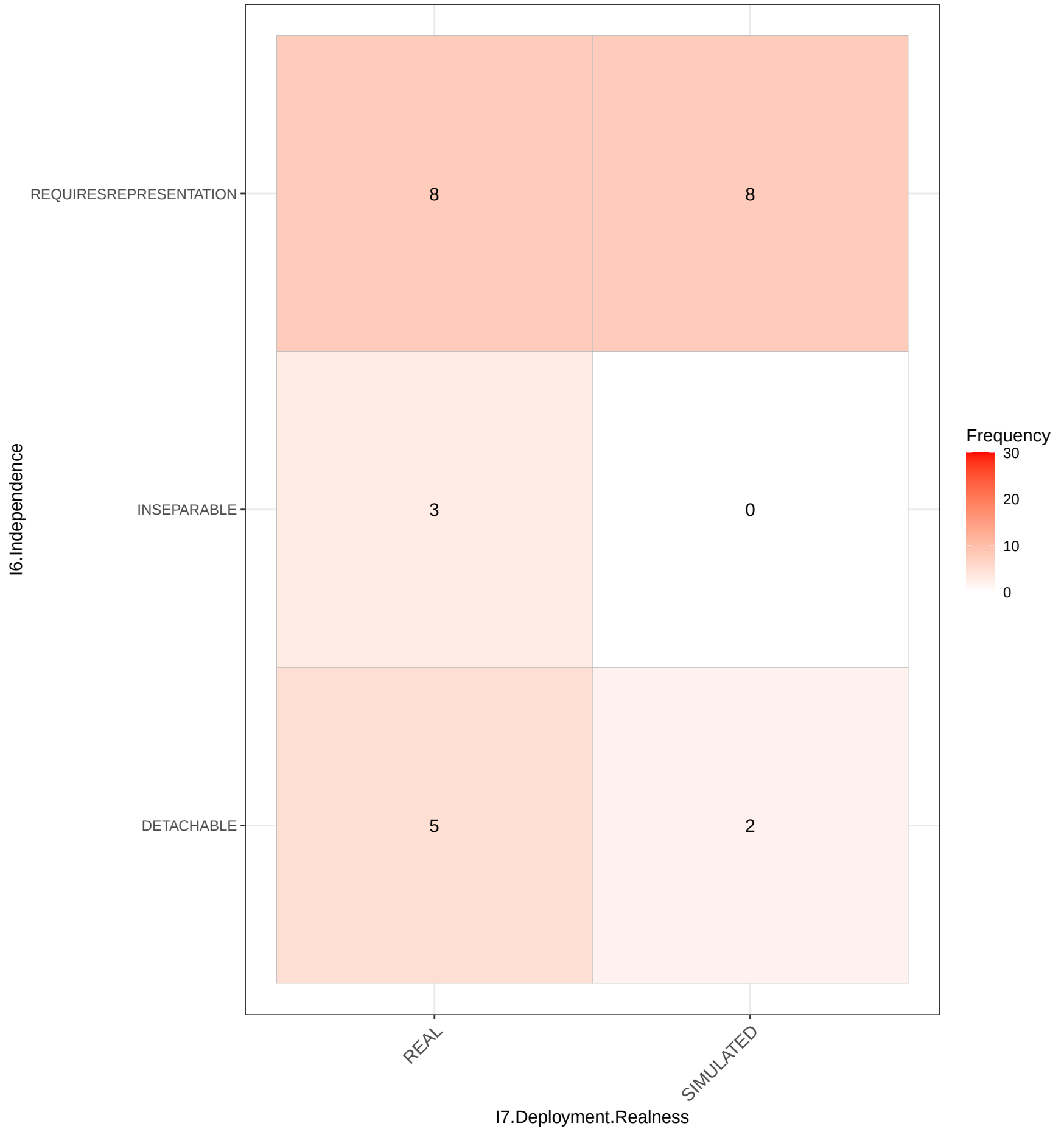
I5.QA



I13.Execute



I6.Independence_____I7.Deployment.Realness



I6.Independence_____I7.Mission.Realness

I6.Independence

REQUIRESREPRESENTATION

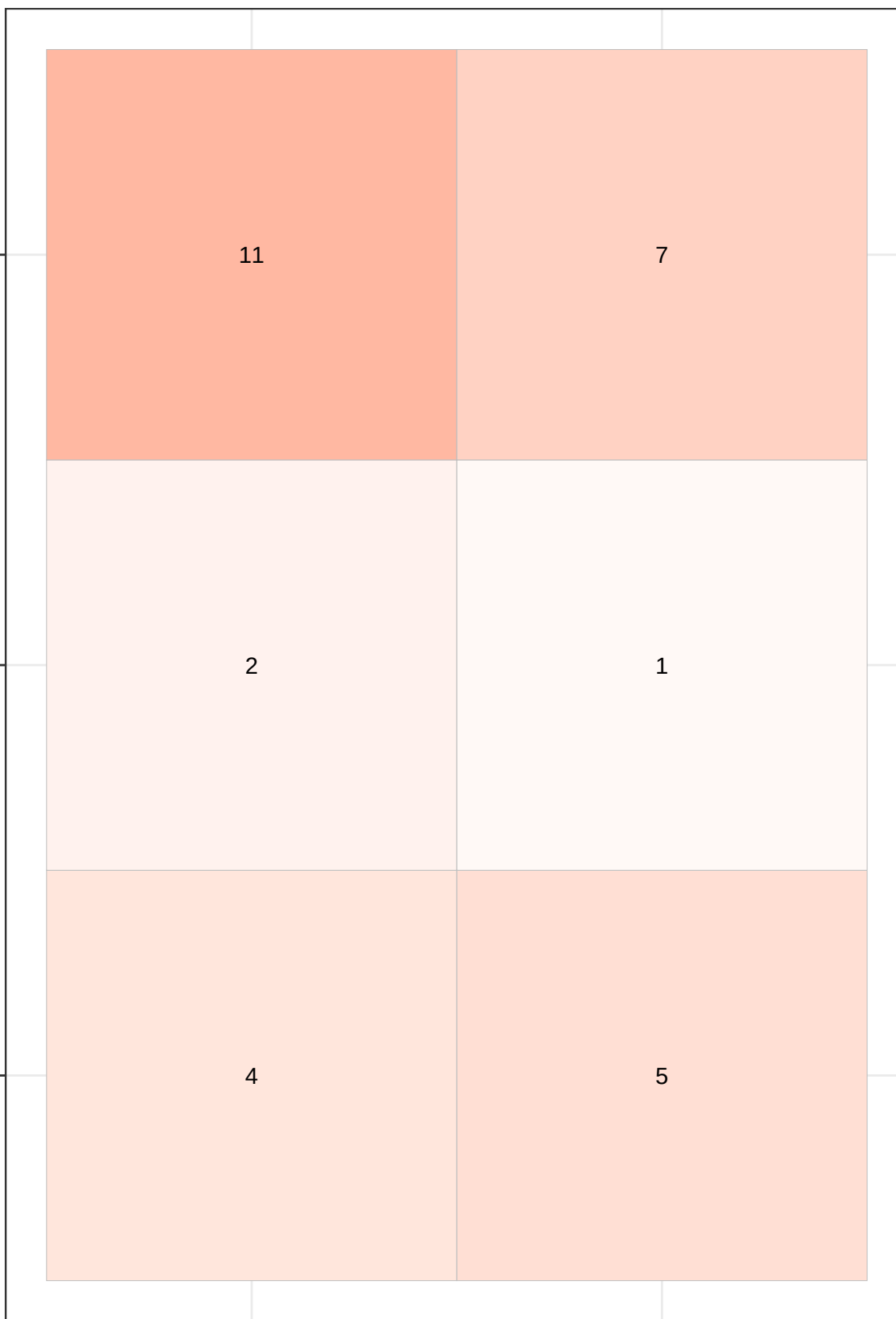
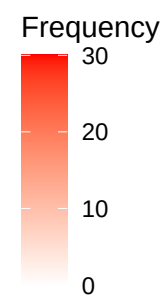
INSEPARABLE

DETACHABLE

REAL

SYNTHETIC

I7.Mission.Realness



I6.Independence

I6.Independence_____Experiment.Method

REQUIRESREPRESENTATION

INSEPARABLE

DETACHABLE

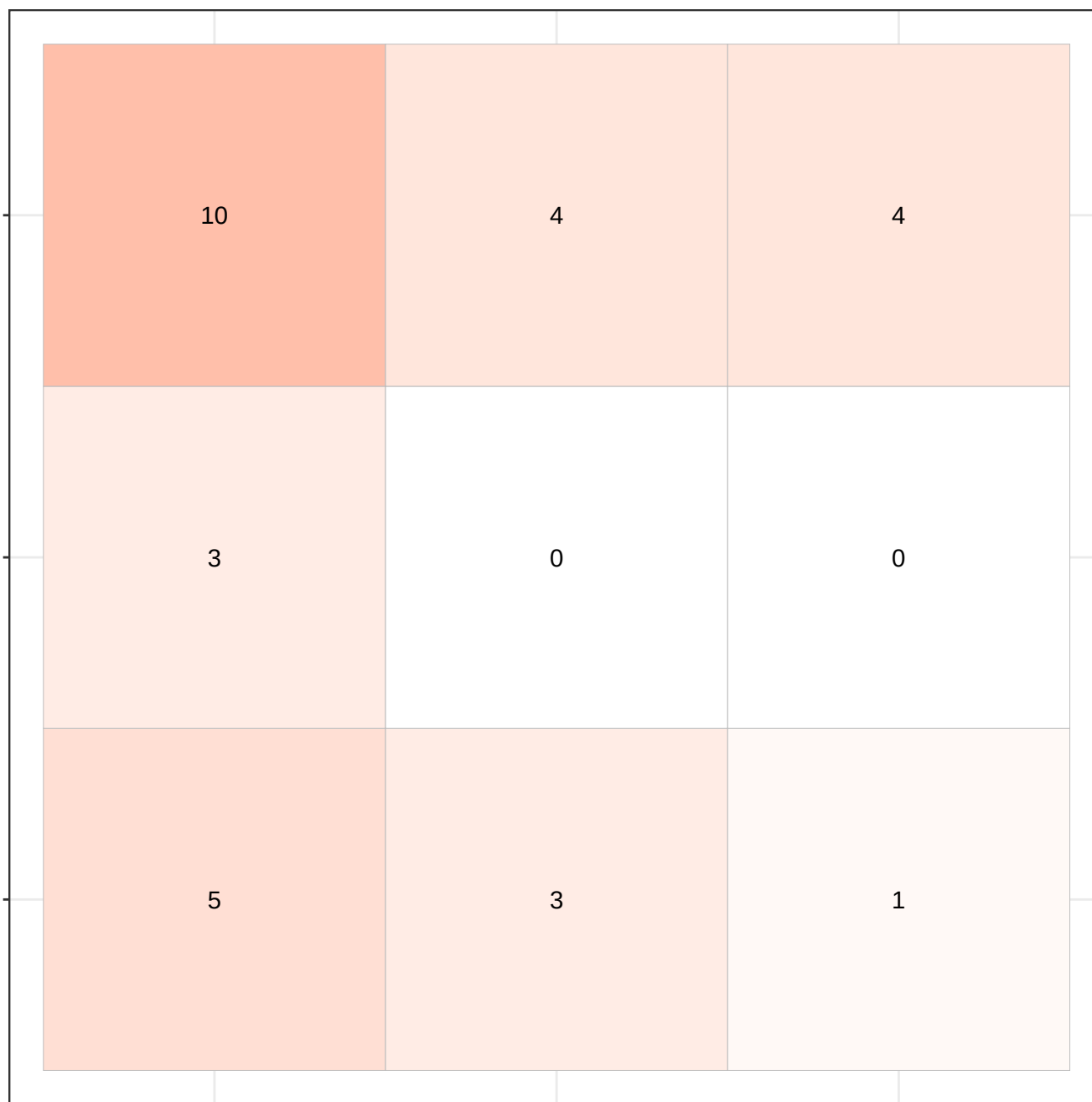
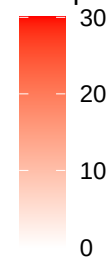
EXPERIMENT

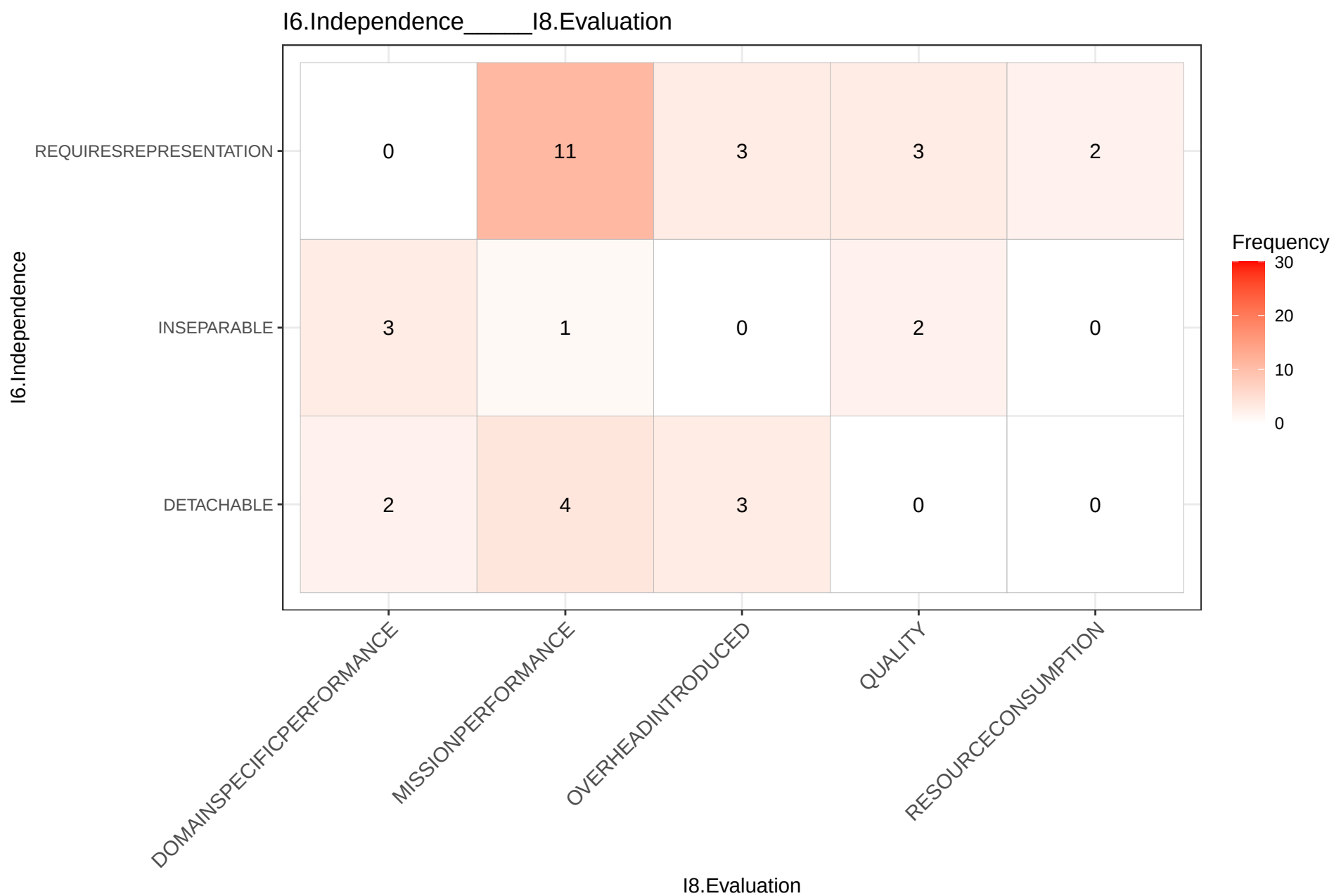
NOEVALUATION

SHOWCASE

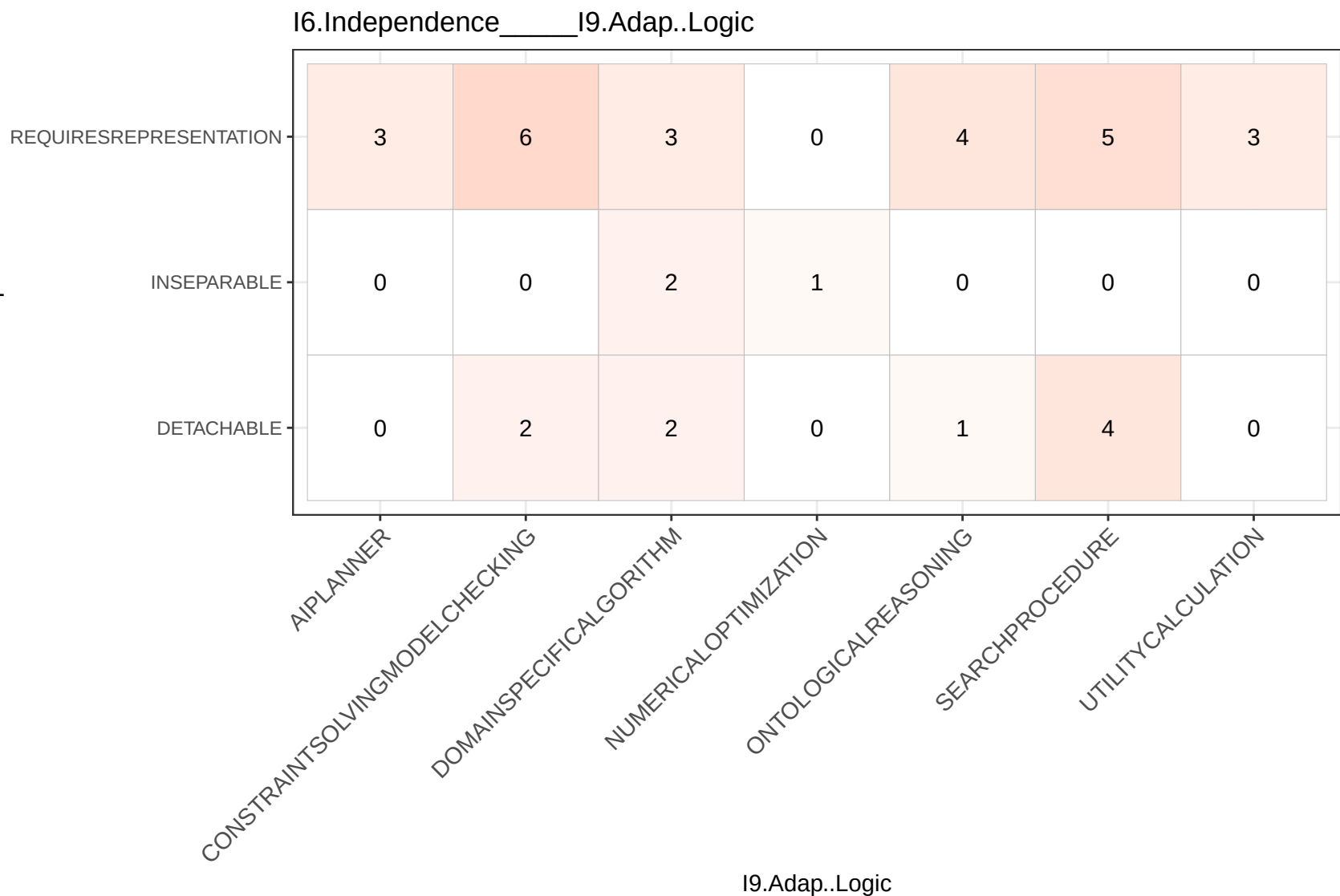
Experiment.Method

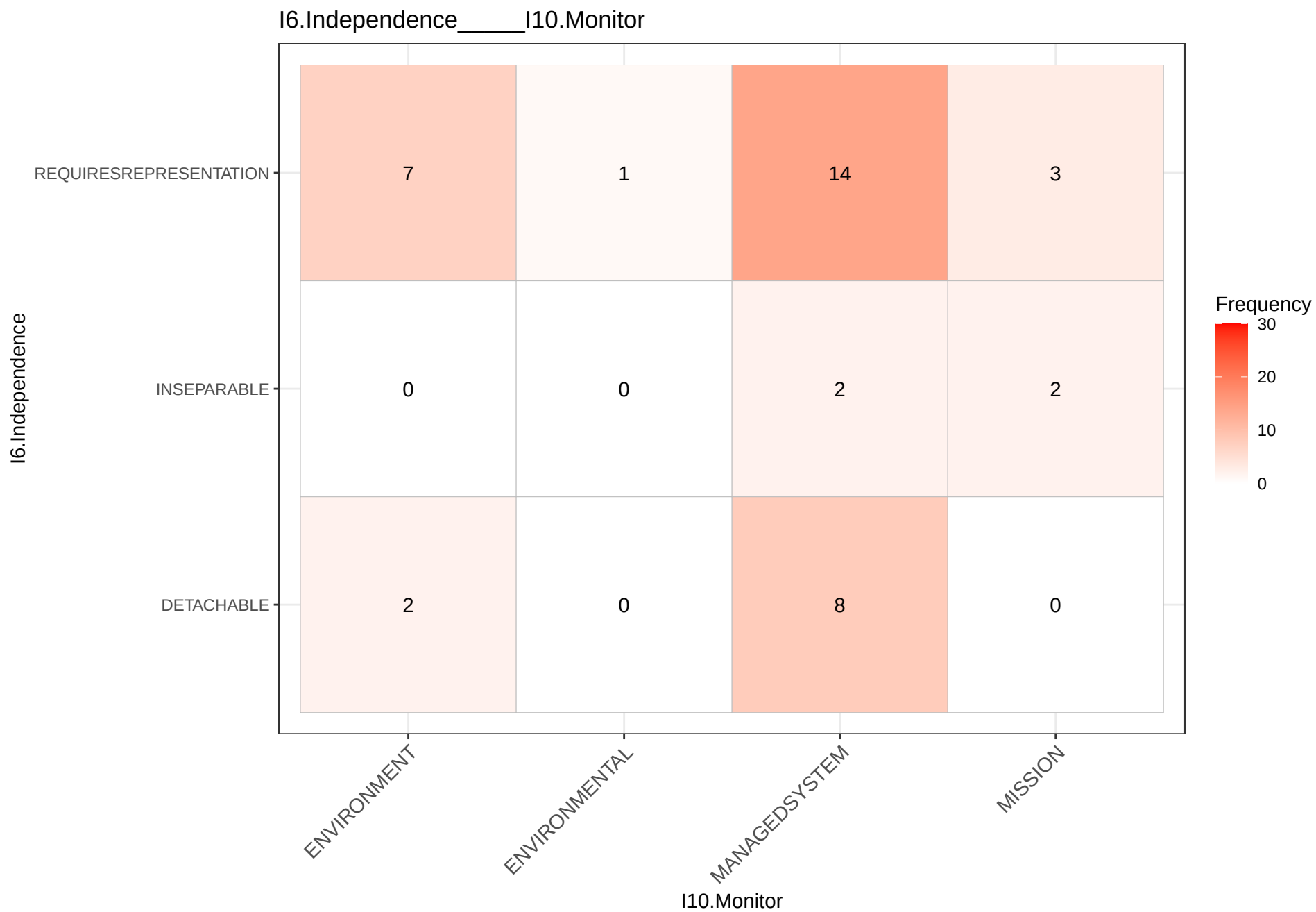
Frequency



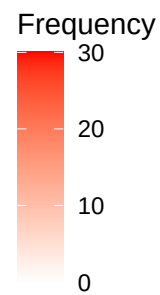
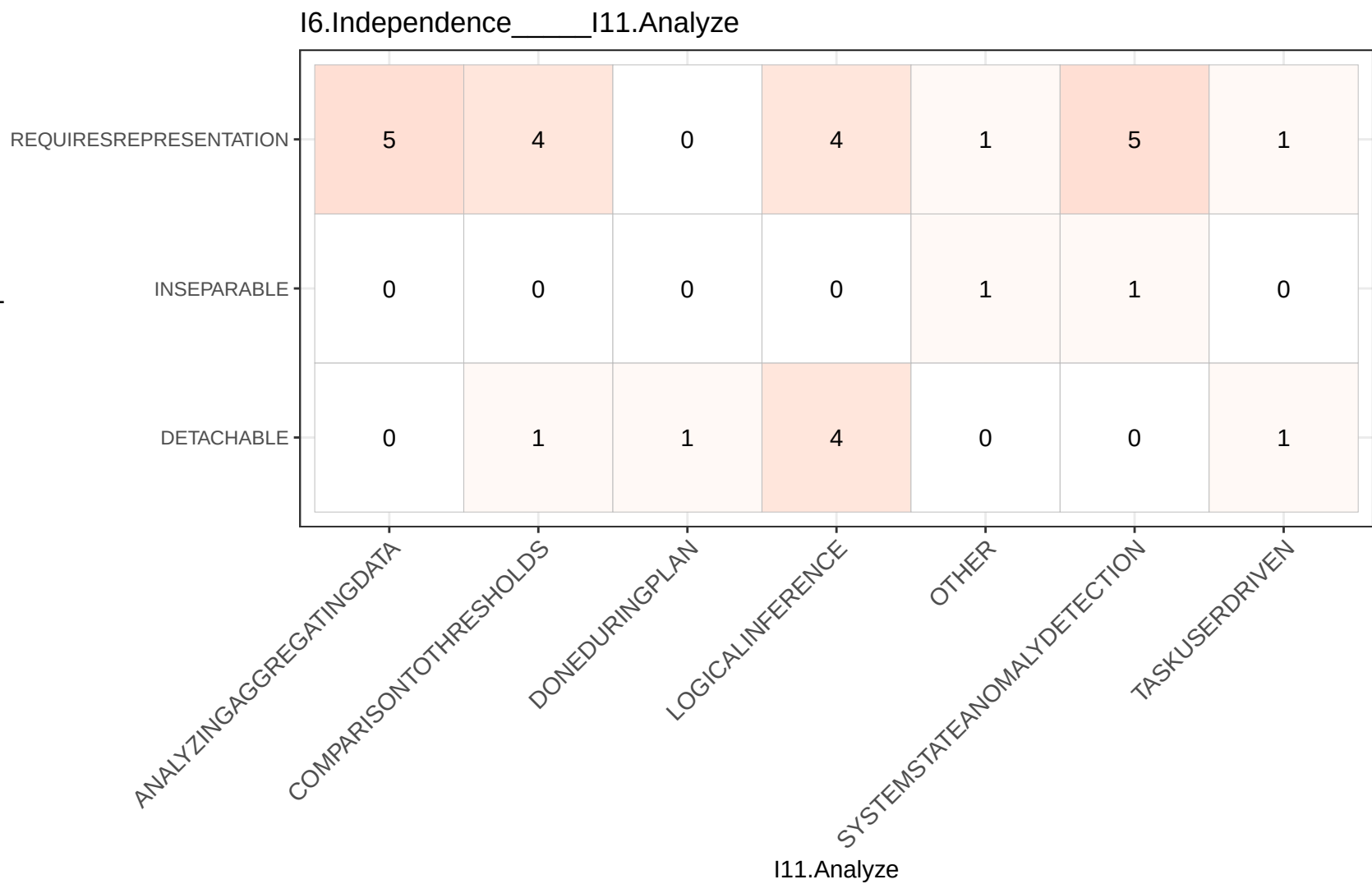


I6.Independence





I6.Independence



I6.Independence_____I12.Plan

I6.Independence

REQUIRESREPRESENTATION

INSEPARABLE

DETACHABLE

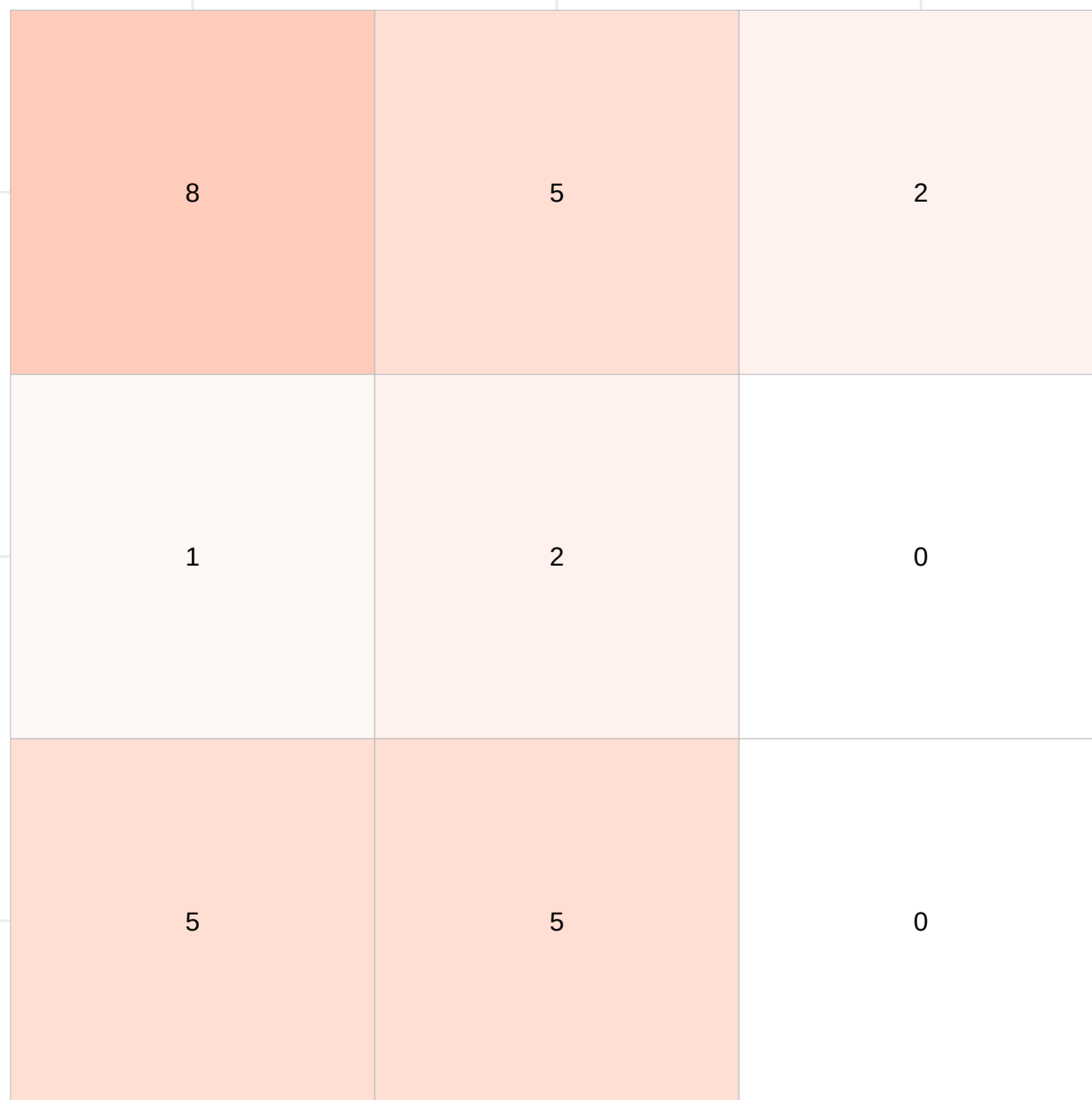
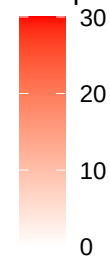
DETERMININGTHEOPTIMALCHOICE

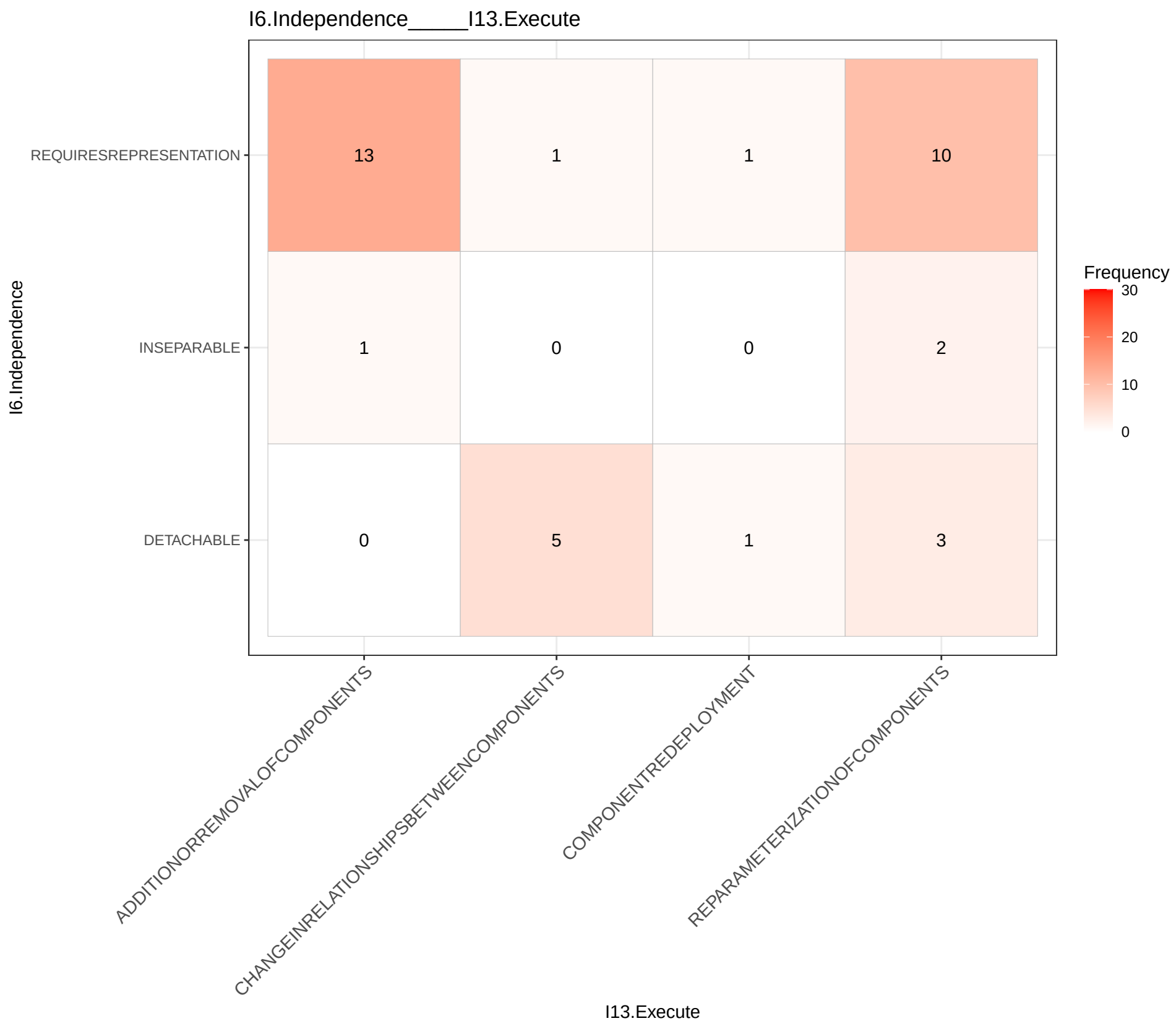
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

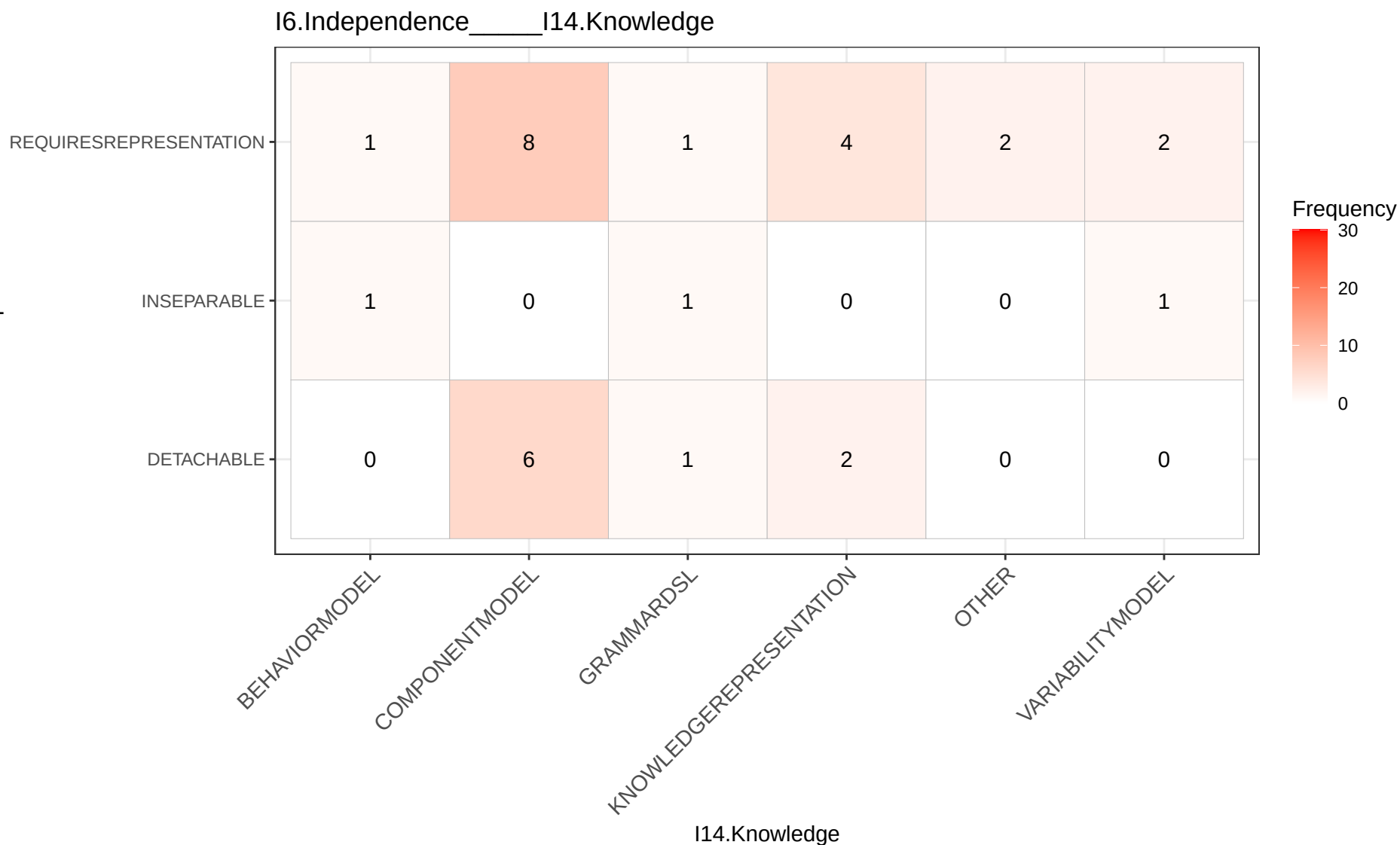
I12.Plan

Frequency





I6.Independence



I7.Deployment.Realness_____I7.Mission.Realness

I7.Deployment.Realness

SIMULATED

6

4

REAL

8

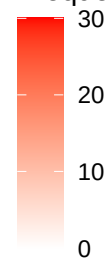
8

REAL

SYNTHETIC

I7.Mission.Realness

Frequency



I7.Deployment.Realness_____Experiment.Method

I7.Deployment.Realness

SIMULATED

6

2

2

REAL

11

2

3

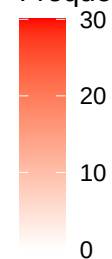
EXPERIMENT

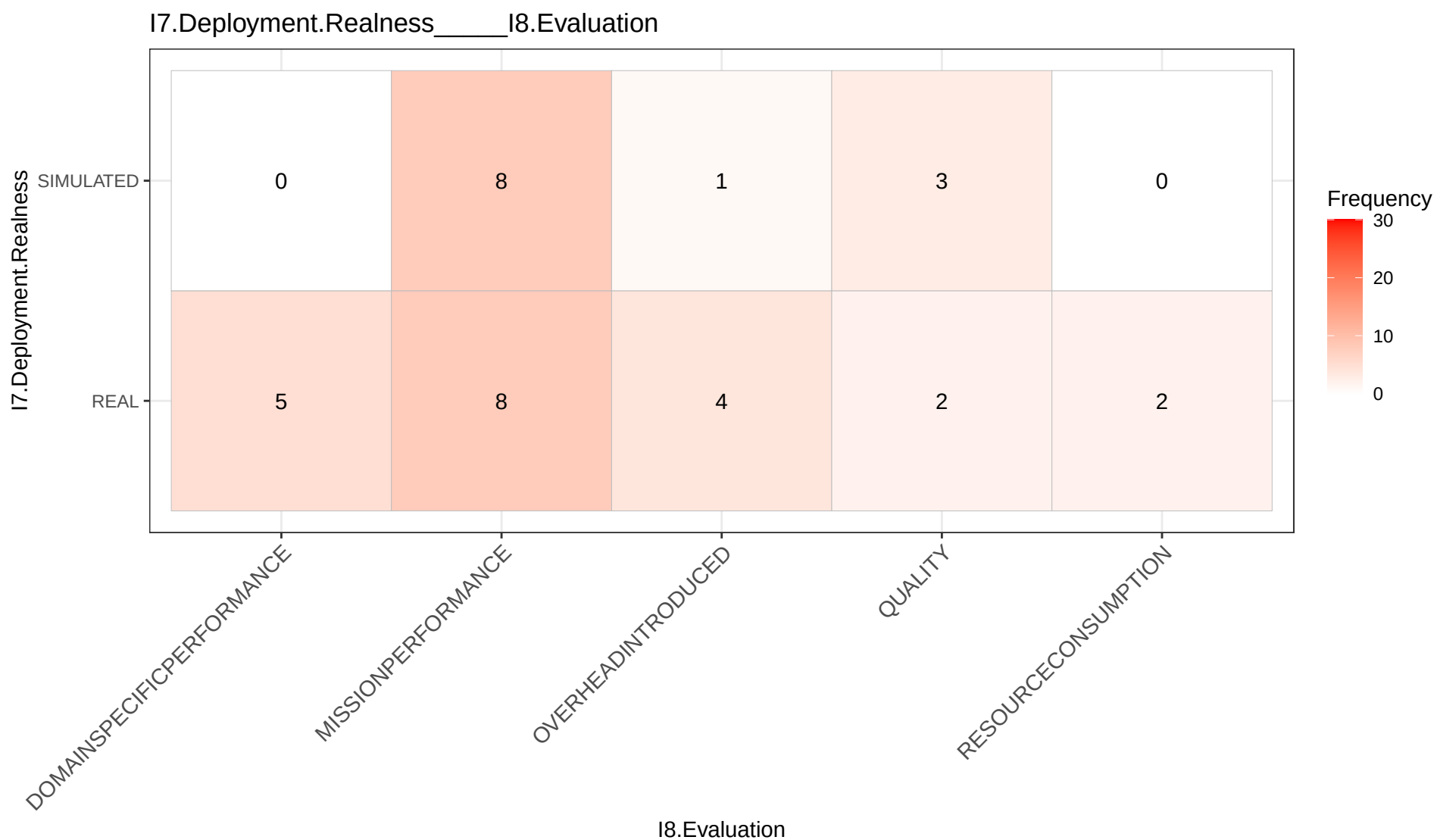
NOEVALUATION

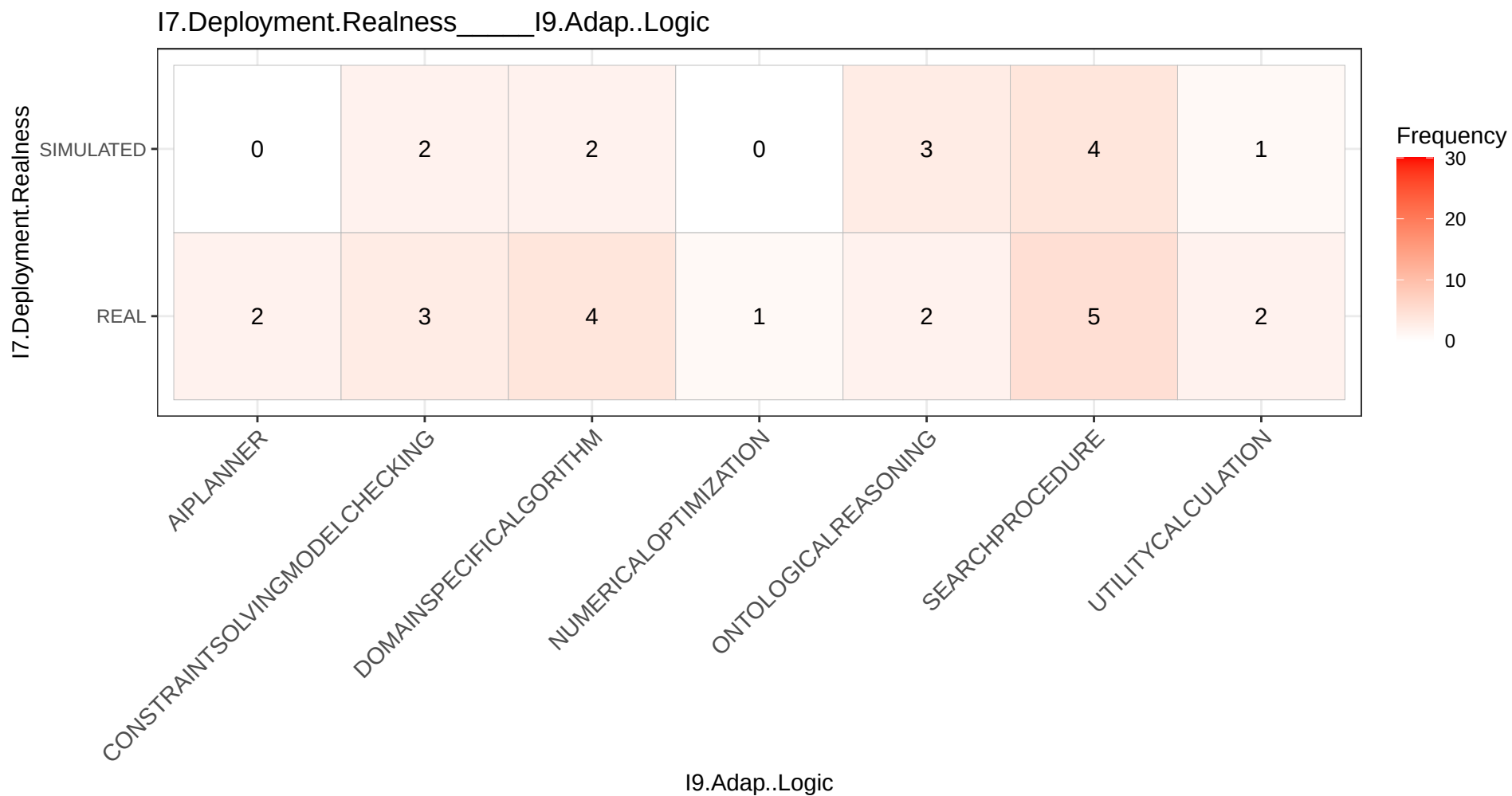
SHOWCASE

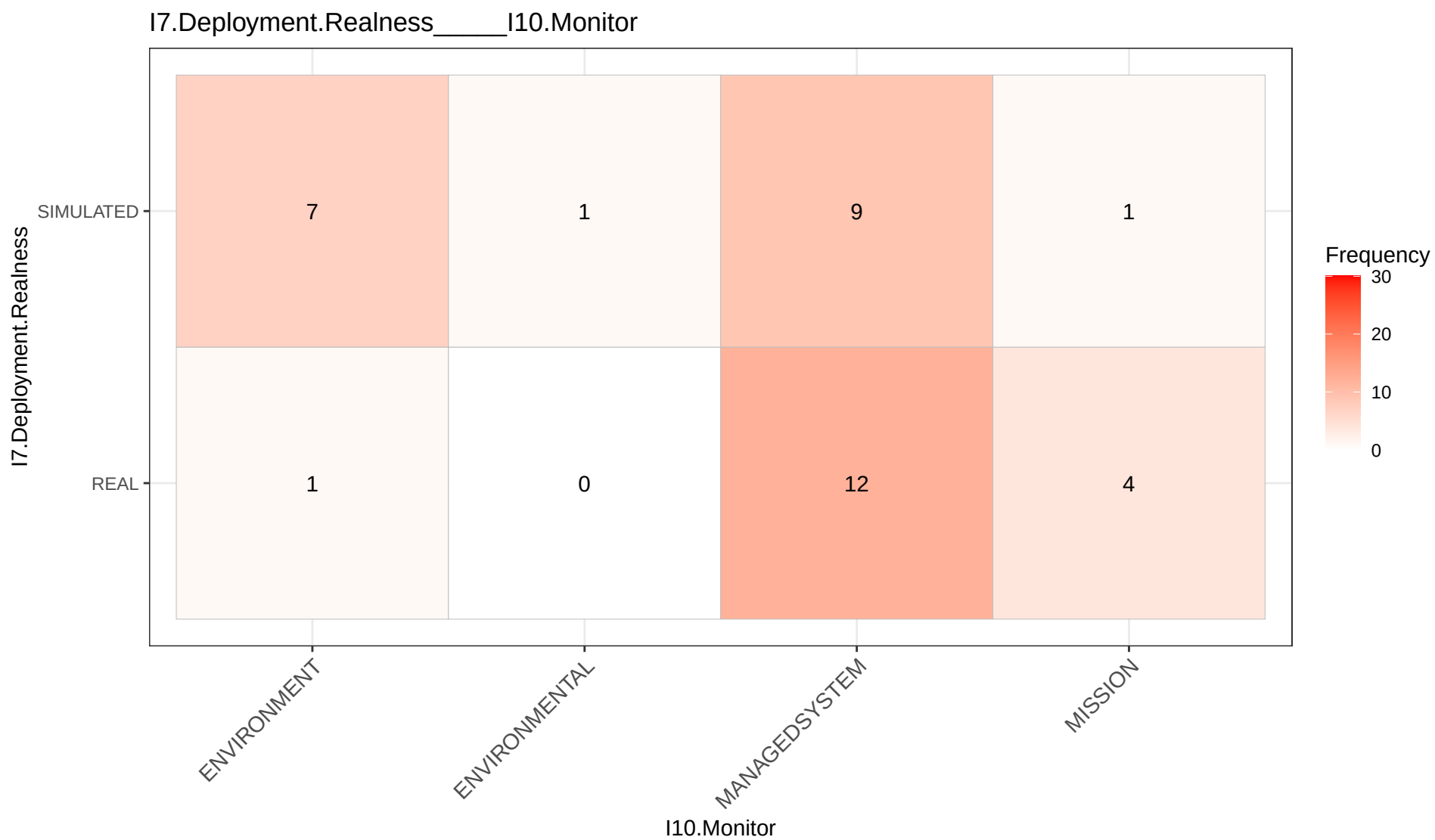
Experiment.Method

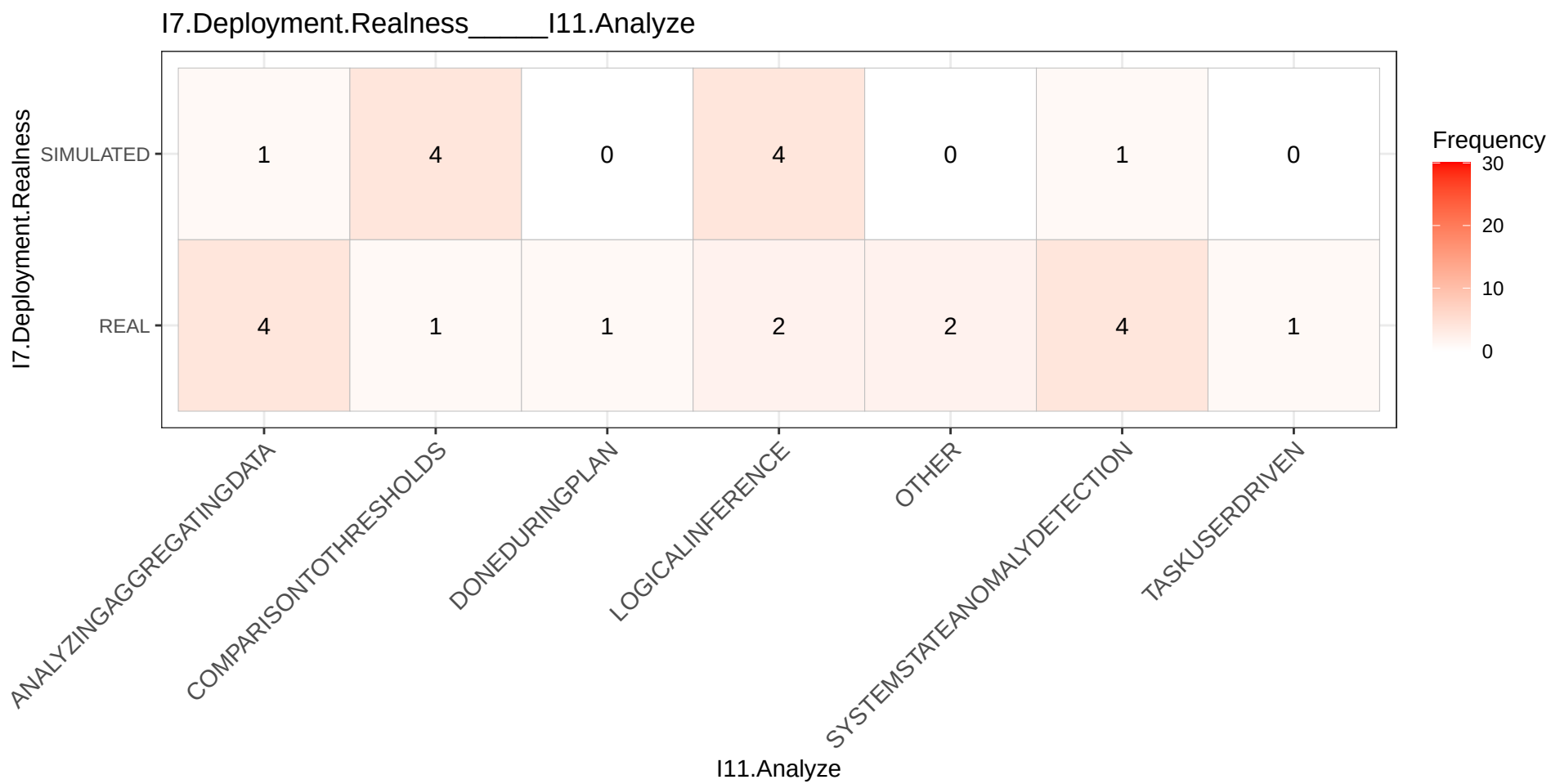
Frequency











I7.Deployment.Realness_____I12.Plan

I7.Deployment.Realness

SIMULATED

5

4

0

REAL

6

5

2

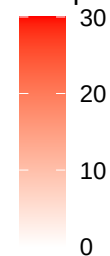
DETERMININGTHEOPTIMALCHOICE

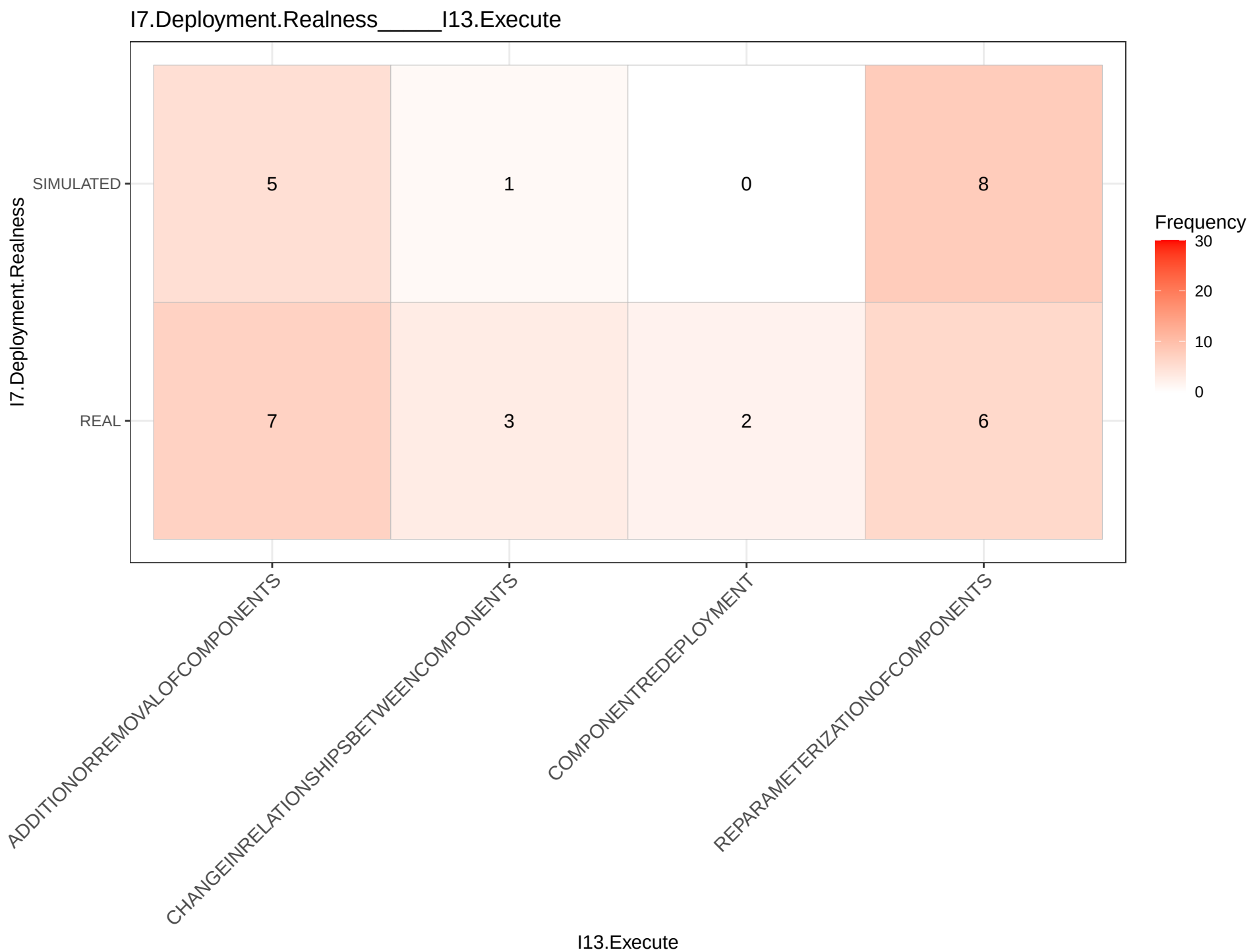
RELYINGONDESIGNTIMERULESMODELS

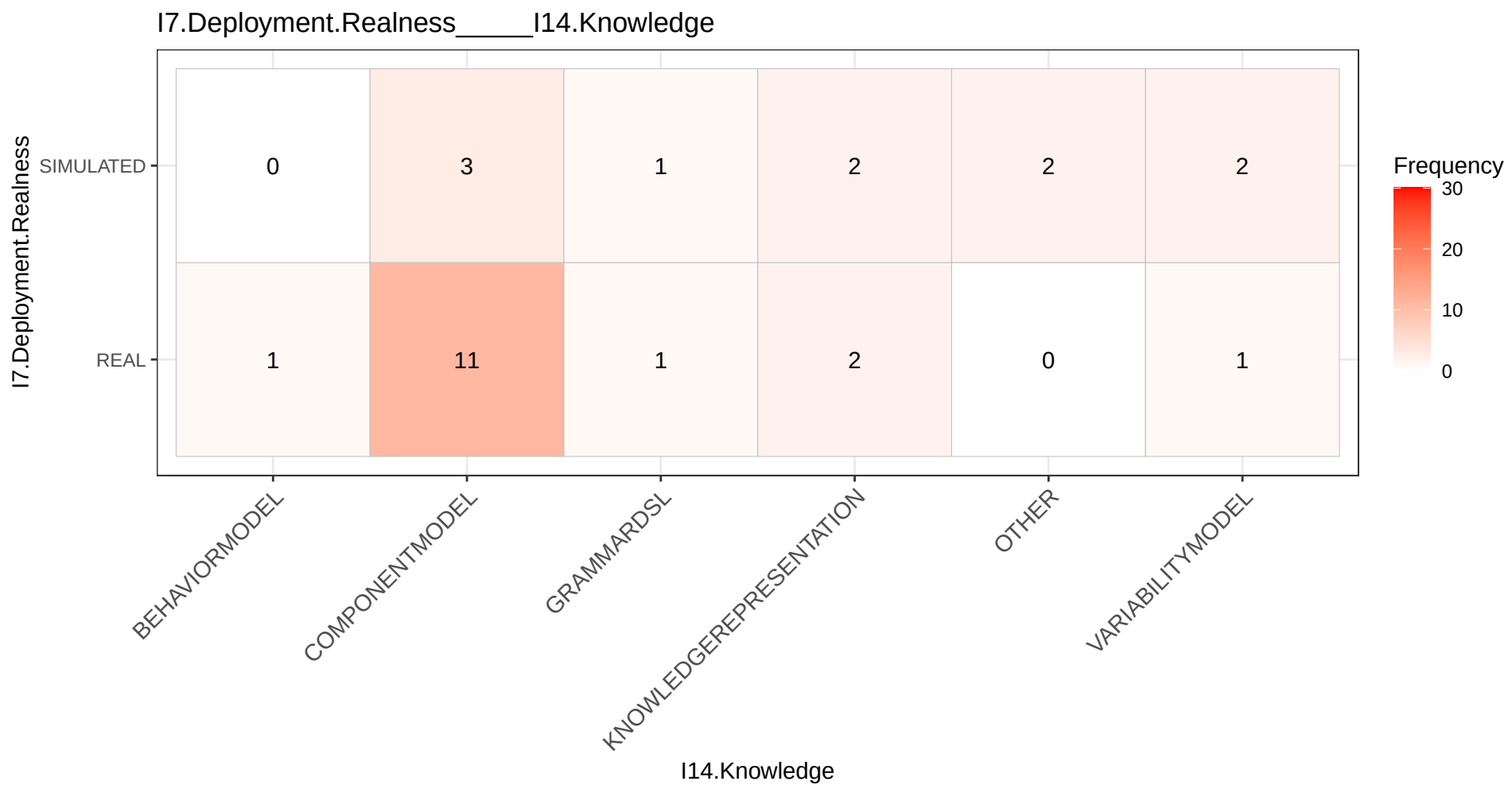
USINGAIPANNINGLANGUAGES

I12.Plan

Frequency







I7.Mission.Realness_____Experiment.Method

I7.Mission.Realness

SYNTHETIC

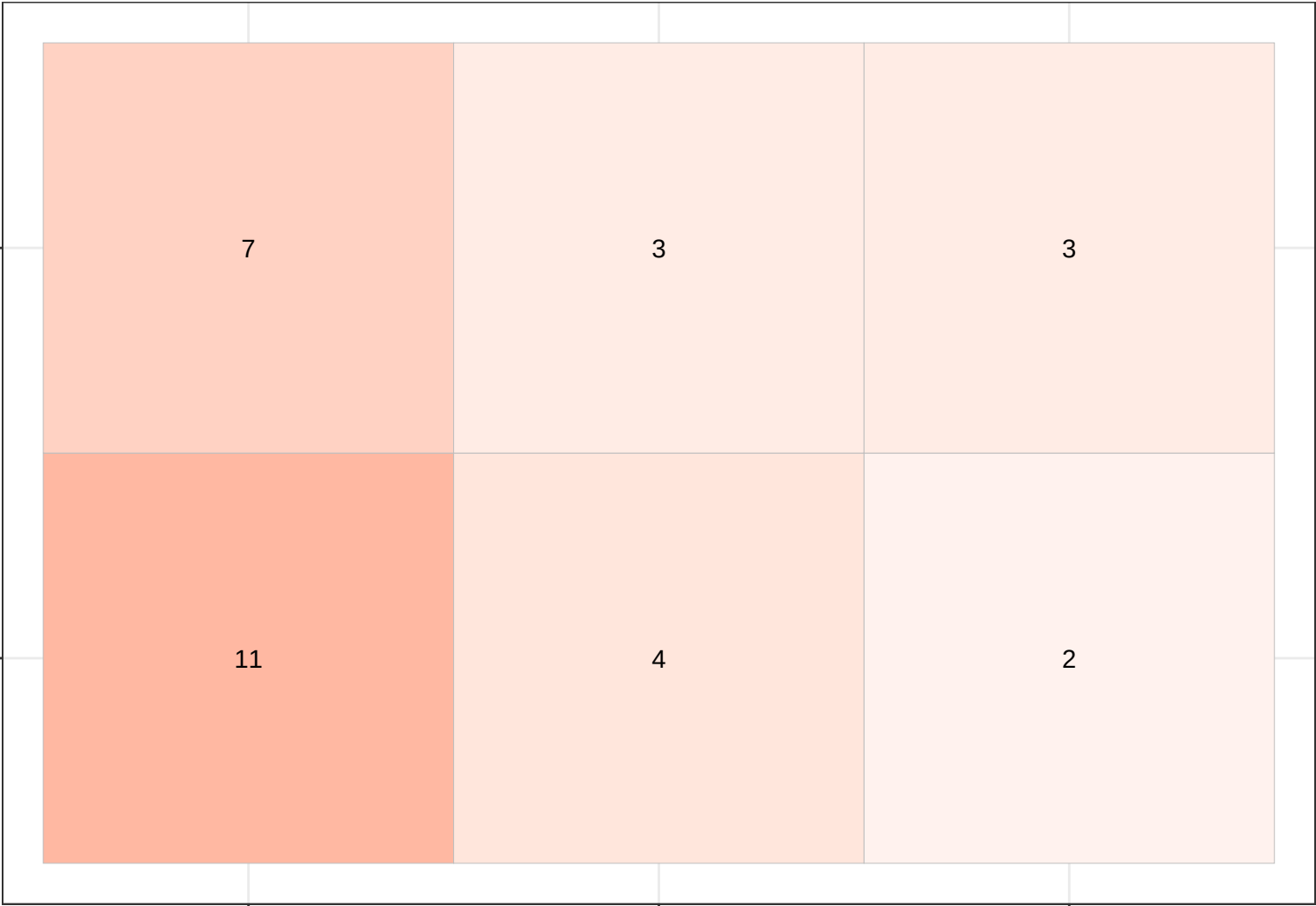
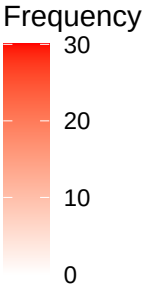
REAL

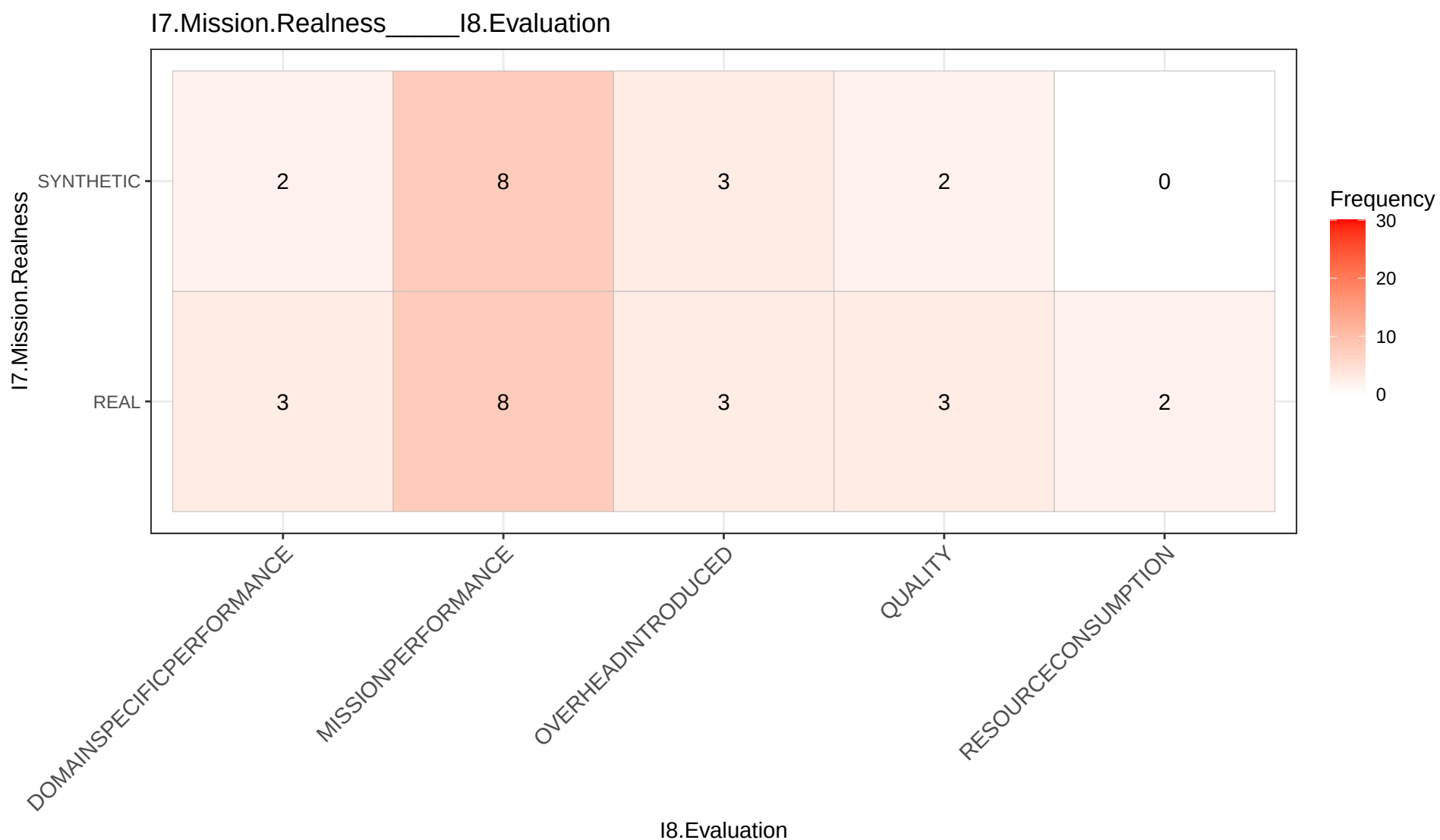
EXPERIMENT

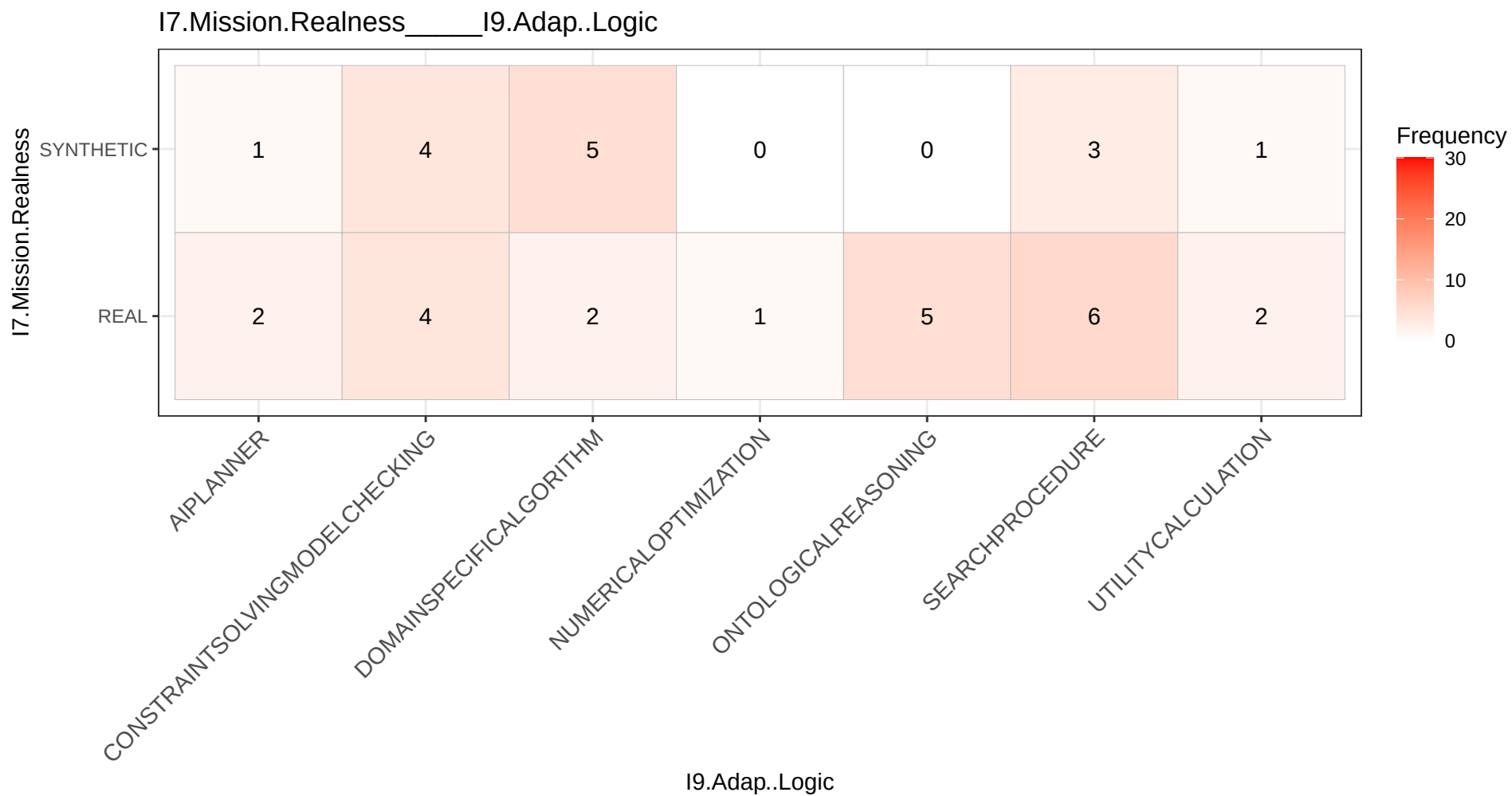
NOEVALUATION

SHOWCASE

Experiment.Method







I7.Mission.Realness

I7.Mission.Realness_____I10.Monitor

SYNTHETIC

3

1

12

2

REAL

6

0

12

3

ENVIRONMENT

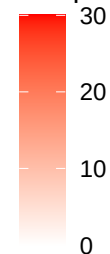
ENVIRONMENTAL

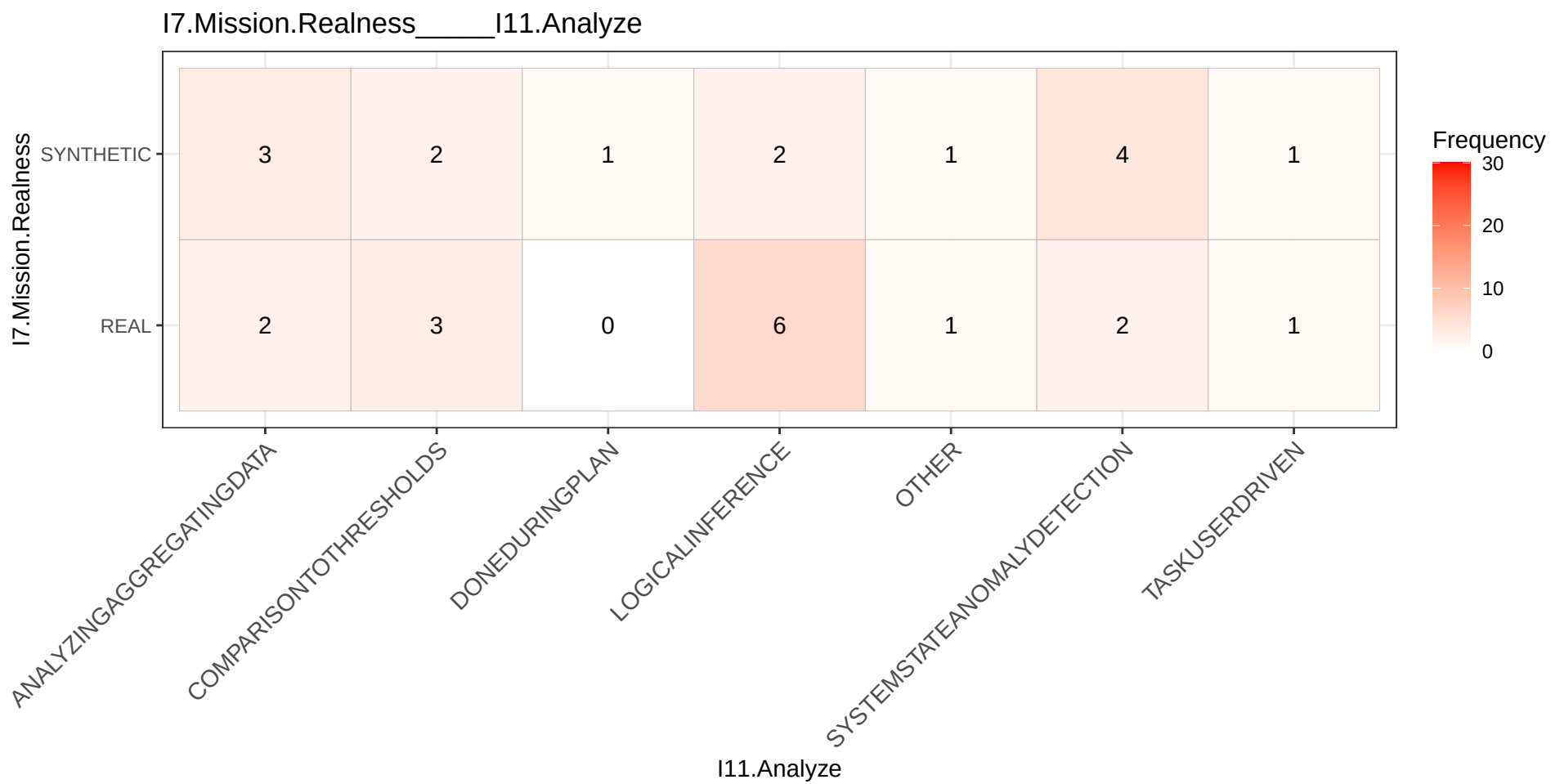
I10.Monitor

MANAGEDSYSTEM

MISSION

Frequency





I7.Mission.Realness _____ I12.Plan

I7.Mission.Realness

SYNTHETIC

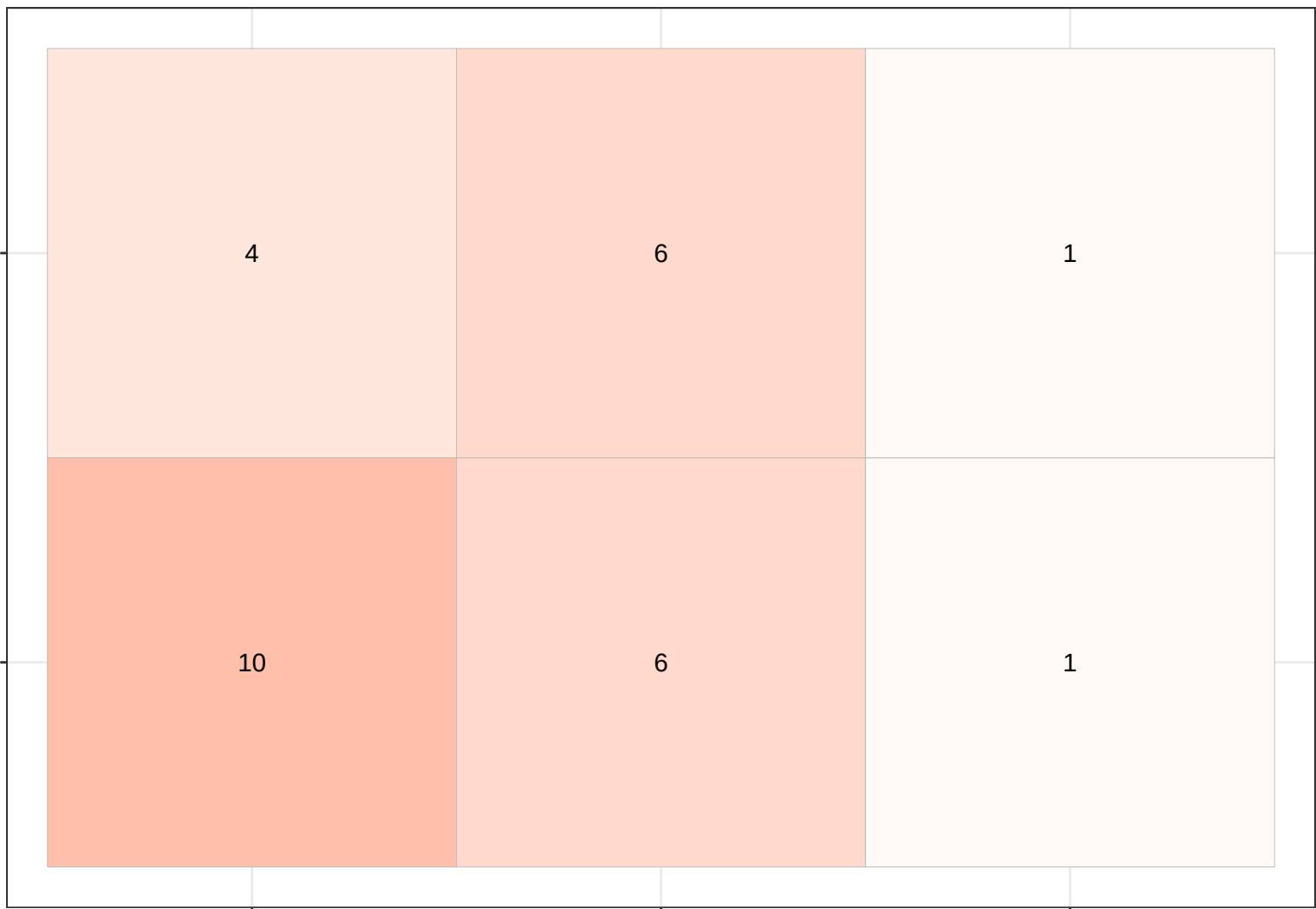
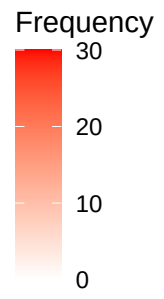
REAL

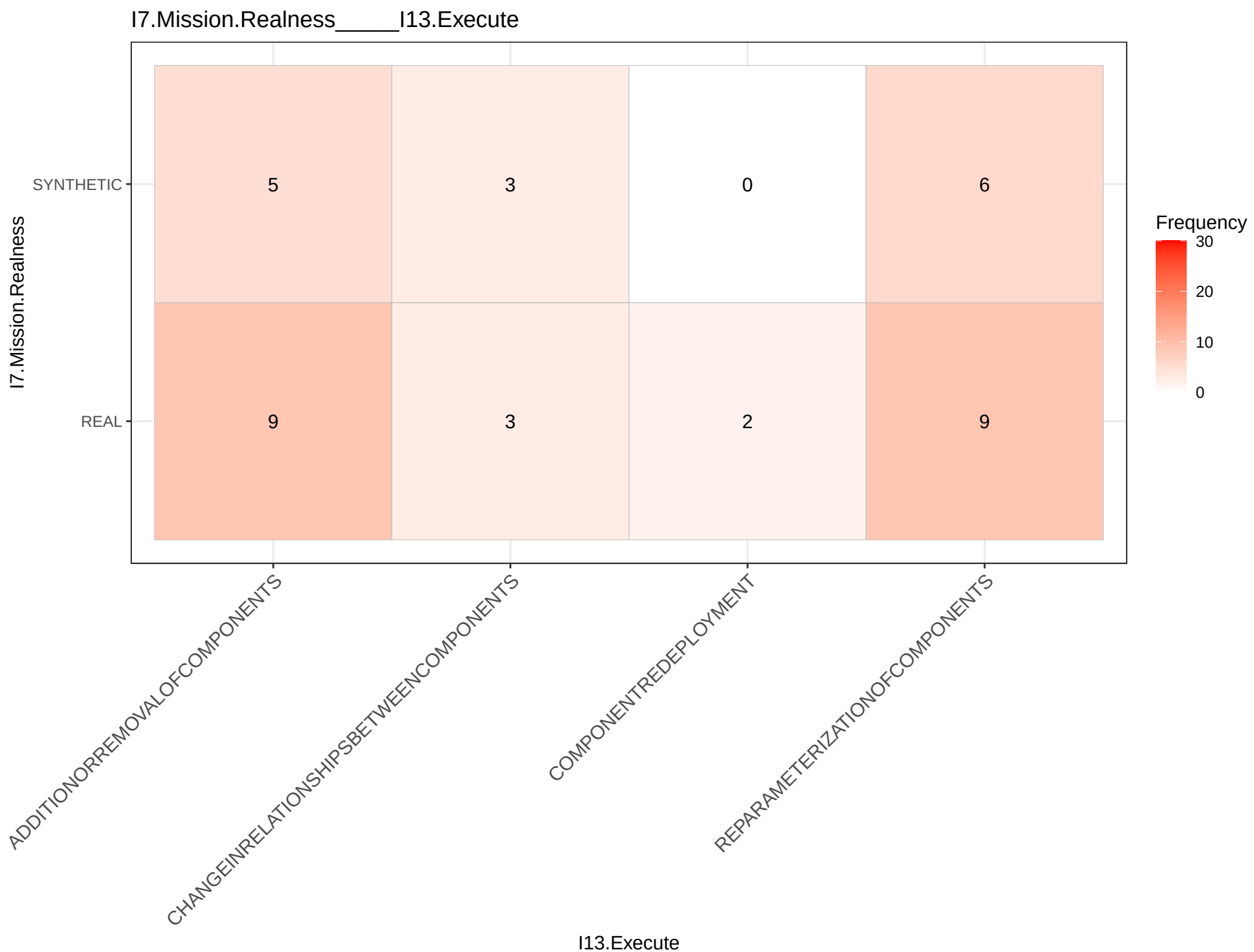
DETERMININGTHEOPTIMALCHOICE

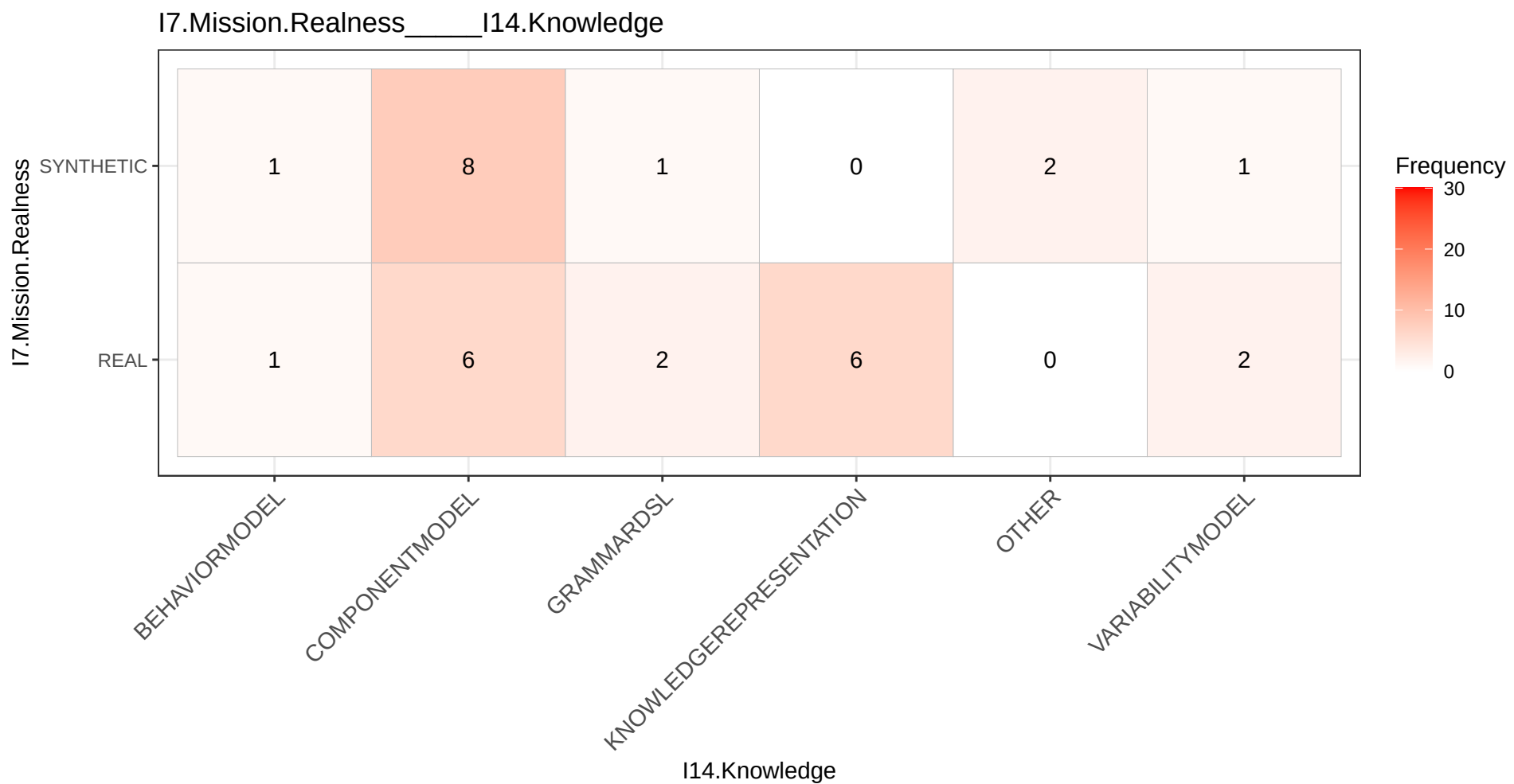
RELYINGONDESIGNTIMERULESMODELS

USINGAIPANNINGLANGUAGES

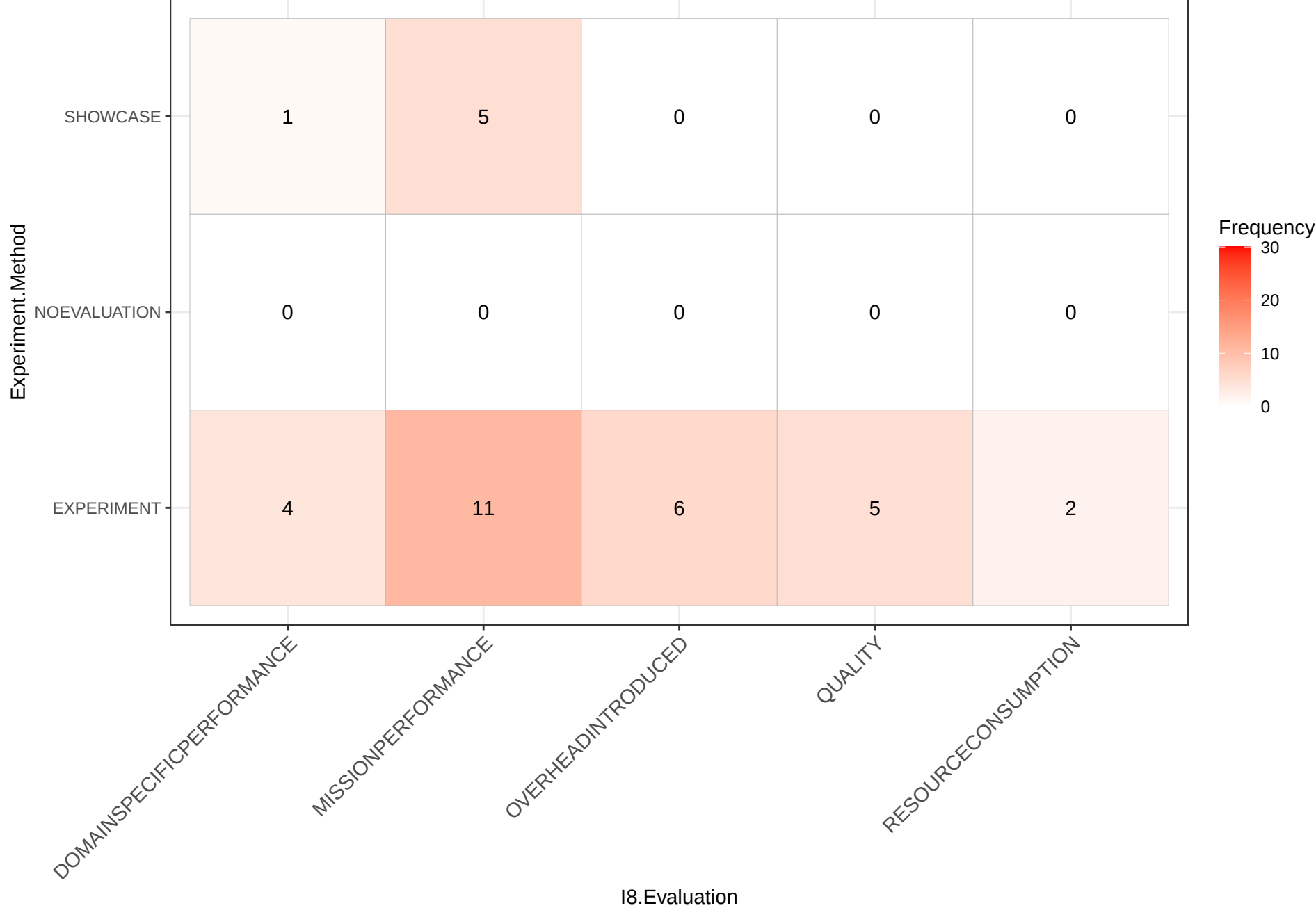
I12.Plan



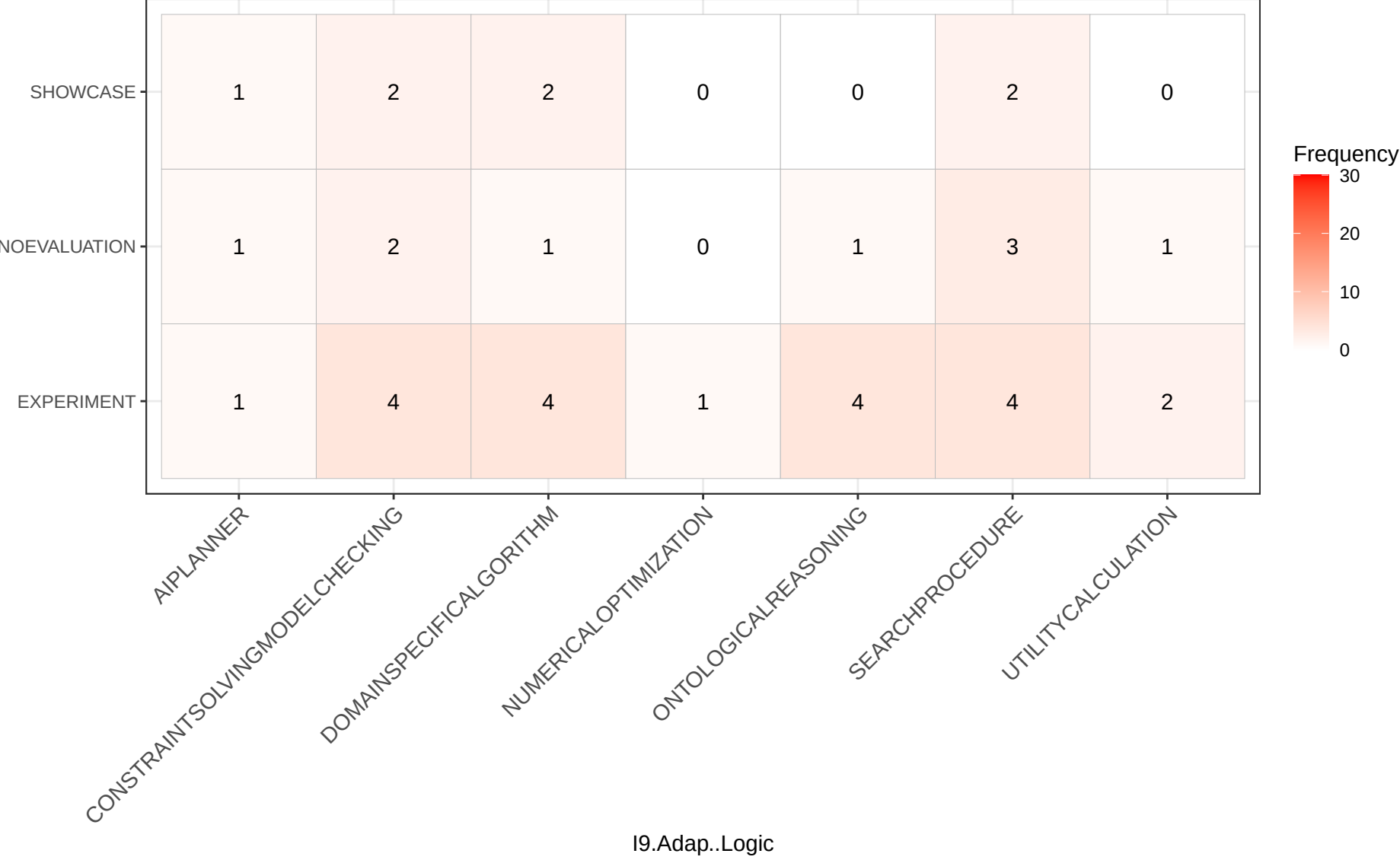




Experiment.Method_____I8.Evaluation



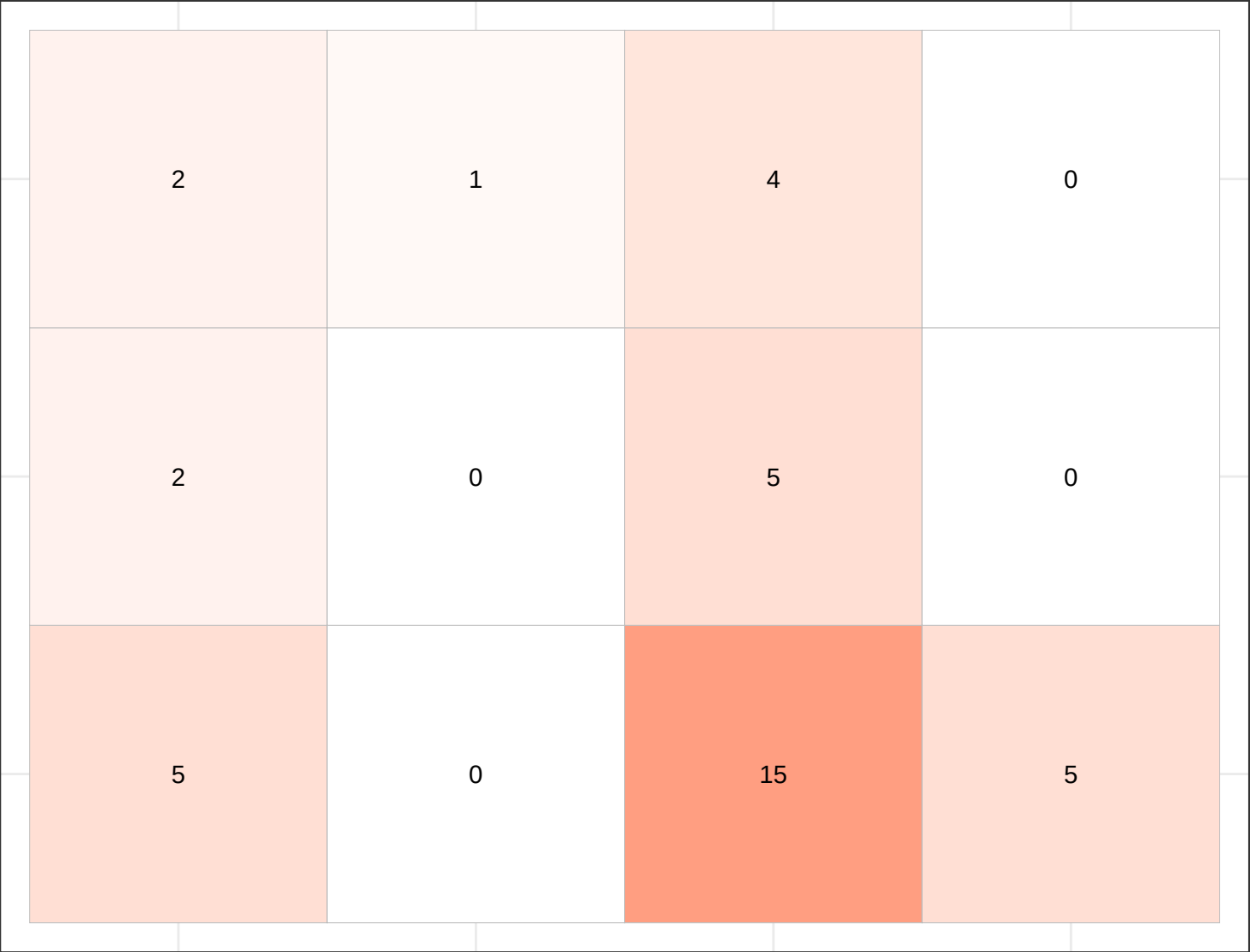
Experiment.Method_____I9.Adap..Logic



Experiment.Method_____I10.Monitor

Experiment.Method

SHOWCASE
NOEVALUATION
EXPERIMENT



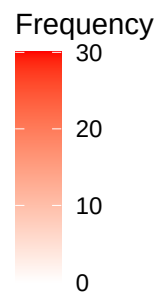
ENVIRONMENT

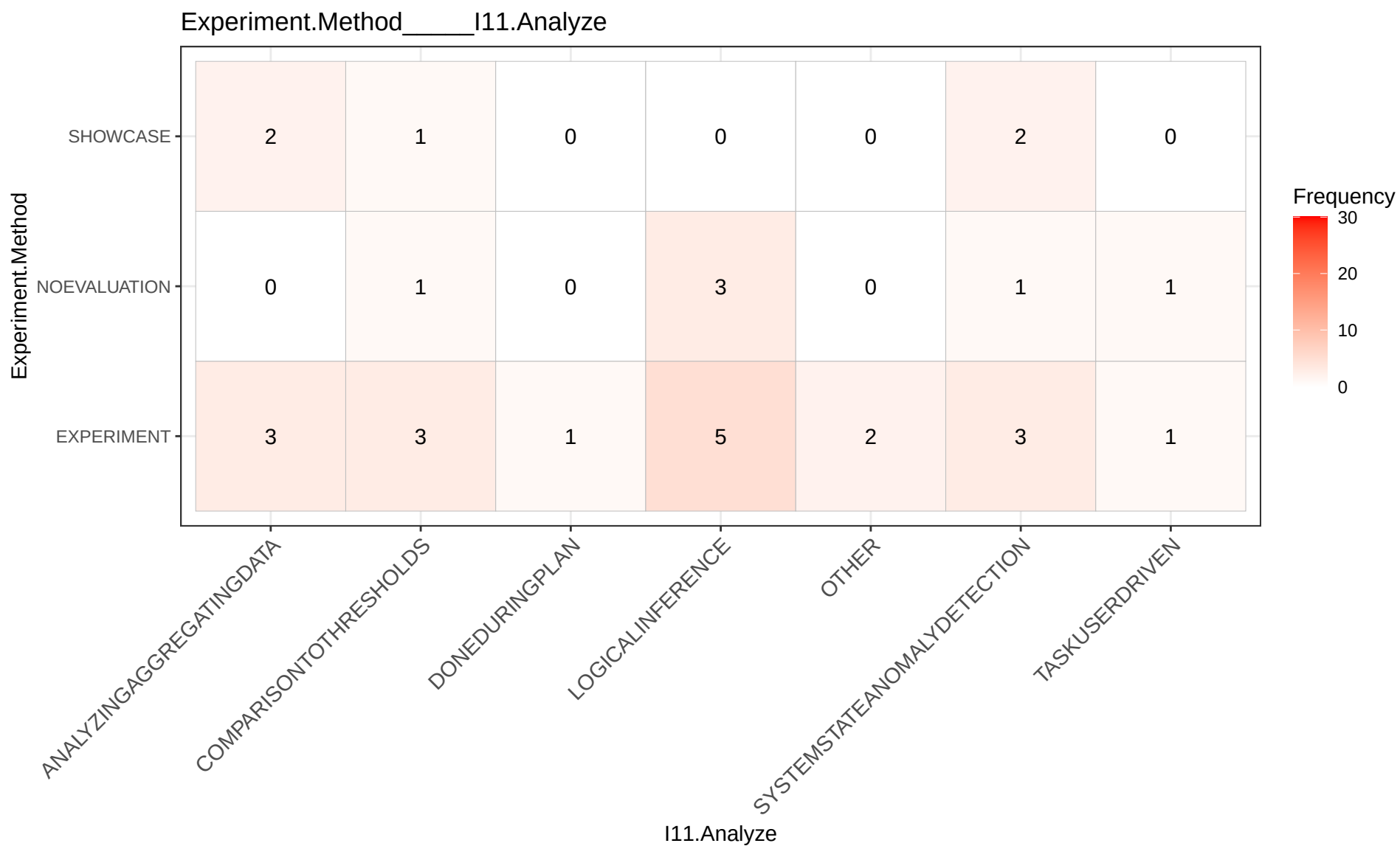
ENVIRONMENTAL

MANAGEDSYSTEM

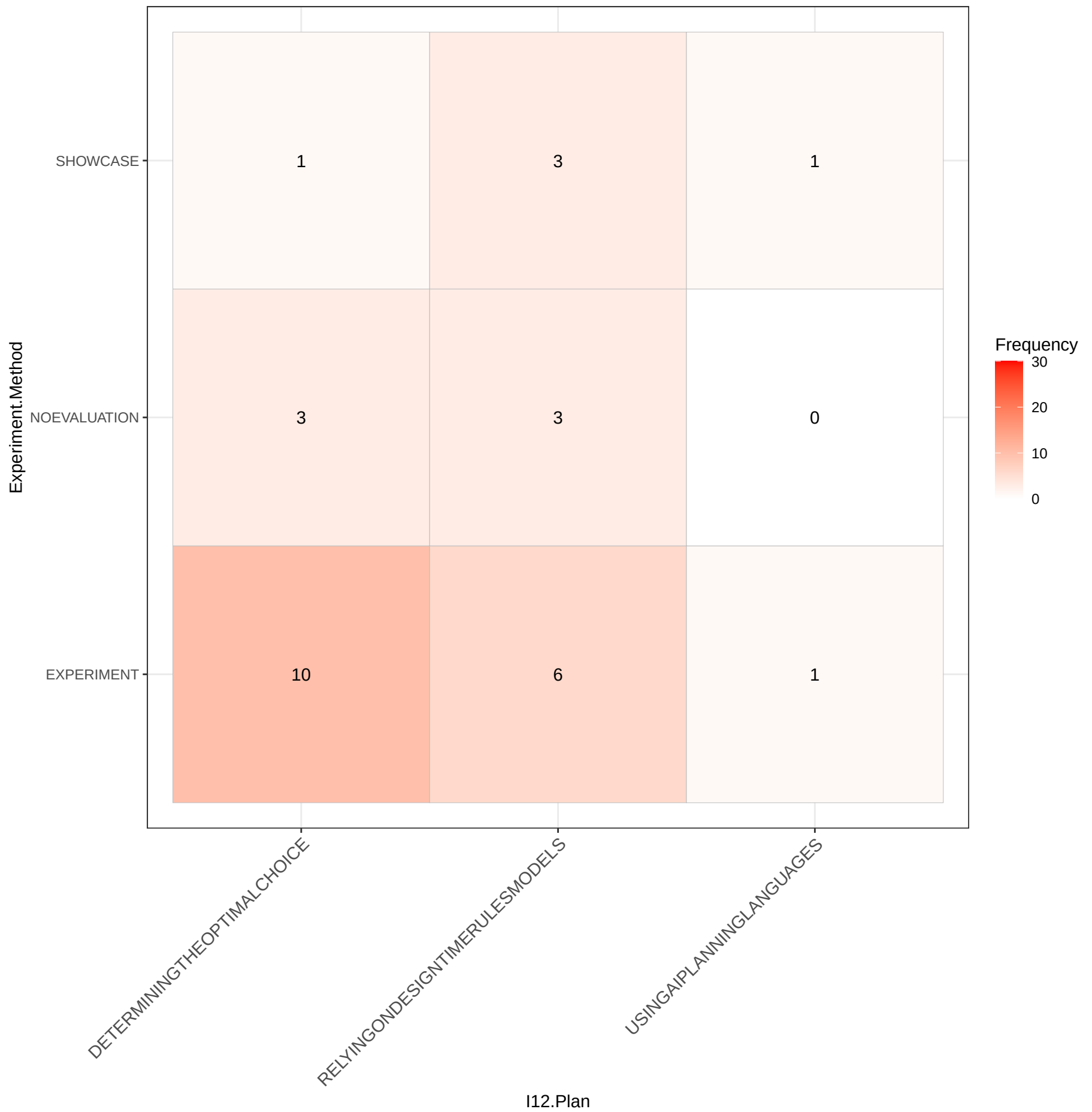
MISSION

I10.Monitor

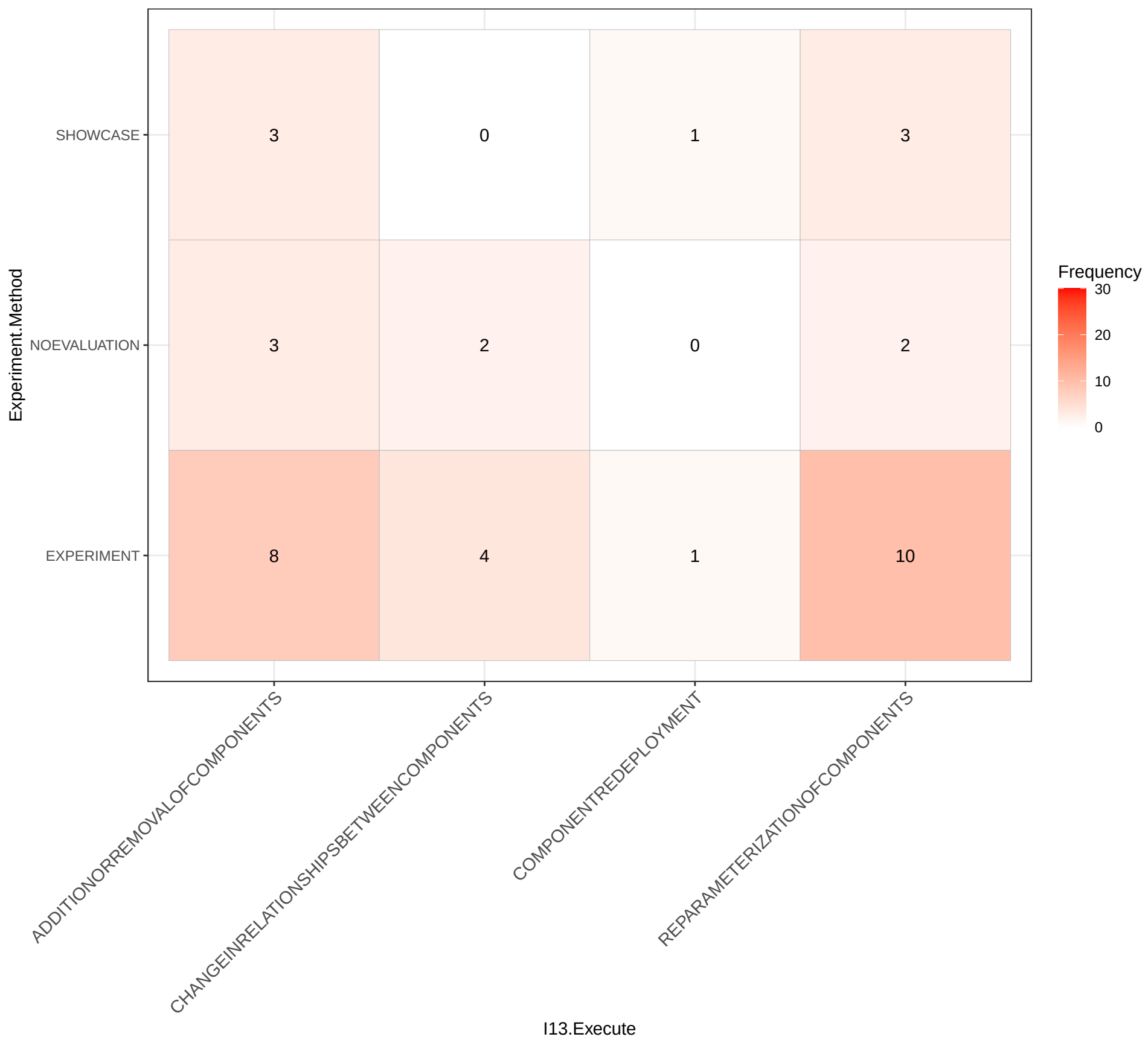




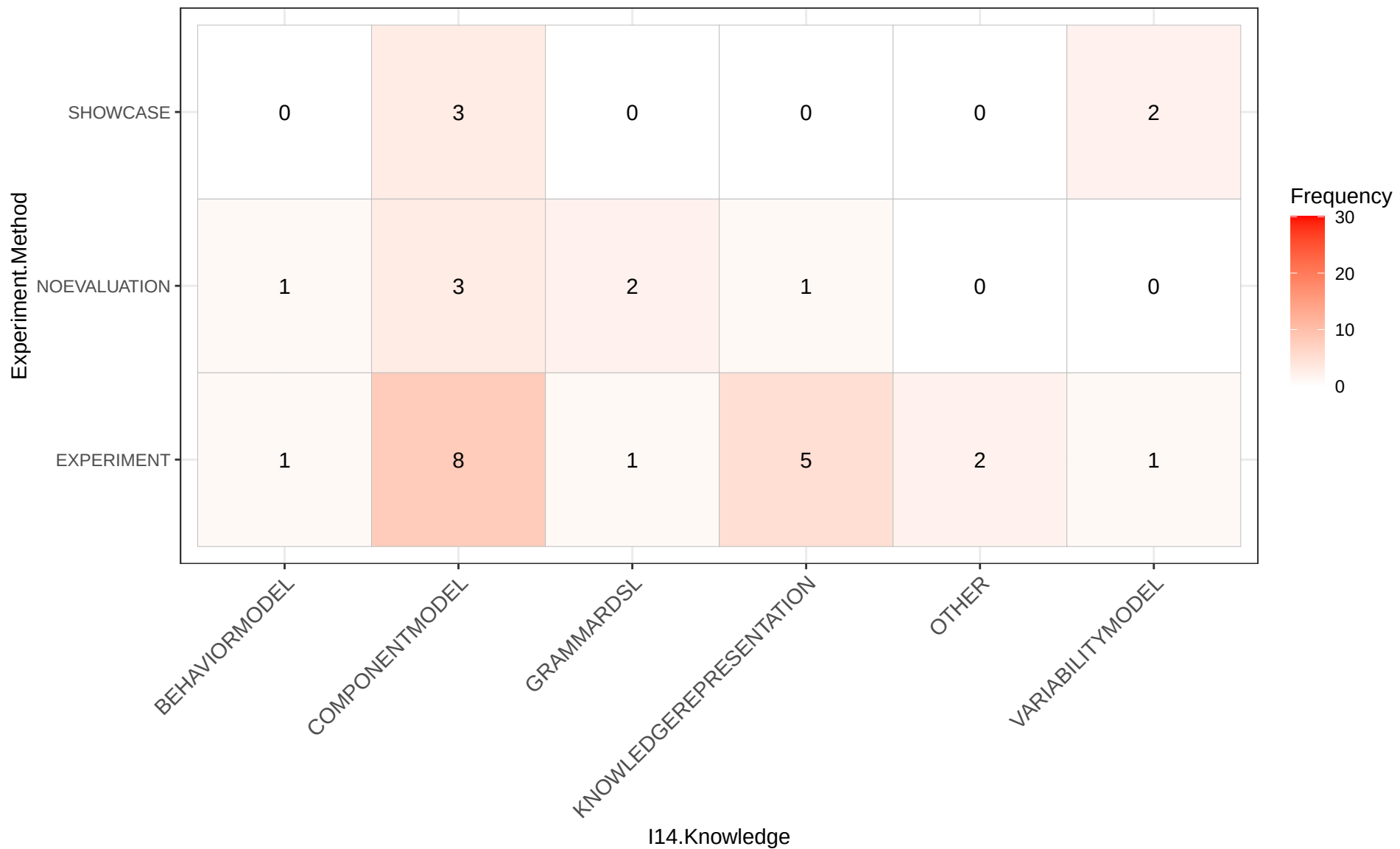
Experiment.Method_____I12.Plan

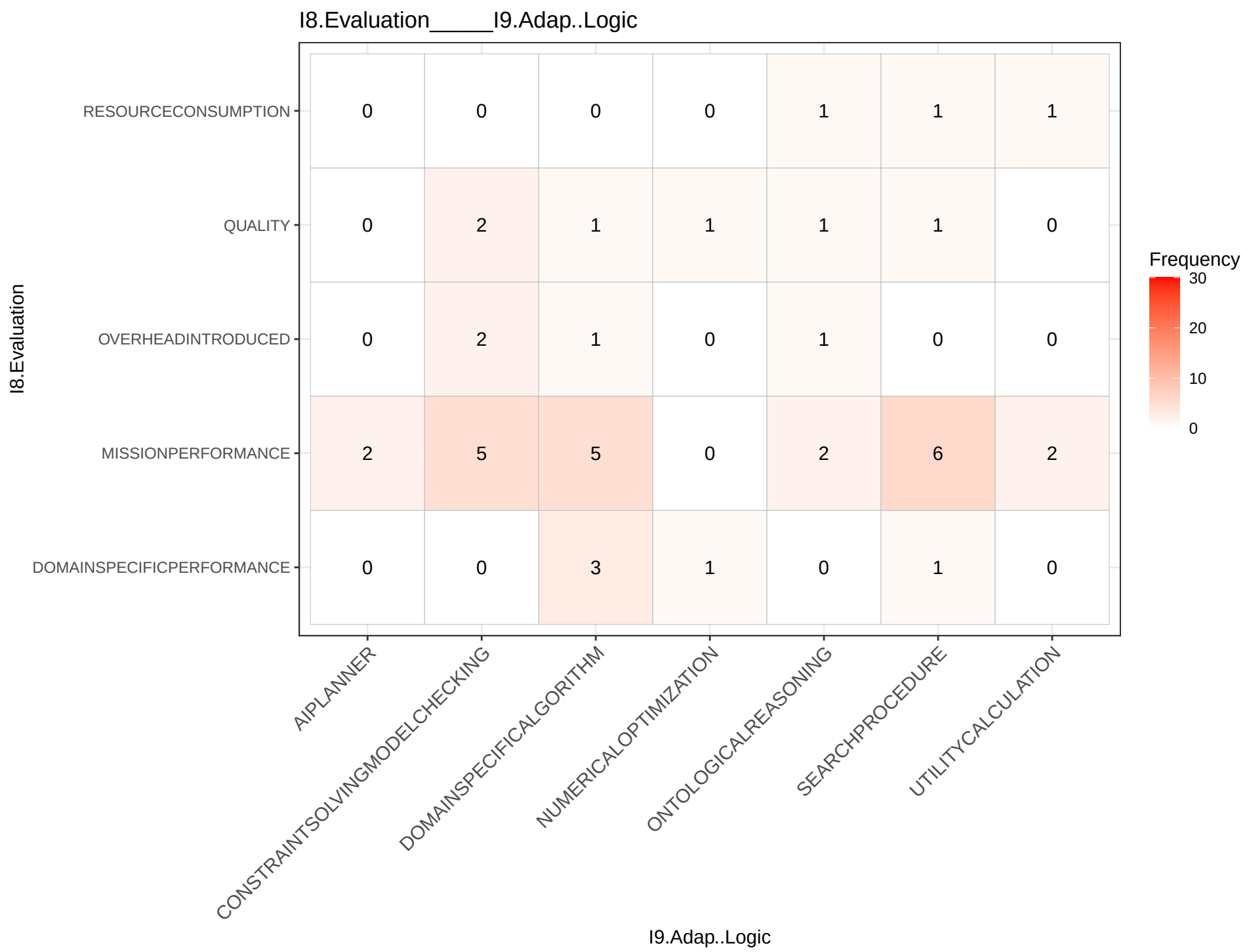


Experiment.Method____I13.Execute



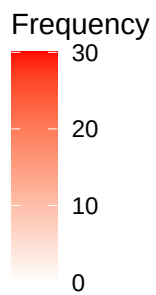
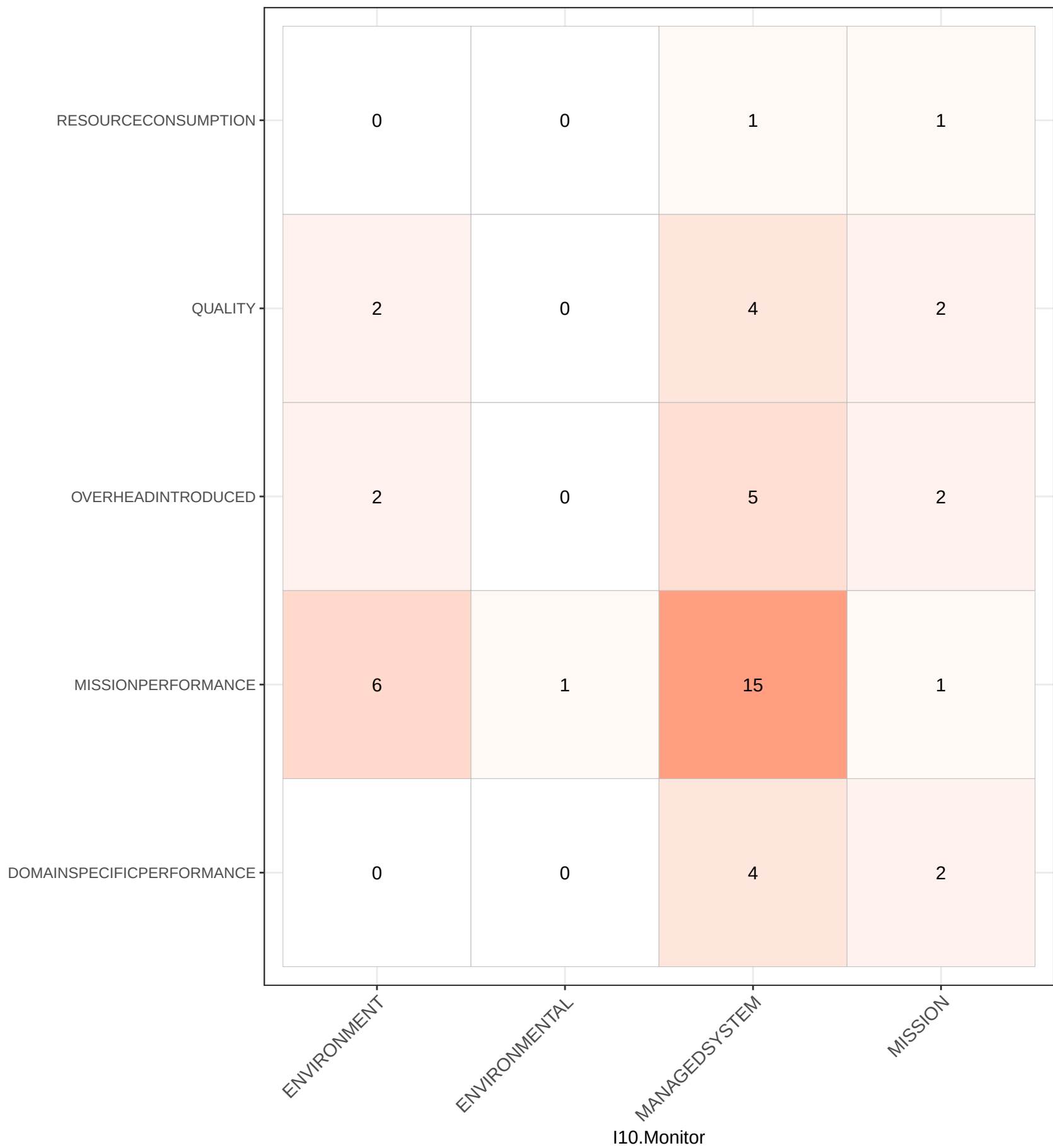
Experiment.Method_____I14.Knowledge





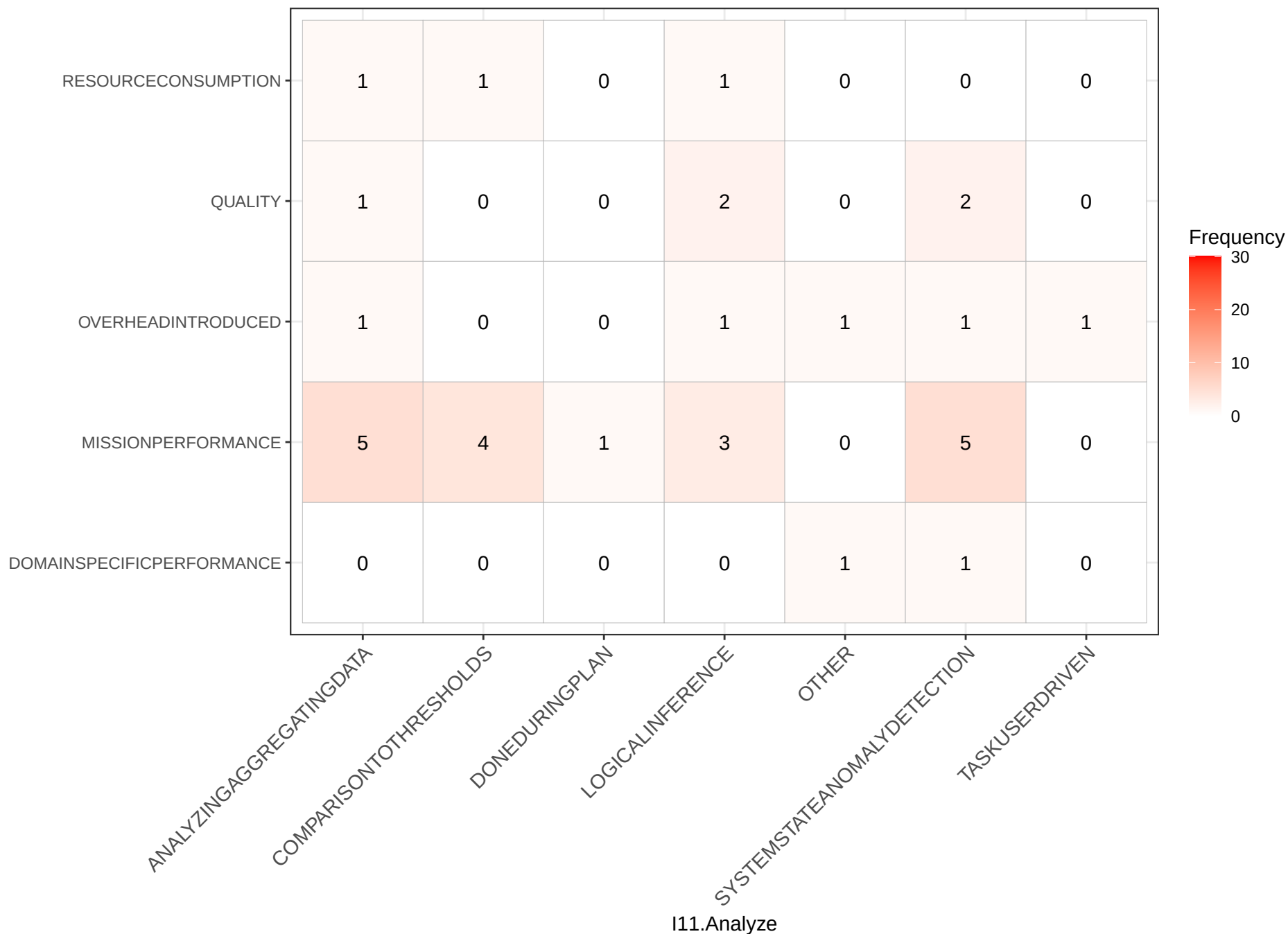
I8.Evaluation_____I10.Monitor

I8.Evaluation

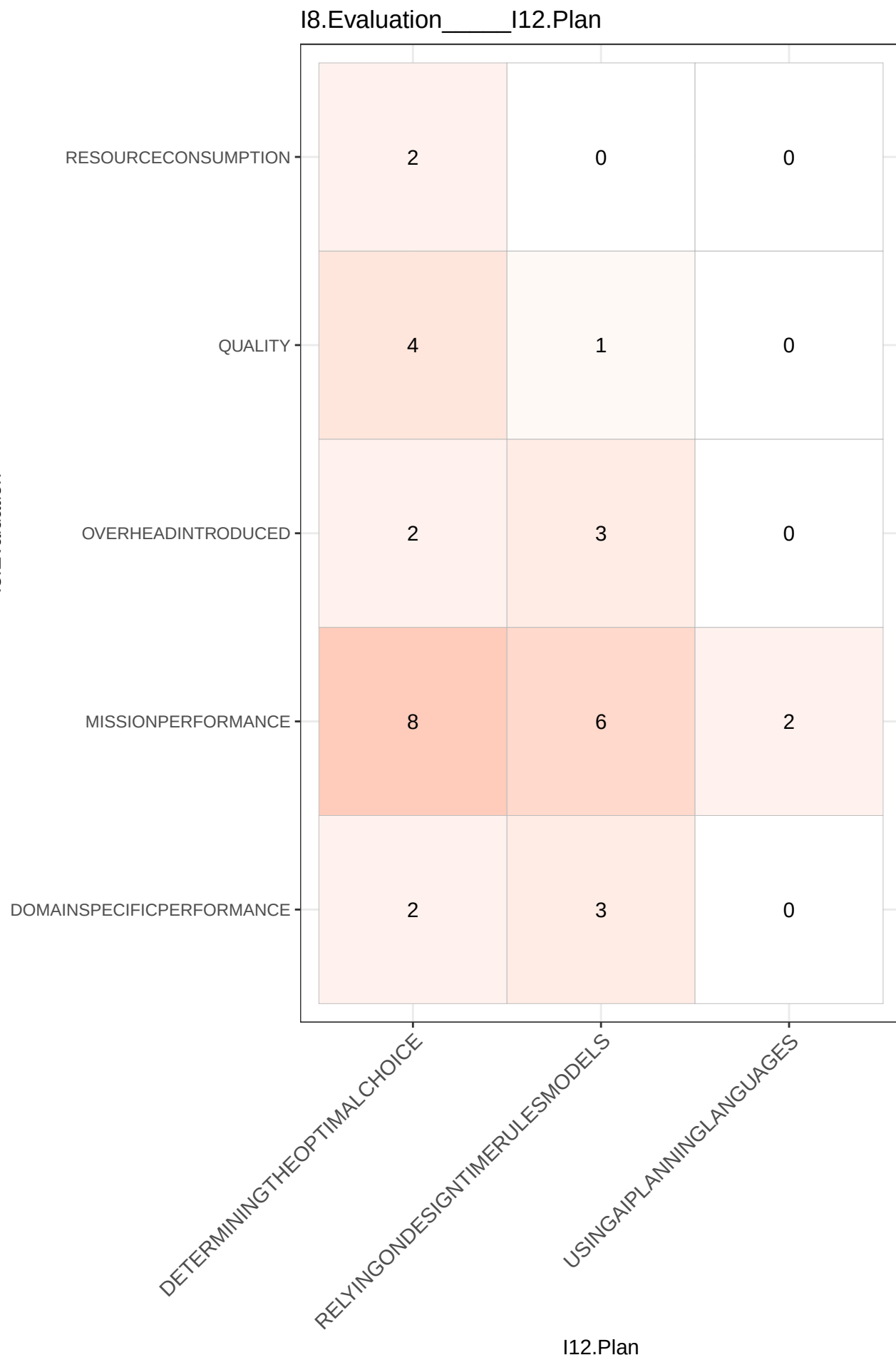


I8.Evaluation

I8.Evaluation_____I11.Analyze



I8.Evaluation



I8.Evaluation_____I13.Execute

I8.Evaluation

RESOURCECONSUMPTION

QUALITY

OVERHEADINTRODUCED

MISSIONPERFORMANCE

DOMAINSPECIFICPERFORMANCE

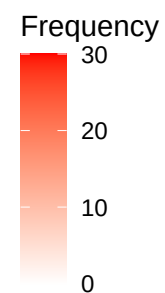
ADDITIONORREMOVALOFCOMPONENTS

CHANGEINRELATIONSHIPS BETWEENCOMPONENTS

COMPONENTREDEPLOYMENT

REPARAMETERIZATIONOFCOMPONENTS

I13.Execute



2

1

0

1

1

0

0

5

2

2

1

2

7

1

1

12

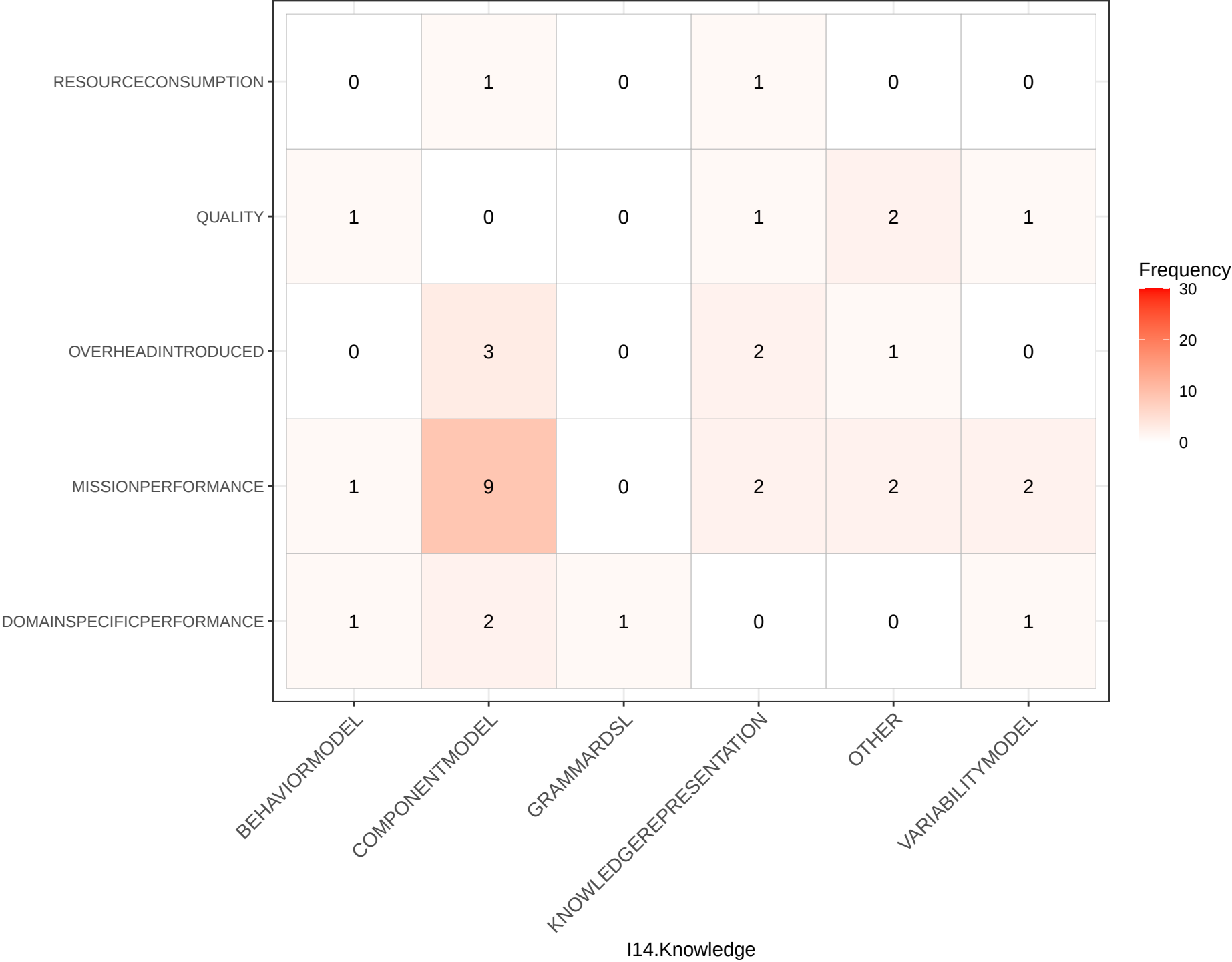
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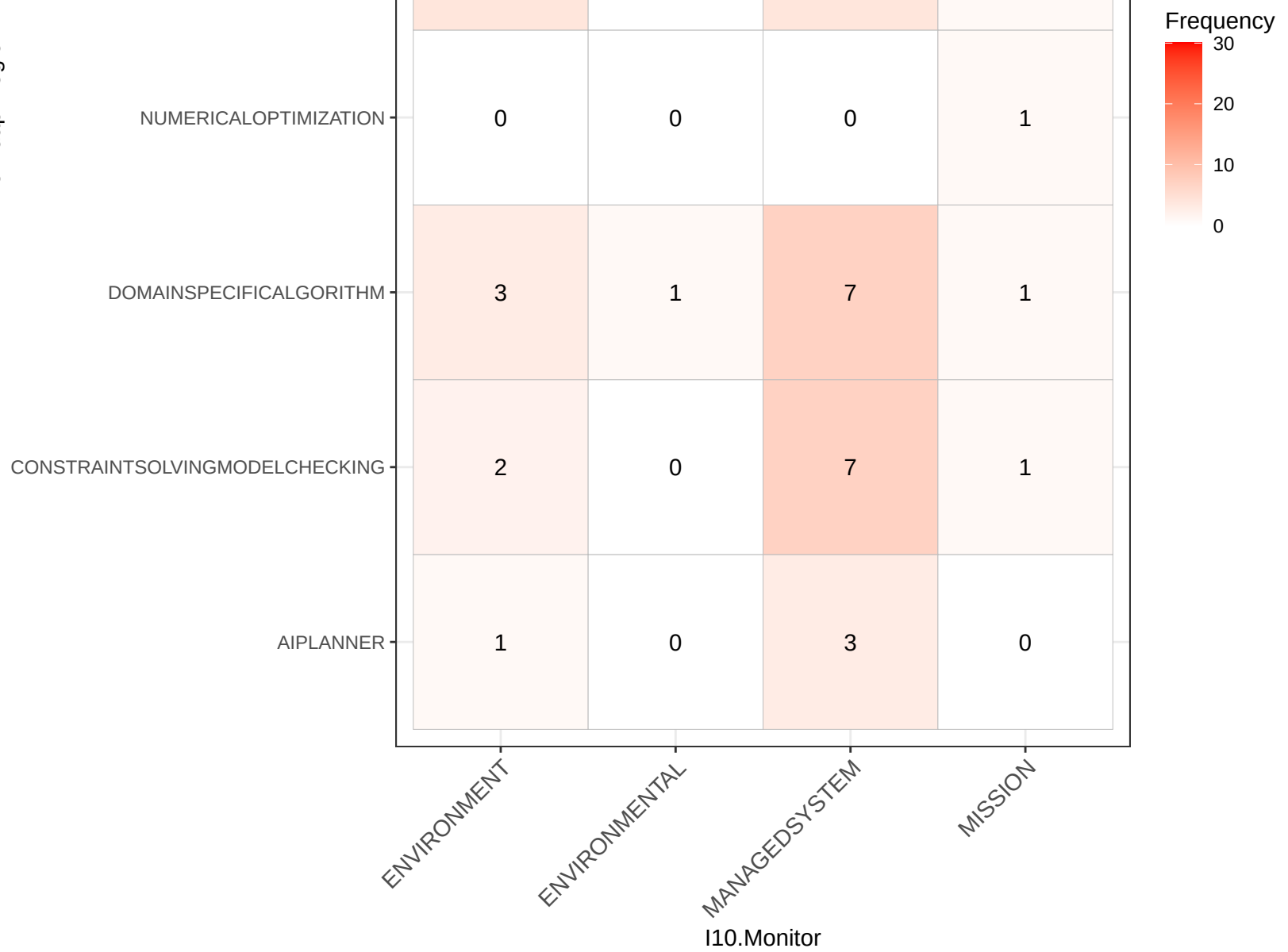
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I8.Evaluation_____I14.Knowledge

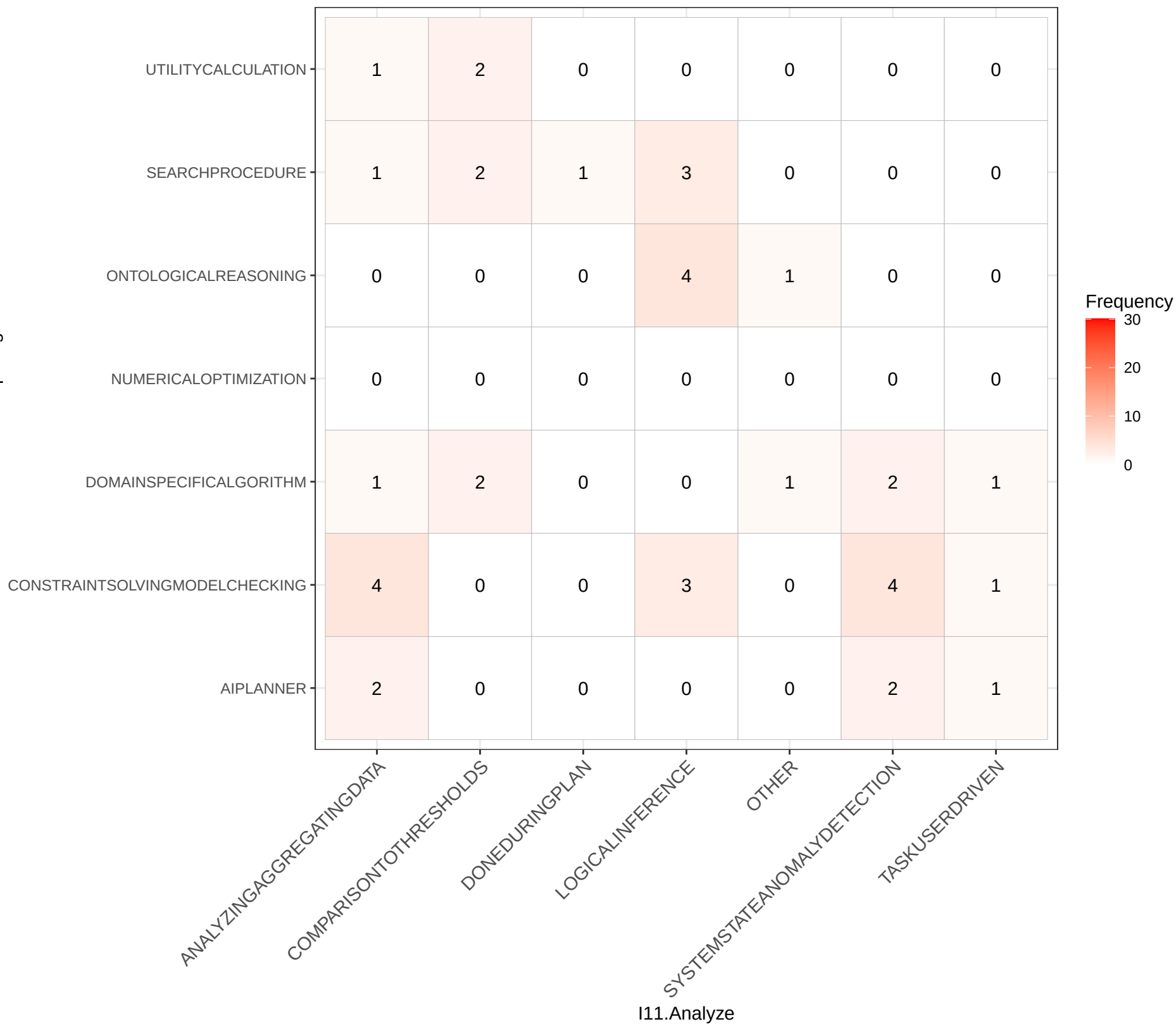


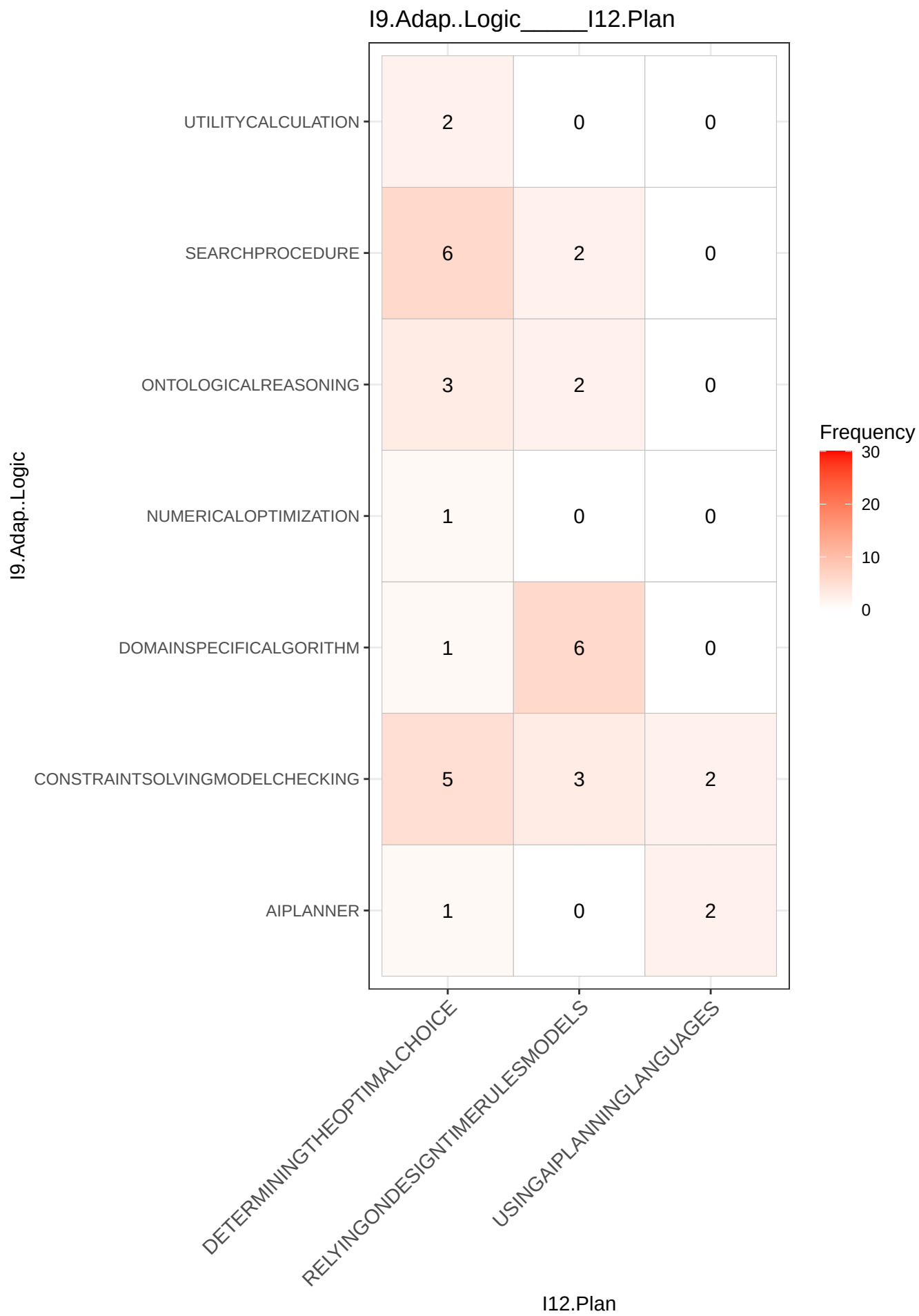
I9.Adap..Logic_____I10.Monitor

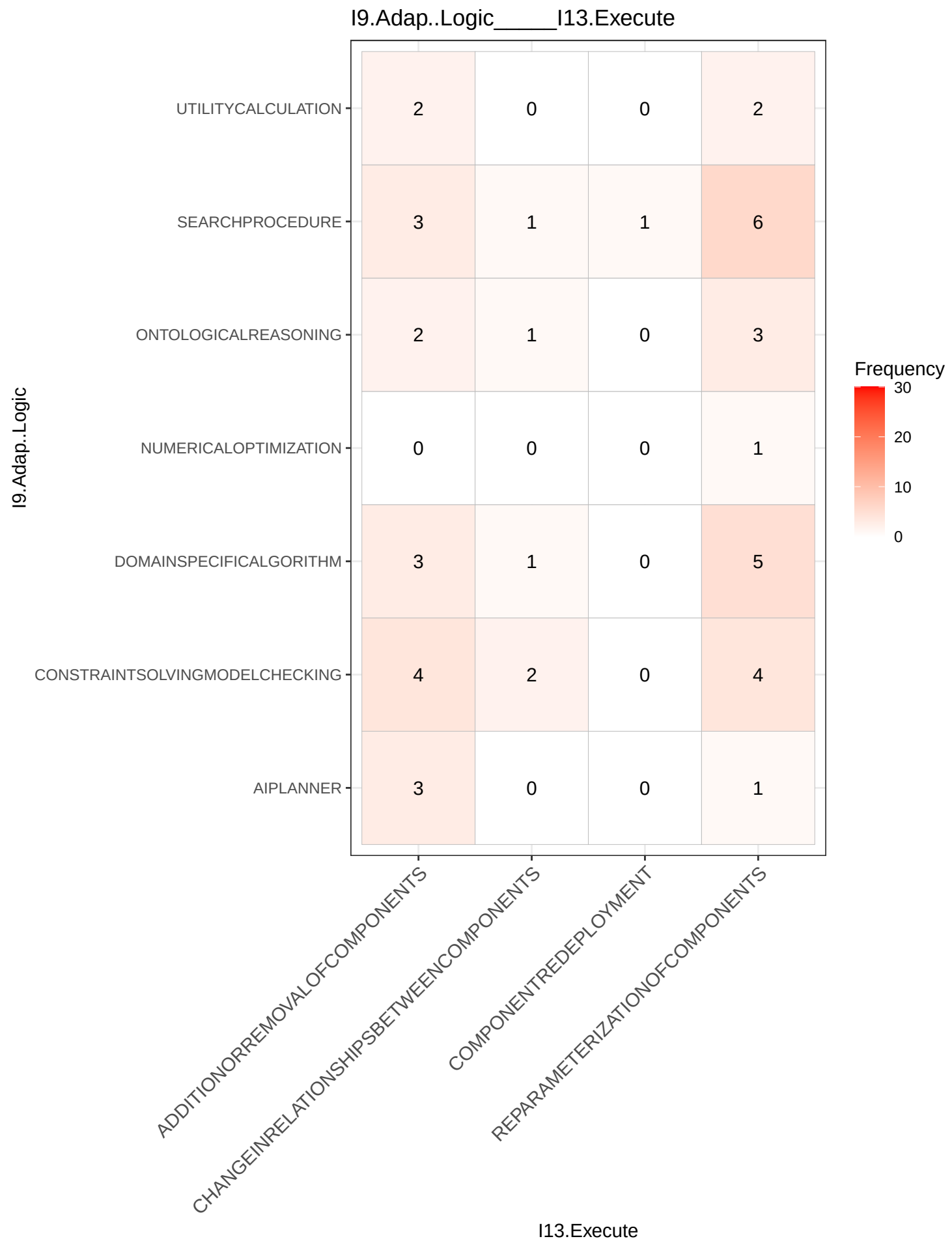
I9.Adap..Logic



I9.Adap..Logic_____I11.Analyze

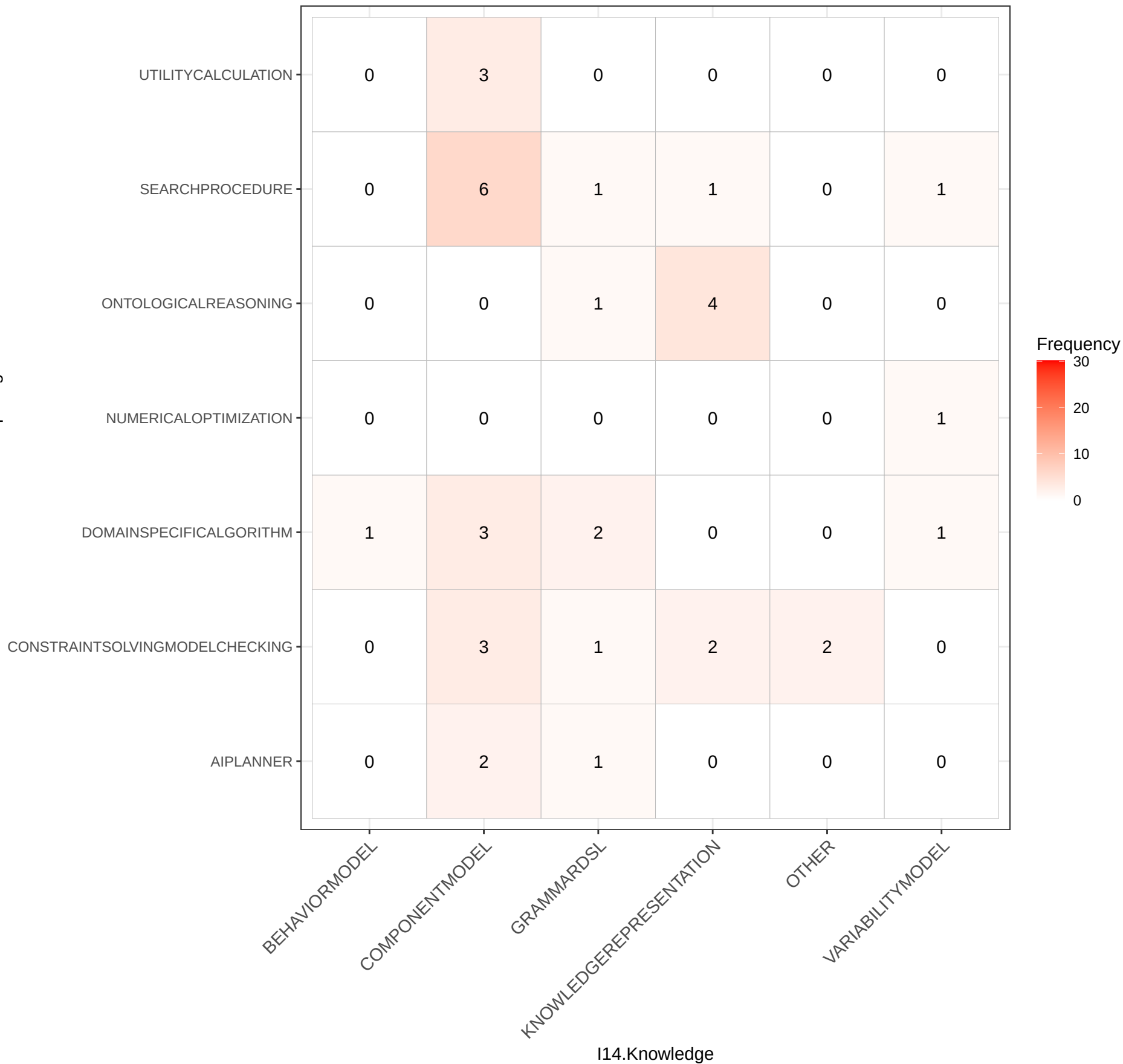






I9.Adap..Logic_____I14.Knowledge

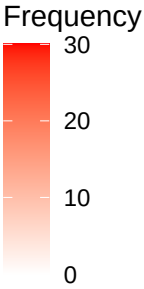
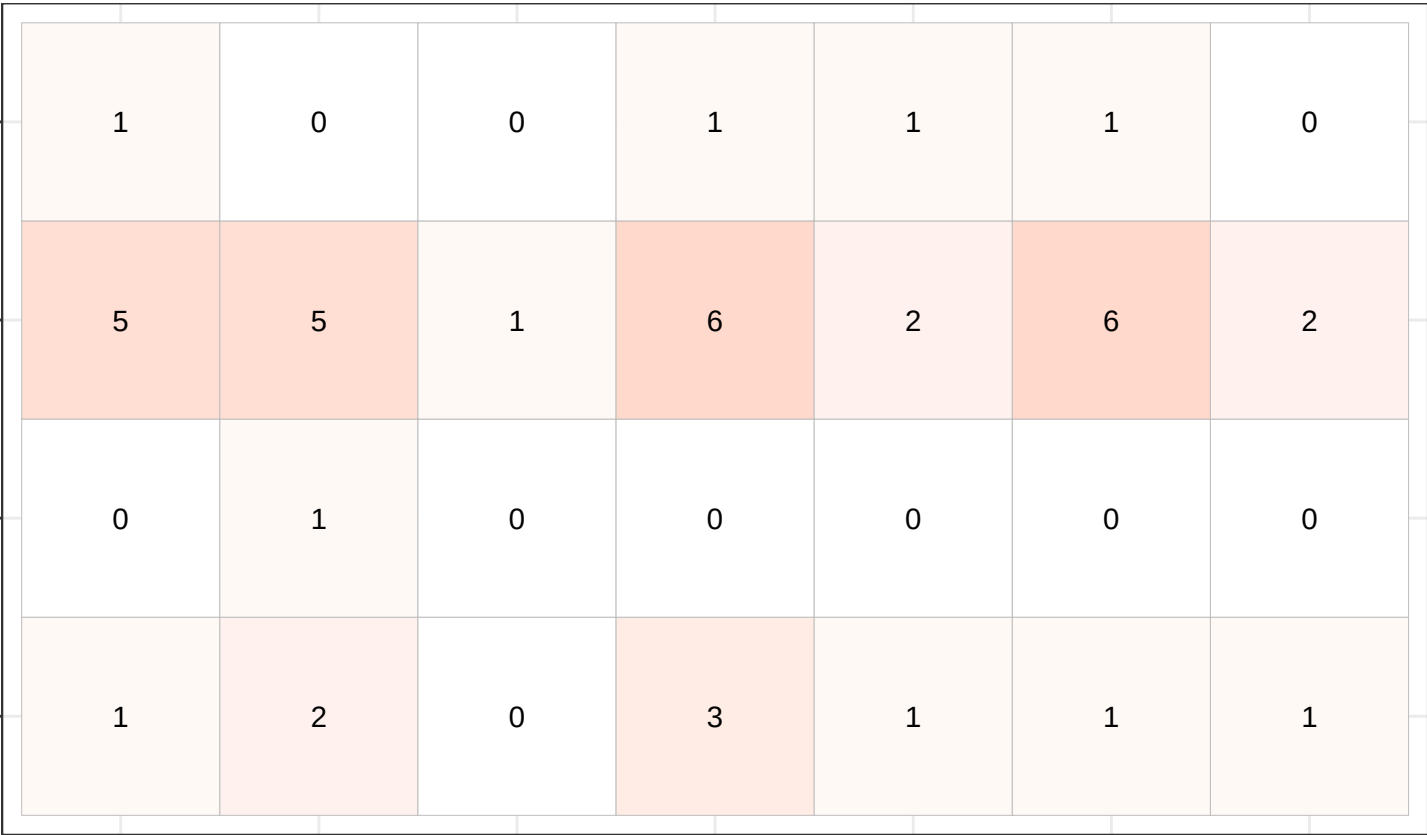
I9.Adap..Logic



I10.Monitor_____I11.Analyze

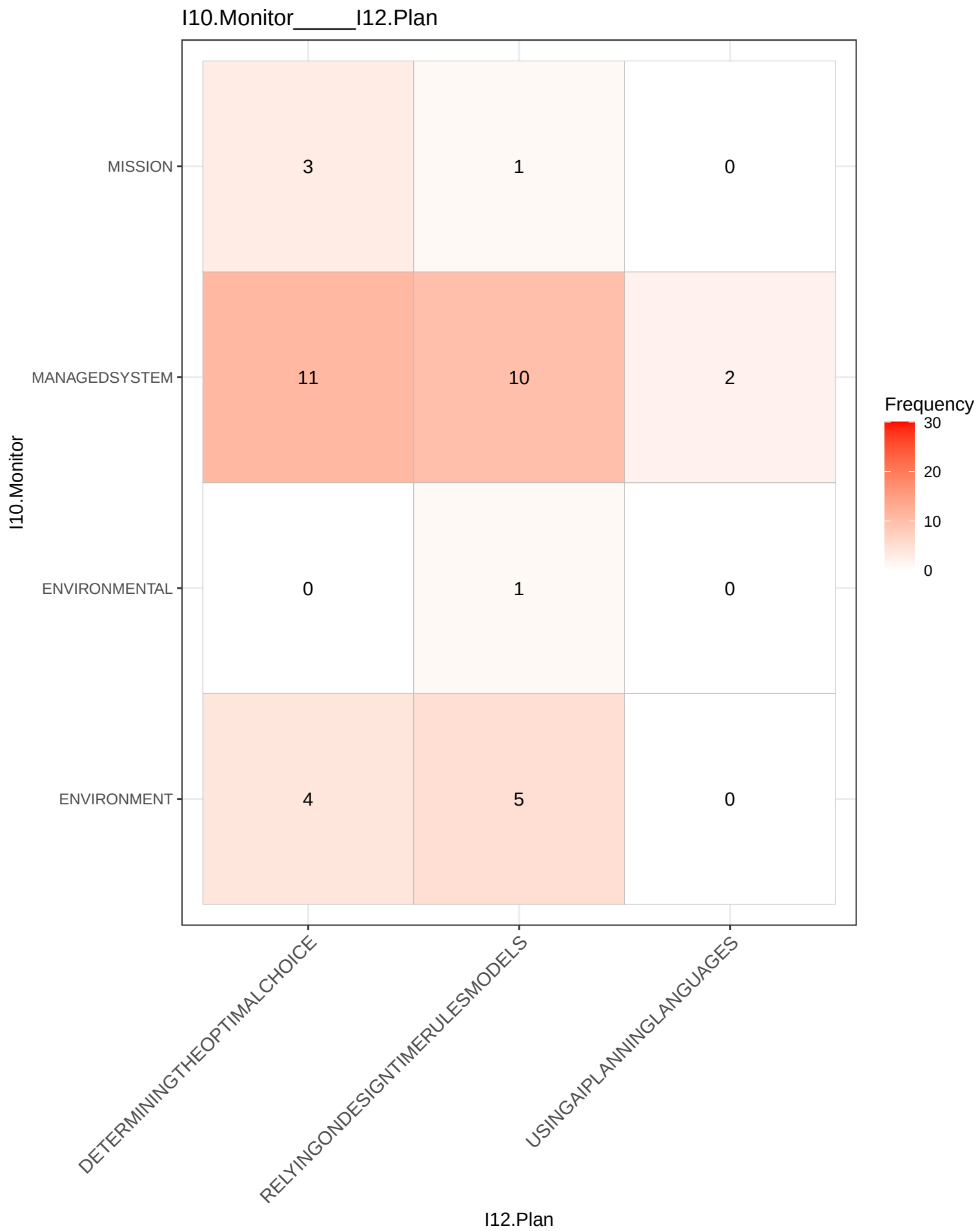
I10.Monitor

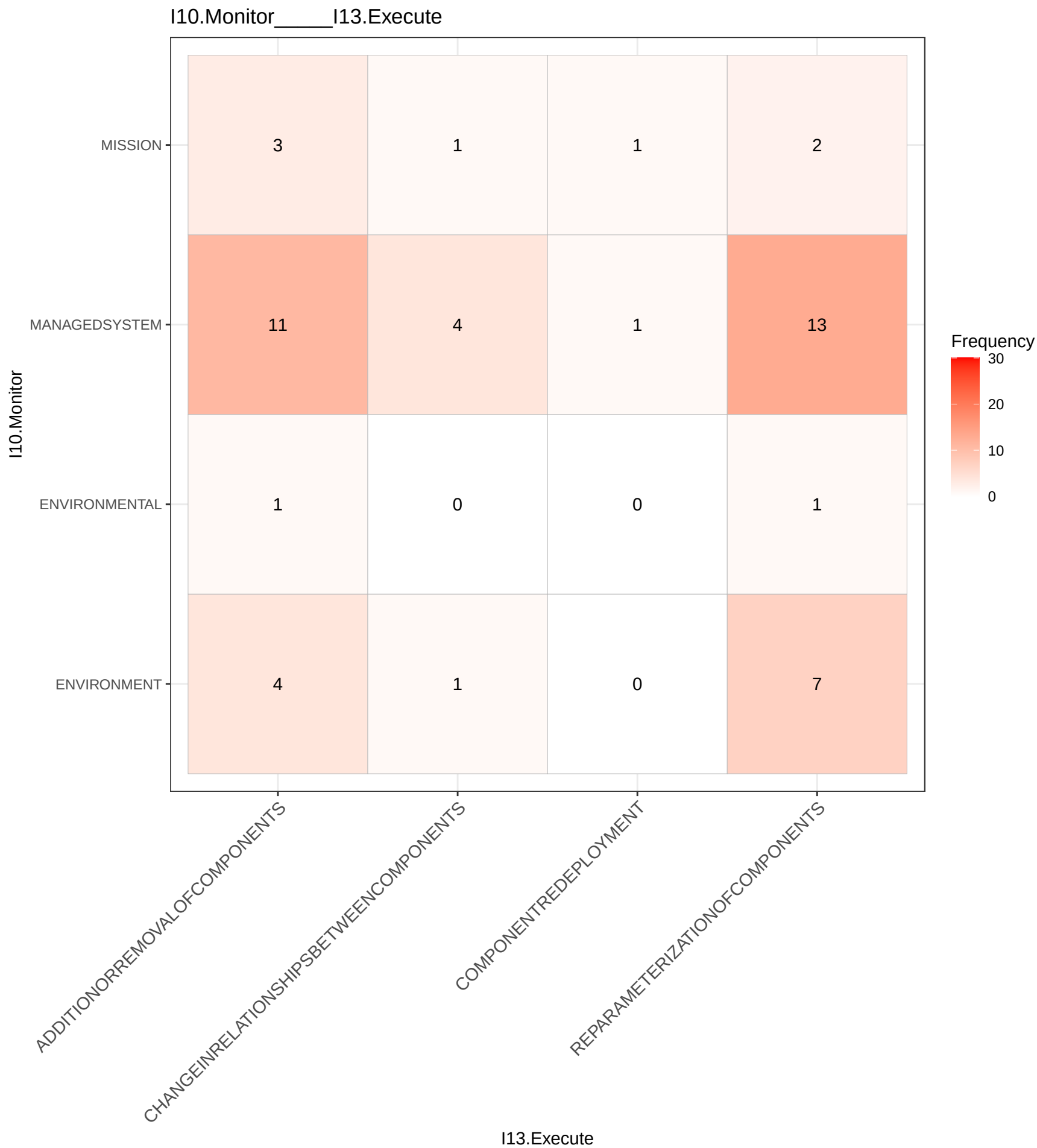
MISSION
MANAGEDSYSTEM
ENVIRONMENTAL
ENVIRONMENT



I11.Analyze

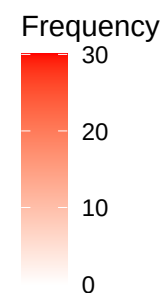
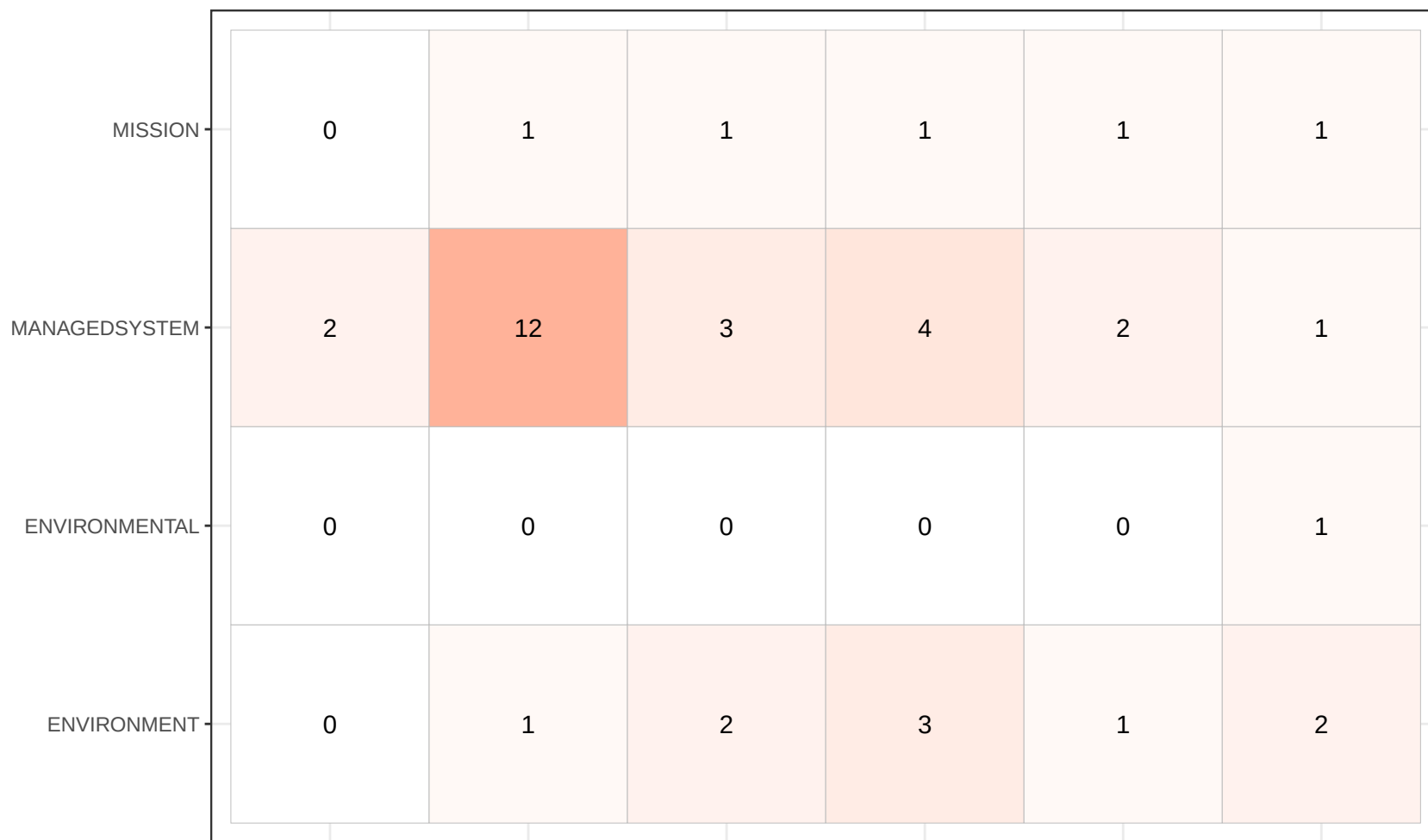
ANALYZINGAGGREGATINGDATA
COMPARISONTOHRESHOLDS
DONEDURINGPLAN
LOGICALINFERENCE
OTHER
SYSTEMSTATEANOMALYDETECTION
TASKUSERDRIVEN



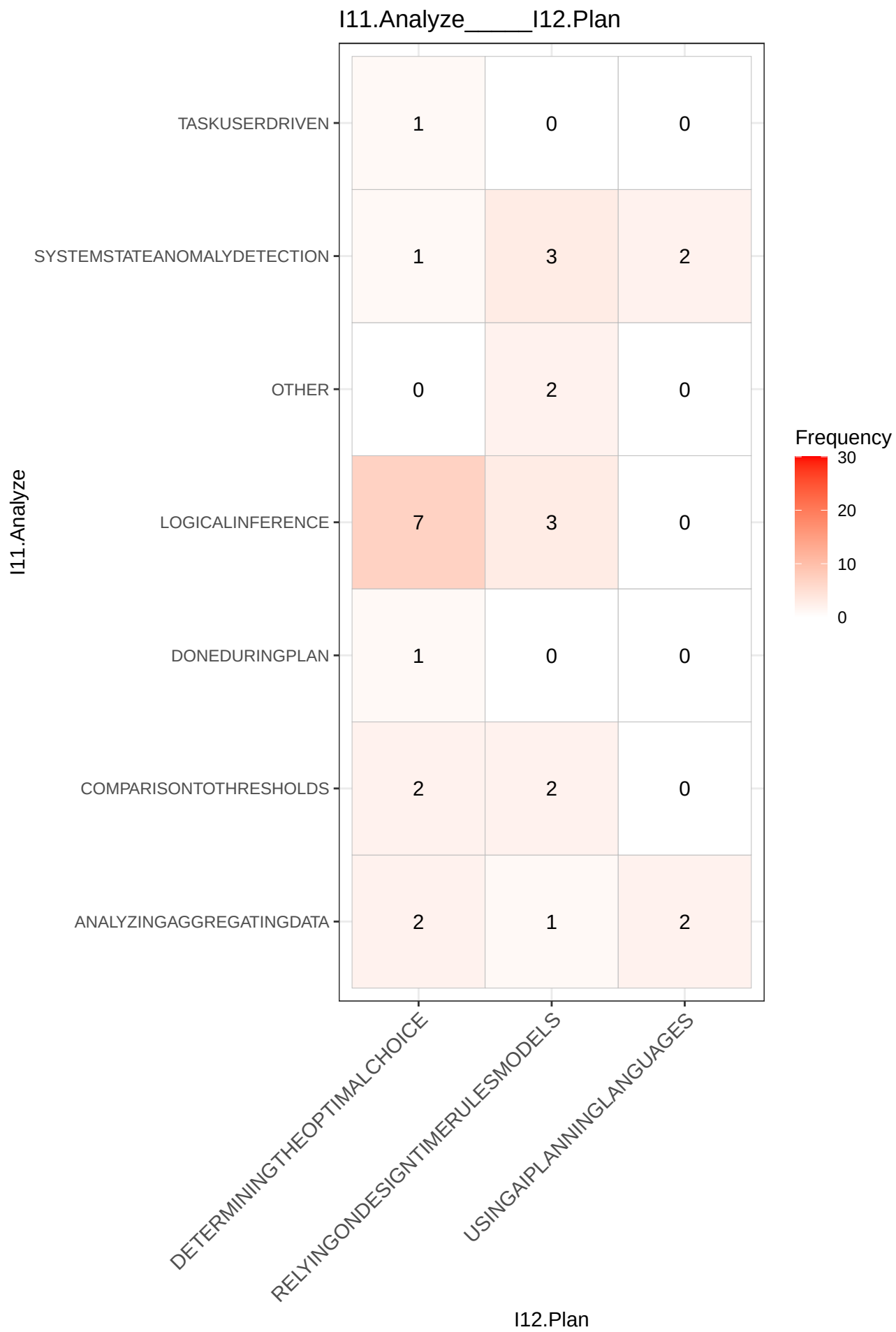


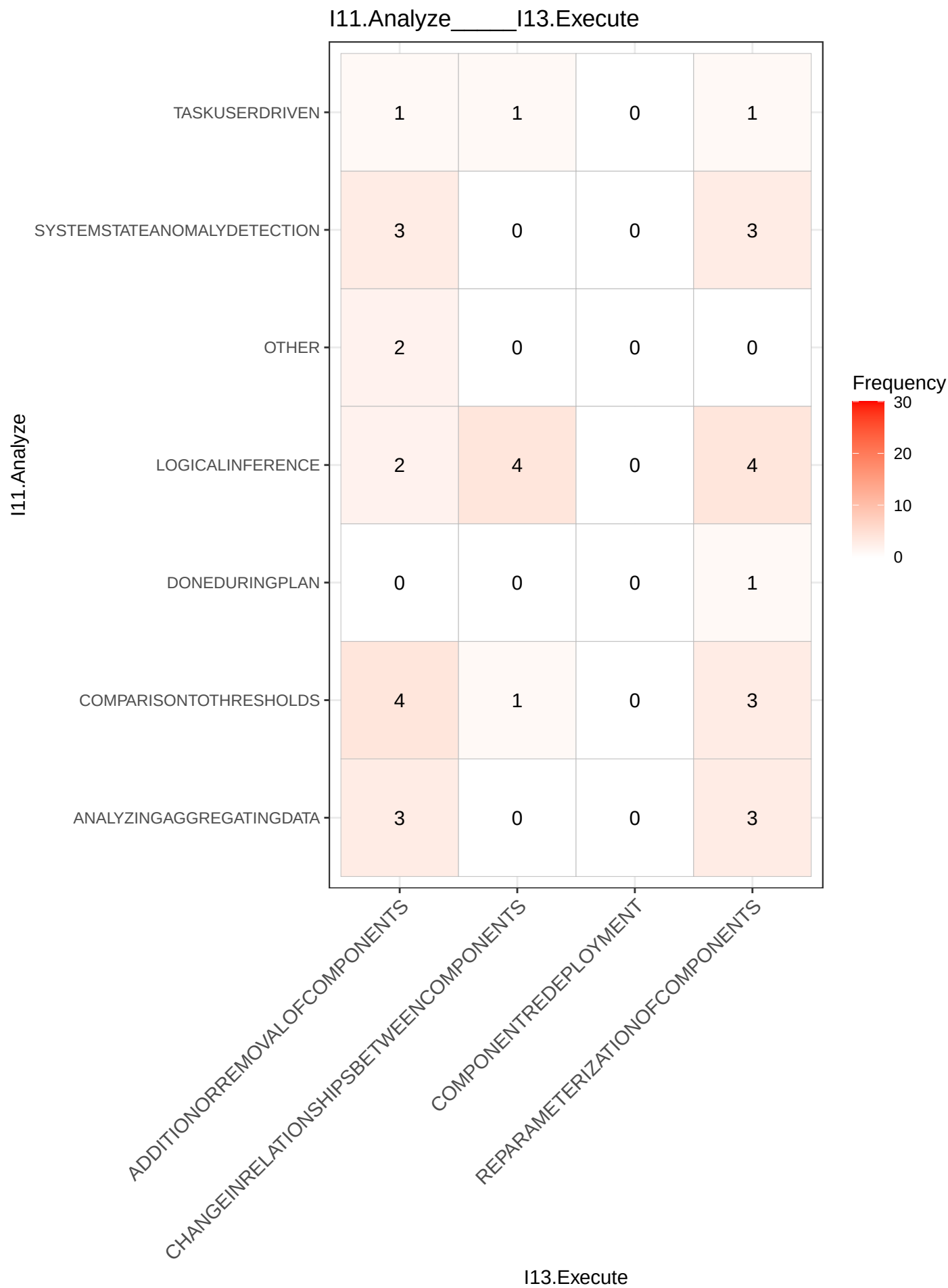
I10.Monitor_____I14.Knowledge

I10.Monitor



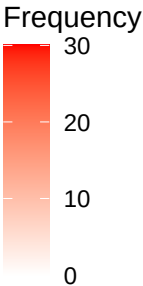
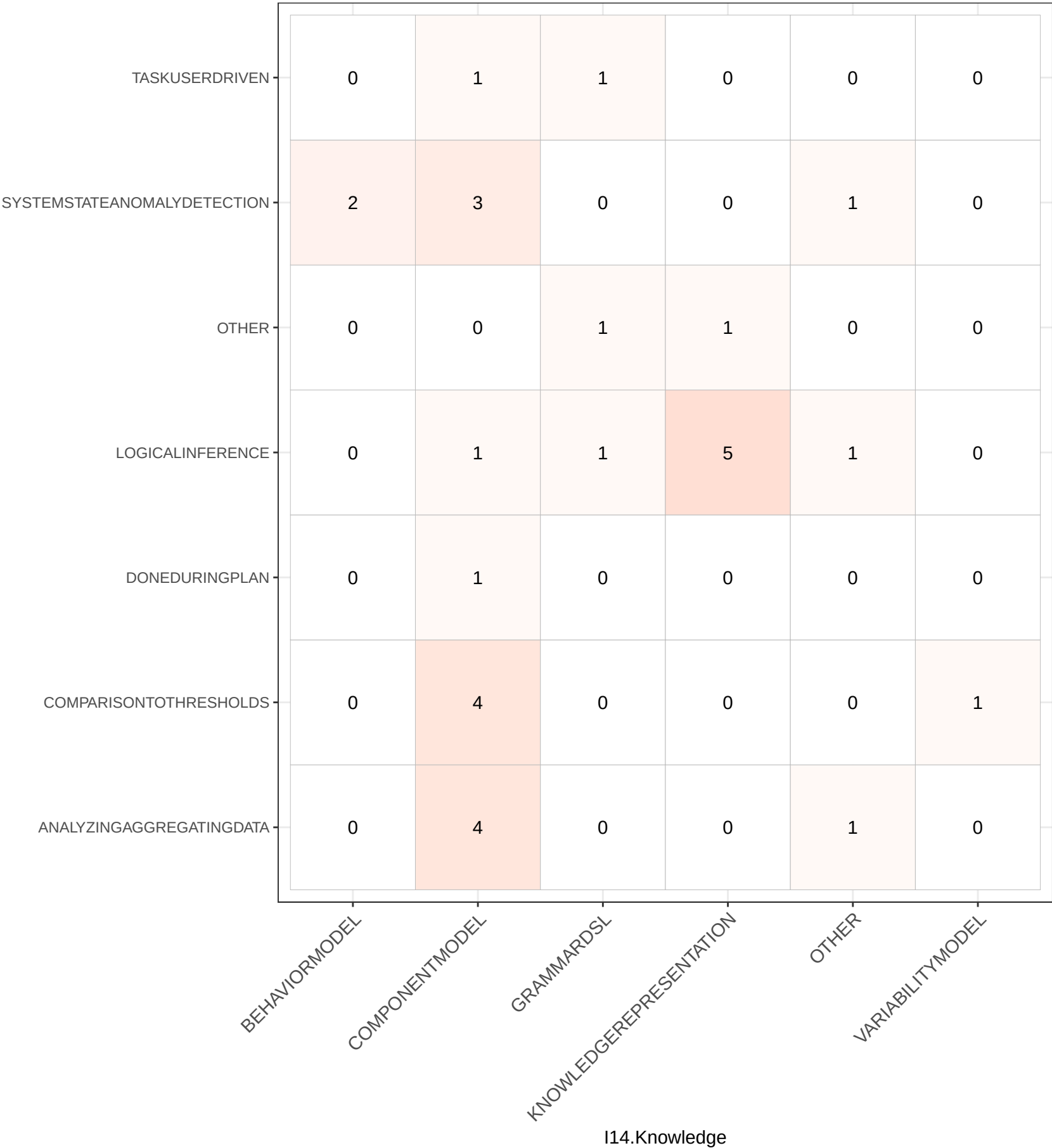
I14.Knowledge

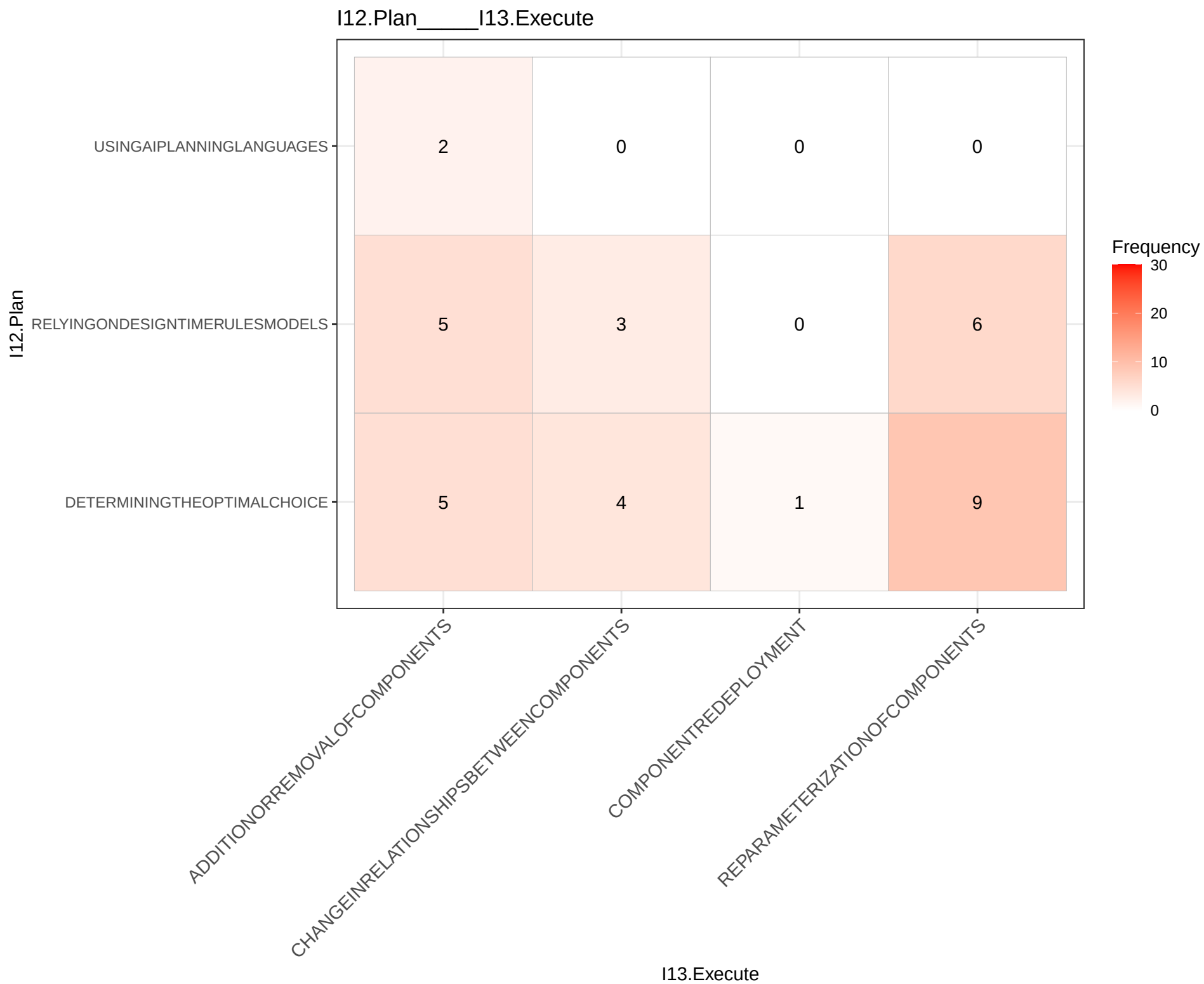


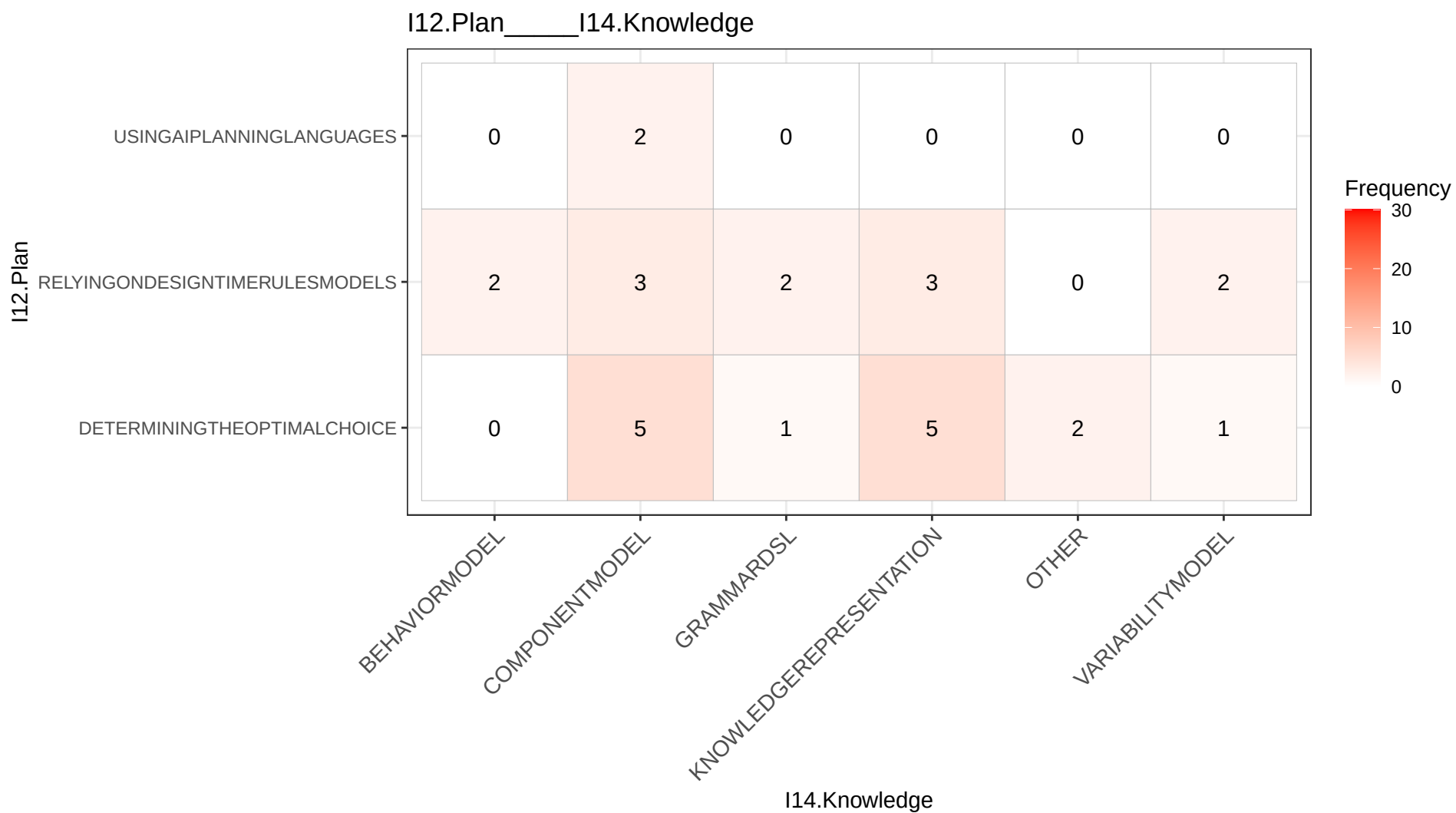


I11.Analyze_____I14.Knowledge

I11.Analyze







I13.Execute

